

Companion I: Ultraviolet Gate and Route 1 Mass-Gap Chain

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This companion contains the detailed proofs for the ultraviolet/public-scope gate and the Route 1 mass-gap chain. It is assembled directly from the canonical March 16, 2026 LaTeX source tree. The document follows the theorem-level organization used by the core paper and incorporates the same-scale Wilson return argument, the Route 1 mass-gap appendix, the compact-simple ultraviolet theorem, and the supporting source appendices used for those results.

GUIDE TO THE DETAILED ARGUMENTS

This companion records the detailed proofs used by the core paper for the compact-simple ultraviolet gate and the Route 1 mass-gap chain. Section III contains the same-scale Wilson return argument, Appendix F contains the main Route 1 mass-gap analysis, Appendix N contains the compact-simple ultraviolet theorem, and Appendices G–K contain the supporting one-plaquette and parameter-schedule material.

Reference orientation. Standard background used across the theorem-level presentation includes Osterwalder–Schrader reconstruction and lattice reflection positivity^{1–3}, Wilson lattice gauge theory and asymptotic freedom^{4–6}, gradient/Wilson flow and heat-kernel tools^{7–9}, cluster expansion and Dobrushin uniqueness^{10–12}, Bessel-ratio and Toeplitz/total-positivity tools used in the one-plaquette comparison arguments^{13–22}, lattice improvement and weak-coupling comparison inputs^{23–25}, the Yang–Mills problem statement²⁶, DHR superselection theory and local quantum physics^{27–31}, and standard real-algebraic geometry background used in auxiliary reductions³².

Cross-reference to detailed proofs.

Core label	Source label	Where proved	First downstream use in the core
III.3	F.317, F.318	Appendix F (QE1 one-shot entrance block)	Core Proposition III.3
IV.1	F.308	Appendix F (QE2 _{lat} closure block)	Core Proposition IV.1
IV.2	III.18	Section III (same-scale Wilson return block)	Core Theorem IV.2
IV.4	F.298	Appendix F (weak-window sufficiency certificate)	Core Theorem IV.4 and then the transport summary in Section VI
VI.1	F.211	Appendix F (bounded Wilson / dyadic OS limit)	Core Theorem VI.1 and then Core Theorem VI.2
VI.2	F.212	Appendix F (continuum-time OS upgrade block)	Core Theorem VI.2 and then Core Theorem VI.7
VI.4	F.213	Appendix F (density export)	Core Corollary VI.4 and then Core Lemma VI.5
VI.5	F.214	Appendix F (graph-core handoff)	Core Lemma VI.5 and then Core Theorem VI.7
VI.7	F.216	Appendix F (continuum vacuum-gap transport)	Core Theorem VI.7 and then Core Corollary VI.8
VI.8	F.5	Appendix F (continuum vacuum-gap corollary)	Core Corollary VI.8 and then Main Theorem I.1(B,D) and the endpoint results

I. ROUTE 1 MEASURE-TRANSPORT SUMMARY

This section isolates the exact measure-theoretic chain consumed later by the same-scale Wilson return theorem. Its role is theorem-level rather than expository: it records

only the conditional-expectation, factorization, and averaged transport nodes actually used downstream on the public Route 1 spine.

Proposition I.1 (Route 1 measure-transport summary). *For the fixed-scale Route 1 test class later consumed by Theorem III.18, the live measure-transport chain is exactly the following.*

- (i) *Theorem III.10 identifies the exact Wilson one-step fluctuation law as a conditional block-product law once the dyadic skeleton variables are frozen.*
- (ii) *Theorem III.11 iterates those conditional expectations and identifies the depth- m Wilson evaluation with the exact blocked conditional-expectation recursion.*
- (iii) *Proposition III.13 proves the one-step averaged transport estimate obtained by conditioning only on the local good-block event on one fixed support hull.*
- (iv) *Proposition III.17 states that the actual chart-truncated Lane C recursion changes only the one-step fluctuation integration operators by the truncation $T_{j+1} \mapsto T_{j+1}^{\text{good}}$, while Proposition III.14 together with Proposition III.8 propagates that one-step averaged estimate through the admissible finite logarithmic depth.*
- (v) *Consequently Theorem III.18 is the sole theorem-level Wilson-to-patched return later consumed on the public Route 1 spine.*

No later Route 1 theorem uses any stronger conditional-expectation, factorization, or measure-transport input than this summary.

Detailed proof. Section III below, in particular Theorem III.10, Theorem III.11, Proposition III.13, Proposition III.14, Proposition III.8, Proposition III.17, and Theorem III.18..

Dependency summary.

Inputs. Theorem III.10, Theorem III.11, Proposition III.13, Proposition III.14, Proposition III.8, Proposition III.17, and Theorem III.18

Imported standard background. none

Conclusion at this node. on the fixed-scale Route 1 test class, the Wilson-to-patched seam uses only the exact conditional-expectation / block-factorization input together with the averaged bad-event transport bound propagated through finite logarithmic depth; no stronger measure-side statement is hidden downstream

Used later in. Theorem III.18, Theorem F.301, and the QE3 transport route in Appendix F

III. REFLECTION POSITIVITY AND DE-PATCHING STABILITY

This section addresses the two non-negotiable referee concerns that arise whenever an “analytic patch” is used inside a constructive RG:

- (i) the *physical* lattice state must remain the unmodified Wilson Yang–Mills measure, hence reflection positivity is available *a priori*;
- (ii) any patched/truncated RG step must be shown to be an *analytic surrogate* that does not change the continuum limit of local/flowed observables.

Throughout this paper we implement the following discipline.

- The lattice states ω_a are always the *unpatched* Wilson Yang–Mills expectations (with gauge fixing used only for coordinates, never for the measure).
- “Patched” refers to a *chart-truncated one-step integration operator* used inside the dyadic RG to obtain uniform analytic constants. The patched operator is never used to *define* the physical lattice state.
- De-patching is proved at the level actually used later: after exact Wilson recursion is identified, the finite-depth difference between unpatched Wilson evaluations and patched/chart-truncated RG evaluations of local or localized flowed observables is $O(u^N)$ for all N on the AF ball; consequently the continuum limits computed with the patched RG agree with the physical Wilson limits.

Let $\Lambda \subset a\mathbb{Z}^4$ be a finite periodic lattice with a time-reflection hyperplane $\Pi = \{x_0 = 0\}$ and reflection map $\theta(x_0, \mathbf{x}) = (-x_0, \mathbf{x})$. Let $\mathcal{A}_+$ denote the algebra generated by bounded cylinder functionals of link variables supported in $\{x_0 > 0\}$. Let $\mathcal{A}_+^{\text{gi}} \subset \mathcal{A}_+$ denote the gauge-invariant subalgebra (the physical observable sector).

Remark III.1 (Gauge averaging and enlargement of the test algebra). Because the Wilson measure is gauge invariant, every bounded cylinder functional F has the same expectation as its local gauge-average $F^{\text{gi}} := \int F(U^g) dg$ (Haar product over gauge transformations on the finitely many sites that meet the support of F). Thus any pointwise convergence statement proved on $\mathcal{A}_+^{\text{gi}}$ automatically extends to $\mathcal{A}_+$. We use $\mathcal{A}_+$ as a technically convenient test algebra (e.g. for quasi-local positive-time representatives of flowed probes), while the reconstructed sharp local theory is generated by the gauge-invariant local sector.

Theorem III.2 (Reflection positivity of Wilson lattice Yang–Mills). *For compact gauge group G , the Wilson lattice Yang–Mills expectation $\omega_a(\cdot)$ is reflection positive: for every $F \in \mathcal{A}_+$,*

$$\omega_a(\Theta F \cdot F) \geq 0,$$

where Θ is the Osterwalder–Schrader reflection involution (time reflection composed with $*$).

Proof. See Appendix A (Theorem A.3); for background on OS reflection positivity and its lattice gauge theory implementation, see e.g.^{1–3}. $\square$

Reflection positivity implies a family of “chessboard” estimates that upgrade one-block bounds to bounds for translated/reflected tilings. We record the version needed for Lemma III.6.

Lemma III.3 (Chessboard estimate from reflection positivity). *Let ω be a reflection-positive, translation-invariant lattice state on a periodic lattice of even side length, with reflections across coordinate hyperplanes. Let E be an event (indicator functional) depending only on variables in a single fundamental reflection block B . Let $E^\#$ denote the event obtained by reflecting/translating E to all blocks in the standard chessboard tiling, and let $N_\#$ be the number of blocks in the tiling. Then*

$$\omega(E) \leq \left(\omega(E^\#) \right)^{1/N_\#}. \quad (1)$$

Proof. See Appendix A (Theorem A.3); for background on OS reflection positivity and its lattice gauge theory implementation, see e.g.^{1–3}. $\square$

Lemma III.4 (Reflected OS matrix elements are stable under tuned limits). *Let $\{\omega_n\}_{n \geq 1}$ be reflection-positive states on $(\mathcal{A}_+, \Theta)$, and let $\mathbb{D} \subset [0, \infty)$ contain 0. Assume that for every bounded cylinder observables $F, G \in \mathcal{A}_+$ and every $s \in \mathbb{D}$ for which $\tau_s G$ is defined on the n th lattice, the limit*

$$K_s(F, G) := \lim_{n \rightarrow \infty} \omega_n(\Theta F \tau_s G)$$

exists. Then K_0 is a positive semidefinite sesquilinear form on bounded positive-time cylinders: for every bounded cylinder $F \in \mathcal{A}_+$,

$$K_0(F, F) \geq 0.$$

If in addition $\omega(F) := \lim_{n \rightarrow \infty} \omega_n(F)$ exists for every bounded cylinder $F \in \mathcal{A}_+$, then ω is a state on the bounded positive-time algebra and, on bounded cylinders,

$$\omega(\Theta F G) := K_0(F, G).$$

Proof. For each bounded cylinder $F \in \mathcal{A}_+$, reflection positivity of ω_n gives

$$\omega_n(\Theta F F) \geq 0 \quad (n \geq 1).$$

Taking $n \rightarrow \infty$ yields $K_0(F, F) \geq 0$. Sesquilinearity of K_0 is inherited from the linearity of the finite- n functionals together with pointwise convergence on finite linear combinations of bounded cylinders. If the one-point limits $\omega(F)$ also exist, then ω is a state on the bounded positive-time algebra by pointwise limits of the states ω_n , and the last display simply records the notation induced by the limiting reflected pairings. $\square$

Lane C uses a chart-truncated one-step integration operator to obtain uniform convexity and cumulant bounds on a fixed exponential chart. We record the only de-patching estimate needed for the continuum limit.

Lemma III.5 (Conditioning/truncation error for bounded observables). *Let $(\Omega, \mathcal{F}, \mu)$ be a probability space and let $E \in \mathcal{F}$ satisfy $\mu(E) > 0$. Define the conditioned measure*

$$\mu^E(d\omega) := \frac{\mathbf{1}_E(\omega)}{\mu(E)} \mu(d\omega).$$

Then for any bounded measurable F with $\|F\|_\infty \leq M$,

$$\left| \mathbb{E}_\mu[F] - \mathbb{E}_{\mu^E}[F] \right| \leq 2M \mu(E^c).$$

Proof. Write $\mathbb{E}_{\mu^E}[F] = \mathbb{E}_\mu[F \mathbf{1}_E] / \mu(E)$. Then

$$\mathbb{E}_\mu[F] - \mathbb{E}_{\mu^E}[F] = \mathbb{E}_\mu[F \mathbf{1}_{E^c}] + \left(1 - \frac{1}{\mu(E)}\right) \mathbb{E}_\mu[F \mathbf{1}_E].$$

Taking absolute values and using $|1 - 1/\mu(E)| = \mu(E^c)/\mu(E)$ gives

$$\left| \mathbb{E}_\mu[F] - \mathbb{E}_{\mu^E}[F] \right| \leq M \mu(E^c) + \frac{\mu(E^c)}{\mu(E)} M \mu(E) = 2M \mu(E^c).$$

$\square$

How Lemma III.5 is used. We apply Lemma III.5 both to a single block fluctuation law and, when needed, to the product law on a finite union of blocks, with E equal to the corresponding conjunction of good-block events. This is the basic measure-theoretic input behind every one-step de-patching estimate below.

What the global tail lemma does – and does not do. Lemma III.6 is a finite-plaquette bound in the *physical Wilson state*. It is the correct large- β input for the later Wilson-to-chart-truncated transport argument once the exact recursion has been identified, because the bad-event contribution is then averaged back in the Wilson state. We do *not* use Lemma III.6 to assert a boundary-uniform pointwise estimate for the exact one-step conditional block factor.

Lemma III.6 (Local plaquette tail bound from chessboard + small-ball lower bound). *Fix $\varepsilon \in (0, 1]$ and consider the Wilson lattice gauge measure μ_β at coupling $\beta > 0$ on a periodic $4D$ torus with compact gauge group G . Let p be any plaquette and write*

$$\theta(U_p) := \text{dist}_G(U_p, \mathbf{1})$$

for the intrinsic distance on G induced by the fixed inner product on $\mathfrak{g}$. Let $d_G := \dim(G)$.

Then there exist constants

$$\varepsilon_{\text{tail}}(G) \in (0, 1], \quad c_{\text{tail}}(G) > 0, \quad C_{\text{tail}}(G) \geq 1,$$

explicit in principle and depending only on (G, ρ) and the fixed lattice dimension $d = 4$, such that for all $\varepsilon \in (0, \varepsilon_{\text{tail}}(G)]$,

$$\mu_\beta(\theta(U_p) > \varepsilon) \leq C_{\text{tail}}(G) \varepsilon^{-2d_G/3} \exp\left(-c_{\text{tail}}(G) \beta \varepsilon^2\right). \quad (2)$$

More generally, for any finite plaquette set Λ ,

$$\mu_\beta(G_\Lambda(\varepsilon)^c) \leq |\Lambda| C_{\text{tail}}(G) \varepsilon^{-2d_G/3} \exp\left(-c_{\text{tail}}(G) \beta \varepsilon^2\right), \quad (3)$$

where $G_\Lambda(\varepsilon) := \bigcap_{p \in \Lambda} \{\theta(U_p) \leq \varepsilon\}$. In particular, taking $\Lambda = \Lambda(\mathcal{Q})$ the plaquettes in a fixed unit coarse block $\mathcal{Q}$ yields the required bound on $\mu_\beta(G_{\mathcal{Q}}^c)$ used in Lemma III.5.

Proof. Let $E := \{\theta(U_{p_0}) > \varepsilon\}$ for a reference plaquette p_0 . Let $E^\#$ denote the full chessboard tiling of E (i.e. the event that *every* plaquette in the torus satisfies $\theta(U_p) > \varepsilon$). By the chessboard inequality (Lemma III.3),

$$\mu_\beta(E) \leq \mu_\beta(E^\#)^{1/|P|}, \quad (4)$$

where $|P|$ is the number of plaquettes.

Wilson action normalization. We write the Wilson action (associated to the fixed Wilson representation ρ of the main text) in the form

$$S_\beta(U) = \beta \sum_{p \in P} A_\rho(U_p), \quad A_\rho(U) := 1 - \frac{1}{\dim(\rho)} \Re \text{Tr}_\rho(U).$$

(Up to the additive constant $\beta|P|$ this is equivalent to the more common normalization $S_\beta = -\frac{\beta}{\dim(\rho)} \sum_p \Re \text{Tr}_\rho(U_p)$.)

Local quadratic control and a global tail lower bound. By Taylor expansion at $\mathbf{1}$ (Appendix L, Lemma L.3) there exist $\delta_0(G) \in (0, 1]$ and constants $0 < m_-(G) \leq m_+(G) < \infty$ such that for all $U = \exp(X)$ with $\|X\| \leq \delta_0(G)$,

$$m_-(G) \|X\|^2 \leq A_\rho(\exp X) \leq m_+(G) \|X\|^2.$$

Shrink $\delta_0(G)$ if needed so that $m_+(G) < 4m_-(G)$.

Since A_ρ is continuous and $A_\rho(U) = 0$ only at $U = \mathbf{1}$, the compact set $\{\theta(U) \geq \delta_0(G)\}$ has strictly positive minimum

$$a_{\text{out}}(G) := \min\{A_\rho(U) : \theta(U) \geq \delta_0(G)\} > 0.$$

Define

$$\varepsilon_{\text{tail}}(G) := \min\left\{\delta_0(G), \sqrt{\frac{a_{\text{out}}(G)}{m_-(G)}}\right\}.$$

Then for $\varepsilon \in (0, \varepsilon_{\text{tail}}(G)]$ and any U with $\theta(U) \geq \varepsilon$ we have the uniform lower bound $A_\rho(U) \geq m_-(G)\varepsilon^2$.

Bad partition function upper bound. On $E^\#$, every plaquette satisfies $\theta(U_p) > \varepsilon$, hence $A_\rho(U_p) \geq m_-(G)\varepsilon^2$. Therefore

$$Z_\beta(E^\#) := \int \mathbf{1}_{E^\#}(U) e^{-S_\beta(U)} dU \leq \exp\left(-\beta m_-(G) \varepsilon^2 |P|\right), \quad (5)$$

using that the total Haar volume of link configurations is 1.

Good partition function lower bound. For the partition function $Z_\beta = \int e^{-S_\beta(U)} dU$ we lower bound by restricting all links to a small ball around $\mathbf{1}$. Fix $\delta := \varepsilon/8$ and let $B(\delta) \subset G$ be the geodesic ball of radius δ around $\mathbf{1}$. Restrict the link variables to $\{U_e \in B(\delta) \forall e\}$. By the triangle inequality for the bi-invariant distance, each plaquette product satisfies $\theta(U_p) \leq 4\delta = \varepsilon/2$. For $\varepsilon \leq \varepsilon_{\text{tail}}(G) \leq \delta_0(G)$ we are within the quadratic regime, hence

$$A_\rho(U_p) \leq m_+(G) \theta(U_p)^2 \leq m_+(G) \varepsilon^2/4.$$

Therefore

$$Z_\beta \geq \mu_{\text{Haar}}(B(\delta))^{|E|} \exp\left(-\beta \frac{m_+(G)}{4} \varepsilon^2 |P|\right), \quad (6)$$

where $|E|$ is the number of edges.

Small-ball Haar lower bound. Since Haar measure has a smooth positive density in exponential coordinates near the identity, there exist constants $c_{\text{ball}}(G) > 0$ and (possibly smaller) $\delta_1(G) \in (0, 1]$ such that for all $\delta \in (0, \delta_1(G)]$,

$$\mu_{\text{Haar}}(B(\delta)) \geq c_{\text{ball}}(G) \delta^{d_G}. \quad (7)$$

Combine. Combining (5)–(6) gives

$$\mu_\beta(E^\#) = \frac{Z_\beta(E^\#)}{Z_\beta} \leq \mu_{\text{Haar}}(B(\delta))^{-|E|} \exp\left(-\beta(m_-(G) - m_+(G)/4)\varepsilon^2 |P|\right).$$

On a periodic $4D$ torus one has $|E|/|P| = 4/6 = 2/3$, hence

$$\mu_\beta(E^\#)^{1/|P|} \leq \mu_{\text{Haar}}(B(\delta))^{-2/3} \exp\left(-\beta(m_-(G) - m_+(G)/4)\varepsilon^2\right).$$

Using (7) with $\delta = \varepsilon/8 \leq \delta_1(G)$ yields

$$\mu_\beta(E^\#)^{1/|P|} \leq c_{\text{ball}}(G)^{-2/3} 8^{2d_G/3} \varepsilon^{-2d_G/3} \exp\left(-\beta(m_-(G) - m_+(G)/4)\varepsilon^2\right).$$

Together with (4) this gives (2) with

$$c_{\text{tail}}(G) := m_-(G) - m_+(G)/4, \quad C_{\text{tail}}(G) := c_{\text{ball}}(G)^{-2/3} 8^{2d_G/3}.$$

The finite-set bound (3) follows by a union bound over $p \in \Lambda$ and translation invariance. $\square$

Lemma III.7 (Flow localization reduces de-patching to finite-support observables). *Fix a flow time $t \in (0, t_*]$ and a bounded flowed observable $F \in \mathcal{A}_{\text{flow}}$. For $R \geq 1$, let $F^{(R)}$ denote the truncation obtained by freezing the initial field outside the union of radius- $R\sqrt{t}$ neighborhoods around the finitely many spacetime regions that enter the definition of F . Then $F^{(R)}$ is a bounded cylinder observable depending only on finitely many unflowed links, contained in at most $C_F(1+R)^4$ unit coarse blocks for a constant $C_F < \infty$ depending only on the geometry of F . Moreover, there exist constants $C_F^{\text{flow}}, c_F^{\text{flow}} > 0$ such that*

$$\|F - F^{(R)}\|_\infty \leq C_F^{\text{flow}} t^{-5} e^{-c_F^{\text{flow}} R^2}.$$

Consequently, for every integer $N \geq 1$ there exists a choice

$$R = R_N(a, F) \asymp \sqrt{\log(u(a)^{-1})}$$

along the AF window such that

$$2\|F - F^{(R)}\|_\infty \leq C_N(F) u(a)^N.$$

Hence finite-support control of the Wilson-to-patched difference reduces to the same comparison for the truncated observable $F^{(R)}$.

Proof. By definition, $\mathcal{A}_{\text{flow}}$ is generated by finite products of smeared flowed curvature composites. Apply the exponential-tail flow quasi-locality theorem for the flowed probe family (Companion II, Appendix B; source Theorem 5.12 in the Lane A proof home) to each flowed factor entering F and combine the resulting bounds with the boundedness of the remaining factors. Because only finitely many insertions and smearing regions occur in F , the truncation depends on finitely many underlying links and occupies at most $C_F(1+R)^4$ unit coarse blocks. The displayed exponential bound follows, after adjusting constants, from the same exponential-tail flow quasi-locality estimate. Finally, along the AF window, choosing R proportional to $\sqrt{\log(u(a)^{-1})}$ makes the exponential tail dominate any fixed power $u(a)^N$. $\square$

Proposition III.8 (Finite-depth surrogate stability). *Fix any quantity later used in the paper that is obtained by composing at most m one-step RG maps, blockwise reblocking/extraction operators, and a final evaluation map that is uniformly Lipschitz on the AF ball. Assume $m \leq C_{\text{depth}} \log(1/u)$ on the asymptotically free window, as happens for the entrance depth and the observation depths used later, and assume that the intermediate observables encountered in this finite-depth architecture remain supported in at most polynomially many unit coarse blocks in the depth parameter (for flowed observables, after the localization step of Lemma III.7, polynomially many also in the localization radius R). Then for every integer $N \geq 1$ there exist $u_0(N) > 0$ and $C_N < \infty$ such that replacing every one-step map in the chosen RG architecture by its patched/chart-truncated counterpart changes the resulting quantity by at most*

$$C_N u^N$$

uniformly in the cutoff a whenever $u \leq u_0(N)$.

Proof. After Theorem III.11, every exact one-step map appearing in the Wilson evaluation is the conditional expectation operator T_{j+1} , while the patched architecture replaces it by

the corresponding good-event truncation T_{j+1}^{good} with the same deterministic post-processing. Write the finite-depth difference as a telescoping sum over the at most m substituted one-step kernels. At the r -th telescoping term one encounters a bounded intermediate observable H_r supported in a finite union Λ_r of unit coarse blocks; by hypothesis, $|\Lambda_r|$ is bounded by a fixed polynomial in r (and, for flowed observables after localization, also polynomially in R).

Because H_r depends only on the blocks in Λ_r , truncating all other block factors is immaterial. Hence Proposition III.13 gives

$$\left| \omega_a(T_{j+1} H_r) - \omega_a(T_{j+1}^{\text{good}} H_r) \right| \leq C(H_r) |\Lambda_r| \exp\left(-c_{\text{tail}}(G) \beta \varepsilon^2\right).$$

The deterministic reblocking/extraction operators and the final evaluation map are uniformly Lipschitz on the AF ball by Theorem 4.25 and Theorem 4.86, so the telescoping sum is bounded by a constant times

$$\left(\sum_{r=0}^{m-1} (1+r)^\nu \right) (1+R)^{\nu_F} \exp\left(-c_{\text{tail}}(G) \beta \varepsilon^2\right)$$

for some fixed exponents ν, ν_F depending only on the chosen quantity. Since $m \leq C_{\text{depth}} \log(1/u)$ and R may be taken $O(\sqrt{\log(1/u)})$, the prefactor is at worst polylogarithmic in $1/u$. Along the AF window one has $\beta \asymp u^{-1}$, so the stretched exponential $\exp(-c/u)$ dominates every polylogarithm. Shrinking $u_0(N)$ if necessary therefore yields an overall bound $C_N u^N$. $\square$

Theorem III.9 (Support propagation for the actual finite-depth Lane C recursion). *Fix a bounded cylinder observable H_0 supported in a finite union Λ_0 of unit coarse blocks at the input scale, and let H_r denote the intermediate observable obtained after r steps of the actual chart-truncated Lane C recursion of Definition 4.59, before the final scalar evaluation. Then there is a canonical dyadic parent-hull operation Par on finite block families such that*

$$\text{supp}(H_r) \subseteq \text{Par}^r(\Lambda_0) \quad (r \geq 0),$$

and

$$|\text{Par}^r(\Lambda_0)| \leq |\Lambda_0|.$$

In particular, for bounded cylinder observables the number of blocks touched by the actual finite-depth Lane C recursion is uniformly bounded in the depth. If $H_0 = F^{(R)}$ is the localization of a flowed observable from Lemma III.7, then

$$|\text{Par}^r(\text{supp } F^{(R)})| \leq C_F (1+R)^4 \quad (r \geq 0),$$

so the finite-depth locality hypothesis of Proposition III.8 is verified concretely for the actual Lane C recursion.

Proof. For one actual Lane C step at scale j , Proposition 4.63 gives the explicit operator identity

$$\text{supp}(\mathcal{R}_j^{\text{good}} H) \subseteq \text{Par}(\Lambda)$$

whenever H is supported in a finite family Λ of j -scale blocks. This is the concrete operator-by-operator support map for the actual chart-truncated fluctuation operator, deterministic reblocking/rescaling, and exact extraction/reparametrization.

Iterating this one-step inclusion starting from H_0 supported in Λ_0 gives

$$\text{supp}(H_r) \subseteq \text{Par}^r(\Lambda_0) \quad (r \geq 0).$$

Because each half-open child block has a unique dyadic parent block, distinct child blocks may merge but no parent block can split; hence

$$|\text{Par}^r(\Lambda_0)| \leq |\Lambda_0|.$$

If $H_0 = F^{(R)}$ is the localization of a flowed observable from Lemma III.7, that lemma gives

$$|\text{supp } F^{(R)}| \leq C_F(1 + R)^4$$

at the input scale, and the same parent-hull monotonicity preserves the bound at every later depth:

$$|\text{Par}^r(\text{supp } F^{(R)})| \leq C_F(1 + R)^4.$$

This is exactly the support-growth statement required in Proposition III.8 for the actual Lane C recursion. $\square$

Architectural clarification. The finite-range / product structure invoked later in Lane C is not an ad hoc surrogate ingredient. For the Wilson theory itself, once the block-skeleton links are held fixed, the complementary interior-link law factorizes exactly over $(j + 1)$ -blocks. Thus the only auxiliary modification used in the analytic RG is the *local good-block chart truncation*; the earlier Wilson-to-product issue is resolved by the exact one-step factorization theorem below.

Theorem III.10 (Exact one-step Wilson block factorization). *Fix a dyadic scale j and partition the j -lattice into half-open $(j+1)$ -blocks of side 2. Let Σ_{j+1} denote the block skeleton, i.e. the set of j -links not strictly contained in the interior of a unique half-open $(j+1)$ -block, and write U_Σ for the link configuration on Σ_{j+1} . Let U_{int} denote the complementary interior-link variables, and let $\mathcal{B}_{j+1}$ denote the family of half-open $(j+1)$ -blocks and, for each block $\mathcal{Q} \in \mathcal{B}_{j+1}$, let $U_{\mathcal{Q}}$ be the collection of interior links belonging to $\mathcal{Q}$. Then the Wilson lattice gauge measure has the conditional product form*

$$\mu_j^{\text{W}}(dU_{\text{int}} \mid U_\Sigma) = \bigotimes_{\mathcal{Q} \in \mathcal{B}_{j+1}} \mu_{j+1, \mathcal{Q}}^{\text{W}}(dU_{\mathcal{Q}} \mid U_\Sigma|_{\partial \mathcal{Q}}), \quad (8)$$

where each block factor depends only on the skeleton data adjacent to $\mathcal{Q}$. After imposing the canonical tree gauge inside $\mathcal{Q}$ with boundary data frozen, each block factor is represented by the finite-dimensional local density used later in Definitions 4.26 and 4.27. In particular, the exact Wilson one-step fluctuation law satisfies the finite-range input (FC1) with $R_{\text{fr}} = 0$ conditional on the skeleton variables, and the chart-truncated law obtained by imposing the good-block event in each block preserves the same block-product structure.

Proof. Write the fine plaquette with anchor x and directions (μ, ν) as $p(x; \mu, \nu)$, and let $\mathcal{Q}(x)$ denote the unique half-open $(j+1)$ -block containing the anchor x . Because the blocks are half-open, every non-skeleton link of $p(x; \mu, \nu)$ is strictly contained in the same owner block $\mathcal{Q}(x)$; any remaining links of the plaquette lie on the block skeleton. Hence no plaquette term can involve interior links from two distinct blocks.

The Wilson measure is the product Haar measure on links multiplied by the local Boltzmann factor

$$\exp\left\{-\beta \sum_p A_\rho(U_p)\right\}, \quad A_\rho(U) := 1 - \frac{1}{\dim(\rho)} \Re \text{Tr}_\rho(U).$$

After U_Σ is held fixed, the conditional action is therefore a sum of block-local terms,

$$S_j(U_{\text{int}}; U_\Sigma) = \sum_{\mathcal{Q} \in \mathcal{B}_{j+1}} S_{\mathcal{Q}}(U_{\mathcal{Q}}; U_\Sigma|_{\partial \mathcal{Q}}),$$

and the product Haar prior on U_{int} already factorizes over blocks. This gives (8). Inside each block one may then impose the canonical tree gauge with the boundary links frozen; by Haar bi-invariance on the finite block, the resulting change of variables is blockwise and produces exactly the finite-dimensional local coordinate description used in Lane C. Finally,

the good-block event is measurable in the corresponding block factor, so truncating by the blockwise indicators preserves the product decomposition. $\square$

Theorem III.11 (Nested Wilson conditioning gives the exact dyadic recursion). *Fix a starting dyadic scale j_0 and a finite depth $m \geq 1$. For each $r \in \{0, \dots, m\}$, let Σ_{j_0+r} be the dyadic skeleton obtained after r repetitions of the half-open 2-blocking rule, and let $\mathcal{F}_{j_0+r}$ be the sigma-algebra generated by the links on Σ_{j_0+r} . Then the skeleton sigma-algebras are nested,*

$$\mathcal{F}_{j_0+m} \subset \dots \subset \mathcal{F}_{j_0+1} \subset \mathcal{F}_{j_0},$$

and for each step $r = 0, \dots, m-1$ the exact one-step conditional expectation onto $\mathcal{F}_{j_0+r+1}$ is given by the block-product kernel of Theorem III.10 applied on the current $(j_0 + r)$ -scale lattice. Consequently, for every bounded observable H measurable with respect to the scale- j_0 field,

$$\omega_a(H) = \omega_a\left(T_{j_0+m} \cdots T_{j_0+1} H\right), \quad (9)$$

where

$$(T_{j+1}G)(U_{\Sigma_{j+1}}) := \int G(U_{\Sigma_{j+1}}, U_{\text{int}}) \mu_j^W(dU_{\text{int}} \mid U_{\Sigma_{j+1}})$$

is the exact Wilson one-step conditional expectation operator. After the deterministic re-blocking/rescaling maps and the exact coordinate re-expression from (62), this is precisely the untruncated Lane C recursion generated by the maps $\mathcal{R}_j$. Hence the chart-truncated recursion differs from the exact Wilson recursion only by replacing each T_{j+1} with its blockwise good-event truncation T_{j+1}^{good} .

Proof. Apply Theorem III.10 at the first step $j_0 \rightarrow j_0 + 1$. Because each $(j+2)$ -block is a union of half-open $(j+1)$ -blocks, the corresponding dyadic skeletons are nested, so the sigma-algebras $\mathcal{F}_{j_0+r}$ form a decreasing filtration. For each r , Theorem III.10 on the current $(j_0 + r)$ -scale lattice identifies T_{j_0+r+1} with the conditional expectation $\mathbb{E}_{\omega_a}[\cdot \mid \mathcal{F}_{j_0+r+1}]$. Iterating conditional expectation therefore gives the tower identity (9).

The reblocking/rescaling maps used in Lane C are deterministic measurable maps of the surviving skeleton variables, so they do not change the underlying conditional-expectation identity. Likewise, the passage from the exact blocked density $\mathcal{Z}_{j+1}$ to the coordinates (u', K') is the exact re-expression described after (62); it introduces no approximation. Therefore the untruncated one-step Lane C map $\mathcal{R}_j$ is exactly the blocked Wilson recursion written in

(u, K) coordinates. By Definition 4.59, the patched recursion keeps the same deterministic post-processing and changes only the one-step conditional expectation $T_{j+1} \mapsto T_{j+1}^{\text{good}}$. $\square$

Corollary III.12 (Terminal blockwise conditional product and disjoint-block independence). *In the setting of Theorem III.11, fix the terminal depth m and the corresponding family $\mathcal{B}_{j_0+m}$ of half-open terminal blocks. Conditional on the terminal skeleton sigma-algebra $\mathcal{F}_{j_0+m}$, the fine Wilson fluctuation law factorizes over the terminal blocks:*

$$\mu_{j_0}^W(dU_{\text{int}} \mid \mathcal{F}_{j_0+m}) = \bigotimes_{\mathcal{Q} \in \mathcal{B}_{j_0+m}} \mu_{\mathcal{Q}}^W(dU_{\mathcal{Q}} \mid U_{\Sigma_{j_0+m}}|_{\partial \mathcal{Q}}), \quad (10)$$

where $U_{\mathcal{Q}}$ denotes the fine links strictly interior to the terminal block $\mathcal{Q}$. Consequently, if $H_1, \dots, H_r$ are bounded observables such that each H_i is measurable with respect to the fine links contained in a family $\mathcal{S}_i \subset \mathcal{B}_{j_0+m}$ and the families $\mathcal{S}_1, \dots, \mathcal{S}_r$ are pairwise disjoint, then

$$\mathbb{E}_{\omega_a} \left[\prod_{i=1}^r H_i \mid \mathcal{F}_{j_0+m} \right] = \prod_{i=1}^r \mathbb{E}_{\omega_a} [H_i \mid \mathcal{F}_{j_0+m}]. \quad (11)$$

Proof. Theorem III.10 gives exact conditional block-factorization at one dyadic step. Iterating that factorization through the nested filtration of Theorem III.11 yields an exact product decomposition of the fine interior-link law over the terminal half-open blocks once the terminal skeleton field is frozen, which is (10). If each H_i depends only on fine links from a block family $\mathcal{S}_i$ and the families are pairwise disjoint, then under the product law (10) the random variables $H_1, \dots, H_r$ are conditionally independent given $\mathcal{F}_{j_0+m}$. This is exactly (11). $\square$

Proposition III.13 (Averaged one-step bad-event transport). *Fix a dyadic step $j \rightarrow j+1$ and let H be a bounded observable depending only on the fluctuation variables in a finite union Λ of unit coarse $(j+1)$ -blocks. Let*

$$G_{\Lambda}(\varepsilon) := \bigcap_{\mathcal{Q} \subset \Lambda} G_{\mathcal{Q}}(\varepsilon)$$

denote the conjunction of the good-block events on those blocks, where $G_{\mathcal{Q}}(\varepsilon)$ is the good-block event of Definition 4.47, and let $T_{j+1}^{\text{good}, \Lambda}$ be the one-step operator obtained from T_{j+1} by conditioning only the block factors for $\mathcal{Q} \subset \Lambda$ on $G_{\mathcal{Q}}(\varepsilon)$ (leaving all other blocks exact). Then

$$\left| \omega_a(T_{j+1}H) - \omega_a(T_{j+1}^{\text{good}, \Lambda}H) \right| \leq 2\|H\|_{\infty} \omega_a(G_{\Lambda}(\varepsilon)^c). \quad (12)$$

Consequently, if $P(\Lambda)$ denotes the finite plaquette set contained in the blocks of Λ and $d_G := \dim(G)$, then

$$\left| \omega_a(T_{j+1}H) - \omega_a(T_{j+1}^{\text{good}, \Lambda}H) \right| \leq 2\|H\|_{\infty} |P(\Lambda)| C_{\text{tail}}(G) \varepsilon^{-2d_G/3} \exp\left(-c_{\text{tail}}(G)\beta\varepsilon^2\right). \quad (13)$$

In particular, on the AF window, for every integer $N \geq 1$ there exists $u_0(N, \Lambda, H) > 0$ such that

$$u \leq u_0(N, \Lambda, H) \implies \left| \omega_a(T_{j+1}H) - \omega_a(T_{j+1}^{\text{good}, \Lambda}H) \right| \leq C_N(\Lambda, H) u^N.$$

Proof. Condition on the skeleton sigma-algebra $\mathcal{F}_{j+1}$. By Theorem III.10, the exact conditional fluctuation law factorizes over the $(j+1)$ -blocks, and on the blocks of Λ the patched law is obtained by conditioning that product law on $G_\Lambda(\varepsilon)$. For each fixed skeleton configuration, apply Lemma III.5 to the product fluctuation law on the blocks of Λ with event $G_\Lambda(\varepsilon)$ to obtain

$$\left| T_{j+1}H - T_{j+1}^{\text{good}, \Lambda}H \right| \leq 2\|H\|_\infty \mu_j^W(G_\Lambda(\varepsilon)^c \mid \mathcal{F}_{j+1}).$$

Now average in the Wilson state and use the tower property:

$$\omega_a \left[\mu_j^W(G_\Lambda(\varepsilon)^c \mid \mathcal{F}_{j+1}) \right] = \omega_a(G_\Lambda(\varepsilon)^c).$$

This proves (12). The explicit estimate (13) follows by applying Lemma III.6 to the finite plaquette set $P(\Lambda)$ and using a union bound. Since along the AF window one has $\beta \asymp u^{-1}$, the exponential term is superpolynomial in u^{-1} and therefore dominates every fixed power u^N . $\square$

Proposition III.14 (Local transport closure for the actual chart-truncated Lane C recursion).

Fix a bounded cylinder observable H supported in a finite union Λ of unit coarse blocks. Let $E_a^W(H)$ denote its evaluation by the exact finite-depth Wilson recursion of Theorem III.11, and let $E_a^{\text{good}}(H)$ denote its evaluation by the actual chart-truncated Lane C recursion, i.e. by the dyadic RG architecture of Definition 4.59 in which each one-step exact conditional expectation T_{j+1} is replaced by the chart-truncated operator T_{j+1}^{good} while all deterministic reblocking/extraction maps are left unchanged. Then for every integer $N \geq 1$ there exist $u_0(N) > 0$ and $C_N(H) < \infty$ such that

$$\left| E_a^W(H) - E_a^{\text{good}}(H) \right| \leq C_N(H) u(a)^N$$

uniformly for a in the AF window. Moreover, if $F \in \mathcal{A}_{\text{low}}$ is fixed and $F^{(R)}$ is the cylinder truncation from Lemma III.7, then for every N one may choose $R = R_N(a, F) \asymp \sqrt{\log(u(a)^{-1})}$ so that

$$\left| \omega_a(F) - \omega_a^{\text{RG}}(F) \right| \leq \left| E_a^W(F^{(R)}) - E_a^{\text{good}}(F^{(R)}) \right| + C_N(F) u(a)^N,$$

and the first term is again $O(u(a)^N)$.

Proof. For bounded cylinder observables, Theorem III.11 identifies the exact Wilson evaluation with a finite-depth composition of the operators T_{j+1} followed by the deterministic reblocking/extraction maps of Lane C. By Definition 4.59 and the construction in Section 4, the actual analytic Lane C recursion uses those same deterministic maps and changes only the one-step fluctuation integration operators by the substitutions $T_{j+1} \mapsto T_{j+1}^{\text{good}}$. Thus the one-step difference between the physical Wilson evaluation and the actual chart-truncated Lane C evaluation is exactly the local good-block truncation treated in Proposition III.13. Theorem III.9 verifies that the actual chart-truncated Lane C recursion keeps the intermediate supports inside a uniformly bounded parent hull (and, for flowed observables after localization, inside at most $C_F(1+R)^4$ blocks). Therefore Proposition III.8 applies to the concrete Lane C recursion and propagates the one-step averaged estimate through the full finite-depth evolution, yielding the stated $O(u^N)$ bound. For flowed observables, apply Lemma III.7 and choose the localization radius so that the deterministic tail from that lemma is itself $O(u(a)^N)$; then apply the cylinder estimate to $F^{(R)}$. $\square$

Remark III.15 (Audit reference for the Wilson-versus-patched seam). Theorem III.18 below is the single theorem-level location where the patched Lane C estimates are returned to the physical Wilson theory. Theorem III.10, Theorem III.11, Proposition III.13, and Proposition III.17 serve only as proof inputs to that fixed-scale closure statement.

Remark III.16 (Phase-V Route 1 measure pipeline ledger). The exact Wilson/Lane C measure-side chain used later is only the following theorem-level packet:

- (1) Theorem III.10 identifies the exact Wilson one-step fluctuation law simultaneously as conditional expectation onto the next skeleton sigma-algebra and as an exact block-product law over the half-open coarse blocks.
- (2) Theorem III.11 iterates those exact one-step conditional expectations through the dyadic filtration and shows that the untruncated Lane C recursion is exactly the blocked Wilson recursion written in (u, K) coordinates.
- (3) Proposition III.13 quantifies the only local measure change later used: conditioning the relevant block factors on the good-block event over one fixed support hull.
- (4) Proposition III.17 states that the actual chart-truncated Lane C recursion makes precisely that truncation-only substitution and no other measure-side modification.

- (5) Theorem III.18 packages items (1)–(4) into the same-scale Wilson-to-patched return used later by the public Route 1 spine.

No later Route 1 theorem reopens the measure-side comparison below Theorem III.18; downstream uses consume only that packaged theorem-facing return.

Proposition III.17 (The actual Lane C chart truncation matches the transport template). *Fix the dyadic RG architecture of Lane C, with chart-truncated one-step map $\mathcal{R}_j^{\text{good}}$ from Definition 4.59. Then, relative to the exact Wilson recursion identified in Theorem III.11, the only modification made by the Lane C analytic recursion is the blockwise good-event truncation*

$$T_{j+1} \mapsto T_{j+1}^{\text{good}}$$

at each one-step fluctuation integration, followed by the same deterministic reblocking/rescaling and exact extraction maps used in the untruncated recursion. Consequently the actual Lane C chart-truncated recursion satisfies the hypotheses of Proposition III.14.

Proof. Theorem III.11 identifies the untruncated Lane C recursion with the exact finite-depth blocked Wilson recursion written in (u, K) coordinates. Definition 4.59 then shows that the chart-truncated map keeps all deterministic post-processing unchanged and alters only the one-step fluctuation expectation by conditioning each block factor on the corresponding good-block event. Because the good-block event factorizes blockwise (Definition 4.55) and the exact one-step Wilson law already has block-product structure by Theorem III.10, this is precisely the truncation-only setting abstracted in Proposition III.14. $\square$

Theorem III.18 (Same-scale Wilson-to-patched transport). *Fix a support budget $M_{\text{supp}} \in \mathbb{N}$ and a depth-growth constant $C_{\text{depth}} > 0$. Use the fixed-scale test norm*

$$\|H\|_{\text{test}} := \|H\|_{L^\infty}$$

on bounded cylinder observables whose input-scale support is contained in a union of at most M_{supp} unit coarse blocks. For such an observable H , let $E_{a,m}^{\text{W}}(H)$ denote the exact Wilson evaluation obtained by the m -step recursion of Theorem III.11, and let $E_{a,m}^{\text{good}}(H)$ denote the parallel m -step chart-truncated evaluation obtained from Definition 4.59. On the finite-volume/admissible-boundary-condition class used later in Route 1, the same notation denotes the identical local construction with the outer boundary data frozen. Then for every

integer $N \geq 1$ there exist $u_0(N, C_{\text{depth}}, M_{\text{supp}}) > 0$ and $C_N(C_{\text{depth}}, M_{\text{supp}}) < \infty$, depending only on the depth-growth constant, the support budget, and the group ledger, such that for all $u(a) \leq u_0$ and all dyadic depths satisfying

$$1 \leq m \leq C_{\text{depth}} \log(1/u(a)),$$

the following hold uniformly in the cutoff a , uniformly in the admissible depths m , and uniformly over the Route 1 finite-volume/admissible-boundary-condition class.

(a) For every bounded cylinder observable H in the fixed-scale test class,

$$\left| E_{a,m}^{\text{W}}(H) - E_{a,m}^{\text{good}}(H) \right| \leq C_N \|H\|_{\text{test}} u(a)^N.$$

(b) For every bounded cylinder pair H_1, H_2 whose common input-scale support hull is contained in a union of at most M_{supp} unit coarse blocks (equivalently, for which H_1 , H_2 , and $H_1 H_2$ all belong to the fixed-scale test class),

$$\left| \text{Cov}_{a,m}^{\text{W}}(H_1, H_2) - \text{Cov}_{a,m}^{\text{good}}(H_1, H_2) \right| \leq C'_N \|H_1\|_{\text{test}} \|H_2\|_{\text{test}} u(a)^N,$$

where

$$\text{Cov}_{a,m}^{\star}(H_1, H_2) := E_{a,m}^{\star}(H_1 H_2) - E_{a,m}^{\star}(H_1) E_{a,m}^{\star}(H_2), \quad \star \in \{\text{W}, \text{good}\}.$$

In particular, if $F, G \in \mathcal{A}_+$ are bounded positive-time cylinder observables and the reflected/translated product $(\Theta F) \tau_s G$ remains in the same fixed-scale test class, then

$$\left| E_{a,m}^{\text{W}}((\Theta F) \tau_s G) - E_{a,m}^{\text{good}}((\Theta F) \tau_s G) \right| \leq C''_N \|F\|_{\infty} \|G\|_{\infty} u(a)^N.$$

If H_1, H_2 are centered with respect to either evaluation, this is the corresponding centered-correlation estimate.

(c) For every fixed flowed local observable $F \in \mathcal{A}_{\text{flow}}$ and every integer $N \geq 1$ there exists $C_N(F) < \infty$ such that

$$\left| \omega_a(F) - \omega_a^{\text{RG}}(F) \right| \leq C_N(F) u(a)^N$$

uniformly for a in the AF window, where ω_a^{RG} denotes the finite- a functional produced by the chart-truncated dyadic RG recursion.

Proof. Part (a) packages the exact Wilson recursion, the actual chart-truncated recursion, and the averaged bad-event transport estimate into one finite-depth argument with logarithmic-depth uniformity. Theorem III.11 identifies $E_{a,m}^W$ as the depth- m composition of the exact one-step conditional expectations T_{j+1} . Proposition III.17 identifies $E_{a,m}^{\text{good}}$ as the parallel depth- m recursion in which the same deterministic reblocking/extraction maps are kept and only the substitutions $T_{j+1} \mapsto T_{j+1}^{\text{good}}$ are made. For an input observable H supported in at most M_{supp} unit coarse blocks, Theorem III.9 shows that every intermediate observable encountered along the patched recursion stays supported in at most M_{supp} blocks. Hence the actual Lane C recursion falls under Proposition III.8 with support budget M_{supp} and with depth bound $m \leq C_{\text{depth}} \log(1/u(a))$. That proposition already absorbs the telescoping sum, the logarithmic depth, and the polylogarithmic prefactors, and yields a bound

$$\left| E_{a,m}^W(H) - E_{a,m}^{\text{good}}(H) \right| \leq C_N(C_{\text{depth}}, M_{\text{supp}}) \|H\|_{\infty} u(a)^N$$

for every prescribed N once $u(a)$ is sufficiently small. Because the argument is local on the support hull and the deterministic support map is unchanged, the same bound is uniform over the finite-volume/admissible-boundary-condition class used in Route 1: boundary data enter only as frozen exterior variables on that hull. This proves part (a).

For part (b), apply part (a) to H_1 , H_2 , and $H_1 H_2$; this is legitimate because the common support-hull hypothesis in part (b) puts all three observables in the same fixed-scale test class. Use then the identity

$$AB - A'B' = (A - A')B + A'(B - B')$$

with $A = E_{a,m}^W(H_1)$, $B = E_{a,m}^W(H_2)$, $A' = E_{a,m}^{\text{good}}(H_1)$, and $B' = E_{a,m}^{\text{good}}(H_2)$. The positive-time pairing is the special case $H = (\Theta F)\tau_s G$. If the observables are centered for either evaluation, the covariance identity reduces immediately to the corresponding centered correlation bound.

For part (c), apply Lemma III.7 and choose the localization radius so that the deterministic truncation tail is $O(u(a)^N)$; then apply Proposition III.14 to the cylinder truncation $F^{(R)}$. The support bound from Lemma III.7 is polynomial in R , so the proposition applies directly to this localized flowed observable. $\square$

Remark III.19 (Phase-IV uniformity note for Theorem III.18). The theorem-facing uniformity is exactly logarithmic in the dyadic depth: $m \leq C_{\text{depth}} \log(1/u(a))$. This is the regime later realized by one-shot entrance, fixed physical comparison scales, and flowed-localization

closures on the public Route 1 spine. The statement should therefore neither be weakened to a merely fixed-depth estimate nor overread as uniform for arbitrary depth families outside that logarithmic window.

Lemma III.20 (De-patching at the level of continuum limits). *Assume the chart-truncated RG outputs converge to a limiting functional ω_{RG} on the flowed algebra $\mathcal{A}_{\text{flow}}$. Then the unpatched Wilson expectations converge with the same limit:*

$$\lim_{a \rightarrow 0} \omega_a(F) = \omega_{\text{RG}}(F), \quad F \in \mathcal{A}_{\text{flow}}.$$

Proof. By assumption, $\omega_a^{\text{RG}}(F) \rightarrow \omega_{\text{RG}}(F)$ for each fixed F . Theorem III.18 gives $|\omega_a(F) - \omega_a^{\text{RG}}(F)| \rightarrow 0$ as $a \rightarrow 0$. Therefore $\omega_a(F) \rightarrow \omega_{\text{RG}}(F)$. $\square$

Consequence. On the tuned dyadic trajectory used on the load-bearing Route 1/Lane B path, Theorem F.211 records the actual bounded positive-time Wilson limit and its reflected OS matrix elements directly. Lemma III.20 identifies the chart-truncated RG continuum state with the Wilson continuum expectations on flowed observables in $\mathcal{A}_{\text{flow}}$. Thus the de-patching module is used only for the flowed/Lane C continuum construction; reflection positivity of the bounded base state is handled separately in Theorem F.211.

Appendix F: Mass gap along the t_0 -defined scaling trajectory

This appendix contains the primary Route 1 mass-gap module along the tuned dyadic scaling trajectory. All constant-tracking and smallness requirements are stated as explicit inequalities between named constants. The load-bearing Route 1 package proved here is

$$\text{QE1, QE2}_{\text{lat}}, \text{QE3},$$

namely one-shot entrance, the tuned-sequence full fixed-lattice OS gap lower bound, and the continuum sharp-local vacuum-gap transfer theorem. A dense-core Euclidean-clustering theorem is recorded later only as a downstream bridge feeding Lane B. For a concise overview of the Route 1 argument, one may begin with the theorem package isolated in this appendix and then read QE1, QE2_{lat}, and QE3 in order.

Route 1 reading guide (where the closure is organized)

This appendix is written so that the *single load-bearing Route 1 entrance mechanism* can be checked directly without hunting across lanes. A concentrated first reading typically focuses on five points: (i) the Plan-A χ^2 gate and its explicit feasibility budget, (ii) the same-scale Wilson-to-patched transport theorem (Theorem III.18) used to return the chart-truncated Lane C estimates to the physical Wilson theory before the one-shot blocked transfer, (iii) the gluing theorem that transfers reference KP-smallness to the *exact* blocked measure at a prescribed block size, (iv) the passage from entrance-scale blocked decay to a full fixed-lattice OS gap theorem, and (v) the conversion of that lower bound into a continuum spectral gap. For quick navigation:

- Primary Route 1 proof spine: see Subsection F 1.
- Plan-A χ^2 gate reduction and explicit transfer-feasibility closure: see §F.17–F.19.8.
- Same-scale Wilson-to-patched transport on the fixed-scale Route 1 test class: Theorem III.18.
- Exact conditional-expectation / factorization / averaged-transport summary underlying that same-scale return: Theorem III.16.
- Weak-window repair certificate and the only live downstream branch from the curvature-sector package: Theorem F.301.
- Local Components 3/8 transport summary for the fixed-lattice-to-continuum gap chain: Theorem F.218.
- One-shot entrance theorem and bounded physical entrance scale (QE1): Theorem F.320 and Corollary F.321.
- QE2_{lat} closure by exact terminal-block covariance transfer with a common rate for all fixed-lattice local observables, followed by the fixed-lattice spectral-gap theorem: Lemma F.308, Corollary F.309, Proposition F.35, Corollary F.310, and Theorem F.311.
- QE3 closure by dyadic-to-continuum OS transport plus non-collapse on the full sharp-local Hilbert space: Theorem F.212, Corollary F.215, Lemma F.216, Theorem F.219, and Corollary F.5.

- The explicit bounded-factor insertion upgrade used in the mixed-channel finite-cap clause: Corollary 5.45A together with Proposition F.245A.
- Appendix M then packages the corresponding OS/Wightman-strength existence statements (Euclidean axioms, reconstruction, locality, positive energy, and field correspondence) in one claim-facing appendix.
- Secondary dense-core Euclidean-clustering / Lane B route: Theorem F.329 and Theorem 6.2.

Reading note. The main Appendix F keeps the theorem-level one-shot entrance chain, the full fixed-lattice gap step, the continuum vacuum-gap transport step, and the blocked covariance / representative input used by the downstream dense-core route. For a first pass, start with Theorem F.218, then read QE1, QE2_{lat}, and QE3 in that order; the dense-core Euclidean-clustering route can be read afterward as a downstream reformulation.

Remark F.1 (Appendix F deliverable). The unique paper-level target theorem is Theorem 2.1. This appendix proves three Route 1 outputs on the tuned dyadic spine. First, it proves the one-shot entrance package at the explicit block scale $B(\beta)$. Second, it closes QE2_{lat} by combining exact terminal-block conditional independence with entrance-scale Dobrushin decay and feeding the resulting uniform local-observable decay into the fixed-lattice OS gap theorem. Third, it closes QE3 by transporting that full fixed-lattice gap through the dyadic bounded-state limits and the continuum-time OS reconstruction furnished by Lane A. A dense-core Euclidean-clustering route is retained later only as a downstream reformulation feeding Lane B.

Notation and constants (quick dictionary).

u_{ren} : fixed weak-window target coupling used to define the tuned dyadic trajectory.

ℓ_0 : fixed physical length unit; $a_n = \ell_0 2^{-n}$ and $\beta_n := \beta(a_n; u_{\text{ren}})$.

q_* : entrance-scale Dobrushin contraction constant (Theorem F.320).

m_{blk} : block-mixing rate, $m_{\text{blk}} := -\log q_*$ (see (F7)).

C_{ent} : entrance bound for the one-shot scale: $a_n B_n \leq C_{\text{ent}} \ell_0$ along the tuned dyadic sequence (Corollary F.321).

m_* : target physical gap constant produced directly from m_{blk} and the entrance bound C_{ent} ; see (F7).

$\Delta_{\text{lat}}(\beta_n)$: full fixed-lattice OS/transfer-matrix gap in lattice units along the tuned dyadic path; physical rate is $\Delta_{\text{lat}}(\beta_n)/a_n$.

$m_{\text{lat}}(\beta_n)$: auxiliary probe-family/channel clustering rate, retained only as a diagnostic and not used on the primary QE2/QE3 spine.

Proof roadmap (load-bearing vs. exploratory)

What this appendix proves. The unique paper-level target theorem is Theorem 2.1. This appendix proves the Route 1 entrance package (QE1), the tuned-sequence full fixed-lattice OS gap lower bound (QE2_{lat}), and the continuum vacuum-gap transport step (QE3) on the tuned dyadic spine. The dense-core Euclidean-clustering / Lane B chain is recorded only as a downstream reformulation of the already established Route 1 gap.

Primary proof path. The front-facing Route 1 package now has the form:

1. one-shot entrance into the KP/Dobrushin basin together with the tuned-sequence physical entrance scale (proved here);
2. exact terminal-block covariance transfer plus entrance-scale block decay, fed into the fixed-lattice OS gap theorem, yielding QE2_{lat} unconditionally;
3. dyadic bounded-state convergence plus Lane A continuum-time OS transport, yielding the continuum spectral gap by non-collapse.

Secondary material. A dense-core Euclidean-clustering route feeding Lane B is retained later in the appendix as an alternative reformulation. Measure-level domination and variance/CLT reductions remain exploratory side-branches and are not part of the front-facing chain unless explicitly cited.

1. Front-facing closure chain (QE1/QE2-lat/QE3)

This subsection isolates the frozen Route 1 proof spine. The front-facing package is

QE1: one-shot entrance + tuned-sequence physical entrance bound (proved here)

$\implies$

QE2_{lat}: full fixed-lattice OS gap from exact entrance-scale block-to-microscopic transfer
(proved here)

$\implies$

QE3: continuum OS spectral gap by non-collapse, closed by dyadic-to-continuum OS
transport through Lane A.

Thus QE1, QE2_{lat}, and QE3 are all proved on the primary Route 1 spine. A dense-core Route 1 / Lane B bridge is recorded later, but it is downstream of the already established gap and is not part of this primary package. Continuum sharp-local existence, universality, and scheme-independence are proved in the Lane A / Lane C modules and are cited directly in Theorem 2.1; they are not repackaged here. Throughout we work in lattice dimension $d = 4$ and with a fixed compact connected simple gauge group G together with the Wilson representation ρ of Theorem 2.1.

a. Standing choices and notational conventions

Wilson lattice action. On each finite lattice region $\Lambda \subset \mathbb{Z}^4$ (torus or box with admissible boundary conditions) we consider the standard Wilson gauge action

$$S_\beta(U) := \beta \sum_{p \in P(\Lambda)} A(U_p), \quad A(g) := 1 - \frac{1}{d_\pi} \Re \operatorname{Tr}_\pi(g), \quad (\text{F1})$$

with $\pi = \rho$ the fixed Wilson representation (equivalently, after descent to the effective quotient $G_{\text{eff}} := G / \ker(\rho)$; cf. Theorems 1.1, 2.4 and 2.5). The corresponding finite-volume Gibbs measure is $d\mu_{\beta,\Lambda}(U) \propto e^{-S_\beta(U)} dU$, with dU Haar on edges and tree gauge used when derivatives are taken.

Remark F.2 (Global form, effective quotient, and line operators). For a fixed pair (G, π) , the Wilson plaquette weight depends on G only through the effective quotient $G_{\text{eff}} := G / \ker(\pi)$ (Theorems 2.4 and 2.5). In particular, for the Spin families with π equal to the defining (vector) representation, G_{eff} is the corresponding SO group. The sharp local algebra treated in this manuscript is generated by strictly local gauge-invariant observables (flowed local composites of curvature), and therefore cannot detect the difference between global forms

with the same Lie algebra once π is fixed (Theorem 2.6). Differences between global forms are encoded in *nonlocal* line/surface operators (e.g. Wilson lines in spinor representations and the dual ’t Hooft defects), which correspond to superselection data for representations of the same local net; see §2.2.

Admissible action class and schemes. We allow, when stated, local gauge actions in the universality class of Definition F.256 and renormalization schemes in the sense of Definition F.272. The Wilson action (F1) is a distinguished member of that class.

Tuned dyadic scaling trajectory. Fix once and for all a weak-window target coupling $u_{\text{ren}} \in (0, u_*]$ and the physical length unit $\ell_0 > 0$. The load-bearing Route 1 path uses the dyadic sequence

$$a_n := \ell_0 2^{-n}, \quad \beta_n := \beta(a_n; u_{\text{ren}}), \quad g_{\text{GF}}^2(\ell_0; a_n, \beta_n) = u(a_n), \quad (\text{F2})$$

where $u(a_n) = u_{-n}$ is the backward step-scaling trajectory from Proposition F.314 and $\beta(a_n; u_{\text{ren}})$ is selected by Theorem F.298. This tuned dyadic sequence is the only renormalization prescription used on the paper-level Route 1 mass-gap spine. Older t_0 -based scale-setting formulas are retained only in auxiliary/archival subsections and are not cited in the final theorem chain.

One-shot entrance scale. For the load-bearing Route 1 chain, the entrance mechanism is the explicit one-shot block size $B(\beta)$ from (F369). Fix once and for all a target value $u_{\text{ren}} \in (0, u_*]$ in the weak window and let

$$a_n := \ell_0 2^{-n}, \quad \beta_n := \beta(a_n; u_{\text{ren}}), \quad B_n := B(\beta_n)$$

be the associated tuned dyadic trajectory from Proposition F.314. Corollary F.321 gives a uniform constant $C_{\text{ent}} < \infty$ and an entry index $N_{\text{ent}}(u_{\text{ren}})$ such that

$$a_n B_n \leq C_{\text{ent}} \ell_0 \quad (n \geq N_{\text{ent}}(u_{\text{ren}})). \quad (\text{F3})$$

Whenever a generic entrance-scale symbol appears in earlier comparison lemmas, the final theorem chain instantiates it with this same one-shot scale $B(\beta)$; no second entrance mechanism is used in the paper-level proof.

b. QE1/QE2_{lat}/QE3 chain used in Theorem 2.1

Proposition F.3 (QE1: one-shot entrance and tuned-sequence physical scale). *Within the standing choices above there exist $\beta_* < \infty$, $q_* \in (0, 1)$, and $C_{\text{ent}} < \infty$ such that:*

(i) *for every $\beta \geq \beta_*$ the exact one-shot blocked specification at the block size $B(\beta)$ satisfies*

$$\delta(\gamma^{(B(\beta))}) \leq q_* < 1; \quad (\text{F4})$$

(ii) *for every fixed target value $u_{\text{ren}} \in (0, u_*]$, along the associated tuned dyadic trajectory $a_n = \ell_0 2^{-n}$, $\beta_n = \beta(a_n; u_{\text{ren}})$, $B_n = B(\beta_n)$, there exists an entry index $N_{\text{ent}}(u_{\text{ren}})$ such that*

$$a_n B_n \leq C_{\text{ent}} \ell_0 \quad (n \geq N_{\text{ent}}(u_{\text{ren}})). \quad (\text{F5})$$

In particular the exact one-shot blocked state is unique and exponentially mixing in entrance-block distance, and Proposition F.3 supplies the scale data used later in the exact block-to-microscopic transfer and non-collapse chain. Only the entry index, not the constant C_{ent} , depends on the chosen target value.

Proof. This is exactly Theorem F.320 together with Corollary F.321. $\square$

Proposition F.4 (QE2_{lat}: unconditional tuned-sequence full fixed-lattice gap). *Within the standing choices above, let*

$$a_n := \ell_0 2^{-n}, \quad \beta_n := \beta(a_n; u_{\text{ren}}), \quad B_n := B(\beta_n).$$

Then there exist $n_ < \infty$ and $\mathfrak{m}_* > 0$ such that for all $n \geq n_*$ the fixed-lattice OS Hamiltonian for the infinite-volume Wilson Gibbs state at bare coupling β_n has full vacuum gap $\Delta_{\text{lat}}(\beta_n) > 0$ and*

$$\frac{\Delta_{\text{lat}}(\beta_n)}{a_n} \geq \mathfrak{m}_*. \quad (\text{F6})$$

One may take

$$\mathfrak{m}_* = \frac{m_{\text{blk}}}{2C_{\text{ent}}\ell_0}, \quad m_{\text{blk}} := -\log q_*. \quad (\text{F7})$$

Proof. This is exactly Theorem F.311. $\square$

Corollary F.5 (QE3: sharp-local continuum vacuum gap on the Route 1 spine). *Assume Proposition F.4 and let $\omega^{(u_{\text{ren}})}$ be the full sharp-local state furnished by Corollary 5.76 over the same tuned dyadic bounded base state. Let $(\mathcal{H}, \Omega, H)$ be its OS reconstruction. Then*

$$\text{Spec}(H) \subset \{0\} \cup [\mathfrak{m}_*, \infty), \quad \ker(H) = \mathbb{C}\Omega, \quad P_{(0, \mathfrak{m}_*)} = 0.$$

Equivalently,

$$\|e^{-tH}\psi\| \leq e^{-\mathfrak{m}_* t} \|\psi\| \quad (t \geq 0, \psi \perp \Omega).$$

This is the continuum sharp-local spectral consequence used in Theorem 2.1(D).

Proof. Proposition F.4 yields

$$\liminf_{n \rightarrow \infty} \frac{\Delta_{\text{lat}}(\beta_n)}{a_n} \geq \mathfrak{m}_*.$$

Apply Theorem F.219 with $m_* = \mathfrak{m}_*$. The semigroup estimate is then the spectral-theorem reformulation of the same gap statement on $\Omega^\perp$. $\square$

Theorem F.6 (Appendix F primary Route 1 proof spine). *Within the standing choices above, QE1, QE2_{lat}, and QE3 hold on the tuned dyadic Route 1 spine. The dense-core Route 1/Lane B bridge retained later is therefore a downstream reformulation rather than an additional load-bearing interface.*

Proof. Proposition F.3 gives QE1, Proposition F.4 gives QE2_{lat}, and Corollary F.5 gives QE3. $\square$

Remark F.7 (Reading guide for a first audit pass). A referee auditing only the primary Route 1 package may start with Theorem F.218, which freezes the fixed-lattice-to-continuum theorem chain, the only live weak-window input, the first Lane A theorem-level dependency, and the fact that Lane B is downstream only. Then verify QE1 from Theorem F.320 together with the tuned-sequence physical-scale corollary, Corollary F.321. Then read Lemma F.308, Proposition F.35, Corollary F.310, and Theorem F.311, which give the unconditional block-to-microscopic transfer and the tuned-sequence full fixed-lattice gap lower bound for QE2_{lat}. Then read Theorem III.18, Theorem F.211, Theorem F.212, Theorem F.219, and Corollary F.5, which close the continuum spectral-gap step. The dense-core route through Theorem F.329 and Lane B is secondary in this organization.

Corollary F.8 (Optional diagnostic: weak-window step scaling has $b_0 > 0$ (not used)). *The gradient-flow step-scaling function admits a quadratic expansion with coefficient $b_0 > 0$ (Proposition F.266). This diagnostic is not used anywhere in the mass-gap proof.*

Organization note. The remainder of this appendix proves QE1 in full detail, closes QE2_{lat} by exact entrance-scale covariance transfer together with the fixed-lattice OS gap theorem, and then closes QE3 by transporting dyadic bounded-state limits through the Lane A continuum-time OS reconstruction. The dense-core Euclidean-clustering route is retained later as a secondary bridge to Lane B. Corollary F.8 records an optional weak-window diagnostic and is not used elsewhere. Any later harmonization of notation or constant names across modules is purely expository and does not affect the stated inequalities or conclusions.

2. Normalization, activity, and the basic envelope $g_b(f)$

Fix lattice dimension $d \geq 2$, compact Lie group G , block size $b \geq 2$, and polymer parameter $\theta > 0$. Set

$$N_b := b^2. \tag{F8}$$

Let dg be Haar *probability* measure on G . For each irrep R with character χ_R and dimension d_R , define the normalized character

$$\varphi_R(g) := \frac{\chi_R(g)}{d_R} \quad \Rightarrow \quad |\varphi_R(g)| \leq 1 \quad \forall g \in G. \tag{F9}$$

A (canonically normalized) plaquette weight is expanded as

$$W(g) = 1 + \sum_{R \neq \mathbf{1}} a_R \varphi_R(g), \tag{F10}$$

with scalar activity

$$f := \sum_{R \neq \mathbf{1}} |a_R|. \tag{F11}$$

Lemma F.9 (Tile-mass envelope). *For all $g \in G$, $|W(g)| \leq 1 + f$. Consequently the tile-mass majorant*

$$g_b(f) := (1 + C_{\text{int}} f)^{N_b} - 1 \tag{F12}$$

holds with $\boxed{C_{\text{int}} = 1}$.

Proof. Immediate from $|\sum a_R \varphi_R(g)| \leq \sum |a_R| = f$. □

Short-distance products and OPE-style bounds (optional refinement) The mass gap objective (existence of an OS-positive continuum Yang–Mills theory with a spectral gap) does not require an operator product expansion at coincident points. However, if one wishes to

promote the inductive-limit algebra $\mathfrak{A}_{\text{loc}}$ to a *fully local* field algebra closed under pointwise products (as distributions), then one needs quantitative short-distance control. In the present paper this is treated as an *additional refinement module* built on the same multiscale inputs already used for dyadic Cauchy convergence.

Lemma F.10 (Uniform moment bounds for renormalized multiplets at fixed truncation). *Fix Δ_{max} and let $\mathcal{O}_{\text{ren}}^{(\leq \Delta_{\text{max}})}$ be the renormalized multiplet from Theorem F.246. For each $p < \infty$ and each test function $\phi \in C_c^\infty(\mathbb{R}^4)$,*

$$\sup_{j \geq 0} \|X_j^{(\leq \Delta_{\text{max}})}(\phi)\|_{L^p(\mathbb{R}^{M(\Delta_{\text{max}})})} < \infty,$$

and the same bound holds uniformly over actions in the universality class of Definition F.256 for β sufficiently large.

Proof. On the small-curvature event Ω_{ε_0} each flowed local observable is a finite sum of local curvature polynomials with uniformly bounded coefficients (heat-kernel norms) at the relevant scale. Curvature Brascamp–Lieb domination (Proposition F.226) implies sub-Gaussian bounds for linear curvature functionals and Hanson–Wright-type bounds for quadratic polynomials; by standard hypercontractivity for polynomial functionals of sub-Gaussian fields, all moments of these finite-degree local polynomials are controlled uniformly. The complement $\Omega_{\varepsilon_0}^c$ contributes an exponentially small tail by Lemma F.224. Finally, $Z^{(\leq \Delta_{\text{max}})}(t_j) = I + O(u_j)$ is uniformly bounded in operator norm along dyadic scales (since $\sum u_j^2 < \infty$), so multiplying by $Z^{(\leq \Delta_{\text{max}})}(t_j)$ preserves moment bounds. $\square$

Proposition F.11 (Renormalized normal products via multiscale subtraction). *Let $A, B \in \mathfrak{A}_{\text{loc}}$ be two renormalized local fields obtained from the dyadic construction. There exists a renormalized product (“normal product”) $A \diamond B \in \mathfrak{A}_{\text{loc}}$ defined as follows:*

For each dyadic scale j , define the bounded approximants A_j, B_j as the renormalized flowed fields at time t_j . Set

$$(A \diamond B)_j := A_j B_j - \Pi_{< \Delta(A) + \Delta(B)}(A_j B_j),$$

where $\Pi_{< D}$ denotes the projection (in L^2) onto the finite-dimensional span of renormalized fields of engineering dimension $< D$ (in a fixed graded basis), with coefficients chosen to keep a fixed reference Gram matrix of two-point functions scale-independent. Then $(A \diamond B)_j$ converges in L^2 as $j \rightarrow \infty$, and its limit is independent of the choice of dyadic sequence.

Moreover, if the supports of the smearing test functions of A and B are disjoint, then $A \diamond B$ coincides with the ordinary product AB .

Proof. The key point is that $A_j B_j$ is itself a bounded local observable at flow time t_j and therefore admits the same one-shell decomposition as in Lemma F.238, with coefficients controlled by curvature domination. The only potentially non-summable part of the shell drift is the projection of $A_j B_j$ onto local operators of dimension $\leq \Delta(A) + \Delta(B)$. By subtracting the projection onto strictly lower dimensions and fixing the remaining dimension- $\Delta(A) + \Delta(B)$ component by normalization (exactly as in the scalar and finite-multiplet cases), the renormalized shell increment becomes $O(u_j^2)$ in L^2 . Summability follows from $\sum_j u_j^2 < \infty$. If the smearings are disjoint at physical distance $\gg \sqrt{t_j}$, locality of the flow implies no coincident-point divergence occurs, so the projection term vanishes for j large and the renormalized product equals the ordinary product. $\square$

Remark F.12 (Associativity and higher products). Higher normal products are defined recursively (or via the standard Zimmermann forest formula) by iterating the dyadic subtraction. Because the subtraction is organized by the engineering-dimension filtration and uses orthogonal projections onto finite-dimensional subspaces, one obtains an associative $*$ -algebra structure on $\mathfrak{A}_{\text{loc}}$ compatible with locality.

Proposition F.13 (OPE-style remainder bound at short distance (finite truncation)). *Fix Δ_{max} and work inside the truncated algebra $\mathfrak{A}_{\text{loc}}^{(\leq \Delta_{\text{max}})}$. Let A, B be two renormalized local fields of engineering dimensions $\Delta_A, \Delta_B \leq \Delta_{\text{max}}$. Then there exists a finite set of renormalized fields $\{\mathcal{O}_k\}$ with dimensions $\Delta_k \leq \Delta_A + \Delta_B$ and coefficient distributions $C_{AB}^k \in \mathcal{D}'(\mathbb{R}^4)$ with singular support contained in $\{0\}$ (allowing contact terms) such that for any test function ϕ supported in the unit ball and any scale $r \in (0, 1]$,*

$$\left\| A(\phi_r) B(\phi_r) - \sum_{\Delta_k \leq \Delta_A + \Delta_B} \langle C_{AB}^k, \phi_r \rangle \mathcal{O}_k(\phi_r) \right\|_{L^2} \leq C_{A,B,\phi} r^\delta,$$

for some $\delta > 0$ depending only on the first omitted dimension. Here $\phi_r(x) := r^{-4} \phi(x/r)$.

Proof. We follow the multiscale power counting argument.

Choose the dyadic scale j so that $\sqrt{t_j} \asymp r$. Write the flowed product $A_j(\phi_r) B_j(\phi_r)$ as a sum over shells using the finite-range decomposition (F298). At scales $< r$ (higher momenta) the two insertions are effectively coincident and produce local counterterms; at scales $> r$ the

insertions are separated and contribute only regular terms. Projecting onto the local operator basis up to dimension $\Delta_A + \Delta_B$ yields the coefficients C_{AB}^k . The first omitted dimension produces an r^δ gain by standard power counting. Curvature domination gives the needed uniform L^2 control on each shell contribution; asymptotic freedom (Lemma F.237) makes the logarithmic corrections summable. $\square$

Wilson coefficients: existence, derivative bounds, and scheme covariance (optional refinement) This subsection upgrades Proposition F.13 by (i) defining the coefficient distributions C_{AB}^k as *limits of dyadic coefficient functionals* extracted from the multiscale subtraction, and (ii) recording quantitative regularity bounds away from the coincidence point. Nothing here is needed for the lattice mass-gap or for the continuum existence statements on bounded macroscopic algebras; it is only required if one wants a fully local field algebra equipped with quantitative OPE control.

Dyadic coefficient extraction. Fix a truncation level $\Delta_{\max}$ and a graded basis $\{\mathcal{O}_k\}$ of renormalized local fields with $\Delta_k \leq \Delta_{\max}$. For a dyadic scale j let $t_j := 2^{-2j}t_0$ and let $\mathcal{O}_{k,j}$ denote the bounded flowed approximant at time t_j after applying the renormalization matrix $Z^{(\leq \Delta_{\max})}(t_j)$ from Theorem F.246. Let A_j, B_j be the corresponding approximants for A, B .

Fix once and for all a reference test function $\phi \in C_c^\infty(\mathbb{R}^4)$ supported in the unit ball and define $\phi_r(x) = r^{-4}\phi(x/r)$. For $r \asymp 2^{-j}$ define the coefficient vector $c_{AB,j}(\phi) \in \mathbb{R}^{m(\Delta_{\max})}$ by the orthogonal-projection identity

$$\Pi_j^{(\leq \Delta_A + \Delta_B)}(A_j(\phi_r)B_j(\phi_r)) = \sum_{\Delta_k \leq \Delta_A + \Delta_B} c_{AB,j}^k(\phi) \mathcal{O}_{k,j}(\phi_r), \quad (\text{F13})$$

where $\Pi_j^{(\leq D)}$ is the L^2 -orthogonal projection onto the span of $\{\mathcal{O}_{\ell,j}(\phi_r) : \Delta_\ell \leq D\}$. Because the Gram matrix of this basis is held scale-independent by the normalization defining $Z^{(\leq \Delta_{\max})}(t_j)$, the coefficients $c_{AB,j}^k(\phi)$ are well-defined and stable under small changes of r within a fixed dyadic shell.

One-shell control of coefficient increments. Write $d_k := \Delta_A + \Delta_B - \Delta_k \geq 0$ for the engineering dimension deficit. For multiindices α we write ∂^α for derivatives in the relative-coordinate variable.

Lemma F.14 (Shell increment bound for dyadic OPE coefficients). *Assume weak-coupling curvature control (Lemma F.224 and Proposition F.226) and dyadic coupling decay*

(Lemma F.237). Then for each fixed truncation $\Delta_{\max}$ and each fixed test function ϕ there exist constants $C_{\phi, \Delta_{\max}} < \infty$ and j_0 such that for all $j \geq j_0$ and all k with $\Delta_k \leq \Delta_A + \Delta_B$,

$$|c_{AB, j+1}^k(\phi) - c_{AB, j}^k(\phi)| \leq C_{\phi, \Delta_{\max}} 2^{jd_k} (u_j + u_j^2) + C_{\phi, \Delta_{\max}} e^{-c_{\text{loc}} \beta}. \quad (\text{F14})$$

Moreover, if ϕ is replaced by $\partial^\alpha \phi$ then the same bound holds with an extra factor $2^{j|\alpha|}$.

Proof. Fix $r \asymp 2^{-j}$. The projected coefficient $c_{AB, j}^k(\phi)$ is a finite linear functional of mixed moments of the local bounded observables $A_j(\phi_r)B_j(\phi_r)$ against the basis fields $\mathcal{O}_{k, j}(\phi_r)$. Replacing scale j by $j+1$ replaces the flow time t_j by $t_{j+1} = t_j/4$ and inserts one additional “shell” in the finite-range decomposition (F298). Exactly as in Lemma F.238, the shell contribution is a finite sum of local curvature polynomials whose coefficients obey the dimensional scaling 2^{jd_k} and (for derivatives) $2^{j|\alpha|}$ by power counting. Curvature Brascamp–Lieb domination bounds the L^2 size of each shell polynomial by $C(u_j^{1/2})^{\text{deg}}$, and the only potentially non-summable part is the order- u_j drift in the dimension- $\Delta_A + \Delta_B$ sector. That drift is precisely absorbed into the renormalization/mixing matrices $Z^{(\leq \Delta_{\max})}(t_j)$ defining the basis $\{\mathcal{O}_{k, j}\}$, leaving a remainder of order u_j^2 ; we record the bound in the safe form $(u_j + u_j^2)$. The tail outside Ω_{ε_0} is exponentially small in β by Lemma F.224. $\square$

Existence and regularity of Wilson coefficients. The coefficients $c_{AB, j}^k(\phi)$ determine a distribution C_{AB}^k by dyadic Littlewood–Paley reconstruction: one defines the action of C_{AB}^k on a general test function ψ by decomposing $\psi = \sum_{j \geq 0} \psi_j$ into dyadic shells of relative scale and setting $\langle C_{AB}^k, \psi_j \rangle$ equal to the coefficient extracted at the matching dyadic scale. The shell increment bound (F14) implies convergence and yields explicit derivative bounds.

Theorem F.15 (Wilson coefficient limits and derivative bounds). *Fix $\Delta_{\max}$ and let $A, B \in \mathfrak{A}_{\text{loc}}^{(\leq \Delta_{\max})}$ with dimensions Δ_A, Δ_B . Assume Lemma F.14. Then for each k with $\Delta_k \leq \Delta_A + \Delta_B$ there exists a distribution $C_{AB}^k \in \mathcal{D}'(\mathbb{R}^4)$ with $\text{sing supp}(C_{AB}^k) \subset \{0\}$ such that:*

1. (OPE remainder) *The bound of Proposition F.13 holds with these coefficients.*
2. (Derivative control on scaled tests) *For every multiindex α there exist constants $C_{\alpha, A, B, k, \phi}$ and an integer $M \geq 0$ such that for all $r \in (0, 1]$,*

$$|\langle C_{AB}^k, \partial^\alpha \phi_r \rangle| \leq C_{\alpha, A, B, k, \phi} r^{-d_k - |\alpha|} \left(1 + \log(1/r)\right)^M. \quad (\text{F15})$$

3. (Smoothness away from coincidence) *There exists a smooth function $c_{AB}^k \in C^\infty(\mathbb{R}^4 \setminus \{0\})$ such that $C_{AB}^k|_{\mathbb{R}^4 \setminus \{0\}}$ equals $c_{AB}^k(x) dx$, and for every multiindex α ,*

$$|\partial^\alpha c_{AB}^k(x)| \leq C_{\alpha,A,B,k} |x|^{-d_k-|\alpha|} \left(1 + \log(1/|x|)\right)^M, \quad 0 < |x| \leq \frac{1}{2}. \quad (\text{F16})$$

Proof. Decompose a test function ψ into dyadic annuli $\psi = \sum_j \psi_j$ with ψ_j supported where $|x| \asymp 2^{-j}$. Define $\langle C_{AB}^k, \psi_j \rangle$ by transporting ψ_j to the unit scale and applying the coefficient functional $c_{AB,j}^k(\cdot)$. The shell increment estimate (F14) and dyadic coupling decay $u_j \lesssim (1+j)^{-1}$ imply the series defining $\langle C_{AB}^k, \psi \rangle$ converges absolutely, with at most a polynomial loss in j giving the logarithmic factor in (F15). Smoothness on $\mathbb{R}^4 \setminus \{0\}$ follows because only finitely many shells interact with any compact set bounded away from 0, and each shell coefficient is given by a smooth kernel (finite-range heat-kernel coefficients). Derivative bounds are obtained by repeating the preceding estimate for $\partial^\alpha \psi$ and tracking the extra factor $2^{j|\alpha|}$. $\square$

Corollary F.16 (Scheme covariance of Wilson coefficients). *Let $\mathbf{S}, \mathbf{S}'$ be two admissible renormalization schemes in the sense of Definition F.272, with scheme-change matrices $R^{(\leq \Delta_{\max})}$ from Theorem F.275. Write the (truncated) renormalized bases as $\{\mathcal{O}_i^{\mathbf{S}}\}$ and $\{\mathcal{O}_i^{\mathbf{S}'}\}$ and assume they are related by*

$$\mathcal{O}_i^{\mathbf{S}'} = \sum_a R_{ia} \mathcal{O}_a^{\mathbf{S}}.$$

Then the Wilson coefficients transform by the standard change-of-basis rule: at the level of coefficient distributions,

$$C_{ij}^{k,\mathbf{S}'} = \sum_{a,b,c} R_{ia} R_{jb} (R^{-1})_{ck} C_{ab}^{c,\mathbf{S}}, \quad (\text{F17})$$

where C_{ij}^k denotes the coefficient in the OPE of $\mathcal{O}_i \cdot \mathcal{O}_j$ onto $\mathcal{O}_k$. In particular, the singular orders and derivative bounds of Theorem F.15 are invariant within the admissible class.

3. KP majorant and admissibility

a. Connected-set counting constant C_{count}

Let $\mathcal{G}_p$ be the plaquette adjacency graph: plaquettes adjacent if they share a link. A plaquette has 4 links. A link in $\mathbb{Z}^d$ belongs to $2(d-1)$ plaquettes, so a plaquette shares a link with at most

$$\Delta_p(d) \leq 4(2(d-1) - 1) = 8d - 12 \quad (\text{F18})$$

other plaquettes. The standard tree-count bound implies

$$\#\{\text{connected plaquette sets of size } n \text{ containing a root}\} \leq (e\Delta_p)^{n-1}. \quad (\text{F19})$$

We therefore take

$$C_{\text{count}} := \lceil e\Delta_p(d) \rceil. \quad (\text{F20})$$

b. Majorant function and ceiling

Define the KP majorant

$$q(f) := C_{\text{count}} e^\theta K_0 g_b(f). \quad (\text{F21})$$

Admissibility is $q(f) < 1$. The maximal admissible activity is the solution of $q(f) = 1$:

$$f_{\text{adm}} = \frac{\left(1 + \frac{1}{C_{\text{count}} e^\theta K_0}\right)^{1/N_b} - 1}{C_{\text{int}}}. \quad (\text{F22})$$

Lemma F.17 (Explicit inverse threshold for the KP majorant). *For any target $\bar{q} \in (0, 1)$ define*

$$f_{\text{KP}}(\bar{q}) := \frac{\left(1 + \frac{\bar{q}}{C_{\text{count}} e^\theta K_0}\right)^{1/N_b} - 1}{C_{\text{int}}}.$$

Then $f \leq f_{\text{KP}}(\bar{q})$ implies $q(f) \leq \bar{q}$. In particular, $\bar{q} = 1$ recovers the maximal admissible activity $f_{\text{adm}} = f_{\text{KP}}(1)$.

Proof. By definition $q(f) = C_{\text{count}} e^\theta K_0 g_b(f)$ and $g_b(f) = (1 + C_{\text{int}} f)^{N_b} - 1$; invert this monotone map. $\square$

4. Projection remainder, drift, and the ratio audit

Stages 1–10 treated the following as explicit inputs:

1. polymer coefficient bound $|J_P| \leq K_2^{|P|} f^{|P|}$;
2. activity Lipschitz bound $|f_1^{\text{exact}} - f_1^\square| \leq C_{\text{act}} \|\Delta(1)\|_\theta$;
3. deterministic projected recursion $f_1^\square \leq \bar{K}_{\text{act}} f^{N_b}$.

a. Canonical choices $K_0 = K_{\text{rep}} = 1$, $K_2 \approx 1$, $C_{\text{act}} = 1$

Lemma F.18 (Contractive Haar averaging). *Working with Haar probability measure and $|\varphi_R| \leq 1$, one may take*

$$\boxed{K_0 = 1 \quad \text{and} \quad K_{\text{rep}}(G) = 1.}$$

Lemma F.19 (Raw coefficient majorant). *Expanding products of (F10) and taking absolute values yields*

$$|J_P| \leq f^{|P|}.$$

So one may take $\boxed{K_2 = 1}$ (raw). For cumulant-safe bookkeeping one may take $\boxed{K_2 = e}$.

Lemma F.20 (Activity Lipschitz constant). *In the ℓ^1 activity norm (F11), coefficient extraction is 1-Lipschitz. Thus $\boxed{C_{\text{act}} = 1}$.*

b. Projection constant and geometric-series bound

Assume

$$|J_P| \leq K_2^{|P|} f^{|P|}. \tag{F23}$$

Define

$$A := C_{\text{count}} K_2 e^\theta. \tag{F24}$$

Then for $Af < 1$, the polymer remainder obeys the standard geometric-series bound

$$\|\Delta(1)\|_\theta \leq K_{\text{proj}}(f) f^2, \quad K_{\text{proj}}(f) := \frac{C_{\text{count}} K_2^2 e^{2\theta}}{1 - Af}. \tag{F25}$$

c. Stage-8 ratio audit

Bernoulli gives $g_b(f) \geq N_b C_{\text{int}} f$ for $f \geq 0$. Hence $q(f) < 1$ implies $C_{\text{count}} e^\theta K_0 N_b C_{\text{int}} f < 1$. Multiplying by $K_2/(N_b C_{\text{int}} K_0)$ yields

$$R_0 := \frac{K_2}{N_b C_{\text{int}} K_0} \quad \Rightarrow \quad q(f) < 1 \quad \Rightarrow \quad Af \leq R_0. \tag{F26}$$

If $R_0 < 1$, define $\eta_0 := 1 - R_0 > 0$. Then admissibility forces the uniform margin $1 - Af \geq \eta_0$.

d. *Exact one-step contraction inside the KP basin*

Inside the KP admissibility window, the projected convolution-power decimation contracts extremely strongly. We record an explicit “once in, stay in” one-step inequality for the *exact* step.

Lemma F.21 (Exact step dominated by the projected step). *Assume the projected step satisfies $f_1^\square \leq \bar{K}_{\text{act}} f^{N_b}$ with $\bar{K}_{\text{act}} = 1$ (true for the convolution-power map in the positive-type sector), and assume the drift bound*

$$|f_1^{\text{exact}} - f_1^\square| \leq C_{\text{act}} \|\Delta(1)\|_\theta, \quad \|\Delta(1)\|_\theta \leq K_{\text{proj}}(u) u^2, \quad u := f_1^\square.$$

If u obeys $C_{\text{act}} K_{\text{proj}}(u) u \leq 1$, then

$$f_1^{\text{exact}} \leq 2u \leq 2f^{N_b}. \quad (\text{F27})$$

Proof. By the drift bound and the geometric-series estimate,

$$f_1^{\text{exact}} \leq u + C_{\text{act}} K_{\text{proj}}(u) u^2 = u(1 + C_{\text{act}} K_{\text{proj}}(u) u) \leq 2u.$$

The last inequality uses $C_{\text{act}} K_{\text{proj}}(u) u \leq 1$. Finally $u = f_1^\square \leq f^{N_b}$ when $\bar{K}_{\text{act}} = 1$. $\square$

Remark F.22 (Baseline sanity check). For $(d, b, \theta) = (4, 2, 1)$ one has $N_b = 4$ and $C_{\text{count}} = 55$. At $f = f_{\text{adm}} \approx 1.668 \times 10^{-3}$, the projected value $u \leq f^4 \approx 7.7 \times 10^{-12}$ is so small that $C_{\text{act}} K_{\text{proj}}(u) u \ll 1$ holds by many orders of magnitude. Thus (F27) is automatic once inside the KP window.

5. Exponential clustering in the KP regime

For finite plaquette sets X, Y , let $\text{dist}(X, Y)$ be the distance in $\mathcal{G}_p$. For bounded observables A, B define the truncated correlation $\langle A; B \rangle := \langle AB \rangle - \langle A \rangle \langle B \rangle$.

Theorem F.23 (Uniform exponential clustering). *Assume $Af < 1$ and (F23). Then for bounded observables supported on plaquette sets X, Y ,*

$$|\langle A; B \rangle| \leq 2 \|A\|_\infty \|B\|_\infty \frac{(Af)^{\text{dist}(X, Y)}}{C_{\text{count}} (1 - Af)}. \quad (\text{F28})$$

Corollary F.24 (Decay rate in the admissible window). *If $R_0 < 1$ and f is admissible, then $Af \leq R_0$. Hence $\mu := -\log(Af) \geq -\log(R_0) = \log\left(\frac{N_b C_{\text{int}} K_0}{K_2}\right) > 0$.*

6. Reflection positivity and transfer matrix

Definition F.25 (Reflection positivity). Fix a time reflection Θ . A measure μ on link configurations is reflection positive if for all F supported in positive times,

$$\langle \Theta F \cdot F \rangle_\mu \geq 0. \quad (\text{F29})$$

Lemma F.26 (Positive type $\Rightarrow$ positive kernel). *If W is of positive type (Definition F.27 below), then*

$$K(g, h) := W(gh^{-1})$$

defines a positive semidefinite kernel on G .

Definition F.27 (Positive type). A central function $W \geq 0$ on G is of *positive type* if

$$W(g) = \sum_R c_R \chi_R(g) \quad \text{with } c_R \geq 0.$$

Definition F.28 (Conditionally negative definite (cnd)). A central function $\psi : G \rightarrow \mathbb{R}$ is *conditionally negative definite* if for all $n \in \mathbb{N}$, all $g_1, \dots, g_n \in G$ and all $c_1, \dots, c_n \in \mathbb{C}$ with $\sum_{i=1}^n c_i = 0$,

$$\sum_{i,j=1}^n \bar{c}_i c_j \psi(g_i^{-1} g_j) \leq 0.$$

Lemma F.29 (Normalized trace defect is cnd). *Let $\pi : G \rightarrow U(d)$ be a unitary representation and set*

$$\phi_\pi(g) := \frac{1}{d} \Re \text{Tr } \pi(g), \quad A_\pi(g) := 1 - \phi_\pi(g).$$

Then A_π is conditionally negative definite.

Proof. Fix an orthonormal basis $(e_k)_{k=1}^d$ of $\mathbb{C}^d$. For each k define

$$\psi_k(g) := \frac{1}{2} \|e_k - \pi(g)e_k\|^2 = 1 - \Re \langle e_k, \pi(g)e_k \rangle.$$

The kernels $(g, h) \mapsto \psi_k(g^{-1}h)$ are conditionally negative definite because ψ_k is a squared Hilbert-space distance under a unitary representation. Since $A_\pi = \frac{1}{d} \sum_{k=1}^d \psi_k$, it is a convex combination of cnd functions, hence cnd. $\square$

Lemma F.30 (Schoenberg). *If ψ is cnd and $t \geq 0$, then the function $g \mapsto e^{-t\psi(g)}$ is positive definite. If in addition ψ is central, then $e^{-t\psi}$ is of positive type (Definition F.27).*

Proof. This is the classical theorem of Schoenberg: the kernels $K_t(g, h) := \exp(-t\psi(g^{-1}h))$ are positive semidefinite for cnd ψ . For central $e^{-t\psi}$, Peter–Weyl then gives a character expansion with nonnegative coefficients. $\square$

Corollary F.31 (Wilson-class weights are positive type). *Let A_π be as in Lemma F.29. Then for every $\beta \geq 0$,*

$$\widehat{W}_{\beta,\pi}(g) := \exp(-\beta A_\pi(g))$$

is a central positive-type function. Moreover, for every $\alpha \in (0, 1]$ the power $\widehat{W}_{\beta,\pi}^\alpha = \exp(-\alpha\beta A_\pi)$ is also positive type.

Proof. By Lemma F.29, A_π is cnd; by Lemma F.30, $\exp(-\beta A_\pi)$ is positive definite. Since it is central, its character expansion has nonnegative coefficients, i.e. it is positive type. The identical character-expansion calculation applies to $\alpha\beta$. $\square$

Corollary F.32 (Reflection positivity for the α -reference family). *The anisotropic α -reference measures $\nu_L^{(\alpha)}$ of (F110) are reflection positive with respect to reflections through coordinate hyperplanes orthogonal to any axis, for every $\alpha \in (0, 1]$.*

Proof. The in-plane plaquette weight is $\widehat{W}_{\beta,\pi}$ and the out-of-plane weight is $\widehat{W}_{\beta,\pi}^\alpha$, both positive type by Corollary F.31. Apply Proposition F.33 plaquette-wise. $\square$

Proposition F.33 (Plaquette-wise positive type $\Rightarrow$ reflection positivity). *Consider a lattice gauge theory with density*

$$d\mu(U) \propto \left(\prod_p W_p(U_p) \right) dU,$$

where each W_p is a central function of positive type (Definition F.27) and the family is reflection-covariant in the sense $W_{\theta p} = W_p$ for the reflection θ under consideration. Then μ is reflection positive with respect to θ in the sense of (F29). Consequently OS reconstruction yields a Hilbert space $\mathcal{H}$, a vacuum Ω , and a positive self-adjoint transfer matrix T .

Proof. Expand each W_p in characters with nonnegative coefficients. Plaquettes crossing the reflection hyperplane produce positive kernels by Lemma F.26; after integrating the reflected half-lattice variables the expectation $\langle \Theta F \cdot F \rangle$ becomes a sum of squares with nonnegative coefficients. $\square$

7. Uniform decay $\Rightarrow$ spectral gap

Theorem F.34 (Uniform decay of all bounded local observables implies a full fixed-lattice gap). *Assume reflection positivity (so T and Ω exist) and Ω is cyclic for bounded gauge-invariant local observables. If there exists $m > 0$ such that for every bounded gauge-invariant local observable $\mathcal{O}$ with $\langle \mathcal{O} \rangle = 0$,*

$$0 \leq \langle \mathcal{O}(t)\mathcal{O}(0) \rangle \leq C_{\mathcal{O}}e^{-mt} \quad (t \in \mathbb{N}), \quad (\text{F30})$$

then $H := -\log T$ has a vacuum gap at least m .

Proof. For every centered bounded gauge-invariant local observable $\mathcal{O}$, OS reconstruction gives

$$\langle \mathcal{O}(t)\mathcal{O}(0) \rangle = \langle [\mathcal{O}], e^{-tH}[\mathcal{O}] \rangle = \int_{[0,\infty)} e^{-t\lambda} d\mu_{\mathcal{O}}(\lambda),$$

where $\mu_{\mathcal{O}}$ is the spectral measure of H associated to $[\mathcal{O}]$. If $\mu_{\mathcal{O}}([0, m - \varepsilon]) > 0$ for some $\varepsilon \in (0, m)$, then the spectral theorem yields the lower bound

$$\langle [\mathcal{O}], e^{-tH}[\mathcal{O}] \rangle \geq e^{-t(m-\varepsilon)}\mu_{\mathcal{O}}([0, m - \varepsilon]),$$

which contradicts (F30) as $t \rightarrow \infty$. Hence $\mu_{\mathcal{O}}([0, m]) = 0$ for every centered local class. Because Ω is cyclic for bounded gauge-invariant local observables, centered local classes are dense in $\Omega^{\perp}$. Therefore the spectral projector $P_{(0,m)}$ vanishes on a dense subspace of $\Omega^{\perp}$ and hence on all of $\Omega^{\perp}$. Thus

$$\text{Spec}(H) \subset \{0\} \cup [m, \infty),$$

as claimed. $\square$

Proposition F.35 (Fixed-lattice positive-time local algebra is cyclic). *Let μ_{β} be a translation-invariant reflection-positive infinite-volume Wilson Gibbs state and let $(\mathcal{H}_{\beta}, \Omega_{\beta}, H_{\beta})$ be its OS reconstruction on the bounded gauge-invariant positive-time cylinder algebra $\mathcal{A}_{+,\beta}^{\text{loc}}$. Then Ω_{β} is cyclic for bounded gauge-invariant local cylinder observables.*

Proof. By construction, $\mathcal{H}_{\beta}$ is the completion of $\mathcal{A}_{+,\beta}^{\text{loc}}/\mathcal{N}_{\beta}$, where $\mathcal{N}_{\beta}$ is the OS null ideal. Hence the classes of bounded gauge-invariant local positive-time cylinder observables are dense in $\mathcal{H}_{\beta}$, and $\Omega_{\beta} = [1]$ is cyclic for that local algebra. $\square$

Corollary F.36 (Gap bound in the admissible window). *Assume W is positive type and $R_0 < 1$. If f is admissible then by Corollary F.24 and Theorem F.34,*

$$\boxed{\Delta_{\text{lat}}(\beta) \geq -\log(R_0) = \log\left(\frac{N_b C_{\text{int}} K_0}{K_2}\right) > 0.} \quad (\text{F31})$$

8. Local non-toy majorant: exact tile integration vs convolution power

This is the hard “non-toy” step on the current critical path: a rigorous one-block comparison between an *actual* $b \times b$ block integration in the full d -dimensional theory and an explicit positivity-preserving convolution/power map.

a. Sup normalization

Lemma F.37 (Positive definite functions are maximized at the identity). *If W is positive definite then $|W(g)| \leq W(e)$ for all g . If also $W \geq 0$ then $0 \leq W(g) \leq W(e)$.*

Definition F.38 (Sup-normalized weight). Assume $W \geq 0$ is positive type and $W(e) > 0$. Define $\widehat{W}(g) := W(g)/W(e)$. Then $\widehat{W}$ is positive type and $0 \leq \widehat{W} \leq 1$.

b. Tile factors

Fix a $b \times b$ tile T of plaquettes in some (μ, ν) plane. Let $U_{\partial T}$ be boundary links and $U_{\text{int}(T)}$ internal links. Define the *full* internal block factor

$$\mathcal{Z}_T^{\text{full}}(U_{\partial T}) := \int \left(\prod_{p \supset \ell, \ell \in \text{int}(T)} \widehat{W}(U_p) \right) dU_{\text{int}(T)}, \quad (\text{F32})$$

where the product runs over all plaquettes containing at least one internal link. Define the 2D-reduced factor (drop all out-of-plane plaquettes)

$$\mathcal{Z}_T^{(2D)}(U_{\partial T}) := \int \left(\prod_{p \subset T} \widehat{W}(U_p) \right) dU_{\text{int}(T)}. \quad (\text{F33})$$

Lemma F.39 (Dropping out-of-plane plaquettes majorizes pointwise). *For every boundary configuration $U_{\partial T}$,*

$$\mathcal{Z}_T^{\text{full}}(U_{\partial T}) \leq \mathcal{Z}_T^{(2D)}(U_{\partial T}).$$

c. 2D tile factor equals a convolution power

For integrable f, h on G , define convolution $(f * h)(x) := \int f(xy^{-1})h(y) dy$ and f^{*n} the n -fold convolution power.

Lemma F.40 (2D tile factor is a convolution power). *Assume $\widehat{W}$ is central and positive type so*

$$\widehat{W}(g) = \sum_R d_R \widehat{F}_R \chi_R(g), \quad \widehat{F}_R \geq 0.$$

Let $\text{Hol}_{\partial T}(U_{\partial T})$ be the boundary holonomy. Then

$$\mathcal{Z}_T^{(2D)}(U_{\partial T}) = \widehat{W}^{*N_b}(\text{Hol}_{\partial T}(U_{\partial T})).$$

Theorem F.41 (Tile majorant by convolution/power map). *Let W be nonnegative, central, and positive type, and let $\widehat{W} = W/W(e)$. Then for every tile T ,*

$$\mathcal{Z}_T^{\text{full}}(U_{\partial T}) \leq \widehat{W}^{*N_b}(\text{Hol}_{\partial T}(U_{\partial T})).$$

Proof. Lemma F.39 and Lemma F.40. □

9. Global positivity-preserving coarse object and the g_b window

a. A global majorant partition function (planar reduction)

Let $\Lambda \subset \mathbb{Z}^d$ be finite. Write

$$Z_\Lambda(W) = W(e)^{|P(\Lambda)|} Z_\Lambda(\widehat{W}), \quad Z_\Lambda(\widehat{W}) = \int \prod_{p \in P(\Lambda)} \widehat{W}(U_p) dU.$$

Fix an orientation $(1, 2)$ and let $P_{12}(\Lambda)$ be the set of $(1, 2)$ plaquettes. Since $0 \leq \widehat{W} \leq 1$,

$$Z_\Lambda(\widehat{W}) \leq \int \prod_{p \in P_{12}(\Lambda)} \widehat{W}(U_p) dU. \tag{F34}$$

The RHS factorizes into 2D slices over fixed transverse coordinates. Exact $b \times b$ blocking on each slice yields the coarse 2D model with plaquette weight $\widehat{W}^{*N_b}$. Thus the explicit positivity-preserving majorant partition function

$$Z_{\Lambda, b}^{\text{maj}}(W) := W(e)^{|P(\Lambda)|} \prod_{x_\perp} Z_{\Lambda_b}^{(2D)}(\widehat{W}^{*N_b}) \tag{F35}$$

satisfies

$$Z_\Lambda(W) \leq Z_{\Lambda, b}^{\text{maj}}(W). \tag{F36}$$

b. Coefficient power map and activity contraction

If W is central positive type in the canonical form (F10) with $a_R \geq 0$, then convolution powers give

$$a_R^{(b)} = \frac{a_R^{N_b}}{d_R^{2N_b-2}} \quad (R \neq \mathbf{1}). \quad (\text{F37})$$

Hence

$$f^{(b)} \leq f^{N_b} \leq (1 + f)^{N_b} - 1 = g_b(f). \quad (\text{F38})$$

This is the precise link between the convolution-power coarse object and the KP admissibility envelope $f \mapsto g_b(f)$.

c. A full-dimensional assembly tool: generalized Hölder/Finner

Lemma F.42 (Generalized Hölder (Finner) inequality). *Let (Ω, μ) be a product probability space with coordinates $(x_v)_{v \in V}$. Let $\{F_i\}_{i \in I}$ be nonnegative functions where F_i depends only on the coordinate subset $S_i \subset V$. If exponents $p_i \in [1, \infty]$ satisfy, for each coordinate $v \in V$,*

$$\sum_{i: v \in S_i} \frac{1}{p_i} \leq 1, \quad (\text{F39})$$

then

$$\int_{\Omega} \prod_{i \in I} F_i d\mu \leq \prod_{i \in I} \|F_i\|_{L^{p_i}(\mu)}. \quad (\text{F40})$$

Remark F.43 (Why this matters). Theorem F.41 is *local*. Lemma F.42 is the clean analytic device to glue such local majorants into a *global* d -dimensional positivity-preserving decimation, with a single geometric price: the maximum overlap number m (how many chosen blocks share a link variable). Choosing a block cover and optimizing $m(d, b)$ is the next concrete step for a Tier-3 entrance attempt.

d. A symmetric all-orientation majorant via Finner

The planar reduction (F34) is extremely lossy (it discards most plaquettes). A less wasteful alternative uses Lemma F.42 across all plaquette orientations.

For each orientation (μ, ν) , set

$$F_{\mu\nu}(U) := \prod_{p \in P_{\mu\nu}(\Lambda)} \widehat{W}(U_p), \quad Z_{\mu\nu} := \int F_{\mu\nu}(U) dU.$$

Then

$$Z_\Lambda(\widehat{W}) = \int \prod_{1 \leq \mu < \nu \leq d} F_{\mu\nu}(U) dU.$$

Fix $p_{\mu\nu} = d - 1$ for all (μ, ν) . Each link variable in direction μ appears only in the factors $F_{\mu\nu}$ with $\nu \neq \mu$, i.e. in exactly $(d - 1)$ such factors. Hence the Finner condition (F39) holds with equality, and Lemma F.42 gives

$$Z_\Lambda(\widehat{W}) \leq \prod_{\mu < \nu} \|F_{\mu\nu}\|_{L^{d-1}}. \quad (\text{F41})$$

Since $0 \leq \widehat{W} \leq 1$, we have $F_{\mu\nu}^{d-1} \leq F_{\mu\nu}$, hence

$$\|F_{\mu\nu}\|_{L^{d-1}} = \left(\int F_{\mu\nu}^{d-1} \right)^{1/(d-1)} \leq Z_{\mu\nu}^{1/(d-1)}.$$

Therefore

$$Z_\Lambda(\widehat{W}) \leq \prod_{\mu < \nu} Z_{\mu\nu}^{1/(d-1)}. \quad (\text{F42})$$

Each $Z_{\mu\nu}$ factorizes into 2D slices and exact $b \times b$ blocking on each slice yields the coarse 2D model with plaquette weight $\widehat{W}^{*N_b}$ (as in (F35)). Thus (F42) gives a global positivity-preserving majorant using *all* orientations, at the cost of fractional exponents.

e. Explicit 2×2 tiling and overlap constant $m(4, 2) = 6$

For $b = 2$, tile each oriented plane (μ, ν) by disjoint 2×2 plaquette blocks whose lower-left corners have even coordinates in directions μ, ν . Let $\mathcal{T}_{\mu\nu}$ be the family of such tiles in the (μ, ν) planes, and let $\mathcal{T} := \bigcup_{\mu < \nu} \mathcal{T}_{\mu\nu}$.

Lemma F.44 (Overlap constant for the 2×2 tiling in $d = 4$). *In $d = 4$ with the above disjoint 2×2 tiling, each fine link variable belongs to the support of at most*

$$m(4, 2) = 6$$

tile factors in $\{\mathcal{Z}_T^{\text{full}} : T \in \mathcal{T}\}$.

Proof. Fix a link ℓ oriented in direction μ . It lies in plaquettes only in planes (μ, ν) with $\nu \neq \mu$ (there are $d - 1 = 3$ such planes in $d = 4$). For each such plane (μ, ν) , the link ℓ is contained in exactly two plaquettes of that plane (one on each side of ℓ), and the 2×2 tiling is disjoint within the plane, so those two plaquettes lie in at most two tiles of $\mathcal{T}_{\mu\nu}$. Hence ℓ meets at most $2(d - 1) = 6$ tiles in total. $\square$

Remark F.45. The overlap constant $m(d, b)$ is the only geometry that enters when gluing local tile bounds via Lemma F.42: taking $p_T = m(d, b)$ for all tile factors makes the Finner condition (F39) hold.

10. Invariant observable algebra and gauge-covariant blocking

Let $\mathcal{U}_\Lambda := G^{E(\Lambda)}$ and let gauge transformations act by $(g \cdot U)_{x \rightarrow y} = g(x)U_{x \rightarrow y}g(y)^{-1}$. Let $\mathfrak{A}_\Lambda$ be the $*$ -algebra of bounded gauge-invariant cylinder functions on $\mathcal{U}_\Lambda$.

Proposition F.46 (Wilson loop generators). *$\mathfrak{A}_\Lambda$ is generated (as a $*$ -algebra) by Wilson loop characters $W_{R, \gamma}(U) = \chi_R(U(\gamma))$.*

Proposition F.47 (Blocking pullback preserves the invariant algebra). *Let $\pi : \mathcal{U}_\Lambda \rightarrow \mathcal{U}_{\Lambda_b}$ be a gauge-covariant blocking map built from parallel transport (coarse links are products of fine links along fixed paths). Then $\pi^* : \mathfrak{A}_{\Lambda_b} \rightarrow \mathfrak{A}_\Lambda$ is a $*$ -algebra homomorphism. In particular, the pullback of a coarse Wilson loop is a gauge-invariant function of fine holonomies and hence lies in the loop-generated algebra.*

Remark F.48 (Uniqueness mechanism). The entrance/non-collapse and UV matching hypotheses imply uniqueness of macroscopic laws on $\mathcal{O}_\ell$ by KP/Dobrushin contraction; see Theorem F.254.

Remark F.49 (Structural “parameter collapse”). Any coupling parameter that changes only the presentation of variables but leaves the induced functional on $\mathfrak{A}_\Lambda$ unchanged is redundant. This is the natural lattice formalization of a *redundant reparameterization*: different parameter choices that induce the same functional on $\mathfrak{A}_\Lambda$ are identified and carry no physical content.

11. A first quantitative remainder variable (KP-basin stability)

To compare exact block integration to the convolution-power local reference, define the defect for the sup-normalized weight $\widehat{W}$:

$$\delta(g) := 1 - \widehat{W}(g) \in [0, 1].$$

If W is positive type in the canonical form (F10) with $a_R \geq 0$, then $W(e) = 1 + f$ and

$$\sup_{g \in G} \delta(g) \leq \frac{2f}{1+f} \leq 2f. \tag{F43}$$

For a $b \times b$ tile T , let $P_\perp(T)$ be the set of out-of-plane plaquettes touching internal links. Then

$$|P_\perp(T)| \leq 4(d-2)b(b-1) \quad (\text{F44})$$

(because T has $2b(b-1)$ internal links and each belongs to $2(d-2)$ out-of-plane plaquettes). A crude bound gives

$$\left| 1 - \prod_{p \in P_\perp(T)} \widehat{W}(U_p) \right| \leq |P_\perp(T)| \sup \delta \leq 8(d-2)b(b-1)f \quad (f \leq 1). \quad (\text{F45})$$

This is the first explicit input for bounding the polymer remainder when upgrading the local majorant to a full KP-norm comparison.

12. From partition-function majorant to measure-level control of coarse observables

The local non-toy majorant (Section F 8) and the global partition-function majorant (Section F 9) support an alternative measure-level domination approach for Tier-3 comparisons of coarse observables; we record it for completeness but do not use it in the primary proof. To compare blocked expectations, one needs either

1. a genuine *one-sided* domination of the blocked *measure* by a positivity-preserving coarse measure (rare and very strong), or
2. a representation of the exact blocked measure as a *local* coarse reference measure times a *KP-small polymer correction*, yielding quantitative stability bounds for local observables.

The second approach is the natural “KP-small correction” upgrade and is what we formalize here.

a. Local coarse reference measure

Let Λ_b be the blocked lattice (spacing b). Let $U^b \in G^{E(\Lambda_b)}$ denote coarse link variables and U_p^b the coarse plaquette holonomy. Define the local (positivity-preserving) coarse plaquette weight

$$W_b(g) := \widehat{W}^{*N_b}(g), \quad (\text{F46})$$

and the corresponding local coarse Gibbs measure

$$d\mu_{\Lambda_b}^{\text{loc}}(U^b) := \frac{1}{Z_{\Lambda_b}^{\text{loc}}} \left(\prod_{p \in P(\Lambda_b)} W_b(U_p^b) \right) dU^b, \quad (\text{F47})$$

with dU^b Haar probability measure on coarse links. If $\widehat{W}$ is positive type, then so is W_b , hence $\mu_{\Lambda_b}^{\text{loc}}$ is reflection positive (Proposition F.33).

b. Exact blocked measure and polymer remainder ansatz

Let $\pi : \mathcal{U}_{\Lambda} \rightarrow \mathcal{U}_{\Lambda_b}$ be a gauge-covariant blocking map (Section F.10). Let μ_{Λ} be the fine-lattice gauge measure with plaquette weight $\widehat{W}$. Define the induced (pushforward) coarse measure

$$\mu_{\Lambda_b}^{\text{ex}} := \pi_{\#} \mu_{\Lambda}. \quad (\text{F48})$$

In general $\mu_{\Lambda_b}^{\text{ex}}$ is *nonlocal* on Λ_b . We package that nonlocality as a polymer remainder around the local reference (F47).

Definition F.50 (Polymer remainder (coarse)). We say that $\mu_{\Lambda_b}^{\text{ex}}$ admits a polymer remainder representation around $\mu_{\Lambda_b}^{\text{loc}}$ if there exist bounded functions $\{R_P\}$ indexed by finite connected plaquette sets $P \subset P(\Lambda_b)$ such that

$$d\mu_{\Lambda_b}^{\text{ex}} = \frac{1}{Z_{\text{rem}}} \left(\prod_{p \in P(\Lambda_b)} W_b(U_p^b) \right) \left(\prod_{P \in \mathcal{P}(\Lambda_b)} (1 + R_P(U_P^b)) \right) dU^b, \quad (\text{F49})$$

where $\mathcal{P}(\Lambda_b)$ is a chosen polymer family (e.g. all connected plaquette sets) and Z_{rem} is the normalizing constant. Define the polymer activity bound

$$\rho(P) := \sup_{U_P^b} |R_P(U_P^b)|. \quad (\text{F50})$$

Remark F.51 (What is non-toy here). The existence of (F49) is a standard consequence of rewriting a nonlocal effective action as a polymer gas. The hard content is to control $\rho(P)$ in a KP norm. The defect bounds in Section F.11 are precisely the first quantitative handle needed for that control.

c. KP majorant for the remainder and observable stability

Given $\rho(P)$, define a KP majorant for the remainder

$$q_{\text{rem}} := \sup_{p_0 \in P(\Lambda_b)} \sum_{P \ni p_0} \rho(P) e^{\theta|P|}. \quad (\text{F51})$$

For a finite plaquette set $S \subset P(\Lambda_b)$ define the localized majorant

$$q_{\text{rem}}(S) := \sup_{p_0 \in S} \sum_{P \ni p_0} \rho(P) e^{\theta|P|}. \quad (\text{F52})$$

Theorem F.52 (KP-small remainder controls coarse observables). *Assume $q_{\text{rem}} < 1$ in (F51) and that (F49) holds. Let $\mathcal{O}$ be any bounded gauge-invariant observable on Λ_b supported on a finite plaquette set S . Then*

$$\left| \langle \mathcal{O} \rangle_{\mu_{\Lambda_b}^{\text{ex}}} - \langle \mathcal{O} \rangle_{\mu_{\Lambda_b}^{\text{loc}}} \right| \leq 2 \|\mathcal{O}\|_{\infty} \frac{q_{\text{rem}}(S)}{1 - q_{\text{rem}}}. \quad (\text{F53})$$

In particular, in a translation-invariant setting one has $q_{\text{rem}}(S) \leq |S| q_{\text{rem}}$, yielding

$$\left| \langle \mathcal{O} \rangle_{\text{ex}} - \langle \mathcal{O} \rangle_{\text{loc}} \right| \leq 2 \|\mathcal{O}\|_{\infty} \frac{|S| q_{\text{rem}}}{1 - q_{\text{rem}}}.$$

Proof. Expand both the numerator and denominator of $\langle \mathcal{O} \rangle_{\mu^{\text{ex}}}$ using the polymer product in (F49). By inclusion–exclusion/Mayer expansion, contributions to the difference from $\langle \mathcal{O} \rangle_{\mu^{\text{loc}}}$ come from polymer clusters that intersect S . The KP condition $q_{\text{rem}} < 1$ implies absolute convergence and bounds the sum of connected clusters touching S by a geometric series with ratio q_{rem} . This yields (F53). $\square$

d. A crude universal bound from the defect δ

The out-of-plane plaquette sector enters through factors $\widehat{W}(U_p) = 1 - \delta(U_p)$ with $\delta \in [0, 2f]$ (Equation (F43)). Treating each such defect insertion as an elementary polymer and bounding by $\sup \delta \leq 2f$ yields the crude activity estimate

$$\rho(P) \leq (2f)^{|P|}, \quad (\text{F54})$$

and hence, using the connected-set count (F19),

$$q_{\text{rem}} \leq \sum_{n \geq 1} C_{\text{count}}^{n-1} (2fe^{\theta})^n = \frac{2fe^{\theta}}{1 - 2fe^{\theta}C_{\text{count}}}. \quad (\text{F55})$$

In particular, a sufficient condition for $q_{\text{rem}} < 1$ is

$$2fe^{\theta}C_{\text{count}} < \frac{1}{2}. \quad (\text{F56})$$

This bound is intentionally conservative.

Eliminating the one-plaquette sector: a quadratic remainder majorant If the local reference $\mu_{\Lambda_b}^{\text{loc}}$ is chosen so that all *single-plaquette* (one-body) terms of the exact blocked weight are absorbed into W_b (equivalently, we take $\rho(P) = 0$ for $|P| = 1$ in the polymerization of the Radon–Nikodým derivative), then the KP majorant starts at $n = 2$ and becomes manifestly quadratic:

$$q_{\text{rem}} \leq \sum_{n \geq 2} C_{\text{count}}^{n-1} (2fe^\theta)^n = \frac{C_{\text{count}} (2fe^\theta)^2}{1 - C_{\text{count}} 2fe^\theta}. \quad (\text{F57})$$

In particular, $q_{\text{rem}} < 1$ follows from $C_{\text{count}} 2fe^\theta < \frac{1}{2}$.

Flux/foam admissibility and a dimension-suppression refinement Write the positive-type expansion in normalized characters $\varphi_R = \chi_R/d_R$:

$$\widehat{W}(g) = 1 + \sum_{R \neq \mathbf{1}} a_R \varphi_R(g), \quad a_R \geq 0, \quad f = \sum_{R \neq \mathbf{1}} a_R.$$

Let $d_{\min} := \min_{R \neq \mathbf{1}} d_R$. Haar orthogonality of matrix coefficients implies, schematically,

$$\int_G \varphi_R(gh) \varphi_R(g^{-1}k) dg = \frac{1}{d_R^2} \varphi_R(hk)$$

(and vanishes unless representation labels match), so each internal link integration that glues two nontrivial plaquette insertions produces at most a factor $1/d_{\min}^2$ in absolute value.

For a connected plaquette polymer P with $|P| = n \geq 2$, choose a spanning tree in the adjacency graph of P (edges correspond to shared links). Integrating sequentially along the tree links yields at least $(n - 1)$ such orthogonality gains. Consequently one has the conservative bound

$$\rho(P) \lesssim d_{\min}^2 \left(\frac{2f}{d_{\min}^2} \right)^{|P|} \quad (|P| \geq 2), \quad (\text{F58})$$

and hence, after summing as in (F57),

$$q_{\text{rem}} \lesssim d_{\min}^2 \frac{C_{\text{count}} (2fe^\theta/d_{\min}^2)^2}{1 - C_{\text{count}} 2fe^\theta/d_{\min}^2}. \quad (\text{F59})$$

For an arbitrary compact connected simple G one still has a strictly positive lower bound $d_{\min}(G) \geq 2$ for the dimension of the smallest nontrivial irreducible representation. Thus the refinement (F59) yields at least as strong a remainder majorant as (F57), and typically a strictly stronger one. For the classical series the explicit values are recorded in the audited appendices.

Remark F.53 (Interpretation). Theorem F.52 is the promised upgrade from a partition-function majorant to *measure-level* control: it compares *expectations of coarse observables*

under the exact blocked measure and the positivity-preserving local reference, provided the multi-loop remainder is KP-small. Closing the remainder bound sharply (improving (F54)) is the next technical battleground for a true Tier-3 entrance theorem.

13. Multiscale iteration inside the KP basin: stability of (q, q_{rem})

This section packages the iteration-friendly consequence of the preceding modules:

Once an exact blocked specification enters the KP admissibility window for the single-plaquette activity (and the one-plaquette sector is absorbed into the local reference), both (i) the KP majorant q and (ii) the measure-level remainder majorant q_{rem} remain valid at all later coarse scales, and in fact collapse extremely fast.

This is the lattice-analytic form of a *certified-domain* viewpoint: the admissibility majorant and its remainder majorant are fixed (regime-locked), and the exact RG step is only invoked inside the domain where these quantitative bounds hold.

a. Two majorantes at scale n

Fix a blocking factor b and write $N_b = b^2$. At scale n (lattice spacing $a_n = b^n a$) suppose the effective single-plaquette weight is positive type and (after canonical normalization) has activity f_n . Define

$$q_n := q(f_n).$$

After one exact block step, choose the local coarse reference to be the convolution-power decimation $W_{n+1}^{\text{loc}} := \widehat{W}_n^{*N_b}$ (Section F 12). Denote by μ_{n+1}^{ex} the exact blocked measure and μ_{n+1}^{loc} the local reference measure. Assume the Radon–Nikodým derivative admits a polymerization in the sense of Definition F.50, and define

$$q_{n+1}^{\text{rem}} := q_{\text{rem}}(\mu_{n+1}^{\text{ex}} \parallel \mu_{n+1}^{\text{loc}}).$$

b. One-step propagation bounds

Proposition F.54 (One-step two-majorant propagation). *Assume $\overline{K}_{\text{act}} = 1$ (convolution-power projection) and the drift hypothesis of Lemma F.21. Let $f_n \in (0, 1)$. Then after one*

exact step:

1. (**KP majorant contracts**) If $C_{\text{act}} K_{\text{proj}}(u_n)u_n \leq 1$ for $u_n := f_n^{N_b}$, then

$$f_{n+1} \leq 2f_n^{N_b}, \quad \text{hence} \quad q_{n+1} = q(f_{n+1}) \leq q(2f_n^{N_b}) < 1. \quad (\text{F60})$$

In particular, if $f_n \leq f_{\text{adm}}$ then $f_{n+1} \ll f_{\text{adm}}$ and the KP regime is preserved.

2. (**Remainder majorant is $O(f_n^2)$**) If the one-plaquette sector is absorbed into the local reference (so the polymer remainder starts at $|P| \geq 2$), then the crude quadratic majorant bound gives

$$q_{n+1}^{\text{rem}} \leq \Phi(f_n) := \frac{C_{\text{count}} (2f_n e^\theta)^2}{1 - C_{\text{count}} 2f_n e^\theta}, \quad (\text{F61})$$

and with the flux/foam (dimension-suppressed) refinement,

$$q_{n+1}^{\text{rem}} \lesssim d_{\min}^2 \frac{C_{\text{count}} (2f_n e^\theta / d_{\min}^2)^2}{1 - C_{\text{count}} 2f_n e^\theta / d_{\min}^2}. \quad (\text{F62})$$

In particular, $f_n \leq f_{\text{adm}}$ implies $q_{n+1}^{\text{rem}} < 1$, so Theorem F.52 applies to local coarse observables at scale $n+1$.

Proof. Item (1) is Lemma F.21 with $u_n = f_n^{N_b}$. Item (2) is (F57) (and (F59)) applied at activity f_n . $\square$

c. **Induction: once in, you stay in (and rapidly deepen)**

Corollary F.55 (Iteration inside the KP basin). *Assume there exists a scale n_0 such that $f_{n_0} \leq f_{\text{adm}}$ and the hypotheses of Proposition F.54 hold for all subsequent steps. Then for all $n \geq n_0$:*

1. $q_n < 1$ (KP admissibility persists) and in fact $f_{n+1} \leq 2f_n^{N_b}$, so $f_n \rightarrow 0$ super-exponentially.
2. The measure-level remainder majorant q_{n+1}^{rem} is well-defined and bounded by $\Phi(f_n) = O(f_n^2)$, hence $q_n^{\text{rem}} \rightarrow 0$.
3. For any bounded gauge-invariant local coarse observable $\mathcal{O}$ supported on a finite plaquette set S at scale $n+1$, one has the explicit comparison bound

$$\left| \langle \mathcal{O} \rangle_{\mu_{n+1}^{\text{ex}}} - \langle \mathcal{O} \rangle_{\mu_{n+1}^{\text{loc}}} \right| \leq 2 \|\mathcal{O}\|_\infty \frac{|S| q_{n+1}^{\text{rem}}}{1 - q_{n+1}^{\text{rem}}}.$$

14. Deterministic global entry for the convolution-power majorant

This section is deliberately *deterministic* and isolates a key simplification: the positivity-preserving convolution-power map $W \mapsto W^{*N_b}$ has a global “mixing-to-uniform” mechanism on compact groups that does *not* rely on small- f . It therefore provides a clean target for a finite-depth entrance transfer argument (Section F 15).

a. Setup: central probability density and its spectral radius

Assume in this section that the plaquette weight is *nonnegative, central, and positive type* and is normalized so that

$$\int_G W(g) dg = 1. \quad (\text{F63})$$

Equivalently, W is a probability density on G . Write its (scalar) Fourier–Peter–Weyl eigenvalues as

$$\lambda_R := \int_G W(g) \overline{\varphi_R(g)} dg \in [0, 1], \quad (R \text{ irrep}), \quad (\text{F64})$$

so that

$$W^{*N}(g) = 1 + \sum_{R \neq \mathbf{1}} d_R^2 \lambda_R^N \varphi_R(g). \quad (\text{F65})$$

In particular, for positive type,

$$f_N := W^{*N}(e) - 1 = \sum_{R \neq \mathbf{1}} d_R^2 \lambda_R^N. \quad (\text{F66})$$

Define the spectral radius

$$\rho(W) := \sup_{R \neq \mathbf{1}} \lambda_R \in [0, 1]. \quad (\text{F67})$$

For any genuinely mixing density (i.e. not supported in a proper closed subgroup and not Haar itself), $\rho(W) < 1$ and $W^{*N} \rightarrow 1$ in L^2 .

b. A sharp L^2 -mixing bound for f_N

Proposition F.56 (Mixing bound). *Let W be as above and assume $\rho(W) < 1$. Then for all integers $N \geq 2$,*

$$f_N \leq \rho(W)^{N-2} (W^{*2}(e) - 1) = \rho(W)^{N-2} \|W - 1\|_2^2. \quad (\text{F68})$$

Proof. From (F66),

$$f_N = \sum_{R \neq 1} d_R^2 \lambda_R^N = \sum_{R \neq 1} d_R^2 \lambda_R^2 \lambda_R^{N-2} \leq \rho(W)^{N-2} \sum_{R \neq 1} d_R^2 \lambda_R^2.$$

By Parseval, $\sum_{R \neq 1} d_R^2 \lambda_R^2 = \|W - 1\|_2^2$, and also $\|W - 1\|_2^2 = W^{*2}(e) - 1$ for central real W . $\square$

Corollary F.57 (Finite-depth deterministic entry for $b = 2$). *Fix $b = 2$ so $N_b = 4$. Let f_{adm} be the KP admissibility ceiling defined by (F22). If $\rho(W) < 1$, then there exists a finite k_{enter} such that*

$$f_{4^{k_{\text{enter}}}} \leq f_{\text{adm}}.$$

A sufficient explicit condition is

$$4^{k_{\text{enter}}} - 2 \geq \frac{\log(f_{\text{adm}}/\|W - 1\|_2^2)}{\log \rho(W)}. \quad (\text{F69})$$

c. Heat-kernel majorant mixing via the Casimir gap (explicit $k(\beta)$ and $\epsilon_n(\beta)$)

This subsection pins an explicit finite-depth schedule for the *majorant mixing* step used in the shortcut viability test: given a fixed target $\delta \in (0, 1)$ (the KP/Plan-A threshold), we construct an explicit integer depth $k(\beta)$ so that

$$D_{\chi^2}(\rho_\beta^{(*L^{2k(\beta)})} \| dg) \leq \delta,$$

for the heat-kernel majorant single-plaquette density ρ_β . Unlike the polynomial bound based on the crude representation-growth envelope $f(\beta) \leq K_G \beta^{m_G+1}$, the present schedule uses the *Casimir gap* and yields a $\log(1/\delta)$ dependence.

Heat-kernel normalization. Let $(K_t)_{t>0}$ denote the (normalized) heat kernel on G for the bi-invariant Laplacian with spectrum $\{C_2(R)\}$. For $\beta > 0$ define the central probability density

$$\rho_\beta(g) := K_{1/\beta}(g), \quad \int_G \rho_\beta dg = 1. \quad (\text{F70})$$

By the semigroup property, for each integer $m \geq 1$,

$$\rho_\beta^{(*m)} = K_{m/\beta}. \quad (\text{F71})$$

Casimir gap constant. Define the (group-dependent) Casimir gap

$$C_{\min} := \min\{C_2(R) : R \neq 1\} > 0. \quad (\text{F72})$$

Lemma F.58 (Exact χ^2 formula for the heat-kernel majorant). *For $m \in \mathbb{N}$,*

$$D_{\chi^2}(\rho_\beta^{(*m)} \parallel dg) = \chi^2(\rho_\beta^{(*m)}) = \sum_{R \neq 1} d_R^2 \exp\left(-\frac{2m}{\beta} C_2(R)\right). \quad (\text{F73})$$

Equivalently,

$$D_{\chi^2}(\rho_\beta^{(*m)} \parallel dg) = f\left(\frac{\beta}{2m}\right), \quad f(x) := \sum_{R \neq 1} d_R^2 e^{-C_2(R)/x}. \quad (\text{F74})$$

Proof. By (F71), $\rho_\beta^{(*m)} = K_{m/\beta}$. In Peter–Weyl coordinates, for central K_t the (scalar) eigenvalues are $\lambda_R(K_t) = e^{-tC_2(R)}$. Thus

$$\chi^2(K_t) = \|K_t - 1\|_2^2 = \sum_{R \neq 1} d_R^2 \lambda_R(K_t)^2 = \sum_{R \neq 1} d_R^2 e^{-2tC_2(R)}.$$

Set $t = m/\beta$ to obtain (F73). The rewriting (F74) is the identity $e^{-2tC_2} = e^{-C_2/(1/(2t))}$. $\square$

Lemma F.59 (L^2 contraction from the Casimir gap). *For all $t \geq s > 0$,*

$$\|K_t - 1\|_2 \leq e^{-C_{\min}(t-s)} \|K_s - 1\|_2. \quad (\text{F75})$$

Proof. Expand $K_t - 1 = \sum_{R \neq 1} d_R e^{-tC_2(R)} \chi_R$ in $L^2(G)$. Then

$$\|K_t - 1\|_2^2 = \sum_{R \neq 1} d_R^2 e^{-2tC_2(R)} = \sum_{R \neq 1} d_R^2 e^{-2(t-s)C_2(R)} e^{-2sC_2(R)} \leq e^{-2C_{\min}(t-s)} \sum_{R \neq 1} d_R^2 e^{-2sC_2(R)},$$

which is exactly (F75) after taking square roots. $\square$

Corollary F.60 (Exponential χ^2 decay). *Fix $s = \frac{1}{2}$ and define the group constant*

$$C_{1/2} := \|K_{1/2} - 1\|_2^2, \quad C_{\text{gap}} := e^{C_{\min}} C_{1/2}. \quad (\text{F76})$$

Then for all integers $m \geq \beta/2$,

$$D_{\chi^2}(\rho_\beta^{(*m)} \parallel dg) = \|K_{m/\beta} - 1\|_2^2 \leq C_{\text{gap}} \exp\left(-\frac{2C_{\min}}{\beta} m\right). \quad (\text{F77})$$

Moreover $C_{1/2} = f(1)$ in (F74) and hence may be bounded using the bound $f(1) \leq K_G$ as $C_{1/2} \leq K_G$ (so $C_{\text{gap}} \leq e^{C_{\min}} K_G$) when an explicit constant is desired.

Proof. Assume $m \geq \beta/2$ so that $t := m/\beta \geq \frac{1}{2}$. Apply Lemma F.59 with $s = \frac{1}{2}$ and square the result:

$$\|K_t - 1\|_2^2 \leq \|K_{1/2} - 1\|_2^2 e^{-2C_{\min}(t-1/2)} = C_{1/2} e^{C_{\min}} e^{-2C_{\min}t}.$$

Substitute $t = m/\beta$ and use (F71). The final inequality $C_{1/2} = f(1) \leq K_G$ uses the definition of f in (F74) and the bound $f(1) \leq K_G$ recorded above. $\square$

Proposition F.61 (Explicit mixing depth $k(\beta)$ for block size L). *Fix a block factor $L \geq 2$ and a target $\delta \in (0, 1)$. Define*

$$m(\beta) := \max\left\{\frac{\beta}{2}, \frac{\beta}{2C_{\min}} \log\left(\frac{C_{\text{gap}}}{\delta}\right)\right\}, \quad k(\beta) := \left\lceil \frac{\log m(\beta)}{2 \log L} \right\rceil. \quad (\text{F78})$$

Then

$$D_{\chi^2}(\rho_\beta^{(*L^{2k(\beta)})} \| dg) \leq \delta. \quad (\text{F79})$$

Proof. Let $m_\beta := L^{2k(\beta)}$. By definition, $m_\beta \geq m(\beta) \geq \beta/2$, so Corollary F.60 applies:

$$D_{\chi^2}(\rho_\beta^{(*m_\beta)} \| dg) \leq C_{\text{gap}} e^{-2C_{\min}m_\beta/\beta}.$$

Also $m_\beta \geq \frac{\beta}{2C_{\min}} \log(\frac{C_{\text{gap}}}{\delta})$ implies $e^{-2C_{\min}m_\beta/\beta} \leq \delta/C_{\text{gap}}$. Multiply to obtain (F79). $\square$

Definition F.62 (Equal-split finite-depth transfer budget). With $k(\beta)$ as in (F78), define

$$\epsilon_n(\beta) := \frac{\delta}{k(\beta)}, \quad n = 0, 1, \dots, k(\beta) - 1. \quad (\text{F80})$$

Then $\sum_{n < k(\beta)} \epsilon_n(\beta) = \delta$.

Remark F.63 (How this plugs into the finite-depth transfer criterion). The finite-depth transfer criterion (F83) requires explicit bounds $\text{Err}_{n+1} \leq \epsilon_{n+1}$ for the first k_{enter} steps. Corollary F.60 provides a concrete *mixing* depth $k(\beta)$ and a canonical slack allocation (F80) satisfying $\sum_{n < k(\beta)} \epsilon_n(\beta) = \delta$. To complete the finite-depth transfer argument one must still bound the actual change-of-measure errors Err_{n+1} by this budget, e.g. by proving per-step L^2 tilt bounds of the form $\text{Err}_{n+1} \lesssim \|\Delta_{n+1}(1)\|_\theta$ and using the already-established shared-edge/outside-factor controls to enforce $\|\Delta_{n+1}(1)\|_\theta \leq \epsilon_{n+1}(\beta)$.

d. Optimizing the Casimir-gap schedule (best reference time s_δ)

The Casimir-gap bound in Corollary F.60 fixed a reference time $s = \frac{1}{2}$ for convenience. Here we record the sharpest version of the same idea for the heat-kernel majorant: choose

the reference time so that the target χ^2 level δ is met exactly at that time. This removes the prefactor C_{gap} from the required convolution power.

Lemma F.64 (Monotonicity and level-set time for $\chi^2(K_t)$). *Let*

$$\Phi(t) := \chi^2(K_t) = \|K_t - 1\|_2^2 = \sum_{R \neq \mathbf{1}} d_R^2 e^{-2tC_2(R)}.$$

Then $\Phi(t)$ is finite for all $t > 0$, continuous, and strictly decreasing in t , with $\Phi(t) \rightarrow \infty$ as $t \downarrow 0$ and $\Phi(t) \rightarrow 0$ as $t \rightarrow \infty$. Consequently, for every $\delta > 0$ there exists a unique $s_\delta > 0$ such that $\Phi(s_\delta) = \delta$.

Proof. Finiteness for $t > 0$ follows from Lemma F.58 and the finiteness of $f(x)$ in (F74) at $x = \frac{1}{2t}$:

$$\Phi(t) = f\left(\frac{1}{2t}\right) < \infty.$$

Termwise continuity of the series is immediate. Strict decrease holds termwise since $C_2(R) > 0$ for $R \neq \mathbf{1}$ and all coefficients are nonnegative. The limit $\Phi(t) \rightarrow 0$ as $t \rightarrow \infty$ follows by dominated convergence from the series representation, because each term decays exponentially and the sum is finite for every fixed $t > 0$. For the $t \downarrow 0$ limit, rewrite using (F74): $\Phi(t) = f(\frac{1}{2t})$, and invoke the recorded growth bound on $f(x)$ as $x \rightarrow \infty$ to conclude $\Phi(t) \rightarrow \infty$. Since Φ is strictly decreasing and continuous with range $(0, \infty)$, the level-set time s_δ exists and is unique. $\square$

Proposition F.65 (Optimal mixing depth via s_δ). *Fix $L \geq 2$ and $\delta \in (0, 1)$. Let s_δ be as in Lemma F.64. Define*

$$m_{\text{opt}}(\beta) := \max\{1, \beta s_\delta\}, \quad k_{\text{opt}}(\beta) := \left\lceil \frac{\log m_{\text{opt}}(\beta)}{2 \log L} \right\rceil. \quad (\text{F81})$$

Then

$$D_{\chi^2}(\rho_\beta^{(*L^{2k_{\text{opt}}(\beta)})} \| dg) \leq \delta. \quad (\text{F82})$$

Moreover, for the heat-kernel majorant one has

$$D_{\chi^2}(\rho_\beta^{(*m)} \| dg) \leq \delta \iff \frac{m}{\beta} \geq s_\delta.$$

Proof. By the semigroup identity (F71), $\rho_\beta^{(*m)} = K_{m/\beta}$. Let $m_\beta := L^{2k_{\text{opt}}(\beta)}$. By definition, $m_\beta \geq m_{\text{opt}}(\beta) \geq \beta s_\delta$, hence $m_\beta/\beta \geq s_\delta$. By monotonicity of $\Phi(t) = \chi^2(K_t)$ (Lemma F.64),

$$D_{\chi^2}(\rho_\beta^{(*m_\beta)} \| dg) = \chi^2(K_{m_\beta/\beta}) \leq \chi^2(K_{s_\delta}) = \delta,$$

proving (F82). The final equivalence is the same strict monotonicity of Φ . $\square$

Definition F.66 (Equal-split budget based on $k_{\text{opt}}(\beta)$). With $k_{\text{opt}}(\beta)$ as in (F81), define

$$\epsilon_n^{\text{opt}}(\beta) := \frac{\delta}{k_{\text{opt}}(\beta)}, \quad n = 0, 1, \dots, k_{\text{opt}}(\beta) - 1.$$

Then $\sum_{n < k_{\text{opt}}(\beta)} \epsilon_n^{\text{opt}}(\beta) = \delta$.

15. Finite-depth entrance transfer: the exact comparison target

This section states (cleanly) what must be proved to transfer the deterministic entry in Section F 14 to the *exact* blocked Yang–Mills action.

a. *The finite-depth reduction*

Let S_0 be the microscopic (fine-lattice) action and let

$$S_{n+1} := B_{\text{exact}}(S_n)$$

denote exact blocking on a fixed scale factor b . Let $S_{n+1}^\square$ denote the local positive-type reference produced by the convolution-power decimation ($W \mapsto W^{*N_b}$) applied to the local plaquette weight extracted from S_n .

Section F 13 already proves the following:

If for some n_0 the scale- n_0 local activity obeys $f_{n_0} \leq f_{\text{adm}}$, then for all $n \geq n_0$ the pair of majorantes (q, q_{rem}) remains admissible, contracts rapidly, and coarse observables are controlled in a KP-small sense relative to the local reference measure.

Thus, the only missing step is to show that such an n_0 exists on the physically relevant trajectory.

Deterministic majorant entry (Section F 14) produces a small integer k_{enter} such that the purely local decimation would have entered:

$$f_{k_{\text{enter}}}^\square \leq f_{\text{adm}}.$$

Therefore the entrance problem reduces to the following finite-depth *transfer* statement: prove that the exact blocked action does not deviate too far from the local decimation during the first k_{enter} steps.

b. A clean transfer criterion (what must be bounded)

A sufficient (and natural) finite-depth criterion is:

$$f_{n+1} \leq f_{n+1}^\square + \text{Err}_{n+1}, \quad \text{Err}_{n+1} \leq \epsilon_{n+1} \text{ for } n = 0, 1, \dots, k_{\text{enter}} - 1, \quad (\text{F83})$$

with a small, explicit error budget ϵ_{n+1} that ensures $f_{k_{\text{enter}}} \leq f_{\text{adm}}$. In the finite-depth certification viewpoint, ϵ_{n+1} is the finite-depth *change-of-measure slack*: it is the maximal permitted deviation before the trajectory loses the admissible envelope.

The work already done in Section F 4 provides a template: the error is controlled by a remainder norm at the *incoming* scale,

$$\text{Err}_{n+1} \leq C_{\text{act}} \|\Delta_{n+1}(1)\|_\theta \lesssim C_{\text{act}} K_{\text{proj}}(f_{n+1}^\square) (f_{n+1}^\square)^2, \quad (\text{F84})$$

once the remainder admits a KP-style polymer bound.

Remark F.67 (Key technical point (handled below)). Outside the KP basin, the remainder does not admit a small-activity expansion *a priori*. The deterministic entry theorem localizes this issue to a *finite* set of early steps. The remainder of this appendix resolves these early steps by replacing the small-activity majorant with the mixing majorant introduced in Section F 16 and by proving the associated transfer bounds, thereby verifying the finite-depth transfer criterion (F83) with explicit constants.

16. An early-step mixing majorant: spectral radius and χ^2 -distance to Haar

The KP majorant q is powerful once f is small, but it is not a good *early-step* majorant in the weak-coupling regime where the plaquette density is sharply peaked at the identity and f is not small. A more structural majorant—suggested by the finite-depth certification viewpoint—is a *mixing majorant* attached to the single-plaquette kernel viewed as a class Markov kernel on G . This section formalizes two such majorantes and records the precise inequalities that would transfer deterministic entry to the exact RG.

Early-step mixing criterion. Under the standing choices above, the *early-step* entrance criterion is formulated in terms of the spectral radius $\rho(W)$ (and the associated conditional χ^2 transfer inequality used to compare exact and reference kernels). Any alternative early-step diagnostic is admissible only if it either (i) is proved to imply this quantitative input with

explicit constants, or (ii) is declared as a genuine scope change with its own transport theorem

a. Central Markov kernels and the spectral-radius majorant

Let $W \geq 0$ be a central (class) function on G with $\int_G W dg = 1$. Define the convolution operator on $L^2(G)$,

$$(T_W f)(x) := \int_G W(y) f(y^{-1}x) dy.$$

If W is positive type, then T_W is self-adjoint, positivity-preserving, and $\|T_W\|_{2 \rightarrow 2} = 1$ with constants as the top eigenspace.

In Peter–Weyl coordinates, for central W the matrix Fourier coefficient is scalar:

$$\widehat{W}(R) = \int_G W(g) \rho_R(g^{-1}) dg = \lambda_R(W) I_{d_R}, \quad \lambda_R(W) \in \mathbb{R}.$$

Equivalently, with our normalization $W(g) = \sum_R d_R \lambda_R(W) \chi_R(g)$ and $\lambda_1(W) = 1$. For positive type, $\lambda_R(W) \geq 0$ for all R .

Definition F.68 (Spectral-radius majorant). For a central probability density W define

$$\rho(W) := \sup_{R \neq 1} \lambda_R(W) \in [0, 1].$$

Lemma F.69 (Convolution powers contract ρ exactly). For central probability densities W ,

$$\lambda_R(W^{*N}) = \lambda_R(W)^N, \quad \rho(W^{*N}) = \rho(W)^N.$$

Proof. The matrix Fourier transform turns convolution into matrix multiplication. For central W , the Fourier coefficients are scalar multiples of the identity, hence multiply as scalars. $\square$

b. χ^2 -distance to Haar and its relation to ρ

Define the χ^2 -distance to Haar (the L^2 distance to the constant density 1):

$$\chi^2(W) := \|W - 1\|_2^2 = \int_G (W(g) - 1)^2 dg.$$

In the character basis,

$$\chi^2(W) = \sum_{R \neq 1} d_R^2 \lambda_R(W)^2. \tag{F85}$$

Lemma F.70 (Dimension lower bound). *Let $d_{\min} := \min\{d_R : R \neq \mathbf{1}\}$. Then for central probability densities W ,*

$$d_{\min}^2 \rho(W)^2 \leq \chi^2(W), \quad \text{hence} \quad \rho(W) \leq \frac{\sqrt{\chi^2(W)}}{d_{\min}}.$$

Proof. By definition $\rho(W) = \sup_{R \neq \mathbf{1}} \lambda_R(W)$. In (F85), choose an R attaining (or approximating) the supremum. Then $\chi^2(W) \geq d_R^2 \rho(W)^2 \geq d_{\min}^2 \rho(W)^2$. $\square$

Proposition F.71 (L^2 -mixing bound for convolution powers). *For $N \geq 2$,*

$$\chi^2(W^{*N}) = \sum_{R \neq \mathbf{1}} d_R^2 \lambda_R(W)^{2N} \leq \rho(W)^{2N-4} \chi^2(W^{*2}).$$

Proof. Factor $\lambda_R(W)^{2N} = \lambda_R(W)^{2N-4} \lambda_R(W)^4 \leq \rho(W)^{2N-4} \lambda_R(W)^4$ and sum over $R \neq \mathbf{1}$. $\square$

c. From L^2 to total variation: a mixing-time bound

For probability densities ν, π on a finite state space, the total variation distance is $d_{\text{TV}}(\nu, \pi) = \frac{1}{2} \sum_x |\nu(x) - \pi(x)|$. In our compact-group setting with Haar probability dg , if W is a density then

$$d_{\text{TV}}(W dg, dg) := \frac{1}{2} \|W - 1\|_{L^1(dg)}. \quad (\text{F86})$$

The Cauchy–Schwarz inequality gives the clean $L^2 \rightarrow L^1$ conversion

$$d_{\text{TV}}(W dg, dg) \leq \frac{1}{2} \|W - 1\|_2 = \frac{1}{2} \sqrt{\chi^2(W)}. \quad (\text{F87})$$

Proposition F.72 (Spectral gap $\Rightarrow$ mixing time on compact groups). *Let W be a central probability density on a compact group G and assume $\rho(W) < 1$. Then for all integers $t \geq 1$,*

$$d_{\text{TV}}(W^{*t} dg, dg) \leq \frac{1}{2} \rho(W)^{t-1} \|W - 1\|_2. \quad (\text{F88})$$

Consequently, for any $\epsilon \in (0, \frac{1}{2})$ the ϵ -mixing time

$$\tau(\epsilon) := \min\{t \geq 1 : d_{\text{TV}}(W^{*t} dg, dg) \leq \epsilon\}$$

satisfies

$$\tau(\epsilon) \leq 1 + \frac{\log(\|W - 1\|_2 / (2\epsilon))}{-\log \rho(W)} \leq 1 + \frac{1}{1 - \rho(W)} \log\left(\frac{\|W - 1\|_2}{2\epsilon}\right). \quad (\text{F89})$$

Proof. The Peter–Weyl expansion gives $\chi^2(W^{*t}) = \sum_{R \neq \mathbf{1}} d_R^2 \lambda_R(W)^{2t}$. For $t \geq 1$, factor $\lambda_R(W)^{2t} = \lambda_R(W)^2 \lambda_R(W)^{2(t-1)} \leq \lambda_R(W)^2 \rho(W)^{2(t-1)}$ and sum:

$$\chi^2(W^{*t}) \leq \rho(W)^{2(t-1)} \sum_{R \neq \mathbf{1}} d_R^2 \lambda_R(W)^2 = \rho(W)^{2(t-1)} \chi^2(W).$$

Take square roots and use (F87). For the final inequality note that $-\log x \geq 1 - x$ for $x \in (0, 1)$. $\square$

Remark F.73 (Relation to the finite-state bound). For a finite, reversible Markov chain with stationary distribution π and spectral gap $\gamma = 1 - \lambda_2$ (or absolute gap $\gamma^* = 1 - \max(|\lambda_2|, |\lambda_n|)$), a standard bound is

$$\tau(\epsilon) \lesssim \frac{1}{\gamma^*} \left(\log \frac{1}{\epsilon} + \frac{1}{2} \log \frac{1}{\pi_*} \right), \quad \pi_* := \min_x \pi(x).$$

The factor involving π_* arises when converting $L^2(\pi)$ control to total variation in the finite-state setting. In our compact-group setting with Haar probability dg , the conversion (F87) is immediate, so the role of the “initial condition” is played by $\|W - 1\|_2$.

d. Why ρ is the right early-step majorant

In a finite-depth certification approach, each admissible RG step is required to preserve a chosen envelope majorant. Inside the KP basin the natural majorant is q ; outside the KP basin q is too crude. The majorant ρ is structural: it measures the maximum nontrivial representation content of the single-plaquette kernel, is bounded in $[0, 1]$, and contracts *globally* under convolution. It therefore provides a concrete, conventional-mathematics target for early-step control.

e. Finite-depth entrance transfer in terms of ρ (exact RG target)

Let W_n^{ex} denote the *single coarse plaquette* density extracted from the exact blocked measure at scale n (by integrating out everything except one coarse plaquette holonomy, normalized to a density). Let $W_n^\square$ denote the deterministic convolution-power decimation at the same scale. Then $W_{n+1}^\square = (W_n^\square)^{*N_b}$ and $\rho(W_{n+1}^\square) = \rho(W_n^\square)^{N_b}$.

Proposition F.74 (A clean ρ -transfer criterion). *Fix a depth k . Assume that for each $n = 0, 1, \dots, k-1$ one has a comparison*

$$\rho(W_{n+1}^{\text{ex}}) \leq \rho(W_{n+1}^{\square}) + \delta_{n+1} \quad \text{with} \quad \delta_{n+1} \leq \frac{1}{2}(\rho(W_n^{\square})^{N_b} - \rho(W_{n+1}^{\square})).$$

Then

$$\rho(W_n^{\text{ex}}) \leq 2\rho(W_n^{\square}) \quad \text{for all } n \leq k.$$

Consequently, by Lemma F.70 and Proposition F.71, the deterministic entry bound of Section F 14 transfers (with explicit constants) to the exact lineage.

Proof. Induct on n . At each step, the hypothesis implies $\rho(W_{n+1}^{\text{ex}}) \leq \rho(W_{n+1}^{\square}) + \frac{1}{2}(\rho(W_n^{\square})^{N_b} - \rho(W_{n+1}^{\square})) \leq \frac{1}{2}\rho(W_n^{\square})^{N_b} + \frac{1}{2}\rho(W_{n+1}^{\square})$. Using $\rho(W_{n+1}^{\square}) = \rho(W_n^{\square})^{N_b}$ gives $\rho(W_{n+1}^{\text{ex}}) \leq \rho(W_{n+1}^{\square})$ up to a factor 2, and the claim follows. $\square$

Remark F.75 (Finite-depth entrance inequality). A key step is to prove, for the first few steps before KP-smallness applies, a bound of the form

$$\rho(W_{n+1}^{\text{ex}}) \leq \rho(W_{n+1}^{\square}) + \delta_{n+1},$$

or a comparable L^2 -mixing bound on $\chi^2(W_{n+1}^{\text{ex}})$. This is a *finite-depth* problem for Wilson-type inputs because Section F 14 shows that k_{enter} is finite and admits an explicit bound. (For illustration only, in baseline Wilson examples at $b = 2$ one typically finds $k_{\text{enter}} \leq 5$, but no numerical value is used.)

17. From measure-level remainder to mixing: an L^2 transfer module

Section F 16 isolates the key entrance goal as a finite-depth control of a mixing majorant (ρ or χ^2) for the *exact* blocked single-plaquette marginal. This section records a concrete, conventional-mathematics bridge:

If one can bound an L^2 change-of-measure between the exact blocked measure and the positivity-preserving local reference, then one gets explicit control of the nontrivial representation eigenvalues, hence of ρ and χ^2 .

The point of this module is to make the early-step target inequality completely explicit and local. In this submission the required non-collapse lower bound on the resulting change-of-measure slack is supplied by the pivot and mixing analysis of Section F 19 and the one-plaquette coefficient discharge (A1) (Proposition F.152, implied by Theorem 1.4).

a. χ^2 divergence and basic transfer inequalities

Let $(X, \mathcal{F})$ be a measurable space and let ν be a probability measure. If $\mu \ll \nu$ with Radon–Nikodým derivative $r := \frac{d\mu}{d\nu}$, define the χ^2 divergence

$$D_{\chi^2}(\mu\|\nu) := \int_X (r - 1)^2 d\nu = \|r - 1\|_{L^2(\nu)}^2. \quad (\text{F90})$$

Lemma F.76 (L^2 change-of-measure control). *Let $\mu \ll \nu$ with density $r = d\mu/d\nu$. Then for any $F \in L^2(\nu)$,*

$$|\mathbb{E}_\mu[F] - \mathbb{E}_\nu[F]| \leq \|F - \mathbb{E}_\nu[F]\|_{L^2(\nu)} \sqrt{D_{\chi^2}(\mu\|\nu)}. \quad (\text{F91})$$

Proof. Write $\mathbb{E}_\mu[F] - \mathbb{E}_\nu[F] = \int (F - \mathbb{E}_\nu[F])(r - 1) d\nu$ and apply Cauchy–Schwarz. $\square$

Lemma F.77 (χ^2 transfer of second moments and variances). *Let $\mu \ll \nu$ with density $r = d\mu/d\nu$ and set $\varepsilon := D_{\chi^2}(\mu\|\nu)$. Then for any $F \in L^4(\nu)$,*

$$|\mathbb{E}_\mu[F] - \mathbb{E}_\nu[F]| \leq \|F - \mathbb{E}_\nu[F]\|_{L^2(\nu)} \sqrt{\varepsilon}, \quad (\text{F92})$$

$$|\mathbb{E}_\mu[F^2] - \mathbb{E}_\nu[F^2]| \leq \|F^2 - \mathbb{E}_\nu[F^2]\|_{L^2(\nu)} \sqrt{\varepsilon} \leq \|F\|_{L^4(\nu)}^2 \sqrt{\varepsilon}. \quad (\text{F93})$$

Consequently, if $\varepsilon \leq 1$ then

$$|\text{Var}_\mu(F) - \text{Var}_\nu(F)| \leq (\|F\|_{L^4(\nu)}^2 + 4\|F\|_{L^2(\nu)}^2) \sqrt{\varepsilon}. \quad (\text{F94})$$

Proof. The mean bound (F92) is Lemma F.76. Apply Lemma F.76 to F^2 to get the first inequality in (F93), and then use $\|F^2 - \mathbb{E}_\nu[F^2]\|_2 \leq \|F^2\|_2 = \|F\|_4^2$.

For the variance, write

$$\text{Var}_\mu(F) - \text{Var}_\nu(F) = (\mathbb{E}_\mu[F^2] - \mathbb{E}_\nu[F^2]) - (\mathbb{E}_\mu[F]^2 - \mathbb{E}_\nu[F]^2).$$

The first difference is bounded by (F93). For the mean term,

$$|\mathbb{E}_\mu[F]^2 - \mathbb{E}_\nu[F]^2| = |\mathbb{E}_\mu[F] - \mathbb{E}_\nu[F]| \cdot |\mathbb{E}_\mu[F] + \mathbb{E}_\nu[F]|.$$

By Cauchy–Schwarz, $|\mathbb{E}_\mu[F]| \leq \|F\|_2 \|r\|_2 = \|F\|_2 \sqrt{1 + \varepsilon}$, and similarly $|\mathbb{E}_\nu[F]| \leq \|F\|_2$. When $\varepsilon \leq 1$, $\sqrt{1 + \varepsilon} \leq \sqrt{2} \leq 2$, so $|\mathbb{E}_\mu[F] + \mathbb{E}_\nu[F]| \leq 3\|F\|_2 \leq 4\|F\|_2$. Combine with (F92) to get (F94). $\square$

Lemma F.78 (χ^2 divergence under conditioning). *Let $\mu \ll \nu$ on $(X, \mathcal{F})$ with density $r = d\mu/d\nu$. Let $\Omega \in \mathcal{F}$ satisfy $\mu(\Omega), \nu(\Omega) > 0$ and write $\mu^\Omega := \mu(\cdot | \Omega)$ and $\nu^\Omega := \nu(\cdot | \Omega)$. Then $\mu^\Omega \ll \nu^\Omega$ with density*

$$\frac{d\mu^\Omega}{d\nu^\Omega} = \frac{\nu(\Omega)}{\mu(\Omega)} r \quad \text{on } \Omega,$$

and

$$D_{\chi^2}(\mu^\Omega \| \nu^\Omega) = \frac{\nu(\Omega)}{\mu(\Omega)^2} \mathbb{E}_\nu[r^2 \mathbf{1}_\Omega] - 1 \leq \frac{\nu(\Omega)}{\mu(\Omega)^2} (1 + D_{\chi^2}(\mu \| \nu)) - 1. \quad (\text{F95})$$

Proof. The conditional density formula is immediate from Bayes' rule. For the divergence, compute

$$D_{\chi^2}(\mu^\Omega \| \nu^\Omega) = \mathbb{E}_{\nu^\Omega} \left[\left(\frac{\nu(\Omega)}{\mu(\Omega)} r - 1 \right)^2 \right] = \frac{1}{\nu(\Omega)} \int_\Omega \left(\frac{\nu(\Omega)}{\mu(\Omega)} r - 1 \right)^2 d\nu = \frac{\nu(\Omega)}{\mu(\Omega)^2} \mathbb{E}_\nu[r^2 \mathbf{1}_\Omega] - 1.$$

The inequality follows from $\mathbb{E}_\nu[r^2 \mathbf{1}_\Omega] \leq \mathbb{E}_\nu[r^2] = 1 + D_{\chi^2}(\mu \| \nu)$. $\square$

b. Application to central single-plaquette marginals

Let w^{ex} and w^{loc} be *central probability densities* on G (e.g. single-plaquette marginals at a given RG scale). Let $\lambda_R(w) := \int_G w(g) \varphi_R(g) dg$ be the normalized character coefficient. Define the spectral-radius majorant $\rho(w) = \sup_{R \neq \mathbf{1}} \lambda_R(w)$ (Definition F.68).

Lemma F.79 (Coefficient transfer from Haar- L^2 density control). *For each irrep $R \neq \mathbf{1}$,*

$$\left| \lambda_R(w^{\text{ex}}) - \lambda_R(w^{\text{loc}}) \right| \leq \frac{1}{d_R} \|w^{\text{ex}} - w^{\text{loc}}\|_{L^2(dg)}. \quad (\text{F96})$$

Consequently,

$$\rho(w^{\text{ex}}) \leq \rho(w^{\text{loc}}) + \frac{1}{d_{\min}} \|w^{\text{ex}} - w^{\text{loc}}\|_2. \quad (\text{F97})$$

Proof. By definition, $\lambda_R(w^{\text{ex}}) - \lambda_R(w^{\text{loc}}) = \int_G \varphi_R(g) (w^{\text{ex}} - w^{\text{loc}})(g) dg$. Apply Cauchy-Schwarz and $\|\varphi_R\|_2^2 = \int \chi_R \overline{\chi_R} / d_R^2 = 1/d_R^2$. Take the supremum over $R \neq \mathbf{1}$. $\square$

The χ^2 divergence relative to w^{loc} controls the Haar- L^2 difference by a trivial bound.

Lemma F.80 (χ^2 divergence implies Haar- L^2 density control). *Assume $w^{\text{ex}} \ll w^{\text{loc}}$ and write $r := w^{\text{ex}}/w^{\text{loc}}$. Then*

$$\|w^{\text{ex}} - w^{\text{loc}}\|_2^2 = \int_G (r - 1)^2 (w^{\text{loc}})^2 dg \leq \|w^{\text{loc}}\|_\infty D_{\chi^2}(w^{\text{ex}} \| w^{\text{loc}}). \quad (\text{F98})$$

Proof. Bound $(w^{\text{loc}})^2 \leq \|w^{\text{loc}}\|_\infty w^{\text{loc}}$. □

Combining Lemmas F.79 and F.80 yields the explicit bound

$$\rho(w^{\text{ex}}) \leq \rho(w^{\text{loc}}) + \frac{\sqrt{\|w^{\text{loc}}\|_\infty}}{d_{\min}} \sqrt{D_{\chi^2}(w^{\text{ex}}\|w^{\text{loc}})}. \quad (\text{F99})$$

c. Bounding χ^2 divergence when the RN factor is bounded by 1

A particularly clean situation occurs when the change of measure is given by a factor R with $0 \leq R \leq 1$ (as happens naturally when “dropping” plaquette factors after sup-normalization).

Lemma F.81 (If $0 \leq R \leq 1$, then χ^2 divergence is controlled by $\mathbb{E}_\nu(R)$). *Let (X, ν) be a probability space and let $R : X \rightarrow [0, 1]$ be measurable with $\mathbb{E}_\nu[R] > 0$. Define the tilted probability measure*

$$d\mu := \frac{R}{\mathbb{E}_\nu[R]} d\nu.$$

Then

$$D_{\chi^2}(\mu\|\nu) = \mathbb{E}_\nu \left[\left(\frac{R}{\mathbb{E}_\nu[R]} - 1 \right)^2 \right] \leq \frac{1}{\mathbb{E}_\nu[R]} - 1. \quad (\text{F100})$$

Proof. Compute $D_{\chi^2}(\mu\|\nu) = \mathbb{E}_\nu[R^2]/\mathbb{E}_\nu[R]^2 - 1$. Since $0 \leq R \leq 1$ implies $R^2 \leq R$, we have $\mathbb{E}_\nu[R^2] \leq \mathbb{E}_\nu[R]$. □

d. Finite-depth entrance transfer rewritten as a local integral bound

Combine (F99) and Lemma F.81. Suppose that at an early RG step $n \mapsto n+1$, the exact blocked single-plaquette marginal w_{n+1}^{ex} can be expressed as a tilt of the local reference marginal w_{n+1}^{loc} by a factor $R_{n+1} \in [0, 1]$:

$$w_{n+1}^{\text{ex}}(g) dg = \frac{R_{n+1}(g)}{\mathbb{E}_{w_{n+1}^{\text{loc}}}[R_{n+1}]} w_{n+1}^{\text{loc}}(g) dg. \quad (\text{F101})$$

Then

$$\rho(w_{n+1}^{\text{ex}}) \leq \rho(w_{n+1}^{\text{loc}}) + \frac{\sqrt{\|w_{n+1}^{\text{loc}}\|_\infty}}{d_{\min}} \sqrt{\frac{1}{\mathbb{E}_{w_{n+1}^{\text{loc}}}[R_{n+1}]} - 1}. \quad (\text{F102})$$

Remark F.82 (Why this is useful). The right-hand side is now expressed in terms of: (i) the deterministic/local contraction $\rho(w_{n+1}^{\text{loc}})$ (which is known exactly for convolution powers), (ii) a mild shape constant $\|w_{n+1}^{\text{loc}}\|_\infty$, and (iii) a single local integral $\mathbb{E}_{w_{n+1}^{\text{loc}}}[R_{n+1}]$.

For Wilson-type inputs, deterministic contraction drives $\rho(w_{n+1}^{\text{loc}})$ to be extremely small after only a few $b = 2$ steps (Section F 14). Thus, to transfer entrance it suffices to prove that the local expectation $\mathbb{E}[R_{n+1}]$ is not absurdly tiny for those few early steps. This is a finite-dimensional problem on a fixed block. In the load-bearing primary chain below it is handled by an analytic representation-theoretic mixing bound; alternatively one may compute the relevant integrals explicitly by finite-dimensional recoupling identities.

18. Local tilt construction for a single coarse plaquette and reduction to a finite block integral

The L^2 transfer module of Section F 17 reduces early-step entrance transfer to verifying, for a *finite depth* of RG steps, a lower bound on a local expectation $\mathbb{E}[R]$ for a factor $R \in [0, 1]$. This section makes that reduction concrete by constructing such a factor for a *single coarse plaquette* and expressing its expectation as a *finite-dimensional block integral*.

a. *Exact vs reduced local measures on a block*

Fix an oriented plane (μ, ν) and let T_L be the $L \times L$ plaquette tile in that plane (with $L = b^{n+1}$ at RG depth $n \mapsto n + 1$).

Definition F.83 (Thin block neighborhood of a coarse plaquette). Let $\mathcal{B}_L$ be the set of fine links needed to support:

- (i) all in-plane plaquettes of T_L , and
- (ii) all out-of-plane plaquettes that share at least one link with an in-plane plaquette of T_L .

Let $P_{\parallel}(\mathcal{B}_L)$ be the set of in-plane plaquettes and $P_{\perp}(\mathcal{B}_L)$ the set of out-of-plane plaquettes in $\mathcal{B}_L$.

Work throughout with a sup-normalized plaquette weight $\widehat{W}$ so that $0 \leq \widehat{W} \leq 1$. (As noted earlier, multiplying each plaquette weight by a constant does not change the Gibbs measure.)

Define two probability measures on the block-link space $G^{E(\mathcal{B}_L)}$:

$$d\nu_L(U) := \frac{1}{Z_L^{\parallel}} \left(\prod_{p \in P_{\parallel}(\mathcal{B}_L)} \widehat{W}(U_p) \right) dU, \quad (\text{F103})$$

$$d\mu_L(U) := \frac{1}{Z_L^{\text{full}}} \left(\prod_{p \in P_{\parallel}(\mathcal{B}_L) \cup P_{\perp}(\mathcal{B}_L)} \widehat{W}(U_p) \right) dU, \quad (\text{F104})$$

where dU is Haar product measure on the block links.

Lemma F.84 (Local change-of-measure factor is in $[0, 1]$). *Let*

$$R_L(U) := \prod_{p \in P_{\perp}(\mathcal{B}_L)} \widehat{W}(U_p) \in [0, 1].$$

Then μ_L is a tilt of ν_L :

$$d\mu_L = \frac{R_L}{\mathbb{E}_{\nu_L}[R_L]} d\nu_L, \quad \text{and} \quad \mathbb{E}_{\nu_L}[R_L] = \frac{Z_L^{\text{full}}}{Z_L^{\parallel}} \in (0, 1]. \quad (\text{F105})$$

Proof. Immediate from the definitions: the only difference between $d\mu_L$ and $d\nu_L$ is the extra factor R_L and the normalizing constants. Since $0 \leq \widehat{W} \leq 1$, one has $0 \leq R_L \leq 1$ and $Z_L^{\text{full}} \leq Z_L^{\parallel}$. $\square$

b. Pushforward to a single coarse plaquette holonomy

Let $\Phi_L : G^{E(\mathcal{B}_L)} \rightarrow G$ be the map that assigns to a block configuration U the holonomy around the *outer boundary* of the tile T_L in the (μ, ν) plane:

$$\Phi_L(U) := \text{Hol}_{\partial T_L}(U).$$

Let w_L^{ex} and $w_L^{\parallel}$ be the pushforward (single-plaquette) densities:

$$w_L^{\text{ex}} dg := (\Phi_L)_{\#} \mu_L, \quad w_L^{\parallel} dg := (\Phi_L)_{\#} \nu_L.$$

Lemma F.85 (Marginal tilt factor is a conditional expectation). *Define*

$$\overline{R}_L(g) := \mathbb{E}_{\nu_L}[R_L \mid \Phi_L = g] \in [0, 1].$$

Then w_L^{ex} is a tilt of $w_L^{\parallel}$:

$$w_L^{\text{ex}}(g) = \frac{\overline{R}_L(g)}{\mathbb{E}_{w_L^{\parallel}}[\overline{R}_L]} w_L^{\parallel}(g), \quad \mathbb{E}_{w_L^{\parallel}}[\overline{R}_L] = \mathbb{E}_{\nu_L}[R_L] = \frac{Z_L^{\text{full}}}{Z_L^{\parallel}}. \quad (\text{F106})$$

Proof. For any bounded measurable F on G ,

$$\int_G F(g) w_L^{\text{ex}}(g) dg = \mathbb{E}_{\mu_L}[F(\Phi_L)] = \frac{\mathbb{E}_{\nu_L}[F(\Phi_L) R_L]}{\mathbb{E}_{\nu_L}[R_L]}.$$

Disintegrate ν_L over Φ_L ; the numerator becomes $\int F(g) \overline{R}_L(g) w_L^{\parallel}(g) dg$. This identifies the density ratio as stated. $\square$

c. A data-processing bound for χ^2 divergence

Lemma F.86 (Data processing for χ^2 divergence). *Let $\mu \ll \nu$ on $(X, \mathcal{F})$ and let $\Phi : X \rightarrow Y$ be measurable. Then*

$$D_{\chi^2}(\Phi_{\#}\mu \parallel \Phi_{\#}\nu) \leq D_{\chi^2}(\mu \parallel \nu).$$

Proof. This is a standard property of f -divergences (here $f(t) = (t - 1)^2$) under Markov kernels. A direct proof is obtained by writing the density ratio of $\Phi_{\#}\mu$ w.r.t. $\Phi_{\#}\nu$ as the conditional expectation of $d\mu/d\nu$ given Φ , and applying Jensen's inequality to the convex function $(t - 1)^2$. $\square$

Combining Lemmas F.84, F.85, F.81, and F.86 yields:

Proposition F.87 (Early-step mixing transfer reduced to a block integral). *With notation as above,*

$$D_{\chi^2}(w_L^{\text{ex}} \parallel w_L^{\parallel}) \leq D_{\chi^2}(\mu_L \parallel \nu_L) \leq \frac{1}{\mathbb{E}_{\nu_L}[R_L]} - 1 = \frac{Z_L^{\parallel}}{Z_L^{\text{full}}} - 1.$$

Consequently,

$$\rho(w_L^{\text{ex}}) \leq \rho(w_L^{\parallel}) + \frac{\sqrt{\|w_L^{\parallel}\|_{\infty}}}{d_{\min}} \sqrt{\frac{Z_L^{\parallel}}{Z_L^{\text{full}}} - 1}. \quad (\text{F107})$$

Remark F.88 (Uniqueness mechanism). The entrance/non-collapse and UV matching hypotheses imply uniqueness of macroscopic laws on $\mathcal{O}_{\ell}$ by KP/Dobrushin contraction; see Theorem F.254.

Remark F.89 (What is genuinely left to bound). Equation (F107) reduces the early-step entrance-transfer inequality to lower-bounding the *finite-dimensional* ratio $Z_L^{\text{full}}/Z_L^{\parallel}$ for the block region $\mathcal{B}_L$ at depths $n = 0, 1, \dots, k_{\text{enter}} - 1$. For Wilson-type inputs, k_{enter} is finite and admits an explicit bound (e.g. Corollary F.57). For illustration (not used in the proof), a numerical evaluation for a baseline Wilson one-plaquette density at block size $b = 2$ typically yields $k_{\text{enter}} \leq 5$ even at large β , so this reduces concretely to a *finite list of finite integrals*. In this submission the required *non-collapse* of the early-step slack ratio is proved analytically by the structural pivot of Section F 19 and the subsequent coefficient-level mixing analysis, culminating in the one-plaquette coefficient discharge (A1) (Proposition F.152, implied by Theorem 1.4). In particular, no numerical verification is used in the proof.

19. Remark: the planar-reference slack collapses at weak coupling (and why we must pivot)

Section F 18 reduced early-step entrance transfer (in the mixing majorant ρ) to a *lower bound* on the block ratio $Z_L^{\text{full}}/Z_L^{\parallel}$. This is a perfectly sharp reduction, but it hides a crucial issue: if the reference ν_L is *purely planar* (i.e. contains no out-of-plane plaquettes), then $Z_L^{\text{full}}/Z_L^{\parallel}$ typically collapses rapidly in weak coupling. This section records an explicit upper bound which already shows that one cannot expect a *uniform-in- β* lower bound for Wilson-type inputs. The correct response is a structural pivot: the reference measure used for the tilt must include out-of-plane plaquettes with *positive exponent*, so that the tilt factor does not lose all mass.

a. A Finner/Hölder upper bound for the planar slack ratio

For $p \geq 1$ define

$$\|\widehat{W}\|_p := \left(\int_G \widehat{W}(g)^p dg \right)^{1/p}.$$

Lemma F.90 (Planar slack has an explicit dimension-free upper bound). *Fix a plane (μ, ν) and tile side L . Let ν_L be the planar reference measure of Section F 18 and $R_L = \prod_{p \in P_{\perp}(\mathcal{B}_L)} \widehat{W}(U_p)$ the out-of-plane product. Then*

$$\mathbb{E}_{\nu_L}[R_L] \leq \|\widehat{W}\|_4^{|P_{\perp}(\mathcal{B}_L)|} = \left(\int_G \widehat{W}(g)^4 dg \right)^{|P_{\perp}(\mathcal{B}_L)|/4}. \quad (\text{F108})$$

In particular, for fixed $d \geq 3$,

$$|P_{\perp}(\mathcal{B}_L)| = 4(d-2)L(L+1),$$

so the right-hand side decays like a stretched power of $\|\widehat{W}\|_4$ with exponent $\asymp L^2$.

Proof. Under ν_L , the out-of-plane link variables appearing in R_L are integrated with Haar product measure. Each out-of-plane plaquette factor depends on at most four such link variables and is bounded by 1. Moreover, each out-of-plane link appears in at most four plaquette factors from $P_{\perp}(\mathcal{B}_L)$ (two with the μ -direction and two with the ν -direction, corresponding to the two “sides” of the link). Thus the incidence hypergraph has maximum degree ≤ 4 . Applying Finner’s inequality (generalized Hölder) with uniform exponents $p_e = 4$

gives

$$\int \prod_{p \in P_\perp} \widehat{W}(U_p) dU_\perp \leq \prod_{p \in P_\perp} \left(\int \widehat{W}(U_p)^4 dU_\perp \right)^{1/4}.$$

For a fixed plaquette p , the holonomy map from the four incident link variables (with Haar product measure) pushes forward to Haar on G , so the L^4 norm equals $\|\widehat{W}\|_4$. This proves (F108). The count $|P_\perp| = 4(d-2)L(L+1)$ follows from: there are $2L(L+1)$ in-plane links in the $L \times L$ tile, and each such link lies in $2(d-2)$ out-of-plane plaquettes. $\square$

b. Implication for Wilson-type weak coupling

For a Wilson-class one-plaquette weight of the form

$$\widehat{W}_\beta(U) := \exp\left(-\beta A(U)\right), \quad A(\mathbf{1}) = 0,$$

one has (by Laplace asymptotics at the identity; see Remark F.94)

$$\int_G \widehat{W}_\beta(U)^4 dU = \int_G \widehat{W}_{4\beta}(U) dU \asymp (4\beta)^{-\dim(G)/2} \quad (\beta \rightarrow \infty). \quad (\text{F109})$$

Combining (F108)–(F109) yields, for fixed L ,

$$\mathbb{E}_{\nu_L}[R_L] \lesssim \beta^{-\frac{\dim(G)}{8}|P_\perp(\mathcal{B}_L)|} \quad (\beta \rightarrow \infty),$$

so the planar slack ratio collapses at weak coupling. In particular, there cannot exist a lower bound on $Z_L^{\text{full}}/Z_L^\parallel$ that is uniform in β for Wilson-type inputs.

Remark F.91 (Structural consequence: planar reference is degenerate at weak coupling). The quantitative “slack” quantity that must remain nondegenerate to transfer entrance is precisely

$$\mathbb{E}_{\nu_L}[R_L] = \frac{Z_L^{\text{full}}}{Z_L^\parallel}.$$

Lemma F.90 shows that the purely planar reference ν_L forces this slack to collapse in weak coupling. Thus the planar reference is *not* a usable comparison reference in the UV scaling regime: it loses too much mass to carry a stable change-of-measure comparison. The appropriate pivot is to use a reference measure that includes out-of-plane plaquettes with positive exponent so that the change-of-measure factor is not catastrophically small.

c. Pivot target: a weighted reference that includes out-of-plane plaquettes

Introduce an interpolation parameter $\alpha \in (0, 1)$ and define a *weighted reference* on the same block:

$$d\nu_L^{(\alpha)}(U) := \frac{1}{Z_L^{(\alpha)}} \left(\prod_{p \in P_{\parallel}(\mathcal{B}_L)} \widehat{W}(U_p) \right) \left(\prod_{p \in P_{\perp}(\mathcal{B}_L)} \widehat{W}(U_p)^{\alpha} \right) dU. \quad (\text{F110})$$

Then the full block measure is the tilt of $\nu_L^{(\alpha)}$ by

$$R_L^{(\alpha)}(U) := \prod_{p \in P_{\perp}(\mathcal{B}_L)} \widehat{W}(U_p)^{1-\alpha} \in [0, 1], \quad d\mu_L = \frac{R_L^{(\alpha)}}{\mathbb{E}_{\nu_L^{(\alpha)}}[R_L^{(\alpha)}]} d\nu_L^{(\alpha)},$$

with

$$\mathbb{E}_{\nu_L^{(\alpha)}}[R_L^{(\alpha)}] = \frac{Z_L^{\text{full}}}{Z_L^{(\alpha)}}.$$

Taking α closer to 1 reduces the exponent $1 - \alpha$ and therefore mitigates slack collapse. The remaining difficulty is to obtain a mixing estimate for the boundary holonomy marginal of $\nu_L^{(\alpha)}$ that is strong enough, over a *finite depth* of steps, to enter the KP basin. This is the next genuinely non-toy entrance-transfer target.

Thin-block leaf factorization: exact slack formula and the right α -scaling A useful (and slightly sobering) structural fact about the specific thin block $\mathcal{B}_L$ defined in Section F 18 is that every out-of-plane plaquette in $P_{\perp}(\mathcal{B}_L)$ carries an *opposite in-plane edge* that is not shared by any other plaquette in the block. Consequently the out-of-plane sector can be integrated out *exactly* as a product of one-plaquette integrals.

For a general nonnegative plaquette weight $\widehat{W} : G \rightarrow (0, 1]$ and exponent $t > 0$ write

$$z(t) := \int_G \widehat{W}(g)^t dg. \quad (\text{F111})$$

Lemma F.92 (Leaf factorization on the thin block). *Let $\mathcal{B}_L$ be the thin block built from the in-plane tile T_L together with the out-of-plane plaquettes that share at least one in-plane link with T_L (Definition F.83). Let $\nu_L^{(\alpha)}$ be the weighted reference (F110) with $\alpha \in (0, 1)$. Then the corresponding block partition functions satisfy the exact identities*

$$Z_L^{(\alpha)} = Z_L^{\parallel} z(\alpha)^{|P_{\perp}(\mathcal{B}_L)|}, \quad Z_L^{\text{full}} = Z_L^{\parallel} z(1)^{|P_{\perp}(\mathcal{B}_L)|}, \quad (\text{F112})$$

and therefore the change-of-measure slack expectation is

$$\mathbb{E}_{\nu_L^{(\alpha)}}[R_L^{(\alpha)}] = \frac{Z_L^{\text{full}}}{Z_L^{(\alpha)}} = \left(\frac{z(1)}{z(\alpha)} \right)^{|P_{\perp}(\mathcal{B}_L)|}. \quad (\text{F113})$$

In particular, for the purely planar reference ($\alpha = 0$, so $z(0) = 1$),

$$\mathbb{E}_{\nu_L}[R_L] = z(1)^{|P_\perp(\mathcal{B}_L)|}. \quad (\text{F114})$$

Proof. Each $p \in P_\perp(\mathcal{B}_L)$ is a plaquette in a plane (μ, ρ) with $\rho \neq \mu, \nu$ that shares a unique in-plane link ℓ of the tile T_L (at perpendicular coordinate 0). Its opposite edge is the translate $\ell^{(\rho, \pm)}$ of that link at perpendicular coordinate ± 1 (in the same in-plane direction as ℓ). By construction of the *thin* block $\mathcal{B}_L$, no plaquettes lying entirely in the shifted in-plane slices are included, hence each translated edge $\ell^{(\rho, \pm)}$ appears in *exactly one* plaquette factor, namely the corresponding p .

Fix all other link variables and integrate first over $\ell^{(\rho, \pm)}$ with Haar measure. Writing the plaquette holonomy as $U_p = A \ell^{(\rho, \pm)} B$ (with A, B depending on the fixed links), Haar invariance gives

$$\int_G \widehat{W}(U_p)^\alpha d\ell^{(\rho, \pm)} = \int_G \widehat{W}(g)^\alpha dg = z(\alpha),$$

independently of A, B . Thus one may integrate out the unique opposite edge for each $p \in P_\perp(\mathcal{B}_L)$ sequentially, factoring out $z(\alpha)^{|P_\perp|}$ and leaving precisely the planar block integral $Z_L^\parallel$. Setting $\alpha = 1$ in the preceding computation gives Z_L^{full} . Taking the ratio yields (F113) and (F114). $\square$

Remark F.93 (Why this matters: it pins down the correct α scaling). Equation (F113) shows that on the thin block the slack depends on α *only through* the one-plaquette integral ratio $z(1)/z(\alpha)$, raised to the power $|P_\perp(\mathcal{B}_L)| \asymp L^2$. Therefore, to keep the χ^2 correction term in (F102) bounded over a finite depth of RG steps with growing L , one needs

$$|P_\perp(\mathcal{B}_L)| \log\left(\frac{z(\alpha)}{z(1)}\right) = O(1), \quad \text{i.e.} \quad 1 - \alpha \lesssim \frac{c}{|P_\perp(\mathcal{B}_L)|} \asymp \frac{c}{L^2}.$$

This is the *reference-stabilizing* choice: keep the slack nondegenerate by letting α approach 1 as the block grows.

A convenient explicit schedule (for $d = 4$) is

$$\alpha(L) := 1 - \frac{\kappa}{|P_\perp(\mathcal{B}_L)|} = 1 - \frac{\kappa}{8L(L+1)}, \quad \kappa \in (0, 1) \text{ fixed.} \quad (\text{F115})$$

Remark F.94 (Weak-coupling scaling of Wilson one-plaquette integrals). Let G be a fixed compact connected Lie group and let $A : G \rightarrow [0, \infty)$ be the Wilson one-plaquette action

density associated to a fixed unitary representation π . By Theorems 1.1 and 2.4 we may and do assume π is faithful by replacing G with $G_{\text{eff}} := G/\ker(\pi)$.

$$A(g) := 1 - \frac{1}{\dim(\pi)} \Re \text{Tr } \pi(g), \quad A(\mathbf{1}) = 0.$$

Define the (sup-normalized) one-plaquette weight and its integral

$$\widehat{W}_\beta(U) := \exp\left(-\beta A(U)\right), \quad z_\beta := \int_G \widehat{W}_\beta(U) dU.$$

By Laplace's method at the unique nondegenerate minimum of A (at $U = \mathbf{1}$, under the effective-group convention), one has as $\beta \rightarrow \infty$

$$z_\beta = c_G \beta^{-\dim(G)/2} \left(1 + O(\beta^{-1})\right), \quad (\text{F116})$$

for an explicit constant $c_G > 0$ depending only on G , π , and the chosen inner product on $\mathfrak{g}$ (equivalently on the Hessian of A at the identity). Consequently, for fixed $\alpha \in (0, 1]$,

$$\frac{z_{\alpha\beta}}{z_\beta} = \alpha^{-\dim(G)/2} \left(1 + o(1)\right) \quad (\beta \rightarrow \infty). \quad (\text{F117})$$

Inserting (F117) into the exact thin-block slack identity (F113) shows that the schedule (F115) yields

$$\mathbb{E}_{\nu_L^{(\alpha(L))}}[R_L^{(\alpha(L))}] \approx \exp\left(-\frac{\dim(G)}{2} \kappa\right) \quad (\beta \text{ large}),$$

uniformly in L .

The asymptotic statements (F116)–(F117) are included only for orientation and are not used in the proofs below; all subsequent bounds are derived from the explicit inequalities proved in this appendix.

Orientation only (not used in the proof). Write $U = \exp(X)$ with $X \in \mathfrak{g}$ near 0, expand $A(\exp X) = \frac{1}{2} \langle X, HX \rangle + O(\|X\|^3)$ with H positive definite, and use the standard Laplace rescaling $X \mapsto X/\sqrt{\beta}$.

Weak-coupling χ^2 -decay for the α -pivot on genuine 4D blocks The thin-block leaf factorization (Lemma F.92) is special to $\mathcal{B}_L$. For a genuine four-dimensional block, the out-of-plane plaquette sector shares edges and does not factor. Nevertheless, in weak coupling one can still prove the required χ^2 decay under the schedule (F115) by combining: (i) *curvature* (rather than edge) Brascamp–Lieb domination, and (ii) a quadratic-form concentration bound (Hanson–Wright) applied to the linearized curvature field.

Transverse energy and RN factor. Define the transverse energy observable

$$X(U) := \sum_{p \in P_\perp(\mathcal{B}_L)} A(U_p), \quad A(g) := 1 - \frac{1}{\dim(\pi)} \Re \text{Tr } \pi(g), \quad (\text{F118})$$

for a fixed unitary representation π (without loss of generality faithful under the effective-group convention). For the sup-normalized Wilson class weights

$$\widehat{W}_\beta(g) := \exp \left(\beta \left(\frac{1}{\dim(\pi)} \Re \text{Tr } \pi(g) - 1 \right) \right) = e^{-\beta A(g)},$$

the α -reference measure (F110) satisfies

$$R_L^{(\alpha)}(U) := \prod_{p \in P_\perp(\mathcal{B}_L)} \widehat{W}_\beta(U_p)^{1-\alpha} = \exp \left(-\beta(1-\alpha) X(U) \right) \in [0, 1]. \quad (\text{F119})$$

Recall the boundary holonomy map $\Phi_L : G^{E(\mathcal{B}_L)} \rightarrow G$ from Section F.18 and define the corresponding α -reference single-plaquette marginal by

$$w_L^{(\alpha)} dg := (\Phi_L)_\# \nu_L^{(\alpha)}.$$

Since $d\mu_L = (R_L^{(\alpha)} / \mathbb{E}_{\nu_L^{(\alpha)}}[R_L^{(\alpha)}]) d\nu_L^{(\alpha)}$, the block-level χ^2 divergence is

$$D_{\chi^2}(\mu_L \| \nu_L^{(\alpha)}) = \frac{\mathbb{E}_{\nu_L^{(\alpha)}}[e^{-2tX}]}{\mathbb{E}_{\nu_L^{(\alpha)}}[e^{-tX}]^2} - 1, \quad t := \beta(1-\alpha).$$

By data processing for χ^2 divergence (Lemma F.86) under Φ_L ,

$$D_{\chi^2}(w_L^{\text{ex}} \| w_L^{(\alpha)}) \leq D_{\chi^2}(\mu_L \| \nu_L^{(\alpha)}) = \frac{\mathbb{E}_{\nu_L^{(\alpha)}}[e^{-2tX}]}{\mathbb{E}_{\nu_L^{(\alpha)}}[e^{-tX}]^2} - 1. \quad (\text{F120})$$

Thus it suffices to control exponential moments of X under $\nu_L^{(\alpha)}$.

Gauge fixing and curvature variables. Fix a spanning tree T on $E(\mathcal{B}_L)$ and impose tree gauge. Let $A = (A_e)_{e \in E_f}$ be exponential coordinates for free edges and let

$$F := dA \in \mathfrak{g}^{P(\mathcal{B}_L)}$$

denote the *linearized curvature* 2-form (cochain differential d). Let Ω_{ε_0} be the small-curvature event of Lemma F.224. On Ω_{ε_0} , Proposition F.226 implies that all centered linear functionals $\langle J, F \rangle$ are sub-Gaussian with variance proxy $(\beta m_-)^{-1}$, uniformly in L .

The key point is that, although the edge Laplacian $\Delta_1 = d^*d$ has small eigenvalues at large L , the curvature covariance involves the projection $\Pi_{\text{ex}} = d\Delta_1^{-1}d^*$ (Lemma F.225), hence has operator norm 1. This removes the long-mode degeneracy and yields uniform control for curvature observables.

Lemma F.95 (Variance scaling for the transverse energy). *Fix $d = 4$ and compact connected G . There exist $\beta_0 = \beta_0(G)$ and $C = C(G)$ such that for all $\beta \geq \beta_0$, all L , and all $\alpha \in [1/2, 1]$,*

$$\text{Var}_{\nu_L^{(\alpha)}}(X) \leq \frac{C}{\beta^2} |P_\perp(\mathcal{B}_L)|. \quad (\text{F121})$$

Proof. Work conditionally on Ω_{ε_0} and then remove the conditioning using Lemma F.224. On Ω_{ε_0} each plaquette holonomy is within a fixed normal neighborhood of the identity, and the class function $A(g)$ admits the Taylor expansion

$$A(\exp \xi) = \frac{1}{2} \langle \xi, \mathcal{Q}\xi \rangle + R(\xi), \quad |R(\xi)| \leq C_R \|\xi\|^3 \quad (\|\xi\| \leq \varepsilon_0), \quad (\text{F122})$$

with $\mathcal{Q} \succeq c_Q I$ depending only on (G, π) . Writing the plaquette logarithm as $\xi_p = \log U_p$, one has $\xi_p = F_p + O(\|A\|^2)$ on Ω_{ε_0} , hence

$$X = \frac{1}{2} \sum_{p \in P_\perp} \langle F_p, \mathcal{Q}F_p \rangle + \text{Rem}, \quad |\text{Rem}| \leq C \sum_{p \in P_\perp} \|F_p\|^3,$$

after enlarging C . The leading term is a quadratic form in the curvature vector F :

$$X_{\text{quad}} := \frac{1}{2} \sum_{p \in P_\perp} \langle F_p, \mathcal{Q}F_p \rangle = \langle F, QF \rangle,$$

where Q is block-diagonal on plaquettes with $\|Q\|_{\text{op}} \leq C$ and $\|Q\|_{\text{HS}}^2 \asymp |P_\perp|$.

By Proposition F.226, F is a sub-Gaussian vector (in the exact 2-form subspace) with parameter $\sigma^2 \asymp (\beta m_-)^{-1}$. The Hanson–Wright inequality for sub-Gaussian vectors therefore yields

$$\text{Var}(X_{\text{quad}} \mid \Omega_{\varepsilon_0}) \leq C_{HW} \sigma^4 \|Q\|_{\text{HS}}^2 \leq \frac{C}{\beta^2} |P_\perp|.$$

For the remainder, sub-Gaussian moment bounds imply $\mathbb{E}[\|F_p\|^6 \mid \Omega_{\varepsilon_0}] \leq C/\beta^3$ uniformly in p , hence by Cauchy–Schwarz,

$$\mathbb{E}[\text{Rem}^2 \mid \Omega_{\varepsilon_0}] \leq C \sum_{p \in P_\perp} \mathbb{E}[\|F_p\|^6 \mid \Omega_{\varepsilon_0}] \leq \frac{C}{\beta^3} |P_\perp|.$$

Thus $\text{Var}(X \mid \Omega_{\varepsilon_0}) \leq (C/\beta^2)|P_\perp|$ after enlarging C . Finally, Lemma F.224 implies $\nu_L^{(\alpha)}(\Omega_{\varepsilon_0}^c) \leq |P(\mathcal{B}_L)|e^{-c\beta}$, so removing the conditioning changes $\text{Var}(X)$ by at most an exponentially small term in β that can be absorbed into C for $\beta \geq \beta_0$. $\square$

Lemma F.96 (Centered mgf bound for the transverse energy). *Under the assumptions of Lemma F.95, there exist constants $\beta_1 = \beta_1(G) \geq \beta_0(G)$ and $C = C(G)$ such that for all $\beta \geq \beta_1$, all L , all $\alpha \in [1/2, 1]$, and all s with $|s| \leq c_1 \beta$,*

$$\log \mathbb{E}_{\nu_L^{(\alpha)}} \left[\exp \left(s(X - \mathbb{E}X) \right) \right] \leq \frac{C}{\beta^2} s^2 |P_\perp(\mathcal{B}_L)|. \quad (\text{F123})$$

Proof. Condition on Ω_{ε_0} and use the decomposition $X = X_{\text{quad}} + \text{Rem}$ from the proof of Lemma F.95. The Hanson–Wright mgf form for sub-Gaussian vectors implies that, for $|s| \leq c_1\beta$,

$$\log \mathbb{E} \left[\exp \left(s(X_{\text{quad}} - \mathbb{E}X_{\text{quad}}) \right) \mid \Omega_{\varepsilon_0} \right] \leq \frac{C}{\beta^2} s^2 |P_{\perp}|.$$

For the remainder, sub-Gaussian moment bounds on F on Ω_{ε_0} imply $\|\text{Rem} - \mathbb{E}\text{Rem}\|_{L^2} \leq C|P_{\perp}|^{1/2}\beta^{-3/2}$, hence by Cauchy–Schwarz

$$\log \mathbb{E} \left[e^{s(\text{Rem} - \mathbb{E}\text{Rem})} \mid \Omega_{\varepsilon_0} \right] \leq \frac{C}{\beta^3} s^2 |P_{\perp}|$$

after restricting to $|s| \leq c_1\beta$ and absorbing constants. Combining these and removing the conditioning as in Lemma F.95 (using Lemma F.224) yields (F123). $\square$

Remark F.97 (Compatibility of the schedule with the mgf radius). Lemma F.96 holds for $|s| \leq c_1\beta$. In the χ^2 -decay application below we take $s = -2t_L$ with $t_L = \beta(1 - \alpha(L)) = \beta\kappa/|P_{\perp}(\mathcal{B}_L)|$. In $d = 4$ one has $|P_{\perp}(\mathcal{B}_L)| = 8L(L+1) \geq 16$ for $L \geq 1$, hence $|s| \leq \beta\kappa/8$. Thus it suffices (and we henceforth assume) that $\kappa \leq \kappa_0(G) := 8c_1(G)$ so that $|s| \leq c_1\beta$ for all L . (If one prefers to allow arbitrary $\kappa \in (0, 1)$, it is enough to check finitely many small L separately and then use the argument below for large L .)

Proposition F.98 (χ^2 decay under the schedule (F115)). *Fix $\kappa \in (0, \kappa_0(G)]$ (cf. Remark F.97) and take the schedule (F115) so that $1 - \alpha(L) = \kappa/|P_{\perp}(\mathcal{B}_L)|$. Set $t_L := \beta(1 - \alpha(L)) = \beta\kappa/|P_{\perp}(\mathcal{B}_L)|$. Then for $\beta \geq \beta_0(G)$,*

$$D_{\chi^2}(w_L^{\text{ex}} \| w_L^{(\alpha(L))}) \leq \frac{C(d, G) \kappa^2}{|P_{\perp}(\mathcal{B}_L)|}, \quad (\text{F124})$$

and in $d = 4$ this is $O(L^{-2})$ since $|P_{\perp}(\mathcal{B}_L)| = 8L(L+1)$.

Proof. By (F120) with $\alpha = \alpha(L)$ it suffices to bound the block-level ratio

$$D_{\chi^2}(\mu_L \| \nu_L^{(\alpha(L))}) = \frac{\mathbb{E}[e^{-2t_L X}]}{\mathbb{E}[e^{-t_L X}]^2} - 1.$$

Write $Y := X - \mathbb{E}_{\nu_L^{(\alpha(L))}} X$. Then

$$\log \frac{\mathbb{E}[e^{-2t_L X}]}{\mathbb{E}[e^{-t_L X}]^2} = \log \mathbb{E}[e^{-2t_L Y}] - 2 \log \mathbb{E}[e^{-t_L Y}] \leq \log \mathbb{E}[e^{-2t_L Y}],$$

since $\log \mathbb{E}[e^{-t_L Y}] \geq 0$ by Jensen (recall $\mathbb{E}Y = 0$ and $e^{-t_L y}$ is convex). Apply Lemma F.96 with $s = -2t_L$ (per Remark F.97) to obtain

$$\log \frac{\mathbb{E}[e^{-2t_L X}]}{\mathbb{E}[e^{-t_L X}]^2} \leq \frac{C}{\beta^2} (2t_L)^2 |P_{\perp}| = \frac{4C \kappa^2}{|P_{\perp}|}.$$

Let $u := 4C\kappa^2/|P_\perp|$. Then

$$D_{\chi^2}(\mu_L \|\nu_L^{(\alpha(L))}) \leq e^u - 1 \leq u e^u.$$

In $d = 4$ one has $|P_\perp| \geq 16$ and $\kappa \leq 1$, hence $u \leq C/4$ and so $e^u \leq e^{C/4}$. Therefore

$$D_{\chi^2}(\mu_L \|\nu_L^{(\alpha(L))}) \leq C'(d, G) \frac{\kappa^2}{|P_\perp|}.$$

Combine with (F120) to conclude (F124). □

The bound (F124) is exactly the local non-toy input required by the χ^2 transfer module in Section F 17 to compare the exact blocked single-plaquette marginal to the α -reference marginal, with a correction of order $|P_\perp|^{-1/2}$ in the spectral-radius majorant ρ and order $|P_\perp|^{-1}$ in χ^2 .

KP-budgeted slack: converting a majorant entry depth into an exact entry depth The deterministic entry mechanism (Proposition F.56 and Corollary F.57) produces, for each microscopic input W , a finite depth k such that the convolution-power reference W^{*4^k} has small activity at the identity,

$$f_{4^k}^{\text{ref}} := W^{*4^k}(e) - 1 \leq f_{\text{adm}}.$$

To transfer this to an exact blocked plaquette marginal, one must control the *normalization loss* coming from the change of measure. The χ^2 transfer bounds in Section F 17 are robust but can be overly conservative for controlling the single scalar $w(e) - 1$ that defines the KP entrance condition in the positive-type sector. A sharper pointwise budget is available because the tilt factor is bounded by 1.

Proposition F.99 (Tilt budget for the activity at the identity). *Let w^{ref} be a probability density on G and let $\bar{R} : G \rightarrow [0, 1]$ be measurable. Define the tilted density*

$$w^{\text{ex}}(g) := \frac{\bar{R}(g)}{\mathbb{E}_{w^{\text{ref}}}[\bar{R}]} w^{\text{ref}}(g), \quad \mathbb{E}_{w^{\text{ref}}}[\bar{R}] := \int_G \bar{R}(g) w^{\text{ref}}(g) dg.$$

Then

$$w^{\text{ex}}(e) - 1 \leq \frac{w^{\text{ref}}(e) - 1}{\mathbb{E}_{w^{\text{ref}}}[\bar{R}]} + \left(\frac{1}{\mathbb{E}_{w^{\text{ref}}}[\bar{R}]} - 1 \right). \quad (\text{F125})$$

Equivalently, with $\delta_0 := \mathbb{E}_{w^{\text{ref}}}[\bar{R}]^{-1} - 1 \geq 0$,

$$w^{\text{ex}}(e) - 1 \leq \delta_0 + (1 + \delta_0)(w^{\text{ref}}(e) - 1). \quad (\text{F126})$$

Proof. Since $\bar{R}(e) \leq 1$ we have $w^{\text{ex}}(e) = \bar{R}(e) w^{\text{ref}}(e) / \mathbb{E}_{w^{\text{ref}}}[\bar{R}] \leq w^{\text{ref}}(e) / \mathbb{E}_{w^{\text{ref}}}[\bar{R}]$. Subtract 1 and split $w^{\text{ref}}(e) = 1 + (w^{\text{ref}}(e) - 1)$. $\square$

Corollary F.100 (A two-line sufficient condition for KP entrance via a reference). *Let f_{adm} be the KP ceiling (F22). Assume $w^{\text{ref}}(e) - 1 \leq f_{\text{ref}}$. If the slack obeys*

$$\delta_0 := \mathbb{E}_{w^{\text{ref}}}[\bar{R}]^{-1} - 1 \leq \frac{f_{\text{adm}} - f_{\text{ref}}}{1 + f_{\text{ref}}}, \quad (\text{F127})$$

then $w^{\text{ex}}(e) - 1 \leq f_{\text{adm}}$, hence (in the positive-type sector) the exact step lies in the KP admissible window. A simpler (slightly stronger) sufficient condition is

$$w^{\text{ref}}(e) - 1 \leq \frac{f_{\text{adm}}}{2} \quad \text{and} \quad \delta_0 \leq \frac{f_{\text{adm}}}{2}.$$

Remark F.101 (What this changes about the α schedule). Remark F.93 focused on keeping the slack *nondegenerate* ($\mathbb{E}[R] = \Theta(1)$) as L grows. Corollary F.100 shows that for an actual entrance transfer into the KP window one needs a much tighter requirement: the normalization loss must be on the same order as the KP ceiling, i.e. $\delta_0 = O(f_{\text{adm}})$. On the thin block, using (F113) and the weak-coupling scaling (F117) from Remark F.94, one has $z(\alpha)/z(1) \sim \alpha^{-\dim(G)/2}$. Thus the slack-stabilizing schedule $\alpha(L) = 1 - \kappa/|P_{\perp}|$ gives

$$\delta_0 = \left(\frac{z(\alpha)}{z(1)} \right)^{|P_{\perp}|} - 1 \approx e^{\frac{\dim(G)}{2} \kappa} - 1.$$

Therefore the KP-budgeted constraint (F127) forces

$$\kappa \lesssim \frac{2}{\dim(G)} \log(1 + O(f_{\text{adm}})) \approx O(f_{\text{adm}}). \quad (\text{F128})$$

In particular, in the baseline window $f_{\text{adm}} \approx 1.668 \times 10^{-3}$ one is in the regime $\kappa \sim 10^{-3}$. (For larger $\dim(G)$ the constraint tightens further through the factor $\dim(G)$.)

Proposition F.102 (Finite-depth slack/mixing bound for a slack-stabilizing reference (FD)). *Fix the α -pivot reference family $\nu_L^{(\alpha)}$ and tilt $R_L^{(\alpha)}$ of (F110), with the α -schedule $\alpha = \alpha(L)$ declared in the manuscript (e.g. $\alpha(L) = 1 - \kappa/L^2$ with κ fixed). Assume there exist constants $\beta_{\text{FD}} < \infty$ and a budget $\delta_{\text{budget}} > 0$ such that for all $\beta \geq \beta_{\text{FD}}$, for all admissible boundary conditions, and for each block size L that occurs in the finite-depth entrance comparison up to depth k_{enter} , the slack denominator is uniformly nondegenerate in the sense that*

$$\delta_{\text{slack}}(L, \beta) := \left(\mathbb{E}_{\nu_L^{(\alpha(L))}}[R_L^{(\alpha(L))}] \right)^{-1} - 1 = \left(\frac{Z_L^{\text{full}}}{Z_L^{(\alpha(L))}} \right)^{-1} - 1 \leq \delta_{\text{budget}}, \quad (\text{F129})$$

with δ_{budget} small enough to close the tilt budget inequality (F126) at the KP threshold f_{adm} .

In addition, assume that the corresponding full 4D α -reference boundary marginal satisfies the early-step mixing estimate needed by the transfer module of Section F.16 (spectral-radius / χ^2 control) with constants uniform in β , volume, and boundary conditions; a concrete sufficient form is Proposition F.103 below.

Comment. This assumption is the “thick-block” analogue of the thin-block surrogate used above: it isolates the remaining mixing input needed for the finite-depth entrance comparison. In the present manuscript, the coefficient-decay form Proposition F.103 is discharged on the load-bearing dyadic path from two theorem-level ingredients: (i) the convolution-dominance/chessboard comparison proved in Proposition F.151 by reflection positivity and bond-moving bookkeeping, and (ii) the one-plaquette coefficient bound recorded in Corollary F.153. All remaining steps in the mass-gap module are independent of N once these coefficient bounds are in place.

Proposition F.103 (Thick-block α -reference mixing (crossover form)). *Fix G and fix $\kappa \in (0, \kappa_0(G)]$ and the schedule (F115). For each L let $w_L^{(\alpha(L))} dg = (\Phi_L)_{\#} \nu_L^{(\alpha(L))}$ be the boundary holonomy marginal of the (thick-block) α -reference family.*

Assume there exists $\beta_{\text{mix}} \geq 1$ such that for all $\beta \geq \beta_{\text{mix}}$, all L in the finite-depth entrance window, and every nontrivial irrep $R \neq \mathbf{1}$,

$$\begin{aligned} \lambda_R(w_L^{(\alpha(L))}) &:= \int_G w_L^{(\alpha(L))}(g) \varphi_R(g) dg, \\ |\lambda_R(w_L^{(\alpha(L))})| &\leq \exp\left(-|T_L| c_{A1}(G, \rho) \frac{C_2(R)}{\alpha(L)\beta + \sqrt{C_2(R)}}\right), \quad |T_L| = L^2. \end{aligned} \quad (\text{F130})$$

Remark. The bound (F130) interpolates between: (i) a heat-kernel regime $C_2(R) \leq \beta^2$, where $|\lambda_R| \leq \exp(-\frac{c_{A1}(G, \rho)}{2} \frac{|T_L|}{\beta} C_2(R))$, and (ii) a large-representation regime $C_2(R) \geq \beta^2$, where $|\lambda_R| \leq \exp(-\frac{c_{A1}(G, \rho)}{2} |T_L| \sqrt{C_2(R)})$.

Remark F.104 (Roadmap for proving Proposition F.103). The coefficient bound (F130) is equivalent to a quantitative area-law / heat-kernel type upper bound for a Wilson loop in the thick α -reference block, since

$$\lambda_R(w_L^{(\alpha(L))}) = \mathbb{E}_{\nu_L^{(\alpha(L))}}[\varphi_R(\Phi_L)].$$

In particular, it is *strictly stronger* than what one gets from the purely abelian / Gaussianized (dA)-approximation: in that linearized model a discrete Stokes identity collapses the surface

flux to a boundary line integral, leading at most to perimeter-scale fluctuations (Lemma F.108 below).

Two complementary closure approaches suggest themselves.

- (A) *Nonabelian Gaussianization/CLT along the tile.* Work on $\Omega_{\varepsilon_0}^{\parallel}$, write Φ_L as a surface-ordered product of plaquette holonomies based at the root in tree gauge, and prove that the resulting random product behaves like the group heat kernel at time $\tau \asymp |T_L|/\beta$. Analytically, one needs a quantitative *area-scale nondegeneracy* in Lie-algebra coordinates that does not collapse to the boundary, together with probabilistic control of the commutator/BCH corrections.
- (B) *Positivity-preserving decimation (bond-moving/Migdal–Kadanoff type comparison).* Use reflection positivity / positivity of the character expansion to compare $\mathbb{E}_{\nu_L^{(\alpha(L))}}[\varphi_R(\Phi_L)]$ to the same expectation in an analytically solvable decimated reference model in which the boundary marginal is an explicit convolution power (hence exhibits the desired $\exp\{-c C_2(R)|T_L|/\beta\}$ coefficient decay).

The remainder of Section F 19 d records one explicit lemma chain of type (A) and isolates the precise analytic inputs that would be required if one pursued Alternative A in the thick-block geometry (the primary proof does not require these inputs). It also records an explicit sufficient closure approach of type (B) in Section F 19 d (chessboard/tiling estimate + one-plaquette coefficient control).

d. Thick-block mixing: explicit reduction to a variance bound and commutator control

Signpost. The variance/CLT reduction recorded in Proposition F.109 and Proposition F.112 is an optional side-branch. The load-bearing inputs used in the primary chain are proved in Subsubsections F 19 d–F 19 d (Lemmas F.138, F.139, F.143, F.145 and Proposition F.146); these discharge $H3 \rightarrow H6 \rightarrow H2$ and feed Theorem F.320.

Definition F.105 (Tile small-field event). For the planar $L \times L$ tile T_L in the $(1, 2)$ -plane define the *tile small-field event*

$$\Omega_{\varepsilon_0}^{\parallel} := \left\{ \|F_p\| \leq \varepsilon_0 \text{ for all } p \in T_L \right\}. \quad (\text{F131})$$

This is the minimal event required to make $F_p = \log U_p$ well-defined for all plaquettes appearing in the surface observable $S_L^{\text{plaq}} = \sum_{p \in T_L} F_p$.

Proposition F.103 is formulated directly in terms of normalized character coefficients of the thick-block α -reference boundary marginal. For completeness and referee navigation we spell out an explicit lemma chain showing *exactly* which analytic ingredients suffice to prove (F130).

Throughout this subsection we work conditionally on the small-curvature event $\Omega_{\varepsilon_0}^{\parallel}$ of Lemma F.224, so that exponential coordinates are valid and curvature variables are uniformly bounded.

Definition F.106 (Linearized flux and plaquette-log surface sum). Let T_L be the oriented $L \times L$ tile in the (μ, ν) -plane spanning the boundary loop ∂T_L used in the definition of Φ_L in Section F 18.

For each plaquette p let $U_p \in G$ be its holonomy (based at the tree-gauge root). On the event $\Omega_{\varepsilon_0}^{\parallel}$, the plaquette logarithm is well-defined and we write

$$U_p = \exp(F_p), \quad F_p \in \mathfrak{g}, \quad \|F_p\| \leq \varepsilon_0.$$

In addition, write the edge variables in exponential coordinates as $U_e = \exp(A_e)$ and let d denote the cochain differential so that $(dA)_p$ is the linearized curvature 2-form on plaquettes.

Define the $\mathfrak{g}$ -valued sums

$$S_L^{\text{plaq}} := \sum_{p \in T_L} F_p, \quad S_L^{\text{lin}} := \sum_{p \in T_L} (dA)_p,$$

and for $v \in \mathfrak{g}$ set $S_L^{\text{plaq}}(v) := \langle v, S_L^{\text{plaq}} \rangle$ and $S_L^{\text{lin}}(v) := \langle v, S_L^{\text{lin}} \rangle$.

Lemma F.107 (Sub-Gaussian upper bound for the linearized flux S_L^{lin} from curvature BL). *Assume Proposition F.226. There exists a constant $C_{\text{BL}} = C_{\text{BL}}(G)$ such that for every $v \in \mathfrak{g}$ and every $s \in \mathbb{R}$,*

$$\log \mathbb{E}_{\nu_L^{(\alpha(L))}} \left[\exp \left(s S_L^{\text{lin}}(v) \right) \mid \Omega_{\varepsilon_0}^{\parallel} \right] \leq C_{\text{BL}} \frac{s^2}{\beta} |T_L| \|v\|^2. \quad (\text{F132})$$

In particular, $S_L^{\text{lin}}(v)$ is sub-Gaussian with variance proxy $\lesssim (\beta^{-1}|T_L|)\|v\|^2$, uniformly in volume and boundary conditions.

Proof. Take the finitely supported $\mathfrak{g}$ -valued 2-form $J^{(v)}$ defined by $J_p^{(v)} = v$ for $p \in T_L$ and $J_p^{(v)} = 0$ otherwise. Then $\langle J^{(v)}, dA \rangle = \sum_{p \in T_L} \langle v, (dA)_p \rangle = S_L^{\text{lin}}(v)$. Applying (F281) with $J = J^{(v)}$ and using $\langle J, \Pi_{\text{ex}} J \rangle \leq \|J\|_2^2 = |T_L| \|v\|^2$ gives (F132) (absorbing $(2m_-)^{-1}$ into C_{BL}). $\square$

Lemma F.108 (Stokes collapse yields perimeter-scale variance in the abelianized Gaussian model). *Consider the Gaussian (abelianized) block measure on tree-gauge 1-forms A with density proportional to $\exp\{-(\beta/2)\|dA\|_2^2\}$. Then for the tile indicator $J^{(v)}$ from Lemma F.107,*

$$S_L^{\text{lin}}(v) = \langle J^{(v)}, dA \rangle = \langle d^* J^{(v)}, A \rangle,$$

so $S_L^{\text{lin}}(v)$ depends only on the boundary 1-current $d^* J^{(v)}$ (supported on ∂T_L). Moreover,

$$\text{Var}\left(S_L^{\text{lin}}(v)\right) = \frac{1}{\beta} \langle d^* J^{(v)}, \Delta_1^{-1} d^* J^{(v)} \rangle \leq C_{\text{per}} \frac{L}{\beta} \|v\|^2,$$

with C_{per} depending only on the lattice geometry and the chosen norm on $\mathfrak{g}$. In particular, an area-scale lower bound of the form $\text{Var} \gtrsim |T_L|/\beta$ cannot hold for the linearized flux in this Gaussian model.

Proof. The identity $S_L^{\text{lin}}(v) = \langle d^* J^{(v)}, A \rangle$ is discrete integration by parts. For the Gaussian measure, $\text{Cov}(A) = \beta^{-1} \Delta_1^{-1}$, hence the variance formula. Finally, standard Green function bounds for the (tree-gauge) 1-form Laplacian in four dimensions imply $(\Delta_1^{-1})(e, e') \leq C(1 + \text{dist}(e, e'))^{-2}$, and since $d^* J^{(v)}$ is supported on a loop of length $\asymp L$ one obtains the stated $O(L)$ bound on $\langle d^* J^{(v)}, \Delta_1^{-1} d^* J^{(v)} \rangle$ by a direct double-sum estimate. $\square$

Proposition F.109 (B9: nondegenerate surface variance for the plaquette-log sum (via Plan A)). *Assume Proposition F.164 at scale L and schedule $\alpha(L)$ (hence with parameters m_L , Δ_X , c_Ψ and $\epsilon_L^{(C)}$). Let $\Omega_{\varepsilon_0}^{\parallel}$ be the tile small-field event from Definition F.105. Then there exist constants $\beta_{\text{var}} = \beta_{\text{var}}(G, \kappa, \varepsilon_0) < \infty$ and $c_{\text{var}} = c_{\text{var}}(G, \kappa, \varepsilon_0) > 0$ such that for all $\beta \geq \beta_{\text{var}}$, all L in the finite-depth entrance window, all admissible boundary conditions, and all $v \in \mathfrak{g}$ with $\|v\| = 1$,*

$$\text{Var}_{\nu_L^{(\alpha(L))}}\left(S_L^{\text{plaq}}(v) \mid \Omega_{\varepsilon_0}^{\parallel}\right) \geq c_{\text{var}} \frac{|T_L|}{\beta}. \quad (\text{F133})$$

Moreover one may take $c_{\text{var}} = c_1/2$, where $c_1(G)$ is the planar one-plaquette constant from Lemma F.111.

Proof. Fix β and L and abbreviate $\Omega := \Omega_{\varepsilon_0}^{\parallel}$. By the planar variance computation in Proposition F.112 (Step 1), using Lemma F.111, one has

$$\text{Var}_{\nu_L^{\parallel}}(S_L^{\text{plaq}}(v) \mid \Omega) \geq c_1 \frac{|T_L|}{\beta}.$$

Let $\nu_{\alpha} := \nu_L^{(\alpha(L))}|_{\mathcal{F}_{\parallel}}$ and $\nu_{\parallel} := \nu_L^{\parallel}|_{\mathcal{F}_{\parallel}}$. Proposition F.164 and Lemma F.181 give the (unconditioned) L^2 -flatness

$$\delta := \left\| \tilde{\Psi}_L^{(\alpha)} - 1 \right\|_{L^2(\nu_{\parallel})} \leq \epsilon_L^{(C)} + \frac{t \Delta_X}{\beta \sqrt{2} c_{\Psi}} \sqrt{m_L}.$$

Since Ω depends only on the in-plane plaquettes, weak-localization (union bound with the one-plaquette tail from Proposition F.226) yields $\nu_{\parallel}(\Omega) \geq 1 - |T_L|e^{-c\beta}$, hence $\nu_{\parallel}(\Omega) \geq 1/2$ for β large. Applying Lemma F.182 with $\nu = \nu_{\parallel}$, $\mu = \nu_{\alpha}$ and event Ω gives

$$D_{\chi^2}(\nu_{\alpha}^{\Omega} \parallel \nu_{\parallel}^{\Omega}) \leq \frac{4}{\nu_{\parallel}(\Omega)} \delta^2 \leq 8 \delta^2,$$

provided $\delta \leq \frac{1}{2} \sqrt{\nu_{\parallel}(\Omega)}$, which holds for β large since $\delta \rightarrow 0$ in the entrance window. Choosing β_{var} large enough so that this χ^2 divergence is at most the threshold required in Proposition F.112, the conclusion (F133) follows from Proposition F.112 with $c_{\text{var}} = c_1/2$. $\square$

Corollary F.110 (B9 via the tilt influence bound (Plan A without a cut surrogate)). *Assume Proposition F.128 at scale L and schedule $\alpha(L)$. Then the conclusion (F133) of Proposition F.109 holds (with possibly different constants β_{var} and c_{var} depending only on G, κ, ε_0).*

Proof. Fix β and L and abbreviate $\Omega := \Omega_{\varepsilon_0}^{\parallel}$. As in the proof of Proposition F.109, one has the planar conditional lower bound

$$\text{Var}_{\nu_L^{\parallel}}(S_L^{\text{plaq}}(v) \mid \Omega) \geq c_1 \frac{|T_L|}{\beta}.$$

Let $\nu_{\alpha} := \nu_L^{(\alpha(L))}|_{\mathcal{F}_{\parallel}}$ and $\nu_{\parallel} := \nu_L^{\parallel}|_{\mathcal{F}_{\parallel}}$. By Corollary F.129, Proposition F.128 implies

$$\delta^2 := \left\| \tilde{\Psi}_L^{(\alpha)} - 1 \right\|_{L^2(\nu_{\parallel})}^2 = D_{\chi^2}(\nu_{\alpha} \parallel \nu_{\parallel}) \leq \frac{|T_L| \Delta_{\Psi}^2}{2\beta^2}.$$

As above, $\nu_{\parallel}(\Omega) \geq 1 - |T_L|e^{-c\beta}$ and hence $\nu_{\parallel}(\Omega) \geq 1/2$ for β large. Applying Lemma F.182 with $\nu = \nu_{\parallel}$, $\mu = \nu_{\alpha}$ and event Ω gives

$$D_{\chi^2}(\nu_{\alpha}^{\Omega} \parallel \nu_{\parallel}^{\Omega}) \leq \frac{4}{\nu_{\parallel}(\Omega)} \delta^2 \leq 8 \delta^2.$$

In the entrance window $|T_L| \lesssim \beta$, this χ^2 divergence tends to 0 as $\beta \rightarrow \infty$, so for β large it is below the threshold required in Proposition F.112. The conclusion (F133) then follows from Proposition F.112 with $c_{\text{var}} = c_1/2$. $\square$

Lemma F.111 (Planar one-plaquette variance lower bound). *Fix $\varepsilon_0 > 0$ within the logarithm chart as in Proposition F.226. There exist $\beta_1 = \beta_1(G) < \infty$ and $c_1 = c_1(G) > 0$ such that for all $\beta \geq \beta_1$ and all $v \in \mathfrak{g}$ with $\|v\| = 1$,*

$$\text{Var}_{\nu_1} \left(\langle v, F_{p_0} \rangle \mid \|F_{p_0}\| \leq \varepsilon_0 \right) \geq \frac{c_1}{\beta}. \quad (\text{F134})$$

Proof. Work in exponential coordinates $g = \exp(X)$ on the ball $\|X\| \leq \varepsilon_0$, where $F = \log g = X$. In these coordinates Haar measure has the form $dg = J(X) dX$ with a smooth Jacobian J and $J(0) = 1$. By Taylor expansion of the Wilson action at the identity and compactness, there exist constants $m_-, m_+ > 0$ and $0 < j_- \leq j_+ < \infty$ (depending only on G and the chosen normalization) such that for all $\|X\| \leq \varepsilon_0$,

$$\frac{m_-}{2} \|X\|^2 \leq A(\exp X) \leq \frac{m_+}{2} \|X\|^2, \quad j_- \leq J(X) \leq j_+. \quad (\text{F135})$$

Since $A(g) = A(g^{-1})$ and Haar is inversion-invariant, the conditional law on $\{\|X\| \leq \varepsilon_0\}$ is symmetric under $X \mapsto -X$, hence $\mathbb{E}[\langle v, X \rangle \mid \|X\| \leq \varepsilon_0] = 0$ and the conditional variance equals the conditional second moment.

Let $R := 1$ and take $\beta \geq \beta_1 := (R/\varepsilon_0)^2$, so that $\{\|X\| \leq R/\sqrt{\beta}\} \subseteq \{\|X\| \leq \varepsilon_0\}$. Lower bound the numerator and upper bound the denominator using (F135):

$$\begin{aligned} \int_{\|X\| \leq \varepsilon_0} \langle v, X \rangle^2 e^{-\beta A(\exp X)} J(X) dX &\geq j_- \int_{\|X\| \leq R/\sqrt{\beta}} \langle v, X \rangle^2 e^{-(\beta m_+/2)\|X\|^2} dX, \\ \int_{\|X\| \leq \varepsilon_0} e^{-\beta A(\exp X)} J(X) dX &\leq j_+ \int_{\mathbb{R}^{\dim G}} e^{-(\beta m_-/2)\|X\|^2} dX. \end{aligned}$$

In the numerator substitute $X = Y/\sqrt{\beta}$ to obtain

$$\begin{aligned} \int_{\|X\| \leq R/\sqrt{\beta}} \langle v, X \rangle^2 e^{-(\beta m_+/2)\|X\|^2} dX &= \beta^{-\frac{\dim G}{2}-1} \int_{\|Y\| \leq R} \langle v, Y \rangle^2 e^{-(m_+/2)\|Y\|^2} dY \\ &=: \beta^{-\frac{\dim G}{2}-1} C_{\text{num}}(G). \end{aligned}$$

where $C_{\text{num}}(G) > 0$ by positivity of the integrand. The Gaussian integral in the denominator equals $\beta^{-\dim G/2} C_{\text{den}}(G)$ with $C_{\text{den}}(G) < \infty$. Therefore the conditional second moment (and hence variance) is at least

$$\frac{j_- C_{\text{num}}(G)}{j_+ C_{\text{den}}(G)} \cdot \frac{1}{\beta}.$$

Set $c_1 := j_- C_{\text{num}}(G)/(j_+ C_{\text{den}}(G)) > 0$. $\square$

Proposition F.112 (Plan A reduction of Proposition F.109). *Let $\mathcal{F}_\parallel := \sigma(\{U_p\}_{p \in T_L})$ be the σ -algebra generated by the in-plane plaquette variables on the tile T_L . Write $\nu_\parallel$ for the purely planar reference on T_L (Lemma F.123) and ν_α for the thick-block $\alpha(L)$ -reference $\nu_L^{(\alpha(L))}$ restricted to $\mathcal{F}_\parallel$. Let $\nu_\parallel^\Omega$ and ν_α^Ω denote these laws further conditioned on $\Omega_{\varepsilon_0}^\parallel$.*

Assume that for some $\varepsilon_ \in (0, 1]$ one has the uniform in-plane χ^2 control*

$$D_{\chi^2}(\nu_\alpha^\Omega \parallel \nu_\parallel^\Omega) \leq \varepsilon_*, \quad (\text{F136})$$

uniformly in admissible boundary conditions. Then there exists a constant $C_A = C_A(G)$ such that for every $v \in \mathfrak{g}$ with $\|v\| = 1$,

$$\text{Var}_{\nu_L^{(\alpha(L))}}(S_L^{\text{plaq}}(v) \mid \Omega_{\varepsilon_0}^\parallel) \geq \text{Var}_{\nu_\parallel}(S_L^{\text{plaq}}(v) \mid \Omega_{\varepsilon_0}^\parallel) - C_A \frac{|T_L|}{\beta} \sqrt{\varepsilon_*}. \quad (\text{F137})$$

In particular, Lemma F.111 provides constants $\beta_1(G)$ and $c_1(G) > 0$ such that for all $\beta \geq \beta_1(G)$,

$$\text{Var}_{\nu_\parallel}(S_L^{\text{plaq}}(v) \mid \Omega_{\varepsilon_0}^\parallel) \geq \frac{c_1(G)}{\beta} |T_L|.$$

Hence if additionally $\sqrt{\varepsilon_} \leq c_1(G)/(2C_A(G))$ then Proposition F.109 holds with $c_{\text{var}} = c_1(G)/2$.*

Proof. We work under the conditional measures ν_α^Ω and $\nu_\parallel^\Omega$ on $\mathcal{F}_\parallel$.

Step 1: planar variance is exactly extensive. By Lemma F.123, in spanning-tree gauge the plaquette variables $\{U_p\}_{p \in T_L}$ are independent under $\nu_\parallel$. The event $\Omega_{\varepsilon_0}^\parallel$ is a product event over plaquettes in this gauge, hence independence is preserved under conditioning. Moreover $A(g) = A(g^{-1})$ and Haar is inversion-invariant, so each $F_p = \log U_p$ is centered under $\nu_\parallel^\Omega$ and

$$\text{Var}_{\nu_\parallel}(S_L^{\text{plaq}}(v) \mid \Omega_{\varepsilon_0}^\parallel) = \sum_{p \in T_L} \text{Var}_{\nu_\parallel}(\langle v, F_p \rangle \mid \|F_p\| \leq \varepsilon_0) = |T_L| \cdot \text{Var}_{\nu_1^\parallel}(\langle v, F_{p_0} \rangle \mid \|F_{p_0}\| \leq \varepsilon_0).$$

In particular, Lemma F.111 implies

$$\text{Var}_{\nu_\parallel}(S_L^{\text{plaq}}(v) \mid \Omega_{\varepsilon_0}^\parallel) \geq \frac{c_1(G)}{\beta} |T_L|. \quad (\text{F138})$$

Step 2: L^4 size of the planar surface sum. On $\Omega_{\varepsilon_0}^\parallel$ one has $|\langle v, F_p \rangle| \leq \varepsilon_0$ for each p , hence for $X_p := \langle v, F_p \rangle$, $\mathbb{E}[(X_p)^4] \leq \varepsilon_0^2 \mathbb{E}[(X_p)^2]$. By independence and the standard fourth-moment expansion for sums of centered independent variables,

$$\mathbb{E}_{\nu_\parallel^\Omega}[(S_L^{\text{plaq}}(v))^4] \leq C(|T_L| \mathbb{E}[X_{p_0}^4] + |T_L|^2 \mathbb{E}[X_{p_0}^2]^2) \leq C'(G) \left(\frac{|T_L|}{\beta^2} + \frac{|T_L|^2}{\beta^2} \right),$$

using the curvature-scale upper bound $\mathbb{E}[X_{p_0}^2] \lesssim 1/\beta$ from Proposition F.226. Thus

$$\|S_L^{\text{plaq}}(v)\|_{L^4(\nu_\parallel^\Omega)}^2 \leq C''(G) \frac{|T_L|}{\beta}. \quad (\text{F139})$$

Similarly $\|S_L^{\text{plaq}}(v)\|_{L^2(\nu_\parallel^\Omega)}^2 = \text{Var}_{\nu_\parallel^\Omega}(S_L^{\text{plaq}}(v)) \lesssim |T_L|/\beta$.

Step 3: χ^2 transfer of variance. Apply Lemma F.77 with $\mu = \nu_\alpha^\Omega$, $\nu = \nu_\parallel^\Omega$, $F = S_L^{\text{plaq}}(v)$ and $\varepsilon = D_{\chi^2}(\nu_\alpha^\Omega \| \nu_\parallel^\Omega) \leq \varepsilon_*$. Using (F139) and the corresponding L^2 bound gives

$$\text{Var}_{\nu_\alpha^\Omega}(S_L^{\text{plaq}}(v)) \geq \text{Var}_{\nu_\parallel^\Omega}(S_L^{\text{plaq}}(v)) - C_A(G) \frac{|T_L|}{\beta} \sqrt{\varepsilon_*},$$

which is (F137).

Finally combine with (F138). If $\sqrt{\varepsilon_*} \leq c_1(G)/(2C_A(G))$ and $\beta \geq \beta_1(G)$, then

$$\text{Var}_{\nu_L^{(\alpha(L))}}(S_L^{\text{plaq}}(v) \mid \Omega_{\varepsilon_0}^\parallel) \geq \frac{c_1(G)}{2\beta} |T_L|,$$

i.e. Proposition F.109 holds with $c_{\text{var}} = c_1(G)/2$. $\square$

Remark F.113 (What the χ^2 hypothesis is). By Lemma F.159, the in-plane marginal of the thick-block α -reference is a tilt of the planar reference: $\nu_\alpha|_{\mathcal{F}_\parallel} = \tilde{\Psi}_L^{(\alpha)} \cdot \nu_\parallel|_{\mathcal{F}_\parallel}$. Therefore the hypothesis (F136) is equivalently the L^2 flatness condition

$$\|\tilde{\Psi}_L^{(\alpha)} - 1\|_{L^2(\nu_\parallel^\Omega)}^2 \leq \varepsilon_*.$$

In the thin-block leaf geometry one has $\tilde{\Psi}_L^{(\alpha)} \equiv 1$ exactly (Lemma F.161), whereas in genuine thick blocks this is precisely the “effective tilt control” problem of Main chain.

Lemma F.114 (Surface representation of Φ_L (lattice nonabelian Stokes; deterministic form)).

In tree gauge, there exists an ordering of plaquettes $p_1, \dots, p_{|T_L|}$ in the tile T_L such that the boundary holonomy satisfies the exact surface-ordered product identity

$$\Phi_L = \prod_{i=1}^{\overrightarrow{|T_L|}} U_{p_i}, \quad (\text{F140})$$

where $U_{p_i} \in G$ is the plaquette holonomy around p_i (with the induced orientation from T_L) and the arrow indicates the chosen surface order.

On the event $\Omega_{\varepsilon_0}^\parallel$ one may write $U_{p_i} = \exp(F_{p_i})$ with $\|F_{p_i}\| \leq \varepsilon_0$. Consequently, there exists a $\mathfrak{g}$ -valued error term E_L such that

$$\Phi_L = \exp(S_L^{\text{plaq}} + E_L), \quad S_L^{\text{plaq}} = \sum_{p \in T_L} F_p, \quad (\text{F141})$$

with the crude deterministic BCH bound

$$\|E_L\| \leq C_{\text{stokes}} |T_L|^2 \varepsilon_0^2, \quad (\text{F142})$$

for a constant $C_{\text{stokes}} = C_{\text{stokes}}(G)$ depending only on the compact group G and the chosen norm on $\mathfrak{g}$.

Proof. The exact product identity (F140) is a standard lattice nonabelian Stokes statement: enumerate the plaquettes of T_L in an order compatible with a spanning tree of the dual tile so that when multiplying the plaquette boundary words, interior edges cancel adjacently (tree gauge is used only to make the resulting surface order canonical; the identity itself is gauge invariant).

On $\Omega_{\varepsilon_0}^{\parallel}$, exponential coordinates are valid and each plaquette holonomy can be written as $U_{p_i} = \exp(F_{p_i})$ with $\|F_{p_i}\| \leq \varepsilon_0$. Write $X_i := F_{p_i}$ and consider the partial products $P_k := \overrightarrow{\prod_{i=1}^k} \exp(X_i)$. For ε_0 fixed below the injectivity radius, the Baker–Campbell–Hausdorff formula implies a uniform estimate

$$\left\| \log(\exp(X) \exp(Y)) - (X + Y) \right\| \leq C_{\text{BCH}} \|X\| \|Y\| \quad (\|X\|, \|Y\| \leq \varepsilon_0),$$

with $C_{\text{BCH}} = C_{\text{BCH}}(G)$. Iterating this bound over $k = 1, \dots, |T_L|$ and using $\|X_i\| \leq \varepsilon_0$ yields the crude estimate $\|\log P_{|T_L|} - \sum_i X_i\| \leq C_{\text{BCH}} |T_L|^2 \varepsilon_0^2$, which is (F142) (absorbing constants into C_{stokes}). $\square$

Remark F.115. Lemma F.114 is the precise form of the “surface representation” step in Remark F.104. Its proof is purely deterministic (surface ordering + BCH commutator bookkeeping) once one works in a normal chart and uses $\|F_p\| \leq \varepsilon_0$ on $\Omega_{\varepsilon_0}^{\parallel}$.

Lemma F.116 (Heat-kernel coefficient decay from Gaussian surface fluctuations). *Assume Lemma F.114 and Proposition F.109. Assume in addition that the one-dimensional marginals $S_L^{\text{plaq}}(v)$ are uniformly strongly log-concave on $\Omega_{\varepsilon_0}^{\parallel}$ in the sense that their conditional densities (with respect to Lebesgue measure on $\mathbb{R}$) have second derivative bounded below by $-C(\beta/|T_L|)$ and above by $-c(\beta/|T_L|)$, with constants uniform in v on the unit sphere.¹ Then there exist*

¹ This hypothesis is a quantitative “Gaussian comparison” statement for the surface sum; it is expected to follow from the curvature domination mechanism plus a nondegeneracy estimate such as (F133), but it is recorded separately here to make the required analytic input explicit.

constants $\beta_{\text{hk}}(G)$ and $c_{\text{mix}}(G) > 0$ such that for all $\beta \geq \beta_{\text{hk}}$ and all L in the finite-depth entrance window,

$$\lambda_R(w_L^{(\alpha(L))}) = \mathbb{E}_{\nu_L^{(\alpha(L))}}[\varphi_R(\Phi_L)] \leq \exp\left(-c_{\text{mix}} \frac{C_2(R)}{\beta} |T_L|\right) \quad \forall R \neq \mathbf{1},$$

i.e. Proposition F.103 holds.

Remark F.117 (What is genuinely new). Proposition F.226 supplies the *upper* sub-Gaussian control (F132) for the *linearized* flux $S_L^{\text{lin}}(v)$ at essentially no additional cost. However, Lemma F.108 shows that in the corresponding Gaussian/abelian model this linearized flux has only perimeter-scale variance, so an area-law coefficient decay cannot come from the linear sector alone.

The genuinely new quantitative content needed for thick-block mixing is therefore twofold: (i) an area-scale variance noncollapse for the *plaquette-log* sum $S_L^{\text{plaq}}(v)$ (Proposition F.109), and (ii) probabilistic control of the commutator/BCH error E_L in Lemma F.114 strong enough to transfer this nondegeneracy to the group-valued holonomy Φ_L . The additional strong log-concavity hypothesis in Lemma F.116 is a convenient way to package the needed Gaussian comparison input.

Alternative approach (B): convolution dominance (bond-moving) and a one-plaquette coefficient bound **Load-bearing (Main chain)**. The inputs proved in Subsubsections F 19 d–F 19 d are used to discharge Proposition F.133, yielding $H3$ for G and feeding the main chain $H3 \rightarrow H6 \rightarrow H2 \rightarrow$ Theorem F.320.

The variance/CLT approach above isolates two genuinely nontrivial analytic inputs (area-scale noncollapse and commutator control). There is, however, a qualitatively different approach that aims to bypass Lie-algebra Gaussianization: compare the thick-block boundary marginal to an analytically solvable *convolution power*. The structural point is that for central densities the normalized character coefficients $\lambda_R(\cdot)$ are precisely the Fourier coefficients, and convolution powers multiply coefficients exactly.

Lemma F.118 (Convolution powers multiply coefficients). *Let $w \, dg$ be a central probability density on G and let w^{*n} denote its n -fold convolution. Then for every irrep R one has*

$$\lambda_R(w^{*n}) = \lambda_R(w)^n, \quad \text{and hence} \quad |\lambda_R(w^{*n})| = |\lambda_R(w)|^n.$$

Proof. This is immediate from Peter–Weyl: convolution corresponds to multiplication of Fourier coefficients. □

Definition F.119 (Coefficient domination order on central densities). Let $u dg$ and $v dg$ be central probability densities on G and write their normalized character coefficients as $\lambda_R(u) = \int_G u(g) \overline{\varphi_R(g)} dg$ as in Section F 16. We write

$$u \preceq_{\text{coeff}} v$$

if for every irrep R one has the coefficient domination (in modulus)

$$|\lambda_R(u)| \leq |\lambda_R(v)|.$$

If u and v are of positive type, then $\lambda_R(u), \lambda_R(v) \geq 0$ for all R and the modulus may be dropped.

Lemma F.120 (Positive type $\Rightarrow$ nonnegative coefficients). *Let $u dg$ be a central probability density of positive type. Then $\lambda_R(u) \geq 0$ for all irreps R and $0 \leq \lambda_R(u) \leq 1$.*

Proof. For a positive-type u , the convolution operator T_u on $L^2(G)$ is positive semidefinite. In Peter–Weyl coordinates one has $\hat{u}(R) = \lambda_R(u) I_{d_R}$ for central u (Section F 16), so positivity of $\hat{u}(R)$ forces $\lambda_R(u) \geq 0$. Moreover, since $u dg$ is a probability measure and $|\varphi_R(g)| \leq 1$ for all $g \in G$,

$$0 \leq \lambda_R(u) = \left| \int_G \varphi_R(g) u(g) dg \right| \leq \int_G |\varphi_R(g)| u(g) dg \leq 1.$$

□

Lemma F.121 (Heat kernel coefficients and semigroup). *Let k_t denote the (bi-invariant) heat kernel on G at time $t > 0$ corresponding to the Casimir/Laplacian normalization used for C_2 , i.e. the unique central probability density whose normalized character coefficients satisfy*

$$\lambda_R(k_t) = e^{-tC_2(R)} \quad \text{for every irrep } R,$$

*equivalently $k_t(g) = \sum_R d_R e^{-tC_2(R)} \chi_R(g)$ in Peter–Weyl form. Then $\{k_t\}_{t \geq 0}$ is a convolution semigroup: $k_t * k_s = k_{t+s}$ and hence $k_t^{*n} = k_{nt}$.*

Proof. In Peter–Weyl, convolution corresponds to multiplication of Fourier coefficients. Thus for each irrep R ,

$$\lambda_R(k_t * k_s) = \lambda_R(k_t) \lambda_R(k_s) = e^{-tC_2(R)} e^{-sC_2(R)} = e^{-(t+s)C_2(R)} = \lambda_R(k_{t+s}).$$

Since the characters form an orthonormal basis for central $L^2(G)$ functions and $\lambda_1(k_t) = 1$, this identifies $k_t * k_s$ with k_{t+s} and shows $\int_G k_t dg = 1$. Iterating gives $k_t^{*n} = k_{nt}$. □

Remark F.122 (One-plaquette bounds and heat-kernel domination). For a compact Lie group G , a Gaussian-in- C_2/β one-plaquette coefficient bound implies coefficient domination by a heat kernel: if $|\lambda_R(w_1^{(\alpha)})| \leq e^{-cC_2(R)/\beta}$ uniformly in $\alpha \in [1/2, 1]$, then $w_1^{(\alpha)} \preceq_{\text{coeff}} k_{c/\beta}$ in the sense of Definition F.119.

In the present manuscript we use instead the crossover bound (F173). It still yields a heat-kernel style decay in the low-representation regime $C_2(R) \leq \beta^2$ via (F174), and a stronger $\sqrt{C_2}$ decay outside that regime.

Lemma F.123 (Planar (2D) exact convolution power). *Consider the purely planar reference on a simply connected $L \times L$ tile T_L with plaquette weight W and no out-of-plane plaquettes. Let $w_L^\parallel dg = (\Phi_L)_\# \nu_L$ be the planar boundary marginal around T_L , and let $w_1^\parallel$ be the corresponding 1×1 marginal. Then one may fix a spanning-tree gauge in which the plaquette holonomies $\{U_p\}_{p \in T_L}$ become independent with marginal density proportional to W , and the boundary holonomy is the ordered product of these plaquette variables. Consequently,*

$$w_L^\parallel = (w_1^\parallel)^{*\lvert T_L \rvert}, \quad \text{and in particular} \quad \lambda_R(w_L^\parallel) = \lambda_R(w_1^\parallel)^{\lvert T_L \rvert} \quad \text{for all } R.$$

Proof. Choose a rooted spanning tree of the tile that contains all boundary edges and all but one edge of each plaquette. Gauge-fix all tree edges to the identity. The remaining (non-tree) edge variables can be chosen one-per-plaquette, and in this gauge each plaquette holonomy U_p is exactly its unique non-tree edge variable. Since the Haar measure factorizes over the remaining edges, the planar reference density $\prod_{p \in T_L} W(U_p)$ becomes a product measure over plaquettes. Moreover, because the boundary edges lie in the tree, the boundary holonomy Φ_L is the ordered product of the plaquette holonomies. Thus Φ_L is the product of $\lvert T_L \rvert$ i.i.d. G -valued variables with density $w_1^\parallel$, and its law is the convolution power $(w_1^\parallel)^{*\lvert T_L \rvert}$. $\square$

Lemma F.124 (Schur insertion bound (uniform deterministic boundary data)). *Let $w dg$ be a central probability density on G and let $U_1, \dots, U_N$ be independent G -valued random variables with law $w dg$. Fix deterministic elements $g_0, \dots, g_N \in G$ and define the interleaved product*

$$\Phi := g_0 U_1 g_1 U_2 \cdots g_{N-1} U_N g_N.$$

Then for every irreducible unitary representation R ,

$$\mathbb{E}[\varphi_R(\Phi)] = \lambda_R(w)^N \cdot \frac{1}{d_R} \text{Tr} \left(\prod_{k=0}^N \pi_R(g_k) \right), \quad (\text{F143})$$

and in particular

$$\left| \mathbb{E}[\varphi_R(\Phi)] \right| \leq \lambda_R(w)^N. \quad (\text{F144})$$

Proof. Let

$$M_R := \int_G \pi_R(u) w(u) du.$$

Since w is central and Haar is conjugation-invariant,

$$\pi_R(h) M_R \pi_R(h)^{-1} = \int_G \pi_R(huh^{-1}) w(u) du = M_R \quad \text{for all } h \in G,$$

so M_R commutes with $\pi_R(h)$ for all $h \in G$. By Schur's lemma $M_R = c_R \text{Id}$ for some scalar c_R . Taking traces yields

$$c_R = \frac{1}{d_R} \text{Tr}(M_R) = \int_G \varphi_R(u) w(u) du = \lambda_R(w).$$

By independence,

$$\mathbb{E}[\pi_R(\Phi)] = \pi_R(g_0) M_R \pi_R(g_1) M_R \cdots M_R \pi_R(g_N) = \lambda_R(w)^N \prod_{k=0}^N \pi_R(g_k).$$

Taking normalized traces gives (F143). Finally, each $\pi_R(g_k)$ is unitary, hence $|\text{Tr}(U)| \leq d_R$ for unitary U , yielding (F144). $\square$

Remark F.125 (Planar power bounds are uniform in boundary conditions). In a planar reference model on a simply connected tile T_L , a spanning-tree gauge expresses the boundary holonomy Φ_L as an interleaved product of the independent plaquette variables (each with marginal $w_1^\parallel$) and deterministic group insertions coming from the exterior boundary data. Lemma F.124 then yields the uniform estimate

$$\left| \lambda_R\left((\Phi_L)_\# \nu_L^\parallel(\cdot \mid \eta)\right) \right| \leq \lambda_R(w_1^\parallel)^{|T_L|} \quad \text{for all boundary conditions } \eta.$$

Thus boundary data can modify only the trace factor in (F143); it cannot increase the coefficient magnitude.

Proposition F.126 (Dyadic bond-moving / decimation inequality (local form)). *There exists $\beta_{\text{BM}} = \beta_{\text{BM}}(G)$ such that for all $\beta \geq \beta_{\text{BM}}$ and all $\alpha \in [1/2, 1]$ the following holds. For each dyadic $L = 2^n$ let $w_L^{(\alpha)} dg = (\Phi_L)_\# \nu_L^{(\alpha)}$ denote the thick-block α -reference boundary marginal around the $L \times L$ tile T_L . Assume that for every dyadic L and every nontrivial irrep $R \neq \mathbf{1}$,*

$$\left| \lambda_R(w_{2L}^{(\alpha)}) \right| \leq \left| \lambda_R(w_L^{(\alpha)}) \right|^4. \quad (\text{F145})$$

Equivalently, by Definition F.119 and Lemma F.118,

$$w_{2L}^{(\alpha)} \preceq_{\text{coeff}} (w_L^{(\alpha)})^{*4}.$$

Hardening note: a direct χ^2 approach via influences of $\tilde{\Psi}$ For Plan A in Section F.19 d, it is enough to control the χ^2 divergence

$$D_{\chi^2} \left(\nu_L^{(\alpha)} \Big|_{\mathcal{F}_{\parallel}} \Big\| \nu_L^{\parallel} \right) = \left\| \tilde{\Psi}_L^{(\alpha)} - 1 \right\|_{L^2(\nu_L^{\parallel})}^2,$$

and the cut-local surrogate mechanism in Proposition F.164 is one way to do so. Independently, one can bound the same quantity by an Efron–Stein influence estimate on the *in-plane plaquette coordinates* under the planar reference (tree gauge / i.i.d. plaquettes).

Lemma F.127 (Conditional Efron–Stein bound for localization). *Let $(\sigma_a)_{a=1}^n$ be independent coordinates under a product probability space, and let $\mathcal{F}_{\text{cut}} = \sigma(\sigma_1, \dots, \sigma_m)$ with $m \leq n$. For any $Y = Y(\sigma_1, \dots, \sigma_n) \in L^2$ and independent resamplings $\sigma'_a \sim \sigma_a$, let $Y^{(a)}$ denote Y with σ_a replaced by σ'_a . Then*

$$\left\| Y - E[Y \mid \mathcal{F}_{\text{cut}}] \right\|_{L^2}^2 = E[\text{Var}(Y \mid \mathcal{F}_{\text{cut}})] \leq \frac{1}{2} \sum_{a=m+1}^n \left\| Y - Y^{(a)} \right\|_{L^2}^2. \quad (\text{F146})$$

Proof. Fix the cut coordinates $(\sigma_1, \dots, \sigma_m)$. Conditionally on $\mathcal{F}_{\text{cut}}$, the remaining coordinates $(\sigma_{m+1}, \dots, \sigma_n)$ remain independent. Applying the standard Efron–Stein inequality to the conditional law gives

$$\text{Var}(Y \mid \mathcal{F}_{\text{cut}}) \leq \frac{1}{2} \sum_{a=m+1}^n E[(Y - Y^{(a)})^2 \mid \mathcal{F}_{\text{cut}}].$$

Taking expectation of both sides yields (F146). $\square$

Proposition F.128 (Main-chain per-plaquette influence bound for the normalized effective tilt). *Under the planar reference $\nu_L^{\parallel}$ in tree gauge, realize the in-plane plaquette variables as independent coordinates $(\sigma_a)_{a=1}^{|T_L|}$. The normalized effective tilt $\tilde{\Psi}_L^{(\alpha)} = \tilde{\Psi}_L^{(\alpha)}(\sigma_1, \dots, \sigma_{|T_L|})$ satisfies the per-plaquette influence bound*

$$\left\| \tilde{\Psi}_L^{(\alpha)} - \left(\tilde{\Psi}_L^{(\alpha)} \right)^{(a)} \right\|_{L^2(\nu_L^{\parallel})} \leq \frac{\Delta_{\Psi}}{\beta} \quad \text{for all } a \in \{1, \dots, |T_L|\}, \quad (\text{F147})$$

with a constant Δ_{Ψ} independent of L and uniform in all admissible boundary conditions.

Corollary F.129 (Sufficient influence bound for Plan A χ^2 -flatness). *Assume Proposition F.128. Then*

$$\left\| \tilde{\Psi}_L^{(\alpha)} - 1 \right\|_{L^2(\nu_L^\parallel)}^2 = D_{\chi^2} \left(\nu_L^{(\alpha)} \Big|_{\mathcal{F}_\parallel} \Big\| \nu_L^\parallel \right) \leq \frac{|T_L| \Delta_\Psi^2}{2\beta^2} = \frac{L^2 \Delta_\Psi^2}{2\beta^2}. \quad (\text{F148})$$

In particular, in the entrance window $L \lesssim \sqrt{\beta}$ this yields $D_{\chi^2} \lesssim \beta^{-1}$.

Proof. Since $\tilde{\Psi}_L^{(\alpha)}$ is a Radon–Nikodým derivative, $E_{\nu_L^\parallel}[\tilde{\Psi}_L^{(\alpha)}] = 1$ and hence $\|\tilde{\Psi}_L^{(\alpha)} - 1\|_2^2 = \text{Var}_{\nu_L^\parallel}(\tilde{\Psi}_L^{(\alpha)})$. Apply Lemma F.127 with $m = 0$ (trivial $\mathcal{F}_{\text{cut}}$) and $Y = \tilde{\Psi}_L^{(\alpha)}$ to obtain

$$\text{Var}_{\nu_L^\parallel}(\tilde{\Psi}_L^{(\alpha)}) \leq \frac{1}{2} \sum_{a=1}^{|T_L|} \left\| \tilde{\Psi}_L^{(\alpha)} - (\tilde{\Psi}_L^{(\alpha)})^{(a)} \right\|_2^2 \leq \frac{|T_L| \Delta_\Psi^2}{2\beta^2},$$

which is (F148). $\square$

Remark F.130 (Thin-block case). In the thin-block model of Lemma F.161, $\tilde{\Psi}_L^{(\alpha)} \equiv 1$, so (F147) holds with $\Delta_\Psi = 0$ and the χ^2 -flatness is exact.

Lemma F.131 (Surrogate approximation $\Rightarrow$ a tilt influence bound). *Assume Proposition F.164. Realize the in-plane plaquette field under $\nu_L^\parallel$ in tree gauge as independent coordinates $(\sigma_a)_{a=1}^{|T_L|}$, and assume without loss that the cut σ -algebra $\mathcal{F}_{\text{cut}}$ is generated by $(\sigma_1, \dots, \sigma_{m_L})$. Then for each coordinate $a \in \{1, \dots, |T_L|\}$ and the corresponding independent resampling σ'_a , one has*

$$\left\| \tilde{\Psi}_L^{(\alpha)} - (\tilde{\Psi}_L^{(\alpha)})^{(a)} \right\|_{L^2(\nu_L^\parallel)} \leq 2\epsilon_L^{(C)} + \frac{t \Delta_X}{\beta c_\Psi} \mathbf{1}_{\{a \leq m_L\}}. \quad (\text{F149})$$

In particular, if $\epsilon_L^{(C)} \leq \Delta'/\beta$ for some constant Δ' independent of L , then the global influence target (F147) holds with

$$\Delta_\Psi = 2\Delta' + \frac{t \Delta_X}{c_\Psi}. \quad (\text{F150})$$

Proof. Write $\tilde{\Psi} := \tilde{\Psi}_L^{(\alpha)}$ and $\tilde{\Psi}_{\text{cut}}$ for the surrogate from (F189). By the triangle inequality and invariance of the $L^2(\nu_L^\parallel)$ norm under a single-coordinate resampling,

$$\begin{aligned} \|\tilde{\Psi} - \tilde{\Psi}^{(a)}\|_2 &\leq \|\tilde{\Psi} - \tilde{\Psi}_{\text{cut}}\|_2 + \|\tilde{\Psi}_{\text{cut}} - (\tilde{\Psi}_{\text{cut}})^{(a)}\|_2 + \|(\tilde{\Psi}_{\text{cut}})^{(a)} - \tilde{\Psi}^{(a)}\|_2 \\ &\leq 2\|\tilde{\Psi} - \tilde{\Psi}_{\text{cut}}\|_2 + \|\tilde{\Psi}_{\text{cut}} - (\tilde{\Psi}_{\text{cut}})^{(a)}\|_2. \end{aligned}$$

The first term is bounded by $2\epsilon_L^{(C)}$ by (F190). For the second term, note that $\tilde{\Psi}_{\text{cut}} = e^{-tX_{\text{cut}}}/E[e^{-tX_{\text{cut}}}]$ and the denominator is nonrandom. If $a > m_L$ then X_{cut} does not depend on σ_a and the difference vanishes. If $a \leq m_L$, then using $|e^{-tx} - e^{-ty}| \leq t|x - y|$ and (F191),

$$\|\tilde{\Psi}_{\text{cut}} - (\tilde{\Psi}_{\text{cut}})^{(a)}\|_2 = \frac{\|e^{-tX_{\text{cut}}} - e^{-tX_{\text{cut}}^{(a)}}\|_2}{E[e^{-tX_{\text{cut}}}]_2} \leq \frac{t}{c_\Psi} \|X_{\text{cut}} - X_{\text{cut}}^{(a)}\|_2 \leq \frac{t \Delta_X}{\beta c_\Psi}$$

by (F188). Combining these bounds gives (F149); the final statement is immediate. $\square$

Corollary F.132 (Influences outside the cut $\Rightarrow L^2$ cut-localization). *In the setting of Lemma F.127, assume*

$$\|Y - Y^{(a)}\|_{L^2} \leq \frac{\Delta}{\beta} \quad \text{for all } a \in \{m+1, \dots, n\}. \quad (\text{F151})$$

Then

$$\|Y - E[Y \mid \mathcal{F}_{\text{cut}}]\|_{L^2} \leq \frac{\Delta}{\beta\sqrt{2}} \sqrt{n-m}. \quad (\text{F152})$$

In particular, if $Y = \tilde{\Psi}_L^{(\alpha)}$ and $n = |T_L|$, this gives a fully explicit sufficient condition for the existence of an $\mathcal{F}_{\text{cut}}$ -measurable L^2 approximation without committing to the exponential form (F189).

Proof. By Lemma F.127 and (F151),

$$\|Y - E[Y \mid \mathcal{F}_{\text{cut}}]\|_2^2 \leq \frac{1}{2} \sum_{a=m+1}^n \|Y - Y^{(a)}\|_2^2 \leq \frac{1}{2} (n-m) \frac{\Delta^2}{\beta^2}.$$

Taking square roots yields (F152). $\square$

The influence bound (F147) is not an additional hypothesis in the main proof chain. It is recorded only as a particularly sharp *sufficient* target for discharging the Plan A χ^2 hypothesis without invoking the cut-local exponential surrogate (F189). Lemma F.131 shows how it can be derived from Proposition F.164 when the surrogate error $\epsilon_L^{(C)}$ improves to $O(\beta^{-1})$.

Proposition F.133 (Dyadic bond-moving inequality on the theorem-level dyadic path). *Fix $\beta \geq \beta_{\text{BM}}$ and $\alpha \in [1/2, 1]$. Then the dyadic bond-moving inequality holds with the explicit partition-correction factor*

$$\lambda_R(w_{2L}^{(\alpha)}) \leq \exp\left(\frac{C}{L\beta}\right) \lambda_R(w_L^{(\alpha)})^4, \quad (\text{F153})$$

for every dyadic L and every nontrivial irrep $R \neq \mathbf{1}$, where $C = C(G)$ is the partition-correction constant of Proposition F.146. In particular, after absorbing the finite number of initial dyadic correction factors into the universal weak-window constants,

$$|\lambda_R(w_{2L}^{(\alpha)})| \leq |\lambda_R(w_L^{(\alpha)})|^4 \quad (\text{F154})$$

for the dyadic scales used on the Route 1 load-bearing path.

Theorem-level ingredients used in the proof.

- *Lemma F.138: channel decomposition on the cut links.*
- *Lemma F.139: channel chessboard inequality (reflection positivity).*
- *Proposition F.146: partition correction under the α -schedule.*

Proof. By Lemma F.138 and Lemma F.139,

$$Z_R^{(2L)} = \sum_c Z_{R,c}^{(2L)} \leq \sum_c \left(Z_{R,c}^{(L)} \right)^4.$$

Since $x \mapsto x^4$ is convex on $\mathbb{R}_{\geq 0}$,

$$\sum_c \left(Z_{R,c}^{(L)} \right)^4 \leq \left(\sum_c Z_{R,c}^{(L)} \right)^4 = \left(Z_R^{(L)} \right)^4. \quad (\text{F155})$$

Divide by the partition functions and use Proposition F.146:

$$\lambda_R(w_{2L}^{(\alpha)}) = \frac{Z_R^{(2L)}}{Z^{(2L)}} \leq \frac{(Z_R^{(L)})^4}{(Z^{(L)})^4} \exp\left(\frac{C}{L\beta}\right) = \exp\left(\frac{C}{L\beta}\right) \lambda_R(w_L^{(\alpha)})^4,$$

which is (F153). The finite dyadic overhead remark following Proposition F.146 then yields the clean step form (F154) on the scales used later in Route 1. $\square$

Lemma F.134 (Corrected dyadic iteration with explicit overhead). *Assume the one-step dyadic bond-moving inequality in the explicit form*

$$\left| \lambda_R(w_{2L}^{(\alpha)}) \right| \leq \exp\left(\frac{C_{\text{pc}}(G) \kappa_{\text{pc}}}{L\beta}\right) \left| \lambda_R(w_L^{(\alpha)}) \right|^4 \quad (\text{F156})$$

for every dyadic $L = 2^m$ and every nontrivial irrep $R \neq \mathbf{1}$. Then for every dyadic $L = 2^n$,

$$\left| \lambda_R(w_L^{(\alpha)}) \right| \leq \exp\left(\frac{2C_{\text{pc}}(G) \kappa_{\text{pc}}}{7\beta} (L^2 - L^{-1})\right) \left| \lambda_R(w_1^{(\alpha)}) \right|^{L^2}. \quad (\text{F157})$$

Proof. Write E_n for the minimal nonnegative number such that

$$\left| \lambda_R(w_{2^n}^{(\alpha)}) \right| \leq \exp(E_n/\beta) \left| \lambda_R(w_1^{(\alpha)}) \right|^{4^n}.$$

Then $E_0 = 0$, and (F156) gives the recurrence

$$E_{n+1} = 4E_n + \frac{C_{\text{pc}}(G) \kappa_{\text{pc}}}{2^n}.$$

Dividing by 4^{n+1} and summing the resulting geometric series yields

$$\frac{E_n}{4^n} = \sum_{m=0}^{n-1} \frac{C_{\text{pc}}(G) \kappa_{\text{pc}}}{2^m 4^{m+1}} = \frac{C_{\text{pc}}(G) \kappa_{\text{pc}}}{4} \sum_{m=0}^{n-1} 8^{-m} = \frac{2C_{\text{pc}}(G) \kappa_{\text{pc}}}{7} (1 - 8^{-n}).$$

Hence

$$E_n = \frac{2C_{\text{pc}}(G) \kappa_{\text{pc}}}{7} 4^n (1 - 8^{-n}) = \frac{2C_{\text{pc}}(G) \kappa_{\text{pc}}}{7} (4^n - 2^{-n}),$$

which is exactly (F157) because $L = 2^n$ and $|T_L| = L^2 = 4^n$. $\square$

Channel decomposition and channel-wise chessboard for a compact group We now discharge items (1)–(2) in Proposition F.133 for a compact group G , using standard reflection positivity. Throughout this subsection, fix a dyadic L and an irreducible unitary representation R of G .

Definition F.135 (Dyadic cut graph and channel basis). For the dyadic split of the $2L \times 2L$ tile into four $L \times L$ quadrants, let $\Gamma_{\text{cut}} = \Gamma_{\text{cut}}(L)$ denote the internal cut graph consisting of the in-plane links lying on the two midlines (together with their cut vertices). Write E_{cut} and V_{cut} for its edge and vertex sets. The gauge group $G^{V_{\text{cut}}}$ acts on $G^{E_{\text{cut}}}$ by the usual lattice gauge transformations.

Define the cut Hilbert space

$$\mathcal{H}_{\text{cut}} := L^2(G^{E_{\text{cut}}})^{G^{V_{\text{cut}}}},$$

the closed subspace of gauge-invariant square-integrable functions of the cut-link variables. A *channel* c is a choice of:

- an irrep label $\lambda_e \in \text{Irr}(G)$ for each cut edge $e \in E_{\text{cut}}$;
- an intertwiner label ι_v at each cut vertex $v \in V_{\text{cut}}$ choosing an element of an orthonormal basis of the invariant space

$$\text{Inv}\left(\bigotimes_{e \ni v} V_{\lambda_e}^{\sigma(v,e)}\right), \quad V_{\lambda}^+ := V_{\lambda}, \quad V_{\lambda}^- := V_{\lambda}^*,$$

where $\sigma(v, e) \in \{+, -\}$ is determined by the orientation of e relative to v (outgoing = +, incoming = -).

The resulting spin-network functions $\psi_c \in \mathcal{H}_{\text{cut}}$ form a countable orthonormal basis of $\mathcal{H}_{\text{cut}}$. We write $\mathcal{C}_L$ for the set of all such channels.

Lemma F.136 (Positive-type factorization of normalized characters). *Let ρ be a unitary irrep of a compact group G with dimension d_{ρ} and normalized character $\varphi_{\rho}(g) = d_{\rho}^{-1} \chi_{\rho}(g)$. Then for all $g, h \in G$,*

$$\varphi_{\rho}(g^{-1}h) = \frac{1}{d_{\rho}} \sum_{a,b=1}^{d_{\rho}} \overline{\rho(g)_{ab}} \rho(h)_{ab}. \quad (\text{F158})$$

In particular, the kernel $K_{\rho}(g, h) := \varphi_{\rho}(g^{-1}h)$ is positive semidefinite.

Proof. Since ρ is unitary, $\rho(g^{-1}) = \rho(g)^*$. Thus $\rho(g^{-1}h) = \rho(g)^* \rho(h)$ and

$$\chi_\rho(g^{-1}h) = \text{Tr}(\rho(g)^* \rho(h)) = \sum_{a,b} \overline{\rho(g)_{ab}} \rho(h)_{ab}.$$

Divide by d_ρ to obtain (F158). Positivity follows because (F158) is a Gram representation. $\square$

Proposition F.137 (Reflection positivity for the thick-block reference). *The thick-block α -reference measure $\nu_{2L}^{(\alpha)}$ is reflection positive with respect to reflections in the two midlines used in the dyadic split, in the standard Osterwalder–Seiler sense.*

Lemma F.138 (Channel decomposition at the cut links (Item (1) in Proposition F.133)). *Assume F.137. Then there exist nonnegative numbers $\{Z_{R,c}^{(2L)}\}_{c \in \mathcal{C}_L}$ and $\{Z_{R,c}^{(L)}\}_{c \in \mathcal{C}_L}$ such that*

$$Z_R^{(2L)} = \sum_{c \in \mathcal{C}_L} Z_{R,c}^{(2L)}, \quad Z_R^{(L)} = \sum_{c \in \mathcal{C}_L} Z_{R,c}^{(L)},$$

and $Z_{R,c}^{(2L)} \geq 0$, $Z_{R,c}^{(L)} \geq 0$ for all c .

Proof (explicit Hilbert-space expansion). Fix the dyadic split and write Θ for the composition of the two midline reflections (so Θ maps the lower-left quadrant to the upper-right). Reflection positivity implies that the sesquilinear form

$$\langle F, G \rangle_{\text{OS}} := \int (\Theta F) G d\nu_{2L}^{(\alpha)}$$

is positive semidefinite on the algebra $\mathcal{A}_{++}$ of observables supported in the lower-left quadrant together with its boundary cut links.

By Lemma F.136, the normalized character insertion on the $2L$ boundary can be written as a finite sum of matrix-coefficient products across the reflection interface. Concretely, there exists a finite family of cut-dependent functions $F_{R,ab} \in \mathcal{A}_{++}$ (built from quadrant plaquette weights and a boundary matrix coefficient of ρ_R) such that

$$Z_R^{(2L)} = \sum_{a,b=1}^{d_R} \langle F_{R,ab}, F_{R,ab} \rangle_{\text{OS}}. \quad (\text{F159})$$

Each inner product is nonnegative by reflection positivity, so the right-hand side is a sum of nonnegative terms.

Now expand each $F_{R,ab}$ in the cut Hilbert space basis of Definition F.135. More precisely: the restriction of $F_{R,ab}$ to the cut link variables defines an element of $\mathcal{H}_{\text{cut}}$, hence admits an expansion $F_{R,ab} = \sum_{c \in \mathcal{C}_L} f_{R,ab}(c) \psi_c$ with $\sum_c |f_{R,ab}(c)|^2 < \infty$. By Parseval,

$$\langle F_{R,ab}, F_{R,ab} \rangle_{\text{OS}} = \sum_{c \in \mathcal{C}_L} |f_{R,ab}(c)|^2.$$

Define

$$Z_{R,c}^{(2L)} := \sum_{a,b=1}^{d_R} |f_{R,ab}(c)|^2 \geq 0.$$

Summing over c and using (F159) yields $Z_R^{(2L)} = \sum_c Z_{R,c}^{(2L)}$.

The same construction applied at scale L (with the corresponding dyadic cut graph in the L -tile reference geometry) yields the decomposition $Z_R^{(L)} = \sum_c Z_{R,c}^{(L)}$ with $Z_{R,c}^{(L)} \geq 0$. $\square$

Lemma F.139 (Channel-wise chessboard estimate (Item (2) in Proposition F.133)). *Assume F.137. With the channel pieces defined in Lemma F.138, for every channel $c \in \mathcal{C}_L$,*

$$Z_{R,c}^{(2L)} \leq \left(Z_{R,c}^{(L)} \right)^4.$$

Proof (two-step RP / Cauchy–Schwarz). Fix c and fix the basis element ψ_c on the cut graph. By construction in Lemma F.138, $Z_{R,c}^{(2L)}$ is a finite sum of Osterwalder–Schrader norms of observables supported in a fundamental quadrant with boundary channel ψ_c . Apply reflection positivity across the vertical midline to double the quadrant to a half-tile, and then across the horizontal midline to double to the full $2L$ tile. At each doubling step, Cauchy–Schwarz bounds the doubled norm by the product of the norms on the two halves. Because the channel insertion ψ_c is reflection-even (spin-network basis elements are real class functions on each cut edge and gauge-invariant at vertices), the same channel label is preserved under reflections. After two doublings one obtains precisely the fourth power bound. $\square$

Corollary F.140 (Discharge of items (1)–(2) and closure of BM_dyadic). *Assume F.137. Then items (1)–(2) of Proposition F.133 hold for G . Since item (3) is discharged in the strengthened branch (Lemma F.143 + Proposition F.146), Proposition F.126 holds for G up to the explicit correction of (F153).*

Partition correction inequality: an explicit attack The analytic input (3) in Proposition F.133 is a statement about how the thick-block partition function scales under dyadic gluing. The purpose of this subsection is to reduce (F169) to a cut-local finite-dimensional tilt bound and then to bound that tilt explicitly in weak coupling.

Definition F.141 (Dyadic cut set and cut-adjacent plaquettes). Let T_{2L} be the $2L \times 2L$ tile in the (μ, ν) -plane and split it into four $L \times L$ quadrants by the coordinate midlines. Let $E_{\text{cut}} = E_{\text{cut}}(L)$ be the set of in-plane links lying on the two midlines (the internal cuts).

Let P_{cut} denote the set of *out-of-plane* plaquettes in the thick block incident to at least one link in E_{cut} . In $d = 4$ one has $|E_{\text{cut}}| \asymp L$ and

$$|P_{\text{cut}}| \leq C_{\text{cut}} L, \quad (\text{F160})$$

since each cut link lies in at most $2(d - 2) = 4$ out-of-plane plaquettes.

Lemma F.142 (Gluing without extra cross weights is supermultiplicative). *Let $Z_{\text{glue}}^{(2L)}$ be the partition function of the geometry obtained by gluing four L -quadrant blocks along E_{cut} (identifying the cut links) without introducing any additional plaquette weights beyond those already present in the quadrants. Then*

$$Z_{\text{glue}}^{(2L)} \geq (Z^{(L)})^4. \quad (\text{F161})$$

Proof. Integrate out interior variables of each quadrant, leaving a nonnegative effective weight $F(U_{E_{\text{cut}}})$ on the cut links. By symmetry the same F appears for each quadrant, and with Haar-product measure on $U_{E_{\text{cut}}}$ normalized to 1 one has

$$Z^{(L)} = \int F dU_{E_{\text{cut}}}, \quad Z_{\text{glue}}^{(2L)} = \int F^4 dU_{E_{\text{cut}}}.$$

Since $x \mapsto x^4$ is convex on $\mathbb{R}_{\geq 0}$, Jensen gives $\int F^4 \geq (\int F)^4$, which is (F161). $\square$

Lemma F.143 (Verification of the cut-local tilt form for the dyadic reference). *Fix G and dyadic L . In the strengthened dyadic construction, the $2L$ thick-block reference partition function is obtained from the glued-quadrant geometry of Lemma F.142 by adding only the cut-adjacent plaquette weights on P_{cut} (with exponent $\theta \in (0, 1]$), i.e. the density ratio is exactly*

$$D^{(2L)}(U) = D_{\text{glue}}^{(2L)}(U) \cdot \prod_{p \in P_{\text{cut}}} \widehat{W}_{\beta}(U_p)^{\theta}. \quad (\text{F162})$$

Consequently the partition functions satisfy the exact identity

$$Z^{(2L)} = Z_{\text{glue}}^{(2L)} \mathbb{E}_{\nu_{\text{glue}}} \left[\prod_{p \in P_{\text{cut}}} \widehat{W}_{\beta}(U_p)^{\theta} \right], \quad (\text{F163})$$

which is the hypothesis (F166) of Proposition F.146. Moreover, under the strengthened schedule of Remark F.147 (taking $\theta = 1 - \alpha_{\text{pc}}(2L, \beta)$), one has $\beta\theta \lesssim 1/L^2$ as required in (F167).

Proof. Equation (F162) is the defining relationship between the glued-quadrant dyadic reference and the true $2L$ reference in the strengthened construction: the only additional factors beyond those present in the quadrants are the cut-adjacent plaquette weights on P_{cut} , inserted with exponent θ . Integrating (F162) against Haar measure and dividing by $Z_{\text{glue}}^{(2L)}$ yields (F163). The schedule statement follows from Remark F.147. $\square$

Lemma F.144 (Weak-coupling plaquette defect bound). *Fix a unitary representation π and set $A(g) := 1 - \frac{1}{\dim(\pi)} \Re \text{Tr } \pi(g)$ as in (F119). As above, we may assume π is faithful after replacing G by $G_{\text{eff}} := G / \ker(\pi)$. There exist $\beta_{\text{plaq}} > 0$ and $C_{\text{plaq}} < \infty$ (depending only on G and π) such that for every weak-coupling α -reference measure of Wilson class on a finite block (including the glued geometry above) and every plaquette p in that block,*

$$\mathbb{E}[A(U_p)] \leq \frac{C_{\text{plaq}}}{\beta} \quad \text{for all } \beta \geq \beta_{\text{plaq}}, \alpha \in [1/2, 1]. \quad (\text{F164})$$

Proof. Work conditionally on the small-curvature event Ω_{ε_0} of Lemma F.224. On Ω_{ε_0} one may write $U_p = \exp(\xi_p)$ with $\|\xi_p\| \leq \varepsilon_0$, and the Taylor expansion (F122) yields $A(U_p) \leq C \|\xi_p\|^2$ after shrinking ε_0 and enlarging C . Moreover, on Ω_{ε_0} the plaquette logarithm is controlled by the curvature 2-form, so $\|\xi_p\| \leq C \|F_p\|$. By Proposition F.226 (curvature Brascamp–Lieb domination), F_p is sub-Gaussian with parameter $\sigma^2 \asymp 1/\beta$, hence $\mathbb{E}[\|F_p\|^2 \mid \Omega_{\varepsilon_0}] \leq C/\beta$ uniformly in p . Therefore $\mathbb{E}[A(U_p) \mid \Omega_{\varepsilon_0}] \leq C/\beta$.

Finally, remove the conditioning using Lemma F.224, which gives $\mathbb{P}(\Omega_{\varepsilon_0}^c) \leq |P(\mathcal{B}_L)|e^{-c\beta}$ and $0 \leq A \leq 2$. Absorbing the exponentially small contribution into C_{plaq} for $\beta \geq \beta_{\text{plaq}}$ yields (F164). $\square$

Lemma F.145 (Weak-coupling plaquette defect L^2 bound). *Under the hypotheses of Lemma F.144, there exist $\beta_{\text{plaq},2} > 0$ and $C_{\text{plaq},2} < \infty$ (depending only on G and π) such that for every weak-coupling α -reference measure of Wilson class on a finite block and every plaquette p ,*

$$\mathbb{E}[A(U_p)^2] \leq \frac{C_{\text{plaq},2}}{\beta^2} \quad \text{for all } \beta \geq \beta_{\text{plaq},2}, \alpha \in [1/2, 1]. \quad (\text{F165})$$

In particular, $\|A(U_p)\|_{L^2} \leq \sqrt{C_{\text{plaq},2}}/\beta$.

Proof. Repeat the argument of Lemma F.144 but keep second moments. On Ω_{ε_0} , the Taylor bound gives $A(U_p) \leq C \|\xi_p\|^2 \leq C \|F_p\|^2$, hence $A(U_p)^2 \leq C \|F_p\|^4$. By Proposition F.226, F_p is sub-Gaussian with parameter $\sigma^2 \asymp 1/\beta$, so $\mathbb{E}[\|F_p\|^4 \mid \Omega_{\varepsilon_0}] \leq C/\beta^2$ uniformly in p . Thus $\mathbb{E}[A(U_p)^2 \mid \Omega_{\varepsilon_0}] \leq C/\beta^2$.

Remove the conditioning as before: $\mathbb{P}(\Omega_{\varepsilon_0}^c) \leq |P(\mathcal{B}_L)|e^{-c\beta}$ and $0 \leq A \leq 2$ imply $\mathbb{E}[A(U_p)^2 \mathbf{1}_{\Omega_{\varepsilon_0}^c}] \leq 4\mathbb{P}(\Omega_{\varepsilon_0}^c)$, which is exponentially small in β and can be absorbed into $C_{\text{plaq},2}$ for $\beta \geq \beta_{\text{plaq},2}$. $\square$

Proposition F.146 (Partition correction from a cut-local tilt). *Assume that, relative to the glued-quadrant geometry of Lemma F.142, the true $2L$ thick-block partition function differs only by the inclusion of additional cut-adjacent plaquette weights on P_{cut} with a (possibly small) exponent $\theta \in (0, 1]$:*

$$Z^{(2L)} = Z_{\text{glue}}^{(2L)} \mathbb{E}_{\nu_{\text{glue}}} \left[\prod_{p \in P_{\text{cut}}} \widehat{W}_\beta(U_p)^\theta \right], \quad (\text{F166})$$

where ν_{glue} is the normalized glued measure (so $\mathbb{E}_{\nu_{\text{glue}}}[\cdot]$ is expectation under the density defining $Z_{\text{glue}}^{(2L)}$), and $\widehat{W}_\beta(g) = e^{-\beta A(g)}$ is the sup-normalized Wilson class weight.

If the exponent satisfies the quantitative smallness condition

$$\beta \theta \leq \frac{\kappa_{\text{pc}}}{L^2} \quad (\text{F167})$$

for some fixed $\kappa_{\text{pc}} > 0$, then with

$$C_{\text{pc}}(G) := C_{\text{cut}} C_{\text{plaq}} \quad (\text{F168})$$

one has the explicit lower bound

$$Z^{(2L)} \geq (Z^{(L)})^4 \exp\left(-\frac{C_{\text{pc}}(G) \kappa_{\text{pc}}}{L\beta}\right). \quad (\text{F169})$$

In particular, (F169) holds with $C = C_{\text{pc}}(G)\kappa_{\text{pc}}$ whenever (F166) and (F167) hold.

Proof. Combine (F166) with Lemma F.142:

$$Z^{(2L)} \geq (Z^{(L)})^4 \mathbb{E}_{\nu_{\text{glue}}} \left[\prod_{p \in P_{\text{cut}}} \widehat{W}_\beta(U_p)^\theta \right].$$

Since $\log$ is concave and $\widehat{W}_\beta(U_p)^\theta \in (0, 1]$, Jensen gives

$$\log \mathbb{E}_{\nu_{\text{glue}}} \left[\prod_{p \in P_{\text{cut}}} \widehat{W}_\beta(U_p)^\theta \right] \geq \mathbb{E}_{\nu_{\text{glue}}} \left[\log \prod_{p \in P_{\text{cut}}} \widehat{W}_\beta(U_p)^\theta \right] = -\beta\theta \sum_{p \in P_{\text{cut}}} \mathbb{E}_{\nu_{\text{glue}}} [A(U_p)].$$

Apply Lemma F.144 to bound each expectation by C_{plaq}/β and use the counting bound (F160):

$$\log \mathbb{E}_{\nu_{\text{glue}}} \left[\prod_{p \in P_{\text{cut}}} \widehat{W}_\beta(U_p)^\theta \right] \geq -\beta\theta \cdot |P_{\text{cut}}| \cdot \frac{C_{\text{plaq}}}{\beta} \geq -C_{\text{cut}} C_{\text{plaq}} \theta L.$$

Under (F167), $\theta L \leq \kappa_{\text{pc}}/(L\beta)$, hence the right-hand side is at least $-C_{\text{pc}}(G)\kappa_{\text{pc}}/(L\beta)$ by (F168). Exponentiating yields (F169). $\square$

Remark F.147 (How the α -schedule can enforce (F167)). Proposition F.146 is designed for the situation where the only genuinely “cross” contributions are cut-local weights with a *small* exponent θ . If these cut-local factors arise from a tilt sector with $\theta = 1 - \alpha$, then (F167) is equivalent to choosing

$$1 - \alpha \leq \frac{\kappa_{\text{pc}}}{\beta L^2}.$$

One convenient sufficient form is the strengthened schedule

$$\alpha_{\text{pc}}(L, \beta) := 1 - \frac{\kappa_{\text{pc}}}{\beta |P_{\perp}(\mathcal{B}_L)|} \asymp 1 - \frac{\kappa_{\text{pc}}}{\beta L^2} \quad (d = 4),$$

which is admissible since it only *reduces* the tilt exponent $\beta(1 - \alpha)$ (hence improves the χ^2 and slack budgets recorded earlier).

Lemma F.148 (Iteration of the dyadic bond-moving inequality). *Assume Proposition F.126. Then for every dyadic $L = 2^n$ and every irrep R ,*

$$|\lambda_R(w_L^{(\alpha)})| \leq |\lambda_R(w_1^{(\alpha)})|^{|T_L|} \quad (|T_L| = L^2). \quad (\text{F170})$$

Equivalently,

$$w_L^{(\alpha)} \preceq_{\text{coeff}} (w_1^{(\alpha)})^{*|T_L|}.$$

Proof. Remark on the partition-correction factor. The $\exp(C/(L\beta))$ correction produced in Proposition F.133 is absorbed into a finite dyadic overhead (at most a constant factor 2^{K_0} as discussed there), so it does not affect the scale-by-scale iteration of the pure step inequality (F154).

For $L = 1$ there is nothing to prove. Assume (F170) holds at scale L and apply (F154):

$$|\lambda_R(w_{2L}^{(\alpha)})| \leq |\lambda_R(w_L^{(\alpha)})|^4 \leq \left(|\lambda_R(w_1^{(\alpha)})|^{|T_L|} \right)^4 = |\lambda_R(w_1^{(\alpha)})|^{4|T_L|}.$$

Since $|T_{2L}| = (2L)^2 = 4L^2 = 4|T_L|$, this is exactly (F170) at scale $2L$. $\square$

Remark F.149 (Local dyadic decimation as a sufficient form of Proposition F.150). Lemma F.148 shows that the global power bound (F171) follows (for dyadic L) from the local dyadic step (F154). Thus, in implementations where the blocking factor L is taken dyadic (cf. the $N = 4^k$ conventions in Section F.16), it suffices to establish Proposition F.126 uniformly in $\alpha \in [1/2, 1]$ in the required weak-coupling window.

Proposition F.150 (Convolution dominance for the thick-block α -reference Wilson loop). *Let $w_L^{(\alpha(L))} dg = (\Phi_L)_\# \nu_L^{(\alpha(L))}$ be the thick-block boundary marginal around the $L \times L$ tile T_L , and let $w_1^{(\alpha(L))}$ be the corresponding 1×1 marginal. Assume that for every nontrivial irrep $R \neq \mathbf{1}$ one has the coefficient bound (in modulus)*

$$\left| \lambda_R(w_L^{(\alpha(L))}) \right| = \left| \mathbb{E}_{\nu_L^{(\alpha(L))}} [\varphi_R(\Phi_L)] \right| \leq \left| \mathbb{E}_{\nu_1^{(\alpha(L))}} [\varphi_R(\Phi_1)] \right|^{|T_L|}. \quad (\text{F171})$$

Equivalently, by Definition F.119 and Lemma F.118,

$$w_L^{(\alpha(L))} \preceq_{\text{coeff}} (w_1^{(\alpha(L))})^{*|T_L|}.$$

Comment. For the planar reference the inequality is an equality by Lemma F.123. Even when exterior boundary data cannot be gauged away, the same *planar power upper bound* remains valid: deterministic boundary insertions can only affect the trace factor, so by Lemma F.124 (Remark F.125) one still has $|\lambda_R| \leq \lambda_R(w_1^{\parallel})^{|T_L|}$ uniformly in boundary conditions. In the thick-block α -reference the out-of-plane plaquettes couple the tile degrees of freedom and destroy exact convolution structure. Proposition F.150 packages the desired bond-moving / domination comparison: it asserts that this coupling can only *reduce* the magnitude of the boundary character coefficient relative to the solvable convolution-power model. Reflection positivity of the α -reference family (Corollary F.32) is a natural structural prerequisite for such comparison inequalities, but does not by itself imply (F171); one needs an additional monotone decimation or domination argument.

Proposition F.151 (Thick-block convolution dominance is proved on the dyadic load-bearing path). *Assume the dyadic bond-moving inequality of Proposition F.133 on the weak-coupling window under consideration. Then for every dyadic block size $L = 2^n$ and every nontrivial irrep $R \neq \mathbf{1}$ one has*

$$\left| \lambda_R(w_L^{(\alpha(L))}) \right| \leq \left| \mathbb{E}_{\nu_1^{(\alpha(L))}} [\varphi_R(\Phi_1)] \right|^{|T_L|}. \quad (\text{F172})$$

Equivalently,

$$w_L^{(\alpha(L))} \preceq_{\text{coeff}} (w_1^{(\alpha(L))})^{*|T_L|}.$$

In particular, Proposition F.150 is no longer an external hypothesis on the load-bearing Route 1 chain.

Proof. Proposition F.133 yields the clean dyadic step inequality (F154) after absorbing the finite number of initial correction factors into the universal weak-window constants. Iterating

that step by Lemma F.148 gives

$$\left| \lambda_R(w_L^{(\alpha(L))}) \right| \leq \left| \lambda_R(w_1^{(\alpha(L))}) \right|^{|T_L|}$$

for every dyadic L . By the definition of the boundary marginal and (F171), this is exactly the desired coefficient bound. The coefficient-domination reformulation follows from Definition F.119 and Lemma F.118. $\square$

Proposition F.152 (One-plaquette weak-coupling coefficient bound (crossover form)). *Assume Theorem 1.4. Then there exist constants $c_{A1}(G, \rho) > 0$ and $\beta_{A1}(G, \rho) \geq 1$ such that for all $\beta \geq \beta_{A1}(G, \rho)$, all $\alpha \in [1/2, 1]$, and all nontrivial irreps $R \neq \mathbf{1}$,*

$$\left| \mathbb{E}_{\nu_1^{(\alpha)}}[\varphi_R(\Phi_1)] \right| \leq \exp\left(-c_{A1}(G, \rho) \frac{C_2(R)}{\alpha\beta + \sqrt{C_2(R)}} \right). \quad (\text{F173})$$

Moreover, in the low-representation heat-kernel regime $C_2(R) \leq \beta^2$ one has the simplified bound

$$\left| \mathbb{E}_{\nu_1^{(\alpha)}}[\varphi_R(\Phi_1)] \right| \leq \exp\left(-\frac{c_{A1}(G, \rho)}{2} \frac{C_2(R)}{\beta} \right). \quad (\text{F174})$$

Corollary F.153 (One-plaquette crossover bound on the Route 1 load-bearing path). *For every compact connected simple G and fixed Wilson representation ρ , the coefficient bounds (F173) and (F174) hold uniformly in $\alpha \in [1/2, 1]$ once $\beta \geq \beta_{A1}(G, \rho)$. In particular, the one-plaquette input used by the thick-block comparison is now available in theorem-level form, with no remaining condition wrapper on the load-bearing Route 1 spine.*

Proof. This is exactly the compact-simple analytic input proved in Appendix N and summarized in Theorem 1.4; the classical-family appendices remain direct audited special cases. $\square$

Proposition F.154 (Tile crossover from theorem-level dyadic inputs). *For every compact connected simple G and fixed Wilson representation ρ , the thick-block coefficient bound (F130) holds for all dyadic block sizes in the weak-coupling window relevant to Route 1.*

Proof. By Proposition F.151, for each nontrivial $R \neq \mathbf{1}$,

$$\left| \lambda_R(w_L^{(\alpha(L))}) \right| \leq \left| \mathbb{E}_{\nu_1^{(\alpha(L))}}[\varphi_R(\Phi_1)] \right|^{|T_L|}.$$

Applying Corollary F.153 to the one-plaquette factor yields

$$\left(\exp\left(-c_{A1}(G, \rho) \frac{C_2(R)}{\alpha(L)\beta + \sqrt{C_2(R)}} \right) \right)^{|T_L|} = \exp\left(-|T_L| c_{A1}(G, \rho) \frac{C_2(R)}{\alpha(L)\beta + \sqrt{C_2(R)}} \right),$$

which is exactly (F130). $\square$

Corollary F.155 (Thick-block crossover coefficient decay is closed on the load-bearing path). *For every compact connected simple G and fixed Wilson representation ρ , the coefficient decay (F130) holds for all dyadic block sizes in the weak-coupling window relevant to Route 1. Equivalently, the thick-block reference satisfies the mixing input packaged in Proposition F.103.*

Proof. Apply Proposition F.154. □

Remark F.156. Proposition F.150 is a global convolution-dominance statement comparing the thick-block boundary marginal to a solvable convolution power. In dyadic implementations it is implied by the local decimation step Proposition F.126 via Lemma F.148 (see Remark F.149). Reflection positivity of the α -reference family (Corollary F.32) is a structural prerequisite, but establishing the local step (F154) requires an additional monotone domination/decimation argument (bond moving).

Proposition F.152 is a separate local weak-coupling inequality. It is proved for arbitrary compact connected simple G in Appendix N; Appendices G–I retain explicit classical-family audits and are the only places where special-function/determinantal formulas enter the manuscript.

Alternative approach (C): convolution-power mixing input for coefficient pinning The bond-moving/chessboard approach above packages the planar comparison into a single hard inequality (BM_dyadic). Main chain is a complementary mechanism: it isolates the *representation-dependent* part of the needed decorrelation into an explicit, finite-dimensional representation-theoretic lemma (Lemma F.179 below). This lemma is the missing line-by-line input behind the heuristic statement “products of many i.i.d. plaquettes look Haar in high representations”, and it is the place where the familiar $1/\sqrt{d_R}$ factor (from Schur orthogonality) meets the additional $\lambda_R(w_1)^k$ damping (from convolution-power mixing).

In Main chain one combines:

- (i) a *covariance identity* for the shift of Fourier/character coefficients under a (small) transverse tilt;
- (ii) a cut-local Lipschitz/influence bound for the tilt observable (already proved elsewhere in the paper);
- (iii) the convolution-power damping supplied by Lemma F.179.

Item (i) is a one-line consequence of the marginal-tilt identity (Lemma F.85) and is recorded explicitly here.

Lemma F.157 (Coefficient shift identity under a positive tilt). *Let w^{ref} be a central probability density on G and let $\bar{R} : G \rightarrow [0, \infty)$ be measurable with $0 < \mathbb{E}_{w^{\text{ref}}}[\bar{R}] < \infty$. Define the tilted density*

$$w^{\text{tilt}}(g) := \frac{\bar{R}(g)}{\mathbb{E}_{w^{\text{ref}}}[\bar{R}]} w^{\text{ref}}(g). \quad (\text{F175})$$

Then for every irrep R with normalized character φ_R ,

$$\lambda_R(w^{\text{tilt}}) = \frac{\mathbb{E}_{w^{\text{ref}}}[\bar{R} \varphi_R]}{\mathbb{E}_{w^{\text{ref}}}[\bar{R}]} = \lambda_R(w^{\text{ref}}) + \frac{\text{Cov}_{w^{\text{ref}}}(\varphi_R, \bar{R})}{\mathbb{E}_{w^{\text{ref}}}[\bar{R}]}. \quad (\text{F176})$$

Proof. The first equality is immediate from (F175):

$$\lambda_R(w^{\text{tilt}}) = \int_G \varphi_R(g) w^{\text{tilt}}(g) dg = \frac{\int_G \bar{R}(g) \varphi_R(g) w^{\text{ref}}(g) dg}{\int_G \bar{R}(g) w^{\text{ref}}(g) dg} = \frac{\mathbb{E}_{w^{\text{ref}}}[\bar{R} \varphi_R]}{\mathbb{E}_{w^{\text{ref}}}[\bar{R}]}.$$

For the covariance form, expand $\mathbb{E}_{w^{\text{ref}}}[\bar{R} \varphi_R] = \mathbb{E}_{w^{\text{ref}}}[\bar{R}] \mathbb{E}_{w^{\text{ref}}}[\varphi_R] + \text{Cov}_{w^{\text{ref}}}(\varphi_R, \bar{R})$ and note that $\mathbb{E}_{w^{\text{ref}}}[\varphi_R] = \lambda_R(w^{\text{ref}})$ by definition. $\square$

Remark F.158 (How (F176) is used for thick blocks). In the thick-block setting one typically has two measures on the same block, e.g. a planar reference block measure $\nu_L^{\parallel}$ (out-of-plane plaquettes turned off) and an α -reference block measure $\nu_L^{(\alpha)}$. Let Φ_L be the induced boundary holonomy map and write $w_L^{\parallel} = \Phi_{L*} \nu_L^{\parallel}$, $w_L^{(\alpha)} = \Phi_{L*} \nu_L^{(\alpha)}$. If $R_L := d\nu_L^{(\alpha)} / d\nu_L^{\parallel}$ exists, then Lemma F.85 gives

$$w_L^{(\alpha)}(g) = \frac{\bar{R}_L(g)}{\mathbb{E}_{w_L^{\parallel}}[\bar{R}_L]} w_L^{\parallel}(g), \quad \bar{R}_L(g) := \mathbb{E}_{\nu_L^{\parallel}}[R_L \mid \Phi_L = g].$$

Applying Lemma F.157 with $\bar{R} = \bar{R}_L$ yields

$$\lambda_R(w_L^{(\alpha)}) = \lambda_R(w_L^{\parallel}) + \frac{\text{Cov}_{w_L^{\parallel}}(\varphi_R, \bar{R}_L)}{\mathbb{E}_{w_L^{\parallel}}[\bar{R}_L]}.$$

Moreover, by conditional expectation, $\text{Cov}_{w_L^{\parallel}}(\varphi_R, \bar{R}_L) = \text{Cov}_{\nu_L^{\parallel}}(\varphi_R(\Phi_L), R_L)$. Thus, bounding the coefficient shift reduces to a covariance estimate on the planar reference probability space, which is exactly what Lemma F.183 provides once a cut-local representation of R_L (or a surrogate) is available.

Lemma F.159 (Effective planar tilt after transverse integration). *Fix L and $\beta > 0$. Let $\nu_L^\parallel$ denote the planar reference block measure: the density includes the in-plane plaquette weights $\widehat{W}_\beta(U_p)$ for $p \in T_L$ and no weights for out-of-plane plaquettes $p \in P_\perp(\mathcal{B}_L)$. Let $\nu_L^{(\alpha)}$ denote the α -reference block measure: the same in-plane weights and out-of-plane weights $\widehat{W}_\beta(U_p)^\alpha$. Define the (unnormalized) transverse tilt on configurations by*

$$T_L^{(\alpha)}(U) := \prod_{p \in P_\perp(\mathcal{B}_L)} \widehat{W}_\beta(U_p)^\alpha. \quad (\text{F177})$$

Then $\nu_L^{(\alpha)}$ is the tilt of $\nu_L^\parallel$ by $T_L^{(\alpha)}$:

$$d\nu_L^{(\alpha)} = \frac{T_L^{(\alpha)}}{E_{\nu_L^\parallel}[T_L^{(\alpha)}]} d\nu_L^\parallel. \quad (\text{F178})$$

Let $\mathcal{F}_\parallel := \sigma(\{U_p\}_{p \in T_L})$ be the σ -algebra generated by the in-plane plaquette variables (e.g. in the spanning-tree gauge of Lemma F.123). Define the transverse-averaged tilt

$$\Psi_L^{(\alpha)} := E_{\nu_L^\parallel}[T_L^{(\alpha)} \mid \mathcal{F}_\parallel]. \quad (\text{F179})$$

Then for every bounded $\mathcal{F}_\parallel$ -measurable observable F ,

$$E_{\nu_L^{(\alpha)}}[F] = \frac{E_{\nu_L^\parallel}[F \Psi_L^{(\alpha)}]}{E_{\nu_L^\parallel}[\Psi_L^{(\alpha)}]}. \quad (\text{F180})$$

In particular, for the boundary holonomy Φ_L (which is $\mathcal{F}_\parallel$ -measurable), the induced boundary marginal $w_L^{(\alpha)} = \Phi_{L*}\nu_L^{(\alpha)}$ is a tilt of $w_L^\parallel = \Phi_{L*}\nu_L^\parallel$ by the function

$$\overline{\Psi}_L^{(\alpha)}(g) := E_{\nu_L^\parallel}[\Psi_L^{(\alpha)} \mid \Phi_L = g] = E_{\nu_L^\parallel}[T_L^{(\alpha)} \mid \Phi_L = g], \quad (\text{F181})$$

so that

$$w_L^{(\alpha)} = \frac{\overline{\Psi}_L^{(\alpha)}}{E_{w_L^\parallel}[\overline{\Psi}_L^{(\alpha)}]} w_L^\parallel. \quad (\text{F182})$$

Moreover, for each irrep R ,

$$\lambda_R(w_L^{(\alpha)}) - \lambda_R(w_L^\parallel) = \frac{\text{Cov}_{\nu_L^\parallel}(\varphi_R(\Phi_L), \Psi_L^{(\alpha)})}{E_{\nu_L^\parallel}[\Psi_L^{(\alpha)}]} = \frac{\text{Cov}_{w_L^\parallel}(\varphi_R, \overline{\Psi}_L^{(\alpha)})}{E_{w_L^\parallel}[\overline{\Psi}_L^{(\alpha)}]}. \quad (\text{F183})$$

Proof. Equation (F178) is immediate from the definitions of $\nu_L^{(\alpha)}$ and $\nu_L^\parallel$ (they share the same in-plane plaquette weights; only the out-of-plane plaquette weights differ, producing the Radon–Nikodym factor (F177)). For any bounded $\mathcal{F}_\parallel$ -measurable F , we have

$$E_{\nu_L^{(\alpha)}}[F] = \frac{E_{\nu_L^\parallel}[F T_L^{(\alpha)}]}{E_{\nu_L^\parallel}[T_L^{(\alpha)}]} = \frac{E_{\nu_L^\parallel}[F E_{\nu_L^\parallel}[T_L^{(\alpha)} \mid \mathcal{F}_\parallel]]}{E_{\nu_L^\parallel}[E_{\nu_L^\parallel}[T_L^{(\alpha)} \mid \mathcal{F}_\parallel]]} = \frac{E_{\nu_L^\parallel}[F \Psi_L^{(\alpha)}]}{E_{\nu_L^\parallel}[\Psi_L^{(\alpha)}]},$$

which is (F180). The identities (F181)–(F182) are the specialization of Lemma F.85 to the tilt (F178), together with the tower property $E[T_L^{(\alpha)} | \Phi_L] = E[E[T_L^{(\alpha)} | \mathcal{F}_\parallel] | \Phi_L]$ (since Φ_L is $\mathcal{F}_\parallel$ -measurable). Finally, (F183) is (F182) tested against φ_R and rewritten as a covariance, exactly as in Lemma F.157. $\square$

Corollary F.160 (Normalized effective tilt). *In the setting of Lemma F.159, define the (in-plane) normalized effective tilt*

$$\tilde{\Psi}_L^{(\alpha)} := \frac{\Psi_L^{(\alpha)}}{E_{\nu_L}[\Psi_L^{(\alpha)}]} = E_{\nu_L} \left[\frac{T_L^{(\alpha)}}{E_{\nu_L}[T_L^{(\alpha)}]} \middle| \mathcal{F}_\parallel \right]. \quad (\text{F184})$$

Then $E_{\nu_L}[\tilde{\Psi}_L^{(\alpha)}] = 1$ and the coefficient shift identity admits the denominator-free form

$$\lambda_R(w_L^{(\alpha)}) - \lambda_R(w_L^\parallel) = \text{Cov}_{\nu_L}(\varphi_R(\Phi_L), \tilde{\Psi}_L^{(\alpha)}) = \text{Cov}_{w_L^\parallel}(\varphi_R, \tilde{\Psi}_L^{(\alpha)}), \quad (\text{F185})$$

where $\tilde{\Psi}_L^{(\alpha)}(g) := E_{\nu_L}[\tilde{\Psi}_L^{(\alpha)} | \Phi_L = g] = \bar{\Psi}_L^{(\alpha)}(g)/E_{w_L^\parallel}[\bar{\Psi}_L^{(\alpha)}]$.

Lemma F.161 (Thin-block degeneration of the effective tilt). *Assume the block is the thin neighborhood $\mathcal{B}_L$ of Definition F.83. Then, in the setting of Lemma F.159, the conditional transverse partition function is exactly constant:*

$$\Psi_L^{(\alpha)} \equiv z(\alpha)^{|P_\perp(\mathcal{B}_L)|}. \quad (\text{F186})$$

Consequently $\tilde{\Psi}_L^{(\alpha)} \equiv 1$ and the boundary holonomy marginal coincides with the planar marginal:

$$w_L^{(\alpha)} = w_L^\parallel. \quad (\text{F187})$$

Proof. By Lemma F.92, each out-of-plane plaquette $p \in P_\perp(\mathcal{B}_L)$ contains an *opposite in-plane edge* $\ell^{(\rho,\pm)}$ which appears in no other plaquette holonomy in the thin block. Fix all link variables except $\ell^{(\rho,\pm)}$ and write $U_p = A \ell^{(\rho,\pm)} B$ for some A, B depending on the fixed links. Haar bi-invariance gives

$$\int \widehat{W}_\beta(U_p)^\alpha d\ell^{(\rho,\pm)} = \int \widehat{W}_\beta(g)^\alpha dg = z(\alpha),$$

independent of A, B . Iterating this integration over all such leaf edges shows that, *for every fixed in-plane configuration,*

$$E_{\nu_L} \left[T_L^{(\alpha)} \middle| \mathcal{F}_\parallel \right] = z(\alpha)^{|P_\perp(\mathcal{B}_L)|},$$

which is (F186). The remaining statements follow immediately from Corollary F.160. $\square$

Corollary F.162 (Thin-block tilt constancy is uniform under exterior conditioning). *Let η be any fixed exterior configuration on the complement links of the thin block $\mathcal{B}_L$. Let $\nu_{L,\eta}^\parallel$ and $\nu_{L,\eta}^{(\alpha)}$ denote the corresponding planar and α -reference conditional measures on the links in $\mathcal{B}_L$ (obtained by freezing the exterior links to η in the local block weight). Then the conclusion of Lemma F.161 holds pointwise in η : the effective tilt is constant and the boundary holonomy marginal is unchanged,*

$$\Psi_{L,\eta}^{(\alpha)} \equiv z(\alpha)^{|P_\perp(\mathcal{B}_L)|}, \quad (\Phi_L)_\# \nu_{L,\eta}^{(\alpha)} = (\Phi_L)_\# \nu_{L,\eta}^\parallel.$$

In particular, for every irrep R ,

$$\lambda_R((\Phi_L)_\# \nu_{L,\eta}^{(\alpha)}) = \lambda_R((\Phi_L)_\# \nu_{L,\eta}^\parallel).$$

Proof. The proof of Lemma F.161 integrates, one by one, the Haar leaf edges $\ell^{(\rho,\pm)}$ that appear in exactly one out-of-plane plaquette holonomy. This integration is local and uses only Haar bi-invariance on those leaf edges. Freezing an exterior configuration η changes only the deterministic group elements carried by edges *outside* $\mathcal{B}_L$; it does not remove the Haar integration over the leaf edges *inside* $\mathcal{B}_L$. Therefore the same leaf integration yields (F186) for $\nu_{L,\eta}^\parallel$, and the remaining conclusions follow exactly as in Lemma F.161. $\square$

Remark F.163. Lemma F.161 highlights that the “thin” neighborhood $\mathcal{B}_L$ is a toy model for transverse integration: all out-of-plane plaquettes are Haar-leaves and hence contribute only a constant factor. The genuine thick-block mixing wall begins when the out-of-plane plaquette sector has a nontrivial “core” (shared transverse edges), so that $\Psi_L^{(\alpha)}$ acquires nontrivial dependence on the in-plane field.

The present subsection records item (iii) in enough detail to upgrade Main chain from a heuristic to a fully checkable inequality chain. Once item (ii) and the geometric instantiation of the effective tilt $\bar{R}$ are supplied, Main chain can be treated as “proved” rather than “standard”.

Proposition F.164 (Main-chain cut-local surrogate for the effective planar tilt). *Fix L and α and let $\tilde{\Psi}_L^{(\alpha)}$ be the normalized effective tilt from (F184). There exist*

- *a cut σ -algebra $\mathcal{F}_{\text{cut}} \subset \mathcal{F}_\parallel$ generated by at most m_L in-plane plaquettes, with $m_L \lesssim L$, and*

- an $\mathcal{F}_{\text{cut}}$ -measurable nonnegative observable $X_{\text{cut}} = X_{\text{cut}}(\sigma_{\text{cut}})$,

such that the following hold.

- (i) **L^2 -influence bound.** There is a constant $\Delta_X > 0$ so that, for each coordinate $a \in \{1, \dots, m_L\}$ and an independent resampling of that coordinate,

$$\|X_{\text{cut}} - X_{\text{cut}}^{(a)}\|_{L^2(\nu_L^\parallel)} \leq \frac{\Delta_X}{\beta}. \quad (\text{F188})$$

- (ii) **Exponential surrogate and L^2 -approximation.** There exists a tilt parameter $t = t_L \geq 0$ (possibly depending on α, β, L) such that the normalized cut-local surrogate

$$\tilde{\Psi}_{\text{cut}} := \frac{e^{-tX_{\text{cut}}}}{E_{\nu_L^\parallel}[e^{-tX_{\text{cut}}}]}$$
(F189)

approximates the true normalized effective tilt in $L^2(\nu_L^\parallel)$:

$$\|\tilde{\Psi}_L^{(\alpha)} - \tilde{\Psi}_{\text{cut}}\|_{L^2(\nu_L^\parallel)} \leq \epsilon_L^{(C)}. \quad (\text{F190})$$

- (iii) **Mean lower bound.** There exists $c_\Psi > 0$ (uniform in L) such that

$$E_{\nu_L^\parallel}[e^{-tX_{\text{cut}}}] \geq c_\Psi. \quad (\text{F191})$$

Lemma F.165 (Main-chain reduction via a cut-local surrogate). Assume Proposition F.164.

Then for every nontrivial irrep R ,

$$\begin{aligned} |\lambda_R(w_L^{(\alpha)}) - \lambda_R(w_L^\parallel)| &= |\text{Cov}_{\nu_L^\parallel}(\varphi_R(\Phi_L), \tilde{\Psi}_L^{(\alpha)})| \quad (\text{by (F185)}) \\ &\leq |\text{Cov}_{\nu_L^\parallel}(\varphi_R(\Phi_L), \tilde{\Psi}_{\text{cut}})| + |\text{Cov}_{\nu_L^\parallel}(\varphi_R(\Phi_L), \tilde{\Psi}_L^{(\alpha)} - \tilde{\Psi}_{\text{cut}})| \\ &\leq \frac{1}{E_{\nu_L^\parallel}[e^{-tX_{\text{cut}}}] |\text{Cov}_{\nu_L^\parallel}(\varphi_R(\Phi_L), e^{-tX_{\text{cut}}})|} + \epsilon_L^{(C)} \|\varphi_R(\Phi_L)\|_{L^2(\nu_L^\parallel)}. \end{aligned}$$

Moreover, $\|\varphi_R(\Phi_L)\|_{L^2(\nu_L^\parallel)} \leq d_R^{-1} \|w_L^\parallel\|_\infty^{1/2}$, and the first covariance term is bounded by Lemma F.183 applied to X_{cut} .

Remark F.166 (The surrogate normalizer is bounded away from 0). Since $X_{\text{cut}} \geq 0$ and $t > 0$, one has $0 < E_{\nu_L^\parallel}[e^{-tX_{\text{cut}}}] \leq 1$. Moreover, Jensen's inequality gives

$$E_{\nu_L^\parallel}[e^{-tX_{\text{cut}}}] \geq \exp(-t E_{\nu_L^\parallel}[X_{\text{cut}}]). \quad (\text{F192})$$

For the explicit strip observable of Lemma F.184, the one-plaquette defect mean bound (F164) yields

$$E_{\nu_L^\parallel}[X_{\text{cut}}] \leq |T_{\text{cut}}| \frac{C_{\text{plaq}}}{\beta} \leq \frac{C_{\text{cut}} L C_{\text{plaq}}}{\beta}. \quad (\text{F193})$$

Consequently,

$$1 \leq \frac{1}{E_{\nu_L^\parallel}[e^{-tX_{\text{cut}}}] } \leq \exp\left(t \frac{C_{\text{cut}} L C_{\text{plaq}}}{\beta}\right), \quad (\text{F194})$$

which is harmless provided $t L/\beta$ stays bounded; for instance this holds under the common choice $t = \beta(1 - \alpha)$ with $1 - \alpha \asymp \kappa/L^2$ at the entrance scale $L^2/\beta \asymp 1$.

Remark F.167 (Diagnostic on shared-edge Haar integrations at weak coupling). A basic obstruction to a “one-plaquette-at-a-time” leaf integration on a *thick* out-of-plane sector is the appearance of *shared* transverse edges. The simplest local diagnostic is the two-plaquette shared-edge integral

$$\mathcal{I}_\alpha(X) := \int_G \widehat{W}_\beta(g)^\alpha \widehat{W}_\beta(gX)^\alpha dg, \quad X \in G, \quad (\text{F195})$$

which arises when two transverse plaquettes share the same integrated edge-variable g . Let $z_\alpha := \int_G \widehat{W}_\beta(g)^\alpha dg$.

By Peter–Weyl and Schur orthogonality, one has the exact character expansion

$$\mathcal{I}_\alpha(X) = \sum_{S \in \widehat{G}} d_S \lambda_S(\widehat{W}_\beta^\alpha)^2 \chi_S(X) = z_\alpha^2 + \sum_{S \neq \mathbf{1}} d_S \lambda_S(\widehat{W}_\beta^\alpha)^2 \chi_S(X),$$

so using $|\chi_S(X)| \leq d_S$ gives the uniform bound

$$|\mathcal{I}_\alpha(X) - z_\alpha^2| \leq \sum_{S \neq \mathbf{1}} d_S^2 \lambda_S(\widehat{W}_\beta^\alpha)^2 = \|\widehat{W}_\beta^\alpha\|_{L^2(G)}^2 - z_\alpha^2 = z_{2\alpha} - z_\alpha^2 \leq z_{2\alpha}. \quad (\text{F196})$$

In particular, each shared-edge integration produces a *polynomially small* absolute error at weak coupling, since $z_{2\alpha} \asymp (2\alpha\beta)^{-\dim(G)/2}$ as $\beta \rightarrow \infty$ (by Laplace expansion in a chart near the identity).

Higher degree. If an integrated transverse edge is incident to $k \geq 2$ plaquette-factors, the same philosophy still applies. Indeed, for any $X_1, \dots, X_k \in G$ and $f(g) := \widehat{W}_\beta(g)^\alpha$ one has $0 \leq \int_G \prod_{j=1}^k f(gX_j) dg \leq \int_G f(g)f(gX) dg \leq \int_G f(g)^2 dg = z_{2\alpha}$ by dropping factors ≤ 1 and using Cauchy–Schwarz (here $X := X_1^{-1}X_2$). Moreover $z_\alpha^k \leq z_\alpha^2 \leq z_{2\alpha}$ for $k \geq 2$. Hence the deviation from the naive “decoupled” value is still uniformly bounded by $z_{2\alpha}$: $|\int \prod_{j=1}^k f(gX_j) dg - z_\alpha^k| \leq z_{2\alpha}$. Consequently, whenever a global shared-edge correction is

expressed as a product over edge-integrals, Lemma F.172 canonically yields an edge-by-edge decomposition with per-edge defect bounded by $z_{2\alpha}$.

This estimate is only a *local* diagnostic: to control a thick-block effective tilt one must still (i) identify the relevant set of shared edges in the transverse core, (ii) count how many such shared-edge integrations survive after all available leaf eliminations, and (iii) propagate the local bound (F196) through the remaining conditional expectations (typically via an L^2 tensor-network estimate rather than a pointwise union bound).

Lemma F.168 (Sufficient conditions for the cut-local surrogate estimate). *Work under $\nu_L^\parallel$ and fix a cut σ -algebra $\mathcal{F}_{\text{cut}} \subset \mathcal{F}_\parallel$. Let $T := T_L^{(\alpha)}$ and $\tilde{\Psi}_L^{(\alpha)}$ be defined as in (F184). Assume that on Ω_{ε_0} there exist a parameter $t \geq 0$, a nonnegative $\mathcal{F}_{\text{cut}}$ -measurable random variable X_{cut} , a nonnegative random variable X_{out} , and an error term Δ_{sh} such that*

$$T = e^{-tX_{\text{cut}}} e^{-tX_{\text{out}}} (1 + \Delta_{\text{sh}}). \quad (\text{F197})$$

Define the $\mathcal{F}_\parallel$ -conditional objects

$$m_\parallel := \mathbb{E}_{\nu_L^\parallel} [e^{-tX_{\text{out}}} \mid \mathcal{F}_\parallel], \quad \text{Err}_\parallel := \mathbb{E}_{\nu_L^\parallel} [e^{-tX_{\text{out}}} \Delta_{\text{sh}} \mid \mathcal{F}_\parallel],$$

and set $\bar{m} := \mathbb{E}_{\nu_L^\parallel} [e^{-tX_{\text{out}}}]$ and $Z_{\text{cut}} := \mathbb{E}_{\nu_L^\parallel} [e^{-tX_{\text{cut}}}]$. Assume the quantitative bounds

- (i) $\|m_\parallel - \bar{m}\|_{L^2(\nu_L^\parallel | \Omega_{\varepsilon_0})} \leq \varepsilon_{\text{out}},$
- (ii) $\|\text{Err}_\parallel\|_{L^2(\nu_L^\parallel | \Omega_{\varepsilon_0})} \leq \varepsilon_{\text{sh}},$
- (iii) $Z_{\text{cut}} \geq c_\Psi > 0$ and $\bar{m} \geq c_m > 0,$
- (iv) $\varepsilon_{\text{out}} + \varepsilon_{\text{sh}} \leq \frac{1}{2} c_\Psi c_m.$

Then the cut-local surrogate $\tilde{\Psi}_{\text{cut}} := e^{-tX_{\text{cut}}} / \mathbb{E}_{\nu_L^\parallel} [e^{-tX_{\text{cut}}}]$ satisfies

$$\|\tilde{\Psi}_L^{(\alpha)} - \tilde{\Psi}_{\text{cut}}\|_{L^2(\nu_L^\parallel | \Omega_{\varepsilon_0})} \leq \frac{8}{c_\Psi c_m} (\varepsilon_{\text{out}} + \varepsilon_{\text{sh}}).$$

Proof. On Ω_{ε_0} , the decomposition (F197) and the $\mathcal{F}_{\text{cut}}$ -measurability of X_{cut} give

$$\Psi_L^{(\alpha)} = \mathbb{E}_{\nu_L^\parallel} [T \mid \mathcal{F}_\parallel] = e^{-tX_{\text{cut}}} \mathbb{E}_{\nu_L^\parallel} [e^{-tX_{\text{out}}} (1 + \Delta_{\text{sh}}) \mid \mathcal{F}_\parallel] = e^{-tX_{\text{cut}}} (m_\parallel + \text{Err}_\parallel).$$

Hence, with $Z := \mathbb{E}_{\nu_L^\parallel} [\Psi_L^{(\alpha)}] = \mathbb{E}_{\nu_L^\parallel} [T]$, we have

$$\tilde{\Psi}_L^{(\alpha)} = \frac{\Psi_L^{(\alpha)}}{Z} = \frac{e^{-tX_{\text{cut}}} (m_\parallel + \text{Err}_\parallel)}{Z}.$$

By the triangle inequality and Cauchy–Schwarz,

$$\left| Z - Z_{\text{cut}} \bar{m} \right| = \left| \mathbb{E}_{\nu_{\parallel}}[e^{-tX_{\text{cut}}}(m_{\parallel} - \bar{m})] + \mathbb{E}_{\nu_{\parallel}}[e^{-tX_{\text{cut}}} \text{Err}_{\parallel}] \right| \leq \varepsilon_{\text{out}} + \varepsilon_{\text{sh}}.$$

Using $Z_{\text{cut}} \geq c_{\Psi}$ and $\bar{m} \geq c_m$ together with (iv) yields the lower bound

$$Z \geq Z_{\text{cut}} \bar{m} - (\varepsilon_{\text{out}} + \varepsilon_{\text{sh}}) \geq \frac{1}{2} c_{\Psi} c_m.$$

Now write

$$\tilde{\Psi}_L^{(\alpha)} - \tilde{\Psi}_{\text{cut}} = e^{-tX_{\text{cut}}} \left(\frac{m_{\parallel} + \text{Err}_{\parallel}}{Z} - \frac{1}{Z_{\text{cut}}} \right),$$

so that $e^{-tX_{\text{cut}}} \leq 1$ gives

$$\left\| \tilde{\Psi}_L^{(\alpha)} - \tilde{\Psi}_{\text{cut}} \right\|_2 \leq \left\| \frac{m_{\parallel} + \text{Err}_{\parallel}}{Z} - \frac{1}{Z_{\text{cut}}} \right\|_2.$$

Decompose the right-hand side as

$$\frac{m_{\parallel} + \text{Err}_{\parallel}}{Z} - \frac{1}{Z_{\text{cut}}} = \frac{(m_{\parallel} - \bar{m}) + \text{Err}_{\parallel}}{Z} + \frac{\bar{m}}{Z} - \frac{1}{Z_{\text{cut}}} = \frac{(m_{\parallel} - \bar{m}) + \text{Err}_{\parallel}}{Z} + \frac{Z_{\text{cut}} \bar{m} - Z}{Z Z_{\text{cut}}}.$$

Using the lower bounds $Z \geq \frac{1}{2} c_{\Psi} c_m$ and $Z_{\text{cut}} \geq c_{\Psi}$ and the estimates above,

$$\left\| \frac{(m_{\parallel} - \bar{m}) + \text{Err}_{\parallel}}{Z} \right\|_2 \leq \frac{2}{c_{\Psi} c_m} (\varepsilon_{\text{out}} + \varepsilon_{\text{sh}}), \quad \left| \frac{Z_{\text{cut}} \bar{m} - Z}{Z Z_{\text{cut}}} \right| \leq \frac{4}{c_{\Psi} c_m} (\varepsilon_{\text{out}} + \varepsilon_{\text{sh}}).$$

Combining the last displays yields the stated bound. $\square$

Remark F.169 (Where the hard analysis lives). Lemma F.168 makes explicit that the quantitative content of Proposition F.164(ii) reduces to two concrete inputs:

- a *geometric/probabilistic* decoupling statement that produces the factorization (F197) with a controlled shared-edge error ε_{sh} , and
- a weak-coupling control of the *in-plane fluctuations* of the outside factor, namely an L^2 bound on $m_{\parallel} - \bar{m}$ where $m_{\parallel} = \mathbb{E}_{\nu_{\parallel}}[e^{-tX_{\text{out}}} \mid \mathcal{F}_{\parallel}]$ and $\bar{m} = \mathbb{E}_{\nu_{\parallel}}[e^{-tX_{\text{out}}}]$. A sufficient approach is: (a) an influence estimate for X_{out} under single-plaquette resampling and (b) a sub-Gaussian/mgf bound for X_{out} , which together give stability of conditional Laplace transforms and hence the desired L^2 control of $m_{\parallel} - \bar{m}$.

In the thin-block model $\mathcal{B}_L$ the shared-edge term vanishes (Lemma F.161), while for genuine thick blocks one needs a quantitative version of the shared-edge diagnostic in Remark F.167 combined with a counting argument for the number of transverse shared edges crossing the cut.

Lemma F.170 (From outside-energy influences to L^2 flatness of $m_{\parallel}$). *Work under $\nu_L^{\parallel}$ and suppose $X_{\text{out}} = X_{\text{out}}(\sigma_{\parallel}, \omega)$ is a nonnegative random variable on a product space where $\sigma_{\parallel} = (\sigma_1, \dots, \sigma_N)$ are the in-plane plaquette variables (independent under $\nu_L^{\parallel}$) and ω collects all non-in-plane edge variables (Haar under $\nu_L^{\parallel}$, independent of $\sigma_{\parallel}$). Fix $t \geq 0$ and define*

$$m_{\parallel}(\sigma_{\parallel}) := \mathbb{E}[e^{-tX_{\text{out}}(\sigma_{\parallel}, \omega)} \mid \sigma_{\parallel}], \quad \bar{m} := \mathbb{E}[e^{-tX_{\text{out}}}].$$

For each $a \in \{1, \dots, N\}$ let $\sigma^{(a)}$ be obtained by resampling only σ_a independently and set $X_{\text{out}}^{(a)} := X_{\text{out}}(\sigma^{(a)}, \omega)$ and $m_{\parallel}^{(a)} := m_{\parallel}(\sigma^{(a)})$. Then

$$\|m_{\parallel} - \bar{m}\|_{L^2(\nu_L^{\parallel})} \leq \frac{1}{\sqrt{2}} \left(\sum_{a=1}^N \|e^{-tX_{\text{out}}} - e^{-tX_{\text{out}}^{(a)}}\|_{L^2(\nu_L^{\parallel})}^2 \right)^{1/2}. \quad (\text{F198})$$

In particular, if there is an index set $\mathcal{I}_{\text{out}} \subset \{1, \dots, N\}$ such that X_{out} does not depend on σ_a for $a \notin \mathcal{I}_{\text{out}}$, and

$$\|X_{\text{out}} - X_{\text{out}}^{(a)}\|_{L^2(\nu_L^{\parallel})} \leq \frac{\Delta_{\text{out}}}{\beta} \quad (a \in \mathcal{I}_{\text{out}}), \quad (\text{F199})$$

then

$$\|m_{\parallel} - \bar{m}\|_{L^2(\nu_L^{\parallel})} \leq \frac{t \Delta_{\text{out}}}{\beta \sqrt{2}} \sqrt{|\mathcal{I}_{\text{out}}|}. \quad (\text{F200})$$

Proof. Apply Lemma F.180 (Efron–Stein) to the function $Y := m_{\parallel}(\sigma_{\parallel})$ of the independent coordinates $\sigma_1, \dots, \sigma_N$. Then

$$\|m_{\parallel} - \bar{m}\|_2^2 = \text{Var}(m_{\parallel}) \leq \frac{1}{2} \sum_{a=1}^N \mathbb{E}[(m_{\parallel} - m_{\parallel}^{(a)})^2].$$

Now fix a and work on the probability space with coordinates $(\sigma_{-a}, \sigma_a, \sigma'_a, \omega)$ where σ'_a is an independent copy of σ_a . Write $Z := e^{-tX_{\text{out}}(\sigma_{-a}, \sigma_a, \omega)}$ and $Z^{(a)} := e^{-tX_{\text{out}}(\sigma_{-a}, \sigma'_a, \omega)}$. Then $m_{\parallel} = \mathbb{E}_{\omega}[Z]$ and $m_{\parallel}^{(a)} = \mathbb{E}_{\omega}[Z^{(a)}]$, so Jensen gives

$$(m_{\parallel} - m_{\parallel}^{(a)})^2 = \left(\mathbb{E}_{\omega}[Z - Z^{(a)}] \right)^2 \leq \mathbb{E}_{\omega}[(Z - Z^{(a)})^2].$$

Taking expectation over $(\sigma_{-a}, \sigma_a, \sigma'_a)$ yields $\mathbb{E}[(m_{\parallel} - m_{\parallel}^{(a)})^2] \leq \mathbb{E}[(Z - Z^{(a)})^2] = \|e^{-tX_{\text{out}}} - e^{-tX_{\text{out}}^{(a)}}\|_2^2$. Inserting this into the Efron–Stein bound and taking square roots gives (F198). If (F199) holds, then $|e^{-tx} - e^{-ty}| \leq t|x - y|$ implies $\|e^{-tX_{\text{out}}} - e^{-tX_{\text{out}}^{(a)}}\|_2 \leq t \|X_{\text{out}} - X_{\text{out}}^{(a)}\|_2 \leq t\Delta_{\text{out}}/\beta$ for $a \in \mathcal{I}_{\text{out}}$ and 0 otherwise. This yields (F200). $\square$

Lemma F.171 (Bounding the shared-edge error contribution). *Work under $\nu_L^{\parallel}$ and let Δ_{sh} be square-integrable. With $X_{\text{out}} \geq 0$ and $t \geq 0$ define*

$$\text{Err}_{\parallel} := \mathbb{E}_{\nu_L^{\parallel}}[e^{-tX_{\text{out}}} \Delta_{\text{sh}} \mid \mathcal{F}_{\parallel}].$$

Then

$$\|\text{Err}_\parallel\|_{L^2(\nu_L^\parallel)} \leq \|\Delta_{\text{sh}}\|_{L^2(\nu_L^\parallel)}. \quad (\text{F201})$$

Moreover, if $\Delta_{\text{sh}} = \sum_{e \in E_{\text{sh}}} \delta_e$ with E_{sh} finite and $\|\delta_e\|_2 \leq \zeta$ for all e , then

$$\|\Delta_{\text{sh}}\|_{L^2(\nu_L^\parallel)} \leq |E_{\text{sh}}| \zeta. \quad (\text{F202})$$

In particular, a sufficient pointwise control is $|\delta_e| \leq z_{2\alpha}$ (cf. (F196)), which implies $\|\delta_e\|_2 \leq z_{2\alpha}$.

Proof. Conditional expectation is a contraction in L^2 , hence $\|\text{Err}_\parallel\|_2 \leq \|e^{-tX_{\text{out}}} \Delta_{\text{sh}}\|_2 \leq \|\Delta_{\text{sh}}\|_2$ since $0 \leq e^{-tX_{\text{out}}} \leq 1$. The counting bound (F202) is Minkowski's inequality: $\|\sum_e \delta_e\|_2 \leq \sum_e \|\delta_e\|_2 \leq |E_{\text{sh}}| \zeta$. $\square$

Lemma F.172 (Edge-by-edge telescoping decomposition for shared-edge products). *Let E be a finite index set and let $(a_e)_{e \in E}$ and $(b_e)_{e \in E}$ be real-valued random variables. Assume that $0 \leq a_e \leq 1$ and $0 \leq b_e \leq 1$ almost surely for every $e \in E$. Define the product difference*

$$\Delta_E := \prod_{e \in E} b_e - \prod_{e \in E} a_e. \quad (\text{F203})$$

Then for any ordering $E = \{e_1, \dots, e_n\}$ one has the telescoping identity

$$\Delta_E = \sum_{j=1}^n (b_{e_j} - a_{e_j}) \prod_{i < j} b_{e_i} \prod_{i > j} a_{e_i}. \quad (\text{F204})$$

In particular, if there is a deterministic $\zeta \geq 0$ such that $|b_e - a_e| \leq \zeta$ pointwise for all $e \in E$, then Δ_E admits a decomposition

$$\Delta_E = \sum_{e \in E} \delta_e \quad \text{with} \quad |\delta_e| \leq \zeta. \quad (\text{F205})$$

Proof. The identity (F204) is the standard expansion $\prod_{j=1}^n b_j - \prod_{j=1}^n a_j = \sum_{j=1}^n (b_j - a_j) \prod_{i < j} b_i \prod_{i > j} a_i$, applied with $b_j = b_{e_j}$ and $a_j = a_{e_j}$. If $0 \leq a_e, b_e \leq 1$, then each product factor in (F204) has absolute value at most 1, so the pointwise bound $|b_e - a_e| \leq \zeta$ implies $|\delta_{e_j}| \leq \zeta$ where $\delta_{e_j} := (b_{e_j} - a_{e_j}) \prod_{i < j} b_{e_i} \prod_{i > j} a_{e_i}$. $\square$

Corollary F.173 (From per-edge product control to a $z_{2\alpha}$ shared-edge decomposition). *Let E be a finite set and suppose that the shared-edge correction satisfies the product form $\Delta_{\text{sh}} = \prod_{e \in E} b_e - \prod_{e \in E} a_e$ with $0 \leq a_e, b_e \leq 1$ and $|b_e - a_e| \leq z_{2\alpha}$ pointwise for all $e \in E$. Then Δ_{sh} admits a decomposition $\Delta_{\text{sh}} = \sum_{e \in E} \delta_e$ with $|\delta_e| \leq z_{2\alpha}$. Consequently, Lemma F.171 applies with $E_{\text{sh}} = E$.*

Proof. Apply Lemma F.172 with $\zeta = z_{2\alpha}$. $\square$

Corollary F.174 (Quantitative surrogate error from outside influences and shared-edge counting). *In the setting of Lemma F.168, assume additionally that:*

(a) *there is an index set $\mathcal{I}_{\text{out}}$ such that (F199) holds and X_{out} does not depend on the remaining in-plane variables, and*

(b) $\Delta_{\text{sh}} = \sum_{e \in E_{\text{sh}}} \delta_e$ *with $|E_{\text{sh}}| < \infty$ and $|\delta_e| \leq z_{2\alpha}$.*

Midline instantiation. For the canonical outside choice (F230) associated to the strip T_{cut} (Lemma F.184), Lemma F.185 supplies (a) with $\mathcal{I}_{\text{out}} = I_{\text{out}} = T_{\text{out}}$ and $\Delta_{\text{out}} = 2\sqrt{C_{\text{plaq},2}}$. Then the surrogate approximation error in Proposition F.164(ii) can be taken as

$$\epsilon_L^{(C)} \leq \frac{6}{c_\Psi c_m} \left(\frac{t \Delta_{\text{out}}}{\beta \sqrt{2}} \sqrt{|\mathcal{I}_{\text{out}}|} + |E_{\text{sh}}| z_{2\alpha} \right). \quad (\text{F206})$$

In particular, if $|\mathcal{I}_{\text{out}}| \lesssim L^2$ and $|E_{\text{sh}}| \lesssim L^2$, then at the entrance scaling $L^2 \asymp \beta$ one has $\epsilon_L^{(C)} = O(\beta^{1-\dim(G)/2})$ as $\beta \rightarrow \infty$.

Proof. In Lemma F.168, the surrogate L^2 error is bounded by $\frac{6}{c_\Psi c_m}(\varepsilon_{\text{out}} + \varepsilon_{\text{sh}})$ with $\varepsilon_{\text{out}} := \|m_{\parallel} - \bar{m}\|_2$ and $\varepsilon_{\text{sh}} := \|\text{Err}_{\parallel}\|_2$. Apply Lemma F.170 to bound ε_{out} by the first term in (F206). Apply Lemma F.171 to bound $\varepsilon_{\text{sh}} \leq \|\Delta_{\text{sh}}\|_2 \leq |E_{\text{sh}}| z_{2\alpha}$, giving (F206). $\square$

Concrete size bounds for the midline cut parameters

Lemma F.175 (Midline cut bookkeeping in $d = 4$). *Work in $d = 4$ and let T_L be the $L \times L$ planar tile in the $(1, 2)$ -plane used in Main chain. Let T_{cut} be the unit-width midline strip from Lemma F.184, and write $m_L := |T_{\text{cut}}| \leq 2L$.*

Let $\mathcal{E}_{\text{cut}}$ be the set of in-plane links that appear in at least one plaquette of T_{cut} . Let $P_{\perp, \text{cut}}$ be the set of out-of-plane plaquettes (in the thick block) whose boundary contains at least one link of $\mathcal{E}_{\text{cut}}$. Let $\mathcal{E}_{\text{sh}}$ be the set of non-in-plane links that are contained in at least two plaquettes of $P_{\perp, \text{cut}}$.

Then there exist absolute constants $C_{\mathcal{E}}, C_{\perp}, C_{\text{sh}} < \infty$ (depending only on the dimension $d = 4$) such that

$$|\mathcal{E}_{\text{cut}}| \leq C_{\mathcal{E}} L, \quad |P_{\perp, \text{cut}}| \leq C_{\perp} L, \quad |\mathcal{E}_{\text{sh}}| \leq C_{\text{sh}} L. \quad (\text{F207})$$

Moreover, in any shared-edge decomposition of Δ_{sh} supported on shared non-in-plane links inside $P_{\perp, \text{cut}}$, one may take the index set E_{sh} in Corollary F.174 to satisfy $|E_{\text{sh}}| \leq C_{\text{sh}} L$.

Proof. The strip T_{cut} contains $m_L \leq 2L$ plaquettes, each plaquette has 4 boundary links, and each in-plane link can be incident to at most $2(d-2) = 4$ out-of-plane plaquettes in $d = 4$ (one for each choice of transverse direction and orientation). Thus,

$$|\mathcal{E}_{\text{cut}}| \leq 4m_L \leq 8L, \quad \text{and} \quad |P_{\perp, \text{cut}}| \leq 4|\mathcal{E}_{\text{cut}}| \leq 32L,$$

so one may take $C_{\mathcal{E}} = 8$ and $C_{\perp} = 32$.

Each plaquette in $P_{\perp, \text{cut}}$ has 4 boundary links, hence the total link-plaquette incidence count satisfies $\sum_e \deg(e) \leq 4|P_{\perp, \text{cut}}|$. Every $e \in \mathcal{E}_{\text{sh}}$ has $\deg(e) \geq 2$ by definition, so

$$|\mathcal{E}_{\text{sh}}| \leq \frac{1}{2} \sum_{e \in \mathcal{E}_{\text{sh}}} \deg(e) \leq \frac{1}{2} \sum_e \deg(e) \leq 2|P_{\perp, \text{cut}}| \leq 64L.$$

Thus one may take $C_{\text{sh}} = 64$. The final statement follows by choosing the shared-edge index set E_{sh} to be any subset of $\mathcal{E}_{\text{sh}}$ that supports the decomposition $\Delta_{\text{sh}} = \sum_{e \in E_{\text{sh}}} \delta_e$.

Lemma F.176 (Explicit shared-edge product for the canonical midline strip cut). *Work in the midline strip cut geometry of Lemma F.175 inside the thick block. Let $P_{\perp, \text{cut}}$ and $\mathcal{E}_{\text{sh}}$ be as there, and set $E_{\text{sh}} := \mathcal{E}_{\text{sh}}$.*

Let $f(g) := \widehat{W}_{\beta}(g)^{\alpha}$ and $z_{\alpha} := \int_G f(g) dg$. For each $e \in E_{\text{sh}}$, define the cut-adjacent transverse incidence number

$$\deg_{\perp}(e) := \left| \{p \in P_{\perp, \text{cut}} : e \subset \partial p\} \right|,$$

so that $\deg_{\perp}(e) \geq 2$ by definition of E_{sh} .

After performing the canonical leaf eliminations on all transverse links in $P_{\perp, \text{cut}}$ that are not in E_{sh} , the remaining cut-adjacent transverse contribution admits the edge-factor form

$$1 + \Delta_{\text{sh}} = \prod_{e \in E_{\text{sh}}} b_e, \quad b_e = \int_G \prod_{j=1}^{\deg_{\perp}(e)} f(gX_{e,j}) dg, \quad (\text{F208})$$

for some $X_{e,1}, \dots, X_{e, \deg_{\perp}(e)} \in G$ measurable with respect to the in-plane field (and any already-integrated transverse leaf data).

Define the decoupled baseline factors $a_e := z_{\alpha}^{\deg_{\perp}(e)}$ and the product difference

$$\Delta_{\text{sh}} = \prod_{e \in E_{\text{sh}}} b_e - \prod_{e \in E_{\text{sh}}} a_e. \quad (\text{F209})$$

Moreover, for every $e \in E_{\text{sh}}$ one has $0 \leq a_e, b_e \leq 1$ and the pointwise defect bound

$$|b_e - a_e| \leq z_{2\alpha}. \quad (\text{F210})$$

Proof. (Edge-factor form.) By construction, $P_{\perp, \text{cut}}$ consists of out-of-plane plaquettes that are incident to the in-plane cut-link set $\mathcal{E}_{\text{cut}}$ (Lemma F.175). The canonical leaf eliminations integrate, one by one, transverse link variables that appear in exactly one remaining plaquette factor; by Haar bi-invariance, each such integration contributes the constant z_α and removes that plaquette-variable dependence. After all such eliminations, the only transverse link variables that remain in $P_{\perp, \text{cut}}$ are precisely the non-in-plane links incident to at least two cut-adjacent transverse plaquettes, i.e. $\mathcal{E}_{\text{sh}}$. In the midline strip geometry, the remaining transverse factors group by shared link $e \in \mathcal{E}_{\text{sh}}$, producing (F208) with $\deg_\perp(e)$ shifted copies of f ; the shifts $X_{e,j}$ are the group-word remainders of the incident plaquette holonomies with the e -variable factored out.

(Pointwise bounds.) Since $0 \leq f \leq 1$, we have $0 \leq a_e = z_\alpha^{\deg_\perp(e)} \leq 1$. Also $0 \leq b_e \leq 1$ because the integrand is a product of factors in $[0, 1]$ and Haar has total mass 1. Finally, (F210) is exactly the higher-degree extension stated in Remark F.167 (take $k = \deg_\perp(e) \geq 2$). $\square$

Corollary F.177 (Shared-edge L^2 bound for the midline strip cut). *In the setting of Lemma F.176, the shared-edge correction admits a decomposition $\Delta_{\text{sh}} = \sum_{e \in E_{\text{sh}}} \delta_e$ with $|\delta_e| \leq z_{2\alpha}$ pointwise. Consequently,*

$$\|\Delta_{\text{sh}}\|_{L^2(\nu_L^\parallel)} \leq |E_{\text{sh}}| z_{2\alpha} \leq C_{\text{sh}} L z_{2\alpha},$$

where C_{sh} is the geometric constant from Lemma F.175 (one may take $C_{\text{sh}} = 64$).

Proof. Combine (F209) with Corollary F.173 (using (F210)) and then apply Lemma F.171. The final inequality uses (F207). $\square$

$\square$

Remark F.178 (Scaling consequence for $\varepsilon_L^{(C)}$). Assume the hypotheses of Corollary F.174. Using (F207) and the trivial bound $|I_{\text{out}}| \leq |T_L| = L^2$, we obtain the explicit estimate

$$\varepsilon_L^{(C)} \leq \frac{t \Delta_{\text{out}}}{\beta c_\Psi} L + \frac{C_{\text{sh}}}{c_\Psi} L z_{2\alpha}. \quad (\text{F211})$$

In the finite-depth entrance window $L^2 \simeq \beta$ and under the weak-coupling asymptotic $z_{2\alpha} \lesssim (\alpha\beta)^{-\frac{1}{2} \dim G}$ (from stationary phase at the identity), the right-hand side satisfies

$$\varepsilon_L^{(C)} \lesssim \frac{t}{\sqrt{\beta}} + \beta^{\frac{1}{2} - \frac{1}{2} \dim G},$$

and in particular $\varepsilon_L^{(C)} = o_{\beta \rightarrow \infty}(1)$ for every compact connected simple Lie group G (where $\dim G \geq 3$).

Lemma F.179 (Convolution-power damping of matrix coefficients). *Let G be a compact group with normalized Haar measure dg , let $w \geq 0$ be a central probability density on G ($\int w dg = 1$ and $w(hgh^{-1}) = w(g)$), and let R be an irreducible unitary representation $\pi_R : G \rightarrow U(d_R)$ with character χ_R and normalized character $\varphi_R := \chi_R/d_R$. For indices $1 \leq i, j \leq d_R$, define the matrix coefficient function*

$$u_{ij}(g) := \left(\pi_R(g) \right)_{ij} \in \mathbb{C}, \quad g \in G.$$

Let T_w be the left-convolution operator on $L^2(G)$,

$$(T_w f)(g) := \int_G f(hg) w(h) dh.$$

Then the following hold.

(a) (**Schur orthogonality / L^2 norm**) One has

$$\int_G |u_{ij}(g)|^2 dg = \frac{1}{d_R}, \quad \text{equivalently} \quad \|u_{ij}\|_{L^2(G)} = \frac{1}{\sqrt{d_R}}. \quad (\text{F212})$$

(b) (**Convolution coefficient identity**) The coefficient u_{ij} is an eigenfunction of T_w :

$$T_w u_{ij} = \lambda_R(w) u_{ij}, \quad \text{where} \quad \lambda_R(w) := \int_G \varphi_R(h) w(h) dh. \quad (\text{F213})$$

(c) (**Convolution powers and variance bound**) For every integer $k \geq 1$,

$$T_w^k u_{ij} = \lambda_R(w)^k u_{ij} \quad \text{and} \quad \|T_w^k u_{ij}\|_{L^2(G)}^2 = \frac{|\lambda_R(w)|^{2k}}{d_R}. \quad (\text{F214})$$

Equivalently, if $A = X_1 \cdots X_k$ with X_m i.i.d. having density w (so $A \sim w^{*k}$), then the averaged matrix coefficient

$$F_{ij}(g) := \mathbb{E}[u_{ij}(Ag)] = \int_G u_{ij}(hg) w^{*k}(h) dh$$

satisfies $F_{ij}(g) = \lambda_R(w)^k u_{ij}(g)$ and therefore has Haar L^2 -variance $\int |F_{ij}(g)|^2 dg = |\lambda_R(w)|^{2k}/d_R$.

Proof. We prove the three parts in order.

(a) Schur orthogonality and (F212). Fix indices j, ℓ and let $E_{j\ell}$ be the $d_R \times d_R$ matrix unit. Define the operator

$$A_{j\ell} := \int_G \pi_R(g) E_{j\ell} \pi_R(g)^{-1} dg.$$

By left–right invariance of Haar measure, for every $h \in G$ we have

$$\pi_R(h) A_{j\ell} \pi_R(h)^{-1} = \int_G \pi_R(hg) E_{j\ell} \pi_R(hg)^{-1} dg = \int_G \pi_R(g) E_{j\ell} \pi_R(g)^{-1} dg = A_{j\ell},$$

so $A_{j\ell}$ commutes with $\pi_R(h)$ for all $h \in G$. Since R is irreducible, Schur's lemma implies $A_{j\ell} = c_{j\ell} I$ for some scalar $c_{j\ell}$. Taking traces and using cyclicity gives

$$c_{j\ell} d_R = \text{tr}(A_{j\ell}) = \int_G \text{tr}(\pi_R(g) E_{j\ell} \pi_R(g)^{-1}) dg = \int_G \text{tr}(E_{j\ell}) dg = \delta_{j\ell}.$$

Hence $c_{j\ell} = \delta_{j\ell}/d_R$. Now take the (i, k) entry of $A_{j\ell}$:

$$(A_{j\ell})_{ik} = \int_G (\pi_R(g))_{ij} (\pi_R(g)^{-1})_{\ell k} dg.$$

Because $\pi_R(g)$ is unitary, $\pi_R(g)^{-1} = \pi_R(g)^*$ and therefore $(\pi_R(g)^{-1})_{\ell k} = \overline{(\pi_R(g))_{k\ell}}$. Thus

$$\int_G (\pi_R(g))_{ij} \overline{(\pi_R(g))_{k\ell}} dg = (A_{j\ell})_{ik} = (c_{j\ell} I)_{ik} = \frac{1}{d_R} \delta_{j\ell} \delta_{ik}.$$

Setting $k = i$ and $\ell = j$ yields $\int |u_{ij}(g)|^2 dg = 1/d_R$, i.e. (F212).

(b) Convolution coefficient identity (F213). Define the $d_R \times d_R$ matrix

$$M := \int_G \pi_R(h) w(h) dh.$$

Using that w is central and Haar is conjugation-invariant, for any $g \in G$,

$$\pi_R(g) M \pi_R(g)^{-1} = \int_G \pi_R(g h g^{-1}) w(h) dh = \int_G \pi_R(h) w(h) dh = M.$$

Thus M commutes with $\pi_R(g)$ for all g , so by Schur's lemma $M = c I$ for some scalar c .

Taking trace gives

$$\begin{aligned} c d_R &= \text{tr}(M) = \int_G \chi_R(h) w(h) dh, \\ \text{so } c &= \frac{1}{d_R} \int_G \chi_R(h) w(h) dh = \int_G \varphi_R(h) w(h) dh = \lambda_R(w). \end{aligned}$$

Now compute, for any $g \in G$,

$$(T_w u_{ij})(g) = \int_G u_{ij}(hg) w(h) dh = \int_G (\pi_R(hg))_{ij} w(h) dh$$

$$\begin{aligned}
&= \int_G \sum_{a=1}^{d_R} \left(\pi_R(h) \right)_{ia} \left(\pi_R(g) \right)_{aj} w(h) dh = \sum_{a=1}^{d_R} M_{ia} \left(\pi_R(g) \right)_{aj} \\
&= \sum_{a=1}^{d_R} (c \delta_{ia}) \left(\pi_R(g) \right)_{aj} = c \left(\pi_R(g) \right)_{ij} = \lambda_R(w) u_{ij}(g),
\end{aligned}$$

which is exactly (F213).

(c) Convolution powers and (F214). Applying T_w repeatedly and using (F213) gives $T_w^k u_{ij} = \lambda_R(w)^k u_{ij}$. Taking $L^2(G)$ norms and using (F212) yields

$$\|T_w^k u_{ij}\|_{L^2(G)}^2 = |\lambda_R(w)|^{2k} \|u_{ij}\|_{L^2(G)}^2 = \frac{|\lambda_R(w)|^{2k}}{d_R}.$$

For the probabilistic restatement, note that if $A \sim w^{*k}$ then $g \mapsto \mathbb{E}[u_{ij}(Ag)]$ equals $(T_{w^{*k}} u_{ij})(g)$, and $T_{w^{*k}} = T_w^k$. $\square$

Lemma F.180 (Efron–Stein variance bound from L^2 influences). *Let $\xi = (\xi_1, \dots, \xi_m)$ be independent random variables on a product probability space, and let $Y = Y(\xi)$ be square-integrable. For each $a \in \{1, \dots, m\}$, let $\xi^{(a)}$ be obtained by resampling only ξ_a (independently of everything else), and set $Y^{(a)} := Y(\xi^{(a)})$. Then*

$$\text{Var}(Y) \leq \frac{1}{2} \sum_{a=1}^m \mathbf{E}[(Y - Y^{(a)})^2]. \quad (\text{F215})$$

In particular, if $X = X(\xi)$ satisfies $\|X - X^{(a)}\|_2 \leq \Delta/\beta$ for all a , then for every $t \geq 0$,

$$\sqrt{\text{Var}(e^{-tX})} \leq \frac{t \Delta}{\beta \sqrt{2}} \sqrt{m}. \quad (\text{F216})$$

Proof. Eq. (F215) is the standard Efron–Stein inequality. For completeness, one may write $\text{Var}(Y) = \frac{1}{2} \mathbf{E}[(Y - Y')^2]$ where Y' is an independent copy built from an independent copy ξ' of ξ , then reveal the coordinates one by one and compare $Y(\xi)$ to $Y(\xi_1, \dots, \xi_{a-1}, \xi'_a, \dots, \xi'_m)$ to obtain the telescoping bound. The consequence (F216) follows by applying (F215) to $Y = e^{-tX}$ and using $|e^{-tX} - e^{-tX^{(a)}}| \leq t|X - X^{(a)}|$. $\square$

Lemma F.181 (Cut-local surrogate $\Rightarrow \chi^2$ flatness of the effective tilt). *Assume Proposition F.164. Then the normalized cut surrogate $\tilde{\Psi}_{\text{cut}}$ from (F189) satisfies*

$$\left\| \tilde{\Psi}_{\text{cut}} - 1 \right\|_{L^2(\nu_L^\parallel)} \leq \frac{t \Delta_X}{\beta \sqrt{2} c_\Psi} \sqrt{m_L}. \quad (\text{F217})$$

Consequently,

$$\left\| \tilde{\Psi}_L^{(\alpha)} - 1 \right\|_{L^2(\nu_L^\parallel)} \leq \epsilon_L^{(C)} + \frac{t \Delta_X}{\beta \sqrt{2} c_\Psi} \sqrt{m_L}. \quad (\text{F218})$$

In particular, the (unconditioned) in-plane χ^2 divergence obeys

$$D_{\chi^2}(\nu_L^{(\alpha)}|_{\mathcal{F}_{\parallel}} \parallel \nu_L^{\parallel}|_{\mathcal{F}_{\parallel}}) = \|\tilde{\Psi}_L^{(\alpha)} - 1\|_{L^2(\nu_L^{\parallel})}^2 \leq \left(\epsilon_L^{(C)} + \frac{t \Delta_X}{\beta \sqrt{2} c_{\Psi}} \sqrt{m_L} \right)^2. \quad (\text{F219})$$

Proof. By definition $\tilde{\Psi}_{\text{cut}} = Y/E_{\nu_L^{\parallel}}[Y]$ with $Y = e^{-tX_{\text{cut}}}$. Since $E_{\nu_L^{\parallel}}[Y] \geq c_{\Psi}$ by (F191),

$$\|\tilde{\Psi}_{\text{cut}} - 1\|_2^2 = \text{Var}\left(\frac{Y}{E[Y]}\right) = \frac{\text{Var}(Y)}{E[Y]^2} \leq \frac{\text{Var}(Y)}{c_{\Psi}^2}.$$

Applying Lemma F.180 with the influence bound (F188) yields $\sqrt{\text{Var}(Y)} \leq (t\Delta_X/(\beta\sqrt{2}))\sqrt{m_L}$, hence (F217). The bound (F218) follows from the triangle inequality and (F190). Finally, since $\tilde{\Psi}_L^{(\alpha)}$ is the Radon–Nikodym derivative of $\nu_L^{(\alpha)}|_{\mathcal{F}_{\parallel}}$ with respect to $\nu_L^{\parallel}|_{\mathcal{F}_{\parallel}}$ (Lemma F.159), one has $D_{\chi^2} = \|\tilde{\Psi}_L^{(\alpha)} - 1\|_2^2$, giving (F219). $\square$

Lemma F.182 (L^2 flatness on an event $\Rightarrow$ conditional χ^2 control). *Let ν be a probability measure on $(X, \mathcal{F})$, let $\Omega \in \mathcal{F}$ satisfy $\nu(\Omega) > 0$, and let $r \geq 0$ satisfy $E_{\nu}[r] = 1$. Define $\mu := r \cdot \nu$. Set $\delta_{\Omega} := \|r - 1\|_{L^2(\nu^{\Omega})}$, where $\nu^{\Omega} := \nu(\cdot | \Omega)$. If $\delta_{\Omega} < 1$, then*

$$D_{\chi^2}(\mu^{\Omega} \parallel \nu^{\Omega}) \leq \frac{\delta_{\Omega}^2}{(1 - \delta_{\Omega})^2}. \quad (\text{F220})$$

In particular, $\delta_{\Omega} \leq \|r - 1\|_{L^2(\nu)}/\sqrt{\nu(\Omega)}$, so if $\|r - 1\|_{L^2(\nu)} \leq \frac{1}{2}\sqrt{\nu(\Omega)}$ then

$$D_{\chi^2}(\mu^{\Omega} \parallel \nu^{\Omega}) \leq \frac{4}{\nu(\Omega)} \|r - 1\|_{L^2(\nu)}^2. \quad (\text{F221})$$

Proof. Write $\nu^{\Omega}(\cdot) = \nu(\cdot \cap \Omega)/\nu(\Omega)$. Since $d\mu^{\Omega}/d\nu^{\Omega} = r/E_{\nu^{\Omega}}[r]$ on Ω ,

$$D_{\chi^2}(\mu^{\Omega} \parallel \nu^{\Omega}) = \text{Var}_{\nu^{\Omega}}\left(\frac{r}{E_{\nu^{\Omega}}[r]}\right) = \frac{\text{Var}_{\nu^{\Omega}}(r)}{E_{\nu^{\Omega}}[r]^2}.$$

Now $\text{Var}_{\nu^{\Omega}}(r) \leq E_{\nu^{\Omega}}[(r-1)^2] = \delta_{\Omega}^2$. Also $|E_{\nu^{\Omega}}[r] - 1| = |E_{\nu^{\Omega}}[r-1]| \leq \delta_{\Omega}$ by Cauchy–Schwarz, so $E_{\nu^{\Omega}}[r] \geq 1 - \delta_{\Omega}$. This gives (F220). Finally, $\delta_{\Omega}^2 = E_{\nu}[(r-1)^2 \mathbf{1}_{\Omega}]/\nu(\Omega) \leq \|r - 1\|_{L^2(\nu)}^2/\nu(\Omega)$, and (F221) follows by inserting this bound into (F220). $\square$

Lemma F.183 (Main-chain covariance pinning under cut-local tilts). *Work on a planar reference tile of side length L with boundary holonomy Φ_L and planar reference single-plaquette density $w_1^{\parallel}$. Let $\mathcal{F}_{\text{cut}}$ be a sub- σ -algebra (the “cut” information) such that:*

1. (Cut size) $\mathcal{F}_{\text{cut}}$ is generated by at most m_L plaquette variables with $m_L \leq C_{\text{cut}}L$.
2. (Tilt locality) The transverse-energy tilt observable X entering $\bar{R} := e^{-tX}$ is $\mathcal{F}_{\text{cut}}$ -measurable.

3. (L^2 influence bound) Writing $X^{(a)}$ for the version of X in which the a th generating cut variable is resampled (independently, holding all other cut variables fixed), assume

$$\|X - X^{(a)}\|_{L^2(\parallel)} \leq \frac{\Delta_X}{\beta} \quad \text{for each } a = 1, \dots, m_L. \quad (\text{F222})$$

Then for any irrep $R \neq \mathbf{1}$ and any $t \geq 0$,

$$\left| \text{Cov}_{\parallel}(\varphi_R(\Phi_L), e^{-tX}) \right| \leq \sqrt{\text{Var}_{\parallel}(e^{-tX})} \cdot \left\| \mathbb{E}_{\parallel}[\varphi_R(\Phi_L) \mid \mathcal{F}_{\text{cut}}] \right\|_{L^2(\parallel)}. \quad (\text{F223})$$

Moreover,

$$\sqrt{\text{Var}_{\parallel}(e^{-tX})} \leq t \frac{\Delta_X}{\beta} \sqrt{m_L}, \quad (\text{F224})$$

and, writing $k_L := |T_L| - m_L$, one has the representation-mixing bound

$$\left\| \mathbb{E}_{\parallel}[\varphi_R(\Phi_L) \mid \mathcal{F}_{\text{cut}}] \right\|_{L^2(\parallel)} \leq \frac{|\lambda_R(w_1^{\parallel})|^{k_L}}{d_R} \sqrt{\|(w_1^{\parallel})^{*m_L}\|_{L^{\infty}(G)}}. \quad (\text{F225})$$

Consequently,

$$\left| \text{Cov}_{\parallel}(\varphi_R(\Phi_L), e^{-tX}) \right| \leq \frac{t \Delta_X}{\beta} \sqrt{m_L} \frac{|\lambda_R(w_1^{\parallel})|^{k_L}}{d_R} \sqrt{\|(w_1^{\parallel})^{*m_L}\|_{L^{\infty}(G)}}. \quad (\text{F226})$$

In particular, if $m_L \leq \frac{1}{2}|T_L|$ (hence $k_L \geq \frac{1}{2}|T_L|$) then

$$\left| \text{Cov}_{\parallel}(\varphi_R(\Phi_L), e^{-tX}) \right| \leq \frac{t \Delta_X}{\beta} \sqrt{m_L} \cdot \frac{|\lambda_R(w_1^{\parallel})|^{|T_L|/2}}{d_R} \sqrt{\|(w_1^{\parallel})^{*m_L}\|_{L^{\infty}(G)}}.$$

Moreover, since $w_1^{\parallel}$ is of positive type, each convolution power $(w_1^{\parallel})^{*m}$ is positive definite, hence $\|(w_1^{\parallel})^{*m}\|_{L^{\infty}(G)} = (w_1^{\parallel})^{*m}(e)$. Using the one-plaquette coefficient bound $\lambda_S(w_1^{\parallel}) \leq e^{-c_{\text{plaq}} C_2(S)/\beta}$ we obtain

$$\|(w_1^{\parallel})^{*m}\|_{L^{\infty}(G)} \leq \sum_S d_S^2 e^{-c_{\text{plaq}} C_2(S) m/\beta} = H_G \left(c_{\text{plaq}} \frac{m}{\beta} \right). \quad (\text{F227})$$

Proof. Since X is $\mathcal{F}_{\text{cut}}$ -measurable, so is $\bar{R} = e^{-tX}$, hence

$$\text{Cov}_{\parallel}(\varphi_R(\Phi_L), \bar{R}) = \text{Cov}_{\parallel}(\mathbb{E}_{\parallel}[\varphi_R(\Phi_L) \mid \mathcal{F}_{\text{cut}}], \bar{R}).$$

Cauchy–Schwarz gives the general bound (F223).

The influence assumption (F222) together with the Efron–Stein inequality (Lemma F.180) yields

$$\|\bar{R} - \mathbb{E}_{\parallel}[\bar{R}]\|_{L^2(\parallel)} \leq \frac{t \Delta_X}{\beta} \sqrt{m_L}, \quad (\text{F228})$$

which is (F224).

It remains to bound $\|\mathbb{E}_{\parallel}[\varphi_R(\Phi_L) \mid \mathcal{F}_{\text{cut}}]\|_{L^2(\parallel)}$. Write $N := |T_L|$ and index the plaquettes in a fixed surface order as $U_1, \dots, U_N$ so that $\Phi_L = \prod_{\ell=1}^N U_{\ell}$. Let $C \subseteq \{1, \dots, N\}$ be the cut indices, $|C| = m_L$, and C^c its complement, $|C^c| = k_L$.

Let π_R be a unitary realization of R . Centrality of $w_1^{\parallel}$ implies that the matrix

$$M_R := \int_G \pi_R(g) w_1^{\parallel}(g) dg$$

commutes with $\pi_R(h)$ for all $h \in G$, hence $M_R = \lambda_R(w_1^{\parallel}) \text{Id}$ by Schur's lemma. Conditioning on $\mathcal{F}_{\text{cut}} = \sigma(U_{\ell} : \ell \in C)$ and integrating the variables $\{U_{\ell} : \ell \in C^c\}$ sequentially therefore gives the exact identity

$$\mathbb{E}_{\parallel}[\pi_R(\Phi_L) \mid \mathcal{F}_{\text{cut}}] = \lambda_R(w_1^{\parallel})^{k_L} \pi_R(\Phi_{\text{cut}}), \quad \Phi_{\text{cut}} := \prod_{\ell \in C}^{(\text{surface order})} U_{\ell}.$$

Taking normalized traces yields $\mathbb{E}_{\parallel}[\varphi_R(\Phi_L) \mid \mathcal{F}_{\text{cut}}] = \lambda_R(w_1^{\parallel})^{k_L} \varphi_R(\Phi_{\text{cut}})$. Since Φ_{cut} is the product of m_L i.i.d. variables with density $w_1^{\parallel}$, it has density $(w_1^{\parallel})^{*m_L}$ with respect to Haar, so

$$\left\| \mathbb{E}_{\parallel}[\varphi_R(\Phi_L) \mid \mathcal{F}_{\text{cut}}] \right\|_{L^2(\parallel)}^2 = \lambda_R(w_1^{\parallel})^{2k_L} \int_G |\varphi_R(g)|^2 (w_1^{\parallel})^{*m_L}(g) dg.$$

Bounding the integral by $\|(w_1^{\parallel})^{*m_L}\|_{L^{\infty}(G)} \int_G |\varphi_R(g)|^2 dg$ and using character orthogonality $\int_G |\chi_R(g)|^2 dg = 1$ together with $\varphi_R = \chi_R/d_R$ (hence $\int_G |\varphi_R(g)|^2 dg = 1/d_R^2$) gives (F225).

Finally, insert (F228) and (F225) into (F223) to obtain (F226). $\square$

A concrete cut σ -algebra and the $1/\beta$ influence bound Lemma F.183 was stated for an abstract cut σ -algebra $\mathcal{F}_{\text{cut}}$. For Main chain we take $\mathcal{F}_{\text{cut}}$ to be generated by a unit-width strip of plaquettes that separates the tile into two large sub-tiles.

Fix $L \geq 2$ and, for notational simplicity, assume L is even. Let T_L be the $L \times L$ set of in-plane plaquettes in the planar reference tile. Write $T_L = T_L^- \cup T_L^+$ for the left/right decomposition by the vertical midline and let

$$T_{\text{cut}} := \{p \in T_L : p \text{ intersects the midline between } T_L^- \text{ and } T_L^+\}.$$

Then $|T_{\text{cut}}| \leq 2L$. Define the cut σ -algebra and the cut-local energy

$$\mathcal{F}_{\text{cut}} := \sigma(\{U_p\}_{p \in T_{\text{cut}}}), \quad X_{\text{cut}} := \sum_{p \in T_{\text{cut}}} A(U_p). \quad (\text{F229})$$

(The choice of a vertical midline is inessential; any rectilinear cut of length $O(L)$ gives the same bounds up to constants.)

Lemma F.184 (Verification of Lemma F.183 hypotheses for X_{cut}). *Under the planar reference product representation of the plaquette variables on T_L :*

1. $\mathcal{F}_{\text{cut}}$ is generated by $m_L := |T_{\text{cut}}| \leq 2L$ plaquette variables, hence $m_L \leq C_{\text{cut}}L$ with $C_{\text{cut}} = 2$;
2. X_{cut} is $\mathcal{F}_{\text{cut}}$ -measurable;
3. the L^2 influence bound (F222) holds with $\Delta_X := 2\sqrt{C_{\text{plaq},2}}$, where $C_{\text{plaq},2}$ is the constant in Lemma F.145.

Proof. Items (1)–(2) are immediate from (F229) and the cardinality estimate $|T_{\text{cut}}| \leq 2L$.

For (3), fix $p \in T_{\text{cut}}$ and let U'_p be an independent resample of U_p (with the same marginal $w_1^\parallel$). Since X_{cut} is a sum over T_{cut} , resampling U_p affects exactly one term:

$$X_{\text{cut}} - X_{\text{cut}}^{(p)} = A(U_p) - A(U'_p).$$

Therefore

$$\|X_{\text{cut}} - X_{\text{cut}}^{(p)}\|_{L^2(\parallel)} \leq \|A(U_p)\|_{L^2(\parallel)} + \|A(U'_p)\|_{L^2(\parallel)} = 2\|A(U_p)\|_{L^2(\parallel)} \leq 2\frac{\sqrt{C_{\text{plaq},2}}}{\beta},$$

where the last inequality is Lemma F.145. This is exactly (F222) with $\Delta_X = 2\sqrt{C_{\text{plaq},2}}$. $\square$

Lemma F.185 (Canonical choice of X_{out} and verification of (F199)). *Fix $T_{\text{cut}} \subset T_L$ as in Lemma F.184 and write the complementary set of in-plane plaquettes*

$$T_{\text{out}} := T_L \setminus T_{\text{cut}} \quad (|T_{\text{out}}| = |T_L| - |T_{\text{cut}}| = L^2 + O(L)).$$

Define the “outside” planar energy

$$X_{\text{out}} := \sum_{p \in T_{\text{out}}} A(U_p). \tag{F230}$$

Then X_{out} depends only on the planar plaquette coordinates indexed by

$$I_{\text{out}} := T_{\text{out}}, \quad |I_{\text{out}}| = |T_{\text{out}}| = L^2 + O(L). \tag{F231}$$

Moreover, for every $a \in I_{\text{out}}$ one has the L^2 influence bound

$$\|X_{\text{out}} - X_{\text{out}}^{(a)}\|_{L^2(\nu_L^\parallel)} \leq \frac{\Delta_{\text{out}}}{\beta}, \quad \Delta_{\text{out}} := 2\sqrt{C_{\text{plaq},2}}, \tag{F232}$$

with $C_{\text{plaq},2}$ from Lemma F.145. In particular, (F199) holds for this choice of X_{out} (and the auxiliary μ_ω is irrelevant here since X_{out} is planar).

Proof. By construction X_{out} is a sum of the in-plane plaquette actions over T_{out} , hence depends only on the coordinates in I_{out} . Fix $a \in I_{\text{out}}$ and let $p_a \in T_{\text{out}}$ denote the corresponding plaquette. Under Efron–Stein resampling at coordinate a we have

$$X_{\text{out}} - X_{\text{out}}^{(a)} = A(U_{p_a}) - A(U'_{p_a}),$$

where U_{p_a} and U'_{p_a} are i.i.d. with the one-plaquette Wilson law $w_1^\parallel$ (tree-gauge independence, Lemma F.123). Therefore, by the triangle inequality in L^2 and Lemma F.145,

$$\|X_{\text{out}} - X_{\text{out}}^{(a)}\|_2 \leq \|A(U_{p_a})\|_2 + \|A(U'_{p_a})\|_2 = 2\|A(U_{p_a})\|_2 \leq \frac{2\sqrt{C_{\text{plaq},2}}}{\beta}.$$

This is (F232). □

With this choice of $\mathcal{F}_{\text{cut}}$ and X_{cut} , the covariance bound (F226) in Lemma F.183 is now completely explicit.

e. Explicit weak-window bound for the per-step transfer defect $\|\Delta_n\|_2$

In the finite-depth entrance/transfer criterion of Section F 15 one needs explicit, scale-indexed bounds on the L^2 -size of the *normalized* effective-tilt error. The Main chain surrogate machinery already provides such bounds in terms of (i) outside/cut influences and (ii) the shared-edge correction. We now package these ingredients into an explicit n -dependent estimate (dyadic instantiation $L = 2^n$) that is directly usable as a transfer budget $\epsilon_n(\beta)$.

Notation. For a tile of side length L (with the midline strip cut), set

$$\Delta_L := \tilde{\Psi}_L^{(\alpha)} - 1. \tag{F233}$$

where $\tilde{\Psi}_L^{(\alpha)}$ is the normalized effective tilt from Lemma F.181 (Main chain).

Proposition F.186 (Master Main chain weak-window bound for $\|\Delta_L\|_2$). *Assume the Main chain setup of Remark F.178, with the midline strip cut and associated parameters $t = \beta(1 - \alpha)$, $m_L = |T_{\text{cut}}|$ and constants $c_\Psi > 0$, $C_{\text{sh}} > 0$, $\Delta_{\text{out}}, \Delta_X > 0$ as in Remark F.178 and Lemma F.185. Then for every $L \geq 2$,*

$$\|\Delta_L\|_2 \leq \frac{t}{\beta c_\Psi} \left(\Delta_{\text{out}} L + \Delta_X \sqrt{L} \right) + \frac{C_{\text{sh}}}{c_\Psi} L z_{2\alpha}(\beta). \tag{F234}$$

Proof. By Lemma F.181,

$$\|\Delta_L\|_2 = \|\tilde{\Psi}_L^{(\alpha)} - 1\|_2 \leq \epsilon_L^{(C)} + \frac{t \Delta_X}{\beta \sqrt{2} c_\Psi} \sqrt{m_L}.$$

In the midline strip geometry one has $m_L \leq 2L$, hence $\frac{1}{\sqrt{2}} \sqrt{m_L} \leq \sqrt{L}$. Insert the midline surrogate bound of Remark F.178,

$$\epsilon_L^{(C)} \leq \frac{t \Delta_{\text{out}}}{\beta c_\Psi} L + \frac{C_{\text{sh}}}{c_\Psi} L z_{2\alpha}(\beta),$$

and combine the two estimates to obtain (F234). $\square$

Corollary F.187 (Dyadic instantiation $L = 2^n$ under the pivot schedule $\alpha(L)$). *Fix $\kappa > 0$ and define the pivot*

$$\alpha(L) := 1 - \frac{\kappa}{|P_\perp(\mathbb{B}_L)|} \quad \text{with} \quad |P_\perp(\mathbb{B}_L)| = 8L(L+1)$$

(as in Remark F.178), so that $t = \beta(1 - \alpha(L)) = \beta\kappa/(8L(L+1))$. For dyadic $L = 2^n$ ($n \in \mathbb{N}$) write $\alpha_n := \alpha(2^n)$ and $\Delta_n := \Delta_{2^n}$. Then, for every $n \geq 1$,

$$\|\Delta_n\|_2 \leq \frac{\kappa \Delta_{\text{out}}}{8 c_\Psi} \frac{1}{2^n + 1} + \frac{\kappa \Delta_X}{8 c_\Psi} \frac{1}{2^{n/2}(2^n + 1)} + \frac{C_{\text{sh}}}{c_\Psi} 2^n z_{2\alpha_n}(\beta). \quad (\text{F235})$$

In particular, using $\frac{1}{2^n + 1} \leq 2^{-n}$ and $\frac{1}{2^{n/2}(2^n + 1)} \leq 2^{-3n/2}$,

$$\|\Delta_n\|_2 \leq \frac{\kappa \Delta_{\text{out}}}{8 c_\Psi} 2^{-n} + \frac{\kappa \Delta_X}{8 c_\Psi} 2^{-3n/2} + \frac{C_{\text{sh}}}{c_\Psi} 2^n z_{2\alpha_n}(\beta). \quad (\text{F236})$$

Proof. Insert $t/\beta = \kappa/(8L(L+1))$ into (F234) and set $L = 2^n$. This yields (F235). The simplified bound (F236) follows from the elementary inequalities $\frac{1}{2^n + 1} \leq 2^{-n}$ and $\frac{1}{2^{n/2}(2^n + 1)} \leq 2^{-3n/2}$. $\square$

Lemma F.188 (An explicit weak-coupling bound for $z_{2\alpha}(\beta)$). *Let G be a compact connected Lie group with $\dim(G) = d_G$ and Haar measure dg . Assume the one-plaquette Wilson action $A_\pi(g)$ has a unique minimum at e and admits:*

1. a coordinate chart $g = \exp X$ on a neighborhood $\mathcal{U}_0$ of e with radius $r_0 > 0$,
2. constants $m_- > 0$ and $J_{\text{max}} < \infty$ such that for all $\|X\| \leq r_0$,

$$A_\pi(\exp X) \geq \frac{m_-}{2} \|X\|^2, \quad dg = J(X) dX, \quad 0 \leq J(X) \leq J_{\text{max}},$$

3. a constant $a_0 > 0$ such that $A_\pi(g) \geq a_0$ for all $g \notin \mathcal{U}_0$.

Then for every $\beta > 0$ and $\alpha \in (0, 1]$,

$$z_{2\alpha}(\beta) = \int_G e^{-2\alpha\beta A_\pi(g)} dg \leq J_{\max} \left(\frac{\pi}{\alpha\beta m_-} \right)^{d_G/2} + e^{-2\alpha\beta a_0}. \quad (\text{F237})$$

Proof. Split the integral over $\mathcal{U}_0$ and its complement. On $\mathcal{U}_0$ write $g = \exp X$ and use $A_\pi(\exp X) \geq \frac{m_-}{2} \|X\|^2$ and $J(X) \leq J_{\max}$:

$$\begin{aligned} \int_{\mathcal{U}_0} e^{-2\alpha\beta A_\pi(g)} dg &\leq \int_{\|X\| \leq r_0} J_{\max} e^{-\alpha\beta m_- \|X\|^2} dX \\ &\leq J_{\max} \int_{\mathbb{R}^{d_G}} e^{-\alpha\beta m_- \|X\|^2} dX \\ &= J_{\max} \left(\frac{\pi}{\alpha\beta m_-} \right)^{d_G/2}. \end{aligned}$$

On $G \setminus \mathcal{U}_0$ use $A_\pi(g) \geq a_0$ and dg has total mass 1 to obtain $\int_{G \setminus \mathcal{U}_0} e^{-2\alpha\beta A_\pi(g)} dg \leq e^{-2\alpha\beta a_0}$. Summing the two contributions yields (F237). $\square$

f. Closing the finite-depth transfer budget using explicit $\|\Delta_n\|_2$ bounds

The dyadic estimate (F235) can be used directly as an admissible transfer budget $\epsilon_n(\beta)$ in the finite-depth entrance criterion (Section F 15), once a mixing depth $k(\beta)$ is chosen (e.g. via Proposition F.65). The key point is that the *sum* of these explicit defects up to the mixing depth is small for large β , with fully transparent dependence on κ and the Wilson-action Laplace constants.

Proposition F.189 (An explicit summable transfer budget up to the mixing depth). *Fix $\delta \in (0, 1)$ and let $k(\beta)$ be any mixing depth satisfying $4^{k(\beta)} \geq m(\beta)$ with $m(\beta)$ as in Proposition F.61. Assume the pivot schedule of Corollary F.187 and set $\alpha_{\min} := 1 - \kappa/48$ (so that $\alpha_n \geq \alpha_{\min}$ for all $n \geq 1$). Define*

$$Z_\star(\beta) := J_{\max} \left(\frac{\pi}{\alpha_{\min}\beta m_-} \right)^{d_G/2} + e^{-2\alpha_{\min}\beta a_0}.$$

Then the transfer defects satisfy, for all $\beta > 0$,

$$\sum_{n < k(\beta)} \|\Delta_n\|_2 \leq \frac{\kappa}{8c_\Psi} \left(2\Delta_{\text{out}} + \frac{\Delta_X}{1 - 2^{-3/2}} \right) + \frac{2C_{\text{sh}}}{c_\Psi} \sqrt{m(\beta)} Z_\star(\beta). \quad (\text{F238})$$

In particular, since $Z_\star(\beta) = O(\beta^{-d_G/2})$ as $\beta \rightarrow \infty$, the right-hand side is $O(\kappa) + O(\beta^{\frac{1}{2} - \frac{d_G}{2}} \sqrt{\log(1/\delta)})$ and hence tends to 0 as $\beta \rightarrow \infty$ for every nonabelian G (where $d_G \geq 3$).

Proof. Sum (F235) over $n < k(\beta)$. For the first two sums use the bounds

$$\sum_{n \geq 1} \frac{1}{2^n + 1} \leq \sum_{n \geq 1} 2^{-n} = 1, \quad \sum_{n \geq 1} \frac{1}{2^{n/2}(2^n + 1)} \leq \sum_{n \geq 1} 2^{-3n/2} = \frac{2^{-3/2}}{1 - 2^{-3/2}} \leq \frac{1}{1 - 2^{-3/2}}.$$

For the shared-edge term, note that $\alpha_n \geq \alpha_{\min}$ for all $n \geq 1$, hence by Lemma F.188 we have $z_{2\alpha_n}(\beta) \leq Z_*(\beta)$ uniformly in n . Therefore,

$$\sum_{n < k(\beta)} 2^n z_{2\alpha_n}(\beta) \leq \left(\sum_{n < k(\beta)} 2^n \right) Z_*(\beta) \leq 2^{k(\beta)} Z_*(\beta).$$

Finally, $4^{k(\beta)} \geq m(\beta)$ implies $2^{k(\beta)} \leq 2\sqrt{m(\beta)}$, giving the last term in (F238). The stated asymptotics follow by inspection of $Z_*(\beta)$ and the explicit form of $m(\beta)$ from Proposition F.61. $\square$

Remark F.190 (How to read Proposition F.189). The first term on the right-hand side of (F238) is purely a κ -budget (pivot slack) contribution. The second term is the explicit weak-coupling decay of the shared-edge correction. For every compact connected simple G one has $d_G = \dim(G) \geq 3$, so this term decays at least as $O(\beta^{-1}\sqrt{\log(1/\delta)})$; the decay is faster in higher dimension. Thus for any fixed KP entrance threshold δ one can choose κ and then choose β large enough so that the full transfer-budget sum is $\leq \delta$.

g. Explicit feasibility of the finite-depth χ^2 transfer budget (closed-form $\kappa(\delta)$ and $\beta_0(\delta)$)

Proposition F.189 shows that the cumulative early-step transfer slack $\sum_{n < k(\beta)} \|\Delta_n\|_2$ is the sum of (i) a *pure pivot-budget* term proportional to κ and (ii) a *shared-edge* term that decays polynomially in β . In this subsection we make the parameter selection completely explicit: for a fixed target $\delta \in (0, 1)$ we define deterministic functions κ_δ and $\beta_0(\delta)$ so that the transfer sum is $\leq \delta$ for all $\beta \geq \beta_0(\delta)$. No new probabilistic assumptions are introduced; only the constants already present in Proposition F.189 are used.

Lemma F.191 (Elementary domination: exponential by a polynomial). *Let $p > 0$. Then for all $x > 0$,*

$$e^{-x} \leq \left(\frac{p}{e}\right)^p x^{-p}. \tag{F239}$$

Proof. Consider $g(x) := x^p e^{-x}$ on $(0, \infty)$. Then $g'(x) = x^{p-1} e^{-x} (p - x)$, so the unique maximizer is $x = p$, where $g(p) = p^p e^{-p}$. Thus $x^p e^{-x} \leq p^p e^{-p}$ for all $x > 0$, and (F239) follows after dividing by x^p . $\square$

Corollary F.192 (Uniform polynomial envelope for $Z_\star(\beta)$). *With $Z_\star(\beta)$ as in Proposition F.189 and $p = d_G/2$, one has for all $\beta > 0$,*

$$Z_\star(\beta) \leq C_Z(\kappa) \beta^{-d_G/2}, \quad (\text{F240})$$

where

$$C_Z(\kappa) := J_{\max} \left(\frac{\pi}{\alpha_{\min} m_-} \right)^{d_G/2} + \left(\frac{d_G}{2e} \right)^{d_G/2} (2\alpha_{\min} a_0)^{-d_G/2}, \quad \alpha_{\min} = 1 - \kappa/48. \quad (\text{F241})$$

Proof. The first term in $Z_\star(\beta)$ already has the form $\text{const} \cdot \beta^{-d_G/2}$. For the second term, apply Lemma F.191 with $p = d_G/2$ and $x = 2\alpha_{\min} \beta a_0$ to obtain $e^{-2\alpha_{\min} \beta a_0} \leq (\frac{d_G}{2e})^{d_G/2} (2\alpha_{\min} a_0)^{-d_G/2} \beta^{-d_G/2}$. Summing the two contributions gives (F240). $\square$

Proposition F.193 (Explicit (κ, β) choice forcing $\sum_{n < k_{\text{opt}}(\beta)} \|\Delta_n\|_2 \leq \delta$). *Fix a target $\delta \in (0, 1)$. Let $k_{\text{opt}}(\beta)$ be the optimal mixing depth from Proposition F.65 for block factor $L = 2$, and write $m_{\text{opt}}(\beta) = \max\{1, \beta s_\delta\}$. Define the deterministic constant*

$$A_{\text{out}} := 2 \Delta_{\text{out}} + \frac{\Delta_X}{1 - 2^{-3/2}}. \quad (\text{F242})$$

Set

$$\kappa_\delta := \min \left\{ 24, \frac{4c_\Psi}{A_{\text{out}}} \delta \right\}, \quad \alpha_{\min} := 1 - \kappa_\delta/48 \geq \frac{1}{2}, \quad (\text{F243})$$

and define $C_Z := C_Z(\kappa_\delta)$ as in (F241). Finally define

$$\beta_0(\delta) := \max \left\{ \frac{1}{s_\delta}, 1, \left(\frac{4C_{\text{sh}}}{c_\Psi} \sqrt{s_\delta} C_Z \delta^{-1} \right)^{\frac{2}{d_G-1}} \right\}. \quad (\text{F244})$$

Then for all $\beta \geq \beta_0(\delta)$, under the pivot schedule $\alpha(L) = 1 - \kappa_\delta/|P_\perp(\mathbb{B}_L)|$, the dyadic defects satisfy

$$\sum_{n < k_{\text{opt}}(\beta)} \|\Delta_n\|_2 \leq \delta. \quad (\text{F245})$$

Proof. Apply Proposition F.189 with $k(\beta) = k_{\text{opt}}(\beta)$ and $\kappa = \kappa_\delta$. The first term on the right-hand side of (F238) equals $\kappa_\delta A_{\text{out}}/(8c_\Psi)$ and is $\leq \delta/2$ by (F243).

For the shared-edge term, note that $\beta \geq 1/s_\delta$ implies $m_{\text{opt}}(\beta) = \beta s_\delta$. Corollary F.192 gives $Z_\star(\beta) \leq C_Z \beta^{-d_G/2}$. Therefore the second term in (F238) is bounded by

$$\frac{2C_{\text{sh}}}{c_\Psi} \sqrt{m_{\text{opt}}(\beta)} Z_\star(\beta) \leq \frac{2C_{\text{sh}}}{c_\Psi} \sqrt{\beta s_\delta} C_Z \beta^{-d_G/2} = \frac{2C_{\text{sh}}}{c_\Psi} \sqrt{s_\delta} C_Z \beta^{-(d_G-1)/2}.$$

With $\beta \geq \beta_0(\delta)$ from (F244), this bound is $\leq \delta/2$. Summing the two contributions yields (F245). $\square$

Corollary F.194 (One-shot extension of the transfer budget). *Under the hypotheses and notation of Proposition F.193, the same choice of $(\kappa_\delta, \beta_0(\delta))$ also yields*

$$\sum_{n \leq k_{\text{opt}}(\beta)} \|\Delta_n\|_2 \leq \delta \quad (\beta \geq \beta_0(\delta)).$$

Equivalently, the dyadic defect control is available at the one-shot scale $L = B(\beta) = 2^{k_{\text{opt}}(\beta)}$ itself.

Proof. Apply Proposition F.189 with the slightly enlarged depth

$$k^\sharp(\beta) := k_{\text{opt}}(\beta) + 1.$$

Because $k_{\text{opt}}(\beta)$ is the smallest integer with $4^{k_{\text{opt}}(\beta)} \geq m_{\text{opt}}(\beta)$, one has

$$2^{k^\sharp(\beta)} = 2^{k_{\text{opt}}(\beta)+1} \leq 4\sqrt{m_{\text{opt}}(\beta)}.$$

Repeating the proof of Proposition F.193 with $k^\sharp$ in place of k_{opt} therefore changes only the shared-edge contribution, which acquires at most an additional factor 2. The definition of $\beta_0(\delta)$ in (F244) already contains this safety factor, so the same choice of $(\kappa_\delta, \beta_0(\delta))$ forces the enlarged sum to be at most δ . Since

$$\sum_{n < k^\sharp(\beta)} \|\Delta_n\|_2 = \sum_{n \leq k_{\text{opt}}(\beta)} \|\Delta_n\|_2,$$

the claim follows. $\square$

Remark F.195 (Scaling with the group dimension). The exponent in (F244) is $2/(d_G - 1)$, hence the entrance scale becomes *easier* as d_G grows. For arbitrary compact connected simple G the exponent in (F244) is always strictly less than 1 because $d_G = \dim(G) \geq 3$. In the classical audited examples the explicit exponents can be read off from the corresponding dimensions recorded in the family appendices.

h. Splicing the explicit transfer-feasibility bound into the Plan-A χ^2 gate

Proposition F.112 reduces the Plan-A variance transfer to the conditional χ^2 gate (F136) with a constant ε_* satisfying $\sqrt{\varepsilon_*} \leq c_1/(2C_A)$. We now connect this gate directly to the explicit weak-window bounds of Section F.19g, eliminating the remaining “choose β large enough” ambiguity and making the (κ, β) entrance choice audit-ready.

Definition F.196 (Plan-A χ^2 gate parameters). Fix the constants $c_1(G)$ from Lemma F.111 and $C_A(G)$ from Proposition F.112. Define

$$\varepsilon_* := \left(\frac{c_1}{2C_A} \right)^2, \quad \delta_{\text{PA}} := \frac{c_1}{4\sqrt{2}C_A}. \quad (\text{F246})$$

Lemma F.197 (L^2 -flatness at level δ_{PA} implies the Plan-A conditional χ^2 gate). *Fix a planar tile of side length L and write $\nu := \nu_L^\parallel|_{\mathcal{F}_\parallel}$, $\mu := \nu_L^{(\alpha)}|_{\mathcal{F}_\parallel}$. Let $\Omega \in \mathcal{F}_\parallel$ satisfy $\nu(\Omega) \geq \frac{1}{2}$. Set $\Delta_L := \tilde{\Psi}_L^{(\alpha)} - 1$ as in (F233). If $\|\Delta_L\|_{L^2(\nu)} \leq \delta_{\text{PA}}$, then*

$$D_{\chi^2}(\mu^\Omega \parallel \nu^\Omega) \leq \varepsilon_*. \quad (\text{F247})$$

Proof. By Lemma F.182, if $\|\Delta_L\|_{L^2(\nu)} \leq \frac{1}{2}\sqrt{\nu(\Omega)}$ then

$$D_{\chi^2}(\mu^\Omega \parallel \nu^\Omega) \leq \frac{4}{\nu(\Omega)} \|\Delta_L\|_{L^2(\nu)}^2.$$

Since $\nu(\Omega) \geq \frac{1}{2}$, the hypothesis $\|\Delta_L\|_{L^2(\nu)} \leq \delta_{\text{PA}}$ implies $\|\Delta_L\|_{L^2(\nu)} \leq 1/(2\sqrt{2})$, hence the lemma applies and yields

$$D_{\chi^2}(\mu^\Omega \parallel \nu^\Omega) \leq 8 \|\Delta_L\|_{L^2(\nu)}^2 \leq 8\delta_{\text{PA}}^2 = \varepsilon_*,$$

using (F246) in the last step. □

Proposition F.198 (A uniform (κ, β) choice meeting the Plan-A χ^2 gate in the weak window). *Assume the weak-window setting and pivot schedule of Proposition F.193. Fix any $\delta \in (0, \delta_{\text{PA}}]$ and choose $\kappa = \kappa_\delta$ and $\beta_0(\delta)$ as in (F243)–(F244). Then for every $\beta \geq \beta_0(\delta)$ and every dyadic scale $L = 2^n$ with $n \leq k_{\text{opt}}(\beta)$,*

$$\|\Delta_L\|_{L^2(\nu_L^\parallel|_{\mathcal{F}_\parallel})} \leq \delta \leq \delta_{\text{PA}}.$$

Consequently, for every such L and every event $\Omega \in \mathcal{F}_\parallel$ with $\nu_L^\parallel(\Omega) \geq \frac{1}{2}$ (in particular for the standard weak-field event Ω_{ε_0}), the Plan-A conditional χ^2 requirement (F136) holds with ε_ from (F246).*

Proof. By Corollary F.194,

$$\sum_{m \leq k_{\text{opt}}(\beta)} \|\Delta_m\|_2 \leq \delta.$$

Each summand is nonnegative, hence in particular $\|\Delta_n\|_2 \leq \delta$ for every $n \leq k_{\text{opt}}(\beta)$. The conditional χ^2 conclusion is Lemma F.197. $\square$

Corollary F.199 (Plan-A gate at the one-shot block size). *Under the hypotheses of Proposition F.198, let*

$$B(\beta) := 2^{k_{\text{opt}}(\beta)}.$$

Then the exact-to-reference comparison at the same block size $L = B(\beta)$ satisfies the Plan-A conditional χ^2 gate (F136) with defect ε_ , uniformly in finite volume, admissible boundary conditions, and all events Ω of reference probability at least $1/2$.*

Proof. Take $n = k_{\text{opt}}(\beta)$ in Proposition F.198. $\square$

Proposition F.200 (Crossover coefficient decay $\Rightarrow$ reference KP-smallness at a computable scale). *Assume the coefficient decay (F130) for the thick-block reference family. Define the nonnegative L^2 activity*

$$f_L^{(\alpha)} := \|w_L^{(\alpha)} - 1\|_{L^2(G)}^2. \quad (\text{F248})$$

Then, using the character orthogonality identity for central functions,

$$f_L^{(\alpha)} = \sum_{R \in \text{Irr}(G) \setminus \{1\}} d_R^2 \left| \lambda_R(w_L^{(\alpha)}) \right|^2. \quad (\text{F249})$$

In particular, for $\beta \geq \beta_{\text{mix}}$ and $\alpha(L) \in [1/2, 1]$,

$$f_L^{(\alpha(L))} \leq \sum_{R \neq 1} d_R^2 \exp\left(-2|T_L| c_{A1}(G, \rho) \frac{C_2(R)}{\alpha(L)\beta + \sqrt{C_2(R)}}\right). \quad (\text{F250})$$

Setting $s := |T_L|/\beta = L^2/\beta$, the sum admits the genuine low-/high-representation split

$$\begin{aligned} f_L^{(\alpha(L))} &\leq \sum_{\substack{R \neq 1 \\ C_2(R) \leq \beta^2}} d_R^2 \exp\left(-c_{A1}(G, \rho) s C_2(R)\right) \\ &\quad + \sum_{\substack{R \neq 1 \\ C_2(R) > \beta^2}} d_R^2 \exp\left(-c_{A1}(G, \rho) |T_L| \sqrt{C_2(R)}\right). \end{aligned} \quad (\text{F251})$$

Define the two summability constants

$$K_{0,G}^{\text{lo}} := \sum_{R \neq 1} d_R^2 \exp\left(-\frac{c_{A1}(G, \rho)}{2} C_2(R)\right) \in (0, \infty), \quad (\text{F252})$$

$$H_{0,G}^{\text{hi}} := \sum_{R \neq \mathbf{1}} d_R^2 \exp\left(-\frac{c_{A1}(G,\rho)}{2} \sqrt{C_2(R)}\right) \in (0, \infty), \quad (\text{F253})$$

and let

$$C_{\min}(G) := \min\{C_2(R) : R \neq \mathbf{1}\}$$

be the first positive Laplace–Beltrami eigenvalue on G . For every target $f_\star \in (0, 1)$ set

$$S_\star := \max\left\{\frac{1}{c_{A1}(G,\rho) C_{\min}(G)} \log\left(\frac{2K_{0,G}^{\text{lo}}}{f_\star}\right) + \frac{1}{2}, \frac{1}{c_{A1}(G,\rho) \sqrt{C_{\min}(G)}} \log\left(\frac{2H_{0,G}^{\text{hi}}}{f_\star}\right) + \frac{1}{2}\right\}. \quad (\text{F254})$$

Then

$$\frac{|T_L|}{\beta} = \frac{L^2}{\beta} \geq S_\star \implies f_L^{(\alpha(L))} \leq f_\star.$$

Proof. Identity (F249) is standard Peter–Weyl orthogonality for central functions: if $w = \sum_R d_R \lambda_R(w) \chi_R$ and $\int w dg = 1$, then $\|w - 1\|_2^2 = \sum_{R \neq \mathbf{1}} d_R^2 |\lambda_R(w)|^2$.

Bound (F250) follows immediately from (F249) and the coefficient decay (F130). To obtain (F251), split the irrep sum into the low-representation regime $C_2(R) \leq \beta^2$ and the high-representation regime $C_2(R) > \beta^2$. If $C_2(R) \leq \beta^2$, then $\sqrt{C_2(R)} \leq \beta$ and $\alpha(L)\beta \leq \beta$, so

$$\alpha(L)\beta + \sqrt{C_2(R)} \leq 2\beta, \\ 2|T_L| c_{A1}(G,\rho) \frac{C_2(R)}{\alpha(L)\beta + \sqrt{C_2(R)}} \geq c_{A1}(G,\rho) \frac{|T_L|}{\beta} C_2(R) = c_{A1}(G,\rho) s C_2(R).$$

If $C_2(R) > \beta^2$, then $\beta < \sqrt{C_2(R)}$ and therefore

$$\alpha(L)\beta + \sqrt{C_2(R)} \leq 2\sqrt{C_2(R)}, \quad 2|T_L| c_{A1}(G,\rho) \frac{C_2(R)}{\alpha(L)\beta + \sqrt{C_2(R)}} \geq c_{A1}(G,\rho) |T_L| \sqrt{C_2(R)}.$$

This yields (F251).

For the low-representation tail and $s \geq 1/2$,

$$e^{-c_{A1}(G,\rho)sC_2(R)} = e^{-\frac{c_{A1}(G,\rho)}{2}C_2(R)} e^{-c_{A1}(G,\rho)(s-\frac{1}{2})C_2(R)} \leq e^{-\frac{c_{A1}(G,\rho)}{2}C_2(R)} e^{-c_{A1}(G,\rho)(s-\frac{1}{2})C_{\min}(G)}.$$

Hence

$$\sum_{\substack{R \neq \mathbf{1} \\ C_2(R) \leq \beta^2}} d_R^2 e^{-c_{A1}(G,\rho)sC_2(R)} \leq K_{0,G}^{\text{lo}} e^{-c_{A1}(G,\rho)(s-\frac{1}{2})C_{\min}(G)}.$$

For the high-representation tail, use $|T_L| = \beta s$, $\beta \geq 1$, $s \geq 1/2$, and $C_2(R) \geq C_{\min}(G)$ to write

$$e^{-c_{A1}(G,\rho)|T_L|\sqrt{C_2(R)}} = e^{-c_{A1}(G,\rho)\beta s\sqrt{C_2(R)}} \leq e^{-\frac{c_{A1}(G,\rho)}{2}\sqrt{C_2(R)}} e^{-c_{A1}(G,\rho)(s-\frac{1}{2})\sqrt{C_{\min}(G)}}.$$

Therefore

$$\sum_{\substack{R \neq \mathbf{1} \\ C_2(R) > \beta^2}} d_R^2 e^{-c_{A1}(G, \rho) |T_L| \sqrt{C_2(R)}} \leq H_{0,G}^{\text{hi}} e^{-c_{A1}(G, \rho) (s - \frac{1}{2}) \sqrt{C_{\min}(G)}}.$$

The finiteness of $H_{0,G}^{\text{hi}}$ follows from Weyl polynomial growth of d_R together with the Casimir asymptotic $C_2(R) \asymp |\lambda_R|^2$, which makes the tail comparable to a polynomial times $e^{-c|\lambda_R|}$.

If $s \geq S_*$, each displayed tail is at most $f_*/2$ by (F254), hence $f_L^{(\alpha(L))} \leq f_*$. $\square$

Corollary F.201 (Reference KP-smallness is closed on the load-bearing Route 1 path). *For every compact connected simple G and every target $f_* \in (0, 1)$, the computable two-tail threshold S_* from (F254) has the following property: whenever L is dyadic and $L^2/\beta \geq S_*$, the exact thick-block reference activity satisfies*

$$f_L^{(\alpha(L))} \leq f_*.$$

This is the theorem-level QE1 reference-budget input used in the one-shot entrance proof.

Proof. Corollary F.155 supplies the coefficient decay (F130); Proposition F.200 then yields the stated bound with the explicit threshold (F254). $\square$

Remark F.202 (Why a thick reference is still needed). Lemma F.92 is exact because $\mathcal{B}_L$ is a *thin* neighborhood: the translated opposite edges of $P_\perp$ plaquettes are leaves, so the out-of-plane sector carries no structural feedback into the in-plane holonomy marginal on this block. In the *true* 4-dimensional lattice, those translated edges are not leaves: they participate in additional plaquettes in shifted slices and in perpendicular cubes. Therefore the thin-block computation is used only as a diagnostic for the slack bookkeeping in (F115); it is not the reference used in the entrance proof. The entrance construction in this appendix is implemented with a positivity-preserving coarse reference in which the translated edges are no longer leaves (see Sections F 9 and F 8), and is packaged in the one-shot KP entrance theorem (Theorem F.320).

20. Infinite-volume exponential clustering at fixed large β via finite-depth KP entrance

This section records the next mathematically forced implication of the already completed modules:

1. the downstream KP module (Dobrushin/KP contraction inside the admissible window);
2. the exact-step contraction and mixing bound (Proposition 12.1 and its corollaries in the current numbering);
3. the α -pivot slack analysis and χ^2 -transfer module, including the weak-coupling χ^2 decay of Section F 19 c.

We do *not* re-derive those results; we only assemble them into an infinite-volume statement.

a. Blocked specifications and influence coefficients

Let $\mathbb{Z}_L^4 := L\mathbb{Z}^4$ be the block lattice. For a finite block region $\Lambda \subset \mathbb{Z}_L^4$, write μ_Λ^{ex} for the exact blocked coarse measure produced by the RG map at block size L (with any fixed admissible boundary condition on $\partial\Lambda$), and $\mu_\Lambda^{(\alpha)}$ for the corresponding α -reference blocked measure. Denote by $\gamma^{\text{ex}} = \{\gamma_i^{\text{ex}}\}_{i \in \mathbb{Z}_L^4}$ (resp. $\gamma^{(\alpha)}$) the associated single-site conditional kernels (specifications).

For sites $i \neq j$, define the total-variation influence coefficient

$$c_{ij}(\gamma) := \sup_{\eta, \eta': \eta_{\mathbb{Z}_L^4 \setminus \{j\}} = \eta'_{\mathbb{Z}_L^4 \setminus \{j\}}} \left\| \gamma_i(\cdot \mid \eta) - \gamma_i(\cdot \mid \eta') \right\|_{\text{TV}}.$$

Write $C(\gamma)$ for the influence matrix $(c_{ij}(\gamma))$ and the Dobrushin coefficient

$$\delta(\gamma) := \sup_i \sum_{j \neq i} c_{ij}(\gamma).$$

Proposition F.203 (Finite-range KP/Dobrushin criterion in oscillation-matrix form). *Let γ be a single-site specification on a graph (V, E) with graph distance dist_G and influence matrix $C(\gamma) = (c_{ij})$. For a bounded local observable F , define the single-site oscillation seminorm*

$$\text{osc}_i(F) := \sup_{\eta, \eta': \eta_{V \setminus \{i\}} = \eta'_{V \setminus \{i\}}} |F(\eta) - F(\eta')|, \quad \text{supp}_{\text{osc}}(F) := \{i \in V : \text{osc}_i(F) \neq 0\}.$$

Assume that

$$c_{ij}(\gamma) = 0 \quad \text{whenever } \text{dist}_G(i, j) > 1, \quad \delta(\gamma) := \sup_i \sum_{j \neq i} c_{ij}(\gamma) \leq q < 1.$$

Then the Gibbs measure specified by γ exists and is unique in infinite volume. Let

$$D := \sum_{m \geq 0} C(\gamma)^m = (I - C(\gamma))^{-1}.$$

For bounded local observables F, G and the unique infinite-volume Gibbs state $\langle \cdot \rangle_\gamma$ one has the standard oscillation-matrix covariance bound

$$|\text{Cov}_{\langle \cdot \rangle_\gamma}(F, G)| \leq \sum_{i \in V} \sum_{j \in V} \text{osc}_i(F) D_{ij} \text{osc}_j(G). \quad (\text{F255})$$

Moreover, the finite-range hypothesis implies the explicit path-sum estimate

$$D_{ij} \leq \sum_{m \geq \text{dist}_G(i, j)} q^m = \frac{e^{-(\log q) \text{dist}_G(i, j)}}{1 - q}. \quad (\text{F256})$$

Consequently, if $S := \text{supp}_{\text{osc}}(F)$ and $T := \text{supp}_{\text{osc}}(G)$, then

$$|\text{Cov}_{\langle \cdot \rangle_\gamma}(F, G)| \leq \frac{e^{-(\log q) \text{dist}_G(S, T)}}{1 - q} \left(\sum_{i \in S} \text{osc}_i(F) \right) \left(\sum_{j \in T} \text{osc}_j(G) \right). \quad (\text{F257})$$

In particular, since $\text{osc}_i(F) \leq 2\|F\|_\infty$ and $\text{osc}_j(G) \leq 2\|G\|_\infty$,

$$|\text{Cov}_{\langle \cdot \rangle_\gamma}(F, G)| \leq \frac{4|S||T|}{1 - q} \|F\|_\infty \|G\|_\infty e^{-(\log q) \text{dist}_G(S, T)}. \quad (\text{F258})$$

Proof. The uniqueness statement is the standard Dobrushin criterion. The covariance estimate (F255) is the standard Dobrushin–Shlosman oscillation-matrix bound for the unique Gibbs state. Because $c_{ij}(\gamma) = 0$ unless $\text{dist}_G(i, j) \leq 1$, every nonzero contribution to $(C(\gamma)^m)_{ij}$ comes from a path of length m joining i to j , so $m \geq \text{dist}_G(i, j)$ whenever $(C(\gamma)^m)_{ij} \neq 0$. Using the row-sum bound $\sup_i \sum_j c_{ij}(\gamma) \leq q$ therefore gives

$$D_{ij} = \sum_{m \geq 0} (C(\gamma)^m)_{ij} \leq \sum_{m \geq \text{dist}_G(i, j)} q^m,$$

which is exactly (F256). Substituting (F256) into (F255) yields (F257), and the cardinality estimate (F258) follows from the trivial bounds on single-site oscillations. $\square$

b. Stability of the influence matrix under χ^2 transfer

The completed χ^2 -transfer module gives, for each finite Λ and each block site i , a uniform bound on the divergence between the exact and reference *single-site* kernels, after the finite entrance depth has been reached. We record the elementary consequence needed here.

Lemma F.204 (χ^2 control implies TV kernel stability). *Let $\mu \ll \nu$ be probability measures on a common measurable space with $D_{\chi^2}(\mu \| \nu) \leq \varepsilon$. Then $\|\mu - \nu\|_{\text{TV}} \leq \frac{1}{2}\sqrt{\varepsilon}$. In particular, for two specifications $\gamma, \tilde{\gamma}$ with*

$$\sup_i \sup_\eta D_{\chi^2}(\gamma_i(\cdot | \eta) \| \tilde{\gamma}_i(\cdot | \eta)) \leq \varepsilon,$$

one has the influence stability bound

$$\delta(\gamma) \leq \delta(\tilde{\gamma}) + \sqrt{\varepsilon}.$$

Proof. The TV bound is (F87) in the present notation. For influence stability, fix $i \neq j$ and boundary conditions η, η' differing only at j . Insert and subtract $\tilde{\gamma}_i(\cdot \mid \eta)$ and $\tilde{\gamma}_i(\cdot \mid \eta')$ and use the triangle inequality:

$$\|\gamma_i(\cdot \mid \eta) - \gamma_i(\cdot \mid \eta')\|_{\text{TV}} \leq \|\tilde{\gamma}_i(\cdot \mid \eta) - \tilde{\gamma}_i(\cdot \mid \eta')\|_{\text{TV}} + 2 \sup_{\xi} \|\gamma_i(\cdot \mid \xi) - \tilde{\gamma}_i(\cdot \mid \xi)\|_{\text{TV}}.$$

Take suprema and apply the TV- χ^2 inequality. $\square$

c. Infinite-volume blocked exponential clustering at fixed β

Theorem F.205 (Blocked infinite-volume uniqueness and clustering after finite entrance depth). *Fix $d = 4$ and compact G . Assume β is in the weak-coupling regime of the entrance construction and that the schedule $\alpha(L) = 1 - \kappa/|P_{\perp}(\mathcal{B}_L)|$ is used. Suppose that at some finite entrance depth (already established in the completed KP module) the α -reference blocked specification satisfies the uniform contraction bound*

$$\delta(\gamma^{(\alpha)}) \leq q < 1,$$

with q independent of L , the finite block volume Λ , and the boundary condition. Then there exists $L_0(\beta, G, \kappa)$ such that for all $L \geq L_0$:

- 1. the exact blocked specification satisfies $\delta(\gamma^{\text{ex}}) < 1$ uniformly in Λ and boundary conditions;*
- 2. the infinite-volume exact blocked Gibbs measure exists and is unique;*
- 3. block-local truncated correlations decay exponentially in block distance.*

Proof. By the weak-coupling χ^2 -decay lemma of Section F 19 c and the χ^2 -transfer module, for L large one has a uniform conditional divergence bound

$$\sup_i \sup_{\eta} D_{\chi^2}(\gamma_i^{\text{ex}}(\cdot \mid \eta) \parallel \gamma_i^{(\alpha)}(\cdot \mid \eta)) \leq \varepsilon_L, \quad \varepsilon_L \rightarrow 0.$$

Lemma F.204 then gives $\delta(\gamma^{\text{ex}}) \leq q + \sqrt{\varepsilon_L}$. Choose L_0 so that $q + \sqrt{\varepsilon_L} < 1$ for all $L \geq L_0$. Apply Proposition F.203 to γ^{ex} . $\square$

d. Descent to microscopic observables

To convert blocked clustering to microscopic clustering, it is enough to control the block-conditional fluctuation $O - \mathbb{E}[O \mid \mathcal{F}_{\text{blocked}}]$ for local gauge-invariant observables O .

Lemma F.206 (Uniform block-conditional Poincaré inequality at weak coupling). *Fix compact G and $d = 4$. There exist $\beta_1(G) < \infty$ and constants $c_{\text{PI}}(G), C_{\text{PI}}(G) < \infty$ such that for all $\beta \geq \beta_1(G)$, for every block B_L and every admissible boundary condition on ∂B_L , the conditional law of the microscopic edge variables in B_L (in spanning-tree gauge) satisfies the Poincaré inequality*

$$\text{Var}_{\mu_{B_L}^{\text{cond}}}(f) \leq \frac{C_{\text{PI}}(G)}{\beta} \sum_{e \in E_f(B_L)} \int \|\nabla_e f\|^2 d\mu_{B_L}^{\text{cond}},$$

for all smooth f , where $E_f(B_L)$ are the free edges after tree gauge. In particular, the Poincaré constant is uniform in L and the boundary condition.

Proof. Work in spanning-tree gauge and exponential coordinates $U_e = \exp(iA_e)$ on a fixed normal neighborhood of the identity in G , with $A_e \in \mathfrak{g}$ for $e \in E_f(B_L)$. For β large, the conditional block density takes the standard weak-coupling form

$$\frac{1}{Z} \exp\left(-\beta \Phi(A)\right) dA, \quad \Phi(A) = \frac{1}{2} \langle A, \mathcal{M}A \rangle + R(A),$$

where $\mathcal{M}$ is the gauge-fixed magnetic quadratic form and $R(A) = O(\|A\|^3)$ is smooth (uniformly in L and boundary condition on the chart). Tree gauge removes the gauge null space; hence there exists $c_0(G) > 0$ such that $\mathcal{M} \succeq c_0(G)I$ uniformly in L and the boundary condition (discrete Hodge theory with fixed gauge).

Choose β_1 so large that on the chart region $\{\|A\| \leq r\}$ (with fixed small r) the Hessian obeys the uniform strong convexity bound

$$\nabla^2 \Phi(A) \succeq \frac{c_0(G)}{2} I.$$

On this region, the Brascamp–Lieb inequality yields

$$\text{Var}(f) \leq \frac{2}{\beta c_0(G)} \int \|\nabla f\|^2 d\mu_{B_L}^{\text{cond}}.$$

Finally, the Wilson density supplies Gaussian tails at scale β outside the chart; a standard localization/cutoff argument extends the bound globally at the price of a multiplicative constant depending only on G (absorbed into $C_{\text{PI}}(G)$). $\square$

Lemma F.207 (Block-conditional Poincaré at weak coupling). *Fix $d = 4$ and compact G . There exist $\beta_{\text{PI}}(G) < \infty$ and $C_{\text{PI}}(G) < \infty$ such that for all $\beta \geq \beta_{\text{PI}}$, all block sizes L , all $\alpha \in [\alpha_*, 1]$, and all gauge-invariant functions F supported on a single block B_L , the conditional law of the edge variables in B_L given the exterior configuration (in spanning-tree gauge) obeys the Poincaré inequality*

$$\text{Var}(F \mid \mathcal{F}_{\text{ext}}) \leq \frac{C_{\text{PI}}(G)}{\beta} \sum_{e \in E_f(B_L)} \mathbb{E}[\|\nabla_e F\|^2 \mid \mathcal{F}_{\text{ext}}],$$

with constants independent of L and of the exterior configuration.

Proof. Work in spanning-tree gauge on B_L so the conditional density is a smooth positive function on $G^{E_f(B_L)}$. For β large, restrict to the exponential chart $U_e = \exp(iA_e)$ on each free edge. The (conditional) negative log-density is

$$H(A) = \frac{\beta}{2} \langle A, \mathcal{M}_{\alpha, \text{cond}} A \rangle + R(A),$$

where $\mathcal{M}_{\alpha, \text{cond}}$ is the gauge-fixed quadratic form induced by the plaquette terms inside B_L (with transverse weights α), and the remainder satisfies $\|\nabla^2 R(A)\| \leq \frac{1}{4}\beta \lambda_{\min}(\mathcal{M}_{\alpha, \text{cond}})$ on a high-probability region $\|A\| \leq c(\beta)$, while the complement has exponentially small conditional mass (localization via the plaquette tail bound). Discrete Hodge theory in tree gauge gives a uniform coercivity bound $\mathcal{M}_{\alpha, \text{cond}} \succeq c_0(G)I$ independent of L and exterior data. Therefore on the localized region, $\nabla^2 H \succeq \frac{\beta c_0(G)}{4}I$. Brascamp–Lieb then yields a Poincaré inequality with constant $\leq 4/(\beta c_0(G))$ for the localized measure. The exterior region is handled by a standard two-region decomposition (Holley–Stroock gluing): the tail probability is $\leq e^{-c\beta}$ and only inflates the constant by a bounded factor. Collecting constants gives the stated bound with $C_{\text{PI}}(G)$ and $\beta_{\text{PI}}(G)$. $\square$

Lemma F.208 (Descent inequality). *Assume Theorem F.205 and Lemma F.206. Let O be a bounded local gauge-invariant observable supported in a microscopic set of diameter $O(1)$. Let $O^{(L)} := \mathbb{E}[O \mid \mathcal{F}_{\text{blocked}}]$ and $\delta O := O - O^{(L)}$. Then for all $x \in \mathbb{Z}^4$,*

$$|\langle O(x)O(0) \rangle_c| \leq C_1 e^{-m_1|x|/L} + C_2 \|\delta O\|_{L^2}^2,$$

where $m_1 > 0$ depends only on the blocked Dobrushin constant and C_1, C_2 depend on O, β, G . Moreover, Lemma F.206 yields $\|\delta O\|_{L^2} \leq C(O, \beta) e^{-cL}$ (or at least a polynomial decay in L), so optimizing in L gives a microscopic exponential bound $|\langle O(x)O(0) \rangle_c| \leq C e^{-m|x|}$ at fixed β .

Proof. Write $O = O^{(L)} + \delta O$ and expand the truncated correlation. The term $\langle O^{(L)}(x)O^{(L)}(0) \rangle_c$ is a truncated correlation of block-measurable observables and decays exponentially in block distance by Theorem F.205. The remaining terms are controlled by Cauchy–Schwarz and the L^2 bound on δO . The final claim follows by applying Lemma F.207 to the block-conditional variance term in the martingale decomposition. $\square$

Remark F.209 (Status of the fixed- β descent lemma). Lemma F.208 remains a qualitative fixed- β diagnostic and is not part of the front-facing QE2_{lat} proof spine. The actual QE2 closure below no longer seeks a separate fluctuation-remainder estimate at the one-shot scale. Instead it uses exact terminal-block conditional independence at that scale, supplied by Corollary III.12 together with Proposition F.316, to eliminate the fluctuation term entirely for separated local probes.

21. Continuum limit and OS reconstruction: interface statements and closure

a. *Continuum scaling setup*

Fix a lattice spacing parameter $a > 0$ and identify the rescaled lattice $a\mathbb{Z}^4$ with $\mathbb{Z}^4$ by the map $x \mapsto x/a$. For each β write $\mu_{\beta,\Lambda}$ for the finite-volume Wilson measure in a box $\Lambda \subset \mathbb{Z}^4$ with a standard reflection-positive boundary condition (e.g. free; or periodic handled via a standard doubling trick). When the thermodynamic limit exists, write μ_β for the infinite-volume Gibbs state.

We distinguish three notions of gap:

- $\Delta_{\text{lat}}(\beta)$: the *full fixed-lattice OS/transfer-matrix gap* in lattice units, i.e.

$$\Delta_{\text{lat}}(\beta) := \inf(\text{Spec}(H_\beta) \setminus \{0\}),$$

where $(\mathcal{H}_\beta, \Omega_\beta, H_\beta)$ is the OS reconstruction of the fixed-lattice Wilson Gibbs state on the bounded gauge-invariant positive-time cylinder algebra. This is the quantity used on the primary QE2/QE3 spine.

- $m_{\text{lat}}(\beta)$: an *auxiliary probe-family/channel decay rate* in lattice units. It is retained only as a diagnostic shorthand for particular local probe families; the primary spine no longer identifies it with the full lattice spectral gap unless a separate density/cyclicity theorem is stated.

- m_{phys} : the physical mass in continuum units, obtained after choosing a scaling trajectory $a = a(\beta) \downarrow 0$ and a renormalization prescription for observables.

b. OS axioms and reconstruction target

Let Θ denote reflection in a time hyperplane and let $\mathcal{A}_+$ be the algebra generated by observables supported in the positive-time half-space. Reflection positivity is the requirement

$$\langle F \Theta F \rangle_{\mu_{\beta, \Lambda}} \geq 0 \quad \forall F \in \mathcal{A}_+.$$

For the Wilson action this holds at finite volume. The continuum construction must preserve it in the infinite-volume and scaling limits so that Osterwalder–Schrader reconstruction yields a Hilbert space, a Hamiltonian with spectrum bounded below, and a vacuum. A *mass gap* in the mass gap sense is a strictly positive spectral gap above the vacuum in this reconstructed Hamiltonian.

c. Interface theorem chain to the continuum mass gap

The mass-gap interfaces are separated as follows: (i) one-shot KP entrance together with a tuned-sequence physical entrance bound is proved unconditionally earlier in Appendix F; (ii) the tuned-sequence full fixed-lattice OS gap is proved below by exact terminal-block covariance transfer at the one-shot scale together with the fixed-lattice spectral-gap theorem; and (iii) QE3 now requires only a continuum-time OS reconstruction theorem stronger than the dyadic statement of Theorem F.211. The dense-core Route 1/Lane B package is recorded later only as a secondary reformulation/consistency check and is not part of the primary front-facing spine.

Theorem F.210 (Provisional fixed- β full-lattice-gap diagnostic (not used on the primary spine)). *Fix $\beta \geq \beta_*$ in the range of the one-shot entrance theorem (Theorem F.320). If the fixed- β fluctuation-control step in Lemma F.208 is upgraded to a genuine rate-preserving microscopic descent theorem for all bounded gauge-invariant local observables, then the lattice OS Hamiltonian of the infinite-volume Gibbs state μ_β has a full vacuum gap $\Delta_{\text{lat}}(\beta) > 0$.*

Proof. This theorem is retained only as a diagnostic side-branch. Concretely, Theorem F.320 yields KP entrance at the one-shot block scale $B(\beta)$ and hence blocked exponential clustering.

A quantitative fixed- β descent theorem for all bounded local observables would then feed Theorem F.34 together with Proposition F.35, yielding a genuine full fixed-lattice OS gap. Because the primary QE2_{lat} route below proves that gap directly along the tuned dyadic path, this diagnostic statement is not used on the paper-level spine. $\square$

Theorem F.211 (Bounded positive-time continuum Wilson state and dyadic OS matrix elements along the tuned trajectory). *Fix a weak-window target value $u_{\text{ren}} \in (0, u_*]$ and let*

$$a_n := \ell_0 2^{-n}, \quad \beta_n := \beta(a_n; u_{\text{ren}}).$$

Define the dense dyadic translation set

$$\mathbb{D}_{\text{dyad}} := \{m\ell_0 2^{-k} : m, k \in \mathbb{N}_0\}.$$

Then the following hold along this fixed tuned dyadic sequence:

(i) *For every bounded cylinder observable $F \in \mathcal{A}_+$ the limit*

$$\omega_{\text{bd}}^{(u_{\text{ren}})}(F) := \lim_{n \rightarrow \infty} \omega_{a_n}(F)$$

exists. The resulting functional $\omega_{\text{bd}}^{(u_{\text{ren}})}$ is a state on the bounded positive-time algebra $\mathcal{A}_+$ and is uniquely determined by the chosen target value u_{ren} .

(ii) *For every bounded $F, G \in \mathcal{A}_+$ and every dyadic shift $s \in \mathbb{D}_{\text{dyad}}$, the OS matrix-element limit*

$$\mathcal{K}_{u_{\text{ren}}}(F, G; s) := \lim_{n \rightarrow \infty} \omega_{a_n}(\Theta F \tau_s G) \tag{F259}$$

exists (for all sufficiently large n such that $s/a_n \in \mathbb{N}$). In particular,

$$\mathcal{K}_{u_{\text{ren}}}(F, F; 0) \geq 0, \tag{F260}$$

so $\omega_{\text{bd}}^{(u_{\text{ren}})}$ is reflection positive on $\mathcal{A}_+$.

Proof. For part (i), Corollary 4.94 gives joint convergence for every finite family of bounded cylinder observables from $\mathcal{A}_+$ along the tuned dyadic sequence. Applying that corollary to singleton families defines

$$\omega_{\text{bd}}^{(u_{\text{ren}})}(F) := \lim_{n \rightarrow \infty} \omega_{a_n}(F) \quad (F \in \mathcal{A}_+).$$

Linearity and unitality are inherited from the finite-volume states ω_{a_n} term by term, and positivity follows by applying the same convergence statement to F^*F . Hence the limit is a

state on $\mathcal{A}_+$, uniquely determined by the chosen target value u_{ren} because the tuned sequence itself is fixed.

For part (ii), Corollary 4.95 gives joint convergence of every finite family of reflected/transformed bounded cylinder observables built from $\mathcal{A}_+$ using dyadic time translations. Applying that corollary to the single observable $\Theta F \tau_s G$ yields the limit (F259). When $s = 0$ and $G = F$, Theorem III.2 gives

$$\omega_{a_n}(\Theta F F) \geq 0 \quad \text{for every } n,$$

and taking the limit gives (F260). Thus the limiting bounded positive-time state is reflection positive on $\mathcal{A}_+$, with the actual OS matrix elements on the dense dyadic translation set recorded explicitly by (F259). $\square$

Dependency summary.

Inputs. Corollary 4.94, Corollary 4.95, and Theorem III.2

Imported standard background. none

Conclusion at this node. along the tuned dyadic trajectory the bounded Wilson state on $\mathcal{A}_+$ and its dyadic reflected OS matrix elements converge; the theorem records reflection positivity on $\mathcal{A}_+$ only and does not yet construct the all-time sharp-local semigroup

Used later in. Theorem F.212 and Theorem F.219

Theorem F.212 (Continuum-time OS upgrade of the bounded base state via Lane A). *Fix a target value $u_{\text{ren}} \in (0, u_*]$ and let $\omega^{(u_{\text{ren}})}$ be the full sharp-local state furnished by Corollary 5.76 over the same tuned dyadic bounded base state. Let $(\mathcal{H}, \Omega, H)$ be its OS reconstruction. Then:*

(i) *the restriction of $\omega^{(u_{\text{ren}})}$ to $\mathcal{A}_+$ is exactly the bounded state of Theorem F.211:*

$$\omega^{(u_{\text{ren}})} \upharpoonright_{\mathcal{A}_+} = \omega_{\text{bd}}^{(u_{\text{ren}})};$$

(ii) *for every bounded $F, G \in \mathcal{A}_+$ and every dyadic shift $s \in \mathbb{D}_{\text{dyad}}$,*

$$\omega^{(u_{\text{ren}})}(\Theta F \tau_s G) = \mathcal{K}_{u_{\text{ren}}}(F, G; s);$$

(iii) *for every bounded $F, G \in \mathcal{A}_+$ and every $t \geq 0$,*

$$\omega^{(u_{\text{ren}})}(\Theta F \tau_t G) = \langle [F], e^{-tH} [G] \rangle_{\mathcal{H}},$$

and the map $t \mapsto \omega^{(u_{\text{ren}})}(\Theta F \tau_t G)$ is continuous on $[0, \infty)$.

Consequently the bounded positive-time state of Theorem F.211 has a canonical continuum-time OS semigroup/reconstruction obtained by transport through the Lane A sharp-local state; no separate non-dyadic lattice-shift convergence theorem is required on the primary Route 1 spine.

Proof. The role split from Lane A must be read literally. First, part (i) is the bounded-state restriction statement isolated in Corollary 5.75: after the full sharp-local state has been assembled, that corollary identifies its restriction to $\mathcal{A}_+$ with the bounded state of Theorem F.211. Second, the dyadic identity in part (ii) is the second clause of the same bounded-state compatibility theorem. In particular, Corollary 5.75 is not the inductive-union theorem.

The finite-cap sharp-local OS structure used for part (iii) comes instead from Theorem 5.74(b) on each finite engineering cap. Corollary 5.76 is the separate passage that transports those compatible capwise OS structures to the full sharp-local inductive union. Hence the sharp-local state $\omega^{(u_{\text{ren}})}$ satisfies the OS axioms on the relevant local positive-time algebra containing $\mathcal{A}_+$, so OS reconstruction yields the Hilbert space $(\mathcal{H}, \Omega, H)$ and the semigroup identity

$$\omega^{(u_{\text{ren}})}(\Theta F \tau_t G) = \langle [F], e^{-tH}[G] \rangle_{\mathcal{H}} \quad (F, G \in \mathcal{A}_+, t \geq 0).$$

Strong continuity of e^{-tH} gives the continuity statement in part (iii). The later passage from bounded classes to the full sharp-local Hilbert space is not used here; it is isolated separately in Corollary F.215 and Lemma F.216. The final sentence records that the bounded base state is the restriction of this OS-positive sharp-local state, so its positive-time semigroup is canonically inherited from the Lane A reconstruction and no extra continuum-time lattice theorem is needed. $\square$

Dependency summary.

Inputs. Theorem F.211, Corollary 5.75, Theorem 5.74(b), and Corollary 5.76

Imported standard background. the standard OS reconstruction theorem for reflection-positive states on a positive-time algebra, applied only after the Lane A seam package has produced the full sharp-local state

Conclusion at this node. the bounded base state is identified as the $\mathcal{A}_+$ restriction of the full sharp-local state, and its dyadic matrix elements extend to the canonical continuum-time semigroup for all $t \geq 0$; the theorem does not yet perform the density/graph-core or gap transfer

Used later in. Corollary F.215, Lemma F.216, and Theorem F.219

Remark F.213 (Phase-II seam note for Theorem F.212). Corollary 5.75 is the bounded-state compatibility theorem only. The finite-cap sharp-local OS structure used above is Theorem 5.74(b), the passage to the full inductive union is Corollary 5.76, and the later density/graph-core handoff is isolated separately in Corollary F.215 and Lemma F.216.

Remark F.214 (Phase-IV dyadic-to-all-time route note). Theorem F.211 is only the bounded dyadic OS-limit theorem on $\mathcal{A}_+$. Theorem F.212 is the only public-spine step that upgrades those dyadic matrix elements to all $t \geq 0$, and it does so only after Theorem 5.74(b) and Corollary 5.76 have produced the sharp-local OS semigroup. No direct all-time lattice theorem is used or needed.

Corollary F.215 (The bounded base algebra is dense in the full sharp-local vacuum sector). *In the setting of Theorem F.212, let $(\mathcal{H}, \Omega, H)$ be the OS reconstruction of the full sharp-local state. Then the image of $\mathcal{A}_+$ is dense in $\mathcal{H}$, and*

$$\overline{\text{span}}\{[F - \omega^{(u_{\text{ren}})}(F)] : F \in \mathcal{A}_+\} = \Omega^\perp.$$

In particular, the non-collapse argument may be carried out on bounded positive-time observables and then extended by density to the full sharp-local Hilbert space.

Proof. Theorem 5.77 proves that, for the full sharp-local OS reconstruction, the image of the bounded positive-time algebra is dense and its centered classes span the vacuum-orthogonal sector. Theorem F.212 singles out exactly the same full sharp-local state as the one whose restriction to $\mathcal{A}_+$ is the tuned bounded base state. Rewriting the conclusions of Theorem 5.77 in the notation of the present theorem gives the stated density and centered-span claims. $\square$

Lemma F.216 (Hilbert/graph-core approximation package used in QE3). *In the setting of Theorem F.212, two approximation facts are used in the final gap transfer.*

- (i) *The bounded positive-time algebra $\mathcal{A}_+$ is dense in $\mathcal{H}$, and its centered classes are dense in $\Omega^\perp$, by Corollary F.215.*
- (ii) *Whenever a finite family of renormalized sharp-local fields belongs to a fixed engineering cap Δ_{max} , the corresponding field-polynomial vectors are realized as limits of vectors from the common invariant core $\mathcal{D}_{\text{flow}}^{(\leq \Delta_{\text{max}})}$ of Lemma F.247; the same approximants converge not only in the OS Hilbert norm but also in every finite graph norm relevant to that cap.*

For each $t \geq 0$ one has $\|e^{-tH}\| \leq 1$, so any semigroup estimate established first on the bounded positive-time core extends by Hilbert-norm continuity to the capwise sharp-local limits from part (ii). Thus the QE3 passage from bounded observables to the full sharp-local Hilbert space uses only the explicit density/core package already isolated in Lane A and no hidden domain argument.

Proof. Part (i) is Corollary F.215, which already identifies the bounded positive-time algebra as dense in $\mathcal{H}$ and its centered classes as dense in $\Omega^\perp$. For part (ii), Lemma F.247 proves on each fixed engineering cap that the field-polynomial vectors lie in the closure of the common invariant core $\mathcal{D}_{\text{flow}}^{(\leq \Delta_{\text{max}})}$ in every finite graph norm relevant to that cap. Corollary 5.76 transports those finite-cap realizations compatibly through the inductive union, so the same approximants may be viewed inside the full sharp-local reconstruction. Finally, because e^{-tH} is a contraction on the OS Hilbert space for each $t \geq 0$, any semigroup estimate first proved on the bounded positive-time core extends by Hilbert-norm continuity to the capwise sharp-local limit vectors. Thus the QE3 passage uses exactly the explicit density and graph-core package stated here and no hidden domain-completion step. $\square$

Remark F.217 (Phase-IV finiteness note for Lemma F.216). Every sharp-local vector later inserted into the QE3 transfer first lies in one fixed engineering cap, and only the finite list of graph norms attached to that cap is used. The lemma therefore does not invoke a graph-norm completion uniform over the full inductive union; the full-union step enters only through Corollary F.215 and the boundedness of the semigroup.

Dependency summary.

Inputs. the tuned full fixed-lattice gap input, Theorem III.18, Theorem F.211, Theorem F.212, and Lemma F.216

Imported standard background. the spectral theorem only, used through the local semigroup and Laplace-transform lemmas isolated in this appendix

Conclusion at this node. the fixed-lattice lower bound transfers to the full sharp-local OS Hamiltonian, yielding $\ker(H) = \mathbb{C}\Omega$ and $\text{Spec}(H) \subset \{0\} \cup [m_*, \infty)$; the theorem is the gap-transfer node itself, not merely a later reformulation

Used later in. Corollary F.5, Theorem M.1 via Corollary M.2, and the main-theorem assembly

Primary non-collapse route from the full fixed-lattice gap. The next theorem compresses the remaining Route 1 continuum transfer into one theorem-level statement. Its inputs are: the tuned-sequence lower bound on the physical full fixed-lattice OS gap; the actual Wilson bounded-state and dyadic reflected limits recorded in Theorem F.211 (whose Wilson character is supplied upstream by the same-scale transport theorem III.18); the continuum-time OS upgrade of Theorem F.212; and the explicit Hilbert/graph-core approximation package just recorded in Lemma F.216. The dense-core Route 1/Lane B package retained later is therefore a downstream reformulation rather than an additional closure interface.

Remark F.218 (Packet 3/8 transport certification ledger). On the frozen Route 1 spine, the fixed-lattice-to-continuum gap route is

$$\begin{aligned}
&\text{Corollary N.21} \implies \text{Proposition F.3} \implies \text{Proposition F.4} \\
&\implies \text{Theorem III.18} \implies \text{Theorem F.211} \\
&\implies \text{Theorem F.212} \implies \text{Theorem F.219} \implies \text{Corollary F.5.}
\end{aligned}$$

The only live weak-window input on this chain is the repaired curvature-sector package certified by Theorem F.301; after that certificate, no edge-Hessian placeholder or auxiliary weak-window branch re-enters the gap proof. The fixed-lattice gap entering the transport step is the tuned-sequence full OS gap package Proposition F.4, whose theorem-level closure is Theorem F.311; Theorem III.18 is the same-scale Wilson-to-patched return, and Theorem F.211 is the bounded Wilson/dyadic OS limit theorem. The first Lane A theorem-level compatibility input actually consumed by this transport route is Corollary 5.75. In Theorem F.212 it supplies only the identification of the sharp-local state with the bounded base state on $\mathcal{A}_+$ and the dyadic OS matrix-element identity. The finite-cap sharp-local OS structure used there is Theorem 5.74(b) on each finite cap, while Corollary 5.76 is only the later inductive-union passage. The later Hilbert/graph-core passage used by Theorem F.219 is exactly Corollary F.215 together with Lemma F.216. Lane B is not part of the proof of the gap: Theorem F.329 and Theorem 6.2 are downstream reformulations of the already established QE3 conclusion.

Theorem F.219 (Continuum gap transfer from the tuned full fixed-lattice Route 1 input).

Fix a target value $u_{\text{ren}} \in (0, u_]$ and let*

$$a_n := \ell_0 2^{-n}, \quad \beta_n := \beta(a_n; u_{\text{ren}})$$

be the tuned dyadic trajectory. Let $\omega^{(u_{\text{ren}})}$ be the full sharp-local state furnished by Corollary 5.76 over the same bounded base state, and let $(\mathcal{H}, \Omega, H)$ be its OS reconstruction on the full sharp-local positive-time algebra. Assume in addition that there exists $m_* > 0$ such that

$$\liminf_{n \rightarrow \infty} \frac{\Delta_{\text{lat}}(\beta_n)}{a_n} \geq m_* > 0.$$

Then the canonical continuum-time OS semigroup on the full sharp-local Hilbert space has spectral gap at least m_* , equivalently

$$\text{Spec}(H) \subset \{0\} \cup [m_*, \infty), \quad \ker(H) = \mathbb{C}\Omega.$$

The only topology used in the passage from bounded positive-time observables to sharp-local vectors is the Hilbert-norm / finite-graph-norm approximation package of Lemma F.216.

OS spectral-gap extraction from Euclidean decay (no hidden constants)

Lemma F.220 (OS semigroup representation of time-separated two-point functions). *Let $\langle \cdot \rangle_{\text{cont}, \ell}$ be an OS-positive continuum Schwinger functional on a translation-invariant algebra $\mathcal{A}$ with positive-time subalgebra $\mathcal{A}_+$ and reflection Θ . Let $(\mathcal{H}, \Omega, H)$ be the OS reconstruction: Ω is the vacuum, $H \geq 0$ the Hamiltonian, and τ_t denotes Euclidean time translation. Then for every $F \in \mathcal{A}_+$ and every $t \geq 0$,*

$$\langle F \Theta \tau_t F \rangle_{\text{cont}, \ell} = \langle [F], e^{-tH} [F] \rangle_{\mathcal{H}}, \quad (\text{F261})$$

where $[F]$ is the OS equivalence class of F in the reconstructed Hilbert space. Moreover, $\langle F \rangle_{\text{cont}, \ell} = \langle \Omega, [F] \rangle_{\mathcal{H}}$, so

$$[F_0] \perp \Omega \quad \text{for } F_0 := F - \langle F \rangle_{\text{cont}, \ell}. \quad (\text{F262})$$

Proof. This is the standard OS reconstruction identity. The pre-Hilbert space is $\mathcal{A}_+/\mathcal{N}$ where $\mathcal{N}$ is the OS null ideal, with inner product $(F, G) := \langle F \Theta G \rangle_{\text{cont}, \ell}$. Time translations act by $T(t)[F] = [\tau_t F]$ and are well-defined and contractive by translation invariance and OS positivity. The OS axioms imply $\{T(t)\}_{t \geq 0}$ is a strongly continuous symmetric contraction semigroup on $\mathcal{H}$; hence $T(t) = e^{-tH}$ for a self-adjoint $H \geq 0$. Then

$$\langle [F], e^{-tH} [F] \rangle = ([F], T(t)[F]) = (F, \tau_t F) = \langle F \Theta \tau_t F \rangle,$$

giving (F261). Finally $\Omega = [1]$, so $\langle F \rangle = (F, 1) = \langle [F], \Omega \rangle$, and (F262) follows. $\square$

Lemma F.221 (Laplace-transform gap criterion). *Let $H \geq 0$ be self-adjoint on a Hilbert space $\mathcal{H}$ and let $\psi \in \mathcal{H}$. Assume there exist constants $m > 0$ and $C < \infty$ such that for all $t \geq 0$,*

$$\langle \psi, e^{-tH} \psi \rangle \leq C e^{-mt}. \quad (\text{F263})$$

Then the spectral measure μ_ψ of H associated to ψ satisfies

$$\mu_\psi([0, m)) = 0. \quad (\text{F264})$$

In particular, if $\psi \perp \ker(H)$ then ψ has spectral support contained in $[m, \infty)$.

Proof. By the spectral theorem there exists a finite positive measure μ_ψ on $[0, \infty)$ such that

$$\langle \psi, e^{-tH} \psi \rangle = \int_0^\infty e^{-t\lambda} d\mu_\psi(\lambda), \quad t \geq 0. \quad (\text{F265})$$

Fix $\varepsilon \in (0, m)$. Then for all $t \geq 0$,

$$\int_{[0, m-\varepsilon]} e^{-t\lambda} d\mu_\psi(\lambda) \geq e^{-t(m-\varepsilon)} \mu_\psi([0, m-\varepsilon]).$$

Combine this lower bound with (F263) and (F265) to obtain

$$\mu_\psi([0, m-\varepsilon]) \leq C e^{-\varepsilon t} \quad \forall t \geq 0.$$

Let $t \rightarrow \infty$ to get $\mu_\psi([0, m-\varepsilon]) = 0$. Since $\varepsilon \downarrow 0$ and $[0, m) = \bigcup_{n \geq 1} [0, m - 1/n]$, we obtain (F264). If $\psi \perp \ker(H)$ then $\mu_\psi(\{0\}) = 0$, hence the spectral support is contained in $[m, \infty)$. $\square$

Lemma F.222 (Centered positive-time classes are dense in the vacuum-orthogonal sector).

Let $(\mathcal{H}, \Omega)$ be the OS reconstruction of a reflection-positive state on $\mathcal{A}_+$. Then

$$\overline{\text{span}}\{[F - \omega(F)] : F \in \mathcal{A}_+\} = \Omega^\perp.$$

Proof. By construction of the OS Hilbert space, the span of classes $[F]$ with $F \in \mathcal{A}_+$ is dense in $\mathcal{H}$. For any such F one has

$$[F] = [F - \omega(F)] + \omega(F) \Omega,$$

so the span of centered classes together with Ω is dense in $\mathcal{H}$. Taking orthogonal projection to $\Omega^\perp$ gives the claim. $\square$

Proof of Theorem F.219. By Theorem F.212, the sharp-local state $\omega^{(u_{\text{ren}})}$ restricts to the bounded base state on $\mathcal{A}_+$ and provides the canonical continuum-time OS semigroup for that base state. Theorem F.211, whose Wilson character is supplied upstream by the same-scale transport closure in Theorem III.18, records the actual Wilson one-point limits and dyadic reflected OS matrix elements needed below. Fix $\varepsilon \in (0, m_*)$. By the hypothesis on the tuned dyadic sequence, there exists $n_\varepsilon < \infty$ such that

$$\frac{\Delta_{\text{lat}}(\beta_n)}{a_n} \geq m_* - \varepsilon \quad (n \geq n_\varepsilon).$$

Step 1 (fixed-lattice OS semigroup estimate on the dyadic time set). Fix a bounded gauge-invariant local observable $F \in \mathcal{A}_+$ and set

$$F_0 := F - \omega^{(u_{\text{ren}})}(F), \quad F_{0,n} := F - \omega_{a_n}(F).$$

For each n , let $(\mathcal{H}_n, \Omega_n, H_n^{\text{lat}})$ be the fixed-lattice OS reconstruction of the Wilson Gibbs state ω_{a_n} on the bounded gauge-invariant positive-time cylinder algebra, so that

$$\Delta_{\text{lat}}(\beta_n) = \inf(\text{Spec}(H_n^{\text{lat}}) \setminus \{0\}).$$

Let $s \in \mathbb{D}_{\text{dyad}}$. For every sufficiently large n one has $s/a_n \in \mathbb{N}$. Since $\Delta_{\text{lat}}(\beta_n)$ is the full fixed-lattice OS gap in lattice units, the spectral theorem gives

$$0 \leq \omega_{a_n}(\Theta F_{0,n} \tau_s F_{0,n}) = \langle [F_{0,n}], e^{-(s/a_n)H_n^{\text{lat}}} [F_{0,n}] \rangle \leq \omega_{a_n}(\Theta F_{0,n} F_{0,n}) e^{-\Delta_{\text{lat}}(\beta_n)s/a_n}.$$

For $n \geq n_\varepsilon$ this yields

$$\omega_{a_n}(\Theta F_{0,n} \tau_s F_{0,n}) \leq \omega_{a_n}(\Theta F_{0,n} F_{0,n}) e^{-(m_* - \varepsilon)s} \quad (s \in \mathbb{D}_{\text{dyad}}). \quad (\text{F266})$$

Step 2 (pass to the continuum on dyadic times). For each fixed dyadic s , Theorem F.211 gives convergence of the one-point functions and of the dyadic OS matrix element. Using Theorem F.212 to identify the limit with $\omega^{(u_{\text{ren}})}$, we may send $n \rightarrow \infty$ in (F266) and obtain

$$\omega^{(u_{\text{ren}})}(\Theta F_0 \tau_s F_0) \leq \omega^{(u_{\text{ren}})}(\Theta F_0 F_0) e^{-(m_* - \varepsilon)s} \quad (s \in \mathbb{D}_{\text{dyad}}). \quad (\text{F267})$$

Step 3 (extend from dyadic times to all $t \geq 0$). By Theorem F.212, for every $t \geq 0$,

$$\omega^{(u_{\text{ren}})}(\Theta F_0 \tau_t F_0) = \langle [F_0], e^{-tH} [F_0] \rangle_{\mathcal{H}},$$

and the right-hand side is continuous in t . Choose any dyadic sequence $s_k \rightarrow t$. Passing to the limit in (F267) gives

$$\langle [F_0], e^{-tH} [F_0] \rangle_{\mathcal{H}} \leq \|[F_0]\|_{\mathcal{H}}^2 e^{-(m_* - \varepsilon)t} \quad (t \geq 0). \quad (\text{F268})$$

Step 4 (spectral support on the bounded positive-time core). Apply Lemma F.221 to (F268) with $\psi = [F_0]$, $C = \|[F_0]\|^2$, and $m = m_* - \varepsilon$. Since $[F_0] \perp \Omega \subset \ker(H)$, we conclude that $[F_0]$ has spectral support contained in $[m_* - \varepsilon, \infty)$. Because $\varepsilon \downarrow 0$ is arbitrary, every centered class $[F_0]$ has spectral support contained in $[m_*, \infty)$.

Step 5 (closure on the full sharp-local Hilbert space). By Lemma F.216(i), centered bounded positive-time classes are dense in $\Omega^\perp$. Hence

$$P_{(0, m_*)} \Big|_{\Omega^\perp} = 0.$$

Moreover, letting $t \rightarrow \infty$ in (F268) gives

$$\|P_{\{0\}}[F_0]\|^2 = \lim_{t \rightarrow \infty} \langle [F_0], e^{-tH}[F_0] \rangle = 0,$$

so $P_{\{0\}}[F_0] = 0$ for every centered class. By density and boundedness of $P_{\{0\}}$, this implies $P_{\{0\}} \Big|_{\Omega^\perp} = 0$. Thus

$$\ker(H) = \mathbb{C}\Omega, \quad \text{Spec}(H) \subset \{0\} \cup [m_*, \infty).$$

If one wants the same conclusion on vectors created by sharp-local field polynomials in a fixed engineering cap rather than only on bounded cylinder classes, Lemma F.216(ii) gives the required Hilbert/graph-norm approximants and the boundedness of e^{-tH} passes the same semigroup estimate to those capwise limits. This is the claimed continuum gap transfer. $\square$

Remark F.223 (Continuum-scaling inputs). The front-facing non-collapse route now uses three continuum-scaling ingredients along the chosen tuned dyadic trajectory: (i) selecting and controlling the tuned sequence $a_n = \ell_0 2^{-n}$ and $\beta_n = \beta(a_n; u_{\text{ren}})$; (ii) dyadic bounded-state convergence together with the continuum-time OS transport furnished by Theorems F.211 and F.212; (iii) non-collapse of the physical mass scale, i.e. a quantitative lower bound on $\Delta_{\text{lat}}(\beta_n)/a_n$. These ingredients are now all closed on the primary Route 1 spine, and Theorem F.219 converts them into the continuum spectral gap.

OS/Wightman packaging pointer The formal translation from the Euclidean OS theory to the Minkowski/Wightman formulation is no longer proved here inline. Instead, Appendix M collects the existence dossier in one place: it packages the Euclidean axioms already established by Lane A, the continuum-time OS reconstruction used in QE3, the positive-time clustering/vacuum-uniqueness consequences of the Route 1 gap, the standard Osterwalder–Schrader analytic continuation to a Poincaré-covariant Wightman/Haag–Kastler

theory, and the field correspondence for the gauge-invariant local differential polynomials represented by the sharp-local algebra. The Minkowski mass-gap corollary used for the final theorem is therefore cited from Appendix M.

d. Step-scaling approach to the physical gap: a concrete plan

The non-collapse condition in Theorem F.219 is best attacked by binding the RG entrance scale (to the KP basin proved above) to a *renormalized, scheme-defined physical length*. A standard and analytically well-behaved choice is the gradient-flow coupling.

A finite-volume coupling and step scaling Fix a finite four-torus $\mathbb{T}_L^4$ of linear size L (in lattice units) with periodic boundary conditions. Let $B_\mu(t, x)$ be the Wilson/gradient flow field at flow time t started from the lattice gauge field. Define the flowed action density $E(t, x)$ and the finite-volume gradient-flow coupling (one standard normalization)

$$g_{\text{GF}}^2(L) := \mathcal{N}^{-1} t^2 \langle E(t, 0) \rangle \Big|_{t=cL^2}, \quad (\text{F269})$$

with fixed $c \in (0, 1)$ and group-dependent constant $\mathcal{N}$ chosen so that $g_{\text{GF}}^2(L) = g_0^2 + O(g_0^4)$ in perturbation theory. Define the step-scaling function

$$\Sigma(u) := g_{\text{GF}}^2(2L) \text{ given } g_{\text{GF}}^2(L) = u. \quad (\text{F270})$$

Entrance to the KP basin as a large-coupling threshold The KP-basin entrance proved in this document is formulated as a deterministic *smallness* condition on a block-local activity norm *after* coarse graining to scale L . In physical terms this corresponds to the blocked single-plaquette law becoming sufficiently *spread* on G (close to Haar in the positive-type sector), which is an *infrared* effect. Accordingly, when the scale-dependent coupling is defined by a flowed energy density, KP admissibility should be triggered once the coupling has grown past a fixed threshold.

A standard way to bind this to (F269) is therefore to prove the one-way implication

$$g_{\text{GF}}^2(L) \geq u_0 \implies \text{KP admissibility holds for the blocked measure at scale } L, \quad (\text{F271})$$

for some universal threshold $u_0 = u_0(d, G, \theta, b)$. In the weak-to-intermediate coupling window, (F271) is expected to follow from a character-expansion bound that converts a lower bound on the local fluctuation scale (as measured by $g_{\text{GF}}^2(L)$) into an upper bound on the nontrivial Fourier–Peter–Weyl eigenvalues of the blocked plaquette density, hence a KP-norm bound.

Scale setting and tuned dyadic normalization. Define the lattice spacing via a flow scale by choosing $t_0 = t_0(\beta)$ through

$$t_0^2 \langle E(t_0, 0) \rangle = c_0 \quad (\text{F272})$$

with fixed c_0 , and set

$$a(\beta) := \frac{\ell_0}{\sqrt{t_0(\beta)}}. \quad (\text{F273})$$

For the final QE1 $\Rightarrow$ QE2 $\Rightarrow$ QE3 chain we do *not* use a separate first-threshold scale $L_*(\beta)$ or a fixed- β coupling-to-KP implication in the main appendix. The only weak-window ingredients retained here are the theorem-level step-scaling control proved below (Proposition F.228), the continuity statement of Lemma F.229, and the explicit bare-to-flow comparison of Theorem F.298, which are used later in the tuned dyadic recursion of Proposition F.314. The older fixed- β coupling-to-KP threshold and $L_*(\beta)/\sqrt{t_0(\beta)}$ comparison package has been moved to the project supplement and is not cited on the load-bearing path.

22. Step scaling and non-collapse: a weak-coupling assembled proof that closes the remaining wall

This section records the exact weak-window input retained on the load-bearing path: a two-sided cubic remainder estimate for the finite-volume step-scaling map together with the local coupling-identification statement used later in the tuned dyadic recursion. The older fixed- β threshold package has been moved to the supplement and is no longer cited in the main Appendix F proof spine.

a. *Finite-volume gradient-flow coupling and its step scaling*

Fix a physical flow scheme (e.g. Wilson flow) and define the finite-volume coupling

$$g_{\text{GF}}^2(L) := \mathcal{N}^{-1} t^2 \langle E(t) \rangle_{\Lambda(L)} \Big|_{t=cL^2}, \quad (\text{F274})$$

with $\Lambda(L)$ a periodic box of linear size L (in lattice units) and fixed $c \in (0, 1)$. Let $\Sigma(u)$ denote the step-scaling function at scale factor 2:

$$\Sigma(u) := g_{\text{GF}}^2(2L) \quad \text{when} \quad g_{\text{GF}}^2(L) = u. \quad (\text{F275})$$

b. Removing the remaining analytic hypothesis: localization and curvature-dominated Brascamp–Lieb

This subsection records the live weak-window package in the exact form used later. Historical discussion of the earlier edge-field formulation has been moved to the supplementary Route 1 archive. The only ingredients retained on the load-bearing path are small-curvature localization and curvature Brascamp–Lieb domination in the plaquette sector—exactly the sector seen by the flowed energy density. The downstream sufficiency certificate is stated later as Theorem F.301.

Spanning-tree gauge and exponential chart. Fix a spanning tree T and impose tree gauge. Parametrize free edges by exponential coordinates $U_e = \exp(iA_e)$ in a normal chart of radius $\varepsilon_0 > 0$. For a plaquette p write $F_p(A) \in \mathfrak{g}$ for the logarithm of the plaquette holonomy in this chart: $U_p = \exp(iF_p(A))$. Define the small-field event

$$\Omega_{\varepsilon_0} := \left\{ A : \|F_p(A)\| \leq \varepsilon_0 \text{ for all plaquettes } p \right\}. \quad (\text{F276})$$

Plaquette potential. Let

$$V(X) := 1 - \frac{1}{\dim} \Re \text{Tr } e^{iX}, \quad X \in \mathfrak{g}, \quad (\text{F277})$$

where $\dim$ is the dimension of the fixed Wilson representation ρ and the inner product on $\mathfrak{g}$ is fixed once and for all. Since G is compact and V is smooth with $\nabla^2 V(0) \succ 0$, there exists $\varepsilon_0 > 0$ and constants $0 < m_- \leq m_+ < \infty$ such that for all $\|X\| \leq \varepsilon_0$,

$$m_- I \preceq \nabla^2 V(X) \preceq m_+ I. \quad (\text{F278})$$

Lemma F.224 (Weak-coupling localization to the exponential chart). *Fix ε_0 as above. There exist constants $c_{\text{loc}}(G, \varepsilon_0) > 0$ and β_{loc} such that for all finite volumes Λ and all $\beta \geq \beta_{\text{loc}}$,*

$$\mu_{\beta, \Lambda}(\Omega_{\varepsilon_0}^c) \leq |P(\Lambda)| e^{-c_{\text{loc}} \beta}. \quad (\text{F279})$$

Proof. If $\|F_p(A)\| > \varepsilon_0$ for some plaquette p , then U_p lies outside a fixed neighborhood of the identity, and hence the plaquette energy satisfies $V(F_p(A)) \geq v_0(G, \varepsilon_0) > 0$. Since the Wilson density is proportional to $\exp\{-\beta \sum_q V(F_q(A))\}$, conditioning on the complement of any single plaquette event and integrating out the remaining variables yields a bound of the form $\mu_{\beta, \Lambda}(\|F_p\| > \varepsilon_0) \leq e^{-\beta v_0}$. Summing over $p \in P(\Lambda)$ gives (F279). $\square$

Curvature Brascamp–Lieb domination. On Ω_{ε_0} , the action is a sum of strictly convex plaquette potentials in the curvature variables $F_p(A)$. The needed uniform estimate is obtained directly in the curvature sector, because the map $A \mapsto dA$ isolates the plaquette modes seen by the flowed energy density.

Let d be the cochain differential mapping (tree-gauge) 1-forms (A_e) to 2-forms on plaquettes (linearized curvature); write d^* for its adjoint in the standard counting inner products on cochains.

Lemma F.225 (Projection identity for the linearized curvature covariance). *Let $\Delta_1 := d^*d$ on 1-forms. Then the operator*

$$\Pi_{\text{ex}} := d\Delta_1^{-1}d^* \quad (\text{F280})$$

is the orthogonal projection from 2-forms onto the exact subspace $\text{im}(d)$. In particular $\|\Pi_{\text{ex}}\|_{2 \rightarrow 2} = 1$.

Proof. On a finite complex, Δ_1 is invertible on tree-gauge 1-forms. The Moore–Penrose identity implies $d\Delta_1^{-1}d^*$ is the orthogonal projection onto $\text{im}(d)$. $\square$

Proposition F.226 (Curvature Brascamp–Lieb domination). *Let $\nu_{\beta,\Lambda}$ be the gauge-fixed Wilson measure in tree gauge on a finite volume Λ . On the event Ω_{ε_0} , for every finitely supported 2-form $J = (J_p)_{p \in P(\Lambda)}$ and every $s \in \mathbb{R}$,*

$$\log \mathbb{E}_{\nu_{\beta,\Lambda}} \left[\exp \left(s \langle J, dA \rangle \right) \mid \Omega_{\varepsilon_0} \right] \leq \frac{s^2}{2\beta m_-} \langle J, \Pi_{\text{ex}} J \rangle \leq \frac{s^2}{2\beta m_-} \|J\|_2^2. \quad (\text{F281})$$

Consequently, all centered linear curvature functionals are sub-Gaussian with variance proxy $(\beta m_-)^{-1}$, uniformly in Λ .

Proof. On Ω_{ε_0} the plaquette potential Hessian bounds (F278) imply that the full gauge-fixed action has (conditional) Hessian bounded below by $\beta m_- d^*d$ in the sense of quadratic forms on the image of d^* . Applying the Brascamp–Lieb inequality to the linear functional $A \mapsto \langle d^*J, A \rangle$ yields

$$\text{Var}(\langle d^*J, A \rangle \mid \Omega_{\varepsilon_0}) \leq \frac{1}{\beta m_-} \langle d^*J, \Delta_1^{-1}d^*J \rangle = \frac{1}{\beta m_-} \langle J, \Pi_{\text{ex}} J \rangle.$$

The sub-Gaussian mgf bound (F281) follows by the usual Herbst argument for convex measures. $\square$

From conditional to unconditional estimates. By Lemma F.224, the complement of Ω_{ε_0} is exponentially small in β . Thus any inequality proved conditionally on Ω_{ε_0} for bounded curvature observables may be upgraded to an unconditional inequality by paying an additive error $O(|P(\Lambda)|e^{-c\beta})$, which is negligible in the step-scaling and non-collapse regime where $\beta \rightarrow \infty$.

Uniform weak-window step-scaling remainder control The entrance/non-collapse package uses one quantitative UV input: a two-scale step-scaling estimate for the chosen finite-volume coupling $g_{\text{GF}}^2(L)$ at scale factor 2. In earlier roadmap form this estimate was recorded via an informal “Laplace/Wick expansion” description. Here we make the dependence on *curvature* Brascamp–Lieb domination explicit. For the mass-gap/non-collapse chain we only need that the quadratic coefficient in the weak-window step-scaling expansion is strictly positive; its closed form is immaterial.

Fixed constants. Fix once and for all: the chart radius ε_0 and convexity constants $m_{\pm}$ from (F278), the flow parameter $c \in (0, 1)$ and normalization $\mathcal{N}$ from (31), and the quadratic step-scaling coefficient $b_0 > 0$ from Lemma F.263 (positivity only; closed form not used below). Set the curvature-variance proxy

$$\sigma_{\beta}^2 := \frac{1}{\beta m_-}. \quad (\text{F282})$$

Lemma F.227 (Weak-window step scaling with a uniform $O(u^3)$ remainder). *Assume Lemma F.224 and Proposition F.226. There exist $\beta_1 \geq \beta_{\text{loc}}$, a weak-window threshold $u_{\text{AF}} > 0$, and a constant $C_{\Sigma} < \infty$, depending only on (d, G, c, ε_0) and the chosen discretization of $E(t)$, such that the following holds.*

For every $\beta \geq \beta_1$ and every L with

$$u := g_{\text{GF}}^2(L) \leq u_{\text{AF}},$$

the step-scaling function Σ from (F275) obeys the two-sided estimate

$$u + b_0 u^2 \log 2 - C_{\Sigma} u^3 \leq \Sigma(u) \leq u + b_0 u^2 \log 2 + C_{\Sigma} u^3. \quad (\text{F283})$$

Equivalently,

$$\Sigma(u) = u + b_0 u^2 \log 2 + R_{\Sigma}(u), \quad |R_{\Sigma}(u)| \leq C_{\Sigma} u^3.$$

All constants are uniform in L and in admissible boundary conditions (after tree-gauge reduction).

Proof. Step 1: curvature cumulant control from Brascamp–Lieb. On the small-curvature event Ω_{ε_0} , Proposition F.226 gives the sub-Gaussian mgf bound (F281) with variance proxy σ_β^2 from (F282). By differentiation of the mgf at the origin and the standard Leonov–Shiryaev cumulant formula, for every $k \geq 2$ and every finitely supported 2-forms $J_1, \dots, J_k$,

$$\left| \kappa_k(\langle J_1, dA \rangle, \dots, \langle J_k, dA \rangle \mid \Omega_{\varepsilon_0}) \right| \leq (k-1)! (\sigma_\beta^2)^{k/2} \prod_{i=1}^k \|J_i\|_2, \quad (\text{F284})$$

where κ_k denotes the joint cumulant.

Step 2: analytic curvature expansion of the flow observable. Fix L and set $t = cL^2$. In tree gauge, the Wilson flow map $A \mapsto V_t(A)$ and the clover energy density $E(t)$ are smooth gauge-covariant functionals of the initial links. Restricted to Ω_{ε_0} these functionals are real-analytic in the finitely many curvature variables $\{F_p(A)\}_{p \in P(\Lambda(L))}$ with an analyticity radius depending only on (G, ε_0) . Consequently $t^2 E(t)$ admits an absolutely convergent expansion on Ω_{ε_0} into curvature polynomials:

$$t^2 E(t) = \sum_{n \geq 1} \mathcal{P}_{2n}(dA), \quad (\text{F285})$$

where each $\mathcal{P}_{2n}$ is a homogeneous polynomial of degree $2n$ in the curvature field dA . Moreover, because the ratio $t/L^2 = c$ is fixed, the coefficients of $\mathcal{P}_{2n}$ are bounded uniformly in L .

Step 3: connected (cumulant) expansion and the $O(u^3)$ remainder. Take expectations in (F285) under the gauge-fixed Wilson measure. Each polynomial expectation can be written as a finite sum of joint cumulants of linear curvature functionals. The cumulant bound (F284), together with $\|\Pi_{\text{ex}}\|_{2 \rightarrow 2} = 1$ (Lemma F.225), implies that for u in a sufficiently small window $u \leq u_{\text{AF}}$ the series (F285) may be termwise integrated and is dominated by a convergent majorant of the form $\sum_{n \geq 1} C_E^n u^n$ with $C_E = C_E(d, G, c, \varepsilon_0)$. In particular, truncating (F285) at $n = 1, 2$ produces an error bounded by $C_\Sigma u^3$ uniformly in L .

Step 4: extraction of the quadratic coefficient. The degree-4 term $\mathcal{P}_4$ in (F285) yields the quadratic contribution to step scaling. The coefficient extraction is isolated below in Theorem F.262: after linearizing the flowed field strength, the connected degree-4 term is written as a positive Lie factor times the explicit squared-kernel shell integral (F309). In particular the coefficient multiplying u^2 at dyadic scale factor 2 is $b_0 \log 2$ with $b_0 > 0$.

Step 5: removal of the small-curvature conditioning. Lemma F.224 gives $\mu_{\beta, \Lambda(L)}(\Omega_{\varepsilon_0}^c) \leq |P(\Lambda(L))| e^{-c_{\text{loc}} \beta}$. On $\Omega_{\varepsilon_0}^c$ the Wilson density is suppressed by at least

$\exp(-\beta v_0)$ for some $v_0 = v_0(G, \varepsilon_0) > 0$, while $t^2 E(t)$ grows at most polynomially in the number of plaquettes for fixed discretization. Because $u \leq u_{\text{AF}}$ forces β into the weak window, one may increase β_1 (equivalently shrink u_{AF}) so that the contribution of $\Omega_{\varepsilon_0}^c$ is absorbed into the $C_\Sigma u^3$ remainder bound. $\square$

c. Step scaling bound (closed in the exact form used later)

Proposition F.228 (Weak-coupling step scaling with a two-sided cubic remainder). *Assume Lemma F.227. Then there exists $u_{\text{AF}} > 0$ and constants $b_0 > 0$, $C_\Sigma < \infty$ such that for all $u \in (0, u_{\text{AF}}]$,*

$$u + b_0 u^2 \log 2 - C_\Sigma u^3 \leq \Sigma(u) \leq u + b_0 u^2 \log 2 + C_\Sigma u^3. \quad (\text{F286})$$

After shrinking u_{AF} so that $C_\Sigma u_{\text{AF}} \leq \frac{1}{2} b_0 \log 2$, one has the forward-increment bounds

$$\frac{1}{2} b_0 u^2 \log 2 \leq \Sigma(u) - u \leq \frac{3}{2} b_0 u^2 \log 2 \quad (0 < u \leq u_{\text{AF}}). \quad (\text{F287})$$

Proof. The two-sided estimate is exactly (F283) from Lemma F.227. Once u_{AF} is chosen so that $C_\Sigma u_{\text{AF}} \leq \frac{1}{2} b_0 \log 2$, the cubic remainder is bounded by $C_\Sigma u^3 \leq \frac{1}{2} b_0 u^2 \log 2$ on $(0, u_{\text{AF}}]$, which yields (F287). $\square$

Lemma F.229 (Weak-window continuity of the step-scaling map). *On the weak window, the step-scaling map Σ is continuous.*

Proof. By Proposition F.286, the exact dyadic blocking map is analytic in the extracted coordinates (u, K) on the small polymer neighborhood. Restricting that analytic map to the invariant graph $K = h(u)$ from Theorem F.293 gives an analytic scalar update

$$F_h(u) := u'(u, h(u)).$$

By the coordinate choice of Proposition 4.12, this scalar update is exactly the weak-window step-scaling map Σ . Hence Σ is analytic, in particular continuous, on the weak window. $\square$

d. Archival fixed- β threshold package (moved to supplement)

The older fixed- β coupling-to-KP threshold and the associated comparison of the first threshold scale $L_*(\beta)$ with $\sqrt{t_0(\beta)}$ are no longer part of the main Appendix F proof spine.

That historical material has been moved to the supplementary Route 1 archive. The load-bearing Route 1 chain now uses only the one-shot entrance theorem at the explicit scale $B(\beta)$ together with the tuned dyadic physical-scale corollary.

23. Bounded macroscopic observables and upgrade to renormalized local fields

This section implements the passage from bounded macroscopic observables to renormalized local fields in a way compatible with reflection positivity. We first take a continuum limit on a separating algebra of bounded macroscopic observables $\mathcal{O}_\ell$ at fixed coarse scale ℓ , and then upgrade to renormalized point fields (operator-valued distributions) by a controlled small-flow-time/shrinking procedure based on the gauge-covariant gradient flow. The uniqueness/universality mechanism used below is contractivity of the exact RG in the KP basin together with the weak-window step-scaling control established earlier.

a. A continuum limit on a bounded observable algebra

The Wilson measure is reflection positive at finite lattice spacing, and bounded gauge-invariant observables automatically satisfy uniform moment bounds. This suggests a clean two-stage construction:

1. First construct the continuum limit on an algebra of *bounded macroscopic observables* (no renormalization needed);
2. Then upgrade to renormalized local fields by a controlled shrinking/small-flow-time limit.

Fix a physical coarse scale $\ell > 0$. For lattice spacing a and a physical point $x \in \mathbb{R}^4$, let $Q_{a,\ell}(x)$ denote the lattice cube of side ℓ/a centered at the nearest lattice point to x . Define the block-averaged plaquette energy density

$$E_{a,\ell}(x) := \frac{1}{|\mathcal{P}(Q_{a,\ell}(x))|} \sum_{p \subset Q_{a,\ell}(x)} \left(1 - \frac{1}{\dim} \Re \operatorname{Tr} U_p \right) \in [0, 2]. \quad (\text{F288})$$

Let $\mathcal{O}_\ell$ be the algebra generated by finite products of the fields $E_{a,\ell}(x)$ at finitely many points.

Lemma F.230 (Polynomial moments determine weak limits on compact cubes). *Let $K \subset \mathbb{R}^m$ be compact and let $(X_n)_{n \geq 1}$ be K -valued random vectors. Assume that for every polynomial*

P on $\mathbb{R}^m$ the limit $\lim_{n \rightarrow \infty} \mathbb{E}[P(X_n)]$ exists. Then the laws of X_n converge weakly to a unique probability measure on K .

Proof. Write $\mathcal{P}(K)$ for the algebra of restrictions to K of polynomials on $\mathbb{R}^m$. By the Stone–Weierstrass theorem, $\mathcal{P}(K)$ is uniformly dense in $C(K)$. Define a linear functional L on $\mathcal{P}(K)$ by $L(P) := \lim_{n \rightarrow \infty} \mathbb{E}[P(X_n)]$. If $P \geq 0$ on K , then $\mathbb{E}[P(X_n)] \geq 0$ for all n , hence $L(P) \geq 0$. Thus L is positive and bounded on $\mathcal{P}(K)$, and extends uniquely by density to a positive bounded linear functional (still denoted L) on $C(K)$.

By the Riesz representation theorem there exists a unique probability measure μ on K such that $L(f) = \int_K f d\mu$ for all $f \in C(K)$. Finally, for any $f \in C(K)$ and $\varepsilon > 0$, choose $P \in \mathcal{P}(K)$ with $\|f - P\|_\infty \leq \varepsilon$. Then for all n ,

$$\left| \mathbb{E}[f(X_n)] - L(f) \right| \leq \left| \mathbb{E}[(f - P)(X_n)] \right| + \left| \mathbb{E}[P(X_n)] - L(P) \right| + \left| L(P - f) \right| \leq 2\varepsilon + \left| \mathbb{E}[P(X_n)] - L(P) \right|.$$

Letting $n \rightarrow \infty$ and then $\varepsilon \downarrow 0$ gives $\mathbb{E}[f(X_n)] \rightarrow \int f d\mu$. This is exactly weak convergence of the laws of X_n to μ . $\square$

Lemma F.231 (Reflection positivity is closed under pointwise convergence of moments). *Let $\langle \cdot \rangle_n$ be expectation functionals on a $*$ -algebra $\mathcal{A}$ with a positive-time subalgebra $\mathcal{A}_+$ and a reflection Θ . Assume each $\langle \cdot \rangle_n$ is reflection positive: $\langle F \Theta F \rangle_n \geq 0$ for all $F \in \mathcal{A}_+$. Assume moreover that for every finite family $F_1, \dots, F_k \in \mathcal{A}_+$, the limits $\lim_{n \rightarrow \infty} \langle F_i \Theta F_j \rangle_n$ exist for all i, j . Then the pointwise limit functional $\langle \cdot \rangle_\infty$ is reflection positive on $\mathcal{A}_+$.*

Proof. Fix $F_1, \dots, F_k \in \mathcal{A}_+$ and define the $k \times k$ matrices $M_{ij}^{(n)} := \langle F_i \Theta F_j \rangle_n$. Reflection positivity of $\langle \cdot \rangle_n$ implies $M^{(n)} \succeq 0$ for all n . By assumption, $M^{(n)} \rightarrow M^{(\infty)}$ entrywise. The cone of positive semidefinite matrices is closed under entrywise limits, hence $M^{(\infty)} \succeq 0$. For any $c \in \mathbb{C}^k$ and $F := \sum_i c_i F_i$ we have

$$\langle F \Theta F \rangle_\infty = c^* M^{(\infty)} c \geq 0,$$

which is reflection positivity on the span of $\{F_i\}$. Since k and $\{F_i\}$ were arbitrary, $\langle \cdot \rangle_\infty$ is reflection positive on $\mathcal{A}_+$. $\square$

Theorem F.232 (Continuum limit on $\mathcal{O}_\ell$ without subsequences). *Fix $\ell > 0$ and a scaling trajectory $a(\beta) \downarrow 0$ as $\beta \rightarrow \infty$ in the fixed scheme (F272). Assume Theorem F.261. Then for every $m \in \mathbb{N}$ and every $x_1, \dots, x_m \in \mathbb{R}^4$, the joint law of*

$$(E_{a(\beta), \ell}(x_1), \dots, E_{a(\beta), \ell}(x_m)) \in [0, 2]^m$$

converges weakly as $\beta \rightarrow \infty$ without extracting subsequences. The limiting Schwinger functional on $\mathcal{O}_\ell$ satisfies reflection positivity and inherits the lattice symmetries.

Proof. Fix m and $x_1, \dots, x_m$. The corresponding random vector is valued in the compact cube $K = [0, 2]^m$. By Theorem F.261, expectations of *all* polynomial observables in these variables converge as $\beta \rightarrow \infty$. Therefore Lemma F.230 yields weak convergence of the joint laws.

For reflection positivity, fix $F_1, \dots, F_k$ in the finitely generated positive-time algebra built from $\mathcal{O}_\ell$. At each β , the Wilson measure satisfies reflection positivity, hence the matrices $M_{ij}^{(\beta)} := \langle F_i \Theta F_j \rangle_{\mu_\beta}$ are positive semidefinite. Since each entry is an expectation of a bounded observable in $\mathcal{O}_\ell$, it has a limit as $\beta \rightarrow \infty$ by Theorem F.261. Lemma F.231 now implies reflection positivity of the limit functional. Lattice translation and hypercubic rotation symmetries are exact at each β (in infinite volume or on the torus), hence are inherited by the limit by the same pointwise-limit argument. $\square$

b. Upgrading to renormalized point fields via gradient flow

The bounded macroscopic algebra $\mathcal{O}_\ell$ provides an OS-positive continuum limit on a separating bounded class and suffices to transfer a positive *mass gap* once non-collapse is known. To state the main theorem on a renormalized local field algebra (operator-valued distributions), we now upgrade from macroscopic observables to renormalized point fields using the gauge-covariant gradient flow. The resulting dyadic Cauchy bounds are recorded in Proposition F.235 and completed in Theorem F.246.

Flow equation. We use the Wilson (gradient) flow on lattice links: given an initial configuration $U = (U_e)$, define flowed links $V_t(e)$ as the solution of

$$\partial_t V_t(e) = -\nabla_e S_W(V_t) V_t(e), \quad V_0(e) = U_e, \quad (\text{F289})$$

where S_W is the Wilson action and ∇_e is the left-invariant gradient on the link variable. The flow is gauge-covariant and smooth for all $t \geq 0$.

Energy density. Let $G_{t,\mu\nu}(x)$ be any standard local discretization of the flowed field strength (e.g. the clover definition) built from the flowed plaquettes. Define the flowed energy density

$$E(t, x) := \frac{1}{2} \text{Tr } G_{t,\mu\nu}(x) G_{t,\mu\nu}(x) \geq 0. \quad (\text{F290})$$

For each fixed $t > 0$, $E(t, x)$ is a bounded, gauge-invariant local functional of V_t and hence a bounded functional of the initial links U .

Definition F.233 (Smearing at fixed physical scale). Let $a = a(\beta)$ be the lattice spacing along the chosen scaling trajectory. For a test function $\phi \in C_c^\infty(\mathbb{R}^4)$ set

$$E_a(t, \phi) := a^4 \sum_{x \in a\mathbb{Z}^4} \phi(x) E_a(t/a^2, x/a), \quad (\text{F291})$$

where $E_a(\tau, \cdot)$ denotes the flowed energy density at lattice spacing a and lattice flow-time τ . We also write the centered variable $\tilde{E}_a(t, \phi) := E_a(t, \phi) - \langle E_a(t, \phi) \rangle$.

Remark F.234. For each fixed physical $t > 0$, the smearing radius is $\sqrt{t}$. In lattice units the flow-time is $\tau = t/a^2 \rightarrow \infty$ as $a \downarrow 0$, so $E_a(t, \phi)$ is automatically a bounded *macroscopic* observable at scale comparable to $\sqrt{t}$. Consequently, the tightness/OS-positivity argument of Theorem F.232 applies verbatim to the family $\{E_a(t, \phi) : \phi \in C_c^\infty\}$ for each fixed $t > 0$.

Remaining UV ingredient: a uniform small-flow-time renormalization bound The non-trivial content is the $t \downarrow 0$ limit. As in standard SFTE/OPE analyses, after subtracting the vacuum piece one expects the leading local operator to be the renormalized energy density $\text{Tr } F_{\mu\nu}^2$ (dimension 4), with higher-dimension operators suppressed by powers of t . The correct Hilbert-space formulation is: the renormalized field exists as an *operator-valued distribution*, meaning that for fixed test functions ϕ the smeared variables converge in L^2 after a multiplicative renormalization.

Proposition F.235 (Dyadic Cauchy estimate for a renormalized $\text{Tr } F^2$ distribution). *Assume the weak-coupling curvature control (Lemma F.224 and Proposition F.226) and the step-scaling control of Proposition F.228. Then there exist:*

1. a positive function $Z_E(t)$ on $(0, t_*)$ with $Z_E(t) = 1 + O(g^2(1/\sqrt{t}))$ as $t \downarrow 0$;
2. constants $\eta > 0$ and, for each $\phi \in C_c^\infty(\mathbb{R}^4)$, a constant $C_\phi < \infty$;

with the property that for dyadic times $t_j := 2^{-2j}t_0$ (with $t_0 \leq t_*$), the centered smeared variables

$$X_j(\phi) := Z_E(t_j) \tilde{E}(t_j, \phi) \quad (\text{F292})$$

obey the summable increment bound

$$\|X_{j+1}(\phi) - X_j(\phi)\|_{L^2} \leq C_\phi (1+j)^{-1-\eta}. \quad (\text{F293})$$

Consequently $X_j(\phi)$ converges in L^2 as $j \rightarrow \infty$ to a limit, denoted

$$\mathcal{E}_{\text{ren}}(\phi) := \lim_{t \downarrow 0} Z_E(t) \left(E(t, \phi) - \langle E(t, \phi) \rangle \right), \quad (\text{F294})$$

which defines an OS field (operator-valued distribution) on the unique continuum limit on $\mathcal{O}_\ell$.

One-shell estimate: explicit dyadic control The increment bound (F293) has two inputs: (i) a *one-shell* L^2 estimate for the flowed-energy difference at scales t_j and t_{j+1} , and (ii) a weak-coupling bound showing that the running coupling along dyadic scales decays like $1/j$, so that the shell contributions are summable.

Lemma F.236 (Two-sided weak-window step scaling). *Assume Lemma F.227. Then there exist $u_{\text{AF}} > 0$ and constants $b_0 > 0, C_\Sigma > 0$ such that for all $u \in (0, u_{\text{AF}}]$,*

$$u + b_0 u^2 \log 2 - C_\Sigma u^3 \leq \Sigma(u) \leq u + b_0 u^2 \log 2 + C_\Sigma u^3. \quad (\text{F295})$$

Proof. Lemma F.227 gives the one-loop expansion for the defining observable of the gradient-flow coupling with a uniform remainder bound of order $O(u^3)$. Substituting into the definition of Σ yields $\Sigma(u) = u + b_0 u^2 \log 2 + R(u)$ with $|R(u)| \leq C_\Sigma u^3$ for u in the weak window, which is exactly (F295). $\square$

Lemma F.237 (Dyadic decay and square-summability of the weak-window coupling). *Fix $t_0 > 0$ and set $t_j := 2^{-2j} t_0$ and $L_j := c^{-1/2} \sqrt{t_j}$ (so that $t_j = c L_j^2$ in the flow-coupling definition). Let $u_j := g_{\text{GF}}^2(L_j)$, so that $u_{j-1} = \Sigma(u_j)$. Assume (F295) holds on $(0, u_{\text{AF}}]$ and that $u_0 \leq u_{\text{AF}}$. Then there exists $c_* > 0$ such that for all $j \geq 0$,*

$$u_j \leq \frac{c_*}{1+j}, \quad \text{and hence} \quad \sum_{j \geq 0} u_j^2 < \infty. \quad (\text{F296})$$

Proof. From $u_{j-1} = \Sigma(u_j)$ and (F295) we have, for u_j small,

$$u_{j-1} - u_j \geq b_0 u_j^2 \log 2 - C_\Sigma u_j^3.$$

Choose u_{AF} smaller if needed so that $C_\Sigma u_{\text{AF}} \leq \frac{1}{2} b_0 \log 2$. Then for all j with $u_j \leq u_{\text{AF}}$,

$$u_{j-1} - u_j \geq \frac{b_0 \log 2}{2} u_j^2.$$

Divide by $u_j u_{j-1}$ and use $u_{j-1} \geq u_j$ to obtain

$$\frac{1}{u_j} - \frac{1}{u_{j-1}} = \frac{u_{j-1} - u_j}{u_j u_{j-1}} \geq \frac{b_0 \log 2}{2}.$$

Summing from 1 to j yields $u_j^{-1} \geq u_0^{-1} + \frac{b_0 \log 2}{2} j$, i.e. (F296) with $c_* := 2/(b_0 \log 2)$ up to adjusting for u_0 . Square-summability follows immediately from $u_j \lesssim (1+j)^{-1}$. $\square$

Lemma F.238 (One-shell L^2 increment bound for the renormalized flowed energy). *Assume the curvature control (Lemma F.224 and Proposition F.226). Let $t_{j+1} = t_j/4$ and $u_j = g_{\text{GF}}^2(L_j)$ as in Lemma F.237. Then for each test function $\phi \in C_c^\infty(\mathbb{R}^4)$ there exist constants $C_\phi < \infty$ and j_ϕ such that for all $j \geq j_\phi$ one can choose $Z_E(t_j)$ (with $Z_E(t) = 1 + O(u_j)$) so that*

$$\|X_{j+1}(\phi) - X_j(\phi)\|_{L^2} \leq C_\phi u_j^2 + C_\phi e^{-c_{\text{loc}}\beta}. \quad (\text{F297})$$

Proof. Work in spanning-tree gauge and on the small-curvature event Ω_{ε_0} . On Ω_{ε_0} the Wilson action is uniformly strictly convex in the curvature sector (Section F.22b), and Proposition F.226 implies sub-Gaussian control for all linear curvature functionals with variance proxy $\sigma^2 \asymp u_j$ at physical scale $\sqrt{t_j}$. The flowed field at times t_j and t_{j+1} depends only on the initial field in a neighborhood of radius $O(\sqrt{t_j})$, and the map $U \mapsto E(t, \phi)$ is smooth on Ω_{ε_0} with derivatives bounded by constants depending on finitely many heat-kernel norms.

Represent the difference as a single “shell” contribution using the semigroup property and a finite-range decomposition of the heat kernel on 2-forms:

$$e^{-t\Delta_2} = \sum_{k \geq 0} K_{t,k}, \quad \text{supp } K_{t,k} \subset \{(x, y) : \text{dist}(x, y) \leq c_{\text{fr}} 2^k \sqrt{t}\}, \quad \|K_{t,k}\|_{\ell^1 \rightarrow \ell^1} \leq C_{\text{fr}} 2^{-Mk}, \quad (\text{F298})$$

for any fixed M (standard on $\mathbb{Z}^4$). Set $S_j := e^{-t_{j+1}\Delta_2} - e^{-t_j\Delta_2}$ and decompose $S_j = \sum_{k \geq 0} S_{j,k}$ with $S_{j,k} := K_{t_{j+1},k} - K_{t_j,k}$.

On Ω_{ε_0} one expands $E(t, \phi)$ in a local Taylor series in the curvature variables, with coefficients given by the kernels $K_{t,k}$ and their discrete derivatives. The t -increment $\tilde{E}(t_{j+1}, \phi) - \tilde{E}(t_j, \phi)$ is therefore a sum of finitely-supported curvature polynomials each containing at least one factor of the shell operator S_j . By Brascamp–Lieb (Proposition F.226) and standard polynomial moment bounds for sub-Gaussian fields (Hanson–Wright for quadratic terms, plus hypercontractive bounds for higher-degree remainders), each such local contribution has L^2 norm bounded by $C_\phi u_j^{r/2}$ where $r \geq 2$ is its curvature degree.

The renormalization factor $Z_E(t)$ is chosen to cancel the *order- u_j* part of the shell drift (the only logarithmically accumulating contribution), leaving a remainder of order u_j^2 . Concretely, choose $Z_E(t_j)$ so that the shell expectation of the leading (dimension-4) curvature polynomial is annihilated, equivalently: fix Z_E by imposing a scale-independent normalization of one nondegenerate two-point function $\langle X_j(\phi_*) X_j(\phi_*) \rangle$ for a fixed reference test function ϕ_* . The

curvature cumulant bounds on Ω_{ε_0} imply that the adjustment $Z_E(t_{j+1}) - Z_E(t_j)$ is $O(u_j)$ and that after this choice the remaining shell contribution to $X_{j+1} - X_j$ is $O(u_j^2)$ in L^2 .

The complement $\Omega_{\varepsilon_0}^c$ contributes at most $C_\phi \mu(\Omega_{\varepsilon_0}^c)^{1/2}$ to the L^2 norm, which is bounded by $C_\phi e^{-c_{\text{loc}}\beta}$ by Lemma F.224. Combining these bounds yields (F297). $\square$

Proof of Proposition F.235. By Lemma F.238 and Lemma F.237,

$$\|X_{j+1}(\phi) - X_j(\phi)\|_{L^2} \leq C_\phi u_j^2 + C_\phi e^{-c_{\text{loc}}\beta} \leq \tilde{C}_\phi (1+j)^{-2}$$

for all j large (absorbing the exponentially small tail into $\tilde{C}_\phi$). This has the form (F293) with $\eta = 1$. Summability implies $(X_j(\phi))_{j \geq 0}$ is Cauchy in L^2 , hence converges to an L^2 limit. $\square$

Remark F.239 (What is *actually* proved vs. what is bookkeeping). The analytic input is entirely local and uses only: (i) small-curvature localization; (ii) curvature Brascamp–Lieb domination; and (iii) a finite-range decomposition for the heat kernel (standard on $\mathbb{Z}^4$). The renormalization factor $Z_E(t)$ is fixed by requiring that $\mathcal{E}_{\text{ren}}$ have finite nonzero two-point function when paired with smooth test functions (equivalently: cancel the logarithmic divergence in the second moment). The precise identification of $\mathcal{E}_{\text{ren}}$ with $(\text{Tr } F^2)_{\text{ren}}$ is then a standard matching statement (using perturbation theory at small running coupling).

Field-algebra completion: a finite operator basis and mixing The single-field construction above extends to any *finite* family of gauge-invariant local fields of a fixed engineering dimension once one allows a *matrix*-valued renormalization to handle operator mixing. This is the standard constructive-QFT upgrade from one distinguished composite field to a finite operator basis.

Fix an integer $\Delta \geq 4$ and let $\{\mathcal{O}_1^{(\Delta)}, \dots, \mathcal{O}_m^{(\Delta)}\}$ be bounded gauge-invariant local lattice observables supported in a fixed unit cube (e.g. a Symanzik basis of local gauge-invariant operators of dimension Δ modulo total derivatives and lattice Bianchi relations). Let $\mathcal{O}(t, \phi) := (\mathcal{O}_1^{(\Delta)}(t, \phi), \dots, \mathcal{O}_m^{(\Delta)}(t, \phi))$ denote their flowed, smeared versions defined exactly as in Definition F.233, and let $\tilde{\mathcal{O}}(t, \phi)$ be the centered vector.

Proposition F.240 (Matrix dyadic Cauchy estimate for a renormalized operator multiplet). *Assume the weak-coupling curvature control (Lemma F.224 and Proposition F.226) and the step-scaling control of Proposition F.228. Then there exist:*

1. an invertible matrix function $Z^{(\Delta)}(t) \in \text{GL}_m(\mathbb{R})$ on $(0, t_*)$ with $Z^{(\Delta)}(t) = I + O(g^2(1/\sqrt{t}))$ as $t \downarrow 0$;

2. constants $\eta > 0$ and, for each $\phi \in C_c^\infty(\mathbb{R}^4)$, a constant $C_\phi < \infty$;

such that for dyadic times $t_j := 2^{-2j}t_0$ (with $t_0 \leq t_*$), the $\mathbb{R}^m$ -valued variables

$$X_j^{(\Delta)}(\phi) := Z^{(\Delta)}(t_j) \tilde{\mathcal{O}}(t_j, \phi)$$

obey the summable increment bound

$$\|X_{j+1}^{(\Delta)}(\phi) - X_j^{(\Delta)}(\phi)\|_{L^2(\mathbb{R}^m)} \leq C_\phi (1+j)^{-1-\eta}.$$

Consequently $X_j^{(\Delta)}(\phi)$ converges in L^2 as $j \rightarrow \infty$, defining an m -component OS field multiplet $\mathcal{O}_{\text{ren}}^{(\Delta)}(\phi)$ on the unique continuum limit on $\mathcal{O}_\ell$.

Remark F.241 (How mixing enters). The proof is identical to the scalar case (Proposition F.235). One applies the one-shell L^2 estimate componentwise (curvature domination controls all local curvature polynomials uniformly), and chooses the renormalization matrix $Z^{(\Delta)}(t_j)$ to cancel the *linear* (order- u_j) shell drift in the multiplet, equivalently by holding a fixed nondegenerate Gram matrix of two-point functions at a reference set of test functions constant across j . After this matrix counterterm is imposed the residual shell increment is $O(u_j^2)$ in operator norm, hence summable.

c. *Field-algebra completion across dimensions (inductive closure)*

The remaining ingredient is to extend the finite-basis construction (Proposition F.240) to a *generating* family of local gauge-invariant fields across engineering dimensions, and to upgrade universality/uniqueness from bounded macroscopic observables to this unbounded local field algebra.

Standard strategy (in standard terminology). Fix a graded local operator basis *dimension by dimension*. At each fixed dimension Δ , allow matrix-valued renormalization to handle operator mixing inside that dimension and with lower dimensions (triangular structure by power counting). Then prove a dyadic Cauchy estimate *uniformly for the whole finite truncation* $\Delta \leq \Delta_{\text{max}}$. Compatibility in Δ_{max} yields an inductive limit defining the renormalized local field algebra.

A graded operator basis

Definition F.242 (Local gauge-invariant operator spaces). Fix an integer $\Delta \geq 0$. Let $\mathcal{V}_\Delta$ be the real vector space spanned by gauge-invariant local lattice observables supported in a unit cube that admit a small-curvature expansion with leading engineering dimension Δ (built from $F_{\mu\nu}$ and covariant derivatives) and that are quotiented by lattice symmetries, total derivatives and lattice Bianchi identities.

Decompose $\mathcal{V}_\Delta$ into its exact-dimension symmetry blocks

$$\mathcal{V}_\Delta = \bigoplus_{(\epsilon, \lambda)} \mathcal{V}_{\Delta, \epsilon, \lambda},$$

where ϵ denotes CP parity and λ the Euclidean tensor type. Choose once and for all the canonical blockwise basis described in Appendix L; in particular, each block is written in a fixed complete-contraction basis and the blocks are ordered by increasing engineering dimension and then by a fixed symmetry-type order. For $\Delta_{\max} \geq 0$ define the truncated operator multiplet

$$\mathcal{O}^{(\leq \Delta_{\max})} := \left(\mathcal{O}_1^{(0)}, \dots, \mathcal{O}_{m_0}^{(0)}; \mathcal{O}_1^{(1)}, \dots; \dots; \mathcal{O}_1^{(\Delta_{\max})}, \dots, \mathcal{O}_{m_{\Delta_{\max}}}^{(\Delta_{\max})} \right),$$

an $M(\Delta_{\max})$ -vector with $M(\Delta_{\max}) = \sum_{\Delta \leq \Delta_{\max}} m_\Delta$.

Remark F.243 (Why a finite basis exists). For each fixed Δ there are only finitely many gauge-invariant monomials of dimension Δ modulo symmetries and identities. On the lattice, one may take a Symanzik-type basis built from small loops and covariant difference operators. Appendix L records the canonical blockwise choice used later in Lane A. The project uses only that each exact-dimension symmetry block is finite dimensional and that every canonical basis element becomes a bounded local observable after positive flow time and smearing.

Triangular mixing and coherent renormalization The gradient-flow smearing at flow time $t > 0$ makes each $\mathcal{O}_i^{(\Delta)}(t, \phi)$ bounded. As $t \downarrow 0$ the family develops UV divergences. Power counting implies that at fixed Δ the flowed observable admits a small-flow-time expansion into local fields of dimensions $\leq \Delta$; equivalently, in the natural ordering by increasing engineering dimension, the renormalization matrix in the graded basis is *block lower triangular*.

Definition F.244 (Graded triangular renormalization matrices). Fix $\Delta_{\max} \geq 0$. A renormalization matrix $Z^{(\leq \Delta_{\max})}(t) \in \text{GL}_{M(\Delta_{\max})}(\mathbb{R})$ is called *graded triangular* if, when written in blocks by engineering dimension,

$$Z^{(\leq \Delta_{\max})}(t) = \left(Z_{\Delta\Delta'}(t) \right)_{0 \leq \Delta, \Delta' \leq \Delta_{\max}},$$

it satisfies $Z_{\Delta\Delta'}(t) = 0$ whenever $\Delta' > \Delta$.

Remark F.245 (Normalization to fix Z). For each $\Delta_{\max}$ the graded triangular matrix is fixed by imposing a finite set of scale-independent normalization conditions on a nondegenerate Gram matrix of two-point functions for a reference set of test functions. This is the same mechanism used in Proposition F.240, but now enforced simultaneously across all dimensions up to $\Delta_{\max}$.

Uniform dyadic Cauchy estimates for finite truncations

Theorem F.246 (Dyadic construction of the renormalized local field algebra up to $\Delta_{\max}$).

Assume:

1. *weak-coupling curvature control* (Lemma F.224 and Proposition F.226);
2. *the theorem-level weak-window package already proved* (Proposition F.228, Lemma F.229, Theorem F.298, and Proposition F.314).

Fix $\Delta_{\max} \geq 0$ and use the canonical blockwise basis of Definition F.242 and Appendix L.

Then there exist:

1. a graded triangular matrix function $Z^{(\leq \Delta_{\max})}(t)$ on $(0, t_*)$;
2. for each test function $\phi \in C_c^\infty(\mathbb{R}^4)$ constants $C_{\phi, \Delta_{\max}} < \infty$ and $\eta > 0$;

such that for dyadic times $t_j = 2^{-2j}t_0$ (with $t_0 \leq t_*$) the renormalized multiplet

$$X_j^{(\leq \Delta_{\max})}(\phi) := Z^{(\leq \Delta_{\max})}(t_j) \tilde{\mathcal{O}}^{(\leq \Delta_{\max})}(t_j, \phi)$$

obeys the summable increment bound

$$\|X_{j+1}^{(\leq \Delta_{\max})}(\phi) - X_j^{(\leq \Delta_{\max})}(\phi)\|_{L^2(\mathbb{R}^{M(\Delta_{\max})})} \leq C_{\phi, \Delta_{\max}} (1+j)^{-1-\eta}.$$

Consequently $X_j^{(\leq \Delta_{\max})}(\phi)$ converges in L^2 as $j \rightarrow \infty$ for each ϕ , defining an OS field multiplet $\mathcal{O}_{\text{ren}}^{(\leq \Delta_{\max})}(\phi)$.

Moreover, if $\mathcal{A}_{+, \text{flow}}^{(\leq \Delta_{\max})}$ denotes the bounded positive-time algebra generated by the canonical flowed lifts in the cap and

$$\mathcal{D}_{\text{flow}}^{(\leq \Delta_{\max})} := \text{span}\{[F] : F \in \mathcal{A}_{+, \text{flow}}^{(\leq \Delta_{\max})}\},$$

then each smeared component $\mathcal{O}_{\text{ren}, i}^{(\leq \Delta_{\max})}(\phi)$ is obtained on $\mathcal{D}_{\text{flow}}^{(\leq \Delta_{\max})}$ as the graph-norm limit of bounded dyadic approximants, and the polynomial $*$ -algebra generated by the components is well defined on this explicit common dense invariant core (see Lemma F.247).

Proof. The proof is the finite-cap version of Proposition F.240, with the earlier basis-by-basis placeholder replaced by the canonical block transport of Corollary 5.40.

(i) *Whole-cap canonical one-shell transport.* Corollary 5.40 provides a block-lower-triangular one-shell update

$$\mathbf{Y}_{j+1}^{(\leq \Delta_{\max})}(\phi) = \left(I + u_j A_j^{(\leq \Delta_{\max})}\right) \mathbf{Y}_j^{(\leq \Delta_{\max})}(\phi) + \mathbf{Q}_j^{(\leq \Delta_{\max})}(\phi),$$

with $\|\mathbf{Q}_j^{(\leq \Delta_{\max})}(\phi)\|_{L^2} \leq C_{\phi, \Delta_{\max}} u_j^2$. This is already basis-free at the block level: Theorem L.8 identifies the exact-dimension quotient intrinsically before any basis is chosen, and the lower-block terms are then forced by power counting in the grading order.

(ii) *Graded triangular renormalization.* Impose the same scale-independent normalization moments used in Proposition F.240, but now recursively in the grading order of the canonical blocks. Because the one-shell transport is block lower triangular, each diagonal block is normalized after the lower blocks have already been fixed, producing a graded triangular matrix $Z^{(\leq \Delta_{\max})}(t_j)$. The order- u_j shell drift is removed block-by-block, leaving only the $O(u_j^2)$ remainder from the canonical one-shell theorem. Compatibility under cap enlargement is ensured by Lemma L.10 and later recorded abstractly in Lemma F.270.

(iii) *Summability.* The step-scaling decay (Lemma F.237) gives $\sum_j u_j^2 < \infty$ in the AF window. Hence the renormalized shell increments are summable in L^2 , yielding the stated dyadic Cauchy property.

(iv) *Algebra/domain statement.* The explicit common-core argument is isolated in Lemma F.247. In brief, the dyadic approximants $X_j^{(\leq \Delta_{\max})}(\phi)$ are bounded flowed observables and therefore preserve the dense span of bounded flowed OS-classes. Theorem 5.45 converts the finite-cap insertion estimates to OS-norm bounds against any fixed bounded flowed vector, and the summable increment bound from part (iii) then gives Cauchy convergence in the graph norms relevant to any fixed finite field polynomial. This yields a common dense invariant core for all finite-cap field polynomials. $\square$

Lemma F.247 (Explicit finite-cap common invariant core). *Fix $\Delta_{\max} \geq 0$ and let $\mathcal{A}_{+, \text{flow}}^{(\leq \Delta_{\max})}$ and $\mathcal{D}_{\text{flow}}^{(\leq \Delta_{\max})}$ be as in Theorem F.246. Then:*

- (i) $\mathcal{D}_{\text{flow}}^{(\leq \Delta_{\max})}$ is dense in the finite-cap OS Hilbert space and is invariant under every bounded dyadic approximant $X_j^{(\leq \Delta_{\max})}(\phi)$.

(ii) For each canonical field component $\mathcal{O}_{\text{ren},i}^{(\leq \Delta_{\max})}(\phi)$ and each $F \in \mathcal{A}_{+, \text{flow}}^{(\leq \Delta_{\max})}$, the classes

$$\left[X_{j,i}^{(\leq \Delta_{\max})}(\phi) F \right]$$

form a Cauchy sequence in the OS Hilbert norm as $j \rightarrow \infty$; moreover the same estimate remains valid after adjoining any fixed finite family of additional field insertions, so the convergence is Cauchy in every finite graph norm relevant to the polynomial field algebra.

(iii) Consequently each $\mathcal{O}_{\text{ren},i}^{(\leq \Delta_{\max})}(\phi)$ is closable, $\mathcal{D}_{\text{flow}}^{(\leq \Delta_{\max})}$ is a common invariant core for the closures, and every finite polynomial in the smeared fields is well defined on that core.

Proof. Density is by construction: before the $j \rightarrow \infty$ limit is taken, the finite-cap OS Hilbert space is generated from bounded flowed positive-time observables. Each $X_j^{(\leq \Delta_{\max})}(\phi)$ is a linear combination of bounded flowed observables at positive flow time, so multiplying by an element of $\mathcal{A}_{+, \text{flow}}^{(\leq \Delta_{\max})}$ stays inside the same algebra; this proves invariance.

Now fix $F \in \mathcal{A}_{+, \text{flow}}^{(\leq \Delta_{\max})}$. Theorem 5.45 upgrades the finite-cap insertion estimates to OS norms against the fixed bounded flowed insertion family determined by F . Applying that upgrade to the dyadic increment bound of Theorem F.246 gives, for each component index i ,

$$\left\| \left[(X_{j+1,i}^{(\leq \Delta_{\max})}(\phi) - X_{j,i}^{(\leq \Delta_{\max})}(\phi)) F \right] \right\| \leq C_{F,\phi,\Delta_{\max}} (1+j)^{-1-\eta}.$$

The same estimate holds after adjoining any fixed finite family of additional dyadic approximants on the left, because those extra factors are again bounded flowed observables and are therefore covered by the same OS-seminorm control. Summability in j yields the asserted Cauchy property in the Hilbert norm and in every finite graph norm used by a fixed field polynomial. Hence the graph-norm limits defining the renormalized fields exist on $\mathcal{D}_{\text{flow}}^{(\leq \Delta_{\max})}$, preserve that core, and are closable there. $\square$

Corollary F.248 (Canonical finite-truncation SFTE and intrinsic remainder control). *Fix $\Delta_{\max} \geq 0$ and the canonical blockwise basis of Definition F.242. Then the canonical flowed lifts and the renormalized local fields produced by Theorem F.246 satisfy, on that finite truncation, the small-flow-time expansion of Lane A with a block-lower-triangular coefficient matrix*

$$\mathbf{Y}_t^{(\leq \Delta_{\max})}(f) = C^{(\leq \Delta_{\max})}(t; \mu) \mathbf{O}_{\text{ren}}^{(\leq \Delta_{\max})}(f; \mu) + \mathbf{R}_t^{(\leq \Delta_{\max})}(f; \mu).$$

Let $\mathcal{H}_{\text{cap}}^{(\leq \Delta_{\max})}$ be the finite-cap OS Hilbert space underlying Theorem F.246, let $\Omega_{\text{cap}} := [1]$ be its vacuum vector, let $\mathcal{D}_{\text{flow}}^{(\leq \Delta_{\max})}$ be the common invariant core of Lemma F.247, and write

$$\mathcal{B}_{+, \text{cap}}^{(\leq \Delta_{\max})} := *(\mathcal{A}_{+, \text{flow}}^{(\leq \Delta_{\max})}, \mathcal{A}_+)$$

for the bounded positive-time capwise algebra generated by the capwise flowed lifts and the bounded positive-time Wilson observables. Then there exists $\eta_{\Delta_{\max}} > 0$ such that for every component index i , every fixed finite insertion family $\{\mathcal{X}_\ell\}$ from the same truncation, and every $\Psi, \Phi \in \mathcal{D}_{\text{flow}}^{(\leq \Delta_{\max})}$,

$$\left| \left\langle \Psi, \left(\mathcal{R}_i^{(\leq \Delta_{\max})}(t; f; \mu) - r_i^{(\leq \Delta_{\max})}(t; f; \mu) \right) \prod_{\ell=1}^k \mathcal{X}_\ell \Phi \right\rangle_{\text{cap}} \right| \leq C_{\Delta_{\max}, f, \Psi, \Phi, \mathcal{X}, \mu} t^{\eta_{\Delta_{\max}}}$$

for all sufficiently small t , where

$$r_i^{(\leq \Delta_{\max})}(t; f; \mu) := \left\langle \Omega_{\text{cap}}, \mathcal{R}_i^{(\leq \Delta_{\max})}(t; f; \mu) \Omega_{\text{cap}} \right\rangle_{\text{cap}}.$$

In particular, taking $\Psi = \Phi = \Omega_{\text{cap}}$ yields the capwise vacuum remainder bound, and the remainder/inverse compatibility condition (157) holds on every finite truncation against any polylogarithmic inverse bound. Equivalently, for every fixed capwise vacuum channel, the centered inverse-flow approximants are Cauchy as $t \downarrow 0$.

Moreover, for every fixed positive-time capwise word

$$W_+ = B_0 \hat{O}_{j_1}^{(\leq \Delta_{\max})}(f_1; \mu) B_1 \cdots \hat{O}_{j_m}^{(\leq \Delta_{\max})}(f_m; \mu) B_m,$$

$$B_r \in \mathcal{B}_{+, \text{cap}}^{(\leq \Delta_{\max})}, \quad \text{supp}(f_r) \subset \{x_0 > 0\},$$

let

$$W_+(\mathbf{t}) := B_0 \mathcal{O}_{j_1}^{\text{app}, (\leq \Delta_{\max})}(t_1; f_1; \mu) B_1 \cdots$$

$$\mathcal{O}_{j_m}^{\text{app}, (\leq \Delta_{\max})}(t_m; f_m; \mu) B_m.$$

Then the compatible finite-observable-set limits packaged one finite family at a time by Corollary F.330A, and later abbreviated locally in Theorem F.334 by $\omega_{\text{cap}, \text{bnd}}^{(u_{\text{ren}})}$, satisfy

$$\omega_{\text{cap}, \text{bnd}}^{(u_{\text{ren}})}(W_+(\mathbf{t}) - W_+(\mathbf{s})) \xrightarrow{\mathbf{t}, \mathbf{s} \downarrow 0} 0.$$

More precisely, Lemma F.245B expands each literal telescoping summand of $W_+(\mathbf{t}) - W_+(\mathbf{s})$ into finitely many bounded-word remainder channels with coefficient envelopes governed by Theorem 5.69; the resulting coefficient envelope is the polylogarithmic one displayed in Lemma F.245B, and Corollary F.248 supplies the finite-cap remainder / inverse compatibility used there. Hence the mixed-channel limit exists for every such W_+ and is independent of the order or relative rates with which the coordinates of $\mathbf{t}$ tend to 0.

Proof. The coefficient matrix is the finite-cap version of the extracted coefficient matrix from Theorem 5.31, written in the canonical block basis and then normalized by Theorem F.246; lower-block mixing is exactly the graded triangular structure already built into both results. The extracted-irrelevant remainder obeys the power bound (115) in insertion seminorm, and Theorem 5.45 upgrades that control to OS matrix elements against any fixed bounded flowed insertion family. Lemma F.247 identifies $\mathcal{D}_{\text{flow}}^{(\leq \Delta_{\max})}$ as the graph-norm closure of such bounded flowed vectors, so the same estimate extends to every pair $\Psi, \Phi \in \mathcal{D}_{\text{flow}}^{(\leq \Delta_{\max})}$ after adjoining any fixed finite capwise insertion family on the right. Because the truncation is finite, one may choose a single exponent $\eta_{\Delta_{\max}}$ and absorb the finitely many vectors and insertions into the constant. Any polylogarithmic inverse factor is dominated by $t^{-\varepsilon}$ for arbitrarily small $\varepsilon > 0$, so the $t^{\eta_{\Delta_{\max}}}$ remainder is compatible with the inverse bounds used later in Lane A. Finally, if $t, t' \leq t_0$, subtracting the SFTE identities at scales t and t' and using cancellation of the common renormalized local term shows that the difference of the centered inverse-flow approximants is a linear combination of the centered remainders with coefficients controlled by Theorem 5.69; the displayed bound therefore gives the announced Cauchy property in every fixed capwise vacuum channel.

For the additional positive-time mixed-channel clause, fix W_+ and two scale vectors $\mathbf{t}, \mathbf{s}$, and write

$$W_+(\mathbf{t}) - W_+(\mathbf{s}) = \sum_{r=1}^m T_r(\mathbf{t}, \mathbf{s})$$

for the multilinear telescoping decomposition. The literal telescoping summands $T_r(\mathbf{t}, \mathbf{s})$ are not fixed-word channels: every non-difference local slot is itself an inverse-flow approximant at one of the scales from $\mathbf{t}$ or $\mathbf{s}$. Lemma F.245B isolates the required reduction. Its displayed bound (F302) is the literal telescoping-summand estimate: (F301) supplies the coefficient envelope coming from Theorem 5.69, and Proposition F.245A, obtained from Corollary 5.45A and hence from Theorem 5.45, supplies the factor $t_r^{\eta_{\Delta_{\max}}} + s_r^{\eta_{\Delta_{\max}}}$ after the r th slot is converted into a centered remainder difference. Equation (F303) is the corresponding slotwise $o(1)$ handoff furnished by the finite-cap remainder / inverse compatibility clause of Corollary F.248 applied to that polylogarithmic envelope. Hence each $\omega_{\text{cap, bnd}}^{(u_{\text{ren}})}(T_r(\mathbf{t}, \mathbf{s}))$ tends to 0 as $\mathbf{t}, \mathbf{s} \downarrow 0$. Summing over the finitely many slots r gives the announced mixed-channel Cauchy property and therefore the order/rate-independent limit, with the single-slot and telescoping inputs supplied by Proposition F.245A and Lemma F.245B. $\square$

Proposition F.245A (State-free positive-time bounded-factor insertion estimate). *Work in the setting and notation of Corollary F.248. Fix a component index i , a test function f with $\text{supp}(f) \subset \{x_0 > 0\}$, bounded factors*

$$B_0, \dots, B_q \in \mathcal{B}_{+, \text{cap}}^{(\leq \Delta_{\max})},$$

and a compact positive-time neighborhood $K_{B,f} \subset \{x_0 > 0\}$ containing $\text{supp}(f)$ together with the supports of every bounded cylinder / flowed factor appearing in the fixed word $B_0, \dots, B_q$. Then for every $\Psi, \Phi \in \mathcal{D}_{\text{flow}}^{(\leq \Delta_{\max})}$ one has

$$\begin{aligned} & \left| \left\langle \Psi, B_0 \left(\mathcal{R}_i^{(\leq \Delta_{\max})}(t; f; \mu) - r_i^{(\leq \Delta_{\max})}(t; f; \mu) \right) B_1 \cdots B_q \Phi \right\rangle_{\text{cap}} \right| \\ & \leq C_{\Psi, \Phi, K_{B,f}, i, \Delta_{\max}, \mu} \left(\prod_{r=0}^q \|B_r\|_{\infty} \right) t^{\eta_{\Delta_{\max}}}. \end{aligned}$$

for all sufficiently small t . After subtracting the centered finite-cap SFTE identities at scales t, s , the same fixed bounded word satisfies

$$\begin{aligned} & \left| \left\langle \Psi, B_0 \left(\mathcal{O}_i^{\text{app}, (\leq \Delta_{\max})}(t; f; \mu) - \mathcal{O}_i^{\text{app}, (\leq \Delta_{\max})}(s; f; \mu) \right) B_1 \cdots B_q \Phi \right\rangle_{\text{cap}} \right| \\ & \leq C'_{\Psi, \Phi, K_{B,f}, i, \Delta_{\max}, \mu} \left(\prod_{r=0}^q \|B_r\|_{\infty} \right) (t^{\eta_{\Delta_{\max}}} + s^{\eta_{\Delta_{\max}}}). \end{aligned}$$

for all sufficiently small t, s . In particular, taking $\Psi = \Phi = \Omega_{\text{cap}}$ gives the vacuum-channel estimate later specialized in the proof of Theorem F.334(i).

Proof. Set

$$F_t := \mathcal{R}_i^{(\leq \Delta_{\max})}(t; f; \mu) - r_i^{(\leq \Delta_{\max})}(t; f; \mu)$$

and let $J_{i,t}$ denote the matching-scale insertion activity whose summed observable is F_t . By the capwise remainder bound (115) at the matching scale,

$$\|J_{i,t}\|_{\text{ins}; \mu; k, K_f, \varepsilon} \leq C_{i, \Delta_{\max}, \mu} t^{\eta_{\Delta_{\max}}}, \quad (\text{F299})$$

where $K_f \subset K_{B,f}$ is the compact neighborhood attached to the test function support. Fix the bounded positive-time word

$$\mathbf{W} := (B_0, \dots, B_q)$$

and let $J_{i,t}^{\mathbf{W}}$ be the modified insertion activity from Corollary 5.45A. Its summed observable is

$$F_{J_{i,t}^{\mathbf{W}}} = B_0 F_t B_1 \cdots B_q.$$

Applying Corollary 5.45A at the finite-observable-set level to the fixed bounded word $\mathbf{W}$ — and only to that positive-time word — then passing through the compatible capwise limit gives

$$\begin{aligned} & \left\langle [F_{J_{i,t}^{\mathbf{W}}}], [F_{J_{i,t}^{\mathbf{W}}}] \right\rangle_{\text{cap}}^{1/2} \\ & \leq C_{\text{OS} \leftarrow \text{ins}} C_{\text{supp}}(K_{B,f}) C_{\text{word}}(K_{B,f}; \mathbf{W}^\#) \left(\prod_{r=0}^q \|B_r\|_\infty \right) \|J_{i,t}\|_{\text{ins}; \mu; k, K_f, \varepsilon}. \end{aligned}$$

Using (F299),

$$\left\langle [F_{J_{i,t}^{\mathbf{W}}}], [F_{J_{i,t}^{\mathbf{W}}}] \right\rangle_{\text{cap}}^{1/2} \leq C_{K_{B,f}, i, \Delta_{\max}, \mu} \left(\prod_{r=0}^q \|B_r\|_\infty \right) t^{\eta_{\Delta_{\max}}} \quad (\text{F300})$$

for all sufficiently small t .

It now suffices to treat core vectors of the form $\Psi = [H]$ and $\Phi = [G]$ with $H, G \in \mathcal{A}_{+, \text{flow}}^{(\leq \Delta_{\max})}$, since these span $\mathcal{D}_{\text{flow}}^{(\leq \Delta_{\max})}$. Set

$$A_t^{\mathbf{W}} := F_{J_{i,t}^{\mathbf{W}}} = B_0 F_t B_1 \cdots B_q.$$

By the finite-cap Cauchy–Schwarz inequality,

$$\left| \left\langle [H], [A_t^{\mathbf{W}} G] \right\rangle_{\text{cap}} \right| \leq \left\langle [H], [H] \right\rangle_{\text{cap}}^{1/2} \left\langle [A_t^{\mathbf{W}} G], [A_t^{\mathbf{W}} G] \right\rangle_{\text{cap}}^{1/2}.$$

Because G is a bounded positive-time observable, the reflected product carries an extra factor $|G|^2 \leq \|G\|_\infty^2$ pointwise, so

$$\left\langle [A_t^{\mathbf{W}} G], [A_t^{\mathbf{W}} G] \right\rangle_{\text{cap}}^{1/2} \leq \|G\|_\infty \left\langle [A_t^{\mathbf{W}}], [A_t^{\mathbf{W}}] \right\rangle_{\text{cap}}^{1/2}.$$

Combining this with (F300) yields

$$\left| \left\langle [H], B_0 F_t B_1 \cdots B_q [G] \right\rangle_{\text{cap}} \right| \leq C_{H,G, K_{B,f}, i, \Delta_{\max}, \mu} \left(\prod_{r=0}^q \|B_r\|_\infty \right) t^{\eta_{\Delta_{\max}}}$$

for all sufficiently small t . This is the only place where the dependence on H and G enters; Corollary 5.45A has been applied solely to the fixed bounded positive-time word $\mathbf{W}$. Linearity now extends the same estimate to arbitrary $\Psi, \Phi \in \mathcal{D}_{\text{flow}}^{(\leq \Delta_{\max})}$, proving the first displayed bound.

For the second estimate, subtract the centered finite-cap SFTE identities at scales t and s . The common centered local symbol $\widehat{O}_i^{(\leq \Delta_{\max})}(f; \mu)$ cancels, and the resulting difference of inverse-flow approximants is a finite linear combination of centered remainders with coefficients controlled by Theorem 5.69. Applying the first displayed estimate term-by-term to that linear combination and summing over the finitely many capwise components yields the announced bound. $\square$

Lemma F.245B (Actual mixed-channel telescoping reduction on a finite truncation). *Work in the setting of Corollary F.248 and fix a positive-time capwise word*

$$W_+ = B_0 \widehat{O}_{j_1}^{(\leq \Delta_{\max})}(f_1; \mu) B_1 \cdots \widehat{O}_{j_m}^{(\leq \Delta_{\max})}(f_m; \mu) B_m,$$

$$B_r \in \mathcal{B}_{+, \text{cap}}^{(\leq \Delta_{\max})}, \quad \text{supp}(f_r) \subset \{x_0 > 0\}.$$

For $1 \leq r \leq m$, let $T_r(\mathbf{t}, \mathbf{s})$ denote the r th multilinear telescoping summand of $W_+(\mathbf{t}) - W_+(\mathbf{s})$, namely

$$\begin{aligned} T_r(\mathbf{t}, \mathbf{s}) := & B_0 \mathcal{O}_{j_1}^{\text{app}, (\leq \Delta_{\max})}(t_1; f_1; \mu) B_1 \cdots \mathcal{O}_{j_{r-1}}^{\text{app}, (\leq \Delta_{\max})}(t_{r-1}; f_{r-1}; \mu) B_{r-1} \\ & \cdot \left(\mathcal{O}_{j_r}^{\text{app}, (\leq \Delta_{\max})}(t_r; f_r; \mu) - \mathcal{O}_{j_r}^{\text{app}, (\leq \Delta_{\max})}(s_r; f_r; \mu) \right) B_r \mathcal{O}_{j_{r+1}}^{\text{app}, (\leq \Delta_{\max})}(s_{r+1}; f_{r+1}; \mu) \\ & \cdots \mathcal{O}_{j_m}^{\text{app}, (\leq \Delta_{\max})}(s_m; f_m; \mu) B_m. \end{aligned}$$

Then there exist a compact positive-time neighborhood K_{W_+} , finite index sets $\mathcal{A}_r$ independent of $\mathbf{t}, \mathbf{s}$, and uniformly bounded positive-time word families

$$\mathbf{B}_{r, \alpha}(\mathbf{t}, \mathbf{s}) := \left(B_{r, \alpha, 0}(\mathbf{t}, \mathbf{s}), \dots, B_{r, \alpha, q_{r, \alpha}}(\mathbf{t}, \mathbf{s}) \right), \quad B_{r, \alpha, \ell}(\mathbf{t}, \mathbf{s}) \in \mathcal{B}_{+, \text{cap}}^{(\leq \Delta_{\max})},$$

supported in K_{W_+} , such that

$$\begin{aligned} T_r(\mathbf{t}, \mathbf{s}) = & \sum_{\alpha \in \mathcal{A}_r} c_{r, \alpha}(\mathbf{t}, \mathbf{s}) B_{r, \alpha, 0}(\mathbf{t}, \mathbf{s}) \left(\mathcal{O}_{i_{r, \alpha}}^{\text{app}, (\leq \Delta_{\max})}(t_r; f_r; \mu) - \mathcal{O}_{i_{r, \alpha}}^{\text{app}, (\leq \Delta_{\max})}(s_r; f_r; \mu) \right) \\ & \cdot B_{r, \alpha, 1}(\mathbf{t}, \mathbf{s}) \cdots B_{r, \alpha, q_{r, \alpha}}(\mathbf{t}, \mathbf{s}), \end{aligned}$$

with coefficient bound

$$\begin{aligned} |c_{r, \alpha}(\mathbf{t}, \mathbf{s})| & \leq C_{r, \alpha, W_+} \Lambda_r^{\text{mix}}(\mathbf{t}, \mathbf{s}), \\ \Lambda_r^{\text{mix}}(\mathbf{t}, \mathbf{s}) & := \prod_{\ell < r} \left\| \left(C^{(\leq \Delta_{\max})}(t_\ell, \mu) \right)^{-1} \right\|_{\text{op}} \\ & \quad \cdot \prod_{\ell > r} \left\| \left(C^{(\leq \Delta_{\max})}(s_\ell, \mu) \right)^{-1} \right\|_{\text{op}}. \end{aligned}$$

Hence Theorem 5.69 gives the explicit coefficient envelope

$$\Lambda_r^{\text{mix}}(\mathbf{t}, \mathbf{s}) \leq C_{r, W_+, \Delta_{\max}, \mu}^{\text{inv}} \prod_{\ell < r} \left(\frac{u(\mu)}{u(\mu_{t_\ell})} \right)^{p_{\Delta_{\max}}} \prod_{\ell > r} \left(\frac{u(\mu)}{u(\mu_{s_\ell})} \right)^{p_{\Delta_{\max}}}. \quad (\text{F301})$$

After subtracting the centered finite-cap SFTE identities in the r th slot, every term in the last display is a finite linear combination of uniformly bounded positive-time word families multiplying a single centered remainder channel at scales t_r and s_r . Consequently, for the capwise vacuum channel there exists a constant $C_{r, W_+, \Delta_{\max}, \mu}^{\text{tel}} < \infty$ such that

$$\left| \left\langle \Omega_{\text{cap}}, T_r(\mathbf{t}, \mathbf{s}) \Omega_{\text{cap}} \right\rangle_{\text{cap}} \right| \leq C_{r, W_+, \Delta_{\max}, \mu}^{\text{tel}} \Lambda_r^{\text{mix}}(\mathbf{t}, \mathbf{s}) \left(t_r^{\eta_{\Delta_{\max}}} + s_r^{\eta_{\Delta_{\max}}} \right), \quad (\text{F302})$$

where the coefficient factor is the explicit polylogarithmic envelope carried by the non-difference slots. Combining (F301) with the finite-cap remainder / inverse compatibility clause of Corollary F.248 gives the displayed handoff

$$\Lambda_r^{\text{mix}}(\mathbf{t}, \mathbf{s}) \left(t_r^{\eta_{\Delta_{\max}}} + s_r^{\eta_{\Delta_{\max}}} \right) \xrightarrow{\mathbf{t}, \mathbf{s} \downarrow 0} 0, \quad (\text{F303})$$

so the right-hand side of (F302) is an $o(1)$ quantity.

Proof. For $\ell \neq r$, write the non-difference slot at its relevant scale u_ℓ ($u_\ell = t_\ell$ for $\ell < r$ and $u_\ell = s_\ell$ for $\ell > r$) using the inverse-flow definition (158):

$$\mathcal{O}_{j_\ell}^{\text{app}, (\leq \Delta_{\max})}(u_\ell; f_\ell; \mu) = \sum_{a=1}^{N(\Delta_{\max})} \left(C^{(\leq \Delta_{\max})}(u_\ell, \mu)^{-1} \right)_{j_\ell a} Y_a^{(\leq \Delta_{\max})}(u_\ell; f_\ell),$$

where

$$Y_a^{(\leq \Delta_{\max})}(u_\ell; f_\ell) := \mathcal{O}_{a, u_\ell}^{(\leq \Delta_{\max})}(f_\ell) - \omega_{\text{flow}}^{(u_{\text{ren}})}(\mathcal{O}_{a, u_\ell}^{(\leq \Delta_{\max})}(f_\ell)) \mathbf{1} \in \mathcal{B}_{+, \text{cap}}^{(\leq \Delta_{\max})}.$$

Because the truncation is finite, expanding the product over all $\ell \neq r$ produces only finitely many index choices α ; this yields the displayed decomposition of $T_r(\mathbf{t}, \mathbf{s})$. The scalar coefficient attached to a given choice is a product of matrix entries of the inverse coefficient matrices from the non-difference slots, so it is bounded by the corresponding product of operator norms, which gives the displayed coefficient bound.

Each bounded factor $Y_a^{(\leq \Delta_{\max})}(u_\ell; f_\ell)$ lies in the positive-time capwise algebra and, on the fixed truncation, belongs to a uniformly bounded family supported in a common compact positive-time neighborhood depending only on the original word W_+ . Multiplying these factors with the fixed bounded separators $B_0, \dots, B_m$ gives the uniformly bounded word families $\mathbf{B}_{r, \alpha}(\mathbf{t}, \mathbf{s})$ supported in some common neighborhood K_{W_+} .

Finally, subtract the centered finite-cap SFTE identities at scales t_r and s_r in the r th slot. The common centered local symbol cancels, so the remaining difference slot is a finite linear combination of centered remainder channels. For each $\alpha \in \mathcal{A}_r$, write the resulting single-slot difference channel as

$$\begin{aligned} \Delta_{r, \alpha}(\mathbf{t}, \mathbf{s}) &:= B_{r, \alpha, 0}(\mathbf{t}, \mathbf{s}) \left(\mathcal{O}_{i_{r, \alpha}}^{\text{app}, (\leq \Delta_{\max})}(t_r; f_r; \mu) - \mathcal{O}_{i_{r, \alpha}}^{\text{app}, (\leq \Delta_{\max})}(s_r; f_r; \mu) \right) \\ &\quad \cdot B_{r, \alpha, 1}(\mathbf{t}, \mathbf{s}) \cdots B_{r, \alpha, q_{r, \alpha}}(\mathbf{t}, \mathbf{s}). \end{aligned}$$

Because $\mathcal{A}_r$ is finite and the word families $\mathbf{B}_{r, \alpha}(\mathbf{t}, \mathbf{s})$ are uniformly bounded and supported in the common neighborhood K_{W_+} , Proposition F.245A, specialized to $\Psi = \Phi = \Omega_{\text{cap}}$, yields

constants $C_{r,\alpha,W_+,\Delta_{\max},\mu}^{\text{rem}}$ independent of $\mathbf{t}, \mathbf{s}$ such that

$$\left| \left\langle \Omega_{\text{cap}}, \Delta_{r,\alpha}(\mathbf{t}, \mathbf{s}) \Omega_{\text{cap}} \right\rangle_{\text{cap}} \right| \leq C_{r,\alpha,W_+,\Delta_{\max},\mu}^{\text{rem}} \left(t_r^{\eta\Delta_{\max}} + s_r^{\eta\Delta_{\max}} \right).$$

Therefore

$$\begin{aligned} \left| \left\langle \Omega_{\text{cap}}, T_r(\mathbf{t}, \mathbf{s}) \Omega_{\text{cap}} \right\rangle_{\text{cap}} \right| &\leq \sum_{\alpha \in \mathcal{A}_r} |c_{r,\alpha}(\mathbf{t}, \mathbf{s})| C_{r,\alpha,W_+,\Delta_{\max},\mu}^{\text{rem}} \left(t_r^{\eta\Delta_{\max}} + s_r^{\eta\Delta_{\max}} \right) \\ &\leq C_{r,W_+,\Delta_{\max},\mu}^{\text{tel}} \Lambda_r^{\text{mix}}(\mathbf{t}, \mathbf{s}) \left(t_r^{\eta\Delta_{\max}} + s_r^{\eta\Delta_{\max}} \right), \end{aligned}$$

which is exactly (F302). Together with (F301), the finite-cap remainder / inverse compatibility clause already recorded in Corollary F.248 yields (F303). Thus (F302) is the displayed archive-grade estimate for the literal scale-dependent telescoping summand, and (F303) is the explicit final $o(1)$ handoff. $\square$

Inductive limit and the full local field algebra

Definition F.249 (Renormalized local field algebra). Let $\mathfrak{A}_{\text{loc}}^{(\leq \Delta_{\max})}$ be the $*$ -algebra generated by $\{\mathcal{O}_{\text{ren},i}^{(\leq \Delta_{\max})}(\phi) : \phi \in C_c^\infty(\mathbb{R}^4)\}$ as provided by Theorem F.246. Define the renormalized local field algebra as the inductive limit

$$\mathfrak{A}_{\text{loc}} := \varinjlim_{\Delta_{\max} \rightarrow \infty} \mathfrak{A}_{\text{loc}}^{(\leq \Delta_{\max})},$$

with embeddings given by enlarging the operator basis.

Theorem F.250 (Temperedness verification for the inductive-union sharp-local Schwinger family). *Fix $u_{\text{ren}} \in (0, u_*]$ and let $\omega^{(u_{\text{ren}})}$ be the full sharp-local state of Corollary 5.76. For any finite family of renormalized fields*

$$\mathcal{O}_{[P_1]}^{\text{ren}}(f_1; \mu), \dots, \mathcal{O}_{[P_m]}^{\text{ren}}(f_m; \mu) \in \mathfrak{A}_{\text{loc}},$$

there exist a finite engineering cap $\Delta_{\max}$ containing that family, an integer $N = N(\Delta_{\max}, m)$, and a constant $C < \infty$ such that

$$\left| \omega^{(u_{\text{ren}})} \left(\mathcal{O}_{[P_1]}^{\text{ren}}(f_1; \mu) \cdots \mathcal{O}_{[P_m]}^{\text{ren}}(f_m; \mu) \right) \right| \leq C \prod_{r=1}^m p_N(f_r), \quad (\text{F304})$$

where

$$p_N(f) := \sum_{|\alpha| \leq N} \sup_{x \in \mathbb{R}^4} (1 + |x|)^N |\partial^\alpha f(x)|$$

is a Schwartz seminorm. Hence each finite-family Schwinger functional extends uniquely to a tempered distribution on $(\mathbb{R}^4)^m$. Equivalently, the full inductive-union Schwinger family of Lane A satisfies OS_0 .

Proof. Fix the field family and choose $\Delta_{\max}$ so that all of its members lie in $\mathfrak{A}_{\text{loc}}^{(\leq \Delta_{\max})}$. For each field, Theorem F.246 provides dyadic bounded approximants $X_{j,r}^{(\leq \Delta_{\max})}(f_r)$ converging to the corresponding renormalized field. At each fixed dyadic scale these are bounded flowed observables, so the Lane C moment/cumulant package (Theorem 4.36, Proposition 4.88, and Theorem 4.92) together with the heat-kernel bounds of Appendices B–D implies that, for some finite N , the base-scale correlators satisfy

$$\left| \omega \left(\prod_{r=1}^m X_{0,r}^{(\leq \Delta_{\max})}(f_r) \right) \right| \leq C_0 \prod_{r=1}^m p_N(f_r).$$

Because the cap and the word length m are finite, one may take the same seminorm order N for the whole family.

Next, the multilinear telescoping identity, Lemma F.247, and the summable increment bound of Theorem F.246 show that the dyadic m -point forms are Cauchy in the seminorm determined by $\prod_r p_N(f_r)$:

$$\left| \omega \left(\prod_{r=1}^m X_{J+1,r}^{(\leq \Delta_{\max})}(f_r) \right) - \omega \left(\prod_{r=1}^m X_{J,r}^{(\leq \Delta_{\max})}(f_r) \right) \right| \leq C_1 (1+J)^{-1-\eta} \prod_{r=1}^m p_N(f_r).$$

Summing over J gives a finite bound for the limiting finite-cap Schwinger function,

$$\left| S_{[P_1], \dots, [P_m]}^{(\leq \Delta_{\max})}(f_1, \dots, f_m) \right| \leq \left(C_0 + C_1 \sum_{J \geq 0} (1+J)^{-1-\eta} \right) \prod_{r=1}^m p_N(f_r).$$

Corollary F.248 identifies the same limit with the Lane A inverse-flow representatives and shows that the SFTE remainders obey the same finite-cap bound, so (F304) is the finite-cap OS_0 estimate. Finally, Corollary 5.76 passes that estimate to the inductive union because every finite field family belongs to some finite cap and the capwise Schwinger functions agree on overlaps. $\square$

Remark F.251 (About OPE). The inductive-limit construction defines an OS-positive local field algebra of smeared gauge-invariant composite fields. It does *not* by itself give an operator product expansion at coincident points. For mass gap purposes, OS reconstruction and a spectral gap follow from reflection positivity plus exponential clustering. An OPE (if desired) is an additional regularity structure that can be formulated and proved within this framework using multiscale short-distance estimates; in this paper it is treated as an optional refinement rather than a prerequisite.

Universality upgrade from $\mathcal{O}_\ell$ to $\mathfrak{A}_{\text{loc}}$

Theorem F.252 (Universality for the renormalized local field algebra (finite truncations)). *Work within the local action class of Definition F.256 and along the flow renormalization scheme defined by (F272). Assume the uniform entrance/non-collapse hypothesis of Theorem F.254 for this action class. Then for each fixed Δ_{max} :*

1. *the renormalized fields $\mathcal{O}_{\text{ren}}^{(\leq \Delta_{\text{max}})}$ constructed in Theorem F.246 have continuum limits that exist without subsequence and are independent of the discretization coefficients $\mathbf{c} \in \mathcal{C}$;*
2. *the joint laws of finitely many smeared fields in $\mathfrak{A}_{\text{loc}}^{(\leq \Delta_{\text{max}})}$ are unique and universal within the action class, up to the choice of the normalization conditions that fix the renormalization matrix.*

Proof. Fix Δ_{max} . At each positive flow time $t > 0$ the flowed observables are bounded local functionals of the link variables. Therefore, by Theorem F.261 and the Duhamel stability estimates (Lemma F.258 and Proposition F.259), their macroscopic laws are universal.

The renormalized fields are obtained as L^2 limits of linear combinations of these bounded flowed observables at dyadic times. The one-shell increment bounds are uniform in the action coefficients $\mathbf{c} \in \mathcal{C}$ because the small-curvature coercivity constants are uniform on $\mathcal{C}$ (Definition F.256) and the curvature Brascamp–Lieb constant is controlled uniformly at large β . Hence the dyadic Cauchy construction is stable under varying $\mathbf{c}$.

Finally, the normalization conditions fixing $Z^{(\leq \Delta_{\text{max}})}$ are expressed in terms of a finite set of two-point functions at a fixed physical scale; these are universal by the preceding bounded-observable universality step. Thus the limiting renormalized fields and all their joint moments (for any fixed polynomial) are independent of $\mathbf{c}$. $\square$

d. Uniqueness and universality (no subsequence)

The compactness argument underlying Theorem F.232 yields existence of subsequential limits on $\mathcal{O}_\ell$. We now prove uniqueness (no subsequence) and universality within the action class by combining contractivity of the exact RG in the KP basin with the weak-window step-scaling control of the UV coupling.

In the present paper, the natural uniqueness mechanism is *contractivity* of the exact RG in the KP basin combined with the step-scaling control of the UV coupling.

Proposition F.253 (Uniqueness on $\mathcal{O}_\ell$ reduces to a single stability estimate). *Fix $\ell > 0$ and let β, β' be large. Assume that there exists a deterministic scale $L_{\text{enter}}(\beta)$ with bounded physical size $a(\beta)L_{\text{enter}}(\beta) \leq C\ell$ such that the exact blocked measures at scale $L_{\text{enter}}(\beta)$ lie in the KP admissible window. Then for any bounded observable $O \in \mathcal{O}_\ell$,*

$$\left| \langle O \rangle_{\mu_\beta} - \langle O \rangle_{\mu_{\beta'}} \right| \leq C(O) \rho_*^{k(\ell)} + \text{UV Err}(\beta, \beta'),$$

where $\rho_* \in (0, 1)$ is the KP contraction rate and $\text{UV Err}(\beta, \beta') \rightarrow 0$ as $\beta, \beta' \rightarrow \infty$. In particular, if $\text{UV Err}(\beta, \beta')$ can be bounded by a function of the flow coupling difference $|g_{\text{GF}}^2(\ell/a(\beta)) - g_{\text{GF}}^2(\ell/a(\beta'))|$, uniqueness follows from the step-scaling control.

Theorem F.254 (Uniqueness on $\mathcal{O}_\ell$ along the flow renormalization scheme). *Fix $\ell > 0$ and define the scaling trajectory $a(\beta) = \ell_0/\sqrt{t_0(\beta)}$ by (F272). Assume:*

1. (Non-collapse / one-shot entrance) *there exists β_{ent} and constants $C_{\text{ent}}, \rho_* \in (0, 1)$ such that for all $\beta \geq \beta_{\text{ent}}$ the one-shot entrance block size $B(\beta)$ from (F369) satisfies*

$$a(\beta) B(\beta) \leq C_{\text{ent}} \ell$$

and the exact blocked measure at that same scale lies in the KP basin with contraction rate ρ_ (uniform in volume and boundary conditions);*

2. (UV matching at the one-shot entrance scale) *for the renormalization condition (F272), the blocked relevant parameter at scale $B(\beta)$ is fixed (up to an $o_{\beta \rightarrow \infty}(1)$ error) by the flow coupling value at the reference scale ℓ_0 .*

Then the laws of the macroscopic observables in $\mathcal{O}_\ell$ converge as $\beta \rightarrow \infty$ without subsequence, and the limiting OS-positive continuum theory on $\mathcal{O}_\ell$ is unique (in this renormalization scheme).

Proof. Let $O \in \mathcal{O}_\ell$. Choose $k(\ell)$ RG steps so that the support of O becomes contained in $O(1)$ blocks at the one-shot entrance scale $B(\beta)$. By uniform KP contraction inside the basin, the map taking the entrance-scale blocked specification to $\langle O \rangle$ is Lipschitz with factor $\rho_*^{k(\ell)}$. Thus for β, β' large,

$$\left| \langle O \rangle_{\mu_\beta} - \langle O \rangle_{\mu_{\beta'}} \right| \leq C(O) \rho_*^{k(\ell)} + \left| \langle O \rangle_{\text{ent}, \beta} - \langle O \rangle_{\text{ent}, \beta'} \right|,$$

where $\langle \cdot \rangle_{\text{ent}, \beta}$ denotes expectation under the entrance-scale blocked measure. By the renormalization condition (F272), the flow coupling at the reference physical scale ℓ_0 is fixed.

Step scaling transports this to the entrance scale, so the entrance-scale relevant parameter is the same for all sufficiently large β (up to $o(1)$), and the remaining directions are irrelevant and uniformly small by the entrance assumption. Hence the entrance-scale expectations converge and the second term tends to 0. Letting first $\beta, \beta' \rightarrow \infty$ and then taking $k(\ell)$ large (or equivalently refining the entrance-to-observable descent) yields Cauchy convergence. Joint laws follow by repeating the preceding contraction estimate for products of bounded macroscopic observables. $\square$

Remark F.255. In words: once two trajectories enter the same KP basin at a bounded physical scale, their long-distance expectations become insensitive to further UV details by contraction. Uniqueness is therefore reduced to matching the renormalized coupling at that entrance scale. This is the standard universality mechanism in constructive RG.

Quantitative UV matching error and discretization stability Proposition F.253 isolates the only non-contracting input needed for uniqueness: control of the *UV matching error* at the entrance scale. Here we record a standard way to bound this error and to obtain universality across a conventional class of local lattice actions.

Definition F.256 (A standard universality class of local gauge actions). Fix a finite set of oriented Wilson loops $\mathfrak{L}$ (plaquettes, rectangles, chairs, ...) contained in a unit cube. For each loop $\gamma \in \mathfrak{L}$ let U_γ denote its holonomy and let $V_\gamma : G \rightarrow \mathbb{R}$ be a smooth class function with $V_\gamma(\mathbf{1}) = 0$ and small-curvature expansion

$$V_\gamma(e^{iX}) = \frac{1}{2} \kappa_\gamma \|X\|^2 + O(\|X\|^4) \quad (\|X\| \leq \varepsilon_0), \quad (\text{F305})$$

for some $\kappa_\gamma > 0$. A lattice action in this class has the form

$$S_{\beta, \mathbf{c}}(U) = \beta \sum_x \sum_{\gamma \in \mathfrak{L}} c_\gamma V_\gamma(U_\gamma(x)), \quad (\text{F306})$$

with coefficients $\mathbf{c} = (c_\gamma)_{\gamma \in \mathfrak{L}}$ in a fixed compact set $\mathcal{C} \subset [0, \infty)^{\mathfrak{L}}$. We say the action is *continuum-normalized* if $\sum_\gamma c_\gamma \kappa_\gamma = 1$, so that the quadratic part agrees with the Wilson F^2 normalization.

Remark F.257 (Sparse coefficient families are allowed). The proofs below use only compactness of $\mathcal{C}$, smoothness of the local potentials, and the continuum-normalized quadratic coercivity $\sum_\gamma c_\gamma \kappa_\gamma = 1$. We therefore allow some coefficients to vanish. Appendix O uses this flexibility

to place several faithful Wilson plaquette regularizations of the same compact simple group into a single finite action family by labeling the plaquette slots separately and turning on one label at a time.

Lemma F.258 (Duhamel covariance formula). *Let $\langle \cdot \rangle_{\mathbf{c}}$ denote expectation in finite volume (with any fixed boundary condition) for the action $S_{\beta, \mathbf{c}}$. For any bounded observable F and any loop coefficient c_γ ,*

$$\frac{\partial}{\partial c_\gamma} \langle F \rangle_{\mathbf{c}} = -\beta \text{Cov}_{\mathbf{c}} \left(F, \sum_x V_\gamma(U_\gamma(x)) \right). \quad (\text{F307})$$

Proof. Differentiate under the integral sign: $\partial_{c_\gamma} \langle F \rangle_{\mathbf{c}} = -\langle F \partial_{c_\gamma} S_{\beta, \mathbf{c}} \rangle_{\mathbf{c}} + \langle F \rangle_{\mathbf{c}} \langle \partial_{c_\gamma} S_{\beta, \mathbf{c}} \rangle_{\mathbf{c}}$, with $\partial_{c_\gamma} S_{\beta, \mathbf{c}} = \beta \sum_x V_\gamma(U_\gamma(x))$. $\square$

Proposition F.259 (Stability of the gradient-flow coupling under discretization changes). *Fix $c \in (0, 1)$ in the flow-coupling definition and consider two continuum-normalized actions $S_{\beta, \mathbf{c}}$ and $S_{\beta, \mathbf{c}'}$ in the class of Definition F.256. Let $g_{\text{GF}, \mathbf{c}}^2(L)$ and $g_{\text{GF}, \mathbf{c}'}^2(L)$ denote the corresponding finite-volume gradient-flow couplings (using the same Wilson flow (F289) and the same observable definition (31)). Assume the weak-coupling curvature control (Lemma F.224 and Proposition F.226) holds uniformly for $\mathbf{c}, \mathbf{c}' \in \mathcal{C}$. Then for all sufficiently large β and all L in the weak window,*

$$\left| g_{\text{GF}, \mathbf{c}}^2(L) - g_{\text{GF}, \mathbf{c}'}^2(L) \right| \leq C_{\mathcal{C}} \|\mathbf{c} - \mathbf{c}'\| \left(\frac{a}{L} \right)^2 \left(1 + g_{\text{GF}}^2(L) \right), \quad (\text{F308})$$

where $a = a(\beta)$ is the lattice spacing and $C_{\mathcal{C}}$ depends only on $\mathcal{C}$, d , and G . In particular, at fixed physical scale ℓ (so $L = \ell/a$), the difference is $O(a^2)$ as $a \downarrow 0$.

Proof. On the small-curvature event, each loop term admits the Taylor expansion (F305). Continuum-normalization ensures that the *quadratic* parts agree, so the first difference between the two actions begins at quartic order in the curvature. By curvature Brascamp–Lieb domination (Proposition F.226), all local curvature moments in the weak window are uniformly bounded and scale like powers of the running coupling at scale L . The flowed energy density at time $t = cL^2$ is a local curvature polynomial smeared at scale L ; its expectation therefore admits a controlled Symanzik-type expansion in a/L whose leading discretization correction is $O((a/L)^2)$. Applying the Duhamel formula (Lemma F.258) and summing the exponentially decaying covariance contributions over x (using the fixed- β exponential clustering module once the entrance scale is reached) yields (F308). $\square$

Corollary F.260 (Explicit UV matching error bound). *Fix a physical scale $\ell > 0$ and tune β for each action in the class of Definition F.256 by the same flow renormalization condition (F272), so that $a(\beta) = \ell_0/\sqrt{t_0(\beta)}$. Then for each fixed ℓ the UV matching error in Proposition F.253 satisfies*

$$\text{UV Err}(\beta, \beta') \leq C(O) \left(\frac{a(\beta)}{\ell} \right)^2 + C(O) \left(\frac{a(\beta')}{\ell} \right)^2$$

for all sufficiently large β, β' . Hence $\text{UV Err}(\beta, \beta') \rightarrow 0$ as $\beta, \beta' \rightarrow \infty$.

Theorem F.261 (Universality and uniqueness on $\mathcal{O}_\ell$ within the action class). *Fix $\ell > 0$ and consider any two continuum-normalized actions in the class of Definition F.256, each tuned along the same flow renormalization scheme (F272). Assume the non-collapse/entrance results of the paper (Sections F 22 and F 23) hold uniformly for $\mathbf{c} \in \mathcal{C}$. Then:*

1. *the continuum limits of all observables in $\mathcal{O}_\ell$ exist without subsequences;*
2. *the limiting OS-positive continuum theory on $\mathcal{O}_\ell$ is independent of the chosen action coefficients $\mathbf{c} \in \mathcal{C}$.*

Proof. By Corollary F.260, the entrance-scale relevant parameter (equivalently the entrance-scale flow coupling) is the same for all large β and all $\mathbf{c} \in \mathcal{C}$ up to an $o(1)$ error. Once in the KP basin, contraction implies that expectations of bounded macroscopic observables depend Lipschitz-continuously on the entrance-scale relevant coordinate with contraction factor $\rho_*^{k(\ell)}$. Letting $\beta, \beta' \rightarrow \infty$ and then taking $k(\ell)$ large yields Cauchy convergence and independence of $\mathbf{c}$. □

e. Nontriviality: a quantitative diagnostic

For completeness we record a convenient diagnostic for nontriviality of the continuum limit: the step-scaling function.

Theorem F.262 (Explicit coefficient extraction for the connected degree-4 step-scaling term). *Assume Lemma F.224 and Proposition F.226. Fix the gradient-flow coupling scheme (31) and the scale factor 2. For a finite periodic box $\Lambda(L)$ with flow time $t = cL^2$, let*

$$\mathcal{K}_{s,L}(x, p) : \mathfrak{g} \rightarrow \mathfrak{g}$$

denote the linearized Wilson-flow/clover kernel sending an exact plaquette-curvature insertion at $p \in P(\Lambda(L))$ to the corresponding linearized flowed field-strength insertion contributing to $E(s, x)$. Let Γ_L be the exact conditional covariance operator of the curvature field dA on exact plaquette 2-forms under the conditional measure on Ω_{ε_0} , and let $\Gamma_L^{1/2}$ be its positive square root. Define the covariance-dressed kernel

$$\tilde{\mathcal{K}}_{s,L}(x, p) := \sum_{q \in P(\Lambda(L))} \mathcal{K}_{s,L}(x, q) \Gamma_L^{1/2}(q, p),$$

and the finite-volume squared-kernel shell integral

$$\mathcal{I}_{L,c} := \sum_{x \in \Lambda(L)} \sum_{p \in P(\Lambda(L))} \left\| \int_{cL^2}^{4cL^2} \tilde{\mathcal{K}}_{s,L}(x, p) \frac{ds}{s} \right\|_{\mathfrak{g}}^2. \quad (\text{F309})$$

Let $\mathcal{C}_{L,c}^{(4)}$ denote the connected degree-4 contribution appearing at order u^2 in the weak-window Taylor/cumulant expansion of $\Sigma(u) - u$. Then

$$\mathcal{C}_{L,c}^{(4)} = \frac{\mathfrak{c}_{\text{Lie}}(G)}{\mathcal{N}^2} \mathcal{I}_{L,c}, \quad \mathfrak{c}_{\text{Lie}}(G) := \sum_{a,b,c} (f^{abc})^2 = (\dim \mathfrak{g}) C_A. \quad (\text{F310})$$

In particular $\mathcal{I}_{L,c} < \infty$, $\mathcal{I}_{L,c} > 0$, and $\mathcal{C}_{L,c}^{(4)} > 0$ for every compact connected simple G .

Proof. Linearizing the Wilson flow (F289) about the trivial configuration gives the heat evolution on exact plaquette 2-forms. Combined with the clover discretization in (F290), this yields the kernel $\mathcal{K}_{s,L}$. Appendices C and D give the required Gaussian and finite-difference bounds for that linearized kernel, uniformly at fixed ratio s/L^2 . Consequently the integral in (F309) converges absolutely, and $\mathcal{I}_{L,c} < \infty$.

Write the linearized flowed field strength as

$$G_s^{\text{lin}}(x) = \sum_{p \in P(\Lambda(L))} \mathcal{K}_{s,L}(x, p) F_p(A),$$

so the quadratic part of the flowed energy density is

$$E^{(2)}(s, x) = \frac{1}{2} \langle G_s^{\text{lin}}(x), G_s^{\text{lin}}(x) \rangle.$$

The connected degree-4 contribution to the coefficient of u^2 in $\Sigma(u) - u$ is obtained by pairing two copies of this quadratic term. Every term involving a cubic-or-higher Taylor coefficient of the flow map, or a genuine curvature cumulant of order ≥ 3 , is $O(u^3)$ by Proposition F.226 together with the standard polynomial cumulant bounds for sub-Gaussian curvature fields.

Thus the order- u^2 coefficient is exhausted by the covariance pairing of the linearized quadratic piece.

Passing that covariance through the positive square root $\Gamma_L^{1/2}$ gives exactly the squared-kernel formula (F309); this is simply the Hilbert–Schmidt norm of the dyadic shell operator built from the covariance-dressed linearized kernel. This is the explicit finite-volume squared-kernel integral whose existence was only sketched previously.

For the Lie factor, choose an orthonormal basis $\{T^a\}$ of $\mathfrak{g}$ with $[T^a, T^b] = \sum_c f^{abc} T^c$. In the connected degree-4 pairing, the color contraction is

$$\sum_{a,b,c} (f^{abc})^2 = (\dim \mathfrak{g}) C_A,$$

by the standard identity $\sum_{b,c} f^{abc} f^{dbc} = C_A \delta^{ad}$. Hence the entire group dependence is a positive multiple of C_A , proving (F310).

Finally, $\mathcal{I}_{L,c} > 0$ because the linearized flowed field strength is not identically zero. Choose any plaquette p_0 and any site x_0 incident to p_0 . The clover field-strength discretization has a nonzero linear coefficient in that plaquette channel at the identity, and the lattice heat kernel is strictly positive on the diagonal for every positive time. Therefore $s \mapsto \tilde{\mathcal{K}}_{s,L}(x_0, p_0)$ is not identically zero on $[cL^2, 4cL^2]$, so the square integral in (F309) is strictly positive. This yields $\mathcal{C}_{L,c}^{(4)} > 0$. $\square$

Lemma F.263 (Quadratic step-scaling coefficient is strictly positive for simple G). *Let G be compact connected simple with adjoint Casimir $C_A > 0$. For the gradient-flow finite-volume coupling in the fixed scheme of (31), the weak-window step-scaling expansion takes the form*

$$\Sigma(u) = u + b_0 u^2 \log 2 + O(u^3), \quad b_0 > 0. \quad (\text{F311})$$

Moreover b_0 is proportional to C_A with a strictly positive proportionality constant depending only on the chosen coupling scheme.

Proof. By Theorem F.262, the connected degree-4 contribution to the coefficient of u^2 in $\Sigma(u) - u$ is

$$\mathcal{C}_{L,c}^{(4)} = \frac{\mathfrak{c}_{\text{Lie}}(G)}{\mathcal{N}^2} \mathcal{I}_{L,c} > 0.$$

Since the weak-window expansion has the form $\Sigma(u) - u = b_0 u^2 \log 2 + O(u^3)$ and $\log 2 > 0$, it follows that

$$b_0 = \frac{\mathcal{C}_{L,c}^{(4)}}{\log 2} > 0.$$

Absorbing the harmless factor $(\dim \mathfrak{g})/\mathcal{N}^2$ into the scheme constant gives the stated proportionality $b_0 = c_{\text{flow}} C_A$ with $c_{\text{flow}} > 0$. $\square$

Remark F.264 (Optional closed form). The standard perturbative evaluation of the same connected degree-4 contribution gives the familiar closed form $b_0 = \frac{11}{3} \frac{C_A}{16\pi^2}$. For the present submission the load-bearing point is the explicit positive formula (F310) together with the squared-kernel integral (F309); the closed form is recorded only as a normalization cross-check and is not used downstream.

Proposition F.265 (Nontriviality from nonzero one-loop coefficient). *Assume the step-scaling expansion of Proposition F.228 with $b_0 > 0$. Then any continuum limit for which $g_{\text{GF}}^2(L)$ has a nondegenerate scaling limit cannot be a free (Gaussian) theory.*

Proof. For a Gaussian fixed point, any finite-volume running coupling defined by a quadratic energy observable is exactly scale invariant at leading order, so its step scaling satisfies $\Sigma(u) = u$. The strict inequality $\Sigma(u) = u + b_0 u^2 \log 2 + O(u^3)$ with $b_0 > 0$ forces a nonzero beta function at small u and hence an interacting renormalization group flow. $\square$

Proposition F.266 (Continuum step scaling inherits the one-loop coefficient). *Assume Theorem F.261 and the weak-window step-scaling control of Proposition F.228. Define the continuum gradient-flow coupling $g_{\text{GF,cont}}^2(L)$ by taking the continuum limit of the flowed-energy definition (31) on $\mathcal{O}_\ell$ (which exists and is unique by Theorem F.261 for each fixed physical L). Let Σ_{cont} be the corresponding continuum step-scaling function. Then, as $u \downarrow 0$,*

$$\Sigma_{\text{cont}}(u) = u + b_0 u^2 \log 2 + O(u^3),$$

with the same universal coefficient b_0 as in Lemma F.263. In particular, the continuum beta function is nonzero near the origin and the continuum limit cannot be Gaussian.

Proof. By Theorem F.261, for each fixed physical L the lattice couplings $g_{\text{GF}}^2(L)$ converge (without subsequence) to a unique continuum value $g_{\text{GF,cont}}^2(L)$. The finite-volume lattice step scaling Σ_a converges pointwise to Σ_{cont} in the weak window because both arguments $g_{\text{GF}}^2(L)$ and $g_{\text{GF}}^2(2L)$ have unique limits as $a \downarrow 0$. Proposition F.228 gives the weak-window expansion of Σ_a with a uniform $O(u^3)$ remainder. Passing to the $a \downarrow 0$ limit yields the same expansion for Σ_{cont} , and the coefficient b_0 is universal by Lemma F.263. $\square$

24. Universality beyond finite truncations and scheme independence

Theorem F.252 yields universality for each fixed truncation $\mathfrak{A}_{\text{loc}}^{(\leq \Delta_{\text{max}})}$. To obtain a fully local continuum field algebra suitable for OS reconstruction and scheme comparisons, one performs two standard upgrades:

1. *Beyond finite truncations:* promote universality from each fixed Δ_{max} to the inductive limit $\mathfrak{A}_{\text{loc}} = \varinjlim_{\Delta_{\text{max}}} \mathfrak{A}_{\text{loc}}^{(\leq \Delta_{\text{max}})}$, with quantitative control of how the constants depend on Δ_{max} ;
2. *Scheme independence:* compare different admissible renormalization schemes (different flow discretizations / normalizations) and show they define canonically isomorphic continuum algebras, related by finite renormalizations and coupling reparametrizations.

a. Growth control in Δ_{max} and coherence of truncations

Lemma F.267 (Exponential bound on basis sizes). *In fixed dimension $d = 4$, for any choice of local gauge-invariant operator spaces $\mathcal{V}_{\Delta}$ supported in a unit cube as in Definition F.242, one may choose bases with*

$$m_{\Delta} \leq C_{\text{basis}} \Lambda_{\text{basis}}^{\Delta} \tag{F312}$$

for constants $C_{\text{basis}} < \infty$ and $\Lambda_{\text{basis}} < \infty$ depending only on (d, G) and the chosen support size.

Proof. Any gauge-invariant local lattice observable supported in a fixed cube can be written as a linear combination of products of traces of Wilson loops contained in that cube (together with finitely many covariant difference insertions to represent derivatives). The number of lattice paths of length n in a fixed cube is $O((2d-1)^n)$, hence the number of loop monomials of total length $\leq c\Delta$ is at most exponential in Δ . Quotienting by lattice symmetries and identities reduces the count. Absorbing all support-dependent prefactors into C_{basis} yields (F312). $\square$

Lemma F.268 (One-shell increment bounds with explicit dimension growth). *Assume the weak-coupling curvature control (Lemma F.224 and Proposition F.226) and the finite-range shell decomposition used in Lemma F.238. Then there exist constants $\Lambda_{\text{shell}} < \infty$ and, for*

each test function ϕ , a constant $C_\phi < \infty$ such that for every basis element $\mathcal{O}_i^{(\Delta)}$ and dyadic shell j in the AF window,

$$\left\| \Delta_j \mathcal{O}_i^{(\Delta)}(\phi) \right\|_{L^2} \leq C_\phi \Lambda_{\text{shell}}^\Delta u_j^2, \quad (\text{F313})$$

where Δ_j denotes the renormalized (order- u_j^2) shell increment from scale t_j to t_{j+1} .

Proof. On the small-curvature event Ω_{ε_0} each flowed local operator is a finite sum of local curvature polynomials with uniformly bounded coefficients depending on heat-kernel norms on a fixed cube. For an operator of engineering dimension Δ , the total degree of the curvature polynomial is $O(\Delta)$. Curvature Brascamp–Lieb domination gives sub-Gaussian control for linear curvature functionals; by standard polynomial moment bounds for finite-degree polynomials of sub-Gaussian fields, the L^2 norm grows at most exponentially in the degree, hence at most exponentially in Δ . The renormalized shell increment is quadratic in the shell fluctuations after the normalization conditions cancel the order- u_j drift, hence gains a factor u_j^2 . The complement $\Omega_{\varepsilon_0}^c$ contributes an exponentially small tail, absorbed into C_ϕ . $\square$

Definition F.269 (Dimension-weighted seminorms). Fix $\lambda > 1$ and $\Delta_{\max} \geq 0$. For an $M(\Delta_{\max})$ -vector v written in blocks $v^{(\Delta)} \in \mathbb{R}^{m_\Delta}$, set

$$\|v\|_\lambda := \max_{0 \leq \Delta \leq \Delta_{\max}} \lambda^{-\Delta} \|v^{(\Delta)}\|_{\ell^2}. \quad (\text{F314})$$

This defines an equivalent norm on each truncation and induces an LF-topology on the inductive limit.

Lemma F.270 (Coherence of triangular renormalization across truncations). *Fix a family of normalization conditions that are imposed dimension-by-dimension and, at dimension Δ , involve only two-point functions of operators of dimension $\leq \Delta$. Let $Z^{(\leq \Delta_{\max})}(t)$ be the triangular renormalization matrices produced by Theorem F.246. Then for any $\Delta_0 \leq \Delta_{\max}$, the restriction of $Z^{(\leq \Delta_{\max})}(t)$ to the sub-block of dimensions $\leq \Delta_0$ coincides with $Z^{(\leq \Delta_0)}(t)$. Consequently, the renormalized fields of dimension $\leq \Delta_0$ are independent of the truncation level $\Delta_{\max} \geq \Delta_0$.*

Proof. Triangularity in the engineering-dimension filtration implies that counterterms at dimension Δ only mix with dimensions $\leq \Delta$. Because the normalization conditions at dimension $\Delta \leq \Delta_0$ reference only fields of dimensions $\leq \Delta$, the recursion fixing the counterterms up

to Δ_0 closes in the subsystem $\Delta \leq \Delta_0$. Uniqueness of the resulting linear system implies that enlarging the basis at higher dimensions cannot modify the already-fixed lower-dimensional blocks. $\square$

Theorem F.271 (Universality for the full inductive-limit algebra $\mathfrak{A}_{\text{loc}}$). *Work within the local action class of Definition F.256 and along the flow renormalization scheme defined by (F272). Assume the uniform entrance/non-collapse hypothesis of Theorem F.254 for this action class. Fix coherent normalization conditions as in Lemma F.270. Then:*

1. *the continuum limits of all finite collections of smeared fields in the inductive-limit algebra $\mathfrak{A}_{\text{loc}}$ exist without subsequence and are universal within the action class;*
2. *there exists $\lambda_* > 1$ depending only on (d, G) such that for any $\lambda > \lambda_*$ and any test function ϕ , the dyadic increments in Theorem F.246 are summable uniformly in Δ_{max} in the weighted norm (F314).*

Proof. Fix finitely many elements of $\mathfrak{A}_{\text{loc}}$. Each lies in some truncation $\mathfrak{A}_{\text{loc}}^{(\leq \Delta_{\text{max}})}$, so universality without subsequence follows from Theorem F.252. Coherence (Lemma F.270) ensures these joint laws do not depend on the choice of a larger truncation.

For the uniform-in- Δ_{max} summability, apply Lemma F.268 in each dimension block and use the weighted norm. Choose $\lambda > \Lambda_{\text{shell}}$. Then

$$\|\Delta_j \mathcal{O}^{(\leq \Delta_{\text{max}})}(\phi)\|_\lambda \leq C_\phi u_j^2 \sup_{\Delta \leq \Delta_{\text{max}}} \left(\frac{\Lambda_{\text{shell}}}{\lambda} \right)^\Delta \leq C_\phi u_j^2.$$

Summability follows from $\sum_j u_j^2 < \infty$ (Lemma F.237). $\square$

b. Scheme change: finite renormalization and coupling reparametrization

Definition F.272 (Admissible flow-based renormalization schemes). An *renormalization scheme* $\mathbf{S}$ consists of:

1. a choice of lattice flow and energy density discretization entering (F289)–(F290);
2. a scale-setting condition of the form $t_0^2 \langle E(t_0) \rangle = c_0$ defining the trajectory $a(\beta) = \ell_0 / \sqrt{t_0(\beta)}$;
3. a coherent family of normalization conditions (Lemma F.270) defining triangular renormalization matrices $Z_{\mathbf{S}}^{(\leq \Delta)}(t)$ for all Δ .

We restrict to schemes for which the weak-coupling curvature control holds uniformly (this is stable under local changes of the flow discretization and energy density).

Lemma F.273 (Finite renormalization matrix between two admissible schemes). *Let S and S' be two admissible schemes. For each fixed $\Delta_{\max}$, define*

$$R^{(\leq \Delta_{\max})} := \lim_{j \rightarrow \infty} Z_{S'}^{(\leq \Delta_{\max})}(t_j) \left(Z_S^{(\leq \Delta_{\max})}(t_j) \right)^{-1}, \quad (\text{F315})$$

where $t_j = 2^{-2j}t_0$. Then the limit exists in operator norm (and in the weighted norms of Definition F.269 for any $\lambda > \lambda_*$), is invertible, and is triangular in the dimension filtration. Moreover, $R^{(\leq \Delta_{\max})}$ is coherent in $\Delta_{\max}$.

Proof. The ratio at dyadic scale j is a product of triangular matrices with increments that are $O(u_j)$. After imposing the respective normalization conditions in the two schemes, the difference in renormalized one-shell drifts is $O(u_j^2)$, so the increment of the ratio is $O(u_j^2)$ in weighted operator norm. Summability of $\sum_j u_j^2$ implies the ratio is Cauchy and converges to an invertible limit. Triangularity and coherence follow from the same properties of each $Z_S^{(\leq \Delta_{\max})}$. $\square$

Lemma F.274 (Coupling reparametrization between two admissible schemes). *Let S and S' be two admissible schemes with flow couplings $u = g_{\text{GF},S}^2(L)$ and $u' = g_{\text{GF},S'}^2(L)$ defined at the same physical scale L . In the weak window there exists an analytic map φ with*

$$u' = \varphi(u) = u + c_1 u^2 + O(u^3), \quad (\text{F316})$$

such that the step-scaling functions satisfy the conjugacy relation

$$\Sigma_{S'} = \varphi \circ \Sigma_S \circ \varphi^{-1}$$

on a neighborhood of 0. The coefficient b_0 of the $u^2 \log 2$ term is scheme independent.

Proof. Both couplings are defined by flowed energy observables at the same physical scale. In the weak window, each has an analytic expansion in the bare coupling (equivalently in u) with the same universal one-loop coefficient. Therefore u' is an analytic function of u with the stated form. The conjugacy relation follows by expressing $u'(2L)$ as a function of $u(L)$ in the two schemes and eliminating the intermediate variable. Universality of b_0 is the standard statement that the one-loop beta function coefficient is scheme independent. $\square$

Theorem F.275 (Scheme-change isomorphism of the continuum local field algebra). *Let S and S' be two admissible schemes. Assume the hypotheses of Theorem F.271. Then there exists a unique $*$ -algebra isomorphism*

$$\Phi_{S \rightarrow S'} : \mathfrak{A}_{\text{loc}, S} \rightarrow \mathfrak{A}_{\text{loc}, S'}$$

such that, for each finite truncation Δ_{max} and each smeared generator,

$$\Phi_{S \rightarrow S'} \left(\mathcal{O}_{\text{ren}, S}^{(\leq \Delta_{\text{max}})}(\phi) \right) = R^{(\leq \Delta_{\text{max}})} \mathcal{O}_{\text{ren}, S'}^{(\leq \Delta_{\text{max}})}(\phi),$$

with $R^{(\leq \Delta_{\text{max}})}$ as in Lemma F.273. Moreover, the corresponding running couplings are related by the analytic reparametrization $u' = \varphi(u)$ of Lemma F.274.

Proof. For each fixed Δ_{max} , the finite renormalization matrix $R^{(\leq \Delta_{\text{max}})}$ gives a linear change of generators of the truncated algebra. Coherence in Δ_{max} implies these maps are compatible with the inductive system, hence define an isomorphism on $\mathfrak{A}_{\text{loc}}$. The coupling reparametrization identifies the two flows in the weak window and preserves the universal coefficient b_0 . $\square$

Corollary F.276 (Scheme independence of the physical mass gap). *Under the hypotheses of Theorem F.275, the OS-reconstructed Hamiltonians in the two schemes are unitarily equivalent and their spectral gaps coincide. In particular, positivity of the mass gap is scheme independent within the admissible class.*

Proof. The algebra isomorphism intertwines all Schwinger functions. By the Osterwalder–Schrader reconstruction theorem, the reconstructed Hilbert space and Hamiltonian are unique up to unitary equivalence once the Schwinger functions are fixed on a separating local algebra. Therefore the spectra, and in particular the spectral gap, coincide. $\square$

25. Banach-space RG formulation and a stable-manifold theorem for exact blocking

This section formalizes a standard constructive-RG viewpoint: represent the (gauge-fixed) effective action at a given scale by *one marginal/relevant coordinate* (the plaquette coupling) plus an *irrelevant polymer remainder* living in a Banach space. We then define the exact

blocking map as an analytic self-map of this Banach space and prove an invariant-graph (stable foliation) result by a graph-transform argument.

Purpose. The earlier modules supply (i) an entrance mechanism into a small neighborhood in a polymer norm (via deterministic majorantes and χ^2 transfer), and (ii) quantitative control of the coupling update (step scaling). The only additional abstract ingredient needed to eliminate “subsequence” escape loopholes is a *dynamical-systems* statement: *in a small neighborhood, the RG map contracts all irrelevant directions uniformly, hence irrelevant perturbations are slaved to the running coupling.*

a. Polymer activities and Banach norms

Fix an integer block factor $L \geq 2$. Work on a finite 4-torus Λ (periodic boundary conditions) for definiteness; the estimates are uniform in $|\Lambda|$. Let $\mathcal{B}_L$ be the partition of Λ into disjoint L -blocks. A *polymer* X is a finite connected union of L -blocks; write $|X|$ for the number of L -blocks in X . Let $E(X)$ be the set of oriented fine edges contained in X .

We work in spanning-tree gauge on each L -block so that gauge orbits are coordinatized by the remaining “free” edges. All functions below are gauge invariant; the gauge fixing is used only to define derivatives and norms.

Definition F.277 (Polymer activity). A polymer activity is an assignment

$$K : \{X \text{ connected polymer}\} \rightarrow \{K(X, \cdot) : G^{E(X)} \rightarrow \mathbb{R}\}$$

with $K(\emptyset) = 0$ and each $K(X, \cdot)$ smooth and gauge invariant.

For $r \in \mathbb{N}$ define the local C^r norm using left-invariant derivatives on G in each edge variable:

$$\|f\|_{C^r(X)} := \max_{0 \leq m \leq r} \max_{e_1, \dots, e_m \in E(X)} \sup_{U \in G^{E(X)}} \|\nabla_{e_1} \cdots \nabla_{e_m} f(U)\|.$$

Fix a polymer-decay parameter $\kappa > 0$ and define the Banach norm

$$\|K\|_{r, \kappa} := \sup_{B \in \mathcal{B}_L} \sum_{\substack{X \ni B \\ X \text{ connected}}} e^{\kappa|X|} \|K(X)\|_{C^r(X)}. \quad (\text{F317})$$

Let $\mathcal{K}_{r, \kappa}$ be the corresponding Banach space.

Lemma F.278 (Algebra property of the polymer norm). *There exists $C_\times = C_\times(r, \kappa, d)$ such that for any two polymer activities K_1, K_2 one has*

$$\|K_1 \cdot K_2\|_{r, \kappa} \leq C_\times \|K_1\|_{r, \kappa} \|K_2\|_{r, \kappa},$$

where $(K_1 \cdot K_2)(X)$ denotes the standard polymer product obtained by summing over decompositions $X = Y \cup Z$ with Y, Z connected and setting $(K_1 \cdot K_2)(X) = \sum_{Y \cup Z = X} K_1(Y) K_2(Z)$.

Proof. This is standard. For fixed base block B and polymer $X \ni B$, the number of decompositions $X = Y \cup Z$ with Y, Z connected is at most exponential in $|X|$; absorb this into $e^{\kappa|X|}$ by reducing κ (or by an explicit combinatorial factor). Derivatives satisfy a Leibniz rule, giving a factor depending on r . Summing over $X \ni B$ yields the stated bound. $\square$

b. Coordinates (u, K) for effective actions and extraction

Let $A(U_p)$ be the standard plaquette energy density (normalized as in the earlier α -pivot section). For a given scale, represent a gauge-fixed effective measure on edges by

$$d\mu_{u, K}(U) = \frac{1}{Z(u, K)} \exp\left(-u^{-1} V(U)\right) \exp\left(\sum_{X \subset \Lambda \text{ conn.}} K(X, U_{E(X)})\right) d\mu_0(U), \quad (\text{F318})$$

where $d\mu_0(U)$ is a fixed gauge-invariant reference measure at that scale (e.g. the gauge-fixed heat-kernel/Gaussian reference from weak-coupling localization), and $V(U) := \sum_{p \in P(\Lambda)} A(U_p)$ is the extracted dimension-4 local functional.

Extraction convention. Write $\mathcal{P}$ for the projection onto the local subspace spanned by the constant functional and the plaquette functional $\sum_p A(U_p)$. We impose

$$\mathcal{P}K = 0, \quad (\text{F319})$$

so that u is the unique plaquette coupling coordinate. (This is the standard “local part extraction” step in constructive RG.)

c. Exact blocking map on (u, K)

Let $\mathcal{R}_L$ denote the exact blocking transformation by factor L followed by rescaling back to unit lattice spacing. Operationally: (i) choose a gauge-covariant block map B_L from

fine edges to coarse edges; (ii) define the coarse marginal by integrating out fine degrees of freedom conditional on the coarse configuration; (iii) re-expand the resulting coarse measure in the form (F318) and apply extraction (F319). This yields

$$\boxed{\mathcal{R}_L : (u, K) \mapsto (u', K').}$$

d. Explicit blocking map and one-step re-expansion

This subsection makes the RG step in Section F 25 c completely explicit. It fixes a canonical gauge-covariant block map B_L and a smoothing kernel $T_{L,\alpha}$ so that the “exact blocked measure” is a well-defined probability measure on coarse links, and it records the concrete re-expansion procedure that produces the extracted coordinates (u', K') .

Block lattice and canonical path-ordered block map Let $\Lambda \subset \mathbb{Z}^d$ be a finite periodic box of side length divisible by $L \geq 2$. Let Λ_L denote the coarse lattice with sites $x^L \in (\mathbb{Z}/L\mathbb{Z})^d$ and edges $E(\Lambda_L)$. For each coarse oriented edge $e^L = (x^L \rightarrow x^L + \hat{\mu})$, define its canonical fine path

$$\text{path}_L(e^L) := \left\{ (Lx^L + j\hat{\mu} \rightarrow Lx^L + (j+1)\hat{\mu}) : j = 0, 1, \dots, L-1 \right\} \subset E(\Lambda).$$

Define the canonical block map $B_L : \mathcal{U}_\Lambda \rightarrow \mathcal{U}_{\Lambda_L}$ by

$$(B_L U)_{e^L} := \prod_{j=0}^{L-1} U_{(Lx^L + j\hat{\mu}) \rightarrow (Lx^L + (j+1)\hat{\mu})}, \quad (\text{F320})$$

where the product is ordered along the path from the tail to the head of e^L .

Lemma F.279 (Gauge covariance of the canonical block map). *Let $g : \Lambda \rightarrow G$ be a fine-lattice gauge transformation and define the induced coarse transformation $g^L : \Lambda_L \rightarrow G$ by $g^L(x^L) := g(Lx^L)$. Then*

$$B_L(g \cdot U) = g^L \cdot (B_L U).$$

Proof. For a fine edge $(x \rightarrow y)$, $(g \cdot U)_{x \rightarrow y} = g(x)U_{x \rightarrow y}g(y)^{-1}$. Along the path $\text{path}_L(e^L)$, the intermediate $g(\cdot)^{-1}g(\cdot)$ factors telescope, leaving $g(Lx^L)$ on the left and $g(Lx^L + L\hat{\mu})^{-1} = g(Lx^L + \hat{\mu})^{-1}$ on the right. $\square$

Smoothing kernel and exact blocked measure To avoid distributional group delta constraints, we define the blocked measure using a positive-type smoothing kernel. Fix $\alpha > 0$ and let $K_\alpha : G \rightarrow (0, \infty)$ be the heat kernel at time α (normalized so $\int_G K_\alpha(g) dg = 1$). Define the gauge-invariant Markov kernel on coarse links by

$$T_{L,\alpha}(U^L | U) := \prod_{e^L \in E(\Lambda_L)} K_\alpha \left(U_{e^L}^L \left((B_L U)_{e^L} \right)^{-1} \right). \quad (\text{F321})$$

Normalization of K_α implies $\int T_{L,\alpha}(U^L | U) dU^L = 1$ for each fixed U .

Given a fine-lattice gauge measure μ_Λ on $\mathcal{U}_\Lambda$, define the exact blocked measure on coarse links by

$$\mu_{\Lambda_L}^{\text{ex}}(dU^L) := \int_{\mathcal{U}_\Lambda} \mu_\Lambda(dU) T_{L,\alpha}(U^L | U). \quad (\text{F322})$$

Lemma F.280 (Gauge invariance of the blocked measure). *If μ_Λ is gauge invariant, then $\mu_{\Lambda_L}^{\text{ex}}$ is gauge invariant on Λ_L .*

Proof. Gauge invariance of μ_Λ and Lemma F.279 reduce the claim to conjugation invariance of the heat kernel: $K_\alpha(hgh^{-1}) = K_\alpha(g)$, which holds because K_α is a class function. $\square$

Exact RG map as re-expansion in the (u, K) coordinates Let the fine measure have the (u, K) form (F318) at unit lattice spacing. Define the blocked (coarse) density by (F322), then rescale Λ_L back to unit spacing (by the canonical identification of Λ_L with a unit lattice). Denote the resulting unit-spacing measure by $\mathcal{B}_{L,\alpha}\mu$.

The *exact RG map* $\mathcal{R}_L$ is the composition:

$$(u, K) \xrightarrow{\text{form } \mu} \mu \xrightarrow{\mathcal{B}_{L,\alpha}} \mathcal{B}_{L,\alpha}\mu \xrightarrow{\text{extract local part}} (u', K'),$$

where the last arrow is the fixed extraction convention (F319) (local plaquette sector absorbed into $u'V$).

Target one-step polymer decomposition The definitions above are unconditional; the nontrivial content is the *quantitative* statement that the blocked measure $\mathcal{B}_{L,\alpha}\mu$ can be re-expanded with a polymer remainder in the Banach norm uniformly in volume. We record the needed statement as a named proposition.

Proposition F.281 (One-step polymer re-expansion of the exact blocked measure). *Fix (d, G, r, κ) and choose $L \geq 2$. There exist $u_0 > 0$, $\varepsilon_0 > 0$ and constants $C_u, C_K < \infty$ (depending only on $(d, G, L, r, \kappa, \alpha)$) such that: whenever $|u| \leq u_0$ and $\|K\|_{r,\kappa} \leq \varepsilon_0$, the*

blocked-and-rescaled measure $\mathcal{B}_{L,\alpha}\mu$ admits a representation (F318) with extracted coordinates (u', K') satisfying

$$|u' - u - b_0 u^2 \log L| \leq C_u u^3 + C_u u \|K\|_{r,\kappa}, \quad (\text{F323})$$

$$\|K'\|_{r,\kappa} \leq C_K (L^{-2} \|K\|_{r,\kappa} + u^2). \quad (\text{F324})$$

Moreover, (u', K') depends analytically on (u, K) on the neighborhood $[-u_0, u_0] \times B_{\varepsilon_0}(0)$.

e. One-step re-expansion proof (small/large field decomposition + polymer cluster expansion)

This subsection completes the proof of Proposition F.281 by supplying two missing ingredients:

1. a *uniform* small-field/large-field decomposition for the conditional block integration defining the exact blocked measure (F322), and
2. a polymer/cluster expansion for the resulting effective interaction, yielding an explicit KP majorant bound on the blocked remainder.

The argument is standard in constructive RG, but we record it here with explicit constants and clear parameter-dependence. No new probabilistic hypotheses are introduced: the only analytic inputs are the already-proved weak-localization tail bound (Lemma F.224) and the curvature Brascamp–Lieb domination on the small-field chart (Proposition F.226), together with the compactness/boundedness of the heat kernel K_α .

(1) *Small-field chart, block gauge-fixing, and conditional measures* Fix the chart radius ε_0 from Lemma F.224 so that on Ω_{ε_0} each plaquette holonomy admits a unique logarithm $\xi_p = \log U_p \in \mathfrak{g}$ with $\|\xi_p\| \leq \varepsilon_0$. Fix once and for all a spanning tree T_B inside each L -block $B \in \mathcal{B}_L$ and impose tree gauge on T_B so that the remaining “free” edges coordinatize the gauge orbit.

For a fixed coarse configuration $U^L \in G^{E(\Lambda_L)}$, define the fine conditional weight

$$W_{U^L}(U) := \exp(-uV(U)) \prod_X (1 + K(X, U_X)) T_{L,\alpha}(U^L | U), \quad (\text{F325})$$

and the corresponding conditional probability measure

$$\mu_{u,K}^{U^L}(dU) := \frac{1}{Z(U^L)} W_{U^L}(U) dU, \quad Z(U^L) := \int W_{U^L}(U) dU. \quad (\text{F326})$$

Lemma F.282 (Uniform boundedness of the conditional density). *Let $M_\alpha := \|K_\alpha\|_\infty < \infty$. Then for every U^L and every (u, K) ,*

$$0 \leq W_{U^L}(U) \leq M_\alpha^{|E(\Lambda_L)|} \exp(-uV(U)) \prod_X (1 + |K(X, U_X)|).$$

In particular, on the small polymer neighborhood $\|K\|_{r,\kappa} \leq \varepsilon_0 \leq 1/4$ one has $\prod_X (1 + |K(X, U_X)|) \leq \exp(C_K^{(0)} \varepsilon_0 |\Lambda|)$ for an explicit $C_K^{(0)}(d, L, \kappa)$.

Proof. The heat kernel is bounded by M_α and the kernel (F321) is a product over coarse edges. The second claim follows by the crude estimate $\log(1 + |K(X, U_X)|) \leq 2|K(X, U_X)|$ for $|K| \leq 1/2$, together with the polymer norm definition and the bound $\sum_{X \ni B} \|K(X)\|_\infty \leq \|K\|_{0,\kappa} \leq \varepsilon_0$ for each block B . Summing over the $O(|\Lambda|/L^4)$ blocks gives the stated volume factor. $\square$

(2) *Large-field suppression at the block-integration level* Let Ω_{ε_0} denote the global small-curvature event (all plaquettes in chart). Write $\Omega_{\varepsilon_0}^c$ for its complement.

Lemma F.283 (Large-field tail for conditional block integration). *There exist constants $c_{\text{LF}} > 0$ and $\beta_{\text{LF}} < \infty$ (depending only on (d, G, ε_0)) such that for all $\beta \geq \beta_{\text{LF}}$, for all U^L and all (u, K) in the small polymer neighborhood,*

$$\mu_{u,K}^{U^L}(\Omega_{\varepsilon_0}^c) \leq C_{\text{LF}} |\Lambda| e^{-c_{\text{LF}}\beta}, \quad (\text{F327})$$

with C_{LF} depending only on (d, G, L, α) . Consequently, the contribution of $\Omega_{\varepsilon_0}^c$ to the blocked measure can be absorbed into the polymer remainder as an error term of order $e^{-c_{\text{LF}}\beta}$.

Proof. On $\Omega_{\varepsilon_0}^c$, at least one plaquette holonomy lies outside the normal neighborhood. The Wilson plaquette energy satisfies a uniform positive lower bound away from identity, hence the action cost is at least $c_{\text{LF}}\beta$ per “bad” plaquette (after fixing normalizations once). Lemma F.224 gives the corresponding unconditional bound $\mu_{u,0}(\Omega_{\varepsilon_0}^c) \leq |P(\Lambda)|e^{-c\beta}$. By Lemma F.282, inserting the bounded kernel and the small K perturbation changes probabilities only by a bounded Radon–Nikodym factor on $\Omega_{\varepsilon_0}^c$; absorb this into C_{LF} . $\square$

(3) *Polymerization of the small-field conditional integral* Work henceforth on Ω_{ε_0} . In block tree gauge, write the free edge variables in exponential coordinates $U_e = \exp(A_e)$ with $A_e \in \mathfrak{g}$ and $\|A_e\| \leq C(\varepsilon_0)$. On Ω_{ε_0} , the plaquette logarithms satisfy $\xi_p = (dA)_p + \text{Rem}_p(A)$ with $\|\text{Rem}_p(A)\| \leq C_{\text{rem}}\|A\|^2$.

Let $\mathcal{N}$ be the Gaussian (quadratic) reference measure on the free edge variables induced by the Hessian lower bound m_- from (F278), so that on Ω_{ε_0} the conditional density is a perturbation of a strictly log-concave Gaussian field on the block graph.

Lemma F.284 (Connected polymer activity bound for one RG step). *Fix the block factor $L \geq 2$. There exist constants C_{pol} and $\kappa_{\text{pol}} > 0$ (depending only on $(d, G, L, \varepsilon_0, r, \kappa, \alpha)$) such that the small-field contribution to the blocked density admits a polymer product representation*

$$\left. \frac{d\mathcal{B}_{L,\alpha}\mu_{u,K}}{d\mu_{u'}^{\text{loc}}}(U^L) \right|_{\Omega_{\varepsilon_0}} = \prod_{\substack{X \subset \mathcal{B}_L \\ X \text{ connected, } |X| \geq 2}} \left(1 + R'_X(U_{E(X)}^L)\right), \quad (\text{F328})$$

where $\mu_{u'}^{\text{loc}}$ is the extracted local reference measure at the coarse scale (one-plaquette sector absorbed into $u'V$), and the polymer remainders satisfy the uniform activity bound

$$\sup_{B \in \mathcal{B}_L} \sum_{X \ni B} e^{\kappa_{\text{pol}}|X|} \|R'_X\|_{C^r(X)} \leq C_{\text{pol}} \left(u^2 + |u| \|K\|_{r,\kappa} + \|K\|_{r,\kappa}^2 \right). \quad (\text{F329})$$

The constants are uniform in the volume and admissible boundary conditions.

Proof. We give a complete cluster/forest-expansion argument with explicit constants.

Throughout this proof we are in the extracted polymer neighborhood $|u| \leq u_0$ and $\|K\|_{r,\kappa} \leq \varepsilon_0$ (with $u_0, \varepsilon_0 > 0$ chosen sufficiently small in the weak window), and we work at fixed finite volume with admissible boundary conditions. In particular, all subset/polymer regroupings below are purely algebraic (finite sums). The polymer product representation is defined by the finite-volume linked-cluster (Ursell) formulas; the smallness of $(u, \|K\|_{r,\kappa})$ is used only to obtain the quantitative activity bound (F329) with uniform constants.

For notational convenience set

$$\mathfrak{f} := u^2 + |u| \|K\|_{r,\kappa} + \|K\|_{r,\kappa}^2.$$

Step 1: local interaction decomposition on the small-field set. On Ω_{ε_0} , after extracting the one-plaquette sector into the coarse local reference measure $\mu_{u'}^{\text{loc}}$, the remaining blocked density ratio can be written in the form

$$\left. \frac{d\mathcal{B}_{L,\alpha}\mu_{u,K}}{d\mu_{u'}^{\text{loc}}}(U^L) \right|_{\Omega_{\varepsilon_0}} = \exp \left(\sum_{b \in \mathcal{J}} W_b(U^L) \right), \quad (\text{F330})$$

where $\mathcal{J}$ is a finite-range collection of *interaction units* b at the L -blocked scale (boundary plaquettes and the finite-range couplings coming from $T_{L,\alpha}$), and each W_b depends only on

U^L restricted to at most $q = q(d, L)$ coarse blocks. After the one-plaquette extraction, every W_b is at least quadratic in the centered curvature coordinates on Ω_{ε_0} . Consequently, by the uniform curvature moment bounds on Ω_{ε_0} (Proposition F.226) and the product/regularity axioms of the local seminorms, there exist constants $C_{W,0}$ and $C_{W,r}$ (depending only on $(d, G, L, \varepsilon_0, r, \kappa, \alpha)$) such that, uniformly in b and in the volume/boundary conditions,

$$\sup_{\Omega_{\varepsilon_0}} |W_b| \leq C_{W,0} \mathfrak{f}, \quad \|W_b\|_{C^r(b)} \leq C_{W,r} \mathfrak{f}. \quad (\text{F331})$$

(Here $\|\cdot\|_{C^r(b)}$ denotes the C^r seminorm in the block variables on the support of b . The point is that, on Ω_{ε_0} , the configuration space is compact in the chart and the interaction units are uniformly local.)

Step 2: polymer expansion by regrouping monomials. For each interaction unit $b \in \mathcal{J}$ define the (bounded) local factor

$$V_b := e^{W_b} - 1.$$

Then

$$\exp\left(\sum_{b \in \mathcal{J}} W_b\right) = \prod_{b \in \mathcal{J}} e^{W_b} = \prod_{b \in \mathcal{J}} (1 + V_b) = \sum_{\Gamma \subset \mathcal{J}} \prod_{b \in \Gamma} V_b.$$

Using $|e^z - 1| \leq e^{|z|}|z|$ and the standard Faà di Bruno bound for derivatives of compositions, and the uniform bounds (F331) on Ω_{ε_0} , there exist explicit constants $C_{V,0}$ and $C_{V,r}$ (depending only on $(d, G, L, \varepsilon_0, r, \kappa, \alpha)$) such that, uniformly in b ,

$$\sup_{\Omega_{\varepsilon_0}} |V_b| \leq C_{V,0} \mathfrak{f}, \quad \|V_b\|_{C^r(b)} \leq C_{V,r} \mathfrak{f}. \quad (\text{F332})$$

Each monomial is supported on the union of coarse blocks touched by Γ ; let $X(\Gamma)$ be this block support. Regroup terms by X and define the *raw polymer weights*

$$A_X(U^L) := \sum_{\Gamma: X(\Gamma)=X} \prod_{b \in \Gamma} V_b(U^L), \quad (\text{F333})$$

so that the density ratio is $1 + \sum_X A_X$. Because each interaction unit crosses a block boundary (after the one-plaquette extraction), we have $A_X \equiv 0$ for $|X| = 1$. Moreover, A_X depends only on the restriction of U^L to $E(X)$.

If X is connected with $|X| = n$, then any Γ with $X(\Gamma) = X$ must contain at least $n - 1$ interaction units (since the interaction units induce an edge set on the block-adjacency graph, and a connected n -vertex graph has at least $n - 1$ edges). Consequently, using

(F332) and the product/regularity axioms of the C^r seminorms, there exists a constant $C_A = C_A(d, G, L, \varepsilon_0, r, \kappa, \alpha)$ such that

$$\|A_X\|_{C^r(X)} \leq (C_A \mathfrak{f})^{|X|-1} \quad \text{for all connected } X \text{ with } |X| \geq 2. \quad (\text{F334})$$

Step 3: connected extraction via the BKAR/forest formula. Apply the standard linked-cluster (Ursell) expansion to the polymer family $\{A_X\}$ to write

$$\log \left(1 + \sum_X A_X \right) = \sum_{\substack{X \text{ connected} \\ |X| \geq 2}} \Phi_X, \quad (\text{F335})$$

where each Φ_X is supported on X and is given by a finite-volume sum over forests (equivalently, spanning trees) via the Brydges–Kennedy–Abdesselam–Rivasseau forest formula for Ursell coefficients.

A key combinatorial point is that for any finite family of polymers $X_1, \dots, X_m$ whose overlap graph is connected and whose union is X , one has

$$\sum_{j=1}^m (|X_j| - 1) \geq |X| - 1, \quad (\text{F336})$$

since $|X| \leq \sum_{j=1}^m |X_j| - (m - 1)$ for a connected overlap graph. Consequently, using (F334) and the product/regularity axioms of the C^r seminorms, every cluster term contributing to Φ_X carries at least $|X| - 1$ factors of $\mathfrak{f}$.

The forest formula bounds the Ursell coefficients by a sum over trees and derivative book-keeping at order r ; this contributes only an explicit combinatorial factor (absorbed below). Therefore there exists a constant $C_\Phi = C_\Phi(d, G, L, \varepsilon_0, r, \kappa, \alpha)$ such that, for all connected X with $|X| \geq 2$,

$$\|\Phi_X\|_{C^r(X)} \leq (C_\Phi \mathfrak{f})^{|X|-1}. \quad (\text{F337})$$

(For instance, one may take $C_\Phi := e C_A C_{\text{count}}^{\text{tree}}(r)$, enlarging constants if desired.)

Step 4: product representation and activity bounds. Exponentiating (F335) gives

$$\exp \left(\sum_{\substack{X \text{ connected} \\ |X| \geq 2}} \Phi_X \right) = \prod_{\substack{X \text{ connected} \\ |X| \geq 2}} \exp(\Phi_X),$$

so (F328) holds with

$$1 + R'_X := \exp(\Phi_X), \quad \text{i.e.} \quad R'_X := \exp(\Phi_X) - 1. \quad (\text{F338})$$

By shrinking the polymer neighborhood (i.e. choosing u_0 and ε_0 sufficiently small), we may assume $C_\Phi \mathfrak{f} \leq \frac{1}{2}$. Since $|X| \geq 2$, this implies $\|\Phi_X\|_{C^0(X)} \leq 1$ for all connected X . Using $|e^z - 1| \leq e^{|z|}|z|$ and the standard Faà di Bruno/Leibniz bounds for derivatives of compositions, there is a constant $C_R = C_R(r) \geq 1$ such that

$$\|R'_X\|_{C^r(X)} \leq C_R e^{\|\Phi_X\|_{C^0(X)}} \|\Phi_X\|_{C^r(X)} \leq (C'_\Phi \mathfrak{f})^{|X|-1}, \quad (\text{F339})$$

where one may take $C'_\Phi := (C_R e) C_\Phi$ (since $C_R e \geq 1$ and $|X| - 1 \geq 1$).

Step 5: polymer counting and the weighted sum bound. Let N_n be the number of connected block-polymers X with $|X| = n$ containing a fixed block B . A standard lattice-animal bound gives $N_n \leq (C_{\text{count}}^{\text{blk}})^n$ with $C_{\text{count}}^{\text{blk}}$ depending only on d .

Fix $\kappa_{\text{pol}} > 0$ (for instance $\kappa_{\text{pol}} = 1$). Then for each block B , using (F339),

$$\sum_{X \ni B} e^{\kappa_{\text{pol}}|X|} \|R'_X\|_{C^r(X)} \leq \sum_{n \geq 2} (C_{\text{count}}^{\text{blk}})^n e^{\kappa_{\text{pol}}n} (C'_\Phi \mathfrak{f})^{n-1}.$$

Write $m = n - 2$ and factor out the $n = 2$ contribution. This is

$$\leq (C_{\text{count}}^{\text{blk}} e^{\kappa_{\text{pol}}})^2 (C'_\Phi \mathfrak{f}) \sum_{m \geq 0} (C_{\text{count}}^{\text{blk}} e^{\kappa_{\text{pol}}} C'_\Phi \mathfrak{f})^m.$$

Choose the weak-window constants u_0 and ε_0 so that

$$C_{\text{count}}^{\text{blk}} e^{\kappa_{\text{pol}}} C'_\Phi \mathfrak{f} \leq \frac{1}{2} \quad \text{throughout the polymer neighborhood.} \quad (\text{F340})$$

Then the geometric series is bounded by 2, and we obtain

$$\sup_{B \in \mathcal{B}_L} \sum_{X \ni B} e^{\kappa_{\text{pol}}|X|} \|R'_X\|_{C^r(X)} \leq 2 (C_{\text{count}}^{\text{blk}} e^{\kappa_{\text{pol}}})^2 C'_\Phi \mathfrak{f}.$$

This yields (F329) upon defining $C_{\text{pol}} := 2 (C_{\text{count}}^{\text{blk}} e^{\kappa_{\text{pol}}})^2 C'_\Phi$. Uniformity in the volume and admissible boundary conditions follows because all bounds were local on Ω_{ε_0} and the counting bound is uniform. $\square$

(4) Completion of Proposition F.281

Proof of Proposition F.281. Fix $(d, G, L, r, \kappa, \alpha)$.

Step 1: split into small-field and large-field parts. Write the blocked density as

$$\mu_{\Lambda_L}^{\text{ex}}(dU^L) = \int_{\Omega_{\varepsilon_0}} \mu_{u,K}(dU) T_{L,\alpha}(U^L \mid U) + \int_{\Omega_{\varepsilon_0}^c} \mu_{u,K}(dU) T_{L,\alpha}(U^L \mid U).$$

By Lemma F.283, the second term contributes an additive error to the blocked density bounded by $O(|\Lambda|e^{-c_{\text{LF}}\beta})$ uniformly in U^L . Choose β (equivalently, choose u_0 small in the weak window) so that $|\Lambda|e^{-c_{\text{LF}}\beta} \leq u^3$ on the neighborhood of interest; absorb this into the $C_u u^3$ term in (F323) and into the u^2 source term in (F324) after enlarging constants.

Step 2: small-field polymer product representation. On Ω_{ε_0} , Lemma F.284 gives the product form (F328) relative to an extracted local reference $\mu_{u'}^{\text{loc}}$ and an activity bound (F329). Choose u_0, ε_0 small so that the RHS of (F329) is $\leq 1/4$. Then the blocked remainder majorant q_{rem} defined by (F51) satisfies $q_{\text{rem}} \leq C_{\text{pol}}(u^2 + |u| \|K\| + \|K\|^2)$. Lemma F.285 converts this into a Banach-norm polymer activity K' with

$$\|K'\|_{r,\kappa} \leq C_r q_{\text{rem}} \leq C_r C_{\text{pol}}(u^2 + |u| \|K\|_{r,\kappa} + \|K\|_{r,\kappa}^2).$$

Shrink ε_0 further so that $\|K\|^2 \leq \|K\|$ on the neighborhood (e.g. $\varepsilon_0 \leq 1$) and absorb the mixed and quadratic terms into a constant multiple of $u^2 + \|K\|$. Finally, the engineering-dimension rescaling bound (Lemma F.290) yields the explicit L^{-2} factor on the inherited irrelevant part of K , giving (F324) after enlarging C_K .

Step 3: coupling update. By construction, u' is the extracted coefficient of the one-plaquette sector in the blocked local reference. At $K = 0$, the weak-window step-scaling control (Proposition F.228) identifies the one-loop coefficient b_0 and yields $u'(u, 0) = u + b_0 u^2 \log L + O(u^3)$ uniformly. The linear dependence of u' on K is controlled by the same polymer activity bound (F329) (a single insertion of K changes the extracted one-plaquette coefficient by at most $C|u|\|K\|$), yielding (F323).

Step 4: analyticity. On Ω_{ε_0} the integrand is real-analytic in the exponential coordinates and the polymer sums converge absolutely under the smallness condition; on $\Omega_{\varepsilon_0}^c$ the contribution is exponentially small and analytic by dominated convergence. Thus (u', K') depends analytically on (u, K) on the neighborhood.

This proves the proposition. □

f. From KP-small polymer remainders to the Banach polymer norm

The stable-manifold theorem below is stated in a polymer Banach norm (F317). Earlier in the paper, measure-level stability inside the KP basin is formulated in terms of the remainder majorant q_{rem} defined from the sup-activity bounds $\rho(P)$ in Definition F.50. This subsection supplies the bridge:

KP-smallness of the remainder (in the sense $q_{\text{rem}} < 1$) implies smallness of the corresponding polymer activity in the Banach norm $\|\cdot\|_{r,\kappa}$, after passing from the product form to an exponential polymer action.

Standing simplification. As in Section F 12, we assume the one-plaquette sector has been absorbed into the local reference measure μ^{loc} , i.e. $\rho(P) = 0$ for $|P| = 1$.

Lemma F.285 (From q_{rem} to a Banach-norm polymer activity). *Assume $q_{\text{rem}} < 1/4$ in (F51) and $\rho(P) = 0$ for $|P| = 1$. Define the logarithmic polymer activity*

$$K(P, U_P) := \log(1 + R_P(U_P)), \quad |P| \geq 2, \quad (\text{F341})$$

and $K(P, \cdot) = 0$ for $|P| \leq 1$. Then for each fixed derivative order $r \in \mathbb{N}$ there exist constants $C_r = C_r(G, r)$ and a polymer-decay parameter $\kappa = \kappa(\theta, r, G) > 0$ such that $K \in \mathcal{K}_{r,\kappa}$ and

$$\boxed{\|K\|_{r,\kappa} \leq C_r q_{\text{rem}}.} \quad (\text{F342})$$

The constants can be chosen uniformly in volume and boundary conditions.

Proof. Since $q_{\text{rem}} < 1/4$, one has $\sup_P \rho(P) \leq q_{\text{rem}} < 1/4$ and therefore $\|R_P\|_\infty \leq 1/4$. For $|z| \leq 1/2$, $|\log(1 + z)| \leq 2|z|$, hence

$$\|K(P)\|_\infty \leq 2\rho(P). \quad (\text{F343})$$

To control derivatives, note that each R_P is a smooth (indeed real-analytic) gauge-invariant function on the compact manifold $G^{E(P)}$, obtained by finitely many Haar integrations of products of smooth single-plaquette weights. Repeated Leibniz rules on products over the $O(|P|)$ local factors yield a uniform bound

$$\|R_P\|_{C^r(P)} \leq C_r^{|P|} \rho(P), \quad (\text{F344})$$

where C_r depends only on (G, r) . By the chain rule for $\log(1 + R_P)$ and the bound $\|R_P\|_\infty \leq 1/4$, the same estimate holds for $K(P)$ up to adjusting C_r .

Choose $\kappa > 0$ so that $e^\kappa C_r \leq e^\theta$. Then, for any base plaquette p_0 ,

$$\sum_{P \ni p_0} e^{\kappa|P|} \|K(P)\|_{C^r(P)} \leq \sum_{P \ni p_0} e^{\theta|P|} \rho(P) \leq q_{\text{rem}},$$

which gives (F342). □

Proposition F.286 (Entrance into the small polymer neighborhood (Banach norm)). *Fix (d, G, b, θ) and choose $r \geq 2$. Assume the effective single-plaquette weight is positive type with activity f and lies in the KP admissibility window (Section F.3). Assume further that the one-plaquette sector has been absorbed into the local reference μ^{loc} (so $\rho(P) = 0$ for $|P| = 1$). Then for all sufficiently small f the exact blocked measure admits a Banach-space representation (F318) with*

$$|u| \leq u_0, \quad \|K\|_{r,\kappa} \leq \varepsilon_0, \quad (\text{F345})$$

where one may take $u_0 = O(f)$ and $\varepsilon_0 = O(f^2)$. In particular, for $f \leq f_{\text{adm}}$ the quadratic remainder majorant (F57) gives $q_{\text{rem}} = O(f^2)$ and Lemma F.285 yields an explicit ε_0 .

Moreover, on the neighborhood (F345), the exact blocking map $\mathcal{R}_L$ is analytic as a map $[-u_0, u_0] \times B_{\varepsilon_0}(0) \subset \mathbb{R} \times \mathcal{K}_{r,\kappa} \rightarrow \mathbb{R} \times \mathcal{K}_{r,\kappa}$.

Proof. Inside the KP admissibility window, f is small. After absorbing the one-plaquette sector, the remainder majorant is quadratic in f as in (F57); hence $q_{\text{rem}} = O(f^2)$. Lemma F.285 converts q_{rem} into a Banach-norm bound $\|K\|_{r,\kappa} \leq C_r q_{\text{rem}} = O(f^2)$. The extracted coupling coordinate u is the coefficient of the absorbed one-plaquette sector; in the activity parameterization it satisfies $|u| = O(f)$. This gives (F345).

Analyticity of $\mathcal{R}_L$ is standard: in the polymer representation the blocked weights are obtained by finite Haar integrations of analytic functions and the polymer sums converge absolutely when $\|K\|_{r,\kappa}$ is sufficiently small. $\square$

Corollary F.287 (Entrance alignment to the invariant graph). *Let h be the invariant graph from Theorem F.293. Under the hypotheses of Proposition F.286,*

$$\|K - h(u)\|_{r,\kappa} \leq \|K\|_{r,\kappa} + \|h(u)\|_{r,\kappa} \leq \varepsilon_0 + C_h u^2,$$

so in particular $\|K - h(u)\|_{r,\kappa} = O(u^2)$ when $u = O(f)$ and $\varepsilon_0 = O(f^2)$.

Lemma F.288 (Local invertibility of the coupling update). *Let $u'(u, K)$ denote the first coordinate of $\mathcal{R}_L(u, K)$. In the weak-coupling window where the step-scaling expansion holds (cf. Proposition F.228), one has on $|u| \leq u_0$, $\|K\|_{r,\kappa} \leq \varepsilon_0$,*

$$\left| \partial_u u'(u, K) - 1 \right| + \|\partial_K u'(u, K)\| \leq C(|u| + \|K\|_{r,\kappa})$$

for a constant C depending only on (d, G, L) and the chosen norms. Consequently, for u_0, ε_0 sufficiently small, for every Lipschitz graph h with $\sup \|h(u)\| \leq \varepsilon_0$ the restricted map

$F_h(u) := u'(u, h(u))$ is a C^1 diffeomorphism of $[-u_0, u_0]$ onto its image and has a uniformly Lipschitz inverse.

Proof. The step-scaling control gives a representation $u'(u, 0) = u + \beta_L(u)$ with $\beta_L(u) = O(u^2 \log L)$ and $\partial_u u'(u, 0) = 1 + O(u)$. The dependence on K enters through cumulant corrections and is analytic with first derivative bounded by $O(|u| + \|K\|)$ in the small polymer neighborhood (Proposition F.286); this is the standard fact that inserting one polymer activity changes only local conditional expectations. Choosing u_0, ε_0 so that $C(u_0 + \varepsilon_0) \leq \frac{1}{2}$ yields strict monotonicity and the inverse function theorem. $\square$

g. Irrelevant contraction: the L^{-2} factor from engineering dimension

The stable-manifold theorem requires a uniform contraction on the *irrelevant* sector. The contraction factor L^{-2} in $d = 4$ is the standard dimensional gap once the dimension- ≤ 4 local operators have been extracted.

Definition F.289 (Dimension filtration on local gauge-invariant actions). A local gauge-invariant operator has *engineering dimension* Δ if in the small-curvature chart it is a polynomial in curvature F and covariant derivatives D with the usual assignment $\dim(F) = 2$ and $\dim(D) = 1$. Let $\mathcal{V}_{\leq 4}$ be the span of $\{\mathbf{1}, \sum_p A(U_p)\}$. Define the irrelevant subspace $\mathcal{V}_{\geq 6}$ to be the (closure of the) span of local gauge-invariant operators with $\Delta \geq 6$.

Lemma F.290 (Linear rescaling bound on irrelevant local terms). *Let $\mathcal{S}_L$ be the rescaling map associated to blocking by factor L (after extraction of $\mathcal{V}_{\leq 4}$). On the irrelevant sector one has*

$$\|\mathcal{S}_L\|_{\mathcal{V}_{\geq 6}} \leq C_{\dim} L^{-2},$$

where $C_{\dim}$ depends only on the choice of local basis and the fixed derivative order r in (F317).

Proof. Write an irrelevant local functional as a finite sum $\sum_i c_i \mathcal{O}_i$ with each $\mathcal{O}_i$ of dimension $\Delta_i \geq 6$. Under the change of lattice spacing $a \mapsto La$, the coefficient rescales as $c_i \mapsto L^{4-\Delta_i} c_i$ by standard power counting in $d = 4$. Since $\Delta_i \geq 6$, $|L^{4-\Delta_i}| \leq L^{-2}$. The C^r norm introduces only a fixed multiplicative constant because left-invariant derivatives commute with group translation and the charts are fixed at each scale; absorb this into $C_{\dim}$. $\square$

Proposition F.291 (Irrelevant linearization of $\mathcal{R}_L$ contracts by $O(L^{-2})$). *Assume the small neighborhood of Proposition F.286. Let $D_K \mathcal{R}_L(u, 0)$ denote the Fréchet derivative in the polymer direction at $K = 0$. Then there exists $C_{\text{irr}} = C_{\text{irr}}(r, \kappa, d, G)$ such that for all $|u| \leq u_0$,*

$$\|D_K \mathcal{R}_L(u, 0)\|_{\mathcal{K}_{r,\kappa} \rightarrow \mathcal{K}_{r,\kappa}} \leq C_{\text{irr}} L^{-2}.$$

Proof. At $K = 0$ the measure (F318) is the reference measure μ_0 tilted by uV . The derivative $D_K \mathcal{R}_L(u, 0)$ corresponds to inserting a single polymer activity into the block integration and propagating it through blocking and extraction. Extraction removes the $\mathcal{V}_{\leq 4}$ components, so only irrelevant local contributions survive in the linearization. These contributions rescale according to Lemma F.290, yielding the factor L^{-2} . The polymer norm (F317) is stable under the finite-range dependence induced by a single block integration; the associated combinatorial constants are absorbed into C_{irr} . $\square$

h. Quadratic remainder bound in the polymer norm

Proposition F.292 (Nonlinear remainder is quadratic in the polymer norm). *Under Proposition F.286, there exists C_{Ξ} such that for all $|u| \leq u_0$ and $\|K\|_{r,\kappa} \leq \varepsilon_0$ one can write*

$$K' = D_K \mathcal{R}_L(u, 0) K + \Xi(u, K),$$

with

$$\|\Xi(u, K)\|_{r,\kappa} \leq C_{\Xi} \left(u^2 + |u| \|K\|_{r,\kappa} + \|K\|_{r,\kappa}^2 \right).$$

Moreover Ξ is locally Lipschitz with the corresponding quadratic Lipschitz constant.

Proof. Expand the blocked partition function and conditional expectations in cumulants with respect to the $K = 0$ reference measure. The linear term is exactly $D_K \mathcal{R}_L(u, 0)K$. All higher cumulants contain at least two insertions of K (or one insertion of K and one insertion from the u -dependence), hence are quadratic in the small parameters. Bounds in the norm (F317) follow from the standard polymer algebra property (Lemma F.278) and finite-range dependence of block integrations. The u^2 term reflects the fact that after extraction the u -flow generates irrelevant operators at order u^2 . $\square$

i. Stable-manifold theorem for $\mathcal{R}_L$ (invariant graph)

We now state and prove a Banach-space invariant-graph theorem tailored to the RG map $\mathcal{R}_L$. The coupling coordinate u is marginally relevant (asymptotic freedom), so the result is a stable foliation only in the irrelevant sector.

Theorem F.293 (Invariant renormalized graph and stable foliation). *Assume Proposition F.286. Choose L so large that $C_{\text{irr}}L^{-2} \leq \frac{1}{4}$, where C_{irr} is from Proposition F.291. Then, for sufficiently small u_0, ε_0 , there exists a unique analytic map*

$$h : [-u_0, u_0] \rightarrow \mathcal{K}_{r,\kappa}, \quad h(0) = 0,$$

with $\|h(u)\|_{r,\kappa} \leq C_h u^2$ such that the graph

$$\mathcal{M} := \{(u, h(u)) : |u| \leq u_0\}$$

is invariant under $\mathcal{R}_L$. Moreover, if $(u_n, K_n) = \mathcal{R}_L^n(u_0, K_0)$ remains in the neighborhood, then

$$\|K_n - h(u_n)\|_{r,\kappa} \leq 2^{-n} \|K_0 - h(u_0)\|_{r,\kappa}.$$

Proof. This is a standard graph-transform argument in the Banach space $\mathbb{R} \times \mathcal{K}_{r,\kappa}$. Define the space $\mathcal{H}$ of Lipschitz maps h with $h(0) = 0$ and $\sup_{|u| \leq u_0} \|h(u)\|/u^2 \leq M$ and $\text{Lip}(h) \leq Mu_0$. Given $h \in \mathcal{H}$, write the u -update along the graph as $u' = F_h(u)$ and denote its local inverse by G_h . Define the graph transform

$$(\mathcal{T}h)(u') := D_K \mathcal{R}_L(G_h(u'), 0) h(G_h(u')) + \Xi(G_h(u'), h(G_h(u'))),$$

where Ξ is from Proposition F.292.

By Proposition F.291, the linear part has operator norm at most $\frac{1}{4}$. By Proposition F.292, the nonlinear remainder is $O(u^2) + O(u\|h\|) + O(\|h\|^2)$. Choosing u_0 and ε_0 so that the quadratic Lipschitz constants are $\leq \frac{1}{4}$ makes $\mathcal{T}$ a contraction on $(\mathcal{H}, \|\cdot\|_*)$. Banach's fixed point theorem gives a unique fixed point h . The contraction estimate for $K_n - h(u_n)$ follows by subtracting the invariance equation from the RG recursion and using the same linear/quadratic bounds. Analyticity follows from analyticity of $\mathcal{R}_L$ in the neighborhood. $\square$

Remark F.294 (Interpretation). The invariant graph $K = h(u)$ is the standard *renormalized trajectory* near the weak-coupling (Gaussian) fixed point. Irrelevant perturbations are

exponentially damped under blocking at rate $\approx L^{-2}$ per step once the local extraction is imposed. This is the mechanism behind universality and uniqueness of continuum limits once the running coupling is fixed by a renormalization condition.

j. Archival finite-depth weak-window branch (moved to supplement)

The earlier finite-depth iterated Banach-space control up to a KP threshold is no longer part of the final one-shot Route 1 proof. That historical material has been moved to the supplementary Route 1 archive. The only retained coupling-identification statement used later in the tuned dyadic normalization is the explicit bare/flow comparison theorem recorded in the next subsection.

k. Coupling identification between the bare coupling and the gradient-flow coupling

The load-bearing tuned dyadic construction uses only an explicit comparison between the bare coupling β^{-1} and the gradient-flow coupling g_{GF}^2 at the same physical scale. No inverse-function theorem on the invariant graph and no finite-depth iterated RG control are used in this subsection.

Bare-to-flow comparison. Fix the physical reference scale ℓ_0 and evaluate the flowed observable directly in the bare Wilson measure at lattice spacing $a = \ell_0 2^{-n}$ and box size $L(a) = \ell_0/a$. The next theorem is the sole weak-window identification used later in the tuned dyadic normalization.

Lemma F.295 (Uniform coefficient majorant for the fixed-scale flowed energy). *Fix the physical reference scale ℓ_0 , a dyadic spacing $a = \ell_0 2^{-n}$, set $L = L(a) = \ell_0/a$, and let $t = cL^2$. Under Lemma F.224 and Proposition F.226, there exist constants $A_{\text{flow}}, R_{\text{flow}} < \infty$ and $\beta_{\text{flow}} \geq \beta_1$, depending only on $(d, G, c, \varepsilon_0, \ell_0)$ and the chosen discretization of $E(t)$, such that on the small-curvature event Ω_{ε_0} one has the absolutely convergent expansion*

$$t^2 E(t) = \mathcal{Q}_2(dA) + \sum_{m \geq 2} \mathcal{Q}_{2m}(dA), \quad (\text{F346})$$

where each $\mathcal{Q}_{2m}$ is homogeneous of degree $2m$ in the curvature field and, uniformly in the

dyadic box size L ,

$$\left| \mathbb{E}_{\beta, \Lambda(L)} \left[\mathcal{Q}_{2m}(dA) \mathbf{1}_{\Omega_{\varepsilon_0}} \right] \right| \leq A_{\text{flow}} \left(\frac{R_{\text{flow}}}{\beta} \right)^m, \quad m \geq 1, \quad (\text{F347})$$

for every $\beta \geq \beta_{\text{flow}}$. In particular, after enlarging β_{flow} so that $R_{\text{flow}}/\beta_{\text{flow}} \leq \frac{1}{2}$,

$$\left| \mathbb{E}_{\beta, \Lambda(L)} \left[\sum_{m \geq 2} \mathcal{Q}_{2m}(dA) \mathbf{1}_{\Omega_{\varepsilon_0}} \right] \right| \leq 2A_{\text{flow}} R_{\text{flow}}^2 \beta^{-2}. \quad (\text{F348})$$

Proof. Step 2 of Lemma F.227 gives the analytic expansion (F346) with an L -independent analyticity radius because the ratio $t/L^2 = c$ is fixed. Thus there exist constants $A_0, R_0 < \infty$, independent of L , such that the coefficient tensor of the degree- $2m$ piece is bounded by $A_0 R_0^m$. Every monomial expectation appearing in $\mathbb{E}[\mathcal{Q}_{2m}(dA) \mathbf{1}_{\Omega_{\varepsilon_0}}]$ is a finite linear combination of joint cumulants of $2m$ linear curvature functionals. By (F284), each such cumulant is bounded by $(2m-1)!(\sigma_\beta^2)^m$ times the product of the corresponding L^2 norms, with $\sigma_\beta^2 = (\beta m_-)^{-1}$. Absorbing the finitely many Wick/contraction patterns and the uniform coefficient bounds into new constants gives (F347) for suitable $A_{\text{flow}}, R_{\text{flow}}$. Summing the geometric series for $m \geq 2$ yields (F348) once $R_{\text{flow}}/\beta \leq \frac{1}{2}$. $\square$

Lemma F.296 (Bad-curvature complement for the fixed-scale flow observable). *Under Lemma F.224, there exist constants $C_{\text{bad}}, c_{\text{bad}} > 0$, depending only on $(d, G, c, \varepsilon_0, \ell_0)$ and the chosen discretization of $E(t)$, such that for every dyadic $a = \ell_0 2^{-n}$ and every $\beta \geq \beta_{\text{flow}}$,*

$$\left| \mathbb{E}_{\beta, \Lambda(L(a))} [t^2 E(t) \mathbf{1}_{\Omega_{\varepsilon_0}^c}] \right| \leq C_{\text{bad}} e^{-c_{\text{bad}} \beta}. \quad (\text{F349})$$

Proof. Lemma F.224 gives an exponentially small bound on $\Omega_{\varepsilon_0}^c$ uniformly in finite volume. For the fixed discretization at flow time $t = cL(a)^2$, the observable $t^2 E(t)$ is a bounded-degree local polynomial in the flowed links, hence on each finite box its absolute value is dominated by a polynomial of uniformly bounded degree in the local curvature variables. Combining this polynomial bound with the exponential suppression of $\Omega_{\varepsilon_0}^c$ yields (F349); after enlarging β_{flow} if necessary, the right-hand side is also $O(\beta^{-2})$. $\square$

Proposition F.297 (Quantitative decomposition of the fixed-scale flow coupling). *Fix the physical reference scale ℓ_0 , a dyadic spacing $a = \ell_0 2^{-n}$, and $t = cL(a)^2$. Under Lemma F.224, Proposition F.226, Lemma F.295, and Lemma F.296, there exist constants $C_{\text{good,GF}}, C_{\text{bad,GF}}, c_{\text{bad,GF}} > 0$, depending only on $(d, G, c, \varepsilon_0, \ell_0)$ and the chosen discretization of $E(t)$, such that*

$$\mathbb{E}_{\beta, \Lambda(L(a))} [t^2 E(t)] = \beta^{-1} + R_{\text{good}}(a, \beta) + R_{\text{bad}}(a, \beta), \quad (\text{F350})$$

with

$$|R_{\text{good}}(a, \beta)| \leq C_{\text{good,GF}} \beta^{-2}, \quad |R_{\text{bad}}(a, \beta)| \leq C_{\text{bad,GF}} e^{-c_{\text{bad,GF}} \beta}, \quad (\text{F351})$$

uniformly in the dyadic box size $L(a)$.

Proof. By Lemma F.295, on Ω_{ε_0} one has the absolutely convergent expansion

$$t^2 E(t) = \mathcal{Q}_2(dA) + \sum_{m \geq 2} \mathcal{Q}_{2m}(dA),$$

with the explicit tail bound (F348). The normalization in (F269) was chosen so that the full expectation of the quadratic term is exactly β^{-1} . Therefore

$$\mathbb{E}[t^2 E(t)] - \beta^{-1} = \mathbb{E}\left[\sum_{m \geq 2} \mathcal{Q}_{2m}(dA) \mathbf{1}_{\Omega_{\varepsilon_0}}\right] + \mathbb{E}\left[(t^2 E(t) - \mathcal{Q}_2(dA)) \mathbf{1}_{\Omega_{\varepsilon_0}^c}\right].$$

The first term is bounded by (F348); this gives $|R_{\text{good}}(a, \beta)| \leq 2A_{\text{flow}} R_{\text{flow}}^2 \beta^{-2}$. For the second term, both $t^2 E(t)$ and the quadratic piece $\mathcal{Q}_2(dA)$ are bounded-degree local curvature polynomials at the fixed ratio $t/L(a)^2 = c$; enlarging the constants in Lemma F.296 therefore yields

$$\left| \mathbb{E}\left[(t^2 E(t) - \mathcal{Q}_2(dA)) \mathbf{1}_{\Omega_{\varepsilon_0}^c}\right] \right| \leq C_{\text{bad,GF}} e^{-c_{\text{bad,GF}} \beta}.$$

This is exactly (F350)–(F351). □

Theorem F.298 (Explicit weak-window comparison of the bare and flow couplings). *Fix the physical reference scale ℓ_0 and, for each dyadic spacing $a = \ell_0 2^{-n}$, write*

$$L(a) := \ell_0/a.$$

Under Lemma F.224, Proposition F.226, Lemma F.295, and Lemma F.296, there exist $\beta_{\text{GF}} \geq \beta_{\text{flow}}$ and a constant $C_{\text{GF}} < \infty$, depending only on $(d, G, c, \varepsilon_0, \ell_0)$ and the chosen discretization of $E(t)$, such that for every dyadic $a = \ell_0 2^{-n}$ and every $\beta \geq \beta_{\text{GF}}$,

$$g_{\text{GF}}^2(\ell_0; a, \beta) = \beta^{-1} + R_{\text{GF}}(a, \beta), \quad |R_{\text{GF}}(a, \beta)| \leq C_{\text{GF}} \beta^{-2}. \quad (\text{F352})$$

After enlarging β_{GF} if necessary so that $C_{\text{GF}} \beta_{\text{GF}}^{-1} \leq \frac{1}{2}$, one has the uniform comparison

$$\frac{1}{2\beta} \leq g_{\text{GF}}^2(\ell_0; a, \beta) \leq \frac{3}{2\beta} \quad (\beta \geq \beta_{\text{GF}}). \quad (\text{F353})$$

Consequently, for every target value $u \in (0, (2\beta_{\text{GF}})^{-1}]$ and every dyadic $a = \ell_0 2^{-n}$, there exists at least one bare coupling

$$\beta(a; u) \in \left[\frac{1}{2u}, \frac{3}{2u} \right] \quad (\text{F354})$$

such that

$$g_{\text{GF}}^2(\ell_0; a, \beta(a; u)) = u. \quad (\text{F355})$$

In particular, on the load-bearing tuned dyadic path one has $\beta(a; u) = \Theta(1/u)$ uniformly in a .

This theorem uses only the explicit bare-coupling expansion, the quantitative tail bounds (F348) and (F349), and continuity in β at fixed finite L ; no inverse-function theorem on the invariant graph is used.

Proof. Fix the physical scale ℓ_0 , write $L = L(a) = \ell_0/a$, and set $t = cL^2$. Proposition F.297 gives the uniform expansion

$$\mathbb{E}_{\beta, \Lambda(L)}[t^2 E(t)] = \beta^{-1} + R_{\text{good}}(a, \beta) + R_{\text{bad}}(a, \beta),$$

with the explicit bounds

$$|R_{\text{good}}(a, \beta)| \leq C_{\text{good,GF}} \beta^{-2}, \quad |R_{\text{bad}}(a, \beta)| \leq C_{\text{bad,GF}} e^{-c_{\text{bad,GF}} \beta},$$

uniformly in the dyadic spacing a . Choose

$$C_{\text{GF}} := C_{\text{good,GF}} + 1$$

and enlarge β_{GF} so that

$$C_{\text{bad,GF}} e^{-c_{\text{bad,GF}} \beta} \leq \beta^{-2} \quad (\beta \geq \beta_{\text{GF}}).$$

Then (F352) follows immediately. After enlarging β_{GF} once more so that $C_{\text{GF}} \beta_{\text{GF}}^{-1} \leq \frac{1}{2}$, the two-sided comparison (F353) follows by

$$\beta^{-1} - C_{\text{GF}} \beta^{-2} \leq g_{\text{GF}}^2(\ell_0; a, \beta) \leq \beta^{-1} + C_{\text{GF}} \beta^{-2}.$$

For the selection statement, fix $u \leq (2\beta_{\text{GF}})^{-1}$ and set

$$\beta_- := \frac{1}{2u}, \quad \beta_+ := \frac{3}{2u}.$$

Then $\beta_{\pm} \geq \beta_{\text{GF}}$, and (F353) gives

$$g_{\text{GF}}^2(\ell_0; a, \beta_-) \geq \frac{1}{2\beta_-} = u, \quad g_{\text{GF}}^2(\ell_0; a, \beta_+) \leq \frac{3}{2\beta_+} = u.$$

For fixed finite L , the map $\beta \mapsto g_{\text{GF}}^2(\ell_0; a, \beta)$ is continuous because it is the expectation of a bounded continuous observable against a finite-volume Gibbs density that depends continuously on β . The intermediate-value theorem therefore supplies at least one $\beta(a; u) \in [\beta_-, \beta_+]$ satisfying (F355), and the interval (F354) yields $\beta(a; u) = \Theta(1/u)$ uniformly in a . $\square$

Remark F.299 (Role in the mass gap path). Theorem F.298 is the load-bearing weak-window identification used later in Proposition F.314 and Corollary F.321. The older invariant-graph matching / bi-Lipschitz coordinate comparison and the iterate-level comparison between (u_n, K_n) and the gradient-flow coupling have both been moved to the project supplement.

Definition F.300 (Repaired weak-window package). Let $\mathbf{WW}_{\text{curv}}$ denote the theorem-level weak-window package consisting of Lemma F.224, Proposition F.226, Lemma F.227, Proposition F.228, Lemma F.229, and Theorem F.298. These are the only weak-window statements retained on the live proof spine.

Theorem F.301 (Weak-window sufficiency certificate). *All live downstream weak-window uses in the manuscript factor through the repaired curvature-sector package $\mathbf{WW}_{\text{curv}}$ of Definition F.300 and through nothing stronger. More precisely:*

1. *the only weak-window export to Section 4.2 is the sign input $b_0 > 0$ from Theorem F.262 and Lemma F.263; the dyadic beta-remainder proof in Section 4 is otherwise internal to the chart-truncated Lane C map;*
2. *Proposition F.314 uses only Proposition F.228, Lemma F.229, and Theorem F.298;*
3. *Corollary F.321 uses Proposition F.314 together with Proposition F.65;*
4. *Theorem F.320 and Theorem F.303 do not use any weak-window convexity statement;*
5. *consequently the load-bearing Route 1 package Proposition F.3, Proposition F.4, Corollary F.5, and Theorem F.6 contains no hidden dependence on any edge-Hessian convexity hypothesis.*

The same dependency information is also recorded in table form in the companion cross-reference tables.

Proof. For the Section 4.2 export, inspect first Theorem F.262 and then Lemma F.263. The theorem uses only Lemma F.224, Proposition F.226, and the lattice heat-kernel / parametrix inputs already isolated in Appendices C and D; Lemma F.263 is then an immediate positivity corollary. The beta-remainder file in Section 4 states explicitly that the remainder estimate is internal to the chart-truncated Lane C map, with the same-scale Wilson-versus-patched return handled separately by Theorem III.18. Item (ii) is the explicit hypothesis list in Proposition F.314. Item (iii) is exactly the proof of Corollary F.321. For item (iv), inspect

the proofs of Theorem F.303 and Theorem F.320: they use the one-shot crossover/transfer package and never cite Lemma F.224, Proposition F.226, or any edge-Hessian statement. Finally, Proposition F.3 cites Theorem F.320 together with Corollary F.321; Proposition F.4, Corollary F.5, and Theorem F.6 then consume Proposition F.3 and later Route 1 closure theorems. This closes the dependency chain and leaves no remaining place where an edge-Hessian convexity hypothesis could re-enter. $\square$

26. Same-scale transfer lemma at the one-shot entrance scale

Remark F.302 (What is and is not load-bearing here). Only Theorem F.303 is used in the final one-shot Route 1 proof. The legacy entrance-scale comparison notes have been moved to the supplementary Route 1 archive and are not invoked in Theorem F.320, Corollary F.321, Proposition F.327, or Theorem F.329. A referee auditing only the final load-bearing chain may read Theorem F.303 and then skip directly to Section F 29.

Archival note. The older $L_{\text{ent}}(\beta)$ comparison statements are no longer carried in the main Appendix F. The one-shot proof uses only the prescribed block size $L = B(\beta)$ appearing in Theorem F.320.

This section retains one generic same-scale transfer lemma. The retained lemma says exactly: *if at some prescribed block size L the reference blocked specification is KP/Dobrushin-contracting and the Plan-A conditional χ^2 gate holds for the exact-to-reference comparison, then the exact blocked specification is KP/Dobrushin-contracting at that same block size.*

a. General transfer/gluing theorem at a prescribed block size

This is a scale-local transfer theorem. In the final load-bearing Route 1 chain it is applied at the one-shot entrance scale $L = B(\beta)$, so it does not define a second entrance mechanism.

Theorem F.303 (Transfer/gluing to the KP basin at a prescribed block size). *Fix $d = 4$ and compact G . Fix a block size $L \geq 2$.*

Assume the α -reference blocked specification at block size L satisfies a uniform Dobrushin contraction bound

$$\delta(\gamma^{(\alpha)}) \leq q_* < 1$$

uniformly in finite block volume and admissible boundary conditions. Assume moreover that the exact-to-reference comparison at the same block size satisfies the Plan-A conditional χ^2 gate with deterministic defect $\varepsilon_L \geq 0$ (as produced in the χ^2 -transfer module), so that the single-site kernels obey the influence diagnostic of Lemma F.204 with error $\sqrt{\varepsilon_L}$.

Then the exact blocked specification satisfies

$$\delta(\gamma^{\text{ex}}) \leq q_* + \sqrt{\varepsilon_L}.$$

In particular, if $q_* + \sqrt{\varepsilon_L} < 1$, the infinite-volume exact blocked Gibbs state at block size L exists and is unique, and block-local truncated correlations decay exponentially in block distance with constants depending only on (q_*, G) .

Proof. By Lemma F.204, the assumed conditional χ^2 gate implies the influence comparison

$$\delta(\gamma^{\text{ex}}) \leq \delta(\gamma^{(\alpha)}) + \sqrt{\varepsilon_L} \leq q_* + \sqrt{\varepsilon_L}.$$

If $q_* + \sqrt{\varepsilon_L} < 1$, Proposition F.203 applies directly to the exact blocked specification and yields the asserted infinite-volume existence, uniqueness, and exponential block-distance decay. $\square$

Corollary F.304 (Infinite-volume uniqueness and boundary-condition uniformity). *If $\delta(\gamma^{\text{ex}}) \leq q < 1$ uniformly in finite volume and admissible boundary conditions, then the infinite-volume Gibbs/DLR state exists and is unique and block-local truncated correlations decay exponentially with constants depending only on (q, G) .*

Proof. Immediate from Proposition F.203. $\square$

b. QE2_{lat} closure module: exact terminal-block transfer at the one-shot scale

Status. The phase-1 reduction isolated a missing fluctuation-remainder estimate in the law of total covariance. At the exact one-shot scale that remainder can in fact be removed completely. The key point is that the blocked sigma-algebra is the terminal skeleton sigma-algebra, and Corollary III.12 gives exact conditional independence of fine observables supported in disjoint entrance-block families once that sigma-algebra is frozen. Combined with entrance-scale Dobrushin decay for the exact blocked specification, this yields QE2_{lat} directly.

The repaired implication now has the form

$$\begin{aligned}
& \text{(one-shot block-scale Dobrushin contraction)} \\
& + \text{(exact terminal-block conditional independence)} \\
& \implies \text{QE2}_{\text{lat}}.
\end{aligned}$$

The block-decay part is already available from the exact one-shot entrance theorem and Proposition F.203. The second input comes from the exact Wilson recursion / terminal-block product structure and is already available elsewhere in the manuscript.

Lemma F.305 (Explicit block-distance decay from finite-range Dobrushin). *Let γ be a single-site specification on a graph with graph distance dist and assume the finite-range Dobrushin hypotheses of Proposition F.203 with coefficient bound $\delta(\gamma) \leq q < 1$. For a bounded local observable F , write*

$$\text{osc}_i(F) := \sup_{\eta, \eta': \eta_{V \setminus \{i\}} = \eta'_{V \setminus \{i\}}} |F(\eta) - F(\eta')|, \quad \text{supp}_{\text{osc}}(F) := \{i : \text{osc}_i(F) \neq 0\}.$$

Then every pair of bounded local functions F, G satisfies the support-weighted covariance estimate

$$\begin{aligned}
|\text{Cov}(F, G)| &\leq \frac{e^{-m_{\text{blk}} \text{dist}(\text{supp}_{\text{osc}}(F), \text{supp}_{\text{osc}}(G))}}{1 - q} \left(\sum_{i \in \text{supp}_{\text{osc}}(F)} \text{osc}_i(F) \right) \\
&\quad \times \left(\sum_{j \in \text{supp}_{\text{osc}}(G)} \text{osc}_j(G) \right), \quad m_{\text{blk}} := -\log q.
\end{aligned} \tag{F356}$$

In particular,

$$|\text{Cov}(F, G)| \leq \frac{4 |\text{supp}_{\text{osc}}(F)| |\text{supp}_{\text{osc}}(G)|}{1 - q} \|F\|_{\infty} \|G\|_{\infty} e^{-m_{\text{blk}} \text{dist}(\text{supp}_{\text{osc}}(F), \text{supp}_{\text{osc}}(G))}. \tag{F357}$$

Proof. This is exactly the support-weighted estimate (F257) from Proposition F.203, specialized to the present graph and with $m_{\text{blk}} := -\log q$. The cardinality estimate (F357) follows from $\text{osc}_i(F) \leq 2\|F\|_{\infty}$ and $\text{osc}_j(G) \leq 2\|G\|_{\infty}$. $\square$

Remark F.306 (Why the fluctuation remainder disappears at the one-shot scale). The relevant blocked sigma-algebra at scale B is the terminal skeleton sigma-algebra by Proposition F.316. For observables supported in disjoint families of entrance blocks, Corollary III.12 therefore gives exact conditional independence once that sigma-algebra is frozen. Equivalently, the fluctuation covariance term in the law of total covariance vanishes identically for separated local probes at the exact one-shot scale.

Lemma F.307 (Entrance-block geometry at a fixed one-shot lattice). *Fix n and let B_n be the one-shot block side on the tuned lattice (a_n, β_n) . For a bounded local cylinder observable H , define*

$$\alpha_n(H) := \left\lceil \frac{\text{diam}_\infty(\text{supp}H)}{B_n} \right\rceil, \quad N_n(H) := (\alpha_n(H) + 2)^4.$$

Then for every translate $H(x)$:

1. *the entrance-block hull satisfies*

$$|\mathcal{B}_n(H(x))| \leq N_n(H);$$

2. *if H_1, H_2 are bounded local cylinder observables and*

$$r := \text{dist}_\infty(\text{supp}H_1(x), \text{supp}H_2(y)),$$

then

$$\text{dist}_{\text{blk}}(\mathcal{B}_n(H_1(x)), \mathcal{B}_n(H_2(y))) \geq \frac{r}{2B_n} - \Gamma_n(H_1, H_2), \quad (\text{F358})$$

where

$$\Gamma_n(H_1, H_2) := \alpha_n(H_1) + \alpha_n(H_2) + 4.$$

In particular, if

$$r \geq 2B_n(\Gamma_n(H_1, H_2) + 2),$$

the two entrance-block hulls are disjoint.

Proof. Fix H . In each coordinate direction, the projection of $\text{supp}H(x)$ meets at most $\alpha_n(H) + 2$ consecutive B_n -blocks, so the full entrance-block hull contains at most $(\alpha_n(H) + 2)^4 = N_n(H)$ blocks. This proves item (1).

For item (2), let d_∞^{idx} be the ℓ^∞ -distance between the corresponding entrance-block index hulls. The support of H_i can enlarge the index hull by at most $\alpha_n(H_i) + 1$ block units in each coordinate, so

$$d_\infty^{\text{idx}} \geq \frac{r}{B_n} - \alpha_n(H_1) - \alpha_n(H_2) - 2.$$

One step in the interaction graph of Definition F.318 changes the block index by at most 2 in ℓ^∞ , hence

$$\text{dist}_{\text{blk}}(\mathcal{B}_n(H_1(x)), \mathcal{B}_n(H_2(y))) \geq \frac{1}{2}(d_\infty^{\text{idx}} - 2) \geq \frac{r}{2B_n} - \Gamma_n(H_1, H_2),$$

which is exactly (F358). If $r \geq 2B_n(\Gamma_n(H_1, H_2) + 2)$, the right-hand side is at least 2, so the two hulls are disjoint. $\square$

Lemma F.308 (Fixed-lattice block-to-microscopic covariance transfer with common rate). *Fix a target value $u_{\text{ren}} \in (0, u_*]$ and let $a_n := \ell_0 2^{-n}$, $\beta_n := \beta(a_n; u_{\text{ren}})$, and $B_n := B(\beta_n)$. Then there exists $n_{\text{mic}}(u_{\text{ren}}) < \infty$ such that for every $n \geq n_{\text{mic}}(u_{\text{ren}})$, every pair of bounded gauge-invariant local cylinder observables H_1, H_2 , and all translates x, y with microscopic support separation*

$$r := \text{dist}_\infty(\text{supp} H_1(x), \text{supp} H_2(y)),$$

one has

$$\left| \text{Cov}_{\omega_{a_n}}(H_1(x), H_2(y)) \right| \leq C_n(H_1, H_2) \exp\left(-\frac{m_{\text{blk}}}{2B_n} r\right), \quad m_{\text{blk}} := -\log q_*, \quad (\text{F359})$$

for some finite constant $C_n(H_1, H_2)$ depending on n, H_1, H_2 but not on the translates. In particular, for each fixed $n \geq n_{\text{mic}}(u_{\text{ren}})$ the rate $m_{\text{blk}}/(2B_n)$ applies simultaneously to every centered bounded gauge-invariant local observable on the fixed-lattice positive-time local algebra.

Proof. Set

$$n_{\text{mic}}(u_{\text{ren}}) := \max\{N_{\text{ent}}(u_{\text{ren}}), n_{\text{repr}}(u_{\text{ren}})\},$$

where $N_{\text{ent}}(u_{\text{ren}})$ is the entry index from Theorem F.3 and $n_{\text{repr}}(u_{\text{ren}})$ comes from Proposition F.327. Fix $n \geq n_{\text{mic}}(u_{\text{ren}})$ and write

$$\alpha_i := \alpha_n(H_i), \quad N_i := N_n(H_i), \quad \Gamma := \Gamma_n(H_1, H_2).$$

Let $\Phi_{H_i(x), n}$ be the exact blocked representatives furnished by Proposition F.327. By item (1) of that proposition and Lemma F.307,

$$|\text{supp}_{\text{osc}}(\Phi_{H_i(x), n})| \leq N_i, \quad \|\Phi_{H_i(x), n}\|_\infty \leq \|H_i\|_\infty.$$

If

$$r \geq 2B_n(\Gamma + 2),$$

the entrance-block hulls are disjoint by Lemma F.307; hence Proposition F.327 gives the exact covariance identity

$$\text{Cov}_{\omega_{a_n}}(H_1(x), H_2(y)) = \text{Cov}_{\omega_{a_n}^{\langle B_n \rangle}}(\Phi_{H_1(x), n}, \Phi_{H_2(y), n}).$$

Apply Proposition F.326 together with (F358) to obtain

$$\left| \text{Cov}_{\omega_{a_n}}(H_1(x), H_2(y)) \right| \leq \frac{4N_1 N_2}{1 - q_*} \|H_1\|_\infty \|H_2\|_\infty e^{-m_{\text{blk}} \text{dist}_{\text{blk}}(\mathbf{B}_n(H_1(x)), \mathbf{B}_n(H_2(y)))}$$

$$\leq \frac{4N_1N_2}{1-q_*} \|H_1\|_\infty \|H_2\|_\infty e^{m_{\text{blk}}\Gamma} e^{-m_{\text{blk}}r/(2B_n)}.$$

If instead $r < 2B_n(\Gamma + 2)$, use the trivial bound

$$\left| \text{Cov}_{\omega_{a_n}}(H_1(x), H_2(y)) \right| \leq 4\|H_1\|_\infty \|H_2\|_\infty \leq 4e^{m_{\text{blk}}(\Gamma+2)} \|H_1\|_\infty \|H_2\|_\infty e^{-m_{\text{blk}}r/(2B_n)}.$$

Combining the two regimes yields (F359) after setting

$$C_n(H_1, H_2) := 4\|H_1\|_\infty \|H_2\|_\infty \max\left\{e^{m_{\text{blk}}(\Gamma+2)}, \frac{N_1N_2}{1-q_*} e^{m_{\text{blk}}\Gamma}\right\}.$$

Because n is now fixed, this applies to every bounded local pair H_1, H_2 on that lattice; only the prefactor changes with the observables. $\square$

Corollary F.309 (Fixed-lattice common-rate decay criterion). *Fix $n \geq n_{\text{mic}}(u_{\text{ren}})$. Then for every centered bounded gauge-invariant local cylinder observable H on the fixed-lattice positive-time algebra there exists $C_{n,H} < \infty$ such that*

$$0 \leq \omega_{a_n}(\Theta H \tau_t H) \leq C_{n,H} e^{-m_{\text{blk}}t/(2B_n)} \quad (t \in \mathbb{N}).$$

Hence the hypotheses of Theorem F.34 hold at the fixed lattice spacing a_n with common decay rate $m_{\text{blk}}/(2B_n)$.

Proof. Let H be an arbitrary centered bounded gauge-invariant local cylinder observable in the fixed-lattice positive-time algebra. Because H has finite temporal diameter, there exists $c_H < \infty$ such that the microscopic support separation between ΘH and $\tau_t H$ is at least $t - c_H$ for every integer $t \geq 0$. Apply Lemma F.308 to the pair $(\Theta H, H)$ and absorb the finite shift c_H into the prefactor:

$$\left| \omega_{a_n}(\Theta H \tau_t H) \right| \leq C_n(\Theta H, H) e^{m_{\text{blk}}c_H/(2B_n)} e^{-m_{\text{blk}}t/(2B_n)}.$$

Reflection positivity gives the nonnegativity of the left-hand side, so the displayed bound is exactly the required fixed-lattice decay estimate. Because H was arbitrary, Theorem F.34 applies with $m = m_{\text{blk}}/(2B_n)$. $\square$

Corollary F.310 (Uniform full fixed-lattice gap from bounded one-shot entrance). *Fix a target value $u_{\text{ren}} \in (0, u_*]$ and let $a_n = \ell_0 2^{-n}$, $\beta_n = \beta(a_n; u_{\text{ren}})$, and $B_n = B(\beta_n)$. Assume the standing one-shot entrance hypotheses, so that the exact blocked specification has Dobrushin*

constant $\leq q_* < 1$ at scale B_n and $a_n B_n \leq C_{\text{ent}} \ell_0$ for all large n . Then for all sufficiently large n , the fixed-lattice OS gap satisfies

$$\Delta_{\text{lat}}(\beta_n) \geq \frac{m_{\text{blk}}}{2B_n}, \quad \frac{\Delta_{\text{lat}}(\beta_n)}{a_n} \geq \frac{m_{\text{blk}}}{2C_{\text{ent}} \ell_0},$$

with $m_{\text{blk}} := -\log q_*$.

Proof. Fix $n \geq n_{\text{mic}}(u_{\text{ren}})$. By Corollary F.309, every centered bounded gauge-invariant local observable on the fixed-lattice positive-time algebra satisfies the hypothesis of Theorem F.34 with the same decay rate $m_{\text{blk}}/(2B_n)$. Proposition F.35 therefore yields

$$\Delta_{\text{lat}}(\beta_n) \geq \frac{m_{\text{blk}}}{2B_n}.$$

Divide by a_n and use $a_n B_n \leq C_{\text{ent}} \ell_0$. □

Theorem F.311 (Full fixed-lattice OS gap from one-shot entrance). *Within the standing choices of Subsection F 1, there exist $n_* < \infty$ and*

$$\mathbf{m}_* := \frac{m_{\text{blk}}}{2C_{\text{ent}} \ell_0} > 0, \quad m_{\text{blk}} := -\log q_*,$$

such that for all $n \geq n_$,*

$$\frac{\Delta_{\text{lat}}(\beta_n)}{a_n} \geq \mathbf{m}_*.$$

Equivalently, at each fixed tuned lattice spacing a_n with $n \geq n_$, the full fixed-lattice OS Hamiltonian has vacuum gap at least $m_{\text{blk}}/(2B_n)$. In particular $QE2_{\text{lat}}$ holds unconditionally along the tuned dyadic trajectory as a theorem about the full fixed-lattice OS gap.*

Proof. Take n_* to be an entry index for Corollary F.310. □

This section upgrades the finite-depth KP entrance argument into a post-entrance monotone safety statement: once the effective plaquette density (and the full interaction) lies in the KP admissibility basin at some scale, subsequent blocking steps preserve (and strictly improve) that basin membership. This removes the need for a classical global stable-manifold theorem for the (u, K) -dynamics beyond entrance; the only remaining global work is to reach entrance along the tuned ultraviolet trajectory and to construct the continuum limit along that trajectory.

c. KP basin definition in the present paper

Fix a block factor $L = 2$ and the canonical polymer norm $\|\cdot\|_{\text{KP}}$ used in the strong-coupling lattice module. Define the KP basin $\mathcal{B}_{\text{KP}}(\varepsilon_{\text{KP}})$ to be the set of effective interactions whose polymer activity satisfies

$$\|K\|_{\text{KP}} \leq \varepsilon_{\text{KP}},$$

with ε_{KP} chosen so that the KP cluster expansion converges and yields exponential clustering (as in the channel-gap module).

d. One-step forward invariance estimate

Proposition F.312 (Forward invariance of the KP basin). *There exist constants $0 < q_{\text{KP}} < 1$ and $C_{\text{KP}} < \infty$ (depending only on G and the fixed blocking kernel parameters) such that the exact blocking map $\mathcal{B}_{2,\alpha}$ satisfies the one-step estimate*

$$\|K'\|_{\text{KP}} \leq q_{\text{KP}} \|K\|_{\text{KP}} + C_{\text{KP}} a_{\text{plaq}}(\beta),$$

where the plaquette activity control parameter is

$$a_{\text{plaq}}(\beta) := \sum_{R \neq \mathbf{1}} d_R^2 \exp\left(-\frac{C_2(R)}{\beta}\right).$$

In particular, for all $\beta \leq \beta_{\text{KP}}$ such that

$$C_{\text{KP}} a_{\text{plaq}}(\beta) \leq (1 - q_{\text{KP}}) \varepsilon_{\text{KP}},$$

one has forward invariance of the KP basin:

$$K \in \mathcal{B}_{\text{KP}}(\varepsilon_{\text{KP}}) \implies K' \in \mathcal{B}_{\text{KP}}(\varepsilon_{\text{KP}}).$$

Proof. Inside the KP basin, the conditional block integral (Section F 25 e) admits an absolutely convergent polymer expansion. The resulting re-expanded interaction can be written as a convergent product over connected polymers, and the KP norm of the re-expanded remainder satisfies a Lipschitz estimate in the incoming KP norm, yielding the contraction term $q_{\text{KP}}\|K\|_{\text{KP}}$ with $q_{\text{KP}} < 1$. The additive forcing term is controlled by the bare plaquette activity $a_{\text{plaq}}(\beta)$ because the block kernel is class-function heat-kernel smoothing and convolution monotonicity improves a_{plaq} under further blocking. Uniformity in volume and admissible boundary conditions follows because the KP norm is local and the block kernel factorizes over coarse edges. $\square$

e. Consequence: post-entrance monotone safety

Corollary F.313 (Post-entrance monotone safety). *Assume $\beta \leq \beta_{\text{KP}}$ and $K_{n_*} \in \mathcal{B}_{\text{KP}}(\varepsilon_{\text{KP}})$ at some entrance step n_* . Then for all $j \geq 0$,*

$$K_{n_*+j} \in \mathcal{B}_{\text{KP}}(\varepsilon_{\text{KP}}),$$

and moreover

$$\|K_{n_*+j}\|_{\text{KP}} \leq q_{\text{KP}}^j \|K_{n_*}\|_{\text{KP}} + \frac{C_{\text{KP}}}{1 - q_{\text{KP}}} a_{\text{plaq}}(\beta).$$

Proof. Iterate Proposition F.312 and sum the resulting geometric series. $\square$

f. Summary: post-entrance monotonicity and one-shot entrance

With Corollary F.313, post-entrance RG control is monotone: once the effective interaction lies in the KP admissibility basin at some scale, all further coarse graining preserves (and improves) that admissibility. Theorem F.320 complements this by supplying a *one-shot* pre-entrance entrance theorem at the explicit scale $B(\beta)$ without invoking an iterated RG dynamical system. Together with the tuned ultraviolet recursion (Proposition F.314) and the sharp local limitOS reconstruction proved in the main manuscript, the QE1 contraction proof uses no infinite-step RG trajectory, while the tuned-sequence physical normalization still uses the separate weak-window coupling-identification input recorded above.

27. Forward UV $\rightarrow$ entrance tuning note (archival; moved to supplement)

The earlier forward-recursion presentation of the UV $\rightarrow$ entrance tuning argument is no longer part of the load-bearing Appendix F spine. That historical material has been moved to the supplementary Route 1 archive. The only load-bearing UV input retained in the main appendix is the discrete one-dimensional tuning result of Subsection F 28, which is written directly in the exact two-sided step-scaling form used later.

a. OS-positive base and sharp local limit

Purpose of this subsection. mass-gap uses a fixed positive flow time only as a *renormalization prescription* to define tuned ultraviolet couplings and a family of bounded probes.

Osterwalder–Schrader (OS) reflection positivity and OS reconstruction are *not* taken from the flowed probe algebra at fixed flow time: Yang–Mills gradient flow is parabolic and mixes degrees of freedom across a Euclidean time reflection plane. Accordingly, this appendix does not invoke (and does not require) reflection positivity for the fixed- t flowed algebra.

OS-positive base and sharp local limit. OS reflection positivity is taken from the unflowed Wilson positive-time algebra (standard Wilson reflection positivity) and passed to the continuum sharp local limit by stability of positivity under pointwise limits. The passage from bounded probes to the renormalized local field algebra and the verification of the OS axioms are carried out in the main manuscript (Lane A closure theorem); we do not duplicate that construction here. In particular, the continuum Hamiltonian whose spectral gap is claimed is the OS Hamiltonian reconstructed from the sharp local state on the *unflowed* local positive-time algebra.

Compatibility with the gap constant. The target *physical* mass gap along the tuned dyadic trajectory (Corollary F.332) is a statement about the spectrum of that reconstructed OS Hamiltonian once the QE2/QE3 interfaces are closed. It is therefore independent of which fixed- t bounded probe family is used to define the tuning scheme.

28. UV tuning trajectory as a one–dimensional recursion

a. *Statement*

Fix the step–scaling map $\Sigma : [0, u_{\text{AF}}] \rightarrow \mathbb{R}$ from Subsubsection F.22 b. On the weak window we use exactly the two ingredients proved above:

1. the two-sided cubic remainder bound (F286) and its forward-increment consequence (F287);
2. continuity of Σ on the weak window (Lemma F.229).

No derivative bound on the remainder is used anywhere in the tuning argument.

Proposition F.314 (Existence of a tuned UV sequence from the exact step-scaling input). *Assume Proposition F.228, Lemma F.229, and the explicit bare-to-flow comparison of Theorem F.298. Then there exists $u_* \in (0, u_{\text{AF}}]$ such that the following holds.*

For every target value $u_0 \in (0, u_]$ there exists a sequence $\{u_{-n}\}_{n \geq 0} \subset (0, u_*]$ with $u_0 = u_0$*

and

$$\Sigma(u_{-(n+1)}) = u_{-n}, \quad n \geq 0, \quad (\text{F360})$$

which is strictly decreasing and converges to 0. Moreover, with

$$c_- := \frac{1}{2}b_0 \log 2, \quad c_+ := \frac{3}{2}b_0 \log 2,$$

one has the explicit reciprocal bounds

$$\frac{1}{u_0^{-1} + c_+ n} \leq u_{-n} \leq \frac{1}{u_0^{-1} + \frac{c_-}{2} n} \quad (n \geq 0). \quad (\text{F361})$$

Consequently, if

$$N_0(u_0) := \left\lceil \frac{1}{c_+ u_0} \right\rceil, \quad (\text{F362})$$

then for every $n \geq N_0(u_0)$,

$$\frac{1}{2c_+ n} \leq u_{-n} \leq \frac{2}{c_- n}. \quad (\text{F363})$$

Writing $a_n := \ell_0 2^{-n}$, one may choose a bare sequence $\beta(a_n; u_0)$ satisfying

$$g_{\text{GF}}^2(\ell_0; a_n, \beta(a_n; u_0)) = u_{-n} \quad (\text{F364})$$

and the explicit comparison bounds

$$\frac{1}{2u_{-n}} \leq \beta(a_n; u_0) \leq \frac{3}{2u_{-n}} \quad (n \geq 0). \quad (\text{F365})$$

Hence for every $n \geq N_0(u_0)$,

$$\frac{c_-}{4} n \leq \beta(a_n; u_0) \leq 3c_+ n. \quad (\text{F366})$$

Consequently, along the tuned dyadic trajectory attached to u_0 ,

$$u(a_n) = \Theta(1/n) = \Theta(1/\log(1/a_n)), \quad \beta(a_n; u_0) = \Theta(n) = \Theta(\log(1/a_n)),$$

with comparison constants depending only on b_0 , C_Σ , and C_{GF} , and with entry index depending on u_0 only through (F362).

b. Proof

Choose $u_* \leq u_{\text{AF}}$ so small that (F287) holds on $(0, u_*]$ and, in addition,

$$u_* \leq (2\beta_{\text{GF}})^{-1},$$

where β_{GF} is the threshold from Theorem F.298. Set

$$c_- := \frac{1}{2}b_0 \log 2, \quad c_+ := \frac{3}{2}b_0 \log 2.$$

Then for every $u \in (0, u_*]$,

$$c_- u^2 \leq \Sigma(u) - u \leq c_+ u^2. \quad (\text{F367})$$

Step 1: existence of a backward preimage inside the weak window. Fix $y \in (0, u_*]$. By Lemma F.229, Σ is continuous on $[0, u_*]$. Since $\Sigma(0) = 0 < y$ and (F367) gives $\Sigma(y) > y$, the intermediate-value theorem yields at least one $x \in (0, y)$ with $\Sigma(x) = y$. Starting from the prescribed target u_0 , choose $u_{-(n+1)} \in (0, u_{-n})$ recursively so that (F360) holds. This produces a strictly decreasing sequence in $(0, u_*]$, hence $u_{-n} \downarrow u_\infty$ for some $u_\infty \geq 0$. Passing to the limit in (F360) and using (F367), one gets $\Sigma(u_\infty) = u_\infty$ and therefore $u_\infty = 0$.

Step 2: reciprocal increments and the correct eventual-in- n bounds. Let $v_n := 1/u_{-n}$. From $u_{-n} = \Sigma(u_{-(n+1)})$ and (F367) we obtain

$$c_- u_{-(n+1)}^2 \leq u_{-n} - u_{-(n+1)} \leq c_+ u_{-(n+1)}^2.$$

Divide by $u_{-n}u_{-(n+1)}$ to get

$$\frac{c_-}{1 + c_+ u_*} \leq v_{n+1} - v_n \leq c_+.$$

After shrinking u_* once more so that $c_+ u_* \leq 1$, this becomes

$$\frac{c_-}{2} \leq v_{n+1} - v_n \leq c_+. \quad (\text{F368})$$

Summing (F368) from 0 to $n - 1$ gives

$$u_0^{-1} + \frac{c_-}{2}n \leq v_n \leq u_0^{-1} + c_+n.$$

Inverting yields the exact bounds (F361). The upper bound in (F363) follows immediately from the right-hand inequality in (F361). For the lower bound, if $n \geq N_0(u_0)$ then $u_0^{-1} \leq c_+n$, hence

$$u_{-n} \geq \frac{1}{u_0^{-1} + c_+n} \geq \frac{1}{2c_+n}.$$

This proves (F363) and makes the dependence on the target value explicit only through the entry index (F362).

Step 3: selection of the bare tuning sequence. Because the backward trajectory stays inside $(0, u_*]$ and $u_* \leq (2\beta_{\text{GF}})^{-1}$, Theorem F.298 applies to every value u_{-n} . Hence for each $n \geq 0$ there exists a bare coupling $\beta(a_n; u_0)$ satisfying (F364), and the interval bound (F354) gives exactly (F365). Combining (F365) with (F363) yields the explicit eventual-in- n comparison

$$\frac{c_-}{4} n \leq \beta(a_n; u_0) \leq 3c_+ n \quad (n \geq N_0(u_0)),$$

which is (F366). Since $n = (\log(\ell_0/a_n))/\log 2$, the final asymptotics follow. $\square$

c. Consequence for the tuning sequence $\beta(a)$

Fix a target value $u_{\text{ren}} \in (0, u_*]$. Let $a_n = \ell_0 2^{-n}$, define $u(a_n) := u_{-n}$ from (F360), and choose $\beta(a_n; u_{\text{ren}})$ by Theorem F.298 so that

$$g_{\text{GF}}^2(\ell_0; a_n, \beta(a_n; u_{\text{ren}})) = u(a_n).$$

Then (F363) and (F366) imply

$$u(a_n) = \Theta(1/n) = \Theta(1/\log(1/a_n)), \quad \beta(a_n; u_{\text{ren}}) = \Theta(n) = \Theta(\log(1/a_n)),$$

with universal comparison constants and with entry index $N_0(u_{\text{ren}})$.

29. One-shot KP entrance from majorant mixing (no infinite-step RG control)

The remaining “global RG” barrier can be eliminated entirely by reframing the pre-entrance argument as a *single* coarse-graining to the entrance scale, rather than an indefinitely iterated Banach-space dynamical system. The key point is that all post-entrance control is already monotone, while pre-entrance control is finite-depth and uniform. Thus it suffices to define the blocked theory at the entrance scale by a one-shot coarse-graining kernel and verify that it lies in the KP/Dobrushin basin.

a. One-shot entrance scale

Fix block factor $L = 2$. Let $k_{\text{opt}}(\beta)$ be the explicit majorant mixing depth from Proposition F.65 (defined by $2^{2k_{\text{opt}}(\beta)} \geq \max\{1, \beta s_\delta\}$). Define the associated entrance block size

$$B(\beta) := 2^{k_{\text{opt}}(\beta)}. \tag{F369}$$

By definition,

$$\frac{B(\beta)^2}{\beta} \geq s_\delta. \quad (\text{F370})$$

b. Exact one-shot blocked variables

Let $\Lambda \subset \mathbb{Z}^4$ be a finite region with admissible boundary conditions and let $B = 2^k$ be dyadic. For the *exact* blocked theory used on the load-bearing Route 1 chain we do not insert any additional smoothing randomness. Instead we take the terminal skeleton field produced by k repetitions of the half-open dyadic blocking rule of Section III and group those surviving links by their owner entrance blocks. The auxiliary smoothing kernel $\mathcal{T}_{B,\alpha}$ belongs only to the α -reference specification already constructed in the transfer module; it does not define the exact blocked sigma-algebra used in Proposition F.327.

Definition F.315 (Exact one-shot blocked variables and blocked measure at scale B). Let Σ_B be the terminal skeleton obtained after $k = \log_2 B$ exact dyadic Wilson blockings, and let $\mathcal{F}_B$ be the sigma-algebra generated by the links on Σ_B . For each half-open entrance block $\mathcal{Q}$ of side B , let $U_{\mathcal{Q}}^{(B)}$ be the collection of terminal skeleton links whose anchor (equivalently: half-open owner block) lies in $\mathcal{Q}$. The exact blocked field is

$$U^{(B)} := \left(U_{\mathcal{Q}}^{(B)} \right)_{\mathcal{Q}},$$

and the exact scale- B blocked measure is the push-forward

$$\mu_{\beta,\Lambda}^{(B)} := (U \mapsto U^{(B)})_{\#} \mu_{\beta,\Lambda}. \quad (\text{F371})$$

We write $\gamma_{\beta,\Lambda}^{(B)}$ for the corresponding exact blocked specification on the entrance-block lattice.

Proposition F.316 (Exact blocked variables coincide with the terminal skeleton sigma-algebra). *Let $B = 2^k$ and set $j_0 := 0$, $m := k$ in Theorem III.11. Then the sigma-algebra generated by the exact blocked field $U^{(B)}$ of Definition F.315 is exactly the terminal skeleton sigma-algebra:*

$$\sigma(U^{(B)}) = \mathcal{F}_B = \mathcal{F}_{j_0+m}.$$

Consequently, for every bounded fine observable H ,

$$\mathbb{E}_{\omega_a}[H \mid U^{(B)}] = \mathbb{E}_{\omega_a}[H \mid \mathcal{F}_B]. \quad (\text{F372})$$

Under this identification, Corollary III.12 applies directly to observables whose fine supports lie in pairwise disjoint families of entrance blocks.

Proof. The terminal skeleton links are partitioned uniquely by the half-open owner entrance blocks. Thus the family $\{U_{\mathcal{Q}}^{\langle B \rangle}\}_{\mathcal{Q}}$ is simply a relabeling of the full terminal skeleton field, so it generates the same sigma-algebra $\mathcal{F}_B$. Equation (F372) is therefore just the identity of conditional expectations onto the same sigma-algebra. The final sentence is Corollary III.12 rewritten in the owner-block coordinates of Definition F.315. $\square$

Remark F.317 (Reference versus exact blocked objects). The α -reference blocked specification entering Proposition F.200 and Theorem F.303 continues to be the smoothed/reference object constructed in the transfer module. Definition F.315 records only the *exact* blocked law on the load-bearing Route 1 path. Both objects are taken at the same prescribed block size B , so no second entrance mechanism is introduced.

Definition F.318 (Entrance-block interaction graph). For a dyadic entrance scale $B = 2^k$, write the half-open entrance blocks as

$$\mathcal{Q}(\mathbf{m}) := B\mathbf{m} + [0, B)^4, \quad \mathbf{m} \in \mathbb{Z}^4.$$

Two entrance blocks are declared adjacent, written $\mathcal{Q}(\mathbf{m}) \sim_{\text{blk}} \mathcal{Q}(\mathbf{n})$, if

$$\|\mathbf{m} - \mathbf{n}\|_{\infty} \leq 2.$$

Let dist_{blk} denote the graph distance on this interaction graph. Equivalently, one step in dist_{blk} moves across at most two entrance blocks in each coordinate direction.

Proposition F.319 (Local Gibbs form and finite-range property of the exact one-shot blocked specification). *Let $B = 2^k$ and consider the exact blocked measure $\mu_{\beta, \Lambda}^{\langle B \rangle}$ of Definition F.315. Then, with respect to the product Haar measure on the terminal skeleton links grouped by owner entrance blocks, the blocked law has the local Gibbs form*

$$d\mu_{\beta, \Lambda}^{\langle B \rangle}(U^{\langle B \rangle}) = \frac{1}{Z_{\beta, \Lambda}^{\langle B \rangle}} \prod_{\mathcal{Q}} W_{\mathcal{Q}}(U^{\langle B \rangle}|_{\text{St}_1(\mathcal{Q})}) \bigotimes_{\mathcal{P}} dH_{\mathcal{P}}(U_{\mathcal{P}}^{\langle B \rangle}), \quad (\text{F373})$$

where

$$\text{St}_1(\mathcal{Q}) := \{\mathcal{P} : \|\mathbf{p} - \mathbf{q}\|_{\infty} \leq 1\}$$

for $\mathcal{Q} = \mathcal{Q}(\mathbf{q})$, each $dH_{\mathcal{P}}$ is the product Haar measure on the terminal skeleton links owned by $\mathcal{P}$, and each local weight $W_{\mathcal{Q}}$ depends only on the blocked variables carried by blocks in $\text{St}_1(\mathcal{Q})$. Consequently the exact blocked specification is finite range on the interaction graph of Definition F.318: for every entrance block $\mathcal{Q}$, the single-site conditional kernel

$$\gamma_{\mathcal{Q}}^{(B)}(\cdot \mid \eta)$$

depends on the exterior configuration only through the values of η on the graph-neighborhood

$$\text{Nb}_{\text{blk}}(\mathcal{Q}) := \{\mathcal{R} : \text{dist}_{\text{blk}}(\mathcal{Q}, \mathcal{R}) \leq 1\},$$

and hence

$$c_{\mathcal{Q}\mathcal{R}}(\gamma_{\beta, \Lambda}^{(B)}) = 0 \quad \text{whenever } \text{dist}_{\text{blk}}(\mathcal{Q}, \mathcal{R}) > 1. \quad (\text{F374})$$

Proof. By Corollary III.12, once the terminal skeleton field is frozen, the fine interior-link law factorizes exactly over the terminal half-open blocks. Integrating the fine interior links block by block therefore yields the push-forward density of the terminal skeleton field in the form (F373), where $W_{\mathcal{Q}}$ is the terminal-block partition function with the skeleton boundary data held fixed. Because a fine plaquette anchored in $\mathcal{Q}$ can touch only terminal skeleton links whose owner block has anchor at max-distance at most 1 from that of $\mathcal{Q}$, each factor $W_{\mathcal{Q}}$ depends only on the blocked variables carried by blocks in $\text{St}_1(\mathcal{Q})$.

Now fix $\mathcal{Q}_0 = \mathcal{Q}(\mathbf{q}_0)$. In the single-site conditional density of $U_{\mathcal{Q}_0}^{(B)}$ given the other blocked variables, only the factors $W_{\mathcal{P}}$ with $\|\mathbf{p} - \mathbf{q}_0\|_{\infty} \leq 1$ can contain $U_{\mathcal{Q}_0}^{(B)}$. Each such factor depends only on blocks $\mathcal{R} = \mathcal{Q}(\mathbf{r})$ with $\|\mathbf{r} - \mathbf{p}\|_{\infty} \leq 1$, hence on blocks satisfying $\|\mathbf{r} - \mathbf{q}_0\|_{\infty} \leq 2$. By Definition F.318, these are exactly the graph-neighbors of $\mathcal{Q}_0$. Therefore the single-site kernel for $\mathcal{Q}_0$ depends on the exterior configuration only through $\text{Nb}_{\text{blk}}(\mathcal{Q}_0)$, which is the finite-range statement (F374). $\square$

c. One-shot entrance theorem

We now show that for β large enough the one-shot blocked measure at scale $B(\beta)$ lies in the KP basin.

Theorem F.320 (One-shot entrance into the KP basin). *Fix the Plan-A / KP entrance threshold $\delta \in (0, \delta_{\text{PA}}]$ small enough that*

$$s_{\delta} \geq S_{\star},$$

where $S_\star$ is the explicit KP-activity budget threshold from Proposition F.200 for the final choice of $f_\star$. (Existence of such a choice follows from Lemma F.64, since $s_\delta \rightarrow \infty$ as $\delta \downarrow 0$.) Choose κ_δ and $\beta_0(\delta)$ as in Proposition F.193. Then for every $\beta \geq \beta_0(\delta)$, with $B(\beta)$ as in (F369), the one-shot blocked specification induced by $\mu_{\beta,\Lambda}^{\langle B(\beta) \rangle}$ lies in the KP/Dobrushin basin: there exists $q_\star \in (0, 1)$ (independent of β, Λ and boundary conditions) such that

$$\delta(\gamma_{\beta,\Lambda}^{\langle B(\beta) \rangle}) \leq q_\star < 1. \quad (\text{F375})$$

In particular, the corresponding infinite-volume blocked state exists and exhibits exponential clustering.

Proof. Step 1 (reference mixing at the one-shot scale; crossover input). By Proposition F.65 and the definition of $B(\beta)$, the majorant convolution power reaches the χ^2 -mixing level δ at time $B(\beta)^2/\beta \geq s_\delta$. Because $s_\delta \geq S_\star$ by the standing choice of δ , the same one-shot scale also meets the explicit KP activity budget of Proposition F.200:

$$\frac{B(\beta)^2}{\beta} = \frac{|T_{B(\beta)}|}{\beta} \geq s_\delta \geq S_\star.$$

Corollary F.201 therefore supplies the theorem-level QE1 reference budget at that same dyadic block size: Proposition F.200 has already converted the crossover coefficient decay into KP-small L^2 activity by an explicit low-/high-representation split at scale $B(\beta)$:

$$f_{B(\beta)}^{(\alpha)} \leq f_\star, \quad (\text{F376})$$

with $f_\star$ chosen below the admissible threshold in the deterministic KP iteration.

Step 2 (exact-to-reference transfer at the one-shot scale). The finite-depth transfer control was strengthened in Proposition F.198 and Corollary F.199. In particular, under the pivot schedule determined by κ_δ , the exact-to-reference comparison at the *same* one-shot block size $L = B(\beta)$ satisfies the Plan-A conditional χ^2 gate with defect $\varepsilon_\star$, uniformly in finite volume, admissible boundary conditions, and all events Ω of reference probability at least $1/2$. The gluing/transfer theorem (Theorem F.303) applies at this same block size and transfers KP-smallness of the reference specification to the exact blocked specification, with constants independent of volume and boundary conditions. Applying Theorem F.303 at $L = B(\beta)$ (where (F376) holds) yields (F375). By Proposition F.319, the exact blocked specification is finite range on the entrance-block interaction graph, so Proposition F.203 gives existence, uniqueness, and exponential clustering of the blocked infinite-volume state. $\square$

Corollary F.321 (Bounded physical one-shot entrance scale along a fixed tuned dyadic trajectory). *Fix a target value $u_{\text{ren}} \in (0, u_*)$ and let*

$$a_n := \ell_0 2^{-n}, \quad \beta_n := \beta(a_n; u_{\text{ren}}), \quad B_n := B(\beta_n)$$

be the associated tuned dyadic trajectory from Proposition F.314. Then there exists a constant $C_{\text{ent}} < \infty$, depending only on the group ledger and the one-shot threshold data, and an entry index $N_{\text{ent}}(u_{\text{ren}}) \in \mathbb{N}$ such that

$$a_n B_n \leq C_{\text{ent}} \ell_0 \quad (n \geq N_{\text{ent}}(u_{\text{ren}})).$$

In particular the one-shot entrance scale remains bounded in physical units along every tuned dyadic trajectory, with a uniform constant and a trajectory-dependent onset index.

Proof. By Proposition F.65 one has

$$B(\beta) \leq 2\sqrt{\max\{1, \beta s_\delta\}}.$$

Fix u_{ren} and apply Proposition F.314 with $u_0 = u_{\text{ren}}$. For every $n \geq N_0(u_{\text{ren}})$, the explicit bound (F366) gives

$$\beta_n \leq 3c_+ n.$$

Hence for all $n \geq N_{\text{ent}}(u_{\text{ren}}) := N_0(u_{\text{ren}})$,

$$a_n B_n \leq 2\ell_0 2^{-n} \sqrt{\max\{1, 3c_+ s_\delta n\}} \leq 2\ell_0 \sqrt{\max\{1, 3c_+ s_\delta\}},$$

using $\sqrt{n} \leq 2^n$ for $n \geq 1$. Thus one may take

$$C_{\text{ent}} := 2\sqrt{\max\{1, 3c_+ s_\delta\}},$$

which is uniform in the chosen target value u_{ren} . □

d. Route 1 internal theorem chain: dense-core Euclidean clustering

The load-bearing Route 1/Lane B input is now recorded as an explicit theorem/proposition chain, rather than only as a front-facing theorem-package claim. We work with the fixed countable positive-time local core from Lane B3 and prove Euclidean exponential clustering directly on that core from the one-shot entrance-scale blocked theory. The stronger fixed- β microscopic descent statements that appear later remain auxiliary and are not used in the argument below.

Definition F.322 (Route 1 dense positive-time probe core). Set

$$\mathcal{G}_{\text{ent}} := \mathcal{A}_{+, \text{poly}} \subset \mathcal{A}_+,$$

where $\mathcal{A}_{+, \text{poly}}$ is the countable positive-time local $*$ -algebra fixed in Lane B3.

Lemma F.323 (Route 1 probe core is OS-dense). *The span of OS-classes $\{[F] : F \in \mathcal{G}_{\text{ent}}\}$ is dense in $\mathcal{H}$. Consequently the centered classes $\{[\tilde{F}] : F \in \mathcal{G}_{\text{ent}}\}$ are dense in $\Omega^\perp$.*

Proof. By Definition F.322, $\mathcal{G}_{\text{ent}} = \mathcal{A}_{+, \text{poly}}$. The first statement is exactly Lemma 6.4; the second follows by subtracting the vacuum component, i.e. by orthogonal projection onto $\Omega^\perp$. $\square$

Lemma F.324 (Entrance mesh shrinks along the tuned sequence). *Fix a target value $u_{\text{ren}} \in (0, u_*]$ and let $a_n := \ell_0 2^{-n}$, $\beta_n := \beta(a_n; u_{\text{ren}})$, and $B_n := B(\beta_n)$. Then*

$$a_n B_n \xrightarrow{n \rightarrow \infty} 0.$$

In particular $B_n \rightarrow \infty$ in lattice units, and every fixed bounded local cylinder observable is eventually contained in a uniformly bounded number of entrance blocks.

Proof. The same estimate as in the proof of Corollary F.321 gives

$$a_n B_n \leq 2\ell_0 2^{-n} \sqrt{\max\{1, C_\beta s_\delta n\}},$$

which tends to 0 as $n \rightarrow \infty$. Since each fixed local cylinder observable depends on finitely many fine links, the number of entrance blocks intersecting its support is therefore uniformly bounded for all sufficiently large n . $\square$

Remark F.325 (Why a further fixed-physical-scale theorem is still needed). Lemma F.324 shows that the physical size of one entrance block tends to zero along the tuned sequence. Accordingly, any final fixed-physical-shift Euclidean clustering theorem must include an additional persistence or upgrade step beyond bare entrance-scale decay on the blocked interaction graph. This is why the dense-core route recorded below is retained as a secondary bridge rather than as the primary front-facing closure path.

Proposition F.326 (Entrance-scale block covariance decay in oscillation form). *Fix a target value $u_{\text{ren}} \in (0, u_*]$, let $a_n := \ell_0 2^{-n}$, $\beta_n := \beta(a_n; u_{\text{ren}})$, and $B_n := B(\beta_n)$, and let $\omega_{a_n}^{(B_n)}$ denote*

the unique infinite-volume exact blocked state at the one-shot entrance scale furnished by Theorem F.320. For a bounded blocked observable A , define the entrance-block oscillation seminorm

$$\text{osc}_{\mathcal{Q}}(A) := \sup_{\eta, \eta': \eta_{\mathbb{Z}_{B_n}^4} \setminus \{\mathcal{Q}\} = \eta'_{\mathbb{Z}_{B_n}^4} \setminus \{\mathcal{Q}\}} |A(\eta) - A(\eta')|, \quad \text{supp}_{\text{osc}}(A) := \{\mathcal{Q} : \text{osc}_{\mathcal{Q}}(A) \neq 0\}.$$

Then every pair of bounded blocked observables A, B satisfies

$$\begin{aligned} \left| \text{Cov}_{\omega_{a_n}^{(B_n)}}(A, B) \right| &\leq \frac{e^{-m_{\text{blk}} \text{dist}_{\text{blk}}(\text{supp}_{\text{osc}}(A), \text{supp}_{\text{osc}}(B))}}{1 - q_*} \left(\sum_{\mathcal{Q} \in \text{supp}_{\text{osc}}(A)} \text{osc}_{\mathcal{Q}}(A) \right) \\ &\quad \times \left(\sum_{\mathcal{R} \in \text{supp}_{\text{osc}}(B)} \text{osc}_{\mathcal{R}}(B) \right), \quad m_{\text{blk}} := -\log q_*. \end{aligned} \quad (\text{F377})$$

where dist_{blk} denotes graph distance on the interaction graph of Definition F.318. In particular,

$$\left| \text{Cov}_{\omega_{a_n}^{(B_n)}}(A, B) \right| \leq \frac{4 |\text{supp}_{\text{osc}}(A)| |\text{supp}_{\text{osc}}(B)|}{1 - q_*} \|A\|_{\infty} \|B\|_{\infty} e^{-m_{\text{blk}} \text{dist}_{\text{blk}}(\text{supp}_{\text{osc}}(A), \text{supp}_{\text{osc}}(B))}. \quad (\text{F378})$$

Proof. By Theorem F.320, the exact blocked specification at scale B_n satisfies

$$\delta(\gamma_{\beta_n, \Lambda}^{(B_n)}) \leq q_* < 1$$

uniformly in finite volume and admissible boundary conditions. Proposition F.319 shows that the exact blocked specification is finite range on the interaction graph of Definition F.318, i.e. its influence matrix vanishes beyond one graph step. Therefore Lemma F.305 applies to the unique blocked infinite-volume state and yields the support-weighted estimate (F377) with $m_{\text{blk}} = -\log q_*$. The cardinality bound (F378) follows from $\text{osc}_{\mathcal{Q}}(A) \leq 2\|A\|_{\infty}$ and $\text{osc}_{\mathcal{R}}(B) \leq 2\|B\|_{\infty}$. $\square$

Proposition F.327 (Entrance-scale blocked representatives for arbitrary local probes). *Fix a target value $u_{\text{ren}} \in (0, u_*]$. Let $a_n := \ell_0 2^{-n}$, $\beta_n := \beta(a_n; u_{\text{ren}})$, and $B_n := B(\beta_n)$. Then there exists $n_{\text{repr}}(u_{\text{ren}}) < \infty$ such that for every $n \geq n_{\text{repr}}(u_{\text{ren}})$ and every bounded gauge-invariant local cylinder observable H (not necessarily positive-time), there exists a bounded blocked observable $\Phi_{H,n}$ on the one-shot blocked state $\omega_{a_n}^{(B_n)}$ such that:*

1. $\Phi_{H,n}$ depends only on the finite entrance-block hull $\mathbf{B}_n(H)$ of the support of H ;
2. $\|\Phi_{H,n}\|_{\infty} \leq \|H\|_{\infty}$;

3. if H_1, H_2 are bounded local cylinder observables with disjoint entrance-block hulls $\mathbf{B}_n(H_1) \cap \mathbf{B}_n(H_2) = \emptyset$, then

$$\omega_{a_n}(H_1 H_2) = \omega_{a_n}^{\langle B_n \rangle}(\Phi_{H_1, n} \Phi_{H_2, n}), \quad \omega_{a_n}(H_i) = \omega_{a_n}^{\langle B_n \rangle}(\Phi_{H_i, n}) \quad (i = 1, 2), \quad (\text{F379})$$

and therefore

$$\text{Cov}_{\omega_{a_n}}(H_1, H_2) = \text{Cov}_{\omega_{a_n}^{\langle B_n \rangle}}(\Phi_{H_1, n}, \Phi_{H_2, n}). \quad (\text{F380})$$

No support-size restriction enters: at each fixed tuned lattice spacing, every bounded local cylinder observable admits such an exact entrance-scale representative.

Proof. Choose $n_{\text{repr}}(u_{\text{ren}})$ so that along the tuned dyadic trajectory the one-shot block depth

$$k_n := \log_2 B_n$$

is realized by the exact dyadic Wilson recursion and Proposition F.316 applies for all $n \geq n_{\text{repr}}(u_{\text{ren}})$. Fix such an n .

By Proposition F.316, the sigma-algebra generated by the exact blocked variables $U^{\langle B_n \rangle}$ is exactly the terminal skeleton sigma-algebra $\mathcal{F}_{B_n}$ from Theorem III.11 at depth k_n . Define

$$\Phi_{H, n} := \mathbb{E}_{\omega_{a_n}}[H \mid U^{\langle B_n \rangle}] = \mathbb{E}_{\omega_{a_n}}[H \mid \mathcal{F}_{B_n}].$$

The L^∞ bound in item (2) is immediate from conditional expectation.

For item (1), Theorem III.10 shows that at each dyadic step the conditional integration factors blockwise once the next skeleton is fixed, so links outside the parent block hull of the current support integrate out trivially. Iterating through the k_n dyadic steps therefore shows that $\Phi_{H, n}$ depends only on the entrance-block hull $\mathbf{B}_n(H)$. Because H is local, that hull is always finite, but its cardinality may now depend on both H and n .

For item (3), fix H_1, H_2 with disjoint entrance-block hulls. By item (1), each H_i is measurable with respect to the fine links contained in the corresponding finite family of half-open entrance blocks $\mathbf{B}_n(H_i)$. Set $j_0 := 0$ and $m := k_n = \log_2 B_n$ in Corollary III.12; Proposition F.316 identifies its terminal skeleton sigma-algebra with $\sigma(U^{\langle B_n \rangle})$. Because the block families $\mathbf{B}_n(H_1)$ and $\mathbf{B}_n(H_2)$ are disjoint, Corollary III.12 gives

$$\mathbb{E}_{\omega_{a_n}}[H_1 H_2 \mid U^{\langle B_n \rangle}] = \mathbb{E}_{\omega_{a_n}}[H_1 \mid U^{\langle B_n \rangle}] \mathbb{E}_{\omega_{a_n}}[H_2 \mid U^{\langle B_n \rangle}] = \Phi_{H_1, n} \Phi_{H_2, n}.$$

Taking expectations gives (F379), and subtracting the one-point identities yields (F380). $\square$

Remark F.328 (Status of the dense-core Route 1 bridge). Proposition F.326 and Proposition F.327 still isolate the microscopic lattice-side ingredients one would need for a *direct* fixed-physical-scale Euclidean-clustering proof from one-shot entrance. However, Lemma F.324 shows that the entrance mesh shrinks in physical units, so such a direct route would require an additional persistence/upgrade theorem beyond bare entrance-scale decay. We therefore use the spectral-calculus bridge below: once QE3 and Lane A base-state compatibility identify the sharp-local OS Hamiltonian with the same gapped Hamiltonian, dense-core Euclidean clustering follows immediately for that same state. Thus Lane B is a downstream reformulation of the front-facing Route 1 non-collapse proof, while the block-covariance propositions remain as diagnostics toward an independent lattice-side derivation of the same input.

Theorem F.329 (Secondary Route 1 dense-core bridge via QE3 and OS spectral calculus). *Fix a target value $u_{\text{ren}} \in (0, u_*]$ and let $\omega^{(u_{\text{ren}})}$ be the sharp-local continuum state furnished by Corollary 5.76 along the tuned dyadic sequence associated with u_{ren} . Let $(\mathcal{H}, \Omega, H)$ be its OS reconstruction. Then the dense positive-time core*

$$\mathcal{G}_{\text{ent}} = \mathcal{A}_{+, \text{poly}}$$

is OS-dense in $\mathcal{H}$ and satisfies Euclidean exponential clustering with rate

$$m = \mathbf{m}_* := \frac{m_{\text{blk}}}{2C_{\text{ent}}\ell_0}. \quad (\text{F381})$$

More precisely, for every $F, G \in \mathcal{G}_{\text{ent}}$ and every $s \geq 0$,

$$\left| \omega^{(u_{\text{ren}})} \left(\Theta \tilde{F} \tau_s \tilde{G} \right) \right| \leq e^{-\mathbf{m}_* s} \|\tilde{F}\|_{\mathcal{H}} \|\tilde{G}\|_{\mathcal{H}}. \quad (\text{F382})$$

In particular, for every $F \in \mathcal{G}_{\text{ent}}$,

$$\left| \omega^{(u_{\text{ren}})} \left(\Theta \tilde{F} \tau_s \tilde{F} \right) \right| \leq \|\tilde{F}\|_{\mathcal{H}}^2 e^{-\mathbf{m}_* s} \quad (s \geq 0). \quad (\text{F383})$$

Hence the Lane B Euclidean clustering input is verified for the same sharp-local state, now as a corollary of QE3 rather than as an independent replacement proof of QE3.

Proof. OS-density of the core is exactly Lemma F.323. By Theorem 6.3, the OS Hamiltonian reconstructed from $\omega^{(u_{\text{ren}})}$ satisfies

$$\sigma(H) \subset \{0\} \cup [\mathbf{m}_*, \infty), \quad \ker(H) = \mathbb{C}\Omega.$$

Therefore, for every $\psi \in \Omega^\perp$ and every $s \geq 0$,

$$\|e^{-sH}\psi\| \leq e^{-\mathfrak{m}_*s} \|\psi\|.$$

Now fix $F, G \in \mathcal{G}_{\text{ent}}$. Because $[\tilde{F}], [\tilde{G}] \in \Omega^\perp$, the OS semigroup identity gives

$$\omega^{(u_{\text{ren}})}(\Theta \tilde{F} \tau_s \tilde{G}) = \langle [\tilde{F}], e^{-sH} [\tilde{G}] \rangle_{\mathcal{H}}.$$

Applying Cauchy–Schwarz and the previous contraction bound yields

$$\left| \omega^{(u_{\text{ren}})}(\Theta \tilde{F} \tau_s \tilde{G}) \right| \leq \|[\tilde{F}]\|_{\mathcal{H}} \|e^{-sH} [\tilde{G}]\|_{\mathcal{H}} \leq e^{-\mathfrak{m}_*s} \|[\tilde{F}]\|_{\mathcal{H}} \|[\tilde{G}]\|_{\mathcal{H}},$$

which is exactly (F382). Taking $G = F$ gives (F383). $\square$

Theorem F.330 (Identification of the Route 1 bounded state and the Lane A sharp-local state). *Fix a target value $u_{\text{ren}} \in (0, u_*]$ and let*

$$a_n := \ell_0 2^{-n}, \quad \beta_n := \beta(a_n; u_{\text{ren}})$$

be the tuned dyadic sequence from Proposition F.314. Let $\omega_{\text{bd}}^{(u_{\text{ren}})}$ and $\mathcal{K}_{u_{\text{ren}}}(\cdot, \cdot; \cdot)$ be the bounded positive-time state and dyadic OS matrix-element limits furnished by Theorem F.211, and let $\omega^{(u_{\text{ren}})}$ be the full sharp-local continuum state from Corollary 5.76 built over that same base state. Then

$$\omega_{\text{bd}}^{(u_{\text{ren}})} = \omega^{(u_{\text{ren}})} \upharpoonright_{\mathcal{A}_+},$$

and, for every bounded $F, G \in \mathcal{A}_+$ and every dyadic shift $s \in \mathbb{D}_{\text{dyad}}$,

$$\omega^{(u_{\text{ren}})}(\Theta F \tau_s G) = \mathcal{K}_{u_{\text{ren}}}(F, G; s).$$

Consequently any Route 1 estimate proved first for the bounded dyadic OS matrix elements transfers to the same sharp-local state that enters Theorem 2.1; by semigroup continuity the transfer then extends from dyadic s to all $s \geq 0$.

Proof. The equality on $\mathcal{A}_+$ and the dyadic matrix-element identity are exactly the two compatibility clauses isolated in Corollary 5.75. In particular, that corollary is not the inductive-union theorem: the full sharp-local state exists by Corollary 5.76, while the capwise OS structure behind the later semigroup continuity comes from Theorem 5.74(b). The final sentence is therefore just Theorem F.212(iii): once OS reconstruction is available for $\omega^{(u_{\text{ren}})}$, the matrix elements $\omega^{(u_{\text{ren}})}(\Theta F \tau_s G)$ extend continuously from dyadic s to all $s \geq 0$. $\square$

Remark F.331 (Route 1 closure summary). Theorem F.320 and Corollary F.321 supply the entrance constants. Lemma F.308, Proposition F.35, Corollary F.310, and Theorem F.311 close QE2_{lat} by exact entrance-scale covariance transfer plus the fixed-lattice OS gap theorem. Theorem F.219 closes QE3 on the primary Route 1 spine, while Corollary F.5 records the resulting continuum vacuum spectral gap. Proposition F.326 and Proposition F.327 are retained as quantitative diagnostics for the downstream dense-core reformulation. Theorem F.329 then converts the already established Route 1 spectral gap into the dense-core Euclidean bound used by Lane B.

e. Consequence: physical mass-scale noncollapse from one-shot entrance

The one-shot entrance theorem, together with the exact entrance-scale covariance transfer proved above, is sufficient to obtain a uniform physical mass-scale lower bound along the tuned UV sequence. We record this as an explicit corollary, emphasizing that no infinite-step RG well-posedness is required once the one-shot entrance constants are available.

Corollary F.332 (Noncollapse of the physical mass scale from one-shot entrance). *Fix a target value $u_{\text{ren}} \in (0, u_*]$ and let $a_n = \ell_0 2^{-n}$, $\beta_n := \beta(a_n; u_{\text{ren}})$, and $B_n := B(\beta_n)$. Then there exists $m_{\text{phys},*} > 0$ such that for all $n \geq N_{\text{ent}}(u_{\text{ren}})$, the (infinite-volume) Yang–Mills theory at parameters (a_n, β_n) has a physical mass gap*

$$m_{\text{phys}}(a_n) \geq m_{\text{phys},*}. \quad (\text{F384})$$

Remark. The finite dyadic overhead coming from the $\exp(C/(L\beta))$ factor in Proposition F.133 is absorbed into the universal constant $m_{\text{phys},*}$; it does not affect uniformity or positivity.

Proof. Theorem F.311 yields a uniform lower bound on $\Delta_{\text{lat}}(\beta_n)/a_n$. By definition of the physical normalization used on the primary Route 1 spine, this is exactly the physical mass gap $m_{\text{phys}}(a_n)$. Take $m_{\text{phys},*} := \mathbf{m}_*$. $\square$

Remark F.333 (Interpretation). Corollary F.332 shows that, once the exact entrance-scale covariance transfer is taken into account, the only scale-dependent input entering the gap constant is the finite-depth one-shot entrance to the KP basin. No global RG dynamical control or Banach-space stable manifold is required beyond the one-shot entrance itself. Combined with Theorem F.212 and Theorem F.219, this closes QE3 on the primary Route 1 spine.

Corollary F.330A (Finite mixed bounded-family packaging on a finite truncation). *Fix $u_{\text{ren}} \in (0, u_*]$ and $\Delta_{\text{max}} \geq 0$. Let*

$$\mathcal{B}_{\text{cap}}^{(\leq \Delta_{\text{max}})} := *(\mathcal{A}_{\text{flow}}^{(\leq \Delta_{\text{max}})}, \mathcal{A}_+)$$

and let $\mathcal{W} \subset \mathcal{B}_{\text{cap}}^{(\leq \Delta_{\text{max}})}$ be finite. Write $\text{Alg}(\mathcal{W})$ for the unital $$ -subalgebra generated by $\mathcal{W}$. Then:*

(i) *For every finite family $\mathcal{F} \subset \text{Alg}(\mathcal{W})$, the tuned Wilson expectations of the members of $\mathcal{F}$ converge jointly along the dyadic trajectory associated with u_{ren} . Equivalently, the finite-cap evaluation functionals on $\mathcal{F}$ satisfy the same finite-observable-set bounds used in Theorem 4.92(a)–(b) and Corollary 4.94.*

(ii) *Consequently there is a unique functional*

$$\omega_{\mathcal{W}, \text{cap}, \text{bnd}}^{(u_{\text{ren}})}$$

on $\text{Alg}(\mathcal{W})$ satisfying

$$\omega_{\mathcal{W}, \text{cap}, \text{bnd}}^{(u_{\text{ren}})}(1) = 1, \quad \omega_{\mathcal{W}, \text{cap}, \text{bnd}}^{(u_{\text{ren}})}(P^*P) \geq 0 \quad (P \in \text{Alg}(\mathcal{W})),$$

hence a state on $\text{Alg}(\mathcal{W})$. Its restrictions to $\text{Alg}(\mathcal{W}) \cap \mathcal{A}_{\text{flow}}^{(\leq \Delta_{\text{max}})}$ and $\text{Alg}(\mathcal{W}) \cap \mathcal{A}_+$ agree with $\omega_{\text{flow}}^{(u_{\text{ren}})}$ and $\omega_{\text{bd}}^{(u_{\text{ren}})}$, respectively.

(iii) *If $\mathcal{W}_1 \subseteq \mathcal{W}_2 \subset \mathcal{B}_{\text{cap}}^{(\leq \Delta_{\text{max}})}$ are finite, then*

$$\omega_{\mathcal{W}_2, \text{cap}, \text{bnd}}^{(u_{\text{ren}})} \Big|_{\text{Alg}(\mathcal{W}_1)} = \omega_{\mathcal{W}_1, \text{cap}, \text{bnd}}^{(u_{\text{ren}})}.$$

In particular, whenever

$$X \in \text{Alg}(\mathcal{W}_1) \cap \text{Alg}(\mathcal{W}_2),$$

one has

$$\omega_{\mathcal{W}_1, \text{cap}, \text{bnd}}^{(u_{\text{ren}})}(X) = \omega_{\mathcal{W}_2, \text{cap}, \text{bnd}}^{(u_{\text{ren}})}(X).$$

(iv) *For every Euclidean time translation τ_T such that the translated family*

$$\tau_T \mathcal{W} := \{\tau_T Y : Y \in \mathcal{W}\}$$

again lies in $\mathcal{B}_{\text{cap}}^{(\leq \Delta_{\text{max}})}$, the packaged state attached to $\tau_T \mathcal{W}$ satisfies

$$\omega_{\tau_T \mathcal{W}, \text{cap}, \text{bnd}}^{(u_{\text{ren}})}(\tau_T X) = \omega_{\mathcal{W}, \text{cap}, \text{bnd}}^{(u_{\text{ren}})}(X) \quad (X \in \text{Alg}(\mathcal{W})).$$

Only this translation-covariance clause is used later in Theorem F.334(i).

Proof. Finite-family algebra step. By definition, $\text{Alg}(\mathcal{W})$ consists of finite linear combinations of noncommutative monomials in $\mathcal{W} \cup \mathcal{W}^*$. Fix a finite family $\mathcal{F} \subset \text{Alg}(\mathcal{W})$. After expanding each element of $\mathcal{F}$ into such monomials, there is a finite bounded observable set

$$\mathcal{G}_{\mathcal{F}} \subset \mathcal{A}_{\text{flow}}^{(\leq \Delta_{\max})} \cup \mathcal{A}_+$$

containing every bounded flowed factor and every bounded positive-time factor appearing in those monomials. The observable-evaluation Lipschitz step and the Wilson-to-chart-truncated transport theorem used in Theorem 4.92(a)–(b) and Corollary 4.94 are formulated for arbitrary finite bounded observable sets. Hence the tuned Wilson expectations satisfy on $\mathcal{G}_{\mathcal{F}}$ the same finite-observable-set bounds used in that packaging mechanism. Because products, adjoints, and finite linear combinations of bounded observables built from a fixed finite list of generators are again bounded observables depending only on that same list, the same joint convergence holds for the members of $\mathcal{F}$ themselves. This proves (i).

Packaging, positivity, and normalization. Varying $\mathcal{F}$ and using compatibility on overlaps packages these dyadic limits into a unique unital functional $\omega_{\mathcal{W}, \text{cap}, \text{bnd}}^{(u_{\text{ren}})}$ on $\text{Alg}(\mathcal{W})$. To record positivity explicitly, let $P \in \text{Alg}(\mathcal{W})$ and choose a finite family $\mathcal{F}_P \subset \text{Alg}(\mathcal{W})$ containing 1, P , and P^*P . For every finite-cap Wilson functional along the tuned dyadic trajectory one has

$$\omega_{a_n}(1) = 1, \quad \omega_{a_n}(P^*P) \geq 0.$$

Passing to the packaged dyadic limit on $\mathcal{F}_P$ gives

$$\omega_{\mathcal{W}, \text{cap}, \text{bnd}}^{(u_{\text{ren}})}(1) = 1, \quad \omega_{\mathcal{W}, \text{cap}, \text{bnd}}^{(u_{\text{ren}})}(P^*P) = \lim_{n \rightarrow \infty} \omega_{a_n}(P^*P) \geq 0.$$

Hence $\omega_{\mathcal{W}, \text{cap}, \text{bnd}}^{(u_{\text{ren}})}$ is a state on $\text{Alg}(\mathcal{W})$. The restriction statement follows because, when an element of $\text{Alg}(\mathcal{W})$ already lies in $\mathcal{A}_{\text{flow}}^{(\leq \Delta_{\max})}$ or in $\mathcal{A}_+$, the packaged limit is exactly the one already fixed by Corollary 4.93 or Corollary 4.94.

Overlap compatibility. If $\mathcal{W}_1 \subseteq \mathcal{W}_2$, then every $X \in \text{Alg}(\mathcal{W}_1)$ is evaluated from the same finite-family tuned Wilson limits whether one packages inside $\text{Alg}(\mathcal{W}_1)$ or inside $\text{Alg}(\mathcal{W}_2)$. Therefore

$$\omega_{\mathcal{W}_2, \text{cap}, \text{bnd}}^{(u_{\text{ren}})} \Big|_{\text{Alg}(\mathcal{W}_1)} = \omega_{\mathcal{W}_1, \text{cap}, \text{bnd}}^{(u_{\text{ren}})}.$$

This is exactly the overlap-compatibility statement in (iii).

Symmetry inheritance. Finally, let τ_T be as in (iv). For every finite family $\mathcal{F} \subset \text{Alg}(\mathcal{W})$, the translated family $\tau_T \mathcal{F}$ lies in $\text{Alg}(\tau_T \mathcal{W})$, and for every $Y \in \mathcal{F}$ the finite-cap Wilson

expectations satisfy

$$\omega_{a_n}(\tau_T Y) = \omega_{a_n}(Y) \quad (Y \in \mathcal{F})$$

by Euclidean covariance of the finite- a_n Wilson expectations together with translation covariance of the flowed representatives. Passing to the packaged dyadic limits yields

$$\omega_{\tau_T \mathcal{W}, \text{cap}, \text{bnd}}^{(u_{\text{ren}})}(\tau_T X) = \omega_{\mathcal{W}, \text{cap}, \text{bnd}}^{(u_{\text{ren}})}(X) \quad (X \in \text{Alg}(\mathcal{W})),$$

which is the translation-covariance clause used later in Theorem F.334(i). $\square$

Theorem F.334 (Finite-cap mixed-correlator extension from finite-family packaged bounded capwise states). *Fix $u_{\text{ren}} \in (0, u_*]$ and $\Delta_{\text{max}} \geq 0$. Let*

$$\mathcal{B}_{\text{cap}}^{(\leq \Delta_{\text{max}})} := *(\mathcal{A}_{\text{flow}}^{(\leq \Delta_{\text{max}})}, \mathcal{A}_+)$$

denote the unital bounded capwise algebra generated by the capwise flowed subalgebra and the bounded positive-time algebra, and write

$$\mathcal{B}_{+, \text{cap}}^{(\leq \Delta_{\text{max}})} := *(\mathcal{A}_{+, \text{flow}}^{(\leq \Delta_{\text{max}})}, \mathcal{A}_+) \subset \mathcal{B}_{\text{cap}}^{(\leq \Delta_{\text{max}})}$$

for its positive-time subalgebra. For every finite family $\mathcal{W} \subset \mathcal{B}_{\text{cap}}^{(\leq \Delta_{\text{max}})}$, Corollary F.330A furnishes a packaged state

$$\omega_{\mathcal{W}, \text{cap}, \text{bnd}}^{(u_{\text{ren}})}$$

on the generated algebra $\text{Alg}(\mathcal{W})$. In every argument below, first fix a finite family $\mathcal{W} \subset \mathcal{B}_{\text{cap}}^{(\leq \Delta_{\text{max}})}$ containing all bounded words appearing in the formula under consideration and work inside $\text{Alg}(\mathcal{W})$. Once this ambient finite family has been fixed, we abbreviate

$$\omega_{\mathcal{W}, \text{cap}, \text{bnd}}^{(u_{\text{ren}})}$$

by

$$\omega_{\text{cap}, \text{bnd}}^{(u_{\text{ren}})}$$

for that calculation. The compatibility statement in Corollary F.330A guarantees that the resulting value is independent of the chosen admissible finite family.

Let $\widehat{\mathfrak{A}}_{\text{loc}}^{(\leq \Delta_{\text{max}})}$ be the unital $$ -algebra generated by $\mathcal{B}_{\text{cap}}^{(\leq \Delta_{\text{max}})}$ and the formal centered capwise local symbols*

$$\widehat{O}_j^{(\leq \Delta_{\text{max}})}(f; \mu) \quad (1 \leq j \leq N, f \in C_c^\infty(\mathbb{R}^4)).$$

For an abstract word

$$W = B_0 \widehat{O}_{j_1}^{(\leq \Delta_{\max})}(f_1; \mu) B_1 \cdots \widehat{O}_{j_m}^{(\leq \Delta_{\max})}(f_m; \mu) B_m \quad (B_r \in \mathcal{B}_{\text{cap}}^{(\leq \Delta_{\max})}),$$

define its flowed representative at scales $\mathbf{t} = (t_1, \dots, t_m)$ by

$$W(\mathbf{t}) := B_0 \mathcal{O}_{j_1}^{\text{app}, (\leq \Delta_{\max})}(t_1; f_1; \mu) B_1 \cdots \mathcal{O}_{j_m}^{\text{app}, (\leq \Delta_{\max})}(t_m; f_m; \mu) B_m \in \mathcal{B}_{\text{cap}}^{(\leq \Delta_{\max})}.$$

Write also

$$\|W\|_{\text{bnd}} := \prod_{r=0}^m \|B_r\|_{\infty},$$

and let $K_W \subset \mathbb{R}^4$ be a fixed compact neighborhood containing the supports of the test functions f_r together with the supports of every bounded cylinder / flowed factor appearing in the chosen representatives of $B_0, \dots, B_m$. For theorem-facing bookkeeping, let C_W^{ins} denote any fixed constant dominating the single-slot bounded-word insertion constants attached to W . Concretely, in each individual application of Corollary 5.45A / Proposition F.245A, one may take

$$C_W^{\text{ins}} = C_{\text{word}}(K_{W,f}; W^\sharp) \|W\|_{\text{bnd}},$$

for the relevant slotwise positive-time neighborhood $K_{W,f}$ and normalized bounded word $W^\sharp$, with the trivial convention $C_W^{\text{ins}} = 0$ if some $\|B_r\|_{\infty} = 0$; when finitely many such channels arise in Lemma F.245B, enlarge C_W^{ins} once and for all to dominate them all. In the mixed-channel proof of part (i), the constants attached to the fixed word W are recorded through K_W , C_W^{ins} , and the finite truncation data $(m, \Delta_{\max}, \mu)$. Then:

(i) For every abstract word $W \in \widehat{\mathfrak{A}}_{\text{loc}}^{(\leq \Delta_{\max})}$ the limit

$$\tilde{\omega}_{\text{cap}}(W) := \lim_{\mathbf{t} \downarrow 0} \omega_{\text{cap}, \text{bnd}}^{(u_{\text{ren}})}(W(\mathbf{t}))$$

exists and is independent of the order or relative rates with which the coordinates of $\mathbf{t}$ tend to 0.

(ii) The rule $W \mapsto \tilde{\omega}_{\text{cap}}(W)$ defines a unique unital linear functional on $\widehat{\mathfrak{A}}_{\text{loc}}^{(\leq \Delta_{\max})}$ extending the compatible finite-family package limits $\omega_{\mathcal{W}, \text{cap}, \text{bnd}}^{(u_{\text{ren}})}$ furnished by Corollary F.330A. In particular,

$$\tilde{\omega}_{\text{cap}} \upharpoonright_{\mathcal{A}_{\text{flow}}^{(\leq \Delta_{\max})}} = \omega_{\text{flow}}^{(u_{\text{ren}})} \upharpoonright_{\mathcal{A}_{\text{flow}}^{(\leq \Delta_{\max})}} \quad \text{and} \quad \tilde{\omega}_{\text{cap}} \upharpoonright_{\mathcal{A}_+} = \omega_{\text{bd}}^{(u_{\text{ren}})}.$$

(iii) For every $P \in \widehat{\mathfrak{A}}_{\text{loc}}^{(\leq \Delta_{\max})}$ one has

$$\tilde{\omega}_{\text{cap}}(P^*P) \geq 0,$$

so $\tilde{\omega}_{\text{cap}}$ is a state on the capwise mixed algebra.

(iv) If $H, G \in \mathcal{A}_{+, \text{flow}}^{(\leq \Delta_{\max})}$ and all test functions f_r have support in $\{x_0 > 0\}$, then, writing

$$\widehat{\mathbf{O}} := \widehat{O}_{j_1}^{(\leq \Delta_{\max})}(f_1; \mu) \cdots \widehat{O}_{j_m}^{(\leq \Delta_{\max})}(f_m; \mu),$$

one has

$$\tilde{\omega}_{\text{cap}}(\Theta H \widehat{\mathbf{O}} G) = \langle [H], \widehat{\mathbf{O}}[G] \rangle_{\text{cap}}.$$

Here the local fields on the right are the graph-norm limits furnished by Theorem F.246, identified with the Lane A inverse-flow representatives by Corollary F.248, on the common core of Lemma F.247. Hence the capwise mixed-correlator extension agrees with the finite-cap positive-time OS operator realization on its natural domain.

Proof. Corollary F.330A is the explicit packaging node for the bounded capwise mixed words used here. Whenever a formula below involves finitely many bounded representatives, fix a finite family $\mathcal{W} \subset \mathcal{B}_{\text{cap}}^{(\leq \Delta_{\max})}$ containing them and evaluate with the packaged state $\omega_{\mathcal{W}, \text{cap}, \text{bnd}}^{(u_{\text{ren}})}$ furnished by that corollary. When convenient we then abbreviate this packaged state by $\omega_{\text{cap}, \text{bnd}}^{(u_{\text{ren}})}$ only for that local calculation; overlap compatibility ensures that this shorthand does not change any value.

For part (i), fix an abstract word W as above and two scale vectors $\mathbf{t}, \mathbf{s}$, and retain the notation K_W , $\|W\|_{\text{bnd}}$, and C_W^{ins} from the theorem statement.

Primary mixed-channel proof chain for part (i). The mixed-channel Cauchy step is proved here once and for all through the theorem chain

$$\text{Theorem 5.45} \implies \text{Corollary 5.45A} \implies \text{Proposition F.245A}$$

together with

$$\text{Theorem 5.69} \implies \text{Lemma F.245B} \implies \text{Theorem F.334(i)}.$$

No later Lane A statement reproves this step; all downstream uses cite the present theorem.

Because the word is finite and every bounded factor / local test function entering it has compact support, one may choose a common Euclidean time translation τ_T with $T \gg 1$

such that, after translating every bounded factor and every test function in W by τ_T , all translated bounded factors lie in $\mathcal{B}_{+, \text{cap}}^{(\leq \Delta_{\max})}$ and all translated test functions are supported in $\{x_0 > 0\}$. Write $(\tau_T W)(\mathbf{t})$ for the corresponding translated bounded representative. Write

$$(\tau_T W)(\mathbf{t}) - (\tau_T W)(\mathbf{s}) = \sum_{r=1}^m T_r^T(\mathbf{t}, \mathbf{s})$$

for the multilinear telescoping decomposition of the translated word. Lemma F.245B, applied to $\tau_T W$, reduces each literal telescoping summand $T_r^T(\mathbf{t}, \mathbf{s})$ to finitely many uniformly bounded positive-time word families multiplying one single-slot difference channel, and its displayed estimate (F302) gives the corresponding vacuum-channel envelope. There the coefficient growth is recorded explicitly in (F301) via Theorem 5.69, while Proposition F.245A, specialized to $\Psi = \Phi = \Omega_{\text{cap}}$, supplies the factor $t_r^{\eta_{\Delta_{\max}}} + s_r^{\eta_{\Delta_{\max}}}$ after subtracting the centered finite-cap SFTE identities in the r th slot. The single-slot bounded-word contribution from Proposition F.245A is first packaged into a fixed insertion constant for the translated word $\tau_T W$; enlarging C_W^{ins} once more if necessary, we record that translated-word contribution again under the shorthand C_W^{ins} , while Theorem 5.69 controls the remaining coefficient growth. Equation (F303) records the corresponding slotwise final $o(1)$ handoff. Thus, for each r , there exists a nonnegative function $\Xi_{W,r}(\mathbf{t}, \mathbf{s})$, depending on the fixed word through K_W , C_W^{ins} , and the finite truncation data, such that

$$\left| \left\langle \Omega_{\text{cap}}, T_r^T(\mathbf{t}, \mathbf{s}) \Omega_{\text{cap}} \right\rangle_{\text{cap}} \right| \leq \Xi_{W,r}(\mathbf{t}, \mathbf{s}), \quad \Xi_{W,r}(\mathbf{t}, \mathbf{s}) \xrightarrow{\mathbf{t}, \mathbf{s} \downarrow 0} 0.$$

Summing over the finitely many slots gives

$$\left| \left\langle \Omega_{\text{cap}}, \left((\tau_T W)(\mathbf{t}) - (\tau_T W)(\mathbf{s}) \right) \Omega_{\text{cap}} \right\rangle_{\text{cap}} \right| \leq \sum_{r=1}^m \Xi_{W,r}(\mathbf{t}, \mathbf{s}), \quad \sum_{r=1}^m \Xi_{W,r}(\mathbf{t}, \mathbf{s}) \xrightarrow{\mathbf{t}, \mathbf{s} \downarrow 0} 0.$$

To expose the capwise packaging step on the page, fix once and for all a finite set

$$\mathcal{F} \subset \mathcal{A}_{+, \text{flow}}^{(\leq \Delta_{\max})}, \quad 1 \in \mathcal{F},$$

and write

$$p_{\mathcal{F}}(X) := \|X\|_{\infty} \quad (X \in \mathcal{F}).$$

For each $X \in \mathcal{F}$, define the finite-cap difference functional

$$\Lambda_{\mathbf{t}, \mathbf{s}}^{W, \mathcal{F}}(X) := \left\langle \Omega_{\text{cap}}, \left((\tau_T W)(\mathbf{t}) - (\tau_T W)(\mathbf{s}) \right) X \Omega_{\text{cap}} \right\rangle_{\text{cap}}.$$

Because $\mathcal{F}$ is finite and each $X \in \mathcal{F}$ belongs to the common core, Proposition F.245A, applied with $\Psi = \Omega_{\text{cap}}$ and $\Phi = [X]$ to each telescoping summand together with Lemma F.245B, gives a constant $C_{W,\mathcal{F}} < \infty$ such that

$$\left| \Lambda_{\mathbf{t},\mathbf{s}}^{W,\mathcal{F}}(X) \right| \leq C_{W,\mathcal{F}} p_{\mathcal{F}}(X) \sum_{r=1}^m \Xi_{W,r}(\mathbf{t}, \mathbf{s}) \quad (X \in \mathcal{F}),$$

where the fixed-word dependence is recorded through K_W , C_W^{ins} , and the finite truncation data. Equivalently,

$$\left| \Lambda_{\mathbf{t},\mathbf{s}}^{W,\mathcal{F}}(X) \right| \leq C_{W,\mathcal{F}} p_{\mathcal{F}}(X) \sum_{r=1}^m \Xi_{W,r}(\mathbf{t}, \mathbf{s}), \quad \sum_{r=1}^m \Xi_{W,r}(\mathbf{t}, \mathbf{s}) \xrightarrow[\mathbf{t},\mathbf{s} \downarrow 0]{} 0, \quad (\text{F385})$$

so the finite-cap functional difference is displayed on the page as a finite sum of literal slotwise $o(1)$ handoffs coming from (F303). Define the finite bounded family

$$\mathcal{W}_{W,\mathcal{F},\mathbf{t},\mathbf{s}} := \left\{ \left((\tau_T W)(\mathbf{t}) - (\tau_T W)(\mathbf{s}) \right) X : X \in \mathcal{F} \right\} \cup \{ (\tau_T W)(\mathbf{t}), (\tau_T W)(\mathbf{s}) \} \subset \mathcal{B}_{+,\text{cap}}^{(\leq \Delta_{\text{max}})}.$$

Corollary F.330A therefore furnishes a packaged state

$$\omega_{W,\mathcal{F},\mathbf{t},\mathbf{s}}^{\text{pkg}} := \omega_{\mathcal{W}_{W,\mathcal{F},\mathbf{t},\mathbf{s}},\text{cap,bnd}}^{(u_{\text{ren}})}$$

on $\text{Alg}(\mathcal{W}_{W,\mathcal{F},\mathbf{t},\mathbf{s}})$. In particular, since $1 \in \mathcal{F}$, this packaged state satisfies

$$\left| \omega_{W,\mathcal{F},\mathbf{t},\mathbf{s}}^{\text{pkg}} \left((\tau_T W)(\mathbf{t}) - (\tau_T W)(\mathbf{s}) \right) \right| \leq C_{W,\mathcal{F}} \sum_{r=1}^m \Xi_{W,r}(\mathbf{t}, \mathbf{s}) \xrightarrow[\mathbf{t},\mathbf{s} \downarrow 0]{} 0.$$

To compare with the unshifted word, set

$$\mathcal{W}_{W,\mathbf{t},\mathbf{s}}^{\text{tr}} := \{ (\tau_T W)(\mathbf{t}), (\tau_T W)(\mathbf{s}) \}, \quad \mathcal{W}_{W,\mathbf{t},\mathbf{s}} := \{ W(\mathbf{t}), W(\mathbf{s}) \}.$$

By the compatibility clause of Corollary F.330A, the restriction of $\omega_{W,\mathcal{F},\mathbf{t},\mathbf{s}}^{\text{pkg}}$ to $\text{Alg}(\mathcal{W}_{W,\mathbf{t},\mathbf{s}}^{\text{tr}})$ is

$$\omega_{\mathcal{W}_{W,\mathbf{t},\mathbf{s}}^{\text{tr}},\text{cap,bnd}}^{(u_{\text{ren}})}.$$

Hence

$$\left| \omega_{\mathcal{W}_{W,\mathbf{t},\mathbf{s}}^{\text{tr}},\text{cap,bnd}}^{(u_{\text{ren}})} \left((\tau_T W)(\mathbf{t}) - (\tau_T W)(\mathbf{s}) \right) \right| \leq C_{W,\mathcal{F}} \sum_{r=1}^m \Xi_{W,r}(\mathbf{t}, \mathbf{s}) \xrightarrow[\mathbf{t},\mathbf{s} \downarrow 0]{} 0.$$

Because $\tau_T \mathcal{W}_{W,\mathbf{t},\mathbf{s}} = \mathcal{W}_{W,\mathbf{t},\mathbf{s}}^{\text{tr}}$, the translation-covariance clause of Corollary F.330A, together with translation covariance of the inverse-flow representatives, gives

$$\omega_{\mathcal{W}_{W,\mathbf{t},\mathbf{s}}^{\text{tr}},\text{cap,bnd}}^{(u_{\text{ren}})} \left((\tau_T W)(\mathbf{t}) - (\tau_T W)(\mathbf{s}) \right) = \omega_{\mathcal{W}_{W,\mathbf{t},\mathbf{s}},\text{cap,bnd}}^{(u_{\text{ren}})} \left(W(\mathbf{t}) - W(\mathbf{s}) \right).$$

Therefore

$$\left| \omega_{\mathcal{W}, \mathbf{t}, \mathbf{s}, \text{cap}, \text{bnd}}^{(u_{\text{ren}})}(W(\mathbf{t}) - W(\mathbf{s})) \right| \leq C_{W, \mathcal{F}} \sum_{r=1}^m \Xi_{W, r}(\mathbf{t}, \mathbf{s}) \xrightarrow[\mathbf{t}, \mathbf{s} \downarrow 0]{} 0.$$

By the local shorthand convention fixed in the theorem statement, this is exactly the Cauchy estimate for $\omega_{\text{cap}, \text{bnd}}^{(u_{\text{ren}})}(W(\mathbf{t}))$. Hence the net $\omega_{\text{cap}, \text{bnd}}^{(u_{\text{ren}})}(W(\mathbf{t}))$ is Cauchy, proving existence of the limit. Since the displayed Cauchy estimate was obtained for an arbitrary pair of scale vectors $\mathbf{t}, \mathbf{s} \downarrow 0$, the limit is independent of the order or relative rates with which the components of $\mathbf{t}$ approach 0.

Local certification for the mixed-channel seam. For theorem-facing use, the mixed-channel seam is certified locally here as follows. The single-slot bounded-word channel is the chain

$$\text{Theorem 5.45} \implies \text{Corollary 5.45A} \implies \text{Proposition F.245A},$$

and the literal telescoping reduction is the chain

$$\text{Theorem 5.69} \implies \text{Lemma F.245B} \implies \text{Theorem F.334(i)}.$$

The mixed bounded-family packaging node used here is Corollary F.330A: for each finite family of mixed bounded words it packages the corresponding capwise limits on the generated finite-family algebra, and the notation $\omega_{\text{cap}, \text{bnd}}^{(u_{\text{ren}})}$ in Theorem F.334 is only the local shorthand for $\omega_{\mathcal{W}, \text{cap}, \text{bnd}}^{(u_{\text{ren}})}$ after the relevant finite family $\mathcal{W}$ has been fixed. This packaging is invoked only after the state-free estimate of Proposition F.245A and the literal summand reduction of Lemma F.245B are in place. The dependence on the fixed bounded word is recorded through the support ledger K_W , the packaged insertion constant C_W^{ins} , and the finite truncation data. Concretely, in any single-slot channel one may take

$$C_W^{\text{ins}} = C_{\text{word}}(K_{W, f}; W^\sharp) \|W\|_{\text{bnd}}, \quad \|W\|_{\text{bnd}} := \prod_r \|B_r\|_\infty,$$

for the relevant slotwise neighborhood $K_{W, f}$ and normalized bounded word $W^\sharp$ from Corollary 5.45A; when finitely many such channels appear, C_W^{ins} is enlarged once and for all to dominate them all. The displayed slotwise-to-functional $o(1)$ handoff is given by (F303) together with (F385): the former converts the literal summand envelope into slotwise $o(1)$, and the latter records the resulting finite-cap functional difference estimate on the page. The companion cross-reference tables record the matching theorem-location guide for this same seam; see also §P.2.

Part (ii) is immediate from the linearity of the approximant words and the uniqueness of the limit just proved. If no formal local symbol is present, then $W(\mathbf{t}) = W \in \mathcal{B}_{\text{cap}}^{(\leq \Delta_{\max})}$ for every $\mathbf{t}$, so the restriction statement follows directly.

For part (iii), let $P \in \widehat{\mathfrak{A}}_{\text{loc}}^{(\leq \Delta_{\max})}$ and choose one scale parameter for every formal local symbol appearing in P . Denote by $P(\mathbf{t}) \in \mathcal{B}_{\text{cap}}^{(\leq \Delta_{\max})}$ the corresponding bounded representative, and set

$$\mathcal{W}_{P,\mathbf{t}} := \{P(\mathbf{t})\} \subset \mathcal{B}_{\text{cap}}^{(\leq \Delta_{\max})}.$$

Corollary F.330A furnishes a state

$$\omega_{\mathcal{W}_{P,\mathbf{t}},\text{cap},\text{bnd}}^{(u_{\text{ren}})}$$

on $\text{Alg}(\mathcal{W}_{P,\mathbf{t}})$, hence

$$\omega_{\mathcal{W}_{P,\mathbf{t}},\text{cap},\text{bnd}}^{(u_{\text{ren}})}(P(\mathbf{t})^*P(\mathbf{t})) \geq 0 \quad \text{for all } \mathbf{t}.$$

The approximant for P^*P is precisely $P(\mathbf{t})^*P(\mathbf{t})$ when the same scale parameters are used on the two factors, so part (i) allows one to pass to the limit and obtain

$$\tilde{\omega}_{\text{cap}}(P^*P) = \lim_{\mathbf{t} \downarrow 0} \omega_{\mathcal{W}_{P,\mathbf{t}},\text{cap},\text{bnd}}^{(u_{\text{ren}})}(P(\mathbf{t})^*P(\mathbf{t})) \geq 0.$$

Thus $\tilde{\omega}_{\text{cap}}$ is positive and normalized, hence a state.

For part (iv), fix $H, G \in \mathcal{A}_{+,\text{flow}}^{(\leq \Delta_{\max})}$ and choose any sequence of dyadic scales $t_r^{(n)} \downarrow 0$ for each slot r . Set

$$\widehat{\mathbf{O}} := \widehat{O}_{j_1}^{(\leq \Delta_{\max})}(f_1; \mu) \cdots \widehat{O}_{j_m}^{(\leq \Delta_{\max})}(f_m; \mu)$$

and

$$\mathbf{O}_n^{\text{app}} := \mathcal{O}_{j_1}^{\text{app}}(t_1^{(n)}; f_1; \mu) \cdots \mathcal{O}_{j_m}^{\text{app}}(t_m^{(n)}; f_m; \mu).$$

Theorem F.246 and Lemma F.247 realize the local fields as graph-norm limits on the common core, and Corollary F.248 identifies those graph-norm limits with the Lane A inverse-flow representatives. Therefore

$$\langle [H], \widehat{\mathbf{O}}[G] \rangle_{\text{cap}} = \lim_{n \rightarrow \infty} \langle [H], \mathbf{O}_n^{\text{app}}[G] \rangle_{\text{cap}}.$$

For each n , set

$$\mathcal{W}_n^{\text{OS}} := \{\Theta H \mathbf{O}_n^{\text{app}} G\} \subset \mathcal{B}_{\text{cap}}^{(\leq \Delta_{\max})}.$$

Corollary F.330A furnishes a packaged state

$$\omega_{\mathcal{W}_n^{\text{OS}},\text{cap},\text{bnd}}^{(u_{\text{ren}})}$$

on $\text{Alg}(\mathcal{W}_n^{\text{OS}})$. Since $\mathbf{O}_n^{\text{app}}G$ is a bounded positive-time observable and $H \in \mathcal{A}_{+, \text{flow}}^{(\leq \Delta_{\text{max}})}$, the definition of the finite-cap OS inner product on bounded flowed positive-time classes gives

$$\left\langle [H], \mathbf{O}_n^{\text{app}}[G] \right\rangle_{\text{cap}} = \omega_{\mathcal{W}_n^{\text{OS}}, \text{cap}, \text{bnd}}^{(u_{\text{ren}})} \left(\Theta H \mathbf{O}_n^{\text{app}} G \right).$$

Passing to the limit and invoking part (i), together with the local shorthand convention fixed in the theorem statement, gives the claimed identity. $\square$

End of Appendix F (primary mass-gap spine). Theorems and propositions in this appendix close QE1, QE2_{lat}, and QE3 on the primary Route 1 spine. Lane B remains a downstream reformulation.

Appendix G: A1 for $\mathrm{Sp}(2n)$: explicit one-plaquette coefficient bounds

This appendix proves Theorem 4.3. The argument follows a fixed three-step pattern (determinantal reduction $\rightarrow$ TN one-box control $\rightarrow$ Bessel-ratio telescope):

- (1) reduce the one-plaquette coefficient to an explicit determinant ratio via Weyl integration, Weyl's character formula, and Andréief's identity;
- (2) use total nonnegativity (TN) and a one-row shift inequality for Vandermonde-normalized minors (Appendix J) to obtain a one-box multiplicative bound;
- (3) telescope over boxes and bound the resulting product using Amos-type bounds for I_{k+1}/I_k .

1. Maximal-torus parametrization and Wilson weight

Fix $n \geq 3$ and set $G = \mathrm{Sp}(2n)$. Every conjugacy class meets the maximal torus

$$T \cong \{\mathrm{diag}(e^{i\theta_1}, \dots, e^{i\theta_n}, e^{-i\theta_1}, \dots, e^{-i\theta_n})\}, \quad \theta_j \in [0, \pi]. \quad (\mathrm{G1})$$

For the defining (complex) representation ρ of dimension $2n$ one has

$$\Re \mathrm{Tr}_\rho(U) = 2 \sum_{j=1}^n \cos \theta_j. \quad (\mathrm{G2})$$

Thus, up to an irrelevant constant factor $e^{-\alpha\beta}$, Equation (22) is the T -class function with symbol

$$W_t(\theta) := \exp\left(t \sum_{j=1}^n \cos \theta_j\right), \quad t := \frac{\alpha\beta}{n}. \quad (\mathrm{G3})$$

2. Weyl character formula and determinantal reduction

Irreducible representations of $\mathrm{Sp}(2n)$ are indexed by partitions $\lambda = (\lambda_1 \geq \dots \geq \lambda_n \geq 0)$. Define the standard shift sequence

$$b_i := n - i + 1 \quad (1 \leq i \leq n), \quad (\mathrm{G4})$$

and the shifted row indices

$$A_i(\lambda) := \lambda_i + b_i = \lambda_i + n - i + 1 \in \mathbb{N}. \quad (\mathrm{G5})$$

Lemma G.1 (Weyl character formula for $\mathrm{Sp}(2n)$ in sine-determinant form). *Let $U \in T$ have eigenangles $\theta = (\theta_1, \dots, \theta_n)$. Then*

$$\chi_\lambda(\theta) = \frac{\det \left(\sin(A_i(\lambda) \theta_j) \right)_{1 \leq i, j \leq n}}{\det \left(\sin(b_i \theta_j) \right)_{1 \leq i, j \leq n}}. \quad (\text{G6})$$

Remark G.2. Formula (G6) is a standard trigonometric form of Weyl's character formula for type C_n .

Lemma G.3 (Weyl integration formula in determinant form). *There exists a constant $c_{\mathrm{Sp}}(n) > 0$ such that for every continuous class function F on $\mathrm{Sp}(2n)$,*

$$\int_{\mathrm{Sp}(2n)} F(U) d\mu_{\mathrm{Haar}}(U) = c_{\mathrm{Sp}}(n) \int_{[0, \pi]^n} F(\theta) \det \left(\sin(b_i \theta_j) \right)^2 d\theta_1 \cdots d\theta_n. \quad (\text{G7})$$

Proof. This is the Weyl integration formula specialized to type C_n written in the determinant presentation of the Weyl denominator. $\square$

Proposition G.4 (Determinantal formula for one-plaquette coefficients on $\mathrm{Sp}(2n)$). *Let λ be a partition of length at most n . Define the moment kernel*

$$M_t^{\mathrm{Sp}}(p, q) := \frac{2}{\pi} \int_0^\pi e^{t \cos \theta} \sin(p\theta) \sin(q\theta) d\theta, \quad p, q \in \mathbb{N}. \quad (\text{G8})$$

Let

$$D_\lambda^{\mathrm{Sp}}(t) := \det \left(M_t^{\mathrm{Sp}}(A_i(\lambda), b_j) \right)_{1 \leq i, j \leq n}. \quad (\text{G9})$$

Then for $t = \alpha\beta/n$ one has

$$a_\lambda(\beta, \alpha) = \mathbb{E}_{\nu_{\beta, \alpha}} \left[\frac{\chi_\lambda(U)}{\dim \lambda} \right] = \frac{1}{\dim \lambda} \frac{D_\lambda^{\mathrm{Sp}}(t)}{D_\emptyset^{\mathrm{Sp}}(t)}. \quad (\text{G10})$$

Moreover, for every $p, q \in \mathbb{N}$,

$$M_t^{\mathrm{Sp}}(p, q) = I_{|p-q|}(t) - I_{p+q}(t). \quad (\text{G11})$$

Proof. Insert Equation (G6) and Equation (G3) into Equation (23) and use Equation (G7). The denominator determinant in Equation (G6) cancels one factor of the squared Weyl denominator, leaving the torus integral

$$\int_{[0, \pi]^n} W_t(\theta) \det \left(\sin(A_i(\lambda) \theta_j) \right) \det \left(\sin(b_i \theta_j) \right) d\theta. \quad (\text{G12})$$

Apply Andréief's identity to Equation (G12) (with the common product measure $d\theta_1 \cdots d\theta_n$) to obtain the determinant Equation (G9). The same reduction with $\lambda = \emptyset$ gives the normalizing factor, yielding Equation (G10).

Finally, for Equation (G11) use $\sin(p\theta)\sin(q\theta) = \frac{1}{2}(\cos((p-q)\theta) - \cos((p+q)\theta))$ and the standard identity $\int_0^\pi e^{t \cos \theta} \cos(m\theta) d\theta = \pi I_m(t)$. $\square$

3. One-box inequality and telescoping product over boxes

We now exploit TN and the one-row shift inequality from Appendix J.

Lemma G.5 (One-box multiplicative bound for $\text{Sp}(2n)$). *Fix $t > 0$. Let μ be a partition of length at most n and let i be an index such that $\mu + e_i$ is still a partition. Write $A_i := A_i(\mu) = \mu_i + b_i$. Then*

$$\frac{a_{\mu+e_i}}{a_\mu} \leq \frac{I_{A_i+1}(t)}{I_{A_i}(t)}. \quad (\text{G13})$$

Proof. Step 1: TN row-shift bound for the determinantal numerator. By Lemma J.6(i) (Appendix J), the matrix $(M_t^{\text{Sp}}(p, q))_{p, q \geq 1}$ is TN. Apply Lemma J.5 (Appendix J) with the fixed increasing column tuple $C = (1, 2, \dots, n)$ (i.e. $b_n < b_{n-1} < \cdots < b_1$) and the increasing row tuple obtained by sorting $(A_1(\mu), \dots, A_n(\mu))$. Shifting $\mu \mapsto \mu + e_i$ increases A_i by 1 and leaves the remaining row indices unchanged. Lemma J.5 gives

$$\frac{D_{\mu+e_i}^{\text{Sp}}(t)}{D_\mu^{\text{Sp}}(t)} \leq \frac{\Delta(A(\mu + e_i))}{\Delta(A(\mu))} \frac{M_t^{\text{Sp}}(A_i + 1, 1)}{M_t^{\text{Sp}}(A_i, 1)}, \quad (\text{G14})$$

where Δ is the Vandermonde factor in the (sorted) row indices.

Denominator-minor convention (TN ratios). At several points below we form Vandermonde-normalized minor ratios (or equivalent determinant ratios) and invoke the TN one-row shift tool from Appendix J. Whenever a would-be denominator minor can vanish, we apply the zero-case clause from Lemma J.5: if the pivot entry (hence the denominator minor) vanishes, then the corresponding numerator determinant also vanishes and the desired inequality is trivial; otherwise we work in the strictly-positive regime and proceed with the ratio bound.

Step 2: cancel the Vandermonde factor using the Weyl dimension ratio. For type C_n one may write the Weyl dimension formula in the form

$$\dim \lambda = \left(\prod_{i=1}^n \frac{A_i(\lambda)}{b_i} \right) \left(\prod_{1 \leq i < j \leq n} \frac{A_i(\lambda)^2 - A_j(\lambda)^2}{b_i^2 - b_j^2} \right). \quad (\text{G15})$$

Consequently, for a one-box growth $\mu \mapsto \mu + e_i$ (so only A_i increases by 1),

$$\frac{\dim(\mu + e_i)}{\dim(\mu)} = \frac{A_i + 1}{A_i} \prod_{j \neq i} \frac{(A_i + 1)^2 - A_j^2}{A_i^2 - A_j^2} = \frac{A_i + 1}{A_i} \prod_{j \neq i} \frac{A_i + 1 - A_j}{A_i - A_j} \cdot \frac{A_i + 1 + A_j}{A_i + A_j}. \quad (\text{G16})$$

Rearranging gives

$$\frac{\dim(\mu)}{\dim(\mu + e_i)} \frac{\Delta(A(\mu + e_i))}{\Delta(A(\mu))} = \frac{A_i}{A_i + 1} \prod_{j \neq i} \frac{A_i + A_j}{A_i + 1 + A_j} \leq \frac{A_i}{A_i + 1}. \quad (\text{G17})$$

Step 3: use the explicit kernel entry ratio at column 1. From Equation (G11) and the Bessel recurrence $I_{\nu-1} - I_{\nu+1} = \frac{2\nu}{t} I_\nu$ (valid for integer $\nu \geq 1$), we obtain

$$M_t^{\text{Sp}}(p, 1) = I_{p-1}(t) - I_{p+1}(t) = \frac{2p}{t} I_p(t) \quad (p \geq 1). \quad (\text{G18})$$

Therefore

$$\frac{M_t^{\text{Sp}}(A_i + 1, 1)}{M_t^{\text{Sp}}(A_i, 1)} = \frac{A_i + 1}{A_i} \frac{I_{A_i+1}(t)}{I_{A_i}(t)}. \quad (\text{G19})$$

Step 4: combine. By Equation (G10), $\frac{a_{\mu+e_i}}{a_\mu} = \frac{\dim(\mu)}{\dim(\mu+e_i)} \cdot \frac{D_{\mu+e_i}^{\text{Sp}}(t)}{D_\mu^{\text{Sp}}(t)}$. Combine this with Equation (G14), Equation (G17), and Equation (G19):

$$\begin{aligned} \frac{a_{\mu+e_i}}{a_\mu} &\leq \left(\frac{\dim \mu}{\dim(\mu + e_i)} \frac{\Delta(A(\mu + e_i))}{\Delta(A(\mu))} \right) \left(\frac{M_t^{\text{Sp}}(A_i + 1, 1)}{M_t^{\text{Sp}}(A_i, 1)} \right) \\ &\leq \left(\frac{A_i}{A_i + 1} \right) \left(\frac{A_i + 1}{A_i} \frac{I_{A_i+1}(t)}{I_{A_i}(t)} \right) = \frac{I_{A_i+1}(t)}{I_{A_i}(t)}. \end{aligned}$$

This is Equation (G13). □

Corollary G.6 (Telescoping product bound over boxes). *Let λ be a nonempty partition of length at most n . Then*

$$0 \leq a_\lambda(\beta, \alpha) \leq \prod_{(i,j) \in \lambda} \frac{I_{b_i+j}(t)}{I_{b_i+j-1}(t)}, \quad t = \frac{\alpha\beta}{n}. \quad (\text{G20})$$

Proof. Start from the empty partition and build λ by adding boxes one at a time. At the step when the j th box in row i is added, the current value of A_i equals $b_i + (j - 1)$. Apply Equation (G13) at each step and telescope. □

4. From the product bound to the A1 crossover form

We now convert Equation (G20) into the canonical A1 form Equation (4).

Lemma G.7 (Casimir in partition coordinates for $\mathrm{Sp}(2n)$). *Let λ be a partition of length at most n . With the root-length convention of Theorem 1.2 (short roots of length $\sqrt{2}$),*

$$C_2(\lambda) = \sum_{i=1}^n \lambda_i (\lambda_i + 2b_i), \quad b_i = n - i + 1. \quad (\text{G21})$$

Proof. For type C_n with short roots of length squared 2, one has $C_2(\lambda) = \langle \lambda, \lambda + 2\rho \rangle$, with $\rho = (b_1, \dots, b_n)$ in the standard orthonormal basis. Expanding gives Equation (G21). $\square$

Proposition G.8 (Crossover bound from the product estimate). *There exists an explicit constant $c_{\mathrm{Sp}}(n) > 0$ such that for every nontrivial λ and every $\alpha \in [1/2, 1]$, $\beta \geq 1$,*

$$a_\lambda(\beta, \alpha) \leq \exp \left(-c_{\mathrm{Sp}}(n) \frac{C_2(\lambda)}{\beta + \sqrt{C_2(\lambda)}} \right). \quad (\text{G22})$$

Proof. Fix $t = \alpha\beta/n$. Take logarithms of Equation (G20) and apply the Amos bound Equation (J3) from Lemma J.2:

$$\log a_\lambda \leq - \sum_{(i,j) \in \lambda} \operatorname{arsinh} \left(\frac{b_i + j - \frac{1}{2}}{t} \right). \quad (\text{G23})$$

Use the elementary inequality $\operatorname{arsinh}(x) \geq x/(1+x)$ for $x \geq 0$ to obtain

$$\sum_{(i,j) \in \lambda} \operatorname{arsinh} \left(\frac{b_i + j - \frac{1}{2}}{t} \right) \geq \sum_{(i,j) \in \lambda} \frac{b_i + j - \frac{1}{2}}{t + b_i + j - \frac{1}{2}} \geq \frac{\sum_{(i,j) \in \lambda} (b_i + j - \frac{1}{2})}{t + K_{\max}}, \quad (\text{G24})$$

where $K_{\max} := \max_{(i,j) \in \lambda} (b_i + j - \frac{1}{2}) = b_1 + \lambda_1 - \frac{1}{2} \leq n + \lambda_1$.

Next,

$$\begin{aligned} \sum_{(i,j) \in \lambda} (b_i + j - \tfrac{1}{2}) &= \sum_{i=1}^n \sum_{j=1}^{\lambda_i} \left(b_i + j - \tfrac{1}{2} \right) = \sum_{i=1}^n \left(\lambda_i (b_i + \tfrac{1}{2}) + \frac{\lambda_i(\lambda_i - 1)}{2} \right) \\ &= \frac{1}{2} \sum_{i=1}^n (\lambda_i^2 + 2b_i \lambda_i) + \frac{1}{2} \sum_{i=1}^n \lambda_i \geq \frac{1}{2} C_2(\lambda), \end{aligned}$$

using Equation (G21). Moreover, $\lambda_1 \leq \sqrt{C_2(\lambda)}$ for nontrivial λ , hence $K_{\max} \leq n + \sqrt{C_2(\lambda)}$.

Substitute these two bounds into Equation (G24) and then into Equation (G23):

$$\log a_\lambda \leq -\frac{1}{2} \frac{C_2(\lambda)}{t + n + \sqrt{C_2(\lambda)}}. \quad (\text{G25})$$

Finally, since $t \leq \beta/n$ and $\beta \geq 1$, we have $t + n + \sqrt{C_2} \leq \beta + n + \sqrt{C_2} \leq (n+1)(\beta + \sqrt{C_2})$.

Thus Equation (G25) implies Equation (G22) with $c_{\mathrm{Sp}}(n) = \frac{1}{2(n+1)}$. $\square$

Remark G.9 (Conclusion). Theorem G.8 is exactly Theorem 4.3 (with an explicit choice of constants).

Appendix H: A1 for $\text{Spin}(2n+1)$: explicit one-plaquette coefficient bounds

This appendix proves Theorem 4.4. The structure parallels Appendix G, with the only genuine differences being: (i) the maximal-torus trace has an extra constant eigenvalue 1, (ii) the Weyl shift is half-integral (type B_n), (iii) the resulting moment kernel has a +1 Hankel shift.

1. Maximal torus and Wilson weight

Fix $n \geq 2$ and set $G = \text{Spin}(2n+1)$. Since the Wilson weight factors through the projection to $\text{SO}(2n+1)$, only the sector $\pi(-1) = \mathbf{1}$ contributes (Theorem 4.1). We therefore work on $\text{SO}(2n+1)$.

Every conjugacy class meets the maximal torus consisting of block-diagonal rotations with eigenvalues

$$\{1, e^{\pm i\theta_1}, \dots, e^{\pm i\theta_n}\}, \quad \theta_j \in [0, \pi]. \quad (\text{H1})$$

In the defining $(2n+1)$ -dimensional representation ρ ,

$$\Re \text{Tr}_\rho(U) = 1 + 2 \sum_{j=1}^n \cos \theta_j. \quad (\text{H2})$$

The constant term 1 only contributes a global factor to the one-plaquette weight. Thus, up to normalization, Equation (22) corresponds on the torus to

$$W_t(\theta) := \exp \left(t \sum_{j=1}^n \cos \theta_j \right), \quad t := \frac{2\alpha\beta}{2n+1}. \quad (\text{H3})$$

2. Type B_n Weyl character formula and Andréief reduction

Irreducible representations of $\text{SO}(2n+1)$ are indexed by dominant integral highest weights, which we realize as partitions $\lambda = (\lambda_1 \geq \dots \geq \lambda_n \geq 0)$. Define integer shifts

$$b_i := n - i \quad (1 \leq i \leq n), \quad (\text{H4})$$

so that the type- B_n Weyl vector has components

$$\rho_i = b_i + \frac{1}{2} = n - i + \frac{1}{2}. \quad (\text{H5})$$

Define the shifted integer indices

$$A_i(\lambda) := \lambda_i + b_i = \lambda_i + n - i \in \mathbb{N}_0. \quad (\text{H6})$$

Lemma H.1 (Weyl character formula for $\text{SO}(2n+1)$ in half-integer sine form). *Let $U \in T$ have eigenangles $\theta = (\theta_1, \dots, \theta_n)$. Then*

$$\chi_\lambda(\theta) = \frac{\det \left(\sin \left((A_i(\lambda) + \frac{1}{2})\theta_j \right) \right)_{1 \leq i, j \leq n}}{\det \left(\sin \left((b_i + \frac{1}{2})\theta_j \right) \right)_{1 \leq i, j \leq n}}. \quad (\text{H7})$$

Lemma H.2 (Weyl integration formula in determinant form for $\text{SO}(2n+1)$). *There exists a constant $c_B(n) > 0$ such that for every continuous class function F on $\text{SO}(2n+1)$,*

$$\int_{\text{SO}(2n+1)} F(U) d\mu_{\text{Haar}}(U) = c_B(n) \int_{[0, \pi]^n} F(\theta) \det \left(\sin \left((b_i + \frac{1}{2})\theta_j \right) \right)^2 d\theta. \quad (\text{H8})$$

Proof. This is the Weyl integration formula for type B_n written using the determinant representation of the Weyl denominator. $\square$

Proposition H.3 (Determinantal formula for the one-plaquette coefficients on $\text{Spin}(2n+1)$). *Let λ be a partition of length at most n . Define the half-integer sine moment kernel*

$$M_t^B(p, q) := \frac{2}{\pi} \int_0^\pi e^{t \cos \theta} \sin \left((p + \frac{1}{2})\theta \right) \sin \left((q + \frac{1}{2})\theta \right) d\theta, \quad p, q \in \mathbb{N}_0. \quad (\text{H9})$$

Let

$$D_\lambda^B(t) := \det \left(M_t^B(A_i(\lambda), b_j) \right)_{1 \leq i, j \leq n}. \quad (\text{H10})$$

Then with $t = 2\alpha\beta/(2n+1)$ one has

$$a_\lambda(\beta, \alpha) = \mathbb{E}_{\nu_{\beta, \alpha}} \left[\frac{\chi_\lambda(U)}{\dim \lambda} \right] = \frac{1}{\dim \lambda} \frac{D_\lambda^B(t)}{D_\emptyset^B(t)}. \quad (\text{H11})$$

Moreover, for every $p, q \in \mathbb{N}_0$,

$$M_t^B(p, q) = I_{|p-q|}(t) - I_{p+q+1}(t). \quad (\text{H12})$$

Proof. The proof is identical to the $\text{Sp}(2n)$ case (Appendix G) after replacing the type- C determinant formulas by Lemmas H.1 and H.2. The cancellation of one Weyl-denominator factor and Andréief's identity produce Equation (H10).

To compute the kernel, use $\sin((p + \frac{1}{2})\theta) \sin((q + \frac{1}{2})\theta) = \frac{1}{2}(\cos((p-q)\theta) - \cos((p+q+1)\theta))$ so that

$$\begin{aligned} M_t^B(p, q) &= \frac{1}{\pi} \int_0^\pi e^{t \cos \theta} \left(\cos((p-q)\theta) - \cos((p+q+1)\theta) \right) d\theta \\ &= I_{|p-q|}(t) - I_{p+q+1}(t). \end{aligned}$$

$\square$

3. One-box inequality and telescoping

Lemma H.4 (One-box multiplicative bound for type B_n). *Fix $t > 0$. Let μ be a partition of length at most n and let i be an index such that $\mu + e_i$ is still a partition. Write $A_i := A_i(\mu) = \mu_i + b_i$. Then*

$$\frac{a_{\mu+e_i}}{a_\mu} \leq \frac{I_{A_i+1}(t)}{I_{A_i}(t)}. \quad (\text{H13})$$

Proof. Step 1: TN row shift bound for determinants. By Lemma J.6(ii) the matrix $(M_t^B(p, q))_{p, q \geq 0}$ is TN. Apply Lemma J.5 with the increasing column tuple $C = (0, 1, \dots, n-1)$ (i.e. $b_n < b_{n-1} < \dots < b_1$). As in Appendix G this yields

$$\frac{D_{\mu+e_i}^B(t)}{D_\mu^B(t)} \leq \frac{\Delta(A(\mu + e_i))}{\Delta(A(\mu))} \frac{M_t^B(A_i + 1, 0)}{M_t^B(A_i, 0)}. \quad (\text{H14})$$

Step 2: cancel the Vandermonde factor using the type- B Weyl dimension ratio.

Denominator-minor convention (TN ratios). At several points below we form Vandermonde-normalized minor ratios (or equivalent determinant ratios) and invoke the TN one-row shift tool from Appendix J. Whenever a would-be denominator minor can vanish, we apply the zero-case clause from Lemma J.5: if the pivot entry (hence the denominator minor) vanishes, then the corresponding numerator determinant also vanishes and the desired inequality is trivial; otherwise we work in the strictly-positive regime and proceed with the ratio bound.

Write $a_i := A_i + \frac{1}{2}$ and $\rho_i := b_i + \frac{1}{2}$. The type- B_n Weyl dimension formula can be written as

$$\dim \lambda = \left(\prod_{i=1}^n \frac{a_i(\lambda)}{\rho_i} \right) \left(\prod_{1 \leq i < j \leq n} \frac{a_i(\lambda)^2 - a_j(\lambda)^2}{\rho_i^2 - \rho_j^2} \right). \quad (\text{H15})$$

Thus, under $\mu \mapsto \mu + e_i$ (so $a_i \mapsto a_i + 1$ and a_j fixed for $j \neq i$),

$$\frac{\dim(\mu + e_i)}{\dim(\mu)} = \frac{a_i + 1}{a_i} \prod_{j \neq i} \frac{(a_i + 1)^2 - a_j^2}{a_i^2 - a_j^2} = \frac{a_i + 1}{a_i} \prod_{j \neq i} \frac{a_i + 1 - a_j}{a_i - a_j} \cdot \frac{a_i + 1 + a_j}{a_i + a_j}. \quad (\text{H16})$$

Since $a_i - a_j = A_i - A_j$, the product over $(a_i + 1 - a_j)/(a_i - a_j)$ is exactly the Vandermonde ratio $\Delta(A(\mu + e_i))/\Delta(A(\mu))$. Therefore

$$\frac{\dim(\mu)}{\dim(\mu + e_i)} \frac{\Delta(A(\mu + e_i))}{\Delta(A(\mu))} = \frac{a_i}{a_i + 1} \prod_{j \neq i} \frac{a_i + a_j}{a_i + 1 + a_j} \leq \frac{a_i}{a_i + 1} \leq 1. \quad (\text{H17})$$

Step 3: bound the kernel entry ratio by an adjacent Bessel ratio. Using Equation (H12) at column 0,

$$M_t^B(p, 0) = I_p(t) - I_{p+1}(t) \quad (p \geq 0). \quad (\text{H18})$$

Hence

$$\frac{M_t^B(A_i + 1, 0)}{M_t^B(A_i, 0)} = \frac{I_{A_i+1} - I_{A_i+2}}{I_{A_i} - I_{A_i+1}}. \quad (\text{H19})$$

By Lemma J.1, the ratio I_{k+1}/I_k is nondecreasing in k . Thus $I_{A_i}I_{A_i+2} \geq I_{A_i+1}^2$, which is equivalent to $I_{A_i}(I_{A_i+1} - I_{A_i+2}) \leq I_{A_i+1}(I_{A_i} - I_{A_i+1})$. Dividing by the positive denominator $I_{A_i}(I_{A_i} - I_{A_i+1})$ yields

$$\frac{M_t^B(A_i + 1, 0)}{M_t^B(A_i, 0)} \leq \frac{I_{A_i+1}(t)}{I_{A_i}(t)}. \quad (\text{H20})$$

Step 4: combine. Using Equation (H11) we have $\frac{a_{\mu+e_i}}{a_\mu} = \frac{\dim \mu}{\dim(\mu+e_i)} \cdot \frac{D_{\mu+e_i}^B}{D_\mu^B}$. Combine Equation (H14), Equation (H17), and Equation (H20) to obtain Equation (H13). $\square$

Corollary H.5 (Product bound over boxes). *Let λ be a nonempty partition of length at most n . Then*

$$0 \leq a_\lambda(\beta, \alpha) \leq \prod_{(i,j) \in \lambda} \frac{I_{b_i+j}(t)}{I_{b_i+j-1}(t)}, \quad t = \frac{2\alpha\beta}{2n+1}, \quad b_i = n - i. \quad (\text{H21})$$

Proof. Iterate Equation (H13) along any one-box growth chain from $\emptyset$ to λ . At the moment the j th box in row i is added, $A_i = b_i + (j - 1)$. $\square$

4. Crossover bound in Casimir form

Lemma H.6 (Casimir in partition coordinates for type B_n). *Let λ be a partition of length at most n . With the root-length convention of Theorem 1.2,*

$$C_2(\lambda) = \sum_{i=1}^n \lambda_i (\lambda_i + 2b_i + 1), \quad b_i = n - i. \quad (\text{H22})$$

Proof. For type B_n , $\rho_i = b_i + \frac{1}{2}$ and $C_2(\lambda) = \langle \lambda, \lambda + 2\rho \rangle$. Expanding yields Equation (H22). $\square$

Proposition H.7 (A1 crossover bound for $\text{Spin}(2n+1)$). *There exists an explicit constant $c_B(n) > 0$ such that for every nontrivial λ and every $\alpha \in [1/2, 1]$, $\beta \geq 1$,*

$$a_\lambda(\beta, \alpha) \leq \exp \left(-c_B(n) \frac{C_2(\lambda)}{\beta + \sqrt{C_2(\lambda)}} \right). \quad (\text{H23})$$

Proof. Fix $t = 2\alpha\beta/(2n+1)$. Take logarithms of Equation (H21) and apply Lemma J.2:

$$\log a_\lambda \leq - \sum_{(i,j) \in \lambda} \text{arsinh} \left(\frac{b_i + j - \frac{1}{2}}{t} \right). \quad (\text{H24})$$

As in Appendix G, $\operatorname{arsinh}(x) \geq x/(1+x)$ and $x/(1+x) \geq x/(t+K_{\max})$ give

$$\sum_{(i,j) \in \lambda} \operatorname{arsinh}\left(\frac{b_i + j - \frac{1}{2}}{t}\right) \geq \frac{\sum_{(i,j) \in \lambda} (b_i + j - \frac{1}{2})}{t + K_{\max}}, \quad (\text{H25})$$

with $K_{\max} = \max_{(i,j) \in \lambda} (b_i + j - \frac{1}{2}) = b_1 + \lambda_1 - \frac{1}{2} \leq (n-1) + \lambda_1$.

Compute the numerator:

$$\sum_{(i,j) \in \lambda} (b_i + j - \frac{1}{2}) = \sum_{i=1}^n \sum_{j=1}^{\lambda_i} (b_i + j - \frac{1}{2}) = \frac{1}{2} \sum_{i=1}^n (\lambda_i^2 + 2b_i \lambda_i) = \frac{1}{2} (C_2(\lambda) - |\lambda|), \quad (\text{H26})$$

using Equation (H22). Since $\lambda_i^2 + \lambda_i \geq 2\lambda_i$ for $\lambda_i \in \mathbb{N}$ and $b_i \geq 0$, we have $C_2(\lambda) = \sum_i (\lambda_i^2 + 2b_i \lambda_i + \lambda_i) \geq 2|\lambda|$. Thus $C_2(\lambda) - |\lambda| \geq \frac{1}{2} C_2(\lambda)$, so

$$\sum_{(i,j) \in \lambda} (b_i + j - \frac{1}{2}) \geq \frac{1}{4} C_2(\lambda). \quad (\text{H27})$$

Also $\lambda_1 \leq \sqrt{C_2(\lambda)}$, hence $K_{\max} \leq (n-1) + \sqrt{C_2(\lambda)}$. Substitute Equation (H27) into Equation (H25) and then into Equation (H24):

$$\log a_\lambda \leq -\frac{1}{4} \frac{C_2(\lambda)}{t + (n-1) + \sqrt{C_2(\lambda)}}. \quad (\text{H28})$$

Finally, since $t \leq 2\beta/(2n+1) \leq \beta$ and $\beta \geq 1$, $t + (n-1) + \sqrt{C_2} \leq \beta + n + \sqrt{C_2} \leq (n+1)(\beta + \sqrt{C_2})$. Therefore Equation (H28) implies Equation (H23) with $c_B(n) = \frac{1}{4(n+1)}$. $\square$

Appendix I: A1 for $\operatorname{Spin}(2n)$: explicit one-plaquette coefficient bounds

This appendix proves Theorem 4.5. As in Appendix H, the Wilson weight factors through $\operatorname{SO}(2n)$, so only representations with $\pi(-1) = \mathbf{1}$ contribute. We therefore work on $\operatorname{SO}(2n)$ and restrict to dominant integral highest weights (the “tensor” sector).

1. Maximal torus and Wilson weight

Fix $n \geq 4$ and set $G = \operatorname{Spin}(2n)$. On $\operatorname{SO}(2n)$, every conjugacy class meets the maximal torus with eigenvalues

$$\{e^{\pm i\theta_1}, \dots, e^{\pm i\theta_n}\}, \quad \theta_j \in [0, \pi]. \quad (\text{I1})$$

For the defining $2n$ -dimensional representation ρ ,

$$\Re \operatorname{Tr}_\rho(U) = 2 \sum_{j=1}^n \cos \theta_j. \quad (\text{I2})$$

Hence, up to normalization, Equation (22) corresponds on the torus to the symbol

$$W_t(\theta) := \exp \left(t \sum_{j=1}^n \cos \theta_j \right), \quad t := \frac{\alpha\beta}{n}. \quad (\text{I3})$$

2. Determinantal coefficient formula

Irreducible representations of $\text{SO}(2n)$ in the tensor sector are indexed by dominant integral highest weights. For the present A1 gate it suffices to label them by partitions $\lambda = (\lambda_1 \geq \dots \geq \lambda_n \geq 0)$; representations on which $-\mathbf{1} \in \text{Spin}(2n)$ acts nontrivially (spinor sector) do not appear in the Wilson expansion.

Define the type- D_n shifts

$$b_i := n - i \quad (1 \leq i \leq n), \quad (\text{I4})$$

so that the Weyl vector has components $\rho_i = b_i$. Define

$$A_i(\lambda) := \lambda_i + b_i = \lambda_i + n - i \in \mathbb{N}_0. \quad (\text{I5})$$

We record the determinant-form Weyl character and Weyl integration formulas for the tensor sector of $\text{SO}(2n)$, since these are the only group-specific inputs needed to reduce the one-plaquette coefficients to moment determinants (and they avoid any reliance on Toeplitz+Hankel machinery).

Lemma I.1 (Weyl character formula for $\text{SO}(2n)$ in cosine-determinant form (tensor sector)). *Let $U \in T$ have eigenangles $\theta = (\theta_1, \dots, \theta_n)$ as in (I1). For a tensor-sector irreducible representation indexed by a partition λ of length at most n ,*

$$\chi_\lambda(\theta) = \frac{\det \left(\cos(A_i(\lambda) \theta_j) \right)_{1 \leq i, j \leq n}}{\det \left(\cos(b_i \theta_j) \right)_{1 \leq i, j \leq n}}, \quad b_i = n - i, \quad A_i(\lambda) = \lambda_i + b_i. \quad (\text{I6})$$

Lemma I.2 (Weyl integration formula in determinant form for $\text{SO}(2n)$). *There exists a constant $c_D(n) > 0$ such that for every continuous class function F on $\text{SO}(2n)$,*

$$\int_{\text{SO}(2n)} F(U) d\mu_{\text{Haar}}(U) = c_D(n) \int_{[0, \pi]^n} F(\theta) \det \left(\cos(b_i \theta_j) \right)_{1 \leq i, j \leq n}^2 d\theta_1 \cdots d\theta_n, \quad b_i = n - i. \quad (\text{I7})$$

Proof. This is the Weyl integration formula for type D_n written using the determinant presentation of the Weyl denominator. Concretely, the type- D_n Weyl denominator is $\prod_{1 \leq i < j \leq n} 2 \sin \left(\frac{\theta_i - \theta_j}{2} \right) 2 \sin \left(\frac{\theta_i + \theta_j}{2} \right)$, and the identity $2 \sin \left(\frac{\theta_i - \theta_j}{2} \right) 2 \sin \left(\frac{\theta_i + \theta_j}{2} \right) = \cos \theta_j - \cos \theta_i$ shows that (up to a constant) it equals the Vandermonde

Denominator-minor convention (TN ratios). At several points below we form Vandermonde-normalized minor ratios (or equivalent determinant ratios) and invoke the TN one-row shift tool from Appendix J. Whenever a would-be denominator minor can vanish, we apply the zero-case clause from Lemma J.5: if the pivot entry (hence the denominator minor) vanishes, then the corresponding numerator determinant also vanishes and the desired inequality is trivial; otherwise we work in the strictly-positive regime and proceed with the ratio bound.

determinant in the variables $\cos \theta_j$. Since $\cos(b\theta)$ is a degree- b polynomial in $\cos \theta$ with nonzero leading coefficient, the determinant $\det(\cos(b_i \theta_j))$ is a nonzero constant multiple of this Vandermonde, and squaring yields (I7). $\square$

Proposition I.3 (Determinantal formula for one-plaquette coefficients on $\text{SO}(2n)$). *Let λ be a partition of length at most n . Define the cosine moment kernel*

$$M_t^D(p, q) := \frac{2}{\pi} \int_0^\pi e^{t \cos \theta} \cos(p\theta) \cos(q\theta) d\theta, \quad p, q \in \mathbb{N}_0. \quad (\text{I8})$$

Let

$$D_\lambda^D(t) := \det \left(M_t^D(A_i(\lambda), b_j) \right)_{1 \leq i, j \leq n}. \quad (\text{I9})$$

Then with $t = \alpha\beta/n$,

$$a_\lambda(\beta, \alpha) = \mathbb{E}_{\nu_{\beta, \alpha}} \left[\frac{\chi_\lambda(U)}{\dim \lambda} \right] = \frac{1}{\dim \lambda} \frac{D_\lambda^D(t)}{D_\emptyset^D(t)}. \quad (\text{I10})$$

Moreover, for every $p, q \in \mathbb{N}_0$,

$$M_t^D(p, q) = I_{|p-q|}(t) + I_{p+q}(t). \quad (\text{I11})$$

Proof. Insert Equation (I6) and Equation (I3) into the general coefficient formula Equation (23) and use the Weyl integration formula Equation (I7). The denominator determinant in Equation (I6) cancels one factor of the squared Weyl denominator, leaving the torus integral

$$\int_{[0, \pi]^n} W_t(\theta) \det \left(\cos(A_i(\lambda)\theta_j) \right)_{1 \leq i, j \leq n} \det \left(\cos(b_i \theta_j) \right)_{1 \leq i, j \leq n} d\theta_1 \cdots d\theta_n. \quad (\text{I12})$$

Apply Andréief's identity to Equation (I12) (with the common product measure $d\theta_1 \cdots d\theta_n$) to obtain the determinant Equation (I9). The same reduction with $\lambda = \emptyset$ gives the normalizing factor $D_\emptyset^D(t)$, yielding Equation (I10).

For Equation (I11), use $\cos(p\theta) \cos(q\theta) = \frac{1}{2}(\cos((p-q)\theta) + \cos((p+q)\theta))$ and $\int_0^\pi e^{t \cos \theta} \cos(m\theta) d\theta = \pi I_m(t)$. $\square$

3. One-box inequality and telescoping

Lemma I.4 (One-box multiplicative bound for type D_n). *Fix $t > 0$. Let μ be a partition of length at most n and let i be an index such that $\mu + e_i$ is still a partition. Write $A_i := A_i(\mu) = \mu_i + b_i$. Then*

$$\frac{a_{\mu+e_i}}{a_\mu} \leq \frac{I_{A_i+1}(t)}{I_{A_i}(t)}. \quad (\text{I13})$$

Proof. Step 1: TN row shift bound for determinants. By Lemma J.6(iii) the matrix $(M_t^D(p, q))_{p, q \geq 0}$ is TN. Apply Lemma J.5 with the increasing column tuple $C = (0, 1, \dots, n-1)$. As before,

$$\frac{D_{\mu+e_i}^D(t)}{D_\mu^D(t)} \leq \frac{\Delta(A(\mu + e_i))}{\Delta(A(\mu))} \frac{M_t^D(A_i + 1, 0)}{M_t^D(A_i, 0)}. \quad (\text{I14})$$

Step 2: cancel the Vandermonde factor using the type- D Weyl dimension ratio.

For type D_n (tensor sector) one may write

$$\dim \lambda = \prod_{1 \leq i < j \leq n} \frac{A_i(\lambda)^2 - A_j(\lambda)^2}{b_i^2 - b_j^2}. \quad (\text{I15})$$

Hence, for $\mu \mapsto \mu + e_i$,

$$\frac{\dim(\mu + e_i)}{\dim(\mu)} = \prod_{j \neq i} \frac{(A_i + 1)^2 - A_j^2}{A_i^2 - A_j^2} = \prod_{j \neq i} \frac{A_i + 1 - A_j}{A_i - A_j} \cdot \frac{A_i + 1 + A_j}{A_i + A_j}. \quad (\text{I16})$$

Therefore

$$\frac{\dim(\mu)}{\dim(\mu + e_i)} \frac{\Delta(A(\mu + e_i))}{\Delta(A(\mu))} = \prod_{j \neq i} \frac{A_i + A_j}{A_i + 1 + A_j} \leq 1. \quad (\text{I17})$$

Step 3: compute the kernel entry ratio at column 0. From Equation (I11),

$$M_t^D(p, 0) = I_p(t) + I_p(t) = 2I_p(t). \quad (\text{I18})$$

Thus

$$\frac{M_t^D(A_i + 1, 0)}{M_t^D(A_i, 0)} = \frac{I_{A_i+1}(t)}{I_{A_i}(t)}. \quad (\text{I19})$$

Step 4: combine. Using Equation (I10) we have $\frac{a_{\mu+e_i}}{a_\mu} = \frac{\dim \mu}{\dim(\mu+e_i)} \cdot \frac{D_{\mu+e_i}^D}{D_\mu^D}$. Combine Equation (I14), Equation (I17), and Equation (I19) to obtain Equation (I13). $\square$

Corollary I.5 (Product bound over boxes). *Let λ be a nonempty partition of length at most n . Then*

$$0 \leq a_\lambda(\beta, \alpha) \leq \prod_{(i,j) \in \lambda} \frac{I_{b_i+j}(t)}{I_{b_i+j-1}(t)}, \quad t = \frac{\alpha\beta}{n}, \quad b_i = n - i. \quad (\text{I20})$$

Proof. Iterate Equation (I13) along a one-box growth chain. At the moment the j th box in row i is added, $A_i = b_i + (j - 1)$. $\square$

4. Crossover bound in Casimir form

Lemma I.6 (Casimir in partition coordinates for type D_n). *Let λ be a partition of length at most n . With the root-length convention of Theorem 1.2,*

$$C_2(\lambda) = \sum_{i=1}^n \lambda_i (\lambda_i + 2b_i), \quad b_i = n - i. \quad (\text{I21})$$

Proof. For type D_n , $\rho_i = b_i$ and $C_2(\lambda) = \langle \lambda, \lambda + 2\rho \rangle$. $\square$

Proposition I.7 (A1 crossover bound for $\text{Spin}(2n)$). *There exists an explicit constant $c_D(n) > 0$ such that for every nontrivial λ and every $\alpha \in [1/2, 1]$, $\beta \geq 1$,*

$$a_\lambda(\beta, \alpha) \leq \exp \left(-c_D(n) \frac{C_2(\lambda)}{\beta + \sqrt{C_2(\lambda)}} \right). \quad (\text{I22})$$

Proof. The proof is identical to Appendix G 4 after replacing $b_i = n - i + 1$ by $b_i = n - i$ (and noting $t = \alpha\beta/n$ here as well). For completeness we record the final chain.

By Equation (I20) and Lemma J.2, $\log a_\lambda \leq -\sum_{(i,j) \in \lambda} \text{arsinh}((b_i + j - \frac{1}{2})/t)$. As in Equation (G24),

$$\sum_{(i,j) \in \lambda} \text{arsinh} \left(\frac{b_i + j - \frac{1}{2}}{t} \right) \geq \frac{\sum_{(i,j) \in \lambda} (b_i + j - \frac{1}{2})}{t + K_{\max}}, \quad (\text{I23})$$

with $K_{\max} = b_1 + \lambda_1 - \frac{1}{2} \leq (n-1) + \lambda_1$. Moreover

$$\sum_{(i,j) \in \lambda} (b_i + j - \frac{1}{2}) = \frac{1}{2} \sum_{i=1}^n (\lambda_i^2 + 2b_i \lambda_i) + \frac{1}{2} \sum_{i=1}^n \lambda_i \geq \frac{1}{2} C_2(\lambda),$$

using Equation (I21) and $\lambda_i \geq 0$. Also $\lambda_1 \leq \sqrt{C_2(\lambda)}$. Combining these gives

$$\log a_\lambda \leq -\frac{1}{2} \frac{C_2(\lambda)}{t + (n-1) + \sqrt{C_2(\lambda)}}. \quad (\text{I24})$$

Finally $t \leq \beta/n \leq \beta$ and $\beta \geq 1$ imply $t + (n-1) + \sqrt{C_2} \leq \beta + n + \sqrt{C_2} \leq (n+1)(\beta + \sqrt{C_2})$. Thus Equation (I22) holds with $c_D(n) = 1/(2(n+1))$. $\square$

Appendix J: Total positivity and Bessel-ratio tools used in Appendices G–I

This appendix collects the two analytic “mechanisms” used repeatedly in the one-plaquette A1 proofs for $\text{Sp}(2n)$, $\text{Spin}(2n+1)$, and $\text{Spin}(2n)$:

- (a) total-nonnegativity (TN) and a one-row shift inequality for Vandermonde-normalized minors;
- (b) explicit Amos-type bounds for adjacent modified-Bessel ratios.

The purpose is (i) to avoid repeating the same TN and Bessel-ratio arguments in each family, (ii) to make the logical dependencies in the A1 proofs explicit.

1. Modified Bessel functions: positivity, log-convexity, and Amos bounds

We write $I_k(t)$ for the modified Bessel function of the first kind. For integer $k \in \mathbb{Z}$ and $t > 0$ one has $I_{-k}(t) = I_k(t) > 0$.

Lemma J.1 (Bessel adjacent ratio monotonicity). *Fix $t > 0$ and define $q_k(t) := I_{k+1}(t)/I_k(t)$ for integers $k \geq 0$. Then $k \mapsto q_k(t)$ is nondecreasing. Equivalently, the sequence $k \mapsto I_k(t)$ is log-convex:*

$$I_k(t)^2 \leq I_{k-1}(t) I_{k+1}(t) \quad (k \geq 1). \quad (\text{J1})$$

Proof. This is a classical property of $I_k(t)$ as a function of the order k for fixed $t > 0$. One route is via total positivity: the bi-infinite Toeplitz matrix $(I_{i-j}(t))_{i,j \in \mathbb{Z}}$ is totally nonnegative (Lemma J.4 below), and 2×2 minors give (J1). Monotonicity of $q_k(t)$ is equivalent to (J1). $\square$

Lemma J.2 (Amos-type upper bound). *For every $t > 0$ and every integer $k \geq 0$,*

$$q_k(t) = \frac{I_{k+1}(t)}{I_k(t)} \leq \frac{t}{k + \frac{1}{2} + \sqrt{t^2 + (k + \frac{1}{2})^2}}. \quad (\text{J2})$$

Equivalently,

$$q_k(t) \leq \exp \left(-\operatorname{arsinh} \left(\frac{k + \frac{1}{2}}{t} \right) \right). \quad (\text{J3})$$

Proof. Inequality (J2) is an explicit uniform upper envelope for Bessel ratios; see e.g.¹³. The identity (J3) follows from $u + \sqrt{1 + u^2} = e^{\operatorname{arsinh}(u)}$ applied with $u = (k + \frac{1}{2})/t$. $\square$

2. TN structure and a one-row shift inequality for normalized minors

We use the following standard definition.

Definition J.3 (Totally nonnegative matrix). A real matrix A (finite or infinite) whose rows and columns are indexed by totally ordered sets is *totally nonnegative (TN)* if every minor of every finite submatrix is nonnegative.

Lemma J.4 (PF_∞ for the Gross–Witten/Bessel coefficients). *Fix $t > 0$. The bi-infinite Toeplitz matrix*

$$\mathcal{T}_t := \left(I_{i-j}(t) \right)_{i,j \in \mathbb{Z}} \quad (\text{J4})$$

is totally nonnegative.

Proof. We give a self-contained reduction to a finite-support Toeplitz criterion.

Step 1 (finite Laurent truncations). Fix $m \in \mathbb{N}$ and define the Laurent polynomial

$$F_m(z) := \left(1 + \frac{t}{2m}z\right)^m \left(1 + \frac{t}{2m}z^{-1}\right)^m = \sum_{k=-m}^m a_k^{(m)} z^k. \quad (\text{J5})$$

Let $\mathcal{T}_t^{(m)} := (a_{i-j}^{(m)})_{i,j \in \mathbb{Z}}$ be the associated bi-infinite Toeplitz matrix.

Step 2 ($\mathcal{T}_t^{(m)}$ is TN for every m). The one-sided factor $P_m(z) := (1 + \frac{t}{2m}z)^m$ has all its zeros equal to $-\frac{2m}{t} < 0$. Hence its coefficient sequence has a totally positive Toeplitz matrix by the finite-support Toeplitz criterion²⁰ (Thm. 4.5). The second factor is the reversed polynomial $z^{-m}P_m(z^{-1})$; its coefficient sequence also has a totally positive Toeplitz matrix. Moreover, the Toeplitz matrix of the product $F_m = P_m \cdot z^{-m}P_m(z^{-1})$ is the product of the two Toeplitz matrices (equivalently, the coefficients $a_k^{(m)}$ are the convolution of the two one-sided coefficient sequences), and products of totally nonnegative matrices are totally nonnegative by Cauchy–Binet (see e.g.²⁰ (Prop. 1.4)). Therefore $\mathcal{T}_t^{(m)}$ is totally nonnegative.

Step 3 (coefficientwise limit to $\mathcal{T}_t$). For $z = e^{i\theta}$ we have

$$F_m(e^{i\theta}) = \left|1 + \frac{t}{2m}e^{i\theta}\right|^{2m} = \left(1 + \frac{t}{m}\cos\theta + \frac{t^2}{4m^2}\right)^m \xrightarrow{m \rightarrow \infty} e^{t\cos\theta} \quad (\theta \in [0, 2\pi]). \quad (\text{J6})$$

Also $0 \leq F_m(e^{i\theta}) \leq (1 + \frac{t}{2m})^{2m} \leq e^t$ for all m and all θ . Since F_m is a finite Laurent polynomial, its coefficients are its Fourier coefficients:

$$a_k^{(m)} = \frac{1}{2\pi} \int_0^{2\pi} F_m(e^{i\theta}) e^{-ik\theta} d\theta. \quad (\text{J7})$$

Dominated convergence applied to (J7) and (J6) gives $a_k^{(m)} \rightarrow \frac{1}{2\pi} \int_0^{2\pi} e^{t\cos\theta} e^{-ik\theta} d\theta$. Using the standard cosine–Fourier representation $I_k(t) = \frac{1}{\pi} \int_0^\pi e^{t\cos\theta} \cos(k\theta) d\theta$ (which is used repeatedly in Appendices G–I), we obtain $a_k^{(m)} \rightarrow I_k(t)$ for every fixed $k \in \mathbb{Z}$.

Finally, any finite minor of $\mathcal{T}_t$ depends on finitely many entries $I_{i-j}(t)$. For m larger than the maximal index difference appearing in that minor, the corresponding minor of $\mathcal{T}_t^{(m)}$ uses only coefficients $a_k^{(m)}$ with $|k| \leq m$ and converges entrywise to the desired minor of $\mathcal{T}_t$. Since each $\mathcal{T}_t^{(m)}$ is TN, all such minors are nonnegative, and therefore the limit minor is nonnegative. This proves that $\mathcal{T}_t$ is totally nonnegative. $\square$

The next lemma is the TN “engine” behind all one-box inequalities in Appendices G–I.

Lemma J.5 (One-row shift inequality for Vandermonde-normalized minors). *Let A be a TN matrix whose rows and columns are indexed by integers. Fix an integer $N \geq 1$ and a strictly increasing N -tuple of column indices*

$$C = (c_1 < c_2 < \cdots < c_N). \quad (\text{J8})$$

For every strictly increasing N -tuple of row indices

$$R = (r_1 < \cdots < r_N) \quad (\text{J9})$$

define the Vandermonde factor

$$\Delta(R) := \prod_{1 \leq p < q \leq N} (r_q - r_p) > 0, \quad (\text{J10})$$

and the normalized minor

$$\Phi_A(R) := \frac{\det(A[R, C])}{\Delta(R)} \geq 0. \quad (\text{J11})$$

Fix an index $i \in \{1, \dots, N\}$ such that replacing r_i by $r_i + 1$ and reordering preserves strict increase. Let R^+ denote the resulting row tuple. Then

$$\Phi_A(R^+) \leq \frac{A[r_i + 1, c_1]}{A[r_i, c_1]} \Phi_A(R). \quad (\text{J12})$$

Proof. If $A[r_i, c_1] = 0$ then $\det(A[R, C]) = 0$ (by TN and expansion along the first column), so (J12) is trivial. Assume henceforth that $A[r, c_1] > 0$ for all rows r appearing in R and R^+ .

Row-scaling by positive factors preserves total nonnegativity. Let D be the diagonal matrix with entries $D[r, r] = 1/A[r, c_1]$ and set $\tilde{A} := DA$. Then $\tilde{A}$ is TN and satisfies $\tilde{A}[\cdot, c_1] \equiv 1$ on the relevant rows. To invoke the standard node-monotonicity statement from total positivity, we reduce to the strictly totally positive case: by the Whitney density construction (see e.g.²⁰ (Thm. 2.6 and its proof)), every finite TN submatrix is the entrywise limit of strictly

totally positive matrices. Since the quantities in (J12) depend continuously on the finitely many involved entries, it suffices to prove the desired inequality for strictly totally positive matrices.

In the strictly totally positive case, for fixed columns C the Vandermonde-normalized minors (J11) are generalized (Popoviciu–Karlin) divided differences associated with the corresponding discrete totally positive (Chebyshev) system, and they are monotone in each node. For an explicit modern statement with numbered results and a proof via Sylvester’s determinant identity, see²¹ (Cor. 3(ii)); this monotonicity is traced there to²² (Thm. 2). Applying this one-step node monotonicity to the single-node shift $r_i \mapsto r_i + 1$ gives

$$\Phi_{\tilde{A}}(R^+) \leq \Phi_{\tilde{A}}(R).$$

Undoing the row scaling uses that for any row tuple R one has

$$\det(A[R, C]) = \left(\prod_{p=1}^N A[r_p, c_1] \right) \det(\tilde{A}[R, C]) \quad \text{and hence} \quad \Phi_A(R) = \left(\prod_{p=1}^N A[r_p, c_1] \right) \Phi_{\tilde{A}}(R),$$

and similarly for R^+ . Therefore

$$\begin{aligned} \Phi_A(R^+) &= \left(\prod_{p=1}^N A[r_p^+, c_1] \right) \Phi_{\tilde{A}}(R^+) \\ &\leq \left(\prod_{p=1}^N A[r_p^+, c_1] \right) \Phi_{\tilde{A}}(R) \\ &= \frac{\prod_{p=1}^N A[r_p^+, c_1]}{\prod_{p=1}^N A[r_p, c_1]} \Phi_A(R) \\ &= \frac{A[r_i + 1, c_1]}{A[r_i, c_1]} \Phi_A(R). \end{aligned}$$

which is (J12). □

3. TN of the classical-group Bessel moment matrices

Appendices G–I use explicit moment matrices of the form

$$M_t(p, q) = \frac{2}{\pi} \int_0^\pi e^{t \cos \theta} \psi_p(\theta) \psi_q(\theta) d\theta, \tag{J13}$$

for specific trigonometric systems ψ_p (sine with integer or half-integer frequencies, or cosine).

Lemma J.6 (TN of the moment matrices). *Fix $t > 0$. Let $\psi_p(\theta)$ be one of the three families:*

- (i) $\psi_p(\theta) = \sin(p\theta)$ for $p \in \mathbb{N}$ (symplectic case);
- (ii) $\psi_p(\theta) = \sin\left((p + \frac{1}{2})\theta\right)$ for $p \in \mathbb{N}_0$ (odd orthogonal case);
- (iii) $\psi_p(\theta) = \cos(p\theta)$ for $p \in \mathbb{N}_0$ (even orthogonal case).

Then the infinite matrix $(M_t(p, q))$ defined by (J13) is totally nonnegative.

Proof. The trigonometric kernels $(p, \theta) \mapsto \psi_p(\theta)$ in (i)–(iii) are strictly sign-regular kernels of all orders on the ordered sets $p \in \mathbb{N}$ or $\mathbb{N}_0$ and $\theta \in (0, \pi)$. A fundamental closure theorem in total positivity states that composing two sign-regular kernels against a strictly positive measure preserves sign-regularity (hence TN at the discrete level). Applying this to the composition $M_t(p, q) = \int \psi_p(\theta)\psi_q(\theta) e^{t\cos\theta} d\theta$ with the strictly positive weight $e^{t\cos\theta}$ yields TN. We refer to¹⁹ (Vol. I, Ch. 3) for the sign-regularity of the sine/cosine kernels and the composition theorem. $\square$

How this appendix is used. In Appendices G–I the matrices whose minors appear after Andréief reduction are precisely of the form (J13), so Lemma J.6 supplies the TN hypothesis needed in Lemma J.5. Lemma J.2 then provides the quantitative decay needed to turn the telescoping product over boxes into the crossover estimate (4).

Appendix K: Parameter schedule and explicit threshold bookkeeping

This appendix records a single explicit schedule of thresholds which subsumes every “take β large” / “take ε small” clause used in the Lane C (RG closure), Lane A (sharp-local reconstruction), and primary Route 1 mass-gap chains. The intent is theorem-level transparency: once the group-dependent inputs are instantiated, all subsequent choices are forced. Lane B introduces no additional threshold beyond the downstream dense-core input already derived in Appendix F.

1. Named group-dependent inputs

Fix a compact connected simple Lie group G together with the auxiliary faithful Wilson representation ρ from Theorem 2.1. The canonical continuum local theory is faithful- ρ independent by Appendix O, but the explicit ultraviolet and Route 1 constants recorded here are attached to this chosen faithful regularization. The following quantities are treated as

the *base input constants* in this manuscript (some group-only, some (G, ρ) -indexed through the chosen auxiliary faithful regularization):

- (P1) $b_0(G) > 0$: dyadic-gradient-flow one-loop coefficient (Section 4.2).
- (P2) $K_{0,G} \geq 1$: the heat-kernel/parametrix constant used in Appendices B–D.
- (P3) $\lambda_* > 0$: the universal tree-gauge coercivity constant of Appendix E.
- (P4) The compact-simple A1 constant chain from Appendix N:

$$\begin{aligned} & \kappa_-(G), \kappa_+(G), c_{\text{hyp}}(G, \rho), r_{\text{hyp}}(G, \rho), \beta_{\text{hyp}}(G, \rho), \\ & r_{\text{strip}}(G, \rho), c_{\text{strip}}(G, \rho), C_{\text{shift}}(G, \rho), c_{\text{shift}}(G, \rho), \beta_{\text{shift}}(G, \rho), \\ & \sigma_{\text{sc}}(G, \rho), \beta_{\text{sc}}(G, \rho), c_{\text{lo}}(G, \rho), \beta_{\text{lo}}(G, \rho), c_{\text{hi}}(G, \rho), \beta_{\text{hi}}(G, \rho). \end{aligned}$$

Here $C_{\text{shift}}(G, \rho)$, $c_{\text{shift}}(G, \rho)$, and $\beta_{\text{shift}}(G, \rho)$ are the explicit constants returned by the shifted-torus Laplace estimate (Theorem N.17); the internal low-Casimir semiclassical theorem Theorem N.9 then returns $\sigma_{\text{sc}}(G, \rho)$ and $\beta_{\text{sc}}(G, \rho)$, with the low-Casimir formulas recorded explicitly in Theorem N.14:

$$c_{\text{lo}}(G, \rho) = \frac{\sigma_{\text{sc}}(G, \rho)}{\kappa_+(G)}, \quad \beta_{\text{lo}}(G, \rho) = \max\{\beta_{\text{hyp}}(G, \rho), \beta_{\text{sc}}(G, \rho)\}.$$

The crossover constants $c_{\text{A1}}(G, \rho)$ and $\beta_{\text{A1}}(G, \rho)$ are then defined explicitly in Theorem N.20; the classical Appendices G–I remain examples rather than load-bearing inputs.

- (P5) $\rho_G \in (0, 1)$: the linear contraction factor in the dyadic seminorm topology (Section F 4 d).

All other constants appearing in the body are defined deterministically from these inputs and from universal inequalities already stated in the cited lemmas/propositions.

2. Master thresholds

Define the master inverse-coupling threshold

$$\beta_{\text{master}}(G, \rho) := \max\left\{\beta_{\text{A1}}(G, \rho), \beta_{\text{HK}}(G), \beta_{\text{RG}}(G), \beta_{\text{KP}}(G, \rho), \beta_{\text{OS}}(G)\right\}. \quad (\text{K1})$$

Here the A1 and KP thresholds remain indexed by the chosen faithful ultraviolet regularization ρ , while the pure kernel/RG/OS bookkeeping thresholds are recorded in the present submission only as group-dependent analytic constants. Each term is defined as follows.

(i) *Kernel threshold* $\beta_{\text{HK}}(G)$. Let $\beta_{\text{HK}}(G)$ be the maximum of the (finite) thresholds introduced in the kernel stack (Appendices B–D) ensuring all stated bounds hold with constant $K_{0,G}$. Concretely, it is the maximum of the thresholds in Lemmas B.3, C.2, and D.2.

(ii) *RG contraction threshold* $\beta_{\text{RG}}(G)$. Let $\beta_{\text{RG}}(G)$ be the smallest β for which the one-step RG map is a strict contraction on the Lane C ball, i.e. for which the quantitative hypothesis of Proposition F.291 is satisfied with the constants produced by λ_* and $K_{0,G}$.

(iii) *KP entrance threshold* $\beta_{\text{KP}}(G, \rho)$. Let $\beta_{\text{KP}}(G, \rho)$ be the smallest β for which the one-shot entrance theorem of Appendix F applies at the prescribed block size $B(\beta; G, \rho)$. Concretely, this threshold is determined by the compact-simple A1 input (Theorem 1.4 and its explicit classical examples in Appendices G–I), the transfer-feasibility threshold of Proposition F.193, and the theorem-level reference-budget closure summarized in Corollary F.201.

(iv) *OS regularity threshold* $\beta_{\text{OS}}(G)$. Let $\beta_{\text{OS}}(G)$ be the maximum of the (finite) thresholds introduced in the Lane A sharp-local/OS closure (Section 5, Appendix A, and the axioms package of Appendix M), guaranteeing that all OS axioms and the quoted reconstruction inputs are verified with the stated constants.

3. Closure statement in threshold form

Assume $\beta \geq \beta_{\text{master}}(G, \rho)$. Then Lane C yields a unique continuum flow-bundle state (no subsequence extraction), Lane A yields the local gauge-invariant sharp-local OS theory, and Appendix F yields the primary Route 1 mass-gap chain: one-shot entrance, a common-rate fixed-lattice local decay theorem at each tuned lattice, the tuned-sequence full fixed-lattice OS gap, and the resulting continuum vacuum spectral gap. Section 6 then rederives that same vacuum-gap statement from the downstream dense-core Euclidean estimate without introducing any additional thresholds.

Remark K.1 (Fail-closed discipline). No additional tuning parameters occur beyond those explicitly named above. Any attempted introduction of an extra degree of freedom (e.g. by altering the seminorm topology, truncation chart, or entrance depth) constitutes a scope

change and must be stated as a separate theorem statement.

Appendix N: Compact-simple A1 theorem and public group scope

This appendix is Component 1 in the public theorem decomposition of Section 2. It supplies the only genuinely Lie-type-dependent input on the primary proof spine at the level of an arbitrary compact connected simple gauge group. The public Route 1/Lane A/Lane B chain therefore no longer depends on the classical-family appendices for its theorem-level scope: those appendices remain in the manuscript as explicit examples for the standard Sp/Spin choices of Wilson representation, but the load-bearing A1 theorem used by the main theorem is the compact-simple statement proved here.

1. Compact-simple A1 theorem and constant map

Theorem chain. For a first reading, this theorem family can be read in the following order. The low-Casimir branch is closed internally by the shifted-torus Laplace theorem Theorem N.9, exported to the Wilson coefficients by Theorem N.13, and traced by Theorem N.14. The complementary high-Casimir branch is closed by the torus-strip Peter–Weyl theorem Theorem N.19. These two regime theorems are combined only in the compact-simple crossover theorem Theorem N.20, and the public Route 1 / Lane A / Lane B discussion then uses this theorem family only through Theorem N.21 together with the synchronized (G, ρ) constant entries recorded in Appendix K.

Exceptional-type closure. No separate exceptional-family appendix is needed on the public spine. The exceptional-group case is closed exactly where the arbitrary compact connected simple hypotheses first appear, namely in the two regime theorems Theorems N.9 and N.19, and hence in the assembled compact-simple theorem Theorem N.20.

Status of the classical-family appendices. Appendices G–I are retained as explicit examples and classical-family sample calculations only. After Theorem N.21 they are explanatory rather than load-bearing for the public compact-simple group scope.

Constant map. The front-end constants used downstream separate naturally into the following groups.

(G)-only data::

$d_G, r_G, N_+, \delta_W, P_W(\delta_W), \kappa_\pm(G)$. These enter the weight/Casimir comparison and the regime-to-crossover assembly in Theorems N.9, N.19 and N.20.

(G, ρ) local geometry::

$r_0, m_\pm, j_\pm, a_0, c_Z^\pm, \beta_Z, c_{\text{hyp}}, r_{\text{hyp}}, \beta_{\text{hyp}}$. These control faithful local geometry, weak-coupling partition bounds, and the local Gaussian input consumed first by Theorem N.9.

(G, ρ) low-Casimir branch::

$r_{\text{ph}}, c_{\text{ph}}, C_{\text{ph}}, \sigma_{\text{sc}}, \beta_{\text{sc}}, c_{\text{lo}}, \beta_{\text{lo}}$. These feed Theorem N.13 and the traced constant node Theorem N.14.

(G, ρ) high-Casimir branch::

$r_{\text{strip}}, c_{\text{strip}}, C_{\text{PW}}, r_{\text{hi}}, c_{\text{hi}}, \beta_{\text{hi}}$. These feed the strip-based decay theorem Theorem N.19.

Public output::

c_{A1}, β_{A1} . These are the public compact-simple theorem output in Theorem N.21 and the explicit Route 1 ultraviolet summary first read by Proposition F.3.

2. Faithful reduction and local Wilson geometry

Fix a compact connected simple Lie group G and a finite-dimensional unitary representation $\rho : G \rightarrow U(d_\rho)$. Set $K_\rho := \ker(\rho)$ and let $G_{\text{eff}} := G/K_\rho$ be the effective quotient. By Theorems 1.1, 2.4 and 2.5, the Wilson plaquette weight and its character coefficients factor through G_{eff} . Accordingly it suffices to prove the A1 bound for the descended *faithful* representation on G_{eff} . In this appendix we therefore work directly in the faithful case and suppress the descent notation.

Recall the Wilson one-plaquette action density

$$A_\rho(g) := 1 - \frac{1}{d_\rho} \Re \text{Tr}_\rho(g), \quad W_{\beta, \alpha}(g) := e^{-\alpha \beta A_\rho(g)}, \quad \nu_{\beta, \alpha} = Z_{\beta, \alpha}^{-1} W_{\beta, \alpha}(g) dg.$$

Lemma N.1 (Faithful Wilson actions have a unique nondegenerate minimum). *Assume ρ is faithful. Then $A_\rho(g) \geq 0$ for all $g \in G$, with equality if and only if $g = e$. Moreover there exist constants*

$$r_0 > 0, \quad 0 < m_- \leq m_+ < \infty, \quad 0 < j_- \leq j_+ < \infty, \quad a_0 > 0,$$

depending only on (G, ρ) , such that in exponential coordinates $g = e^X$ with $\|X\| \leq r_0$ one has

$$\frac{m_-}{2} \|X\|^2 \leq A_\rho(e^X) \leq \frac{m_+}{2} \|X\|^2, \quad j_- \leq J(X) \leq j_+, \quad (\text{N1})$$

where $dg = J(X) dX$, and such that

$$A_\rho(g) \geq a_0 \quad (g \notin e^{B_{r_0}(0)}). \quad (\text{N2})$$

Proof. Because $\rho(g)$ is unitary, $|\text{Tr}_\rho(g)| \leq d_\rho$, hence $A_\rho(g) \geq 0$. If $A_\rho(g) = 0$, then $\Re \text{Tr}_\rho(g) = d_\rho$; for a unitary matrix this forces every eigenvalue to be 1, hence $\rho(g) = I$. Faithfulness therefore gives $g = e$. Expanding $\rho(e^X) = I + d\rho(X) + \frac{1}{2}d\rho(X)^2 + O(\|X\|^3)$ and using that $d\rho(X)$ is skew-Hermitian, one obtains

$$A_\rho(e^X) = \frac{1}{2d_\rho} \text{Tr}((-d\rho(X)^2)) + O(\|X\|^3).$$

The quadratic form $X \mapsto -\text{Tr}(d\rho(X)^2)$ is positive definite because ρ is faithful, and it is Ad-invariant because ρ is a representation. Since G is simple, every positive Ad-invariant quadratic form is a positive scalar multiple of the fixed metric from Theorem 1.2. This yields the quadratic sandwich in sufficiently small exponential coordinates. The Jacobian bounds are standard on a compact manifold, and the positive lower bound away from the identity follows from continuity and the uniqueness of the minimum. $\square$

Lemma N.2 (Wilson partition function in the weak-coupling window). *Under the hypotheses of Theorem N.1, there exist constants $c_Z^\pm(G, \rho) > 0$ and $\beta_Z(G, \rho) \geq 1$ such that for all $\beta \geq \beta_Z(G, \rho)$ and all $\alpha \in [1/2, 1]$,*

$$c_Z^-(G, \rho) \beta^{-\dim G/2} \leq Z_{\beta, \alpha} \leq c_Z^+(G, \rho) \beta^{-\dim G/2}. \quad (\text{N3})$$

Proof. The upper bound is immediate from Equations (N1) and (N2) exactly as in Theorem F.188. For the lower bound, integrate only over the ball $\|X\| \leq r_0/\sqrt{\beta}$ and use the upper quadratic bound in Equation (N1) together with $J(X) \geq j_-$. Both estimates are uniform in $\alpha \in [1/2, 1]$ because the factor α is bounded above and below away from 0. $\square$

3. Low-Casimir input

Write

$$f_{\beta, \alpha}(g) := Z_{\beta, \alpha}^{-1} W_{\beta, \alpha}(g), \quad d_G := \dim G.$$

Fix once and for all a maximal torus $T \subset G$ with Lie algebra $\mathfrak{t}$, a choice of positive roots, the corresponding half-sum δ_W , and the Weyl denominator

$$\Delta(\theta) := \prod_{\alpha > 0} (e^{i\alpha(\theta)/2} - e^{-i\alpha(\theta)/2}), \quad \theta \in \mathfrak{t}/(2\pi\Gamma).$$

Set

$$r_G := \dim T, \quad N_+ := |\Phi^+| = \frac{d_G - r_G}{2}.$$

For every irreducible π with highest weight λ_π , Weyl's character formula reads

$$\chi_\pi(\theta) \Delta(\theta) = \sum_{w \in W} \det(w) e^{i\langle w(\lambda_\pi + \delta_W), \theta \rangle}. \quad (\text{N4})$$

Lemma N.3 (Coefficient extraction for normalized Wilson coefficients). *Let F be a central real-analytic class function on G with Peter–Weyl expansion*

$$F = \sum_{\pi} d_{\pi} c_{\pi} \chi_{\pi}, \quad d_{\pi} := \dim \pi,$$

converging in the real-analytic topology. Then on the maximal torus,

$$F(\theta) \Delta(\theta) = \sum_{\pi} d_{\pi} c_{\pi} \sum_{w \in W} \det(w) e^{i\langle w(\lambda_{\pi} + \delta_W), \theta \rangle}. \quad (\text{N5})$$

In particular, the Fourier coefficient of $F\Delta$ at the dominant weight $\lambda_{\pi} + \delta_W$ is exactly $d_{\pi} c_{\pi}$. Applied to $F = f_{\beta, \alpha}$ and $c_{\pi} = a_{\pi}(\beta, \alpha)$, this coefficient is $d_{\pi} a_{\pi}(\beta, \alpha)$.

Proof. Multiply the Peter–Weyl expansion of F by the Weyl denominator and use Equation (N4) term by term. Because $\lambda_{\pi} + \delta_W$ is strictly dominant, no other Weyl orbit contains it, so its Fourier coefficient occurs only in the π -summand and with coefficient $d_{\pi} c_{\pi}$. $\square$

Lemma N.4 (Highest-weight/Casimir comparison). *Fix the norm $|\cdot|$ on dominant weights induced by the invariant metric from Theorem 1.2. Then there exist constants $0 < \kappa_- \leq \kappa_+ < \infty$, depending only on G , such that for every nontrivial irreducible representation π with highest weight λ_{π} ,*

$$\kappa_- |\lambda_{\pi}|^2 \leq C_2(\pi) \leq \kappa_+ |\lambda_{\pi}|^2. \quad (\text{N6})$$

Proof. For the chosen invariant inner product, the quadratic Casimir eigenvalue is

$$C_2(\pi) = \langle \lambda_{\pi}, \lambda_{\pi} + 2\delta_W \rangle,$$

where δ_W is the half-sum of positive roots. Because λ_π and δ_W lie in the closed dominant chamber, the cross term is nonnegative, so $C_2(\pi) \geq |\lambda_\pi|^2$; this gives the lower bound with $\kappa_- = 1$. For the upper bound,

$$C_2(\pi) \leq |\lambda_\pi|^2 + 2|\delta_W| |\lambda_\pi|.$$

The dominant integral weights form a lattice, so

$$\ell_0 := \min\{|\lambda| : \lambda \in \Lambda_+ \setminus \{0\}\} > 0.$$

Hence $|\lambda_\pi| \geq \ell_0$ for every nontrivial π , and therefore $2|\delta_W| |\lambda_\pi| \leq 2|\delta_W| \ell_0^{-1} |\lambda_\pi|^2$. This yields the upper bound with $\kappa_+ := 1 + 2|\delta_W| \ell_0^{-1}$. $\square$

Lemma N.5 (Semiclassical hypotheses for the normalized Wilson family). *There exist constants $c_{\text{hyp}}(G, \rho) > 0$, $r_{\text{hyp}}(G, \rho) \in (0, r_0]$, and $\beta_{\text{hyp}}(G, \rho) \geq 1$ with the following property. For every $\beta \geq \beta_{\text{hyp}}(G, \rho)$ and every $\alpha \in [1/2, 1]$:*

- (i) $f_{\beta, \alpha}$ is a central real-analytic probability density on G ;
- (ii) in exponential coordinates $g = e^X$ with $\|X\| \leq r_{\text{hyp}}$,

$$f_{\beta, \alpha}(e^X) \leq C_0(G, \rho) \beta^{d_G/2} \exp(-c_{\text{hyp}}(G, \rho) \beta \|X\|^2); \quad (\text{N7})$$

- (iii) for every multi-index γ there is a constant $C_\gamma(G, \rho) < \infty$ such that the rescaled profile

$$F_{\beta, \alpha}(Y) := \beta^{-d_G/2} f_{\beta, \alpha}(e^{Y/\sqrt{\beta}})$$

satisfies

$$|\partial_Y^\gamma F_{\beta, \alpha}(Y)| \leq C_\gamma(G, \rho) \exp(-c_{\text{hyp}}(G, \rho) \|Y\|^2) \quad (\text{N8})$$

whenever $\|Y\| \leq \frac{1}{2} r_{\text{hyp}} \sqrt{\beta}$;

- (iv) the tail outside the local chart is exponentially small:

$$\int_{G \setminus e^{B_{r_{\text{hyp}}}(0)}} f_{\beta, \alpha}(g) dg \leq C_0(G, \rho) e^{-c_{\text{hyp}}(G, \rho) \beta}. \quad (\text{N9})$$

Proof. Centrality and real analyticity are immediate because A_ρ is a central real-analytic class function and $f_{\beta, \alpha}$ is obtained from it by normalization. The probability-mass statement is the definition of $\nu_{\beta, \alpha}$.

Choose $r_{\text{hyp}} \leq r_0$ so that the quadratic bounds of Theorem N.1 hold throughout $B_{r_{\text{hyp}}}(0)$. Then for $\alpha \in [1/2, 1]$ and $\|X\| \leq r_{\text{hyp}}$,

$$\alpha \beta A_\rho(e^X) \geq \frac{\beta}{2} \cdot \frac{m_-}{2} \|X\|^2 = \frac{m_-}{4} \beta \|X\|^2.$$

Combining this with the lower bound $Z_{\beta,\alpha} \geq c_Z^- \beta^{-d_G/2}$ from Theorem N.2 gives (N7) with $c_{\text{hyp}} := m_-/4$ and $C_0 := (c_Z^-)^{-1}$.

For the rescaled profile,

$$F_{\beta,\alpha}(Y) = \beta^{-d_G/2} Z_{\beta,\alpha}^{-1} \exp\left(-\alpha \beta A_\rho(e^{Y/\sqrt{\beta}})\right).$$

The prefactor $\beta^{-d_G/2} Z_{\beta,\alpha}^{-1}$ is uniformly bounded by Theorem N.2. Because $A_\rho(e^X)$ is real-analytic in X on $B_{r_{\text{hyp}}}(0)$, every derivative of $\beta A_\rho(e^{Y/\sqrt{\beta}})$ is bounded on $\|Y\| \leq \frac{1}{2} r_{\text{hyp}} \sqrt{\beta}$ by a polynomial in $1 + \|Y\|$ whose coefficients depend only on (G, ρ) and the derivative order. Applying Faà di Bruno's formula to the exponential therefore gives

$$|\partial_Y^\gamma F_{\beta,\alpha}(Y)| \leq P_\gamma(1 + \|Y\|) \exp\left(-\frac{m_-}{8} \|Y\|^2\right)$$

for a polynomial P_γ depending only on (G, ρ, γ) . Absorbing the polynomial into the Gaussian yields (N8) after slightly decreasing c_{hyp} .

Finally, on $G \setminus e^{B_{r_{\text{hyp}}}(0)}$ one has $A_\rho \geq a_0$ by Equation (N2), hence

$$f_{\beta,\alpha}(g) \leq (c_Z^-)^{-1} \beta^{d_G/2} e^{-\alpha \beta a_0} \leq (c_Z^-)^{-1} \beta^{d_G/2} e^{-a_0 \beta/2}.$$

Integrating over the compact group and absorbing the polynomial factor in β into the exponential by enlarging the weak-coupling threshold yields (N9). $\square$

Lemma N.6 (Weyl dimension formula and local denominator factorization). *Let*

$$P_W(\xi) := \prod_{\alpha > 0} \langle \xi, \alpha^\vee \rangle,$$

where roots and coroots are identified using the invariant metric from Theorem 1.2. Then for every irreducible representation π ,

$$d_\pi = \frac{P_W(\lambda_\pi + \delta_W)}{P_W(\delta_W)}. \quad (\text{N10})$$

Consequently there exist constants $0 < d_{\text{dim}}^- \leq d_{\text{dim}}^+ < \infty$, depending only on G , such that

$$d_{\text{dim}}^- (1 + |\lambda_\pi|)^{N_+} \leq d_\pi \leq d_{\text{dim}}^+ (1 + |\lambda_\pi|)^{N_+}. \quad (\text{N11})$$

Moreover there exist $r_\Delta(G) \in (0, \pi]$ and a Weyl-invariant holomorphic function B_Δ on $|\zeta| < r_\Delta(G)$ such that

$$\Delta(hY) = h^{N_+} P_W(Y) B_\Delta(hY) \quad (\text{N12})$$

for every $0 < h \leq 1$ and every $Y \in \mathfrak{t}_\mathbb{C}$ with $|hY| < r_\Delta(G)$. Furthermore there are constants $0 < c_\Delta^- \leq c_\Delta^+ < \infty$ such that

$$c_\Delta^- \leq |B_\Delta(hY)| \leq c_\Delta^+ \quad (|hY| \leq r_\Delta(G)/2). \quad (\text{N13})$$

Proof. Formula (N10) is Weyl's dimension formula. The polynomial growth bounds (N11) are immediate because P_W is a homogeneous polynomial of degree N_+ and δ_W is fixed. For the denominator, each factor satisfies

$$e^{i\alpha(hY)/2} - e^{-i\alpha(hY)/2} = ih \alpha(Y) \left(\frac{\sin(\alpha(hY)/2)}{\alpha(hY)/2} \right),$$

and the product of the analytic sine-quotients is a Weyl-invariant holomorphic function. After the metric identification between roots and coroots, all fixed normalization factors are absorbed into B_Δ . Because $B_\Delta(0) \neq 0$, shrinking the radius if necessary yields (N13). $\square$

Lemma N.7 (Weyl anti-invariant divisibility). *Let Ψ be a holomorphic Weyl anti-invariant function on a tube neighborhood of the origin in $\mathfrak{t}_\mathbb{C}$. Then there exists a unique Weyl-invariant holomorphic function H on the same neighborhood such that*

$$\Psi(\xi) = P_W(\xi) H(\xi). \quad (\text{N14})$$

If, on a smaller tube, Ψ satisfies a Gaussian bound

$$|\Psi(\xi)| \leq C(1 + |\xi|)^m e^{-\sigma|\xi|^2},$$

then H satisfies a bound of the same form with possibly different constants depending only on (G, m, σ) .

Proof. For every simple reflection $s_\alpha \in W$, anti-invariance gives $\Psi(s_\alpha \xi) = -\Psi(\xi)$, so Ψ vanishes on the reflection hyperplane $\langle \xi, \alpha^\vee \rangle = 0$. Because these zeros are simple and the defining linear forms are pairwise coprime, their product $P_W(\xi)$ divides Ψ holomorphically. The quotient is automatically Weyl-invariant. The Gaussian bound for H follows by dividing on a smaller tube and using the lower bound for $P_W(\xi)$ away from the reflection hyperplanes together with removability across the hyperplanes. $\square$

Lemma N.8 (Complexified local phase control). *There exist constants*

$$r_{\text{ph}}(G, \rho) \in \left(0, \min\{r_{\text{hyp}}(G, \rho), r_{\Delta}(G), r_{\text{strip}}(G, \rho)\}\right], \quad c_{\text{ph}}(G, \rho) > 0, \quad C_{\text{ph}}(G, \rho) < \infty,$$

with the following property. For every $\alpha \in [1/2, 1]$, every $0 < h \leq 1$, and every $Y, \eta \in \mathfrak{t}$ satisfying $|Y|, |\eta| \leq r_{\text{ph}}(G, \rho)h^{-1}$, the local phase

$$Q_{h,\alpha}(Y + i\eta) := \alpha h^{-2} \tilde{A}_{\rho}(h(Y + i\eta)) \quad (\text{N15})$$

is holomorphic and obeys

$$\Re Q_{h,\alpha}(Y + i\eta) \geq c_{\text{ph}}(G, \rho)|Y|^2 - C_{\text{ph}}(G, \rho)|\eta|^2. \quad (\text{N16})$$

Moreover the local amplitude

$$E_{h,\alpha}(Z) := h^{d_G} Z_{h^{-2},\alpha}^{-1} B_{\Delta}(hZ) \quad (\text{N17})$$

is holomorphic on the same tube and satisfies the uniform bound

$$|E_{h,\alpha}(Y + i\eta)| \leq C_{\text{ph}}(G, \rho). \quad (\text{N18})$$

Proof. The torus extension $\tilde{A}_{\rho}$ is holomorphic by Equation (N28) below. Near the origin,

$$\tilde{A}_{\rho}(Z) = \frac{1}{2} \langle H_{\rho} Z, Z \rangle + O(|Z|^3),$$

with H_{ρ} positive definite on $\mathfrak{t}$ by Theorem N.1. Shrinking the radius and using $\alpha \in [1/2, 1]$ gives

$$\Re \tilde{A}_{\rho}(h(Y + i\eta)) \geq \frac{c_{\text{ph}}}{2} h^2 |Y|^2 - C_{\text{ph}} h^2 |\eta|^2,$$

which is exactly (N16) after dividing by h^2 . The bound (N18) follows from Theorem N.2 and Equation (N13) because $h^{d_G} Z_{h^{-2},\alpha}^{-1} = \beta^{-d_G/2} Z_{\beta,\alpha}^{-1}$ stays uniformly bounded and B_{Δ} is bounded on the chosen tube. $\square$

Theorem N.9 (Internal low-Casimir kernel-to-multiplier theorem). *Set*

$$L_0 := 2\kappa_-^{-1/2}.$$

There exist constants $\sigma_{\text{sc}}(G, \rho) > 0$ and $\beta_{\text{sc}}(G, \rho) \geq 1$ such that for every $\beta \geq \beta_{\text{sc}}(G, \rho)$, every $\alpha \in [1/2, 1]$, and every nontrivial irreducible representation π with

$$|\lambda_{\pi}| \leq L_0 \beta, \quad (\text{N19})$$

one has

$$|a_\pi(\beta, \alpha)| \leq \exp\left(-\sigma_{\text{sc}}(G, \rho) |\lambda_\pi|^2 / \beta\right). \quad (\text{N20})$$

Thus the parabolic low-Casimir window is closed internally, with no remaining imported compact-group harmonic-analysis theorem on Component 1.

Lemma N.10 (Trivial-mode normalization for the local Weyl quotient). *Fix*

$$L_0 := 2\kappa_-^{-1/2}.$$

Let $h \in (0, 1]$, let $\alpha \in [1/2, 1]$, and let $\widehat{\Psi}_{h,\alpha}$ be holomorphic on $|\xi| \leq 2L_0 h^{-1}$. Assume that

$$|\widehat{\Psi}_{h,\alpha}(\xi)| \leq C \exp(-2\sigma|\xi|^2) \quad (|\xi| \leq 2L_0 h^{-1})$$

for some $\sigma > 0$, and that there is a Weyl-invariant holomorphic function $M_{h,\alpha}$ on the same ball with

$$P_W(i\partial_\xi) \widehat{\Psi}_{h,\alpha}(\xi) = P_W(\xi) M_{h,\alpha}(\xi).$$

Assume moreover that the localized coefficient extraction step has produced

$$a_\pi(\beta, \alpha) = P_W(\delta_W) M_{h,\alpha}(h\nu_\pi) + \mathcal{E}_{\pi,h,\alpha}, \quad |\mathcal{E}_{\pi,h,\alpha}| \leq e^{-c\beta}/d_\pi, \quad (\text{N21})$$

for every representation in the parabolic window, with $c > 0$ independent of (h, α, π) . Then, for all sufficiently large β , the trivial representation fixes the normalization explicitly:

$$P_W(\delta_W) M_{h,\alpha}(h\delta_W) = 1 + O(e^{-c\beta}). \quad (\text{N22})$$

Consequently the normalized quotient

$$M_{h,\alpha}^{\text{norm}}(\xi) := \frac{M_{h,\alpha}(\xi)}{P_W(\delta_W) M_{h,\alpha}(h\delta_W)}$$

is well defined, satisfies

$$P_W(\delta_W) M_{h,\alpha}^{\text{norm}}(h\delta_W) = 1,$$

and, after shrinking σ if necessary,

$$|P_W(\delta_W) M_{h,\alpha}^{\text{norm}}(\xi)| \leq \exp(-\sigma(|\xi|^2 - |h\delta_W|^2)) \quad (|\xi| \leq 2L_0 h^{-1}). \quad (\text{N23})$$

Proof. For the trivial representation $\mathbf{1}$ one has $\lambda_1 = 0$, $\nu_1 = \delta_W$, $d_1 = 1$, and $a_1(\beta, \alpha) = 1$ because $f_{\beta,\alpha}$ is a probability density. Evaluating (N21) at $\mathbf{1}$ yields (N22), so the denominator

in the definition of $M_{h,\alpha}^{\text{norm}}$ is nonzero once β is large. To estimate the quotient, run the same shifted-box contour argument used for the Gaussian transform bound, but anchor it at the trivial mode $h\delta_W$. The shift $Y \mapsto Y + i\tau(\xi - h\delta_W)$ contributes the phase gain $-\tau(|\xi|^2 - |h\delta_W|^2)$, while the derivatives entering $P_W(i\partial_\xi)$ only insert the fixed polynomial $P_W(Y)$ in the Fourier integral. Since the factorization has already removed the linear Weyl-wall vanishing, Theorem N.7 promotes the same anchored Gaussian estimate to the invariant quotient, after shrinking σ if necessary. Finally divide by the trivial-mode value from (N22); the normalization makes the anchor value exactly 1 and yields (N23). $\square$

Proof. Set $h := \beta^{-1/2}$ and, for the chosen π , write

$$\nu_\pi := \lambda_\pi + \delta_W.$$

By Theorem N.3, the Fourier coefficient of $f_{\beta,\alpha}\Delta$ at ν_π is exactly $d_\pi a_\pi(\beta, \alpha)$. Choose

$$r_* := \frac{1}{2}r_{\text{ph}}(G, \rho).$$

The contribution of $T \setminus B_{r_*}(0)$ to that Fourier coefficient is $O(e^{-c\beta})$, uniformly in α , by Equation (N9) and the boundedness of Δ on the compact torus. On the local ball $|\theta| \leq r_*$, write $\theta = hY$. Using Equation (N12), the identity $d_G = r_G + 2N_+$, and the flat torus measure in exponential coordinates, the local coefficient becomes

$$d_\pi a_\pi(\beta, \alpha) = h^{-N_+} \left[P_W(i\partial_\xi) \widehat{\Psi}_{h,\alpha}(\xi) \right]_{\xi=h\nu_\pi} + O(e^{-c\beta}), \quad (\text{N24})$$

where

$$\Psi_{h,\alpha}(Y) := E_{h,\alpha}(Y) e^{-Q_{h,\alpha}(Y)}$$

is Weyl-invariant and holomorphic on $|Y| < r_{\text{ph}} h^{-1}$. Its Fourier transform on the local box,

$$\widehat{\Psi}_{h,\alpha}(\xi) := \int_{|Y| \leq r_* h^{-1}} \Psi_{h,\alpha}(Y) e^{-i\langle \xi, Y \rangle} dY,$$

extends holomorphically to $|\xi| \leq 2L_0 h^{-1}$. Indeed, if $|\xi| \leq 2L_0 h^{-1}$ and

$$\eta := \tau\xi, \quad \tau := \min \left\{ \frac{r_{\text{ph}}(G, \rho)}{8L_0}, \frac{1}{4C_{\text{ph}}(G, \rho)} \right\},$$

then $|\eta| \leq r_* h^{-1}$ and the contour shift $Y \mapsto Y + i\eta$ is admissible. Using Equations (N16) and (N18), the side faces contribute only $O(e^{-c\beta})$, while on the shifted box one gets

$$|\widehat{\Psi}_{h,\alpha}(\xi)| \leq C \exp\left(-(\tau - C_{\text{ph}}\tau^2)|\xi|^2\right).$$

After shrinking the coefficient, we may therefore record the Gaussian transform bound

$$|\hat{\Psi}_{h,\alpha}(\xi)| \leq C \exp(-2\sigma_{\text{sc}}(G, \rho)|\xi|^2) \quad (|\xi| \leq 2L_0 h^{-1}).$$

Because $\Psi_{h,\alpha}$ is Weyl-invariant, $P_W(i\partial_\xi)\hat{\Psi}_{h,\alpha}(\xi)$ is Weyl anti-invariant. By Theorem N.7, there exists a Weyl-invariant holomorphic function $M_{h,\alpha}$ on $|\xi| \leq 2L_0 h^{-1}$ such that

$$P_W(i\partial_\xi)\hat{\Psi}_{h,\alpha}(\xi) = P_W(\xi) M_{h,\alpha}(\xi). \quad (\text{N25})$$

Evaluating (N24) at $\xi = h\nu_\pi$ and using $h^{-N_+}P_W(h\nu_\pi) = P_W(\nu_\pi)$ together with Weyl's dimension formula (N10), we obtain (N21). Apply Theorem N.10 to the quotient above. After replacing $M_{h,\alpha}$ by the trivially normalized quotient furnished there (and absorbing the resulting scalar $1+O(e^{-c\beta})$ back into the tail term of (N21)), we may assume $P_W(\delta_W)M_{h,\alpha}(h\delta_W) = 1$ and Equation (N23). Then Equation (N23) gives

$$|a_\pi(\beta, \alpha)| \leq \exp(-\sigma_{\text{sc}}(G, \rho)h^2(|\nu_\pi|^2 - |\delta_W|^2)) + O(e^{-c\beta}).$$

Since

$$|\nu_\pi|^2 - |\delta_W|^2 = |\lambda_\pi|^2 + 2\langle \lambda_\pi, \delta_W \rangle \geq |\lambda_\pi|^2,$$

and the window (N19) implies $|\nu_\pi| \leq 2L_0 h^{-2}$, the tail term is smaller than $\exp(-2\sigma_{\text{sc}}L_0^2\beta)$ after shrinking σ_{sc} once more. Enlarging $\beta_{\text{sc}}(G, \rho)$ if necessary therefore absorbs the tail and yields (N20). $\square$

Corollary N.11 (Wilson-family entry and traced low-Casimir data). *The constants $\sigma_{\text{sc}}(G, \rho)$ and $\beta_{\text{sc}}(G, \rho)$ of Theorem N.9 depend only on the Wilson-family data*

$$\kappa_\pm(G), \ c_Z^\pm(G, \rho), \ c_{\text{hyp}}(G, \rho), \ r_{\text{hyp}}(G, \rho), \ r_{\text{ph}}(G, \rho), \ c_{\text{ph}}(G, \rho), \ C_{\text{ph}}(G, \rho), \ P_W(\delta_W),$$

all of which are internal to Appendix N.

Proof. Every constant entering the proof of Theorem N.9 was defined in Theorems N.1, N.2, N.5, N.6, N.8 and N.10. No external compact-group multiplier theorem is invoked anywhere in that proof. $\square$

Remark N.12 (Status of the low-Casimir node on the final public spine). The low-Casimir branch is now fully internal. Its load-bearing chain is: faithful Wilson local geometry, weak-coupling partition control, Weyl coefficient extraction, local denominator factorization

plus Weyl divisibility, complexified local phase control, explicit trivial-mode multiplier normalization, and the internal kernel-to-multiplier theorem Theorem N.9. The compact-simple A1 theorem therefore no longer contains any imported harmonic-analysis theorem.

Theorem N.13 (Low-Casimir Wilson coefficient estimate). *Let G be compact connected simple and let ρ be faithful. Then there exist constants $c_{\text{lo}}(G, \rho) > 0$ and $\beta_{\text{lo}}(G, \rho) \geq 1$ such that for all $\beta \geq \beta_{\text{lo}}(G, \rho)$, all $\alpha \in [1/2, 1]$, and all irreducibles π satisfying $C_2(\pi) \leq \beta^2$,*

$$|a_\pi(\beta, \alpha)| \leq \exp\left(-c_{\text{lo}}(G, \rho) \frac{C_2(\pi)}{\beta}\right). \quad (\text{N26})$$

Proof. Set

$$c_{\text{lo}}(G, \rho) := \frac{\sigma_{\text{sc}}(G, \rho)}{\kappa_+(G)}, \quad \beta_{\text{lo}}(G, \rho) := \max\{\beta_{\text{hyp}}(G, \rho), \beta_{\text{sc}}(G, \rho)\}.$$

If $C_2(\pi) \leq \beta^2$, then Theorem N.4 yields $|\lambda_\pi| \leq \kappa_-^{-1/2} \beta < 2\kappa_-^{-1/2} \beta$. Therefore Theorem N.9 implies

$$|a_\pi(\beta, \alpha)| \leq \exp\left(-\sigma_{\text{sc}}(G, \rho) \frac{|\lambda_\pi|^2}{\beta}\right).$$

Using again Theorem N.4, namely $C_2(\pi) \leq \kappa_+(G)|\lambda_\pi|^2$, gives exactly (N26). $\square$

Corollary N.14 (Low-Casimir constant chain). *The low-Casimir constants recorded in Appendix K are explicitly*

$$c_{\text{lo}}(G, \rho) = \frac{\sigma_{\text{sc}}(G, \rho)}{\kappa_+(G)}, \quad \beta_{\text{lo}}(G, \rho) = \max\{\beta_{\text{hyp}}(G, \rho), \beta_{\text{sc}}(G, \rho)\},$$

where $\sigma_{\text{sc}}(G, \rho)$ and $\beta_{\text{sc}}(G, \rho)$ are the fully internal constants returned by Theorem N.9.

Proof. This is exactly how $c_{\text{lo}}(G, \rho)$ and $\beta_{\text{lo}}(G, \rho)$ were defined in the proof of Theorem N.13. $\square$

4. Torus-strip estimates and high-Casimir decay

Lemma N.15 (Holomorphic torus extension of the Wilson weight with quadratic strip loss). *The torus restriction of A_ρ admits the finite Fourier presentation*

$$A_\rho(\theta) = 1 - \frac{1}{d_\rho} \sum_{\mu \in \text{Wt}(\rho)} m_\mu \cos\langle \mu, \theta \rangle, \quad \theta \in \mathfrak{t}/(2\pi\Gamma), \quad (\text{N27})$$

where $\text{Wt}(\rho)$ is the finite weight multiset of ρ and m_μ its multiplicities. Consequently

$$\tilde{A}_\rho(\zeta) := 1 - \frac{1}{d_\rho} \sum_{\mu \in \text{Wt}(\rho)} m_\mu \cos\langle \mu, \zeta \rangle, \quad \zeta \in \mathfrak{t}_\mathbb{C}/(2\pi\Gamma), \quad (\text{N28})$$

is a Weyl-invariant holomorphic extension of A_ρ . Moreover there exist constants $r_{\text{strip}}(G, \rho) > 0$ and $c_{\text{strip}}(G, \rho) < \infty$ such that for all $\zeta = \theta + iy$ with $|y| \leq r_{\text{strip}}(G, \rho)$,

$$\Re \tilde{A}_\rho(\theta + iy) \geq -c_{\text{strip}}(G, \rho) |y|^2. \quad (\text{N29})$$

Therefore the normalized density extends holomorphically to the same tube by

$$\tilde{f}_{\beta, \alpha}(\zeta) := Z_{\beta, \alpha}^{-1} e^{-\alpha \beta \tilde{A}_\rho(\zeta)},$$

and for every $0 < r \leq r_{\text{strip}}(G, \rho)$ one has the strip bound

$$\sup_{|\Im \zeta| \leq r} |\tilde{f}_{\beta, \alpha}(\zeta)| \leq (c_Z^-)^{-1} \beta^{d_G/2} \exp(c_{\text{strip}}(G, \rho) \beta r^2). \quad (\text{N30})$$

Proof. On the maximal torus, the character of ρ is the finite Fourier sum $\text{Tr}_\rho(e^{i\theta}) = \sum_\mu m_\mu e^{i\langle \mu, \theta \rangle}$, and taking real parts yields (N27). Formula (N28) is therefore the evident holomorphic continuation obtained by replacing the real cosine by the complex cosine.

For $\zeta = \theta + iy$,

$$\Re \cos\langle \mu, \theta + iy \rangle = \cos\langle \mu, \theta \rangle \cosh\langle \mu, y \rangle.$$

Hence

$$\Re \tilde{A}_\rho(\theta + iy) = A_\rho(\theta) - \frac{1}{d_\rho} \sum_\mu m_\mu \cos\langle \mu, \theta \rangle (\cosh\langle \mu, y \rangle - 1).$$

Using $|\cos\langle \mu, \theta \rangle| \leq 1$ and the finite weight set of ρ , choose $r_{\text{strip}} > 0$ so small that $\cosh u - 1 \leq 2u^2$ for $|u| \leq \max_{\mu \in \text{Wt}(\rho)} |\mu| r_{\text{strip}}$. Then

$$\Re \tilde{A}_\rho(\theta + iy) \geq -\frac{2}{d_\rho} \sum_\mu m_\mu |\langle \mu, y \rangle|^2 \geq -c_{\text{strip}}(G, \rho) |y|^2$$

for a suitable constant $c_{\text{strip}}(G, \rho)$. This proves (N29). Finally,

$$|\tilde{f}_{\beta, \alpha}(\zeta)| = Z_{\beta, \alpha}^{-1} \exp(-\alpha \beta \Re \tilde{A}_\rho(\zeta)) \leq (c_Z^-)^{-1} \beta^{d_G/2} e^{c_{\text{strip}} \beta r^2},$$

where we used $\alpha \leq 1$ and the partition-function lower bound from Theorem N.2. This is (N30). $\square$

Lemma N.16 (Strip Fourier bound for torus frequencies). *Let g be a $2\pi\Gamma$ -periodic holomorphic function on the tube*

$$\mathcal{T}_r := \{\zeta \in \mathfrak{t}_{\mathbb{C}}/(2\pi\Gamma) : |\Im \zeta| < r\}.$$

Then every Fourier coefficient $\widehat{g}(\nu)$ of g satisfies

$$|\widehat{g}(\nu)| \leq \sup_{\mathcal{T}_r} |g| e^{-r|\nu|} \quad (\nu \in \Gamma^*). \quad (\text{N31})$$

Proof. Write

$$\widehat{g}(\nu) = \frac{1}{\text{vol}(T)} \int_T g(\theta) e^{-i\langle \nu, \theta \rangle} d\theta.$$

For $\nu \neq 0$, shift the contour to $\theta + ir\nu/|\nu|$; periodicity and holomorphy justify the shift, and the exponential factor contributes $e^{-r|\nu|}$. The case $\nu = 0$ is immediate. $\square$

Lemma N.17 (Shifted torus integral bound for the Wilson family). *Let*

$$r_G := \text{rank}(G), \quad m_G := |\Phi_+| = \frac{\dim G - r_G}{2}.$$

There exist constants

$$C_{\text{shift}}(G, \rho) < \infty, \quad c_{\text{shift}}(G, \rho) < \infty, \quad \beta_{\text{shift}}(G, \rho) \geq 1,$$

with the following property. For every $\beta \geq \beta_{\text{shift}}(G, \rho)$, every $\alpha \in [1/2, 1]$, and every $y \in \mathfrak{t}$ with $|y| \leq r_{\text{strip}}(G, \rho)/4$,

$$\int_T |\tilde{f}_{\beta, \alpha}(\theta + iy) \tilde{\Delta}(\theta + iy)| d\theta \leq C_{\text{shift}}(G, \rho) (1 + \sqrt{\beta} |y|)^{m_G} \exp(c_{\text{shift}}(G, \rho) \beta |y|^2). \quad (\text{N32})$$

Proof. Choose torus coordinates $\theta \in \mathfrak{t}/(2\pi\Gamma)$ centered at the identity and shrink the local radius, still denoted r_{strip} , so that three standard local facts hold simultaneously on $|\theta| \leq r_{\text{strip}}$ and $|y| \leq r_{\text{strip}}$:

(i) the torus restriction of A_ρ has a nondegenerate quadratic minimum at 0, hence

$$\Re \tilde{A}_\rho(\theta + iy) \geq c_{\text{tor}}(G, \rho) |\theta|^2 - c'_{\text{tor}}(G, \rho) |y|^2; \quad (\text{N33})$$

(ii) away from the local chart one retains a positive action gap,

$$\Re \tilde{A}_\rho(\theta + iy) \geq \frac{a_0(G, \rho)}{2} - c_{\text{strip}}(G, \rho) |y|^2 \quad (|\theta| \geq r_{\text{strip}}); \quad (\text{N34})$$

(iii) the Weyl denominator satisfies the polynomial local bound

$$|\tilde{\Delta}(\theta + iy)| \leq C_{\Delta}(G) (|\theta| + |y|)^{m_G}. \quad (\text{N35})$$

Here (N33) is the torus restriction of the faithful local Wilson geometry, (N34) combines the positive gap away from the identity with continuity of the holomorphic extension, and (N35) is immediate from the product formula for Δ .

Split the torus integral into the local region $|\theta| \leq r_{\text{strip}}$ and its complement. On the complement, Equation (N34) together with the partition-function lower bound of Theorem N.2 gives

$$\int_{|\theta| \geq r_{\text{strip}}} |\tilde{f}_{\beta, \alpha}(\theta + iy) \tilde{\Delta}(\theta + iy)| d\theta \leq C \beta^{d_G/2} \exp\left(-\frac{a_0}{4}\beta + c_{\text{strip}}\beta|y|^2\right),$$

which is absorbed into the right-hand side of (N32) after enlarging the threshold because $|y| \leq r_{\text{strip}}/4$ and $(1 + \sqrt{\beta}|y|)^{m_G} \geq 1$.

On the local chart, use Equations (N33) and (N35) and again Theorem N.2:

$$\int_{|\theta| \leq r_{\text{strip}}} |\tilde{f}_{\beta, \alpha}(\theta + iy) \tilde{\Delta}(\theta + iy)| d\theta \leq C \beta^{d_G/2} \exp(c'_{\text{tor}}\beta|y|^2) \int_{|\theta| \leq r_{\text{strip}}} e^{-c_{\text{tor}}\beta|\theta|^2} (|\theta| + |y|)^{m_G} d\theta.$$

Rescale $\Theta := \sqrt{\beta}\theta$. Since $d_G = r_G + 2m_G$, the Jacobian cancellation is exactly

$$\beta^{d_G/2} d\theta = \beta^{m_G/2} d\Theta, \quad (|\theta| + |y|)^{m_G} = \beta^{-m_G/2} (|\Theta| + \sqrt{\beta}|y|)^{m_G}.$$

Therefore

$$\int_{|\theta| \leq r_{\text{strip}}} |\tilde{f}_{\beta, \alpha}(\theta + iy) \tilde{\Delta}(\theta + iy)| d\theta \leq C \exp(c'_{\text{tor}}\beta|y|^2) \int_{\mathbb{R}^{r_G}} e^{-c_{\text{tor}}|\Theta|^2} (|\Theta| + \sqrt{\beta}|y|)^{m_G} d\Theta.$$

The Gaussian moment integral is bounded by $C'(1 + \sqrt{\beta}|y|)^{m_G}$, so the local contribution is bounded by the right-hand side of (N32). Combining the local and off-chart estimates proves the lemma. $\square$

Corollary N.18 (Strip estimate for the Wilson coefficients). *There exists a constant $C_{\text{PW}}(G, \rho) < \infty$ such that for every $0 < r \leq r_{\text{strip}}(G, \rho)$, all $\beta \geq \beta_Z(G, \rho)$, all $\alpha \in [1/2, 1]$, and all irreducibles π ,*

$$|a_{\pi}(\beta, \alpha)| \leq C_{\text{PW}}(G, \rho) (\dim \pi)^{-1} \beta^{d_G/2} \exp\left(c_{\text{strip}}(G, \rho)\beta r^2 - r|\lambda_{\pi}|\right). \quad (\text{N36})$$

Proof. Apply Theorem N.3 with $F = f_{\beta, \alpha}$. The Fourier coefficient of $f_{\beta, \alpha}\Delta$ at $\lambda_{\pi} + \delta_W$ is therefore $d_{\pi}a_{\pi}(\beta, \alpha)$. By Theorem N.16, this coefficient is bounded by the strip sup of

$\tilde{f}_{\beta,\alpha}(\zeta)\tilde{\Delta}(\zeta)$ times $e^{-r|\lambda_\pi+\delta_W|}$. Because $\tilde{\Delta}$ is a fixed trigonometric polynomial, its strip sup on $|\Im \zeta| \leq r_{\text{strip}}$ is a finite constant depending only on G . Insert the strip bound (N30), divide by d_π , and use $|\lambda_\pi + \delta_W| \geq |\lambda_\pi|$. $\square$

Theorem N.19 (High-Casimir analytic Peter–Weyl decay). *Let G be compact connected simple and let ρ be faithful. There exist constants $c_{\text{hi}}(G, \rho) > 0$ and $\beta_{\text{hi}}(G, \rho) \geq 1$ such that for all $\beta \geq \beta_{\text{hi}}(G, \rho)$, all $\alpha \in [1/2, 1]$, and all irreducibles π with $C_2(\pi) \geq \beta^2$,*

$$|a_\pi(\beta, \alpha)| \leq \exp\left(-c_{\text{hi}}(G, \rho) \sqrt{C_2(\pi)}\right). \quad (\text{N37})$$

Proof. Choose once and for all

$$r_{\text{hi}}(G, \rho) := \min\left\{r_{\text{strip}}(G, \rho), \frac{1}{4c_{\text{strip}}(G, \rho)\sqrt{\kappa_+(G)}}\right\} > 0. \quad (\text{N38})$$

Applying Theorem N.18 with $r = r_{\text{hi}}$ gives

$$|a_\pi(\beta, \alpha)| \leq C_{\text{PW}}(G, \rho) \beta^{d_G/2} \exp(c_{\text{strip}} \beta r_{\text{hi}}^2 - r_{\text{hi}} |\lambda_\pi|).$$

If $C_2(\pi) \geq \beta^2$, then Theorem N.4 yields

$$|\lambda_\pi| \geq \kappa_+(G)^{-1/2} \sqrt{C_2(\pi)} \geq \kappa_+(G)^{-1/2} \beta.$$

By the choice of r_{hi} in (N38),

$$c_{\text{strip}} \beta r_{\text{hi}}^2 \leq \frac{r_{\text{hi}}}{4\sqrt{\kappa_+}} \beta \leq \frac{r_{\text{hi}}}{4} |\lambda_\pi|.$$

Therefore

$$|a_\pi(\beta, \alpha)| \leq C_{\text{PW}}(G, \rho) \beta^{d_G/2} \exp\left(-\frac{3}{4} r_{\text{hi}} |\lambda_\pi|\right).$$

Enlarge the threshold to $\beta_{\text{hi}}(G, \rho)$ so that for all $\beta \geq \beta_{\text{hi}}(G, \rho)$,

$$\log C_{\text{PW}}(G, \rho) + \frac{d_G}{2} \log \beta \leq \frac{r_{\text{hi}}}{4\sqrt{\kappa_+(G)}} \beta.$$

Since $|\lambda_\pi| \geq \beta/\sqrt{\kappa_+(G)}$ in the high-Casimir regime, the prefactor is absorbed and we get

$$|a_\pi(\beta, \alpha)| \leq \exp\left(-\frac{1}{2} r_{\text{hi}} |\lambda_\pi|\right).$$

Using again Theorem N.4, namely $|\lambda_\pi| \geq \kappa_+^{-1/2} \sqrt{C_2(\pi)}$, yields (N37) with

$$c_{\text{hi}}(G, \rho) := \frac{r_{\text{hi}}(G, \rho)}{2\sqrt{\kappa_+(G)}}.$$

$\square$

5. The compact-simple A1 theorem

Theorem N.20 (Compact-simple A1 theorem). *Let G be compact connected simple and let ρ be a finite-dimensional unitary Wilson representation. Then, after descent to $G_{\text{eff}} = G / \ker(\rho)$, there exist constants $c_{\text{A1}}(G, \rho) > 0$ and $\beta_{\text{A1}}(G, \rho) \geq 1$ such that for all $\beta \geq \beta_{\text{A1}}(G, \rho)$, all $\alpha \in [1/2, 1]$, and all nontrivial irreducible representations π of G_{eff} ,*

$$|a_\pi(\beta, \alpha)| \leq \exp\left(-c_{\text{A1}}(G, \rho) \frac{C_2(\pi)}{\beta + \sqrt{C_2(\pi)}}\right). \quad (\text{N39})$$

Proof. By the faithful-reduction discussion above, it suffices to treat the faithful descended representation. If $C_2(\pi) \leq \beta^2$, then Theorem N.13 gives

$$|a_\pi(\beta, \alpha)| \leq \exp\left(-c_{\text{lo}}(G, \rho) \frac{C_2(\pi)}{\beta}\right).$$

Since in this regime $\beta + \sqrt{C_2(\pi)} \leq 2\beta$, the right-hand side is bounded by

$$\exp\left(-2c_{\text{lo}}(G, \rho) \frac{C_2(\pi)}{\beta + \sqrt{C_2(\pi)}}\right).$$

If instead $C_2(\pi) \geq \beta^2$, then Theorem N.19 gives

$$|a_\pi(\beta, \alpha)| \leq \exp\left(-c_{\text{hi}}(G, \rho) \sqrt{C_2(\pi)}\right).$$

Because $\beta + \sqrt{C_2(\pi)} \leq 2\sqrt{C_2(\pi)}$ in this regime, this is at most

$$\exp\left(-2c_{\text{hi}}(G, \rho) \frac{C_2(\pi)}{\beta + \sqrt{C_2(\pi)}}\right).$$

Choose

$$c_{\text{A1}}(G, \rho) := \min\{2c_{\text{lo}}(G, \rho), 2c_{\text{hi}}(G, \rho)\}$$

and let $\beta_{\text{A1}}(G, \rho)$ dominate the thresholds from the two regime theorems. This yields (N39), which is exactly the statement used on the primary Route 1 spine. $\square$

Corollary N.21 (Public compact-simple group scope). *Every place on the Route 1 / Lane A / Lane B proof spine that previously invoked the classical-family theorem Theorem 1.4 may now invoke the compact-simple theorem Theorem N.20 instead. Consequently the ultraviolet theorem component of the manuscript is valid for every compact connected simple gauge group for any chosen auxiliary Wilson regularization ρ . Appendix O later upgrades this auxiliary scope statement to the group-only theorem level by proving universality in the faithful choice of ρ and descent for nonfaithful ρ .*

Proof. The rest of the manuscript uses the one-plaquette sector only through the crossover constants $c_{A1}(G, \rho)$ and $\beta_{A1}(G, \rho)$ attached to the chosen auxiliary Wilson regularization. Those constants are supplied for arbitrary compact connected simple G by Theorem N.20; Component 10 later upgrades the continuum local theory to faithful- ρ universality, but the explicit ultraviolet constant schedule remains indexed by the chosen faithful regularization. $\square$

Remark N.22 (Status of the classical appendices on the public theorem component). Appendices G–I are no longer load-bearing for the public group scope of the paper. They remain useful for two reasons: (i) they provide direct worked proofs of the compact-simple A1 bound in the standard Sp/Spin representations used in the earlier branches, and (ii) they keep the determinantal/total-positivity mechanism available explicitly in the classical series.

DATA AVAILABILITY STATEMENT

Data sharing is not applicable to this article as no new data were created or analyzed in this study.

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